

Project Data Page:





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DESIGN CALCULATION

In Accordance with ASME Section VIII Division 2

ASME Code Version : 2025

Analysis Performed by : SL User

Job File : ...\\122D1 & 163-D0peRev-06.pvdb

Date of Analysis : Aug 25, 2026 5:17pm

PV Elite 28 U2, April 2026

Vessel Design Summary:

ASME Code, Section VIII Division 2, 2025

Diameter Spec : 6700.000 x 6706.000 x 7100.000 mm ID
 Vessel Design Length, Tangent to Tangent 44200.00 mm

Distance of Bottom Tangent above Grade 7500.00 mm
 Distance of Base above Grade 300.00 mm
 Specified Datum Line Distance 7200.00 mm

Internal Design Temperature 130 °C
 Internal Design Pressure 0.380 MPa

External Design Temperature 130 °C
 External Design Pressure 0.052 MPa

Maximum Allowable Working Pressure 0 MPa
 External Max. Allowable Working Pressure 0.1 MPa
 Shop Test Pressure 0.602 MPa

Required Minimum Design Metal Temperature -12.0 °C
 Warmest Computed Minimum Design Metal Temperature -12.0 °C

Wind Design Code ASCE/SEI 7-16
 Earthquake Design Code ASCE/SEI 7-16

Materials of Construction:

Component Type	Material	Class	Thickness	UNS #	Material Treatment Options
Shell	SA-516 70	...	...	K02700	+IT
Head	SA-516 70	...	...	K02700	+IT
Head	SA-516 70	...	...	K02700	+IT
Skirt	SA-516 70	...	...	K02700	
Nozzle	SA-266 2	...	...	K03506	+IT
Nozzle	SA-516 70	...	...	K02700	+IT
Nozzle	SA-105	...	...	K03504	+IT
Nozzle	SA-106 B	...	...	K03006	+IT
Re-Pad	SA-516 70	...	...	K02700	
Nozzle Flg	SA-105	...	...	K03504	+IT
Rings	SA-516 70	...	...	K02700	+IT
Basering	SA-516 70	...	...	K02700	
Base Bolting	SA-36	...	...	K02600	+IT

+FG - Fine Grained, +R - rolled, +T - Tempered, +N - Normalized
 +Q - quenched, +CBV - Cb & V added, +IT - Impact tested

Element Pressures and MAWP (MPa & mm):

Element Description or Type	Design Pressure + Stat. head	Ext. Press.	Element M.A.W.P	Total Corrosion Allowance	Str. Flg. Gov.	In Creep Range
Skirt-3	...	...	...	1.6000	N/A	No
Skirt-2	...	...	...	1.6000	N/A	No
Skirt-1	...	...	...	1.6000	N/A	No
163-D- BTM dish	4.208	0.00	5.000	6.0000	N/A	No
163-D-Shell-1	4.173	0.00	4.000	6.0000	N/A	No
163-D-Shell-2	4.142	0.00	4.000	6.0000	N/A	No
163-D-Shell-3	4.111	0.00	4.000	6.0000	N/A	No
163-D-Shell-4	4.105	0.00	4.000	1.0000	N/A	No
163-D-Shell-5	4.105	0.00	4.000	1.0000	N/A	No
163-D-Shell-6	4.105	0.00	4.000	1.0000	N/A	No
163-D-Shell-7	4.105	0.00	4.000	1.0000	N/A	No
163-D-Top head	4.105	0.00	5.000	1.0000	N/A	No



Int skirt 1	...	...	...	1.6000	N/A	No
Int skirt 2	...	...	...	1.6000	N/A	No
Int skirt 3	...	...	...	1.6000	N/A	No
Int skirt 4	...	...	...	1.6000	N/A	No
122-D1-Btm head	0.470	0.05	1.000	6.0000	No	No
122-D1-Shell-1	0.451	0.05	1.000	6.0000	N/A	No
122-D1-Shell-2	0.420	0.05	1.000	6.0000	N/A	No
122-D1-Shell-3	0.389	0.05	1.000	6.0000	N/A	No
122-D1-Shell-4	0.388	0.05	1.000	6.0000	N/A	No
122-D1-Shell-5	0.388	0.05	1.000	6.0000	N/A	No
122-D1-Shell-6	0.388	0.05	1.000	1.0000	N/A	No
122-D1-Shell-7	0.388	0.05	1.000	1.0000	N/A	No
122-D1-Shell-8	0.388	0.05	0.000	1.0000	N/A	No
122-D1-Shell-9	0.388	0.05	0.000	1.0000	N/A	No
122-D1-Top head	0.388	0.05	1.000	1.0000	No	No

Liquid Level: 17894.00 mm Dens.: 0.001 kg/cm³ Sp. Gr.: 1.060

Stiffener Ring Specifications:

Elevation mm	Selected Type	Ring Description
28150.05	Bar 300.0 x 28.0	3
30900.05	Bar 300.0 x 28.0	Ring R2
33400.05	Bar 250.0 x 28.0	Ring R3
35899.95	Bar 250.0 x 28.0	Ring R4
38399.95	Bar 250.0 x 28.0	Ring R5
40899.95	Bar 250.0 x 28.0	Ring R6
43399.95	Bar 250.0 x 28.0	Ring R7
45899.96	Bar 250.0 x 28.0	Ring R8
48399.96	Bar 250.0 x 28.0	Ring R9
50900.00	Bar 250.0 x 28.0	Ring R10

Element Types and Properties:

Element Type	"To" Elev mm	Element Length mm	Nominal Thickness mm	Finished Thickness mm	Reqd Thk Internal mm	Reqd Thk External mm	Long Eff	Circ Eff
Skirt	-4200.0	3000.0	42.0	42.0	...	...	0.70	0.70
Skirt	-1500.0	2700.0	36.0	36.0	...	...	0.70	0.70
Skirt	0.0	1500.0	42.0	42.0	...	...	0.70	0.55
Sphere	0.0	0.0	68.0	60.0	51.7	...	1.00	1.00
Cylinder	3000.0	3000.0	98.0	98.0	96.8	...	1.00	1.00
Cylinder	6000.0	3000.0	98.0	98.0	96.1	...	1.00	1.00
Cylinder	7900.0	1900.0	96.0	96.0	95.4	...	1.00	1.00
Cylinder	10900.0	3000.0	92.0	92.0	90.2	...	1.00	1.00
Cylinder	13900.0	3000.0	92.0	92.0	90.2	...	1.00	1.00
Cylinder	15900.0	2000.0	92.0	92.0	90.2	...	1.00	1.00
Cylinder	17400.0	1500.0	92.0	92.0	90.2	...	1.00	1.00
Sphere	17400.0	0.0	58.0	52.0	45.5	...	1.00	1.00
Skirt	18650.0	1250.0	26.0	26.0	...	...	1.00	1.00
Skirt	21650.0	3000.0	24.0	24.0	...	...	1.00	1.00
Skirt	24650.0	3000.0	24.0	24.0	...	...	1.00	1.00
Skirt	25900.0	1250.0	24.0	24.0	...	...	1.00	1.00
Ellipse	25950.0	50.0	44.0	36.0	16.8	...	1.00	1.00
Cylinder	28950.0	3000.0	24.0	24.0	16.3	...	1.00	1.00
Cylinder	31950.0	3000.0	24.0	24.0	15.6	...	1.00	1.00
Cylinder	34950.0	3000.0	22.0	22.0	14.9	...	1.00	1.00
Cylinder	37950.0	3000.0	22.0	22.0	14.8	...	1.00	1.00
Cylinder	40750.0	2800.0	20.0	20.0	14.8	...	1.00	1.00
Cylinder	43750.0	3000.0	16.0	16.0	11.4	...	0.85	0.85
Cylinder	46750.0	3000.0	16.0	16.0	11.4	...	0.85	0.85
Cylinder	49750.0	3000.0	16.0	16.0	12.8	...	0.85	0.85
Cylinder	52650.0	2900.0	14.0	14.0	11.4	...	0.85	0.85
Ellipse	52700.0	50.0	20.0	16.0	10.8	...	0.85	0.85

External Pressure Calculations:

From	To	External Actual T.	External Required T.	External Design Pressure	External M.A.W.P.
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		mm	mm	MPa	MPa
10	20	...	No Calc	...	No Calc
20	30	...	No Calc	...	No Calc
30	40	...	No Calc	...	No Calc
40	50	60	No Calc	...	2.60545
50	60	98	No Calc	...	1.45355
60	70	98	No Calc	...	1.45355
70	80	96	No Calc	...	1.38435
80	90	92	No Calc	...	1.42762
90	100	92	No Calc	...	1.42762
100	110	92	No Calc	...	1.42762
110	120	92	No Calc	...	1.42762
120	130	52	No Calc	...	2.41201
130	140	...	No Calc	...	No Calc
140	150	...	No Calc	...	No Calc
150	160	...	No Calc	...	No Calc
160	170	...	No Calc	...	No Calc
170	180	36	17.9883	0.051733	0.31953
180	Ring	24	17.1675	0.051733	0.17017
Ring	190	24	17.0156	0.051733	0.17573
190	Ring	24	17.0156	0.051733	0.17573
Ring	200	24	16.5959	0.051733	0.19249
200	Ring	22	16.5921	0.051733	0.14517
Ring	210	22	16.5922	0.051733	0.14517
210	Ring	22	16.5922	0.051733	0.14517
Ring	220	22	16.5921	0.051733	0.14517
220	Ring	20	16.5885	0.051733	0.10438
Ring	230	20	16.5885	0.051733	0.10438
230	Ring	16	11.5816	0.051733	0.12413
Ring	Ring	16	11.5816	0.051733	0.12413
Ring	240	16	11.5818	0.051733	0.12413
240	Ring	16	11.5818	0.051733	0.12413
Ring	250	16	11.5816	0.051733	0.12413
250	Ring	16	11.5816	0.051733	0.12413
Ring	260	16	11.5818	0.051733	0.12413
260	Ring	14	11.5785	0.051733	0.086916
Ring	270	14	11.3855	0.051733	0.091013
270	280	16	12.9865	0.051733	0.080995

External Pressure Calculations:

From	To	Actual Length Bet. Stiffeners mm	Allowable Length Bet. Stiffeners mm	Ring Inertia Required mm^4	Ring Inertia Available mm^4
10	20	15700	No Calc	No Calc	No Calc
20	30	15700	No Calc	No Calc	No Calc
30	40	15700	No Calc	No Calc	No Calc
40	50	No Calc	No Calc	No Calc	No Calc
50	60	19646.3	No Calc	No Calc	No Calc
60	70	19646.3	No Calc	No Calc	No Calc
70	80	19646.3	No Calc	No Calc	No Calc
80	90	19646.3	No Calc	No Calc	No Calc
90	100	19646.3	No Calc	No Calc	No Calc
100	110	19646.3	No Calc	No Calc	No Calc
110	120	19646.3	No Calc	No Calc	No Calc
120	130	No Calc	No Calc	No Calc	No Calc
130	140	15700	No Calc	No Calc	No Calc
140	150	15700	No Calc	No Calc	No Calc
150	160	15700	No Calc	No Calc	No Calc
160	170	15700	No Calc	No Calc	No Calc
170	180	No Calc	No Calc	No Calc	No Calc
180	Ring	2841.72	8896.97	No Calc	No Calc
Ring	190	2750	8896.93	123852144	160482688
190	Ring	2750	8896.93	No Calc	No Calc
Ring	200	2500	8897.1	124300688	160482688
200	Ring	2500	6718.92	No Calc	No Calc
Ring	210	2499.9	6719.17	91227984	93478560



210	Ring	2499.9	6719.17	No Calc	No Calc
Ring	220	2500	6718.94	91227968	93478560
220	Ring	2500	4886.65	No Calc	No Calc
Ring	230	2499.99	4886.55	65035676	86572312
230	Ring	2499.99	5771.89	No Calc	No Calc
Ring	240	2500	5771.9	77300416	90059568
Ring	240	2500.01	5771.92	77300392	90059568
240	Ring	2500.01	5771.92	No Calc	No Calc
Ring	250	2500	5771.9	77300416	90059568
250	Ring	2500	5771.9	No Calc	No Calc
Ring	260	2500.04	5771.99	77300368	90059568
260	Ring	2500.04	4102.43	No Calc	No Calc
Ring	270	2391.67	4102.41	53860156	82931680
270	280	No Calc	No Calc	No Calc	No Calc

Local Stress Analysis Results:

Description	Analysis Type	Max Stress Ratio	High Stress Location	Pass Fail
D 36"	WRC-107/537	0.848	n/a	Passed
A 36"	WRC-297	0.978	Shell	Passed
E 8"	WRC-107/537	0.873	n/a	Passed
C 48"	WRC-107/537	0.574	n/a	Passed
B 36"	WRC-297	0.936	Nozzle	Passed
K 6"	WRC-297	0.550	Shell	Passed
I 28"	WRC-297	0.882	Shell	Passed
H1 42"	WRC-297	0.872	Nozzle	Passed
H2 42"	WRC-297	0.872	Nozzle	Passed
F 42"	WRC-107/537	0.549	n/a	Passed

Wind/Earthquake Shear, Bending:

From	To	Distance to Support mm	Cumulative Wind Shear N	Earthquake Shear N	Wind Bending N-mm	Earthquake Bending N-mm
10	20	1500	307612	1879828	10.33E+09	76.57E+09
20	30	4350	295994	1879167	9.424E+09	70.93E+09
30	40	6450	284567	1877391	8.640E+09	65.86E+09
40	50	7199.98	278758	1875688	8.217E+09	63.04E+09
50	60	8700	278758	1853003	8.217E+09	63.04E+09
60	70	11700	264950	1817505	7.401E+09	57.53E+09
70	80	14150	250868	1768895	6.627E+09	52.15E+09
80	90	16600	240784	1740310	6.160E+09	48.82E+09
90	100	19600	229093	1705204	5.455E+09	43.65E+09
100	110	22100	214950	1674998	4.789E+09	38.57E+09
110	120	23850	204526	1650475	4.369E+09	35.25E+09
120	130	24600	198680	1618236	4.066E+09	32.79E+09
130	140	25225	195908	1585299	4.066E+09	32.79E+09
140	150	27350	191064	1580529	3.824E+09	30.82E+09
150	160	30350	179571	1561586	3.268E+09	26.10E+09
160	170	32475	167889	1549543	2.747E+09	21.43E+09
170	180	33125	160305	1539340	2.542E+09	19.50E+09
180	190	34650	160103	1396880	2.534E+09	19.43E+09
190	200	37650	145430	1206984	2075159808	15.52E+09
200	210	40650	130693	1019103	1660807680	12.18E+09
210	220	43650	118580	876459	1286747264	9.335E+09
220	230	46550	103977	731587	952777152	6.922E+09
230	240	49450	90125.4	623324	680923904	5.024E+09
240	250	52450	78032.6	534504	428584800	3.287E+09
250	260	55450	63481.3	500020	216228064	1734147072
260	270	58400	30094.1	223511	75808312	648412480
270	280	59875	14177.1	142938	11589157	116845904

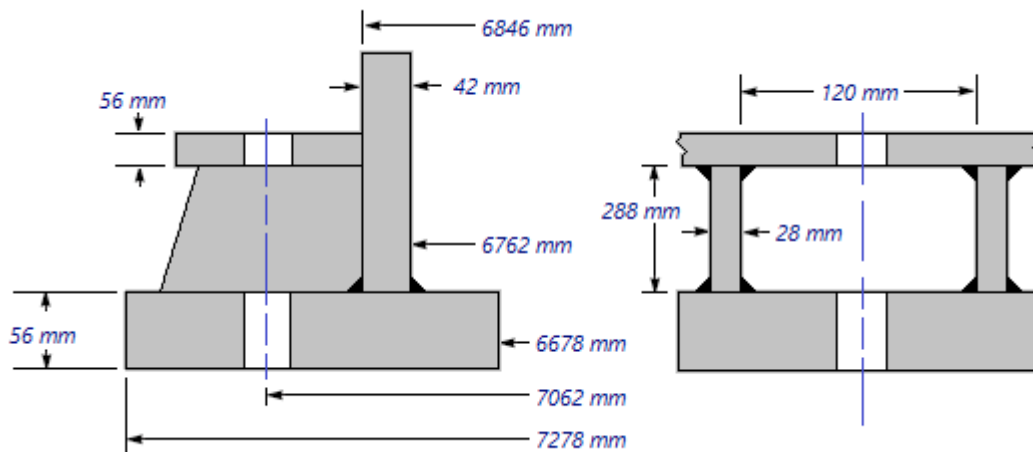
Abs Max of the all of the Stress Ratio's : 0.9311



Basering Data : Continuous Top Ring W/Gussets

Thickness of Basering	56.0000	mm
Inside Diameter of Basering	6678.0000	mm
Outside Diameter of Basering	7278.0000	mm
Nominal Diameter of Bolts	64.0000	mm
Diameter of Bolt Circle	7062.0000	mm
Number of Bolts	48	
Thickness of Gusset Plates	28.0000	mm
Average Width of Gusset Plates	206.0000	mm
Height of Gussets	288.0000	mm
Distance between Gussets	120.0000	mm
Thickness of Top Plate or Ring	56.0000	mm
Radial Width of the Top Plate	216.0000	mm
Number of Gussets	96	

Basering Sketch for Cylindrical Skirts:



Basering Cross Section View

Loads for Foundation/Support Design:

Total Wind Shear on Support	409470.	N
Total Earthquake Shear on Support	1981687.	N
Wind Moment on Support	22335354880.	N-mm
Earthquake Moment on Support	88577687552.	N-mm
Maximum Initial Bolt Load	659911.	N

Wind and Earthquake moments include the effects of user defined forces and moments if any exist in the job and were specified to act (compute loads and stresses) during these cases. Also included are moment effects due to eccentric weights if any are present in the input.

Weights:

Fabricated - Bare w/o removable internals	682287.5	kgm
Shop Test - Fabricated + water (full)	2637955.8	kgm
Shipping - Fab. + rem. intls.+ shipping app.	868768.2	kgm
Erected - Fab. + rem. intls.+ insul. (etc)	897988.2	kgm
Empty - Fab. + intls. + details + wghts.	897988.2	kgm
Operating - Empty + operating liquid (no ca)	1933890.0	kgm
Field Test - Empty weight + test medium (full)	2760660.0	kgm

Nozzle Calculation Summary (MPa & mm):

Description	MAWP	Ext	MAPNC	Tbl 4.5.2	Weld	Pass/Fail
D 36"	4	...	...	OK	OK	Passed
L12 2"	4	...	...	OK	OK	Passed
L14 1.5"	4	...	...	OK	OK	Passed
MH-6 24"	4	...	...	...	OK	Passed
L11 2"	4	...	...	OK	OK	Passed
L13 1.5"	4	...	...	OK	OK	Passed
UC2 24"	4	...	...	OK	OK	Passed
MH-5 24"	4	...	...	...	OK	Passed
MH-4 24"	4	...	...	...	OK	Passed
A 36"	4	...	...	OK	OK	Passed
E 8"	5	...	...	OK	OK	Passed
C 48"	1	OK	...	OK	OK	Passed
MH-3 24"	1	OK	...	...	OK	Passed
L2 2"	55	OK	...	OK	OK	Passed
L4 2"	55	OK	...	OK	OK	Passed
L6 2"	55	OK	...	OK	OK	Passed
L8 2"	55	OK	...	OK	OK	Passed
L10 2"	55	OK	...	OK	OK	Passed
B 36"	1	OK	...	OK	OK	Passed
K 6"	24	OK	...	OK	OK	Passed
I 28"	1	OK	...	OK	OK	Passed
L1 2"	55	OK	...	OK	OK	Passed
L3 2"	55	OK	...	OK	OK	Passed
L5 2"	55	OK	...	OK	OK	Passed
L7 2"	55	OK	...	OK	OK	Passed
L9 2"	55	OK	...	OK	OK	Passed
MH-2 24"	5	OK	...	...	OK	Passed
UC1 24"	1	OK	...	OK	OK	Passed
MH-1 24"	5	OK	...	...	OK	Passed
S1 1"	42	OK	...	OK	OK	Passed
H1 42"	1	OK	...	OK	OK	Passed
H2 42"	1	OK	...	OK	OK	Passed
F 42"	1	OK	...	OK	OK	Passed

Nozzle MAWP Summary:

Minimum MAWP Nozzles : 1 Nozzle : F 42"
 Minimum MAWP Shells/Flanges : 0 Element : 122-D1-Shell-9
 Minimum MAPnc Shells/Flanges : 1 Element : 122-D1-Shell-8

Computed Vessel M.A.W.P. : 0 MPa

Note: MAWPs (Internal Case) shown above are at the High Point.

VIII-2 Nozzle to Gross Structural Discontinuity Check:

- Error: Nozzle UC2 24", was too close to a gross structural discontinuity.
- Error: Nozzle A 36", was too close to a gross structural discontinuity.

Please see ASME VIII-2 Section 4.5.13.1 for more information.

Multiple output lines for the same nozzle indicates required Code calculations in both the longitudinal and circumferential planes of reinforcement where applicable.

Check the spatial relationship between the nozzles:

From Node	Nozzle Description	Y Coordinate mm	Layout Angle deg	Inside Radius mm
40	D 36"	0.000	0.000	463.200
50	L12 2"	150.001	0.000	31.400
50	L14 1.5"	150.001	30.000	25.050
60	MH-6 24"	4399.976	225.000	304.800
70	L11 2"	6750.005	0.000	31.400
70	L13 1.5"	6750.005	30.000	25.050
70	UC2 24"	6899.995	90.000	310.800



90	MH-5 24"	12399.976	225.000	304.800
100	MH-4 24"	14899.980	225.000	304.800
110	A 36"	16649.982	180.000	463.200
120	E 8"	0.000	0.000	88.600
170	C 48"	0.000	0.000	599.600
190	MH-3 24"	29550.053	225.000	294.800
200	L2 2"	33249.953	0.000	28.400
200	L4 2"	33249.953	30.000	28.400
200	L6 2"	33249.953	60.000	28.400
200	L8 2"	33249.953	90.000	28.400
200	L10 2"	33249.953	120.000	28.400
200	B 36"	34250.055	180.000	439.200
200	K 6"	34450.059	0.000	71.180
210	I 28"	36450.051	0.000	361.600
220	L1 2"	39349.961	0.000	28.400
220	L3 2"	39349.961	30.000	28.400
220	L5 2"	39349.961	60.000	28.400
220	L7 2"	39349.961	90.000	28.400
220	L9 2"	39349.961	120.000	28.400
230	MH-2 24"	42850.055	225.000	294.800
240	UC1 24"	43949.957	90.000	310.800
240	MH-1 24"	45149.961	225.000	294.800
250	S1 1"	47949.957	225.000	18.700
250	H1 42"	47550.055	180.000	513.400
250	H2 42"	47550.055	0.000	513.400
270	F 42"	0.000	0.000	515.400

The nozzle spacing is computed by the following:

$$= \sqrt{Il^2 + Ic^2} \text{ where}$$

Il - Arc length along the inside vessel surface in the long. direction.

Ic - Arc length along the inside vessel surface in the circ. direction

If any interferences/violations are found, they will be noted below.

4.5.13 spacing violation detected between : K 6" and L2 2".

The Centerline Dist. is 1200.10 while the sum of the inside Radii is 1499.16 mm.

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Nozzle Schedule:

Description	Nominal or Actual Size	Schd or FVC Type	Flg Type	Nozzle O/Dia in	Wall Thk mm	Reinforcing Pad Diameter	Pad Thk mm	Cut Length mm	Flg Class
S1 1"	1.000 in	Actual	LWN	2.000	12.700	...	...	230.40	150
L14 1.5"	1.500 in	Actual	WNF	2.287	10.000	...	...	348.71	600
L13 1.5"	2.000 in	LWN	WNF	2.287	10.000	...	...	346.71	600
L12 2"	2.000 in	Actual	WNF	2.945	12.000	...	...	295.70	600
L11 2"	2.000 in	LWN	WNF	2.945	12.000	...	...	293.70	600
L2 2"	2.000 in	Actual	LWN	3.320	16.764	...	...	228.63	300
L4 2"	2.000 in	Actual	LWN	3.320	16.764	...	...	228.63	300
L6 2"	2.000 in	Actual	LWN	3.320	16.764	...	...	228.63	300
L8 2"	2.000 in	Actual	LWN	3.320	16.764	...	...	228.63	300
L10 2"	2.000 in	Actual	LWN	3.320	16.764	...	...	228.63	300
L1 2"	2.000 in	Actual	LWN	3.320	16.764	...	...	226.63	300
L3 2"	2.000 in	Actual	LWN	3.320	16.764	...	...	226.63	300
L5 2"	2.000 in	Actual	LWN	3.320	16.764	...	...	226.63	300
L7 2"	2.000 in	Actual	LWN	3.320	16.764	...	...	226.63	300
L9 2"	2.000 in	Actual	LWN	3.320	16.764	...	...	226.63	300
K 6"	6.000 in	160	WNF	6.625	18.237	322.40	18.00	233.00	150
E 8"	8.000 in	Actual	WNF	8.000	16.000	...	...	314.89	600
MH-6 24"	24.000 in	Actual	WNF	25.260	16.000	...	...	431.94	600
UC2 24"	24.000 in	Actual	WNF	25.260	16.000	...	...	620.55	600
MH-5 24"	24.000 in	Actual	WNF	25.260	16.000	...	...	425.92	600
MH-4 24"	24.000 in	Actual	WNF	25.260	16.000	...	...	425.92	600
MH-3 24"	24.000 in	Actual	WNF	24.000	16.000	1209.60	20.00	347.11	150
MH-2 24"	24.000 in	Actual	WNF	24.000	16.000	1209.60	12.00	339.11	150
UC1 24"	24.000 in	Actual	WNF	25.260	16.000	1041.60	20.00	338.52	150
MH-1 24"	24.000 in	Actual	WNF	24.000	16.000	1209.60	16.00	339.11	150
I 28"	28.000 in	Actual	WNF	29.732	22.000	1395.20	22.00	344.14	150
D 36"	36.000 in	Actual	WNF	37.575	20.000	...	...	309.81	600
A 36"	36.000 in	Actual	WNF	37.575	20.000	...	...	505.61	600
B 36"	36.000 in	Actual	WNF	36.000	24.000	1614.40	18.00	365.56	150
H1 42"	42.000 in	Actual	WNF	42.000	26.000	1766.80	16.00	376.30	150
H2 42"	42.000 in	Actual	WNF	42.000	26.000	1824.40	16.00	376.30	150
F 42"	42.000 in	Actual	WNF	42.000	24.000	2126.80	16.00	538.26	150
C 48"	48.000 in	Actual	WNF	48.000	16.000	2439.20	24.00	375.01	150

General Notes for the above table:

The cut length is the outside projection + inside projection + drop + in-plane shell thickness. This value does not include weld gaps, nor does it account for shrinkage.

In the case of oblique nozzles, the outside diameter must be increased. The re-pad width around the nozzle is calculated as follows:

Width of Pad = (Pad Outside Dia. (per above) - Nozzle Outside Dia.)/2

For hub nozzles, the thickness and diameter shown are those of the smaller and thinner section.

Nozzle Material and Weld Fillet Leg Size Details (mm):

Description	Material	Shl Grve Weld	Noz Shl/Pad Weld	Pad OD Weld	Pad Grve Weld	Inside Weld
S1 1"	SA-105	12.000	16.000	...	...	...
L14 1.5"	SA-266 2	96.000	10.000	...	...	...
L13 1.5"	SA-266 2	96.000	10.000	...	...	...
L12 2"	SA-266 2	96.000	10.000	...	...	...
L11 2"	SA-266 2	96.000	10.000	...	...	...
L2 2"	SA-105	16.000	16.000	...	...	...
L4 2"	SA-105	16.000	16.000	...	...	...
L6 2"	SA-105	16.000	16.000	...	...	...
L8 2"	SA-105	16.000	16.000	...	...	...
L10 2"	SA-105	16.000	16.000	...	...	...
L1 2"	SA-105	16.000	16.000	...	...	...
L3 2"	SA-105	16.000	16.000	...	...	...
L5 2"	SA-105	16.000	16.000	...	...	...



L7 2"	SA-105	16.000	16.000	...	...	...
L9 2"	SA-105	16.000	16.000	...	...	...
K 6"	SA-106 B	16.000	16.000	16.000	18.000	...
E 8"	SA-266 2	14.500	20.000	...	...	...
MH-6 24"	SA-266 2	94.000	10.000	...	...	...
UC2 24"	SA-516 70	96.000	10.000	...	...	...
MH-5 24"	SA-266 2	92.000	10.000	...	...	...
MH-4 24"	SA-266 2	92.000	10.000	...	...	...
MH-3 24"	SA-516 70	16.000	16.000	16.000	20.000	...
MH-2 24"	SA-516 70	12.000	16.000	12.000	12.000	...
UC1 24"	SA-516 70	18.000	10.000	18.000	20.000	...
MH-1 24"	SA-516 70	14.000	16.000	16.000	16.000	...
I 28"	SA-516 70	22.000	10.000	18.000	22.000	...
D 36"	SA-266 2	53.000	10.000	...	...	...
A 36"	SA-266 2	10.000	10.000	...	...	...
B 36"	SA-516 70	16.000	16.000	16.000	18.000	...
H1 42"	SA-516 70	16.000	16.000	16.000	16.000	...
H2 42"	SA-516 70	16.000	16.000	16.000	16.000	...
F 42"	SA-516 70	14.000	16.000	16.000	16.000	...
C 48"	SA-516 70	16.000	16.000	16.000	24.000	...

Note: The Outside projections below do not include the flange thickness.

Nozzle Miscellaneous Data:

Description	Elev/Distance From Datum mm	Layout Angle deg	Proj Outside mm	Proj Inside mm	Installed in Component
S1 1"	47949.953	225.0	214.31	0.00	122-D1-Shell-8
L14 1.5"	150.001	30.0	250.00	0.00	163-D-Shell-1
L13 1.5"	6750.005	30.0	250.00	0.00	163-D-Shell-3
L12 2"	150.001	0.0	196.85	0.00	163-D-Shell-1
L11 2"	6750.005	0.0	196.85	0.00	163-D-Shell-3
L2 2"	33249.953	0.0	206.38	0.00	122-D1-Shell-3
L4 2"	33249.953	30.0	206.38	0.00	122-D1-Shell-3
L6 2"	33249.953	60.0	206.38	0.00	122-D1-Shell-3
L8 2"	33249.953	90.0	206.38	0.00	122-D1-Shell-3
L10 2"	33249.953	120.0	206.38	0.00	122-D1-Shell-3
L1 2"	39349.961	0.0	206.38	0.00	122-D1-Shell-5
L3 2"	39349.961	30.0	206.38	0.00	122-D1-Shell-5
L5 2"	39349.961	60.0	206.38	0.00	122-D1-Shell-5
L7 2"	39349.961	90.0	206.38	0.00	122-D1-Shell-5
L9 2"	39349.961	120.0	206.38	0.00	122-D1-Shell-5
K 6"	34450.055	0.0	210.00	0.00	122-D1-Shell-3
E 8"	...	0.0	260.00	0.00	163-D-Top head
MH-6 24"	4399.976	225.0	310.00	0.00	163-D-Shell-2
UC2 24"	6899.995	90.0	500.00	0.00	163-D-Shell-3
MH-5 24"	12399.976	225.0	310.00	0.00	163-D-Shell-5
MH-4 24"	14899.980	225.0	310.00	0.00	163-D-Shell-6
MH-3 24"	29550.053	225.0	310.00	0.00	122-D1-Shell-2
MH-2 24"	42850.051	225.0	310.00	0.00	122-D1-Shell-6
UC1 24"	43949.957	90.0	300.00	0.00	122-D1-Shell-7
MH-1 24"	45149.957	225.0	310.00	0.00	122-D1-Shell-7
I 28"	36450.051	0.0	300.00	0.00	122-D1-Shell-4
D 36"	...	0.0	200.00	0.00	163-D- BTM dish
A 36"	16649.980	180.0	360.00	0.00	163-D-Shell-7
B 36"	34250.055	180.0	310.00	0.00	122-D1-Shell-3
H1 42"	47550.055	180.0	310.00	0.00	122-D1-Shell-8
H2 42"	47550.055	0.0	310.00	0.00	122-D1-Shell-8
F 42"	...	0.0	500.00	0.00	122-D1-Top head
C 48"	...	0.0	310.00	0.00	122-D1-Btm head

Bill of Materials:

QTY	DESCRIPTION	MATERIAL
1	SPHERICAL HEAD: 68.0mm THK X 6738.0mm ID X 0.0mm	SA-516 70
1	SPHERICAL HEAD: 58.0mm THK X 6740.0mm ID X 0.0mm	SA-516 70
1	ELLIPTICAL HEAD: 2.0 X 1, 44.0mm THK X 7100.0mm ID X 50.0mm	SA-516 70
1	ELLIPTICAL HEAD: 2.0 X 1, 20.0mm THK X 7100.0mm ID X 50.0mm	SA-516 70
1	INSULATION: 3000mm X 80mm THK	FP
15	PLATFORM: 1000mm X 1500mm WIDE	
1	INSULATION: 2700mm X 80mm THK	FP
1	INSULATION: 1500mm X 80mm THK	FP
1	PACKING: 3350.000mm	...
1	INSULATION: 3350mm X 80mm THK	
2	PACKING: 1000.000mm	...
11	INSULATION: 3000mm X 80mm THK	
1	PACKING: 1000.020mm	...
1	INSULATION: 1899mm X 80mm THK	
1	PACKING: 999.976mm	...
5	CLADDING: 3000.000mm X 3.000mm THK	...
1	INSULATION: 2000mm X 80mm THK	
1	CLADDING: 2000.000mm X 3.000mm THK	...
1	INSULATION: 1500mm X 80mm THK	
1	CLADDING: 1500.000mm X 3.000mm THK	...
1	INSULATION: 3353mm X 80mm THK	
1	CLADDING: 3370.000mm X 3.000mm THK	...
2	INSULATION: 1250mm X 80mm THK	
1	INSULATION: 1824mm X 80mm THK	
2	BAR RING STIFFENER: 7148.0mm ID X 7748.0mm OD X 28.0mm	SA-516 70
2	BAR RING STIFFENER: 7144.0mm ID X 7644.0mm OD X 28.0mm	SA-516 70
1	PACKING: 1199.950mm	...
1	PACKING: 2800.000mm	...
1	INSULATION: 2800mm X 80mm THK	
1	BAR RING STIFFENER: 7140.0mm ID X 7640.0mm OD X 28.0mm	SA-516 70
4	BAR RING STIFFENER: 7134.0mm ID X 7634.0mm OD X 28.0mm	SA-516 70
1	LADDER: 54800.0mm	...
3	TRAYS # 3 SPACING 35.433071 mm	



1	LADDER: 3000.0mm	...
1	INSULATION: 2900mm X 80mm THK	
1	CLADDING: 2900.000mm X 3.000mm THK	...
1	BAR RING STIFFENER: 7138.0mm ID X 7638.0mm OD X 28.0mm	SA-516 70
1	PLATFORM: 1000mm X 7500mm WIDE	
1	LADDER: 4500.0mm	...
1	INSULATION: 1851mm X 80mm THK	
1	CLADDING: 1825.000mm X 3.000mm THK	...
4	CLASS 600 GR 1.1, 24.0" BLIND FLANGE(S)	SA-105
4	CLASS 150 GR 1.1, 24.0" BLIND FLANGE(S)	SA-105
1	CLASS 150 GR 1.1, 1.0" BLIND FLANGE(S)	SA-105
1	BASERING: 7278.0mm OD X 6678.0 ID X 56.0mm	SA-516 70
48	BASERING BOLTS: 64.0mm DIA	SA-36
48	BASERING BOLT NUTS: 64.0mm DIA	...
1	BASERING TOP RING: 7278.0mm OD X 6846.0 ID X 56.0mm	...
96	BASERING GUSSET PLATES: 206.0mm X 206.0mm X 288.0mm X 28.0mm	...
4	CYLINDRICAL SKIRT SEGMENT 42.0mm THK X 3009.4mm X 3000.0mm	SA-516 70
2	CYLINDRICAL SKIRT SEGMENT 42.0mm THK X 3068.4mm X 3000.0mm	SA-516 70
1	CYLINDRICAL SKIRT SEGMENT 42.0mm THK X 1534.2mm X 3000.0mm	SA-516 70
1	CYLINDRICAL SKIRT SEGMENT 42.0mm THK X 19708.9mm X 3000.0mm	SA-516 70
4	CYLINDRICAL SKIRT SEGMENT 36.0mm THK X 3012.1mm X 2700.0mm	SA-516 70
2	CYLINDRICAL SKIRT SEGMENT 36.0mm THK X 3071.2mm X 2700.0mm	SA-516 70
1	CYLINDRICAL SKIRT SEGMENT 36.0mm THK X 1535.6mm X 2700.0mm	SA-516 70
1	CYLINDRICAL SKIRT SEGMENT 36.0mm THK X 19726.4mm X 2700.0mm	SA-516 70
4	CYLINDRICAL SKIRT SEGMENT 42.0mm THK X 3009.4mm X 1500.0mm	SA-516 70
2	CYLINDRICAL SKIRT SEGMENT 42.0mm THK X 3068.4mm X 1500.0mm	SA-516 70
1	CYLINDRICAL SKIRT SEGMENT 42.0mm THK X 1534.2mm X 1500.0mm	SA-516 70
1	CYLINDRICAL SKIRT SEGMENT 42.0mm THK X 19708.9mm X 1500.0mm	SA-516 70
6	CYLINDRICAL SEGMENT 98.0mm THK X 2981.8mm X 3000.0mm	SA-516 70
4	CYLINDRICAL SEGMENT 98.0mm THK X 3040.3mm X 3000.0mm	SA-516 70
2	CYLINDRICAL SEGMENT 98.0mm THK X 18066.5mm X 3000.0mm	SA-516 70
4	CYLINDRICAL SEGMENT 98.0mm THK X 1520.2mm X 3000.0mm	SA-516 70
4	CYLINDRICAL SEGMENT 96.0mm THK X 2981.8mm X 1900.0mm	SA-516 70
1	CYLINDRICAL SEGMENT 96.0mm THK X 3040.3mm X 1900.0mm	SA-516 70
1	CYLINDRICAL SEGMENT 96.0mm THK X 18008.0mm X 1900.0mm	SA-516 70



2	CYLINDRICAL SEGMENT 96.0mm THK X 1520.2mm X 1900.0mm	SA-516 70
7	CYLINDRICAL SEGMENT 92.0mm THK X 2984.5mm X 3000.0mm	SA-516 70
4	CYLINDRICAL SEGMENT 92.0mm THK X 3043.0mm X 3000.0mm	SA-516 70
1	CYLINDRICAL SEGMENT 92.0mm THK X 18082.7mm X 3000.0mm	SA-516 70
3	CYLINDRICAL SEGMENT 92.0mm THK X 1521.5mm X 3000.0mm	SA-516 70
1	CYLINDRICAL SEGMENT 92.0mm THK X 19545.6mm X 3000.0mm	SA-516 70
4	CYLINDRICAL SEGMENT 92.0mm THK X 2984.5mm X 2000.0mm	SA-516 70
2	CYLINDRICAL SEGMENT 92.0mm THK X 3043.0mm X 2000.0mm	SA-516 70
1	CYLINDRICAL SEGMENT 92.0mm THK X 19545.6mm X 2000.0mm	SA-516 70
1	CYLINDRICAL SEGMENT 92.0mm THK X 1521.5mm X 2000.0mm	SA-516 70
4	CYLINDRICAL SEGMENT 92.0mm THK X 2984.5mm X 1500.0mm	SA-516 70
2	CYLINDRICAL SEGMENT 92.0mm THK X 3043.0mm X 1500.0mm	SA-516 70
1	CYLINDRICAL SEGMENT 92.0mm THK X 1521.5mm X 1500.0mm	SA-516 70
1	CYLINDRICAL SEGMENT 92.0mm THK X 19545.6mm X 1500.0mm	SA-516 70
7	CON SKIRT SEGMENT 26.0mm THK X 2888.4mm X 1250.0mm X 2868.0mm	SA-516 70
1	CON SKIRT SEGMENT 26.0mm THK X 20218.5mm X 1250.0mm X 20076.3mm	SA-516 70
7	CON SKIRT SEGMENT 24.0mm THK X 2935.2mm X 3000.0mm X 2887.0mm	SA-516 70
1	CON SKIRT SEGMENT 24.0mm THK X 20546.4mm X 3000.0mm X 20208.8mm	SA-516 70
7	CON SKIRT SEGMENT 24.0mm THK X 2986.1mm X 3000.0mm X 2937.5mm	SA-516 70
1	CON SKIRT SEGMENT 24.0mm THK X 20903.0mm X 3000.0mm X 20562.2mm	SA-516 70
7	CON SKIRT SEGMENT 24.0mm THK X 3006.5mm X 1250.0mm X 2986.1mm	SA-516 70
1	CON SKIRT SEGMENT 24.0mm THK X 21045.3mm X 1250.0mm X 20903.0mm	SA-516 70
13	CYLINDRICAL SEGMENT 24.0mm THK X 3005.6mm X 3000.0mm	SA-516 70
1	CYLINDRICAL SEGMENT 24.0mm THK X 21039.3mm X 3000.0mm	SA-516 70
1	CYLINDRICAL SEGMENT 24.0mm THK X 19299.3mm X 3000.0mm	SA-516 70
1	CYLINDRICAL SEGMENT 24.0mm THK X 1265.6mm X 3000.0mm	SA-516 70
12	CYLINDRICAL SEGMENT 22.0mm THK X 3005.6mm X 3000.0mm	SA-516 70
2	CYLINDRICAL SEGMENT 22.0mm THK X 19299.3mm X 3000.0mm	SA-516 70
2	CYLINDRICAL SEGMENT 22.0mm THK X 1265.6mm X 3000.0mm	SA-516 70
6	CYLINDRICAL SEGMENT 20.0mm THK X 3005.6mm X 2800.0mm	SA-516 70
1	CYLINDRICAL SEGMENT 20.0mm THK X 19299.3mm X 2800.0mm	SA-516 70
1	CYLINDRICAL SEGMENT 20.0mm THK X 1265.6mm X 2800.0mm	SA-516 70
18	CYLINDRICAL SEGMENT 16.0mm THK X 3005.6mm X 3000.0mm	SA-516 70
3	CYLINDRICAL SEGMENT 16.0mm THK X 19299.3mm X 3000.0mm	SA-516 70
3	CYLINDRICAL SEGMENT 16.0mm THK X 1265.6mm X 3000.0mm	SA-516 70



1	CYLINDRICAL SEGMENT 14.0mm THK X 19299.3mm X 2900.0mm	SA-516 70
6	CYLINDRICAL SEGMENT 14.0mm THK X 3005.6mm X 2900.0mm	SA-516 70
1	CYLINDRICAL SEGMENT 14.0mm THK X 1265.6mm X 2900.0mm	SA-516 70
1	NAMEPLATE	...



Minimum Design Metal Temperature Results Summary (°C):

Description	Curve	Basic MDMT	Reduced MDMT	PWHT	Thickness ratio	Gov Thk mm	E*
Notes							
163-D- BTM dis[9]	!	-12		Yes			
163-D-Shell-1 [8]	!	-12		Yes			
163-D-Shell-2 [8]	!	-12		Yes			
163-D-Shell-3 [8]	!	-12		Yes			
163-D-Shell-4 [8]	!	-12		Yes			
163-D-Shell-5 [8]	!	-12		Yes			
163-D-Shell-6 [8]	!	-12		Yes			
163-D-Shell-7 [8]	!	-12		Yes			
163-D-Top head[9]	!	-12		Yes			
122-D1-Btm he[10]	!	-12		Yes			
122-D1-Btm hea[7]	!	-12		Yes			
122-D1-Shell-1[8]	!	-12		No			
122-D1-Shell-2[8]	!	-12		No			
122-D1-Shell-3[8]	!	-12		No			
122-D1-Shell-4[8]	!	-12		No			
122-D1-Shell-5[8]	!	-12		No			
122-D1-Shell-6[8]	!	-12		No			
122-D1-Shell-7[8]	!	-12		No			
122-D1-Shell-8[8]	!	-12		No			
122-D1-Shell-9[8]	!	-12		No			
122-D1-Top he[10]	!	-12		No			
122-D1-Top hea[7]	!	-12		No			
3 [23]	!	-12		No			
Ring R2 [23]	!	-12		No			
Ring R3 [23]	!	-12		No			
Ring R4 [23]	!	-12		No			
Ring R5 [23]	!	-12		No			
Ring R6 [23]	!	-12		No			
Ring R7 [23]	!	-12		No			
Ring R8 [23]	!	-12		No			
Ring R9 [23]	!	-12		No			
Ring R10 [23]	!	-12		No			
D 36" [1]	!	-12		Yes			
Nozzle Flg [5]	!	-18	-65	Yes			
L12 2" [1]	!	-12		Yes			
Nozzle Flg [5]	!	-18	-66	Yes			
L14 1.5" [1]	!	-12		Yes			
Nozzle Flg [5]	!	-18	-66	Yes			
MH-6 24" [1]	!	-12		Yes			
Nozzle Flg [5]	!	-18	-67	Yes			
L11 2" [1]	!	-12		Yes			
Nozzle Flg [5]	!	-17	-17	Yes			
L13 1.5" [1]	!	-12		Yes			
Nozzle Flg [5]	!	-17	-17	Yes			
UC2 24" [1]	!	-12		Yes			
Nozzle Flg [5]	!	-18	-68	Yes			
MH-5 24" [1]	!	-12		Yes			
Nozzle Flg [5]	!	-18	-68	Yes			
MH-4 24" [1]	!	-12		Yes			
Nozzle Flg [5]	!	-18	-68	Yes			
A 36" [1]	!	-12		Yes			
Nozzle Flg [5]	!	-18	-68	Yes			
E 8" [1]	!	-12		Yes			
Nozzle Flg [5]	!	-18	-68	Yes			
C 48" [1]	!	-12		Yes			
Nozzle Flg [5]	!	-18	-104	Yes			
MH-3 24" [1]	!	-12		No			
Nozzle Flg [5]	!	-18	-104	No			
L2 2" [1]	!	-12		No			
Nozzle Flg [5]	!	-17	-17	No			
L4 2" [1]	!	-12		No			
Nozzle Flg [5]	!	-17	-17	No			
L6 2" [1]	!	-12		No			



Nozzle Flg	[5]	!	-17	-17	No			
L8 2"	[1]	!	-12		No			
Nozzle Flg	[5]	!	-17	-17	No			
L10 2"	[1]	!	-12		No			
Nozzle Flg	[5]	!	-17	-17	No			
B 36"	[1]	!	-12		No			
Nozzle Flg	[5]	!	-18	-104	No			
K 6"	[1]	!	-12		No			
Nozzle Flg	[5]	!	-18	-104	No			
I 28"	[1]	!	-12		No			
Nozzle Flg	[5]	!	-18	-104	No			
L1 2"	[1]	!	-12		No			
Nozzle Flg	[5]	!	-17	-17	No			
L3 2"	[1]	!	-12		No			
Nozzle Flg	[5]	!	-17	-17	No			
L5 2"	[1]	!	-12		No			
Nozzle Flg	[5]	!	-17	-17	No			
L7 2"	[1]	!	-12		No			
Nozzle Flg	[5]	!	-17	-17	No			
L9 2"	[1]	!	-12		No			
Nozzle Flg	[5]	!	-17	-17	No			
MH-2 24"	[1]	!	-12		No			
Nozzle Flg	[5]	!	-18	-104	No			
UC1 24"	[1]	!	-12		No			
Nozzle Flg	[5]	!	-18	-104	No			
MH-1 24"	[1]	!	-12		No			
Nozzle Flg	[5]	!	-18	-104	No			
S1 1"	[1]	!	-12		No			
Nozzle Flg	[5]	!	-17	-17	No			
H1 42"	[1]	!	-12		No			
Nozzle Flg	[5]	!	-18	-104	No			
H2 42"	[1]	!	-12		No			
Nozzle Flg	[5]	!	-18	-104	No			
F 42"	[1]	!	-12		No			
Nozzle Flg	[5]	!	-18	-104	No			
Nozl. Bolting[22]			-48		No	1.000	64.000	1.00

Warmest MDMT: -12 -17

Required Minimum Design Metal Temperature -12.0 °C
 Warmest Computed Minimum Design Metal Temperature -12.0 °C

Notes:

- [!] - This was an impact tested material.
- [1] - Governing Nozzle Weld.
- [4] - ASME Flange MDMT Calcs; Thickness ratio per equation 3.1.
- [5] - ASME Flange MDMT Calcs; Thickness ratio per equation 3.3.
- [6] - MDMT Calculations at the Shell/Head Joint.
- [7] - MDMT Calculations for the Straight Flange.
- [8] - Cylinder/Cone/Flange Junction MDMT.
- [9] - Calculations in the Spherical Portion of the Head.
- [10] - Calculations in the Knuckle Portion of the Head.
- [11] - Calculated (Body Flange) Flange MDMT.
- [12] - Calculated Flat Head MDMT per 3.11.2.3
- [13] - Tubesheet MDMT, shell side, if applicable
- [14] - Tubesheet MDMT, tube side, if applicable
- [15] - Nozzle Material
- [16] - Shell or Head Material
- [17] - Impact Testing required
- [21] - Body Flange Bolting Material
- [22] - Nozzle Flange Bolting Material
- [23] - Stiffening Ring to Shell Weld
- [24] - Saddle to Shell Weld



Class From To : Basic Element Checks.

Warn.	130	140	Skirt diameter at base is < at the top
Warn.	140	150	Skirt diameter at base is < at the top
Warn.	150	160	Skirt diameter at base is < at the top
Warn.	160	170	Skirt diameter at base is < at the top

Class From To: Check of Additional Element Data

PV Elite performs all calculations internally in Imperial Units to remain compliant with the ASME Code and any built in assumptions in the ASME Code formulas. The finalized results are reflected to show the set of selected units for this analysis.

Information Regarding Nozzle Loads

Please note that nozzle loadings, if included, are assumed to be local in nature and will not contribute to or create a net section bending moment. Therefore, the addition of nozzle loads will not affect the support load calculation. If you wish to create a load on the support(s) from nozzle loads, you can enable that feature in tools->configuration->Job Specific Settings.



One or more nozzles in this model was specified using the actual thickness basis and a standard flange is also specified. In many cases a matching nominal diameter cannot be determined and the weight of the flange and optional blind flange cannot be determined. When actual thickness basis is specified, you must calculate the weight of the assembly (by hand) and enter it into the 'Overriding Weight' field in the nozzle dialog when there is a standard flange specified. However, for FVC nozzles that have their nominal diameter specified in the FVC dialog, entering the overriding weight for that nozzle is not required.



The level of precision is set to a value of less than 3. At this level you may not obtain proper resolution on some results. This value can be changed in Tools->Set Configuration Parameters->Default Values->Level of Precision.

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Units used in this Analysis (SIASME):

Name	System Unit	Constant	User Unit
Length	Feet	304.8000	mm
Force	Pounds	4.4480	N
Mass	Pounds	0.4536	kgm
Area	sq. inches	645.1600	mm ²
Moment	ft. lbs.	1356.3000	N-mm
Stress	lbs./sq.in.	0.0069	MPa
Temperature	Degrees F	0.5556	°C
Pressure	psig	0.0069	MPa
Elast. Modulus	lbs./sq.in.	0.0069	MPa
Pipe Density	lbs./cu.in.	0.0277	kg/cm ³
Ins. Density	lbs./cu.ft.	0.1600E-04	kg/cm ³
Fluid Density	lbs./cu.ft.	0.1600E-04	kg/cm ³
Wind Speed	miles/hr	1.6093	km/hr
Tray Weight	lbs./sq.ft.	0.0005	kg/cm ²
Inertia	in.4	416231.0000	mm ⁴
G Load	G's	1.0000	g's
Wind Load	lbs./sq.ft.	0.0479	kPa
Elevation	Feet	304.8000	mm
Volume	in.3	0.0164	ltr
Diameter	inches	25.4000	mm
Thickness	inches	25.4000	mm

PV Elite Vessel Analysis Program: Input Data

Design Internal Pressure (for Hydrotest)	0.38	MPa
Design Internal Temperature	130.0	°C
Projection of Nozzle from Vessel Top	0	mm
Projection of Nozzle from Vessel Bottom	0	mm
Minimum Design Metal Temperature	-12.0	°C
Type of Construction	Welded	
Special Service	None	
Use Higher Longitudinal Stresses (Flag)	N	
Select t for Internal Pressure (Flag)	N	
Select t for External Pressure (Flag)	N	
Select t for Axial Stress (Flag)	N	
Select Location for Stiff. Rings (Flag)	N	
Consider Vortex Shedding	N	

Shop Pressure Test:

Type of Pressure Test	Hydrostatic
Pressure Test Position	Horizontal
Test Performed in Corroded Condition	No

Field Pressure Test:

Type of Pressure Test	Not Applied
Pressure Test Position	Vertical
Test Performed in Corroded Condition	Yes

Load Case 1	NP+EW+1.1WI+FW+BW
Load Case 2	NP+EW+EE+FS+BS
Load Case 3	NP+OW+1.1WI+FW+BW
Load Case 4	NP+OW+EQ+FS+BS
Load Case 5	NP+HW+HI
Load Case 6	NP+HW+HE
Load Case 7	IP+OW+1.1WI+FW+BW
Load Case 8	IP+OW+EQ+FS+BS
Load Case 9	EP+OW+1.1WI+FW+BW
Load Case 10	EP+OW+EQ+FS+BS
Load Case 11	HP+HW+HI
Load Case 12	HP+HW+HE
Load Case 13	IP+WE+EW
Load Case 14	IP+WF+CW
Load Case 15	IP+VO+OW
Load Case 16	IP+VE+EW



Load Case 17		NP+VO+OW	
Load Case 18		FS+BS+IP+OW	
Load Case 19		FS+BS+EP+OW	
Load Case 20		BL+IP+OW	
Wind Design Code		ASCE/SEI 7-16	
Wind Load Reduction Scale Factor		1.100	
Basic Wind Speed	[V]	97.2	km/hr
Surface Roughness Category		C: Open Terrain	
Type of Surface		Moderately Smooth	
Base Elevation		300	mm
Percent Wind for Hydrotect		50.0	
Using User defined Wind Press. Vs Elev.		N	
Height of Hill or Escarpment	H or Hh	0	mm
Distance Upwind of Crest	Lh	0	mm
Distance from Crest to the Vessel	x	0	mm
Type of Terrain (Hill, Escarpment)		Flat	
Damping Factor (Beta) for Wind (Ope)		0.0100	
Damping Factor (Beta) for Wind (Empty)		0.0000	
Damping Factor (Beta) for Wind (Filled)		0.0000	
Site Elevation above Sea Level		0	mm
Seismic Design Code		ASCE/SEI 7-16	
Seismic Load Reduction Scale Factor		1.000	
Importance Factor		1.250	
Table Value Fa		1.600	
Table Value Fv		2.400	
Max. Mapped Res. Acceleration	[Ss]	0.147	
Max. Eff. Ground Acceleration	[S]	0.065	
Force Modification Factor R		1.250	
Site Class		D	
Component Elevation Ratio	z/h	0.000	
Amplification Factor	Ap	0.000	
Force Factor		0.000	
Consider Vertical Acceleration		Yes	
Minimum Acceleration Multiplier		0.000	
User Value of Sds (used if > 0)		0.000	
User Value of Sd1 (used if > 0)		0.000	
Moment Reduction Factor Tau		1.000	
Design Pressure + Static Head		Y	
Consider MAP New and Cold in Noz. Design		N	
Consider External Loads for Nozzle Des.		Y	
Use ASME VIII-1 Appendix 1-9		N	
Perform Blast Load Analysis		No	
Material Database Year	Current w/Addenda or Code Year		

Configuration Directives:

g0 is taken as min(g0, attached shell thk)	Yes
Using Metric Material Databases, ASME II D	Yes
Using ASME VIII-1 Appendix 46	No
Calculate B31.3 type stress for Nozzles with Loads	Yes
Reduce the MDMT due to lower membrane stress	Yes
Consider Longitudinal Stress in MDMT Calculations	Yes

Complete Listing of Vessel Elements and Details:

Element#	1/27	
Element From Node	10	
Element To Node	20	
Element Type	Skirt Support	
Description	Skirt-3	
Distance "FROM" to "TO"	3000	mm
Skirt Outside Diameter	6846	mm
Diameter of Skirt at Base	6846	mm
Skirt Thickness	42	mm
Internal Corrosion Allowance	1.6	mm



Nominal Thickness	42	mm
External Corrosion Allowance	0	mm
Design Temperature Internal Pressure	82	°C
Design Temperature External Pressure	0	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-516 70	
Allowable Stress, Ambient	175	MPa
Allowable Stress, Operating	161.57	MPa
Allowable Stress, Hydrotest	247	MPa
Material Density	0.00775	kg/cm³
P Number Thickness	0	mm
Yield Stress, Operating	242.6	MPa
MDMT Curve	B	
External Pressure Chart Name	CS-2	
UNS Number	K02700	
Product Form	Plate	
Efficiency, Longitudinal Seam	0.7	
Efficiency, Head-to-Skirt or Circ. Seam	0.7	
Weld is pre-Heated	Yes	
Element From Node	10	
Detail Type	Insulation	
Detail ID	FP	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	3000	mm
Thickness of Insulation	80	mm
Density	0.0008	kg/cm³

Element#	2/27	
Element From Node	20	
Element To Node	30	
Element Type	Skirt Support	
Description	Skirt-2	
Distance "FROM" to "TO"	2700	mm
Skirt Outside Diameter	6840	mm
Diameter of Skirt at Base	6840	mm
Skirt Thickness	36	mm
Internal Corrosion Allowance	1.6	mm
Nominal Thickness	36	mm
External Corrosion Allowance	0	mm
Design Temperature Internal Pressure	82	°C
Design Temperature External Pressure	0	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-516 70	
Efficiency, Longitudinal Seam	0.7	
Efficiency, Head-to-Skirt or Circ. Seam	0.7	
Weld is pre-Heated	Yes	

Element From Node	20	
Detail Type	Platform	
Detail ID	SKRT PLAT-1	
Dist. from "FROM" Node / Offset dist	300	mm
Platform Start Angle (degrees)	180.0	
Platform End Angle (degrees)	270.0	
Platform Wind Area	1499999	mm²
Platform Weight	16981	N
Platform Railing Weight	0	N/mm
Platform Grating Weight	0.017	kg/cm²
Platform Width	1500	mm
Platform Height	1000	mm
Platform Clearance or End Offset	115	mm
Platform Force Coefficient	1.0	
Ladder Layout Angle	0.0	
Ladder Start Elevation	0	mm
Ladder End Elevation	0	mm
Unit Weight of Ladder	0	N/mm
Platform Length (top head platform)	0	mm



Element From Node	20	
Detail Type	Insulation	
Detail ID	FP	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	2700	mm
Thickness of Insulation	80	mm
Density	0.0008	kg/cm ³

Element#	3/27	
Element From Node	30	
Element To Node	40	
Element Type	Skirt Support	
Description	Skirt-1	
Distance "FROM" to "TO"	1500	mm
Skirt Outside Diameter	6846	mm
Diameter of Skirt at Base	6846	mm
Skirt Thickness	42	mm
Internal Corrosion Allowance	1.6	mm
Nominal Thickness	42	mm
External Corrosion Allowance	0	mm
Design Temperature Internal Pressure	130	°C
Design Temperature External Pressure	0	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-516 70 +IT	
Efficiency, Longitudinal Seam	0.7	
Efficiency, Head-to-Skirt or Circ. Seam	0.55	
Weld is pre-Heated	Yes	

Element From Node	30	
Detail Type	Insulation	
Detail ID	FP	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	1500	mm
Thickness of Insulation	80	mm
Density	0.0008	kg/cm ³

Element#	4/27	
Element From Node	40	
Element To Node	50	
Element Type	Spherical	
Description	163-D- BTM dish	
Distance "FROM" to "TO"	0	mm
Inside Diameter	6738	mm
Element Thickness	60	mm
Internal Corrosion Allowance	6	mm
Nominal Thickness	68	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	4.1007	MPa
Design Temperature Internal Pressure	130	°C
Design External Pressure	0	MPa
Design Temperature External Pressure	0	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-516 70 +IT	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	Yes	

Element From Node	40	
Detail Type	Packing	
Detail ID	HPPack: 11	
Dist. from "FROM" Node / Offset dist	-3350	mm
Height of Packed Section	3350	mm
Density	0.000166	kg/cm ³
Percent Volume Holdup	0.0	
Specific Gravity of Packing Liquid	0.0	



Element From Node 40
 Detail Type Liquid
 Detail ID Liquid 40
 Dist. from "FROM" Node / Offset dist -3369 mm
 Height/Length of Liquid 3369 mm
 Liquid Density 0.001058 kg/cm³

Element From Node 40
 Detail Type Insulation
 Detail ID Ins: 70
 Dist. from "FROM" Node / Offset dist -3350 mm
 Height/Length of Insulation 3350 mm
 Thickness of Insulation 80 mm
 Density 0.000352 kg/cm³

Element From Node 40
 Detail Type Nozzle
 Detail ID D 36"
 Dist. from "FROM" Node / Offset dist 0 mm
 Nozzle Diameter 36 in.
 Nozzle Schedule None
 Nozzle Class 600
 Layout Angle 0.0
 Blind Flange (Y/N) N
 Weight of Nozzle (Used if > 0) 39594 N
 Grade of Attached Flange GR 1.1
 Nozzle Matl SA-266 2 +IT

Element From Node 40
 Detail Type Weight
 Detail ID HPVB
 Dist. from "FROM" Node / Offset dist -3360 mm
 Miscellaneous Weight 1667 N
 Offset from Element Centerline 0 mm

Element From Node 40
 Detail Type Weight
 Detail ID 163-DBtm SF wt
 Dist. from "FROM" Node / Offset dist 0 mm
 Miscellaneous Weight 19612 N
 Offset from Element Centerline 0 mm

Element# 5/27
 Element From Node 50
 Element To Node 60
 Element Type Cylinder
 Description 163-D-Shell-1
 Distance "FROM" to "TO" 3000 mm
 Inside Diameter 6700 mm
 Element Thickness 98 mm
 Internal Corrosion Allowance 6 mm
 Nominal Thickness 98 mm
 External Corrosion Allowance 0 mm
 Design Internal Pressure 4.1007 MPa
 Design Temperature Internal Pressure 130 °C
 Design External Pressure 0 MPa
 Design Temperature External Pressure 0 °C
 Effective Diameter Multiplier 1.2
 Material Name SA-516 70 +IT
 Efficiency, Longitudinal Seam 1.0
 Efficiency, Circumferential Seam 1.0
 Weld is pre-Heated No

Element From Node 50
 Detail Type Platform
 Detail ID PLAT:1
 Dist. from "FROM" Node / Offset dist 0 mm
 Platform Start Angle (degrees) 0.0



Platform End Angle (degrees)	360.0	
Platform Wind Area	2999999	mm ²
Platform Weight	67798	N
Platform Railing Weight	0	N/mm
Platform Grating Weight	0.017	kg/cm ²
Platform Width	1500	mm
Platform Height	1000	mm
Platform Clearance or End Offset	115	mm
Platform Force Coefficient	1.0	
Ladder Layout Angle	0.0	
Ladder Start Elevation	0	mm
Ladder End Elevation	0	mm
Unit Weight of Ladder	0	N/mm
Platform Length (top head platform)	0	mm
Element From Node	50	
Detail Type	Packing	
Detail ID	HPpack: 12	
Dist. from "FROM" Node / Offset dist	0	mm
Height of Packed Section	1000	mm
Density	0.000166	kg/cm ³
Percent Volume Holdup	0.0	
Specific Gravity of Packing Liquid	0.0	
Element From Node	50	
Detail Type	Liquid	
Detail ID	Liquid 50	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Liquid	3000	mm
Liquid Density	0.001058	kg/cm ³
Element From Node	50	
Detail Type	Insulation	
Detail ID	Ins: 70	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	3000	mm
Thickness of Insulation	80	mm
Density	0.000352	kg/cm ³
Element From Node	50	
Detail Type	Nozzle	
Detail ID	L12 2"	
Dist. from "FROM" Node / Offset dist	150	mm
Nozzle Diameter	2	in.
Nozzle Schedule	None	
Nozzle Class	600	
Layout Angle	0.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	296.96	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-266 2 +IT	
Element From Node	50	
Detail Type	Nozzle	
Detail ID	L14 1.5"	
Dist. from "FROM" Node / Offset dist	150	mm
Nozzle Diameter	1.5	in.
Nozzle Schedule	None	
Nozzle Class	600	
Layout Angle	30.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	259.66	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-266 2 +IT	
Element From Node	50	
Detail Type	Weight	
Detail ID	D Fy wt	
Dist. from "FROM" Node / Offset dist	0.004069	mm
Miscellaneous Weight	29124	N



Offset from Element Centerline 5048 mm

Element#	6/27
Element From Node	60
Element To Node	70
Element Type	Cylinder
Description	163-D-Shell-2
Distance "FROM" to "TO"	3000 mm
Inside Diameter	6700 mm
Element Thickness	98 mm
Internal Corrosion Allowance	6 mm
Nominal Thickness	98 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	4.1007 MPa
Design Temperature Internal Pressure	130 °C
Design External Pressure	0 MPa
Design Temperature External Pressure	0 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70 +IT
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No
Element From Node	60
Detail Type	Platform
Detail ID	PLAT-2
Dist. from "FROM" Node / Offset dist	799.97 mm
Platform Start Angle (degrees)	90.0
Platform End Angle (degrees)	270.0
Platform Wind Area	2999999 mm ²
Platform Weight	33899 N
Platform Railing Weight	0 N/mm
Platform Grating Weight	0.017 kg/cm ²
Platform Width	1500 mm
Platform Height	1000 mm
Platform Clearance or End Offset	115 mm
Platform Force Coefficient	1.0
Ladder Layout Angle	0.0
Ladder Start Elevation	0 mm
Ladder End Elevation	0 mm
Unit Weight of Ladder	0 N/mm
Platform Length (top head platform)	0 mm
Element From Node	60
Detail Type	Liquid
Detail ID	Liquid 60
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Liquid	3000 mm
Liquid Density	0.001058 kg/cm ³
Element From Node	60
Detail Type	Insulation
Detail ID	Ins: 70
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Insulation	3000 mm
Thickness of Insulation	80 mm
Density	0.000352 kg/cm ³
Element From Node	60
Detail Type	Nozzle
Detail ID	MH-6 24"
Dist. from "FROM" Node / Offset dist	1400 mm
Nozzle Diameter	24 in.
Nozzle Schedule	None
Nozzle Class	600
Layout Angle	225.0
Blind Flange (Y/N)	Y
Weight of Nozzle (Used if > 0)	13306 N



Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-266 2 +IT
Element From Node	60
Detail Type	Weight
Detail ID	MH6 davit
Dist. from "FROM" Node / Offset dist	1400 mm
Miscellaneous Weight	490.3 N
Offset from Element Centerline	5000 mm

Element#	7/27
Element From Node	70
Element To Node	80
Element Type	Cylinder
Description	163-D-Shell-3
Distance "FROM" to "TO"	1900 mm
Inside Diameter	6700 mm
Element Thickness	96 mm
Internal Corrosion Allowance	6 mm
Nominal Thickness	96 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	4.1007 MPa
Design Temperature Internal Pressure	130 °C
Design External Pressure	0 MPa
Design Temperature External Pressure	0 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70 +IT
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No

Element From Node	70
Detail Type	Platform
Detail ID	PLAT-3
Dist. from "FROM" Node / Offset dist	1800 mm
Platform Start Angle (degrees)	270.0
Platform End Angle (degrees)	90.0
Platform Wind Area	2999999 mm ²
Platform Weight	33883 N
Platform Railing Weight	0 N/mm
Platform Grating Weight	0.017 kg/cm ²
Platform Width	1500 mm
Platform Height	1000 mm
Platform Clearance or End Offset	115 mm
Platform Force Coefficient	1.0
Ladder Layout Angle	0.0
Ladder Start Elevation	0 mm
Ladder End Elevation	0 mm
Unit Weight of Ladder	0 N/mm
Platform Length (top head platform)	0 mm

Element From Node	70
Detail Type	Packing
Detail ID	HPpack: 21
Dist. from "FROM" Node / Offset dist	899.97 mm
Height of Packed Section	1000 mm
Density	0.000166 kg/cm ³
Percent Volume Holdup	25.0
Specific Gravity of Packing Liquid	1.0599999

Element From Node	70
Detail Type	Liquid
Detail ID	Liquid: 40
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Liquid	600 mm
Liquid Density	0.001058 kg/cm ³

Element From Node	70
-------------------	----



Detail Type	Insulation
Detail ID	Ins: 70
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Insulation	1900 mm
Thickness of Insulation	80 mm
Density	0.000352 kg/cm ³

Element From Node	70
Detail Type	Nozzle
Detail ID	L11 2"
Dist. from "FROM" Node / Offset dist	750 mm
Nozzle Diameter	2 in.
Nozzle Schedule	None
Nozzle Class	600
Layout Angle	0.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	296.96 N
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-266 2 +IT

Element From Node	70
Detail Type	Nozzle
Detail ID	L13 1.5"
Dist. from "FROM" Node / Offset dist	750 mm
Nozzle Diameter	1.5 in.
Nozzle Schedule	None
Nozzle Class	600
Layout Angle	30.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	257.56 N
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-266 2 +IT

Element From Node	70
Detail Type	Nozzle
Detail ID	UC2 24"
Dist. from "FROM" Node / Offset dist	899.99 mm
Nozzle Diameter	24 in.
Nozzle Schedule	None
Nozzle Class	600
Layout Angle	90.0
Blind Flange (Y/N)	Y
Weight of Nozzle (Used if > 0)	13685 N
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-516 70 +IT

Element From Node	70
Detail Type	Weight
Detail ID	HPB#1btm grd
Dist. from "FROM" Node / Offset dist	899.97 mm
Miscellaneous Weight	30399 N
Offset from Element Centerline	0 mm

Element From Node	70
Detail Type	Weight
Detail ID	HPB#1btm grd
Dist. from "FROM" Node / Offset dist	899.97 mm
Miscellaneous Weight	2941.8 N
Offset from Element Centerline	0 mm

Element#	8/27
Element From Node	80
Element To Node	90
Element Type	Cylinder
Description	163-D-Shell-4
Distance "FROM" to "TO"	3000 mm
Inside Diameter	6706 mm
Element Thickness	92 mm



Internal Corrosion Allowance	1	mm
Nominal Thickness	92	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	4.1007	MPa
Design Temperature Internal Pressure	130	°C
Design External Pressure	0	MPa
Design Temperature External Pressure	0	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-516 70	+IT
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	
Element From Node	80	
Detail Type	Packing	
Detail ID	HPpack: 22	
Dist. from "FROM" Node / Offset dist	0.004651	mm
Height of Packed Section	999.98	mm
Density	0.000166	kg/cm³
Percent Volume Holdup	25.0	
Specific Gravity of Packing Liquid	1.0599999	
Element From Node	80	
Detail Type	Insulation	
Detail ID	Ins: 70	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	3000	mm
Thickness of Insulation	80	mm
Density	0.000352	kg/cm³
Element From Node	80	
Detail Type	Lining	
Detail ID	SS304Lclad	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Lining	3000	mm
Thickness of Lining	3	mm
Density	0.008	kg/cm³
Element From Node	80	
Detail Type	Weight	
Detail ID	HPB#1top HDG	
Dist. from "FROM" Node / Offset dist	999.97	mm
Miscellaneous Weight	9413.8	N
Offset from Element Centerline	0	mm
Element From Node	80	
Detail Type	Weight	
Detail ID	HPB#1dis. dea	
Dist. from "FROM" Node / Offset dist	1600	mm
Miscellaneous Weight	57855	N
Offset from Element Centerline	0	mm
Element From Node	80	
Detail Type	Weight	
Detail ID	HPB#1dis. TA	
Dist. from "FROM" Node / Offset dist	1600	mm
Miscellaneous Weight	2941.8	N
Offset from Element Centerline	0	mm
Element From Node	80	
Detail Type	Weight	
Detail ID	HPSHroud	
Dist. from "FROM" Node / Offset dist	2800	mm
Miscellaneous Weight	58836	N
Offset from Element Centerline	0	mm



Element To Node	100
Element Type	Cylinder
Description	163-D-Shell-5
Distance "FROM" to "TO"	3000 mm
Inside Diameter	6706 mm
Element Thickness	92 mm
Internal Corrosion Allowance	1 mm
Nominal Thickness	92 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	4.1007 MPa
Design Temperature Internal Pressure	130 °C
Design External Pressure	0 MPa
Design Temperature External Pressure	0 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70 +IT
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No

Element From Node	90
Detail Type	Platform
Detail ID	PLAT-4
Dist. from "FROM" Node / Offset dist	899.97 mm
Platform Start Angle (degrees)	90.0
Platform End Angle (degrees)	270.0
Platform Wind Area	2999999 mm ²
Platform Weight	33875 N
Platform Railing Weight	0 N/mm
Platform Grating Weight	0.017 kg/cm ²
Platform Width	1500 mm
Platform Height	1000 mm
Platform Clearance or End Offset	115 mm
Platform Force Coefficient	1.0
Ladder Layout Angle	0.0
Ladder Start Elevation	0 mm
Ladder End Elevation	0 mm
Unit Weight of Ladder	0 N/mm
Platform Length (top head platform)	0 mm

Element From Node	90
Detail Type	Insulation
Detail ID	Ins: 70
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Insulation	3000 mm
Thickness of Insulation	80 mm
Density	0.000352 kg/cm ³

Element From Node	90
Detail Type	Lining
Detail ID	SS304Lclad
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Lining	3000 mm
Thickness of Lining	3 mm
Density	0.008 kg/cm ³

Element From Node	90
Detail Type	Nozzle
Detail ID	MH-5 24"
Dist. from "FROM" Node / Offset dist	1500 mm
Nozzle Diameter	24 in.
Nozzle Schedule	None
Nozzle Class	600
Layout Angle	225.0
Blind Flange (Y/N)	Y
Weight of Nozzle (Used if > 0)	13242 N
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-266 2 +IT

Element From Node	90
Detail Type	Weight



Detail ID	MH5	davit	
Dist. from "FROM" Node / Offset dist	1500	mm	
Miscellaneous Weight	490.3	N	
Offset from Element Centerline	5000	mm	

Element#	10/27	
Element From Node	100	
Element To Node	110	
Element Type	Cylinder	
Description	163-D-Shell-6	
Distance "FROM" to "TO"	2000	mm
Inside Diameter	6706	mm
Element Thickness	92	mm
Internal Corrosion Allowance	1	mm
Nominal Thickness	92	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	4.1007	MPa
Design Temperature Internal Pressure	130	°C
Design External Pressure	0	MPa
Design Temperature External Pressure	0	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-516 70	+IT
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	

Element From Node	100	
Detail Type	Platform	
Detail ID	PLAT-5	
Dist. from "FROM" Node / Offset dist	1900	mm
Platform Start Angle (degrees)	270.0	
Platform End Angle (degrees)	90.0	
Platform Wind Area	2999999	mm ²
Platform Weight	33875	N
Platform Railing Weight	0	N/mm
Platform Grating Weight	0.017	kg/cm ²
Platform Width	1500	mm
Platform Height	1000	mm
Platform Clearance or End Offset	115	mm
Platform Force Coefficient	1.0	
Ladder Layout Angle	0.0	
Ladder Start Elevation	0	mm
Ladder End Elevation	0	mm
Unit Weight of Ladder	0	N/mm
Platform Length (top head platform)	0	mm

Element From Node	100	
Detail Type	Insulation	
Detail ID	Ins: 70	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	2000	mm
Thickness of Insulation	80	mm
Density	0.000352	kg/cm ³

Element From Node	100	
Detail Type	Lining	
Detail ID	SS304Lclad	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Lining	2000	mm
Thickness of Lining	3	mm
Density	0.008	kg/cm ³

Element From Node	100	
Detail Type	Nozzle	
Detail ID	MH-4 24"	
Dist. from "FROM" Node / Offset dist	999.97	mm
Nozzle Diameter	24	in.
Nozzle Schedule	None	



Nozzle Class	600
Layout Angle	225.0
Blind Flange (Y/N)	Y
Weight of Nozzle (Used if > 0)	13242 N
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-266 2 +IT

Element From Node	100
Detail Type	Weight
Detail ID	MH4 davit
Dist. from "FROM" Node / Offset dist	999.97 mm
Miscellaneous Weight	490.3 N
Offset from Element Centerline	5000 mm

Element#	11/27
Element From Node	110
Element To Node	120
Element Type	Cylinder
Description	163-D-Shell-7
Distance "FROM" to "TO"	1500 mm
Inside Diameter	6706 mm
Element Thickness	92 mm
Internal Corrosion Allowance	1 mm
Nominal Thickness	92 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	4.1007 MPa
Design Temperature Internal Pressure	130 °C
Design External Pressure	0 MPa
Design Temperature External Pressure	0 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70 +IT
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No

Element From Node	110
Detail Type	Insulation
Detail ID	Ins: 70
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Insulation	1500 mm
Thickness of Insulation	80 mm
Density	0.000352 kg/cm³

Element From Node	110
Detail Type	Lining
Detail ID	SS304Lclad
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Lining	1500 mm
Thickness of Lining	3 mm
Density	0.008 kg/cm³

Element From Node	110
Detail Type	Nozzle
Detail ID	A 36"
Dist. from "FROM" Node / Offset dist	749.98 mm
Nozzle Diameter	36 in.
Nozzle Schedule	None
Nozzle Class	600
Layout Angle	180.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	16971 N
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-266 2 +IT

Element From Node	110
Detail Type	Weight
Detail ID	HPGallary
Dist. from "FROM" Node / Offset dist	1100 mm



Miscellaneous Weight	66681	N
Offset from Element Centerline	0	mm
Element From Node	110	
Detail Type	Weight	
Detail ID	A Fy wt	
Dist. from "FROM" Node / Offset dist	749.98	mm
Miscellaneous Weight	115181	N
Offset from Element Centerline	5059	mm

Element#	12/27	
Element From Node	120	
Element To Node	130	
Element Type	Spherical	
Description	163-D-Top head	
Distance "FROM" to "TO"	0	mm
Inside Diameter	6740	mm
Element Thickness	52	mm
Internal Corrosion Allowance	1	mm
Nominal Thickness	58	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	4.1007	MPa
Design Temperature Internal Pressure	130	°C
Design External Pressure	0	MPa
Design Temperature External Pressure	0	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-516 70 +IT	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	

Element From Node	120	
Detail Type	Platform	
Detail ID	PLAT-6	
Dist. from "FROM" Node / Offset dist	2400	mm
Platform Start Angle (degrees)	270.0	
Platform End Angle (degrees)	90.0	
Platform Wind Area	2999999	mm ²
Platform Weight	33694	N
Platform Railing Weight	0	N/mm
Platform Grating Weight	0.017	kg/cm ²
Platform Width	1500	mm
Platform Height	1000	mm
Platform Clearance or End Offset	115	mm
Platform Force Coefficient	1.0	
Ladder Layout Angle	0.0	
Ladder Start Elevation	0	mm
Ladder End Elevation	0	mm
Unit Weight of Ladder	0	N/mm
Platform Length (top head platform)	0	mm

Element From Node	120	
Detail Type	Insulation	
Detail ID	Ins: 70	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	3353	mm
Thickness of Insulation	80	mm
Density	0.000352	kg/cm ³

Element From Node	120	
Detail Type	Lining	
Detail ID	SS304Lclad	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Lining	3370	mm
Thickness of Lining	3	mm
Density	0.008	kg/cm ³

Element From Node	120	
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Detail Type	Nozzle
Detail ID	E 8"
Dist. from "FROM" Node / Offset dist	0 mm
Nozzle Diameter	8 in.
Nozzle Schedule	None
Nozzle Class	600
Layout Angle	0.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	1021.1 N
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-266 2 +IT

Element From Node	120
Detail Type	Weight
Detail ID	163-DTop SF wt
Dist. from "FROM" Node / Offset dist	0 mm
Miscellaneous Weight	19612 N
Offset from Element Centerline	0 mm

Element From Node	120
Detail Type	Weight
Detail ID	E Fy wt
Dist. from "FROM" Node / Offset dist	-0.02558 mm
Miscellaneous Weight	17543 N
Offset from Element Centerline	4686 mm

Element From Node	120
Detail Type	For./Mom.
Detail ID	HPEFxFz
Dist. from "FROM" Node / Offset dist	3403 mm
Force in X Direction	14294 N
Force in Y Direction	-19792 N
Force in Z Direction	14294 N
Moment about X Axis	8800470 N-mm
Moment about Y Axis	0 N-mm
Moment about Z Axis	8800470 N-mm
Force/Moment Combination Method	SRSS

Element#	13/27
Element From Node	130
Element To Node	140
Element Type	Skirt Support
Description	Int skirt 1
Distance "FROM" to "TO"	1250 mm
Skirt Outside Diameter	6875 mm
Diameter of Skirt at Base	6827 mm
Skirt Thickness	26 mm
Internal Corrosion Allowance	1.6 mm
Nominal Thickness	26 mm
External Corrosion Allowance	0 mm
Design Temperature Internal Pressure	130 °C
Design Temperature External Pressure	130 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70 +IT
Efficiency, Longitudinal Seam	1.0
Efficiency, Head-to-Skirt or Circ. Seam	1.0
Weld is pre-Heated	No

Element From Node	130
Detail Type	Insulation
Detail ID	Ins: 70
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Insulation	1250 mm
Thickness of Insulation	80 mm
Density	0.000352 kg/cm³



Element#	14/27
Element From Node	140
Element To Node	150
Element Type	Skirt Support
Description	Int skirt 2
Distance "FROM" to "TO"	3000 mm
Skirt Outside Diameter	6987 mm
Diameter of Skirt at Base	6873 mm
Skirt Thickness	24 mm
Internal Corrosion Allowance	1.6 mm
Nominal Thickness	24 mm
External Corrosion Allowance	0 mm
Design Temperature Internal Pressure	130 °C
Design Temperature External Pressure	130 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70 +IT
Efficiency, Longitudinal Seam	1.0
Efficiency, Head-to-Skirt or Circ. Seam	1.0
Weld is pre-Heated	No
Element From Node	140
Detail Type	Weight
Detail ID	Misc Margin
Dist. from "FROM" Node / Offset dist	1650 mm
Miscellaneous Weight	98060 N
Offset from Element Centerline	0 mm

Element#	15/27
Element From Node	150
Element To Node	160
Element Type	Skirt Support
Description	Int skirt 3
Distance "FROM" to "TO"	3000 mm
Skirt Outside Diameter	7102 mm
Diameter of Skirt at Base	6987 mm
Skirt Thickness	24 mm
Internal Corrosion Allowance	1.6 mm
Nominal Thickness	24 mm
External Corrosion Allowance	0 mm
Design Temperature Internal Pressure	130 °C
Design Temperature External Pressure	130 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70 +IT
Efficiency, Longitudinal Seam	1.0
Efficiency, Head-to-Skirt or Circ. Seam	1.0
Weld is pre-Heated	No

Element#	16/27
Element From Node	160
Element To Node	170
Element Type	Skirt Support
Description	Int skirt 4
Distance "FROM" to "TO"	1250 mm
Skirt Outside Diameter	7150 mm
Diameter of Skirt at Base	7102 mm
Skirt Thickness	24 mm
Internal Corrosion Allowance	1.6 mm
Nominal Thickness	24 mm
External Corrosion Allowance	0 mm
Design Temperature Internal Pressure	130 °C
Design Temperature External Pressure	130 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70 +IT
Efficiency, Longitudinal Seam	1.0
Efficiency, Head-to-Skirt or Circ. Seam	1.0
Weld is pre-Heated	No



Element From Node	160	
Detail Type	Platform	
Detail ID	PLAT-7	
Dist. from "FROM" Node / Offset dist	899.95	mm
Platform Start Angle (degrees)	90.0	
Platform End Angle (degrees)	270.0	
Platform Wind Area	2999999	mm ²
Platform Weight	35086	N
Platform Railing Weight	0	N/mm
Platform Grating Weight	0.017	kg/cm ²
Platform Width	1500	mm
Platform Height	1000	mm
Platform Clearance or End Offset	115	mm
Platform Force Coefficient	1.0	
Ladder Layout Angle	0.0	
Ladder Start Elevation	0	mm
Ladder End Elevation	0	mm
Unit Weight of Ladder	0	N/mm
Platform Length (top head platform)	0	mm

Element From Node	160	
Detail Type	Insulation	
Detail ID	Ins: 70	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	1250	mm
Thickness of Insulation	80	mm
Density	0.000352	kg/cm ³

Element#	17/27	
Element From Node	170	
Element To Node	180	
Element Type	Elliptical	
Description	122-D1-Btm head	
Distance "FROM" to "TO"	50	mm
Inside Diameter	7100	mm
Element Thickness	36	mm
Internal Corrosion Allowance	6	mm
Nominal Thickness	44	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	0.384	MPa
Design Temperature Internal Pressure	130	°C
Design External Pressure	0.05173	MPa
Design Temperature External Pressure	130	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-516 70 +IT	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Elliptical Head Factor	2.0	
Weld is pre-Heated	No	

Element From Node	170	
Detail Type	Liquid	
Detail ID	Liquid: 10	
Dist. from "FROM" Node / Offset dist	-1775	mm
Height/Length of Liquid	1825	mm
Liquid Density	0.001058	kg/cm ³

Element From Node	170	
Detail Type	Insulation	
Detail ID	Ins: 70	
Dist. from "FROM" Node / Offset dist	-1775	mm
Height/Length of Insulation	1825	mm
Thickness of Insulation	80	mm
Density	0.000352	kg/cm ³

Element From Node	170	
Detail Type	Nozzle	



Detail ID	C 48"
Dist. from "FROM" Node / Offset dist	0 mm
Nozzle Diameter	48 in.
Nozzle Schedule	None
Nozzle Class	150
Layout Angle	0.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	38438 N
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-516 70 +IT

Element From Node	170
Detail Type	Weight
Detail ID	LPVB
Dist. from "FROM" Node / Offset dist	-1800 mm
Miscellaneous Weight	2451.5 N
Offset from Element Centerline	0 mm

Element From Node	170
Detail Type	Weight
Detail ID	C Fy wt
Dist. from "FROM" Node / Offset dist	49.95 mm
Miscellaneous Weight	499616 N
Offset from Element Centerline	5350 mm

Element#	18/27
Element From Node	180
Element To Node	190
Element Type	Cylinder
Description	122-D1-Shell-1
Distance "FROM" to "TO"	3000 mm
Inside Diameter	7100 mm
Element Thickness	24 mm
Internal Corrosion Allowance	6 mm
Nominal Thickness	24 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	0.384 MPa
Design Temperature Internal Pressure	130 °C
Design External Pressure	0.05173 MPa
Design Temperature External Pressure	130 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70 +IT
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No

Element From Node	180
Detail Type	Platform
Detail ID	PLAT-8
Dist. from "FROM" Node / Offset dist	1849.9 mm
Platform Start Angle (degrees)	90.0
Platform End Angle (degrees)	270.0
Platform Wind Area	2999999 mm ²
Platform Weight	34889 N
Platform Railing Weight	0 N/mm
Platform Grating Weight	0.017 kg/cm ²
Platform Width	1500 mm
Platform Height	1000 mm
Platform Clearance or End Offset	115 mm
Platform Force Coefficient	1.0
Ladder Layout Angle	0.0
Ladder Start Elevation	0 mm
Ladder End Elevation	0 mm
Unit Weight of Ladder	0 N/mm
Platform Length (top head platform)	0 mm

Element From Node	180
Detail Type	Liquid



Detail ID	Liquid: 20	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Liquid	3000	mm
Liquid Density	0.001058	kg/cm ³
Element From Node	180	
Detail Type	Insulation	
Detail ID	Ins: 70	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	3000	mm
Thickness of Insulation	80	mm
Density	0.000352	kg/cm ³
Element From Node	180	
Detail Type	Ring	
Detail ID	3	
Dist. from "FROM" Node / Offset dist	2200.1	mm
Inside Diameter of Ring	7148	mm
Thickness of Ring	28	mm
Outside Diameter of Ring	7748	mm
Material Name	SA-516 70	+IT
Height of Section Ring	0	mm
Using Custom Stiffener Section	No	
Element From Node	180	
Detail Type	Weight	
Detail ID	Cleat wt	
Dist. from "FROM" Node / Offset dist	249.95	mm
Miscellaneous Weight	179450	N
Offset from Element Centerline	0	mm
Element From Node	180	
Detail Type	Weight	
Detail ID	Nozzles clad wt	
Dist. from "FROM" Node / Offset dist	49.95	mm
Miscellaneous Weight	2941.8	N
Offset from Element Centerline	0	mm

Element#	19/27	
Element From Node	190	
Element To Node	200	
Element Type	Cylinder	
Description	122-D1-Shell-2	
Distance "FROM" to "TO"	3000	mm
Inside Diameter	7100	mm
Element Thickness	24	mm
Internal Corrosion Allowance	6	mm
Nominal Thickness	24	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	0.384	MPa
Design Temperature Internal Pressure	130	°C
Design External Pressure	0.05173	MPa
Design Temperature External Pressure	130	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-516 70	+IT
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	

Element From Node	190	
Detail Type	Platform	
Detail ID	PLAT-9	
Dist. from "FROM" Node / Offset dist	2849.9	mm
Platform Start Angle (degrees)	270.0	
Platform End Angle (degrees)	90.0	
Platform Wind Area	2999999	mm ²
Platform Weight	34889	N
Platform Railing Weight	0	N/mm



Platform Grating Weight	0.017	kg/cm ²
Platform Width	1500	mm
Platform Height	1000	mm
Platform Clearance or End Offset	115	mm
Platform Force Coefficient	1.0	
Ladder Layout Angle	0.0	
Ladder Start Elevation	0	mm
Ladder End Elevation	0	mm
Unit Weight of Ladder	0	N/mm
Platform Length (top head platform)	0	mm

Element From Node	190	
Detail Type	Liquid	
Detail ID	Liquid: 30	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Liquid	3000	mm
Liquid Density	0.001058	kg/cm ³

Element From Node	190	
Detail Type	Insulation	
Detail ID	Ins: 70	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	3000	mm
Thickness of Insulation	80	mm
Density	0.000352	kg/cm ³

Element From Node	190	
Detail Type	Ring	
Detail ID	Ring R2	
Dist. from "FROM" Node / Offset dist	1950.1	mm
Inside Diameter of Ring	7148	mm
Thickness of Ring	28	mm
Outside Diameter of Ring	7748	mm
Material Name	SA-516 70 +IT	
Height of Section Ring	0	mm
Using Custom Stiffener Section	No	

Element From Node	190	
Detail Type	Nozzle	
Detail ID	MH-3 24"	
Dist. from "FROM" Node / Offset dist	600.05	mm
Nozzle Diameter	24	in.
Nozzle Schedule	None	
Nozzle Class	150	
Layout Angle	225.0	
Blind Flange (Y/N)	Y	
Weight of Nozzle (Used if > 0)	5111.9	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-516 70 +IT	

Element From Node	190	
Detail Type	Weight	
Detail ID	MH3 davit	
Dist. from "FROM" Node / Offset dist	600.05	mm
Miscellaneous Weight	490.3	N
Offset from Element Centerline	5000	mm

Element#	20/27	
Element From Node	200	
Element To Node	210	
Element Type	Cylinder	
Description	122-D1-Shell-3	
Distance "FROM" to "TO"	3000	mm
Inside Diameter	7100	mm
Element Thickness	22	mm
Internal Corrosion Allowance	6	mm
Nominal Thickness	22	mm
External Corrosion Allowance	0	mm



Design Internal Pressure	0.384	MPa
Design Temperature Internal Pressure	130	°C
Design External Pressure	0.05173	MPa
Design Temperature External Pressure	130	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-516 70	+IT
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	
Element From Node	200	
Detail Type	Liquid	
Detail ID	Liquid: 40	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Liquid	100	mm
Liquid Density	0.001058	kg/cm³
Element From Node	200	
Detail Type	Insulation	
Detail ID	Ins: 70	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	3000	mm
Thickness of Insulation	80	mm
Density	0.000352	kg/cm³
Element From Node	200	
Detail Type	Ring	
Detail ID	Ring R3	
Dist. from "FROM" Node / Offset dist	1450.1	mm
Inside Diameter of Ring	7144	mm
Thickness of Ring	28	mm
Outside Diameter of Ring	7644	mm
Material Name	SA-516 70	+IT
Height of Section Ring	0	mm
Using Custom Stiffener Section	No	
Element From Node	200	
Detail Type	Nozzle	
Detail ID	L2 2"	
Dist. from "FROM" Node / Offset dist	1299.9	mm
Nozzle Diameter	2	in.
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	0.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	103.72	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-105	+IT
Element From Node	200	
Detail Type	Nozzle	
Detail ID	L4 2"	
Dist. from "FROM" Node / Offset dist	1299.9	mm
Nozzle Diameter	2	in.
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	30.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	103.72	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-105	+IT
Element From Node	200	
Detail Type	Nozzle	
Detail ID	L6 2"	
Dist. from "FROM" Node / Offset dist	1299.9	mm
Nozzle Diameter	2	in.
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	60.0	



Blind Flange (Y/N) N
 Weight of Nozzle (Used if > 0) 103.72 N
 Grade of Attached Flange GR 1.1
 Nozzle Matl SA-105 +IT

Element From Node 200
 Detail Type Nozzle
 Detail ID L8 2"
 Dist. from "FROM" Node / Offset dist 1299.9 mm
 Nozzle Diameter 2 in.
 Nozzle Schedule None
 Nozzle Class 300
 Layout Angle 90.0
 Blind Flange (Y/N) N
 Weight of Nozzle (Used if > 0) 103.72 N
 Grade of Attached Flange GR 1.1
 Nozzle Matl SA-105 +IT

Element From Node 200
 Detail Type Nozzle
 Detail ID L10 2"
 Dist. from "FROM" Node / Offset dist 1299.9 mm
 Nozzle Diameter 2 in.
 Nozzle Schedule None
 Nozzle Class 300
 Layout Angle 120.0
 Blind Flange (Y/N) N
 Weight of Nozzle (Used if > 0) 103.72 N
 Grade of Attached Flange GR 1.1
 Nozzle Matl SA-105 +IT

Element From Node 200
 Detail Type Nozzle
 Detail ID B 36"
 Dist. from "FROM" Node / Offset dist 2300.1 mm
 Nozzle Diameter 36 in.
 Nozzle Schedule None
 Nozzle Class 150
 Layout Angle 180.0
 Blind Flange (Y/N) N
 Weight of Nozzle (Used if > 0) 4917.7 N
 Grade of Attached Flange GR 1.1
 Nozzle Matl SA-516 70 +IT

Element From Node 200
 Detail Type Nozzle
 Detail ID K 6"
 Dist. from "FROM" Node / Offset dist 2500 mm
 Nozzle Diameter 6 in.
 Nozzle Schedule 160
 Nozzle Class 150
 Layout Angle 0.0
 Blind Flange (Y/N) N
 Weight of Nozzle (Used if > 0) 415.97 N
 Grade of Attached Flange GR 1.1
 Nozzle Matl SA-106 B +IT

Element From Node 200
 Detail Type Weight
 Detail ID K Fy wt
 Dist. from "FROM" Node / Offset dist 2500 mm
 Miscellaneous Weight 51923 N
 Offset from Element Centerline 4808 mm

Element From Node 200
 Detail Type Weight
 Detail ID B Fy wt
 Dist. from "FROM" Node / Offset dist 2300 mm
 Miscellaneous Weight 742853 N
 Offset from Element Centerline 5189 mm



Element#	21/27
Element From Node	210
Element To Node	220
Element Type	Cylinder
Description	122-D1-Shell-4
Distance "FROM" to "TO"	3000 mm
Inside Diameter	7100 mm
Element Thickness	22 mm
Internal Corrosion Allowance	6 mm
Nominal Thickness	22 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	0.384 MPa
Design Temperature Internal Pressure	130 °C
Design External Pressure	0.05173 MPa
Design Temperature External Pressure	130 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70 +IT
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No

Element From Node	210
Detail Type	Platform
Detail ID	PLAT-10
Dist. from "FROM" Node / Offset dist	849.96 mm
Platform Start Angle (degrees)	90.0
Platform End Angle (degrees)	270.0
Platform Wind Area	2999999 mm ²
Platform Weight	34873 N
Platform Railing Weight	0 N/mm
Platform Grating Weight	0.017 kg/cm ²
Platform Width	1500 mm
Platform Height	1000 mm
Platform Clearance or End Offset	115 mm
Platform Force Coefficient	1.0
Ladder Layout Angle	0.0
Ladder Start Elevation	0 mm
Ladder End Elevation	0 mm
Unit Weight of Ladder	0 N/mm
Platform Length (top head platform)	0 mm

Element From Node	210
Detail Type	Packing
Detail ID	LPPack1
Dist. from "FROM" Node / Offset dist	1800.1 mm
Height of Packed Section	1199.9 mm
Density	0.000166 kg/cm ³
Percent Volume Holdup	25.0
Specific Gravity of Packing Liquid	1.0599999

Element From Node	210
Detail Type	Insulation
Detail ID	Ins: 70
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Insulation	3000 mm
Thickness of Insulation	80 mm
Density	0.000352 kg/cm ³

Element From Node	210
Detail Type	Ring
Detail ID	Ring R4
Dist. from "FROM" Node / Offset dist	949.95 mm
Inside Diameter of Ring	7144 mm
Thickness of Ring	28 mm
Outside Diameter of Ring	7644 mm
Material Name	SA-516 70 +IT
Height of Section Ring	0 mm



Using Custom Stiffener Section	No
Element From Node	210
Detail Type	Nozzle
Detail ID	I 28"
Dist. from "FROM" Node / Offset dist	1500.1 mm
Nozzle Diameter	28 in.
Nozzle Schedule	None
Nozzle Class	150
Layout Angle	0.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	3662.7 N
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-516 70 +IT
Element From Node	210
Detail Type	Weight
Detail ID	LPB#1btm grid
Dist. from "FROM" Node / Offset dist	1800.1 mm
Miscellaneous Weight	39224 N
Offset from Element Centerline	0 mm
Element From Node	210
Detail Type	Weight
Detail ID	LPB#1TA
Dist. from "FROM" Node / Offset dist	1800.1 mm
Miscellaneous Weight	3432.1 N
Offset from Element Centerline	0 mm
Element From Node	210
Detail Type	Weight
Detail ID	I Fy wt
Dist. from "FROM" Node / Offset dist	1500.1 mm
Miscellaneous Weight	488319 N
Offset from Element Centerline	5080 mm

Element#	22/27
Element From Node	220
Element To Node	230
Element Type	Cylinder
Description	122-D1-Shell-5
Distance "FROM" to "TO"	2800 mm
Inside Diameter	7100 mm
Element Thickness	20 mm
Internal Corrosion Allowance	6 mm
Nominal Thickness	20 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	0.384 MPa
Design Temperature Internal Pressure	130 °C
Design External Pressure	0.05173 MPa
Design Temperature External Pressure	130 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70 +IT
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No

Element From Node	220
Detail Type	Platform
Detail ID	PLAT-11
Dist. from "FROM" Node / Offset dist	1849.9 mm
Platform Start Angle (degrees)	270.0
Platform End Angle (degrees)	90.0
Platform Wind Area	2999999 mm ²
Platform Weight	34858 N
Platform Railing Weight	0 N/mm
Platform Grating Weight	0.017 kg/cm ²
Platform Width	1500 mm



Platform Height	1000	mm
Platform Clearance or End Offset	115	mm
Platform Force Coefficient	1.0	
Ladder Layout Angle	0.0	
Ladder Start Elevation	0	mm
Ladder End Elevation	0	mm
Unit Weight of Ladder	0	N/mm
Platform Length (top head platform)	0	mm
Element From Node	220	
Detail Type	Packing	
Detail ID	LPPack2	
Dist. from "FROM" Node / Offset dist	0	mm
Height of Packed Section	2800	mm
Density	0.000166	kg/cm ³
Percent Volume Holdup	25.0	
Specific Gravity of Packing Liquid	1.0599999	
Element From Node	220	
Detail Type	Insulation	
Detail ID	Ins: 70	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	2800	mm
Thickness of Insulation	80	mm
Density	0.000352	kg/cm ³
Element From Node	220	
Detail Type	Ring	
Detail ID	Ring R5	
Dist. from "FROM" Node / Offset dist	449.95	mm
Inside Diameter of Ring	7140	mm
Thickness of Ring	28	mm
Outside Diameter of Ring	7640	mm
Material Name	SA-516 70 +IT	
Height of Section Ring	0	mm
Using Custom Stiffener Section	No	
Element From Node	220	
Detail Type	Nozzle	
Detail ID	L1 2"	
Dist. from "FROM" Node / Offset dist	1400	mm
Nozzle Diameter	2	in.
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	0.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	103.72	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-105 +IT	
Element From Node	220	
Detail Type	Nozzle	
Detail ID	L3 2"	
Dist. from "FROM" Node / Offset dist	1400	mm
Nozzle Diameter	2	in.
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	30.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	103.72	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-105 +IT	
Element From Node	220	
Detail Type	Nozzle	
Detail ID	L5 2"	
Dist. from "FROM" Node / Offset dist	1400	mm
Nozzle Diameter	2	in.
Nozzle Schedule	None	
Nozzle Class	300	



Layout Angle	60.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	103.72	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-105	+IT
Element From Node	220	
Detail Type	Nozzle	
Detail ID	L7 2"	
Dist. from "FROM" Node / Offset dist	1400	mm
Nozzle Diameter	2	in.
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	90.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	103.72	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-105	+IT
Element From Node	220	
Detail Type	Nozzle	
Detail ID	L9 2"	
Dist. from "FROM" Node / Offset dist	1400	mm
Nozzle Diameter	2	in.
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	120.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	103.72	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-105	+IT

Element#	23/27
Element From Node	230
Element To Node	240
Element Type	Cylinder
Description	122-D1-Shell-6
Distance "FROM" to "TO"	3000 mm
Inside Diameter	7100 mm
Element Thickness	16 mm
Internal Corrosion Allowance	1 mm
Nominal Thickness	16 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	0.384 MPa
Design Temperature Internal Pressure	130 °C
Design External Pressure	0.05173 MPa
Design Temperature External Pressure	130 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70 +IT
Efficiency, Longitudinal Seam	0.85
Efficiency, Circumferential Seam	0.85
Weld is pre-Heated	No

Element From Node	230
Detail Type	Packing
Detail ID	LPPack3
Dist. from "FROM" Node / Offset dist	0 mm
Height of Packed Section	1000 mm
Density	0.000166 kg/cm³
Percent Volume Holdup	25.0
Specific Gravity of Packing Liquid	1.059999

Element From Node	230
Detail Type	Insulation
Detail ID	Ins: 70
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Insulation	3000 mm
Thickness of Insulation	80 mm



Density	0.000352	kg/cm ³
Element From Node	230	
Detail Type	Lining	
Detail ID	Clad SS304L	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Lining	3000	mm
Thickness of Lining	3	mm
Density	0.008	kg/cm ³
Element From Node	230	
Detail Type	Ring	
Detail ID	Ring R6	
Dist. from "FROM" Node / Offset dist	149.95	mm
Inside Diameter of Ring	7134	mm
Thickness of Ring	28	mm
Outside Diameter of Ring	7634	mm
Material Name	SA-516 70	+IT
Height of Section Ring	0	mm
Using Custom Stiffener Section	No	
Element From Node	230	
Detail Type	Ring	
Detail ID	Ring R7	
Dist. from "FROM" Node / Offset dist	2649.9	mm
Inside Diameter of Ring	7134	mm
Thickness of Ring	28	mm
Outside Diameter of Ring	7634	mm
Material Name	SA-516 70	+IT
Height of Section Ring	0	mm
Using Custom Stiffener Section	No	
Element From Node	230	
Detail Type	Nozzle	
Detail ID	MH-2 24"	
Dist. from "FROM" Node / Offset dist	2100.1	mm
Nozzle Diameter	24	in.
Nozzle Schedule	None	
Nozzle Class	150	
Layout Angle	225.0	
Blind Flange (Y/N)	Y	
Weight of Nozzle (Used if > 0)	4577.1	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-516 70	+IT
Element From Node	230	
Detail Type	Weight	
Detail ID	LPB#1HDG	
Dist. from "FROM" Node / Offset dist	999.95	mm
Miscellaneous Weight	8825.4	N
Offset from Element Centerline	0	mm
Element From Node	230	
Detail Type	Weight	
Detail ID	LPB#1dist. de	
Dist. from "FROM" Node / Offset dist	1599.9	mm
Miscellaneous Weight	56384	N
Offset from Element Centerline	0	mm
Element From Node	230	
Detail Type	Weight	
Detail ID	LPB#1dist.TA	
Dist. from "FROM" Node / Offset dist	1599.9	mm
Miscellaneous Weight	3432.1	N
Offset from Element Centerline	0	mm
Element From Node	230	
Detail Type	Weight	
Detail ID	LPSHroud	
Dist. from "FROM" Node / Offset dist	2800.1	mm



Miscellaneous Weight	96099	N
Offset from Element Centerline	0	mm
Element From Node	230	
Detail Type	Weight	
Detail ID	MH2 davit	
Dist. from "FROM" Node / Offset dist	2100.1	mm
Miscellaneous Weight	490.3	N
Offset from Element Centerline	5000	mm

Element#	24/27	
Element From Node	240	
Element To Node	250	
Element Type	Cylinder	
Description	122-D1-Shell-7	
Distance "FROM" to "TO"	3000	mm
Inside Diameter	7100	mm
Element Thickness	16	mm
Internal Corrosion Allowance	1	mm
Nominal Thickness	16	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	0.384	MPa
Design Temperature Internal Pressure	130	°C
Design External Pressure	0.05173	MPa
Design Temperature External Pressure	130	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-516 70 +IT	
Efficiency, Longitudinal Seam	0.85	
Efficiency, Circumferential Seam	0.85	
Weld is pre-Heated	No	

Element From Node	240	
Detail Type	Platform	
Detail ID	PLAT-12	
Dist. from "FROM" Node / Offset dist	49.955	mm
Platform Start Angle (degrees)	90.0	
Platform End Angle (degrees)	270.0	
Platform Wind Area	2999999	mm ²
Platform Weight	34826	N
Platform Railing Weight	0	N/mm
Platform Grating Weight	0.017	kg/cm ²
Platform Width	1500	mm
Platform Height	1000	mm
Platform Clearance or End Offset	115	mm
Platform Force Coefficient	1.0	
Ladder Layout Angle	0.0	
Ladder Start Elevation	0	mm
Ladder End Elevation	0	mm
Unit Weight of Ladder	0	N/mm
Platform Length (top head platform)	0	mm

Element From Node	240	
Detail Type	Insulation	
Detail ID	Ins: 70	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	3000	mm
Thickness of Insulation	80	mm
Density	0.000352	kg/cm ³

Element From Node	240	
Detail Type	Lining	
Detail ID	Clad SS304L	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Lining	3000	mm
Thickness of Lining	3	mm
Density	0.008	kg/cm ³

Element From Node	240	
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Detail Type	Ring	
Detail ID	Ring R8	
Dist. from "FROM" Node / Offset dist	2150	mm
Inside Diameter of Ring	7134	mm
Thickness of Ring	28	mm
Outside Diameter of Ring	7634	mm
Material Name	SA-516 70	+IT
Height of Section Ring	0	mm
Using Custom Stiffener Section	No	
Element From Node	240	
Detail Type	Nozzle	
Detail ID	UC1 24"	
Dist. from "FROM" Node / Offset dist	199.96	mm
Nozzle Diameter	24	in.
Nozzle Schedule	None	
Nozzle Class	150	
Layout Angle	90.0	
Blind Flange (Y/N)	Y	
Weight of Nozzle (Used if > 0)	2725.8	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-516 70	+IT
Element From Node	240	
Detail Type	Nozzle	
Detail ID	MH-1 24"	
Dist. from "FROM" Node / Offset dist	1400	mm
Nozzle Diameter	24	in.
Nozzle Schedule	None	
Nozzle Class	150	
Layout Angle	225.0	
Blind Flange (Y/N)	Y	
Weight of Nozzle (Used if > 0)	4837.7	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-516 70	+IT
Element From Node	240	
Detail Type	Weight	
Detail ID	LPdwncm PP dea	
Dist. from "FROM" Node / Offset dist	2750	mm
Miscellaneous Weight	784.48	N
Offset from Element Centerline	0	mm
Element From Node	240	
Detail Type	Weight	
Detail ID	MH1 davit	
Dist. from "FROM" Node / Offset dist	1400	mm
Miscellaneous Weight	490.3	N
Offset from Element Centerline	5000	mm

Element#	25/27	
Element From Node	250	
Element To Node	260	
Element Type	Cylinder	
Description	122-D1-Shell-8	
Distance "FROM" to "TO"	3000	mm
Inside Diameter	7100	mm
Element Thickness	16	mm
Internal Corrosion Allowance	1	mm
Nominal Thickness	16	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	0.384	MPa
Design Temperature Internal Pressure	130	°C
Design External Pressure	0.05173	MPa
Design Temperature External Pressure	130	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-516 70	+IT
Efficiency, Longitudinal Seam	0.85	



Efficiency, Circumferential Seam	0.85	
Weld is pre-Heated	No	
Element From Node	250	
Detail Type	Platform	
Detail ID	PLAT-13	
Dist. from "FROM" Node / Offset dist	1500.1	mm
Platform Start Angle (degrees)	270.0	
Platform End Angle (degrees)	90.0	
Platform Wind Area	25270700	mm ²
Platform Weight	54709	N
Platform Railing Weight	0	N/mm
Platform Grating Weight	0.017	kg/cm ²
Platform Width	1500	mm
Platform Height	1000	mm
Platform Clearance or End Offset	115	mm
Platform Force Coefficient	1.0	
Ladder Layout Angle	270.0	
Ladder Start Elevation	0	mm
Ladder End Elevation	54800	mm
Unit Weight of Ladder	0.3628	N/mm
Platform Length (top head platform)	0	mm
Element From Node	250	
Detail Type	Insulation	
Detail ID	Ins: 70	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	3000	mm
Thickness of Insulation	80	mm
Density	0.000352	kg/cm ³
Element From Node	250	
Detail Type	Lining	
Detail ID	Clad SS304L	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Lining	3000	mm
Thickness of Lining	3	mm
Density	0.008	kg/cm ³
Element From Node	250	
Detail Type	Ring	
Detail ID	Ring R9	
Dist. from "FROM" Node / Offset dist	1650	mm
Inside Diameter of Ring	7134	mm
Thickness of Ring	28	mm
Outside Diameter of Ring	7634	mm
Material Name	SA-516 70	+IT
Height of Section Ring	0	mm
Using Custom Stiffener Section	No	
Element From Node	250	
Detail Type	Nozzle	
Detail ID	S1 1"	
Dist. from "FROM" Node / Offset dist	1200	mm
Nozzle Diameter	1	in.
Nozzle Schedule	None	
Nozzle Class	150	
Layout Angle	225.0	
Blind Flange (Y/N)	Y	
Weight of Nozzle (Used if > 0)	48.694	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-105	+IT
Element From Node	250	
Detail Type	Nozzle	
Detail ID	H1 42"	
Dist. from "FROM" Node / Offset dist	800.05	mm
Nozzle Diameter	42	in.
Nozzle Schedule	None	
Nozzle Class	150	



Layout Angle 180.0
 Blind Flange (Y/N) N
 Weight of Nozzle (Used if > 0) 5933.7 N
 Grade of Attached Flange GR 1.1
 Nozzle Matl SA-516 70 +IT

Element From Node 250
 Detail Type Nozzle
 Detail ID H2 42"
 Dist. from "FROM" Node / Offset dist 800.05 mm
 Nozzle Diameter 42 in.
 Nozzle Schedule None
 Nozzle Class 150
 Layout Angle 0.0
 Blind Flange (Y/N) N
 Weight of Nozzle (Used if > 0) 6131.2 N
 Grade of Attached Flange GR 1.1
 Nozzle Matl SA-516 70 +IT

Element From Node 250
 Detail Type Tray
 Detail ID Tray#1-3
 Dist. from "FROM" Node / Offset dist 650.05 mm
 Tray Spacing 900 mm
 Number of Trays 3
 Tray Weight per Unit Area 0.0067 kg/cm²
 Height of Liquid on Tray 0 mm
 Density of Liquid on Tray 0 kg/cm³
 Support Ring and Bolting Bar Weight 2059.3 N

Element From Node 250
 Detail Type Weight
 Detail ID LPGallary
 Dist. from "FROM" Node / Offset dist 650.05 mm
 Miscellaneous Weight 30399 N
 Offset from Element Centerline 0 mm

Element From Node 250
 Detail Type Weight
 Detail ID LPFPGdead
 Dist. from "FROM" Node / Offset dist 2450.1 mm
 Miscellaneous Weight 1470.9 N
 Offset from Element Centerline 0 mm

Element From Node 250
 Detail Type Weight
 Detail ID LPFPGTA
 Dist. from "FROM" Node / Offset dist 2450.1 mm
 Miscellaneous Weight 196.12 N
 Offset from Element Centerline 0 mm

Element From Node 250
 Detail Type Weight
 Detail ID H1/H2 Fy wt
 Dist. from "FROM" Node / Offset dist 799.95 mm
 Miscellaneous Weight 973990 N
 Offset from Element Centerline 5257 mm

Element# 26/27
 Element From Node 260
 Element To Node 270
 Element Type Cylinder
 Description 122-D1-Shell-9
 Distance "FROM" to "TO" 2900 mm
 Inside Diameter 7100 mm
 Element Thickness 14 mm
 Internal Corrosion Allowance 1 mm
 Nominal Thickness 14 mm



External Corrosion Allowance	0	mm
Design Internal Pressure	0.384	MPa
Design Temperature Internal Pressure	130	°C
Design External Pressure	0.05173	MPa
Design Temperature External Pressure	130	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-516 70	+IT
Efficiency, Longitudinal Seam	0.85	
Efficiency, Circumferential Seam	0.85	
Weld is pre-Heated	No	

Element From Node	260	
Detail Type	Platform	
Detail ID	PL-14	
Dist. from "FROM" Node / Offset dist	1000	mm
Platform Start Angle (degrees)	90.0	
Platform End Angle (degrees)	270.0	
Platform Wind Area	5063040	mm ²
Platform Weight	35920	N
Platform Railing Weight	0	N/mm
Platform Grating Weight	0.017	kg/cm ²
Platform Width	1500	mm
Platform Height	1000	mm
Platform Clearance or End Offset	115	mm
Platform Force Coefficient	1.2	
Ladder Layout Angle	180.0	
Ladder Start Elevation	0	mm
Ladder End Elevation	3000	mm
Unit Weight of Ladder	0.37	N/mm
Platform Length (top head platform)	0	mm

Element From Node	260	
Detail Type	Insulation	
Detail ID	Ins: 70	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	2900	mm
Thickness of Insulation	80	mm
Density	0.000352	kg/cm ³

Element From Node	260	
Detail Type	Lining	
Detail ID	Clad SS304L	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Lining	2900	mm
Thickness of Lining	3	mm
Density	0.008	kg/cm ³

Element From Node	260	
Detail Type	Ring	
Detail ID	Ring R10	
Dist. from "FROM" Node / Offset dist	1150	mm
Inside Diameter of Ring	7138	mm
Thickness of Ring	28	mm
Outside Diameter of Ring	7638	mm
Material Name	SA-516 70	+IT
Height of Section Ring	0	mm
Using Custom Stiffener Section	No	

Element From Node	260	
Detail Type	Weight	
Detail ID	Lifting wt	
Dist. from "FROM" Node / Offset dist	1950	mm
Miscellaneous Weight	200042	N
Offset from Element Centerline	0	mm



Element Type	Elliptical
Description	122-D1-Top head
Distance "FROM" to "TO"	50 mm
Inside Diameter	7100 mm
Element Thickness	16 mm
Internal Corrosion Allowance	1 mm
Nominal Thickness	20 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	0.384 MPa
Design Temperature Internal Pressure	130 °C
Design External Pressure	0.05173 MPa
Design Temperature External Pressure	130 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70 +IT
Efficiency, Longitudinal Seam	0.85
Efficiency, Circumferential Seam	0.85
Elliptical Head Factor	2.0
Weld is pre-Heated	No
Element From Node	270
Detail Type	Platform
Detail ID	TOP PLT
Dist. from "FROM" Node / Offset dist	2300.1 mm
Platform Start Angle (degrees)	0.0
Platform End Angle (degrees)	270.0
Platform Wind Area	9328800 mm ²
Platform Weight	95451 N
Platform Railing Weight	0 N/mm
Platform Grating Weight	0.017 kg/cm ²
Platform Width	7500 mm
Platform Height	1000 mm
Platform Clearance or End Offset	3750 mm
Platform Force Coefficient	1.0
Ladder Layout Angle	315.0
Ladder Start Elevation	0 mm
Ladder End Elevation	4500 mm
Unit Weight of Ladder	0.3628 N/mm
Platform Length (top head platform)	7500 mm
Element From Node	270
Detail Type	Insulation
Detail ID	Ins: 70
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Insulation	1851.5 mm
Thickness of Insulation	80 mm
Density	0.000352 kg/cm ³
Element From Node	270
Detail Type	Lining
Detail ID	Clad SS304L
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Lining	1825 mm
Thickness of Lining	3 mm
Density	0.008 kg/cm ³
Element From Node	270
Detail Type	Nozzle
Detail ID	F 42"
Dist. from "FROM" Node / Offset dist	0 mm
Nozzle Diameter	42 in.
Nozzle Schedule	None
Nozzle Class	150
Layout Angle	0.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	8210.9 N
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-516 70 +IT
Element From Node	270
Detail Type	Weight



Detail ID	LPDemister	
Dist. from "FROM" Node / Offset dist	0	mm
Miscellaneous Weight	9413.8	N
Offset from Element Centerline	0	mm
Element From Node	270	
Detail Type	Weight	
Detail ID	PSV	
Dist. from "FROM" Node / Offset dist	1825	mm
Miscellaneous Weight	73545	N
Offset from Element Centerline	3750	mm
Element From Node	270	
Detail Type	Weight	
Detail ID	LPDemisterTA	
Dist. from "FROM" Node / Offset dist	0	mm
Miscellaneous Weight	1470.9	N
Offset from Element Centerline	0	mm
Element From Node	270	
Detail Type	Weight	
Detail ID	F Fy wt	
Dist. from "FROM" Node / Offset dist	0.04186	mm
Miscellaneous Weight	275666	N
Offset from Element Centerline	5259	mm
Element From Node	270	
Detail Type	Weight	
Detail ID	LPDavit	
Dist. from "FROM" Node / Offset dist	0.04186	mm
Miscellaneous Weight	8825.4	N
Offset from Element Centerline	3600	mm
Element From Node	270	
Detail Type	For./Mom.	
Detail ID	FFxFz	
Dist. from "FROM" Node / Offset dist	1800	mm
Force in X Direction	57730	N
Force in Y Direction	-86592	N
Force in Z Direction	57730	N
Moment about X Axis	169792560	N-mm
Moment about Y Axis	0	N-mm
Moment about Z Axis	169792560	N-mm
Force/Moment Combination Method	SRSS	



XY Coordinate Calculations:

From	To	X (Horiz.) mm	Y (Vert.) mm	DX (Horiz.) mm	DY (Vert.) mm
Skirt-3	...	...	3000	...	3000
Skirt-2	...	...	5700	...	2700
Skirt-1	...	...	7200	...	1500
163-D- BTM dish	...	...	7200	...	...
163-D-Shell-1	...	...	10200	...	3000
163-D-Shell-2	...	...	13200	...	3000
163-D-Shell-3	...	...	15100	...	1900
163-D-Shell-4	...	...	18100	...	3000
163-D-Shell-5	...	...	21100	...	3000
163-D-Shell-6	...	...	23100	...	2000
163-D-Shell-7	...	...	24600	...	1500
163-D-Top head	...	...	24600	...	...
Int skirt 1	...	...	25850	...	1250
Int skirt 2	...	...	28850	...	3000
Int skirt 3	...	...	31850	...	3000
Int skirt 4	...	...	33100	...	1250
122-D1-Btm head	...	...	33150	...	50
122-D1-Shell-1	...	...	36150	...	3000
122-D1-Shell-2	...	...	39150	...	3000
122-D1-Shell-3	...	...	42150	...	3000
122-D1-Shell-4	...	...	45150	...	3000
122-D1-Shell-5	...	...	47950	...	2800
122-D1-Shell-6	...	...	50950	...	3000
122-D1-Shell-7	...	...	53950	...	3000
122-D1-Shell-8	...	...	56950	...	3000
122-D1-Shell-9	...	...	59850	...	2900
122-D1-Top head	...	...	59900	...	50

The calculated dimensions are based on the given element lengths. Due to variability in manufacturing (weld gaps etc.), the Tangent to Tangent length may not be exact. The length includes gasket thicknesses and raised face heights.

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Internal Pressure Results Summary:

Element Thickness, Pressure, Diameter and Allowable Stress :

From	To	Int. Press + Liq. Hd MPa	Nominal Thickness mm	Total Corr Allowance mm	Element Diameter mm	Allowable Stress(SE) MPa	Element#
Skirt-3	...		42	1.6	6762	...	1
Skirt-2	...		36	1.6	6768	...	2
Skirt-1	...		42	1.6	6762	...	3
163-D- BTM dish	4.2083		68	6	6738	156.4	4
163-D-Shell-1	4.1733		98	6	6700	156.4	5
163-D-Shell-2	4.1421		98	6	6700	156.4	6
163-D-Shell-3	4.1109		96	6	6700	156.4	7
163-D-Shell-4	4.1047		92	1	6706	156.4	8
163-D-Shell-5	4.1047		92	1	6706	156.4	9
163-D-Shell-6	4.1047		92	1	6706	156.4	10
163-D-Shell-7	4.1047		92	1	6706	156.4	11
163-D-Top head	4.1047		58	1	6740	156.4	12
Int skirt 1	...		26	1.6	6823	...	13
Int skirt 2	...		24	1.6	6939	...	14
Int skirt 3	...		24	1.6	7054	...	15
Int skirt 4	...		24	1.6	7102	...	16
122-D1-Btm head	0.4703		44	6	7100	156.4	17
122-D1-Shell-1	0.4514		24	6	7100	156.4	18
122-D1-Shell-2	0.4202		24	6	7100	156.4	19
122-D1-Shell-3	0.389		22	6	7100	156.4	20
122-D1-Shell-4	0.388		22	6	7100	156.4	21
122-D1-Shell-5	0.388		20	6	7100	156.4	22
122-D1-Shell-6	0.388		16	1	7100	132.94	23
122-D1-Shell-7	0.388		16	1	7100	132.94	24
122-D1-Shell-8	0.388		16	1	7100	117.3	25
122-D1-Shell-9	0.388		14	1	7100	132.94	26
122-D1-Top head	0.388		20	1	7100	132.94	27

Element Required Thickness and MAWP :

From	To	Design Pressure MPa	M.A.W.P. Corroded MPa	M.A.P. New & Cold MPa	Minimum Thickness mm	Required Thickness mm	Element#
Skirt-3	...		No Calc	No Calc	42	No Calc	1
Skirt-2	...		No Calc	No Calc	36	No Calc	2
Skirt-1	...		No Calc	No Calc	42	No Calc	3
163-D- BTM dish	4.1007	4.85757	6.17843	60	51.7129		4
163-D-Shell-1	4.1007	4.15719	5.04595	98	96.7547		5
163-D-Shell-2	4.1007	4.18837	5.04595	98	96.0677		6
163-D-Shell-3	4.1007	4.12878	4.94441	96	95.3812		7
163-D-Shell-4	4.1007	4.18287	4.73696	92	90.1908		8
163-D-Shell-5	4.1007	4.18287	4.73696	92	90.1908		9
163-D-Shell-6	4.1007	4.18287	4.73696	92	90.1908		10
163-D-Shell-7	4.1007	4.18287	4.73696	92	90.1908		11
163-D-Top head	4.1007	4.69294	5.35933	52	45.5271		12
Int skirt 1	...	No Calc	No Calc	26	No Calc		13
Int skirt 2	...	No Calc	No Calc	24	No Calc		14
Int skirt 3	...	No Calc	No Calc	24	No Calc		15
Int skirt 4	...	No Calc	No Calc	24	No Calc		16
122-D1-Btm head	0.384	1.37739	1.96629	36	16.7779		17
122-D1-Shell-1	0.384	0.7223	1.17912	24	16.2777		18
122-D1-Shell-2	0.384	0.75347	1.17912	24	15.5672		19
122-D1-Shell-3	0.384	0.69709	1.08115	22	14.8563		20
122-D1-Shell-4	0.384	0.69813	1.08115	22	14.8326		21
122-D1-Shell-5	0.384	0.61054	0.98315	20	14.8326		22
122-D1-Shell-6	0.384	0.55638	0.66892	16	11.3792		23
122-D1-Shell-7	0.384	0.55638	0.66892	16	11.3792		24
122-D1-Shell-8	0.384	0.49045	0.52749	16	12.7651		25
122-D1-Shell-9	0.384	0.48179	0.58547	14	11.3792		26



122-D1-Top head	0.384	0.6193	0.74398	16	10.8361	27
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Minimum	0	1
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Warning: The flange MAWP was less than the design pressure!

MAWP: 0 MPa, limited by: 122-D1-Shell-9.

Warning: The computed MAWP is less than the design pressure.

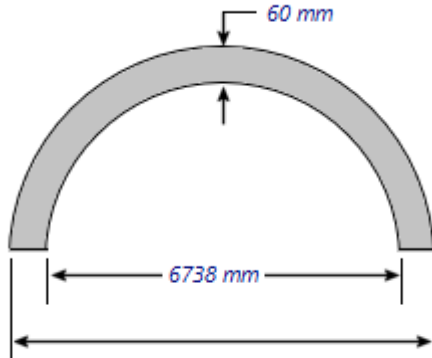
Elements Suitable for Design Internal Pressure.

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Internal Pressure Calculation Results:

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Internal Pressure Results for: 163-D- BTM dish

Spherical Head Design Information:					
Design Pressure	P	4.208 MPa	Int. Design Temperature	130 °C	
Material		SA-516 70	Ext. Design Temperature	0 °C	
Length		0.000 mm	External Pressure Chart	CS-2	
			UNS Number	K02700	
Int. Corr. All.	c	6.0000 mm	Allowable Stress (ope)	S	156.4 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	175.0 MPa
Inside Diameter	Di	6738.000 mm	Actual Stress	130.5 MPa	
Spec. Min. Thk t,e, or sa		60.0000 mm	Spec. Nominal Thk.	68.000 mm	
Surface Area		74223088.0 mm²	SG of Contents	1.060	
Element Volume		80101.3 ltr	Weight of Contents	832090.2 N	
Empty Weight		376035.0 N	Operating Weight	1208125.1 N	
			Comp. All. Ope. Stress	103.4 MPa	

Radiography Information:			
...			...
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Tolerance for Formed Heads per 4.3.2.2:

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 84.225 mm
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 42.113 mm

Material treatment options: +IT

Spherical Shell Calculation - Section 4.3.5.1

Computed Minimum Required Thickness [t]:
 $= D/2(\exp(0.5P/(S \cdot E)) - 1) + ci + co$
 $= 6750.0/2(\exp(0.5 \cdot 4.208/(156.4 \cdot 1.0)) - 1) + 6.0 + 0.0$
 $= 45.7129 + 6.0 + 0.0 = 51.7129 \text{ mm}$

Computed Maximum Allowable Working Pressure [MAWP]:
 $= 2 \cdot S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$
 $= 2 \cdot 156.4 \cdot 1.0 \cdot \ln((2 \cdot 54.0 + 6750.0) / 6750.0) - 0.108$
 $= 4.965 - 0.108 = 4.8576 \text{ MPa}$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:
 $= 2 \cdot S \cdot E \cdot \ln((2 \cdot t + D) / D)$
 $= 2 \cdot 175.0 \cdot 1.0 \cdot \ln((2 \cdot 60.0 + 6738.0) / 6738.0)$
 $= 6.1784 \text{ MPa}$



Computed Stress Intensity at Design Pressure [Si]:

$$\begin{aligned} &= P \cdot D^2 / (E (D_o^2 - D^2)) \\ &= 4.208 \cdot 6750.0^2 / (1.0 (6858.0^2 - 6750.0^2)) \\ &= 130.466 \text{ MPa} \end{aligned}$$

Computed Membrane Stress at Design Pressure [Sm]:

$$\begin{aligned} &= 0.5 \cdot P / (E \cdot \ln((2 \cdot t + D) / D)) \\ &= 0.5 \cdot 4.208 / (1.0 \cdot \ln((2 \cdot 54.0 + 6750.0) / 6750.0)) \\ &= 132.559 \text{ MPa} \end{aligned}$$

Percent Fiber Elongation per Table 6.1

For Heads formed from Segments $(75 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$	1.499 %
For one Piece doubly Curved Heads $100 \cdot \ln(D_b / (D_f - 2t))$	36.656 %

Where D_b is the blank diameter for one piece doubly curved surfaces [D_b]:

$$\begin{aligned} &= \text{square root}(4 \cdot \text{Head Surface Area} / 3.14159) \\ &= \text{square root}(4 \cdot 74223088 / 3.14159) \\ &= 9721.305 \text{ mm} \end{aligned}$$

MDMT Calculations in the Spherical Portion:

Note: This Element/Detail was specified as being Impact Tested.

Note:

Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

Tolerance for Formed Heads per 4.3.2.2:

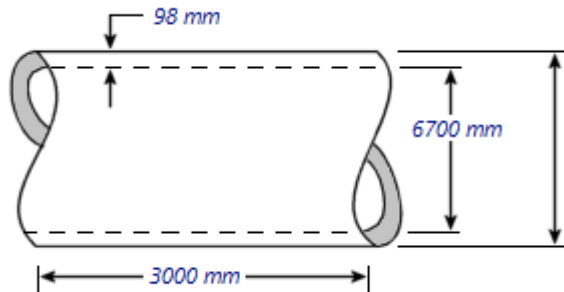
Head inner surface maximum deviation outside the specified shape, 1.25% of D: 84.225 mm
Head inner surface maximum deviation inside the specified shape, 0.625% of D: 42.113 mm

Material treatment options: +IT

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Internal Pressure Calculation Results:

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Internal Pressure Results for: 163-D-Shell-1

Cylindrical Shell Design Information:					
Design Pressure	P	4.173 MPa	Int. Design Temperature		130 °C
Material		SA-516 70	Ext. Design Temperature		0 °C
Length		3000.000 mm	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	6.0000 mm	Allowable Stress (ope)	S	156.4 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	175.0 MPa
Inside Diameter	Di	6700.000 mm	Actual Stress		152.2 MPa
Spec. Min. Thk t,e, or sa		98.0000 mm	Spec. Nominal Thk.		98.000 mm
Surface Area		64993268.0 mm²	SG of Contents		1.060
Element Volume		105788.5 ltr	Weight of Contents		1098928.9 N
Empty Weight		477173.2 N	Operating Weight		1576102.1 N
			Comp. All. Ope. Stress		113.7 MPa

Radiography Information:			
...		...	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Material treatment options: +IT

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 6712.0 \cdot (\exp(4.173/(156.4 \cdot 1.0)) - 1) + 6.0 + 0.0$$

$$= 90.7547 + 6.0 + 0.0 = 96.7547 \text{ mm Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 156.4 \cdot 1.0 \cdot \ln((2 \cdot 92.0 + 6712.0) / 6712.0) - 0.073$$

$$= 4.23 - 0.073 = 4.1572 \text{ MPa}$$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D)$$

$$= 175.0 \cdot 1.0 \cdot \ln((2 \cdot 98.0 + 6700.0) / 6700.0)$$

$$= 5.0459 \text{ MPa}$$

Computed Stress Intensity at Design Pressure [Si]:

$$= P \cdot D / (E (D_o - D))$$

$$= 4.173 \cdot 6712.0 / (1.0 (6896.0 - 6712.0))$$

$$= 152.2343 \text{ MPa}$$

Computed Membrane Stress at Design Pressure [Sm]:

$$= P / (E \cdot \ln((2 \cdot t + D) / D))$$



= 4.173/(1.0 • Ln((2•92.0 + 6712.0)/6712.0))
= 154.3111 MPa

Percent Fiber Elongation per Table 6.1

Cylinders formed from Plate $(50 \cdot t_{nom}/R_f) \cdot (1 - R_f/R_o)$ 1.442 %

Minimum Design Metal Temperature Results:

Note: This Element/Detail was specified as being Impact Tested.

Note:

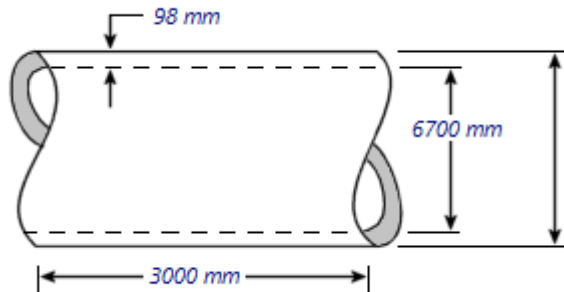
Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

Material treatment options: +IT

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 2, 2025



Internal Pressure Results for: 163-D-Shell-2

Cylindrical Shell Design Information:					
Design Pressure	P	4.142 MPa	Int. Design Temperature		130 °C
Material		SA-516 70	Ext. Design Temperature		0 °C
Length		3000.000 mm	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	6.0000 mm	Allowable Stress (ope)	S	156.4 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	175.0 MPa
Inside Diameter	Di	6700.000 mm	Actual Stress		151.1 MPa
Spec. Min. Thk t _e , or s _a		98.0000 mm	Spec. Nominal Thk.		98.000 mm
Surface Area		64993268.0 mm ²	SG of Contents		1.060
Element Volume		105788.5 ltr	Weight of Contents		1098928.9 N
Empty Weight		477173.2 N	Operating Weight		1576102.1 N
			Comp. All. Ope. Stress		113.7 MPa

Radiography Information:			
...		...	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Material treatment options: +IT

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 6712.0 \cdot (\exp(4.142/(156.4 \cdot 1.0)) - 1) + 6.0 + 0.0$$

$$= 90.0677 + 6.0 + 0.0 = 96.0677 \text{ mm Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 156.4 \cdot 1.0 \cdot \ln((2 \cdot 92.0 + 6712.0) / 6712.0) - 0.041$$

$$= 4.23 - 0.041 = 4.1884 \text{ MPa}$$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D)$$

$$= 175.0 \cdot 1.0 \cdot \ln((2 \cdot 98.0 + 6700.0) / 6700.0)$$

$$= 5.0459 \text{ MPa}$$

Computed Stress Intensity at Design Pressure [Si]:

$$= P \cdot D / (E (D_o - D))$$

$$= 4.142 \cdot 6712.0 / (1.0 (6896.0 - 6712.0))$$

$$= 151.0972 \text{ MPa}$$

Computed Membrane Stress at Design Pressure [Sm]:

$$= P / (E \cdot \ln((2 \cdot t + D) / D))$$



$$= 4.142 / (1.0 \cdot \ln((2 \cdot 92.0 + 6712.0) / 6712.0))$$
$$= 153.1585 \text{ MPa}$$

Percent Fiber Elongation per Table 6.1

Cylinders formed from Plate $(50 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 1.442 %

Minimum Design Metal Temperature Results:

Note: This Element/Detail was specified as being Impact Tested.

Note:

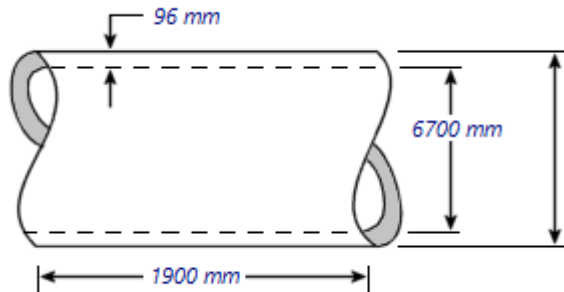
Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

Material treatment options: +IT

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Internal Pressure Calculation Results:

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Internal Pressure Results for: 163-D-Shell-3

Cylindrical Shell Design Information:					
Design Pressure	P	4.111 MPa	Int. Design Temperature		130 °C
Material		SA-516 70	Ext. Design Temperature		0 °C
Length		1900.000 mm	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	6.0000 mm	Allowable Stress (ope)	S	156.4 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	175.0 MPa
Inside Diameter	Di	6700.000 mm	Actual Stress		153.3 MPa
Spec. Min. Thk t _e , or s _a		96.0000 mm	Spec. Nominal Thk.		96.000 mm
Surface Area		41138528.0 mm ²	SG of Contents		1.060
Element Volume		66999.4 ltr	Weight of Contents		219785.8 N
Empty Weight		295955.8 N	Operating Weight		515741.5 N
			Comp. All. Ope. Stress		113.4 MPa

Radiography Information:			
...		...	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Material treatment options: +IT

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 6712.0 \cdot (\exp(4.111/(156.4 \cdot 1.0)) - 1) + 6.0 + 0.0$$

$$= 89.3812 + 6.0 + 0.0 = 95.3812 \text{ mm Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 156.4 \cdot 1.0 \cdot \ln((2 \cdot 90.0 + 6712.0) / 6712.0) - 0.01$$

$$= 4.139 - 0.01 = 4.1288 \text{ MPa}$$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D)$$

$$= 175.0 \cdot 1.0 \cdot \ln((2 \cdot 96.0 + 6700.0) / 6700.0)$$

$$= 4.9444 \text{ MPa}$$

Computed Stress Intensity at Design Pressure [Si]:

$$= P \cdot D / (E (D_o - D))$$

$$= 4.111 \cdot 6712.0 / (1.0 (6892.0 - 6712.0))$$

$$= 153.2923 \text{ MPa}$$

Computed Membrane Stress at Design Pressure [Sm]:

$$= P / (E \cdot \ln((2 \cdot t + D) / D))$$



= 4.111/(1.0 • Ln((2•90.0 + 6712.0)/6712.0))
= 155.3391 MPa

Percent Fiber Elongation per Table 6.1

Cylinders formed from Plate $(50 \cdot t_{nom}/R_f) \cdot (1 - R_f/R_o)$ 1.413 %

Minimum Design Metal Temperature Results:

Note: This Element/Detail was specified as being Impact Tested.

Note:

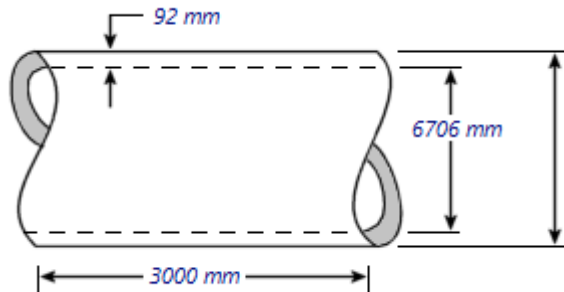
Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

Material treatment options: +IT

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Internal Pressure Calculation Results:

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Internal Pressure Results for: 163-D-Shell-4

Cylindrical Shell Design Information:					
Design Pressure	P	4.105 MPa	Int. Design Temperature		130 °C
Material		SA-516 70	Ext. Design Temperature		0 °C
Length		3000.000 mm	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	1.0000 mm	Allowable Stress (ope)	S	156.4 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	175.0 MPa
Inside Diameter	Di	6706.000 mm	Actual Stress		151.3 MPa
Spec. Min. Thk t _e , or s _a		92.0000 mm	Spec. Nominal Thk.		92.000 mm
Surface Area		64936716.0 mm ²	SG of Contents		0.000
Element Volume		105978.1 ltr	Weight of Contents		0.0 N
Empty Weight		447959.2 N	Operating Weight		447959.2 N
			Comp. All. Ope. Stress		113.5 MPa

Radiography Information:			
...		...	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Material treatment options: +IT

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 6708.0 \cdot (\exp(4.105/(156.4 \cdot 1.0)) - 1) + 1.0 + 0.0$$

$$= 89.1908 + 1.0 + 0.0 = 90.1908 \text{ mm Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 156.4 \cdot 1.0 \cdot \ln((2 \cdot 91.0 + 6708.0) / 6708.0) - 0.004$$

$$= 4.187 - 0.004 = 4.1829 \text{ MPa}$$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D)$$

$$= 175.0 \cdot 1.0 \cdot \ln((2 \cdot 92.0 + 6706.0) / 6706.0)$$

$$= 4.737 \text{ MPa}$$

Computed Stress Intensity at Design Pressure [Si]:

$$= P \cdot D / (E (D_o - D))$$

$$= 4.105 \cdot 6708.0 / (1.0 (6890.0 - 6708.0))$$

$$= 151.2877 \text{ MPa}$$

Computed Membrane Stress at Design Pressure [Sm]:

$$= P / (E \cdot \ln((2 \cdot t + D) / D))$$



= 4.105/(1.0 • Ln((2•91.0 + 6708.0)/6708.0))
= 153.3306 MPa

Percent Fiber Elongation per Table 6.1

Cylinders formed from Plate $(50 \cdot t_{nom}/R_f) \cdot (1 - R_f/R_o)$ 1.353 %

Minimum Design Metal Temperature Results:

Note: This Element/Detail was specified as being Impact Tested.

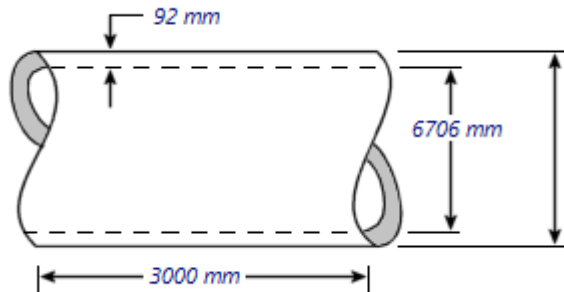
Note:
Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

Material treatment options: +IT

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 2, 2025



Internal Pressure Results for: 163-D-Shell-5

Cylindrical Shell Design Information:					
Design Pressure	P	4.105 MPa	Int. Design Temperature		130 °C
Material		SA-516 70	Ext. Design Temperature		0 °C
Length		3000.000 mm	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	1.0000 mm	Allowable Stress (ope)	S	156.4 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	175.0 MPa
Inside Diameter	Di	6706.000 mm	Actual Stress		151.3 MPa
Spec. Min. Thk t _e , or s _a		92.0000 mm	Spec. Nominal Thk.		92.000 mm
Surface Area		64936716.0 mm ²	SG of Contents		0.000
Element Volume		105978.1 ltr	Weight of Contents		0.0 N
Empty Weight		447959.2 N	Operating Weight		447959.2 N
			Comp. All. Ope. Stress		113.5 MPa

Radiography Information:			
...		...	
Circ. Joint Efficiency		Long. Joint Eff.	E or v
	100 %		100 %

Material treatment options: +IT

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 6708.0 \cdot (\exp(4.105/(156.4 \cdot 1.0)) - 1) + 1.0 + 0.0$$

$$= 89.1908 + 1.0 + 0.0 = 90.1908 \text{ mm Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 156.4 \cdot 1.0 \cdot \ln((2 \cdot 91.0 + 6708.0) / 6708.0) - 0.004$$

$$= 4.187 - 0.004 = 4.1829 \text{ MPa}$$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D)$$

$$= 175.0 \cdot 1.0 \cdot \ln((2 \cdot 92.0 + 6706.0) / 6706.0)$$

$$= 4.737 \text{ MPa}$$

Computed Stress Intensity at Design Pressure [Si]:

$$= P \cdot D / (E (D_o - D))$$

$$= 4.105 \cdot 6708.0 / (1.0 (6890.0 - 6708.0))$$

$$= 151.2877 \text{ MPa}$$

Computed Membrane Stress at Design Pressure [Sm]:

$$= P / (E \cdot \ln((2 \cdot t + D) / D))$$



= 4.105/(1.0 • Ln((2•91.0 + 6708.0)/6708.0))
= 153.3306 MPa

Percent Fiber Elongation per Table 6.1

Cylinders formed from Plate $(50 \cdot t_{nom}/R_f) \cdot (1 - R_f/R_o)$ 1.353 %

Minimum Design Metal Temperature Results:

Note: This Element/Detail was specified as being Impact Tested.

Note:

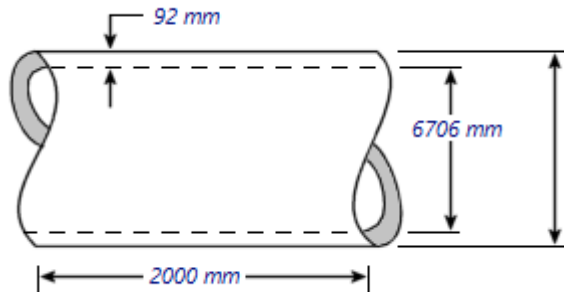
Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

Material treatment options: +IT

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 2, 2025



Internal Pressure Results for: 163-D-Shell-6

Cylindrical Shell Design Information:					
Design Pressure	P	4.105 MPa	Int. Design Temperature		130 °C
Material		SA-516 70	Ext. Design Temperature		0 °C
Length		2000.000 mm	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	1.0000 mm	Allowable Stress (ope)	S	156.4 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	175.0 MPa
Inside Diameter	Di	6706.000 mm	Actual Stress		151.3 MPa
Spec. Min. Thk t _e , or s _a		92.0000 mm	Spec. Nominal Thk.		92.000 mm
Surface Area		43291148.0 mm ²	SG of Contents		0.000
Element Volume		70652.1 ltr	Weight of Contents		0.0 N
Empty Weight		298639.2 N	Operating Weight		298639.2 N
			Comp. All. Ope. Stress		113.5 MPa

Radiography Information:			
...		...	
Circ. Joint Efficiency		Long. Joint Eff.	E or v
	100 %		100 %

Material treatment options: +IT

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 6708.0 \cdot (\exp(4.105/(156.4 \cdot 1.0)) - 1) + 1.0 + 0.0$$

$$= 89.1908 + 1.0 + 0.0 = 90.1908 \text{ mm Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 156.4 \cdot 1.0 \cdot \ln((2 \cdot 91.0 + 6708.0) / 6708.0) - 0.004$$

$$= 4.187 - 0.004 = 4.1829 \text{ MPa}$$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D)$$

$$= 175.0 \cdot 1.0 \cdot \ln((2 \cdot 92.0 + 6706.0) / 6706.0)$$

$$= 4.737 \text{ MPa}$$

Computed Stress Intensity at Design Pressure [Si]:

$$= P \cdot D / (E (D_o - D))$$

$$= 4.105 \cdot 6708.0 / (1.0 (6890.0 - 6708.0))$$

$$= 151.2877 \text{ MPa}$$

Computed Membrane Stress at Design Pressure [Sm]:

$$= P / (E \cdot \ln((2 \cdot t + D) / D))$$



$$= 4.105 / (1.0 \cdot \ln((2 \cdot 91.0 + 6708.0) / 6708.0))$$
$$= 153.3306 \text{ MPa}$$

Percent Fiber Elongation per Table 6.1

Cylinders formed from Plate $(50 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 1.353 %

Minimum Design Metal Temperature Results:

Note: This Element/Detail was specified as being Impact Tested.

Note:

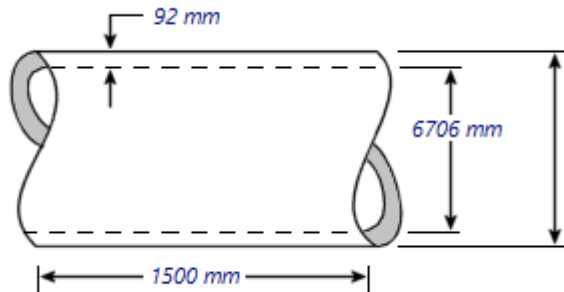
Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

Material treatment options: +IT

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 2, 2025



Internal Pressure Results for: 163-D-Shell-7

Cylindrical Shell Design Information:					
Design Pressure	P	4.105 MPa	Int. Design Temperature		130 °C
Material		SA-516 70	Ext. Design Temperature		0 °C
Length		1500.000 mm	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	1.0000 mm	Allowable Stress (ope)	S	156.4 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	175.0 MPa
Inside Diameter	Di	6706.000 mm	Actual Stress		151.3 MPa
Spec. Min. Thk t _e , or s _a		92.0000 mm	Spec. Nominal Thk.		92.000 mm
Surface Area		32468358.0 mm ²	SG of Contents		0.000
Element Volume		52989.0 ltr	Weight of Contents		0.0 N
Empty Weight		223979.6 N	Operating Weight		223979.6 N
			Comp. All. Ope. Stress		113.5 MPa

Radiography Information:			
...		...	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Material treatment options: +IT

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 6708.0 \cdot (\exp(4.105/(156.4 \cdot 1.0)) - 1) + 1.0 + 0.0$$

$$= 89.1908 + 1.0 + 0.0 = 90.1908 \text{ mm Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 156.4 \cdot 1.0 \cdot \ln((2 \cdot 91.0 + 6708.0) / 6708.0) - 0.004$$

$$= 4.187 - 0.004 = 4.1829 \text{ MPa}$$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D)$$

$$= 175.0 \cdot 1.0 \cdot \ln((2 \cdot 92.0 + 6706.0) / 6706.0)$$

$$= 4.737 \text{ MPa}$$

Computed Stress Intensity at Design Pressure [Si]:

$$= P \cdot D / (E (D_o - D))$$

$$= 4.105 \cdot 6708.0 / (1.0 (6890.0 - 6708.0))$$

$$= 151.2877 \text{ MPa}$$

Computed Membrane Stress at Design Pressure [Sm]:

$$= P / (E \cdot \ln((2 \cdot t + D) / D))$$



= 4.105/(1.0 • Ln((2•91.0 + 6708.0)/6708.0))
= 153.3306 MPa

Percent Fiber Elongation per Table 6.1

Cylinders formed from Plate $(50 \cdot t_{nom}/R_f) \cdot (1 - R_f/R_o)$ 1.353 %

Minimum Design Metal Temperature Results:

Note: This Element/Detail was specified as being Impact Tested.

Note:

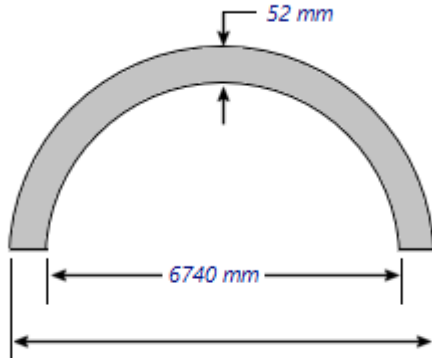
Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

Material treatment options: +IT

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 2, 2025



Internal Pressure Results for: 163-D-Top head

Spherical Head Design Information:					
Design Pressure	P	4.105 MPa	Int. Design Temperature		130 °C
Material		SA-516 70	Ext. Design Temperature		0 °C
Length		0.000 mm	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	1.0000 mm	Allowable Stress (ope)	S	156.4 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	175.0 MPa
Inside Diameter	Di	6740.000 mm	Actual Stress		134.6 MPa
Spec. Min. Thk t,e, or sa		52.0000 mm	Spec. Nominal Thk.		58.000 mm
Surface Area		73834872.0 mm²	SG of Contents		0.000
Element Volume		80172.6 ltr	Weight of Contents		0.0 N
Empty Weight		319978.1 N	Operating Weight		319978.1 N
			Comp. All. Ope. Stress		102.1 MPa

Radiography Information:			
...		...	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Tolerance for Formed Heads per 4.3.2.2:

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 84.250 mm
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 42.125 mm

Material treatment options: +IT

Spherical Shell Calculation - Section 4.3.5.1

Computed Minimum Required Thickness [t]:
 $= D/2 \cdot (\exp(0.5P/(S \cdot E)) - 1) + ci + co$
 $= 6742.0/2 \cdot (\exp(0.5 \cdot 4.105/(156.4 \cdot 1.0)) - 1) + 1.0 + 0.0$
 $= 44.5271 + 1.0 + 0.0 = 45.5271 \text{ mm}$

Computed Maximum Allowable Working Pressure [MAWP]:
 $= 2 \cdot S \cdot E \cdot \ln((2 \cdot t + D)/D) - P_{liq}$
 $= 2 \cdot 156.4 \cdot 1.0 \cdot \ln((2 \cdot 51.0 + 6742.0)/6742.0) - 0.004$
 $= 4.697 - 0.004 = 4.6929 \text{ MPa}$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:
 $= 2 \cdot S \cdot E \cdot \ln((2 \cdot t + D)/D)$
 $= 2 \cdot 175.0 \cdot 1.0 \cdot \ln((2 \cdot 52.0 + 6740.0)/6740.0)$
 $= 5.3593 \text{ MPa}$



Computed Stress Intensity at Design Pressure [Si]:

$$\begin{aligned} &= P \cdot D^2 / (E (D_o^2 - D^2)) \\ &= 4.105 \cdot 6742.0^2 / (1.0 (6844.0^2 - 6742.0^2)) \\ &= 134.638 \text{ MPa} \end{aligned}$$

Computed Membrane Stress at Design Pressure [Sm]:

$$\begin{aligned} &= 0.5 \cdot P / (E \cdot \ln((2 \cdot t + D) / D)) \\ &= 0.5 \cdot 4.105 / (1.0 \cdot \ln((2 \cdot 51.0 + 6742.0) / 6742.0)) \\ &= 136.679 \text{ MPa} \end{aligned}$$

Percent Fiber Elongation per Table 6.1

For Heads formed from Segments $(75 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$	1.280 %
For one Piece doubly Curved Heads $100 \cdot \ln(D_b / (D_f - 2t))$	36.364 %

Where D_b is the blank diameter for one piece doubly curved surfaces [D_b]:

$$\begin{aligned} &= \text{square root}(4 \cdot \text{Head Surface Area} / 3.14159) \\ &= \text{square root}(4 \cdot 73834872 / 3.14159) \\ &= 9695.849 \text{ mm} \end{aligned}$$

MDMT Calculations in the Spherical Portion:

Note: This Element/Detail was specified as being Impact Tested.

Note:

Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

Tolerance for Formed Heads per 4.3.2.2:

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 84.250 mm
Head inner surface maximum deviation inside the specified shape, 0.625% of D: 42.125 mm

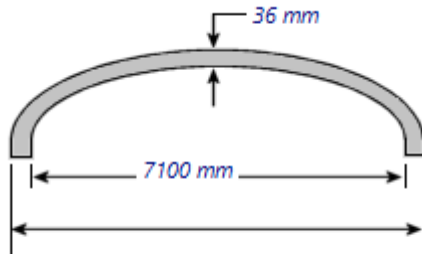
Material treatment options: +IT

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 2, 2025



Internal Pressure Results for: 122-D1-Btm head

Elliptical Head Design Information:					
Design Pressure	P	0.470 MPa	Int. Design Temperature		130 °C
Material		SA-516 70	Ext. Design Temperature		130 °C
Length		50.000 mm	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	6.0000 mm	Allowable Stress (ope)	S	156.4 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	175.0 MPa
Inside Diameter	Di	7100.000 mm	Actual Stress		50.3 MPa
Spec. Min. Thk t _e or s _a		36.0000 mm	Spec. Nominal Thk.		44.000 mm
Surface Area		57135736.0 mm ²	SG of Contents		1.060
Element Volume		48838.8 ltr	Weight of Contents		507336.1 N
Empty Weight		208008.7 N	Operating Weight		715344.9 N
			Comp. All. Ope. Stress		86.7 MPa
			Aspect Ratio		2.00

Radiography Information:			
...		...	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Tolerance for Formed Heads per 4.3.2.2:

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 88.750 mm
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 44.375 mm

Material treatment options: +IT

Head Calculation - Section 4.3.6 (and Section 4.3.7 for Ellipsoidal Head)

Computed Minimum Required Thickness [t]:
 $t_r = 10.7779 \text{ mm}$

$$t = t_r + c_i + c_o$$

$$= 10.7779 + 6.0 + 0.0$$

$$= 16.7779 \text{ mm} - \text{see below for the derivation}$$

For Elliptical heads: Compute [h, k, r] per 4.3.7.1

$$h = D / (2 \cdot \text{Aspect Ratio})$$

$$= 7100.0 / (2 \cdot 2.0)$$

$$= 1775.000 \text{ mm}$$

$$k = D / (2 \cdot h)$$

$$= 7100.0 / (2 \cdot 1775.0)$$

$$= 2.000$$

$$r = D (0.5/k - 0.08) + c_a$$

$$= 7100.0(0.5/2.0 - 0.08) + 6.0$$

$$= 1213.000 \text{ mm}$$

$$L = D(0.44 \cdot k + 0.02) + ca$$

$$= 7100.0(0.44 \cdot 2.0 + 0.02) + 6.0$$

$$= 6396.000 \text{ mm}$$

Step 3: Geometric Constants [βTh , ϕTh , Rth]:
 Equations 4.3.8 to 4.3.11 as appropriate
 t (corroded) and P for the calc.: $t = 10.7779 \text{ mm}$, $P = 0.4703 \text{ MPa}$

Angle [βTh]:

$$= \arccos[(0.5 \cdot D - r)/(L - r)]$$

$$= \arccos[(0.5 \cdot 7112.0 - 1213.0)/(6396.0 - 1213.0)]$$

$$= 1.1017 \text{ radians - Equation 4.3.8}$$

$$\phi Th = \sqrt{(L \cdot t) / r}$$

$$= \sqrt{(6396.0 \cdot 10.778) / 1213.0}$$

$$= 0.2165 \text{ radians - Equation 4.3.9}$$

$\phi Th < \beta Th$, Thus:

$$Rth = (0.5 \cdot D - r) / \cos(\beta Th - \phi Th) + r$$

$$= (0.5 \cdot 7112.0 - 1213.0) / \cos(1.102 - 0.216) + 1213.0$$

$$= 4913.9829 \text{ mm - Equation 4.3.10}$$

Step 4: Coefficients: Equations 4.3.12 to 4.3.15 [$C1$ and $C2$]:

$$r/D > 0.08, \text{ Thus:}$$

$$C1 = 0.692(r/D) + 0.605$$

$$= 0.692(1213.0/7112.0) + 0.605$$

$$= 0.723 \text{ - Equation 4.3.13}$$

$$C2 = 1.46 - 2.6(r/D) = 1.46 - 2.6(1213.0/7112.0)$$

$$= 1.0166, \text{ Equation 4.3.15}$$

Step 5: Pressure to produce knuckle elastic buckling [$Peth$]:

$$= C1 \cdot Et \cdot t^2 / (C2 \cdot Rth(0.5 \cdot Rth - r))$$

$$= 0.723 \cdot 196301.391 \cdot 10.778^2 / (1.017 \cdot 4913.983(0.5 \cdot 4913.983 - 1213.0))$$

$$= 2.65 \text{ MPa, Equation 4.3.16}$$

Step 6: Internal Pressure for max stress in knuckle equals yield [Py]:

$C3 = Sy = 234.4 \text{ MPa}$, Not in the creep range - 4.3.6.1(f)

$$Py = C3 \cdot t / (C2 \cdot Rth(0.5 \cdot Rth / r - 1))$$

$$= 234.4 \cdot 10.778 / (1.017 \cdot 4913.983(0.5 \cdot 4913.983 / 1213.0 - 1))$$

$$= 0.493 \text{ MPa, Equation 4.3.17}$$

Step 7: Value of Internal Press. resulting in buckling of knuckle [G and Pck]:

$$G = Peth/Py = 2.653/0.493 = 5.38$$

$G > 1$, Thus:

Buckling Pressure [Pck]:

$$= ((0.77508 \cdot G - 0.20354 \cdot G^2 + 0.019274 \cdot G^3) /$$

$$(1 + 0.19014 \cdot G - 0.089534 \cdot G^2 + 0.0093965 \cdot G^3)) Py$$

$$= ((0.77508 \cdot 5.38 - 0.20354 \cdot 5.38^2 + 0.019274 \cdot 5.38^3) /$$

$$(1 + 0.19014 \cdot 5.38 - 0.089534 \cdot 5.38^2 + 0.0093965 \cdot 5.38^3)) \cdot 0.493$$

$$= 0.71 \text{ MPa, Equation 4.3.19}$$

Step 8: Compute Allowable Pressure base on buckling of knuckle [Pak]:

$$= Pck/1.5 = 0.71/1.5 = 0.47 \text{ MPa - Equation 4.3.21}$$

Step 9: Compute allowable pressure based on rupture of crown [Pac]:

$$= 2 \cdot S \cdot E / (L/t + 0.5)$$

$$= 2 \cdot 156.4 \cdot 1.0 / (6396.0/10.778 + 0.5)$$

$$= 0.53 \text{ MPa, Equation 4.3.22}$$

Step 10: Compute final pressure [Pa]:

$$= \min(Pak, Pac) = \min(0.47, 0.527)$$

$$= 0.47 \text{ MPa, Equation 4.3.23: (same as original design pressure)}$$



Allowable pressure in Crown at given thickness [Pac]:

$$\begin{aligned} &= 2 \cdot S \cdot E / (L / t + 0.5) \\ &= 2 \cdot 156.4 \cdot 1.0 / (6396.0 / 30.0 + 0.5) \\ &= 1.464 \text{ MPa} \end{aligned}$$

Computed Maximum Allowable Pressure [MAWP]:

$$= \text{MaxPressure} - \text{Pliq} = 1.4637 - 0.086 = 1.3774 \text{ MPa}$$

Computed Maximum Allowable Pressure New and Cold [MAPnc]:

$$= 1.9663 \text{ MPa}$$

Computed Membrane Stress Intensity at Design Pressure [Sm]:

$$= 50.3 \text{ MPa}$$

Required Thickness of Straight Flange = 16.710 mm

Percent Fiber Elongation per Table 6.1

$$\begin{aligned} \text{For Heads formed from Segments } (75 \cdot t_{\text{nom}} / R_f) \cdot (1 - R_f / R_o) & \quad 2.685 \% \\ \text{For one Piece doubly Curved Heads } 100 \cdot \ln(D_b / (D_f - 2t)) & \quad 18.340 \% \end{aligned}$$

Where D_b is the blank diameter for one piece doubly curved surfaces [D_b]:

$$\begin{aligned} &= \text{square root}(4 \cdot \text{Head Surface Area} / 3.14159) \\ &= \text{square root}(4 \cdot 57135736 / 3.14159) \\ &= 8529.213 \text{ mm} \end{aligned}$$

MDMT Calculations in the Knuckle Portion:

Note: This Element/Detail was specified as being Impact Tested.

MDMT Calculations in the Head Straight Flange:

Note: This Element/Detail was specified as being Impact Tested.

Note:

Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

Tolerance for Formed Heads per 4.3.2.2:

$$\begin{aligned} \text{Head inner surface maximum deviation outside the specified shape, } 1.25\% \text{ of } D: & 88.750 \text{ mm} \\ \text{Head inner surface maximum deviation inside the specified shape, } 0.625\% \text{ of } D: & 44.375 \text{ mm} \end{aligned}$$

Material treatment options: +IT

EXTERNAL PRESSURE RESULTS for: 122-D1-Btm head

Elliptical Head From 170 to 180 Ext. Chart: CS-2 at 130 °C

Maximum External Pressure per ASME Division 2 Section 4.4:

Elastic Hoop Compressive Membrane Failure Stress [Fhe]:

$$\begin{aligned} &= 0.075 \cdot E_y (t / R_o) \\ &= 0.075 \cdot 196301 (30.0 / 6394.931) \\ &= 69.067 \text{ MPa} \end{aligned}$$

Ratio of Elastic Hoop Compressive Membrane Failure Stress to Yield [Fr]:

$$\begin{aligned} &= F_{he} / S_y \\ &= 69.1 / 234.4 \\ &= 0.295 \end{aligned}$$

Engineering Yield to Tensile Stress ratio [R]:

$$\begin{aligned} &= \sigma_{ys} / \sigma_{uts} \\ &= 234.4 / 482.6 \\ &= 0.486 \end{aligned}$$

Ratio [AE]:

$$\begin{aligned} &= F_{he} / E \\ &= 69.1 / 196301.4 \end{aligned}$$

= 0.000352

Factor [m1]:

$$= (\ln(r) + (\text{epsp} - \text{epsys})) / \ln(\ln(1 + \text{epsp}) / \ln(1 + \text{epsys}))$$

$$= (\ln(0.486) + (0.00002 - 0.002)) / \ln(\ln(1 + 0.00002) / \ln(1 + 0.002))$$

$$= 0.157$$

Curve Fitting Constant for the Elastic Region of the Stress-Strain Curve [A1]:

$$= \sigma_{ys} (1 + \text{epsys}) / \ln(1 + \text{epsys})^{m1}$$

$$= 234.4 (1 + 0.002) / \ln(1 + 0.002)^{0.157301}$$

$$= 624.374 \text{ MPa}$$

Curve Fitting Exponent for, Ferritic Steel [m2]:

$$= m2f(m2f2 - R)$$

$$= 0.6(1.0 - 0.486)$$

$$= 0.309$$

Curve Fitting Constant [A2]:

$$= (\sigma_{uts} \cdot \exp(m2)) / (m2^{m2})$$

$$= (482.6 \cdot \exp(0.309)) / (0.309^{0.309})$$

$$= 944.566 \text{ MPa}$$

Material Parameter for Stress-Strain Curve Model [K]:

$$= 1.5 \cdot R1.5 - 0.5 \cdot R2.5 - R3.5$$

$$= 1.5 \cdot 0.4861.5 - 0.5 \cdot 0.4862.5 - 0.4863.5$$

$$= 0.346$$

Compute the Allowable Stress per 4.4.3

	Fic	H	D1	D2	D3	D4	Tangent Modulus	Iteration Diff
1	241	-1.839	0.313E-04	0.271E-04	0.807E-04	-0.632E-04	0.124E+05	0.192E-01
2	120	-4.651	0.763E-06	0.763E-06	0.171E-04	-0.171E-04	0.151E+06	0.448E-03
3	60	-6.057	0.186E-07	0.186E-07	0.361E-05	-0.361E-05	0.195E+06	0.423E-04
4	90	-5.354	0.163E-06	0.163E-06	0.896E-05	-0.896E-05	0.184E+06	0.139E-03
5	75	-5.706	0.615E-07	0.615E-07	0.596E-05	-0.596E-05	0.192E+06	0.416E-04
6	67	-5.881	0.350E-07	0.350E-07	0.470E-05	-0.470E-05	0.194E+06	0.133E-05
7	71	-5.793	0.467E-07	0.467E-07	0.531E-05	-0.531E-05	0.193E+06	0.198E-04
8	69	-5.837	0.405E-07	0.405E-07	0.500E-05	-0.500E-05	0.193E+06	0.918E-05
9	68	-5.859	0.377E-07	0.377E-07	0.485E-05	-0.485E-05	0.193E+06	0.391E-05
10	68	-5.870	0.363E-07	0.363E-07	0.478E-05	-0.478E-05	0.194E+06	0.129E-05
11	68	-5.876	0.356E-07	0.356E-07	0.474E-05	-0.474E-05	0.194E+06	0.197E-07
12	68	-5.873	0.360E-07	0.360E-07	0.476E-05	-0.476E-05	0.194E+06	0.634E-06
13	68	-5.874	0.358E-07	0.358E-07	0.475E-05	-0.475E-05	0.194E+06	0.307E-06
14	68	-5.875	0.357E-07	0.357E-07	0.474E-05	-0.474E-05	0.194E+06	0.144E-06
15	68	-5.875	0.357E-07	0.357E-07	0.474E-05	-0.474E-05	0.194E+06	0.620E-07
16	68	-5.876	0.357E-07	0.357E-07	0.474E-05	-0.474E-05	0.194E+06	0.211E-07
17	68	-5.876	0.356E-07	0.356E-07	0.474E-05	-0.474E-05	0.194E+06	0.704E-09

Factor of Safety [FS]:

$$\text{Since Fic [68.11]} < 0.55 \cdot 234.4, \text{FS} = 2$$

Allowable Circumferential Compressive Stress [Fha]:

$$= \text{Fic} / \text{FS}$$

$$= 68.112 / 2.0$$

$$= 34.056 \text{ MPa}$$

Allowable External Pressure at the given Thickness [Pa]:

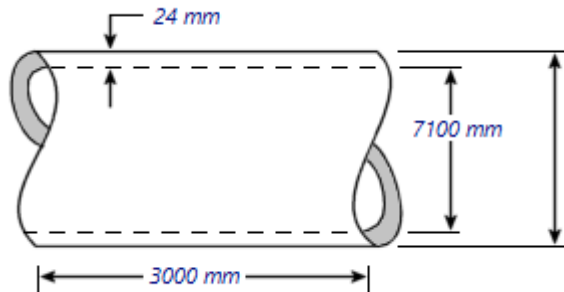
$$= 2 \cdot \text{Fha} [t / R_o]$$

$$= 2 \cdot 34.056 [30.0 / 6394.931]$$

$$= 0.320 \text{ MPa}$$

Internal Pressure Calculation Results:

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Internal Pressure Results for: 122-D1-Shell-1

Cylindrical Shell Design Information:					
Design Pressure	P	0.451 MPa	Int. Design Temperature		130 °C
Material		SA-516 70	Ext. Design Temperature		130 °C
Length		3000.000 mm	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	6.0000 mm	Allowable Stress (ope)	S	156.4 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	175.0 MPa
Inside Diameter	Di	7100.000 mm	Actual Stress		89.2 MPa
Spec. Min. Thk t _e , or s _a		24.0000 mm	Spec. Nominal Thk.		24.000 mm
Surface Area		67368312.0 mm ²	SG of Contents		1.060
Element Volume		118797.0 ltr	Weight of Contents		1234060.8 N
Empty Weight		122463.4 N	Operating Weight		1356524.2 N
			Comp. All. Ope. Stress		62.9 MPa
Radiography Information:					
...			...		
Circ. Joint Efficiency		100 %	Long. Joint Eff.	E or v	100 %

Material treatment options: +IT

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 7112.0 \cdot (\exp(0.451/(156.4 \cdot 1.0)) - 1) + 6.0 + 0.0$$

$$= 10.2777 + 6.0 + 0.0 = 16.2777 \text{ mm Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 156.4 \cdot 1.0 \cdot \ln((2 \cdot 18.0 + 7112.0) / 7112.0) - 0.067$$

$$= 0.79 - 0.067 = 0.7223 \text{ MPa}$$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D)$$

$$= 175.0 \cdot 1.0 \cdot \ln((2 \cdot 24.0 + 7100.0) / 7100.0)$$

$$= 1.1791 \text{ MPa}$$

Computed Stress Intensity at Design Pressure [Si]:

$$= P \cdot D / (E (D_o - D))$$

$$= 0.451 \cdot 7112.0 / (1.0 (7148.0 - 7112.0))$$

$$= 89.1730 \text{ MPa}$$

Computed Membrane Stress at Design Pressure [Sm]:

$$= P / (E \cdot \ln((2 \cdot t + D) / D))$$



$$= 0.451 / (1.0 \cdot \ln((2 \cdot 18.0 + 7112.0) / 7112.0))$$
$$= 89.3989 \text{ MPa}$$

Percent Fiber Elongation per Table 6.1

$$\text{Cylinders formed from Plate} \quad (50 \cdot t_{\text{nom}} / R_f) \cdot (1 - R_f / R_o) \quad 0.337 \%$$

Minimum Design Metal Temperature Results:

Note: This Element/Detail was specified as being Impact Tested.

Material treatment options: +IT

EXTERNAL PRESSURE RESULTS for: 122-D1-Shell-1

Cylindrical Shell From 180 to 3 Ext. Chart: CS-2 at 130 °C

Maximum External Pressure per ASME Division 2 Section 4.4:

Determine Parameter [Mx]:

$$= L / \sqrt{(R_o \cdot t)}$$
$$= 2841.717 / \sqrt{(3574.0 \cdot 18.0)}$$
$$= 11.204$$

Compute the Factor [Ch]:

Since $M_x > 1.5$ and $M_x < 13$, $Ch = 0.92 / (M_x - 0.579)$

$$= 0.92 / (11.204 - 0.579)$$
$$= 0.0866$$

Elastic Hoop Compressive Membrane Failure Stress [Fhe]:

$$= 1.6 \cdot Ch \cdot E_y (t / D_o)$$
$$= 1.6 \cdot 0.087 \cdot 196301 (18.0 / 7148.0)$$
$$= 68.485 \text{ MPa}$$

Ratio of Elastic Hoop Compressive Membrane Failure Stress to Yield [Fr]:

$$= F_{he} / F_y$$
$$= 68.5 / 234.4$$
$$= 0.292$$

Predicted Buckling Stress [Fic]:

Engineering Yield to Tensile Stress ratio [R]:

$$= \sigma_{ys} / \sigma_{uts}$$
$$= 234.4 / 482.6$$
$$= 0.486$$

Ratio [AE]:

$$= F_{he} / E$$
$$= 68.5 / 196301.4$$
$$= 0.000349$$

Factor [m1]:

$$= (\ln(r) + (\epsilon_{psp} - \epsilon_{sys})) / \ln(\ln(1 + \epsilon_{psp}) / \ln(1 + \epsilon_{sys}))$$
$$= (\ln(0.486) + (0.00002 - 0.002)) / \ln(\ln(1 + 0.00002) / \ln(1 + 0.002))$$
$$= 0.157$$

Curve Fitting Constant for the Elastic Region of the Stress-Strain Curve [A1]:

$$= \sigma_{ys} (1 + \epsilon_{sys}) / \ln(1 + \epsilon_{sys})^{m1}$$
$$= 234.4 (1 + 0.002) / \ln(1 + 0.002)^{0.157301}$$
$$= 624.374 \text{ MPa}$$

Curve Fitting Exponent for, Ferritic Steel [m2]:

$$= m2f(m2f2 - R)$$
$$= 0.6(1.0 - 0.486)$$
$$= 0.309$$

Curve Fitting Constant [A2]:

$$= (\sigma_{uts} \cdot \exp(m2)) / (m2^{m2})$$
$$= (482.6 \cdot \exp(0.309)) / (0.309^{0.309})$$



= 944.566 MPa

Material Parameter for Stress-Strain Curve Model [K]:

$$= 1.5 \cdot R1.5 - 0.5 \cdot R2.5 - R3.5$$

$$= 1.5 \cdot 0.4861.5 - 0.5 \cdot 0.4862.5 - 0.4863.5$$

$$= 0.346$$

Compute the Allowable Stress per 4.4.3

	Fic	H	D1	D2	D3	D4	Tangent Modulus	Iteration Diff
1	241	-1.839	0.313E-04	0.271E-04	0.807E-04	-0.632E-04	0.124E+05	0.192E-01
2	120	-4.651	0.763E-06	0.763E-06	0.171E-04	-0.171E-04	0.151E+06	0.451E-03
3	60	-6.057	0.186E-07	0.186E-07	0.361E-05	-0.361E-05	0.195E+06	0.393E-04
4	90	-5.354	0.163E-06	0.163E-06	0.896E-05	-0.896E-05	0.184E+06	0.142E-03
5	75	-5.706	0.615E-07	0.615E-07	0.596E-05	-0.596E-05	0.192E+06	0.446E-04
6	67	-5.881	0.350E-07	0.350E-07	0.470E-05	-0.470E-05	0.194E+06	0.164E-05
7	64	-5.969	0.257E-07	0.257E-07	0.414E-05	-0.414E-05	0.194E+06	0.190E-04
8	65	-5.925	0.301E-07	0.301E-07	0.442E-05	-0.442E-05	0.194E+06	0.875E-05
9	66	-5.903	0.325E-07	0.324E-07	0.456E-05	-0.456E-05	0.194E+06	0.357E-05
10	67	-5.892	0.337E-07	0.337E-07	0.463E-05	-0.463E-05	0.194E+06	0.970E-06
11	67	-5.887	0.343E-07	0.343E-07	0.467E-05	-0.467E-05	0.194E+06	0.333E-06
12	67	-5.890	0.340E-07	0.340E-07	0.465E-05	-0.465E-05	0.194E+06	0.319E-06
13	67	-5.888	0.342E-07	0.342E-07	0.466E-05	-0.466E-05	0.194E+06	0.693E-08

Factor of Safety [FS]:

$$\text{Since } F_{ic} [67.58] < 0.55 \cdot 234.4, FS = 2$$

Allowable Circumferential Compressive Stress [Fha]:

$$= F_{ic}/FS$$

$$= 67.578/2.0$$

$$= 33.789 \text{ MPa}$$

Allowable External Pressure at the given Thickness [Pa]:

$$= 2 \cdot F_{ha} (t / D_o)$$

$$= 2 \cdot 33.789 (18.0/7148.0)$$

$$= 0.170 \text{ MPa}$$

Cylindrical Shell From 3 to 190 Ext. Chart: CS-2 at 130 °C

Maximum External Pressure per ASME Division 2 Section 4.4:

Determine Parameter [Mx]:

$$= L / \sqrt{(R_o \cdot t)}$$

$$= 2750.0 / \sqrt{(3574.0 \cdot 18.0)}$$

$$= 10.842$$

Compute the Factor [Ch]:

$$\text{Since } M_x > 1.5 \text{ and } M_x < 13, Ch = 0.92 / (M_x - 0.579)$$

$$= 0.92 / (10.842 - 0.579)$$

$$= 0.0896$$

Elastic Hoop Compressive Membrane Failure Stress [Fhe]:

$$= 1.6 \cdot Ch \cdot E_y (t / D_o)$$

$$= 1.6 \cdot 0.09 \cdot 196301 (18.0/7148.0)$$

$$= 70.898 \text{ MPa}$$

Ratio of Elastic Hoop Compressive Membrane Failure Stress to Yield [Fr]:

$$= F_{he} / F_y$$

$$= 70.9/234.4$$

$$= 0.302$$

Predicted Buckling Stress [Fic]:

Engineering Yield to Tensile Stress ratio [R]:

$$= \sigma_{ys} / \sigma_{ts}$$

$$= 234.4/482.6$$

$$= 0.486$$

Ratio [AE]:
 = Fhe / E
 = 70.9/196301.4
 = 0.000361

Factor [m1]:
 = (ln(r) + (epsp - epsys)) / ln(ln(1 + epsp) / ln(1 + epsys))
 = (ln(0.486) + (0.00002 - 0.002)) / ln(ln(1 + 0.00002) / ln(1 + 0.002))
 = 0.157

Curve Fitting Constant for the Elastic Region of the Stress-Strain Curve [A1]:
 = $\sigma_{ys}(1 + \epsilon_{psys}) / \ln(1 + \epsilon_{psys})$
 = $234.4(1 + 0.002) / \ln(1 + 0.002)$ 0.157301
 = 624.374 MPa

Curve Fitting Exponent for, Ferritic Steel [m2]:
 = $m2f(m2f2 - R)$
 = $0.6(1.0 - 0.486)$
 = 0.309

Curve Fitting Constant [A2]:
 = $(\sigma_{uts} \cdot \exp(m2)) / (m2 \cdot m2)$
 = $(482.6 \cdot \exp(0.309)) / (0.309 \cdot 0.309)$
 = 944.566 MPa

Material Parameter for Stress-Strain Curve Model [K]:
 = $1.5 \cdot R1.5 - 0.5 \cdot R2.5 - R3.5$
 = $1.5 \cdot 0.4861.5 - 0.5 \cdot 0.4862.5 - 0.4863.5$
 = 0.346

Compute the Allowable Stress per 4.4.3

	Fic	H	D1	D2	D3	D4	Tangent Modulus	Iteration Diff
1	241	-1.839	0.313E-04	0.271E-04	0.807E-04	-0.632E-04	0.124E+05	0.192E-01
2	120	-4.651	0.763E-06	0.763E-06	0.171E-04	-0.171E-04	0.151E+06	0.439E-03
3	60	-6.057	0.186E-07	0.186E-07	0.361E-05	-0.361E-05	0.195E+06	0.516E-04
4	90	-5.354	0.163E-06	0.163E-06	0.896E-05	-0.896E-05	0.184E+06	0.129E-03
5	75	-5.706	0.615E-07	0.615E-07	0.596E-05	-0.596E-05	0.192E+06	0.323E-04
6	67	-5.881	0.350E-07	0.350E-07	0.470E-05	-0.470E-05	0.194E+06	0.107E-04
7	71	-5.793	0.467E-07	0.467E-07	0.531E-05	-0.531E-05	0.193E+06	0.105E-04
8	69	-5.837	0.405E-07	0.405E-07	0.500E-05	-0.500E-05	0.193E+06	0.145E-06
9	70	-5.815	0.435E-07	0.435E-07	0.515E-05	-0.515E-05	0.193E+06	0.516E-05
10	70	-5.826	0.420E-07	0.420E-07	0.508E-05	-0.508E-05	0.193E+06	0.250E-05
11	69	-5.832	0.412E-07	0.412E-07	0.504E-05	-0.504E-05	0.193E+06	0.118E-05
12	69	-5.835	0.409E-07	0.409E-07	0.502E-05	-0.502E-05	0.193E+06	0.516E-06
13	69	-5.836	0.407E-07	0.407E-07	0.501E-05	-0.501E-05	0.193E+06	0.186E-06
14	69	-5.837	0.406E-07	0.406E-07	0.501E-05	-0.501E-05	0.193E+06	0.204E-07
15	69	-5.837	0.406E-07	0.405E-07	0.500E-05	-0.500E-05	0.193E+06	0.622E-07
16	69	-5.837	0.406E-07	0.406E-07	0.500E-05	-0.500E-05	0.193E+06	0.209E-07
17	69	-5.837	0.406E-07	0.406E-07	0.501E-05	-0.501E-05	0.193E+06	0.247E-09

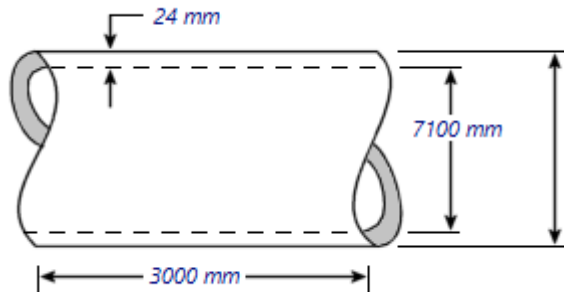
Factor of Safety [FS]:
 Since Fic [69.78] < 0.55 * 234.4, FS = 2

Allowable Circumferential Compressive Stress [Fha]:
 = Fic/FS
 = 69.784/2.0
 = 34.892 MPa

Allowable External Pressure at the given Thickness [Pa]:
 = $2 \cdot Fha(t / Do)$
 = $2 \cdot 34.892(18.0/7148.0)$
 = 0.176 MPa

Internal Pressure Calculation Results:

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Internal Pressure Results for: 122-D1-Shell-2

Cylindrical Shell Design Information:					
Design Pressure	P	0.420 MPa	Int. Design Temperature		130 °C
Material		SA-516 70	Ext. Design Temperature		130 °C
Length		3000.000 mm	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	6.0000 mm	Allowable Stress (ope)	S	156.4 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	175.0 MPa
Inside Diameter	Di	7100.000 mm	Actual Stress		83.0 MPa
Spec. Min. Thk t _e , or s _a		24.0000 mm	Spec. Nominal Thk.		24.000 mm
Surface Area		67368312.0 mm ²	SG of Contents		1.060
Element Volume		118797.0 ltr	Weight of Contents		1234060.8 N
Empty Weight		122463.4 N	Operating Weight		1356524.2 N
			Comp. All. Ope. Stress		62.9 MPa
Radiography Information:					
...			...		
Circ. Joint Efficiency		100 %	Long. Joint Eff.	E or v	100 %

Material treatment options: +IT

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 7112.0 \cdot (\exp(0.42/(156.4 \cdot 1.0)) - 1) + 6.0 + 0.0$$

$$= 9.5672 + 6.0 + 0.0 = 15.5672 \text{ mm Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 156.4 \cdot 1.0 \cdot \ln((2 \cdot 18.0 + 7112.0) / 7112.0) - 0.036$$

$$= 0.79 - 0.036 = 0.7535 \text{ MPa}$$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D)$$

$$= 175.0 \cdot 1.0 \cdot \ln((2 \cdot 24.0 + 7100.0) / 7100.0)$$

$$= 1.1791 \text{ MPa}$$

Computed Stress Intensity at Design Pressure [Si]:

$$= P \cdot D / (E (D_o - D))$$

$$= 0.42 \cdot 7112.0 / (1.0 (7148.0 - 7112.0))$$

$$= 83.0148 \text{ MPa}$$

Computed Membrane Stress at Design Pressure [Sm]:

$$= P / (E \cdot \ln((2 \cdot t + D) / D))$$



$$= 0.42 / (1.0 \cdot \ln((2 \cdot 18.0 + 7112.0) / 7112.0))$$
$$= 83.2251 \text{ MPa}$$

Percent Fiber Elongation per Table 6.1

$$\text{Cylinders formed from Plate} \quad (50 \cdot t_{\text{nom}} / R_f) \cdot (1 - R_f / R_o) \quad 0.337 \%$$

Minimum Design Metal Temperature Results:

Note: This Element/Detail was specified as being Impact Tested.

Material treatment options: +IT

EXTERNAL PRESSURE RESULTS for: 122-D1-Shell-2

Cylindrical Shell From 190 to Ring R2 Ext. Chart: CS-2 at 130 °C

Maximum External Pressure per ASME Division 2 Section 4.4:

Determine Parameter [Mx]:

$$= L / \sqrt{(R_o \cdot t)}$$
$$= 2750.0 / \sqrt{(3574.0 \cdot 18.0)}$$
$$= 10.842$$

Compute the Factor [Ch]:

$$\text{Since } M_x > 1.5 \text{ and } M_x < 13, \text{ Ch} = 0.92 / (M_x - 0.579)$$
$$= 0.92 / (10.842 - 0.579)$$
$$= 0.0896$$

Elastic Hoop Compressive Membrane Failure Stress [Fhe]:

$$= 1.6 \cdot \text{Ch} \cdot E_y (t / D_o)$$
$$= 1.6 \cdot 0.09 \cdot 196301 (18.0 / 7148.0)$$
$$= 70.898 \text{ MPa}$$

Ratio of Elastic Hoop Compressive Membrane Failure Stress to Yield [Fr]:

$$= F_{he} / F_y$$
$$= 70.9 / 234.4$$
$$= 0.302$$

Predicted Buckling Stress [Fic]:

Engineering Yield to Tensile Stress ratio [R]:

$$= \sigma_{ys} / \sigma_{uts}$$
$$= 234.4 / 482.6$$
$$= 0.486$$

Ratio [AE]:

$$= F_{he} / E$$
$$= 70.9 / 196301.4$$
$$= 0.000361$$

Factor [m1]:

$$= (\ln(r) + (\epsilon_{psp} - \epsilon_{sys})) / \ln(\ln(1 + \epsilon_{psp}) / \ln(1 + \epsilon_{sys}))$$
$$= (\ln(0.486) + (0.00002 - 0.002)) / \ln(\ln(1 + 0.00002) / \ln(1 + 0.002))$$
$$= 0.157$$

Curve Fitting Constant for the Elastic Region of the Stress-Strain Curve [A1]:

$$= \sigma_{ys} (1 + \epsilon_{sys}) / \ln(1 + \epsilon_{sys})^{m1}$$
$$= 234.4 (1 + 0.002) / \ln(1 + 0.002)^{0.157301}$$
$$= 624.374 \text{ MPa}$$

Curve Fitting Exponent for, Ferritic Steel [m2]:

$$= m2f(m2f2 - R)$$
$$= 0.6(1.0 - 0.486)$$
$$= 0.309$$

Curve Fitting Constant [A2]:

$$= (\sigma_{uts} \cdot \exp(m2)) / (m2^{m2})$$
$$= (482.6 \cdot \exp(0.309)) / (0.309^{0.309})$$



= 944.566 MPa

Material Parameter for Stress-Strain Curve Model [K]:

$$= 1.5 \cdot R1.5 - 0.5 \cdot R2.5 - R3.5$$

$$= 1.5 \cdot 0.4861.5 - 0.5 \cdot 0.4862.5 - 0.4863.5$$

$$= 0.346$$

Compute the Allowable Stress per 4.4.3

	Fic	H	D1	D2	D3	D4	Tangent Modulus	Iteration Diff
1	241	-1.839	0.313E-04	0.271E-04	0.807E-04	-0.632E-04	0.124E+05	0.192E-01
2	120	-4.651	0.763E-06	0.763E-06	0.171E-04	-0.171E-04	0.151E+06	0.439E-03
3	60	-6.057	0.186E-07	0.186E-07	0.361E-05	-0.361E-05	0.195E+06	0.516E-04
4	90	-5.354	0.163E-06	0.163E-06	0.896E-05	-0.896E-05	0.184E+06	0.129E-03
5	75	-5.706	0.615E-07	0.615E-07	0.596E-05	-0.596E-05	0.192E+06	0.323E-04
6	67	-5.881	0.350E-07	0.350E-07	0.470E-05	-0.470E-05	0.194E+06	0.107E-04
7	71	-5.793	0.467E-07	0.467E-07	0.531E-05	-0.531E-05	0.193E+06	0.105E-04
8	69	-5.837	0.405E-07	0.405E-07	0.500E-05	-0.500E-05	0.193E+06	0.145E-06
9	70	-5.815	0.435E-07	0.435E-07	0.515E-05	-0.515E-05	0.193E+06	0.516E-05
10	70	-5.826	0.420E-07	0.420E-07	0.508E-05	-0.508E-05	0.193E+06	0.250E-05
11	69	-5.832	0.412E-07	0.412E-07	0.504E-05	-0.504E-05	0.193E+06	0.118E-05
12	69	-5.835	0.409E-07	0.409E-07	0.502E-05	-0.502E-05	0.193E+06	0.516E-06
13	69	-5.836	0.407E-07	0.407E-07	0.501E-05	-0.501E-05	0.193E+06	0.186E-06
14	69	-5.837	0.406E-07	0.406E-07	0.501E-05	-0.501E-05	0.193E+06	0.204E-07
15	69	-5.837	0.406E-07	0.405E-07	0.500E-05	-0.500E-05	0.193E+06	0.622E-07
16	69	-5.837	0.406E-07	0.406E-07	0.500E-05	-0.500E-05	0.193E+06	0.209E-07
17	69	-5.837	0.406E-07	0.406E-07	0.501E-05	-0.501E-05	0.193E+06	0.247E-09

Factor of Safety [FS]:

$$\text{Since } Fic [69.78] < 0.55 \cdot 234.4, FS = 2$$

Allowable Circumferential Compressive Stress [Fha]:

$$= Fic/FS$$

$$= 69.784/2.0$$

$$= 34.892 \text{ MPa}$$

Allowable External Pressure at the given Thickness [Pa]:

$$= 2 \cdot Fha(t / Do)$$

$$= 2 \cdot 34.892(18.0/7148.0)$$

$$= 0.176 \text{ MPa}$$

Cylindrical Shell From Ring R2 to 200 Ext. Chart: CS-2 at 130 °C

Maximum External Pressure per ASME Division 2 Section 4.4:

Determine Parameter [Mx]:

$$= L / \sqrt{(Ro \cdot t)}$$

$$= 2500.0 / \sqrt{(3574.0 \cdot 18.0)}$$

$$= 9.857$$

Compute the Factor [Ch]:

$$\text{Since } Mx > 1.5 \text{ and } Mx < 13, Ch = 0.92 / (Mx - 0.579)$$

$$= 0.92 / (9.857 - 0.579)$$

$$= 0.0992$$

Elastic Hoop Compressive Membrane Failure Stress [Fhe]:

$$= 1.6 \cdot Ch \cdot Ey(t / Do)$$

$$= 1.6 \cdot 0.099 \cdot 196301(18.0/7148.0)$$

$$= 78.430 \text{ MPa}$$

Ratio of Elastic Hoop Compressive Membrane Failure Stress to Yield [Fr]:

$$= Fhe / Fy$$

$$= 78.4/234.4$$

$$= 0.335$$

Predicted Buckling Stress [Fic]:

Engineering Yield to Tensile Stress ratio [R]:

$$= \sigma_{ys} / \sigma_{uts}$$

$$= 234.4/482.6$$

$$= 0.486$$

Ratio [AE]:

$$= F_{he} / E$$

$$= 78.4/196301.4$$

$$= 0.000400$$

Factor [m1]:

$$= (\ln(r) + (\text{epsps} - \text{epsys})) / \ln(\ln(1 + \text{epsps}) / \ln(1 + \text{epsys}))$$

$$= (\ln(0.486) + (0.00002 - 0.002)) / \ln(\ln(1 + 0.00002) / \ln(1 + 0.002))$$

$$= 0.157$$

Curve Fitting Constant for the Elastic Region of the Stress-Strain Curve [A1]:

$$= \sigma_{ys}(1 + \text{epsys}) / \ln(1 + \text{epsys}) m1$$

$$= 234.4(1 + 0.002) / \ln(1 + 0.002) 0.157301$$

$$= 624.374 \text{ MPa}$$

Curve Fitting Exponent for, Ferritic Steel [m2]:

$$= m2f(m2f2 - R)$$

$$= 0.6(1.0 - 0.486)$$

$$= 0.309$$

Curve Fitting Constant [A2]:

$$= (\sigma_{uts} \cdot \exp(m2)) / (m2 m2)$$

$$= (482.6 \cdot \exp(0.309)) / (0.309 0.309)$$

$$= 944.566 \text{ MPa}$$

Material Parameter for Stress-Strain Curve Model [K]:

$$= 1.5 \cdot R1.5 - 0.5 \cdot R2.5 - R3.5$$

$$= 1.5 \cdot 0.4861.5 - 0.5 \cdot 0.4862.5 - 0.4863.5$$

$$= 0.346$$

Compute the Allowable Stress per 4.4.3

	Fic	H	D1	D2	D3	D4	Tangent Modulus	Iteration Diff
1	241	-1.839	0.313E-04	0.271E-04	0.807E-04	-0.632E-04	0.124E+05	0.191E-01
2	120	-4.651	0.763E-06	0.763E-06	0.171E-04	-0.171E-04	0.151E+06	0.400E-03
3	60	-6.057	0.186E-07	0.186E-07	0.361E-05	-0.361E-05	0.195E+06	0.900E-04
4	90	-5.354	0.163E-06	0.163E-06	0.896E-05	-0.896E-05	0.184E+06	0.911E-04
5	75	-5.706	0.615E-07	0.615E-07	0.596E-05	-0.596E-05	0.192E+06	0.607E-05
6	82	-5.530	0.102E-06	0.102E-06	0.737E-05	-0.737E-05	0.189E+06	0.401E-04
7	79	-5.618	0.799E-07	0.799E-07	0.664E-05	-0.664E-05	0.190E+06	0.165E-04
8	77	-5.662	0.702E-07	0.702E-07	0.629E-05	-0.629E-05	0.191E+06	0.512E-05
9	76	-5.684	0.657E-07	0.657E-07	0.612E-05	-0.612E-05	0.191E+06	0.502E-06
10	76	-5.673	0.679E-07	0.679E-07	0.621E-05	-0.621E-05	0.191E+06	0.230E-05
11	76	-5.678	0.668E-07	0.668E-07	0.617E-05	-0.617E-05	0.191E+06	0.897E-06
12	76	-5.681	0.663E-07	0.663E-07	0.614E-05	-0.614E-05	0.191E+06	0.197E-06
13	76	-5.682	0.660E-07	0.660E-07	0.613E-05	-0.613E-05	0.191E+06	0.153E-06
14	76	-5.682	0.661E-07	0.661E-07	0.614E-05	-0.614E-05	0.191E+06	0.222E-07
15	76	-5.682	0.661E-07	0.661E-07	0.614E-05	-0.614E-05	0.191E+06	0.652E-07
16	76	-5.682	0.661E-07	0.661E-07	0.614E-05	-0.614E-05	0.191E+06	0.215E-07
17	76	-5.682	0.661E-07	0.661E-07	0.614E-05	-0.614E-05	0.191E+06	0.338E-09

Factor of Safety [FS]:

$$\text{Since } F_{ic} [76.44] < 0.55 \cdot 234.4, FS = 2$$

Allowable Circumferential Compressive Stress [Fha]:

$$= F_{ic}/FS$$

$$= 76.441/2.0$$

$$= 38.221 \text{ MPa}$$

Allowable External Pressure at the given Thickness [Pa]:

$$= 2 \cdot F_{ha}(t / D_o)$$

$$= 2 \cdot 38.221(18.0/7148.0)$$

$$= 0.192 \text{ MPa}$$

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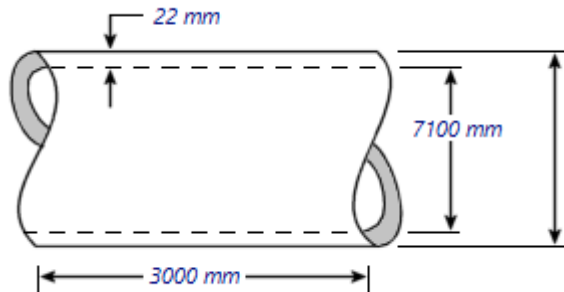
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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 2, 2025



Internal Pressure Results for: 122-D1-Shell-3

Cylindrical Shell Design Information:					
Design Pressure	P	0.389 MPa	Int. Design Temperature		130 °C
Material		SA-516 70	Ext. Design Temperature		130 °C
Length		3000.000 mm	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	6.0000 mm	Allowable Stress (ope)	S	156.4 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	175.0 MPa
Inside Diameter	Di	7100.000 mm	Actual Stress		86.5 MPa
Spec. Min. Thk t,e, or sa		22.0000 mm	Spec. Nominal Thk.		22.000 mm
Surface Area		67330608.0 mm²	SG of Contents		1.060
Element Volume		118797.0 ltr	Weight of Contents		41135.4 N
Empty Weight		112225.8 N	Operating Weight		153361.1 N
			Comp. All. Ope. Stress		56.0 MPa

Radiography Information:			
...		...	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Material treatment options: +IT

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 7112.0 \cdot (\exp(0.389/(156.4 \cdot 1.0)) - 1) + 6.0 + 0.0$$

$$= 8.8563 + 6.0 + 0.0 = 14.8563 \text{ mm Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 156.4 \cdot 1.0 \cdot \ln((2 \cdot 16.0 + 7112.0) / 7112.0) - 0.005$$

$$= 0.702 - 0.005 = 0.6971 \text{ MPa}$$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D)$$

$$= 175.0 \cdot 1.0 \cdot \ln((2 \cdot 22.0 + 7100.0) / 7100.0)$$

$$= 1.0811 \text{ MPa}$$

Computed Stress Intensity at Design Pressure [Si]:

$$= P \cdot D / (E (D_o - D))$$

$$= 0.389 \cdot 7112.0 / (1.0 (7144.0 - 7112.0))$$

$$= 86.4651 \text{ MPa}$$

Computed Membrane Stress at Design Pressure [Sm]:

$$= P / (E \cdot \ln((2 \cdot t + D) / D))$$



$$= 0.389 / (1.0 \cdot \ln((2 \cdot 16.0 + 7112.0) / 7112.0))$$
$$= 86.6584 \text{ MPa}$$

Percent Fiber Elongation per Table 6.1

$$\text{Cylinders formed from Plate} \quad (50 \cdot t_{\text{nom}} / R_f) \cdot (1 - R_f / R_o) \quad 0.309 \%$$

Minimum Design Metal Temperature Results:

Note: This Element/Detail was specified as being Impact Tested.

Material treatment options: +IT

EXTERNAL PRESSURE RESULTS for: 122-D1-Shell-3

Cylindrical Shell From 200 to Ring R3 Ext. Chart: CS-2 at 130 °C

Maximum External Pressure per ASME Division 2 Section 4.4:

Determine Parameter [Mx]:

$$= L / \sqrt{(R_o \cdot t)}$$
$$= 2500.0 / \sqrt{(3572.0 \cdot 16.0)}$$
$$= 10.457$$

Compute the Factor [Ch]:

$$\text{Since } M_x > 1.5 \text{ and } M_x < 13, \text{ Ch} = 0.92 / (M_x - 0.579)$$
$$= 0.92 / (10.457 - 0.579)$$
$$= 0.0931$$

Elastic Hoop Compressive Membrane Failure Stress [Fhe]:

$$= 1.6 \cdot \text{Ch} \cdot E_y (t / D_o)$$
$$= 1.6 \cdot 0.093 \cdot 196301 (16.0 / 7144.0)$$
$$= 65.512 \text{ MPa}$$

Ratio of Elastic Hoop Compressive Membrane Failure Stress to Yield [Fr]:

$$= F_{he} / F_y$$
$$= 65.5 / 234.4$$
$$= 0.279$$

Predicted Buckling Stress [Fic]:

Engineering Yield to Tensile Stress ratio [R]:

$$= \sigma_{ys} / \sigma_{uts}$$
$$= 234.4 / 482.6$$
$$= 0.486$$

Ratio [AE]:

$$= F_{he} / E$$
$$= 65.5 / 196301.4$$
$$= 0.000334$$

Factor [m1]:

$$= (\ln(r) + (\epsilon_{ps} - \epsilon_{sys})) / \ln(\ln(1 + \epsilon_{ps}) / \ln(1 + \epsilon_{sys}))$$
$$= (\ln(0.486) + (0.00002 - 0.002)) / \ln(\ln(1 + 0.00002) / \ln(1 + 0.002))$$
$$= 0.157$$

Curve Fitting Constant for the Elastic Region of the Stress-Strain Curve [A1]:

$$= \sigma_{ys} (1 + \epsilon_{sys}) / \ln(1 + \epsilon_{sys})^{m1}$$
$$= 234.4 (1 + 0.002) / \ln(1 + 0.002)^{0.157301}$$
$$= 624.374 \text{ MPa}$$

Curve Fitting Exponent for, Ferritic Steel [m2]:

$$= m2f(m2f2 - R)$$
$$= 0.6(1.0 - 0.486)$$
$$= 0.309$$

Curve Fitting Constant [A2]:

$$= (\sigma_{uts} \cdot \exp(m2)) / (m2^{m2})$$
$$= (482.6 \cdot \exp(0.309)) / (0.309^{0.309})$$



= 944.566 MPa

Material Parameter for Stress-Strain Curve Model [K]:

$$= 1.5 \cdot R1.5 - 0.5 \cdot R2.5 - R3.5$$

$$= 1.5 \cdot 0.4861.5 - 0.5 \cdot 0.4862.5 - 0.4863.5$$

$$= 0.346$$

Compute the Allowable Stress per 4.4.3

	Fic	H	D1	D2	D3	D4	Tangent Modulus	Iteration Diff
1	241	-1.839	0.313E-04	0.271E-04	0.807E-04	-0.632E-04	0.124E+05	0.192E-01
2	120	-4.651	0.763E-06	0.763E-06	0.171E-04	-0.171E-04	0.151E+06	0.466E-03
3	60	-6.057	0.186E-07	0.186E-07	0.361E-05	-0.361E-05	0.195E+06	0.241E-04
4	90	-5.354	0.163E-06	0.163E-06	0.896E-05	-0.896E-05	0.184E+06	0.157E-03
5	75	-5.706	0.615E-07	0.615E-07	0.596E-05	-0.596E-05	0.192E+06	0.597E-04
6	67	-5.881	0.350E-07	0.350E-07	0.470E-05	-0.470E-05	0.194E+06	0.168E-04
7	64	-5.969	0.257E-07	0.257E-07	0.414E-05	-0.414E-05	0.194E+06	0.388E-05
8	65	-5.925	0.301E-07	0.301E-07	0.442E-05	-0.442E-05	0.194E+06	0.640E-05
9	65	-5.947	0.278E-07	0.278E-07	0.428E-05	-0.428E-05	0.194E+06	0.125E-05
10	64	-5.958	0.268E-07	0.268E-07	0.421E-05	-0.421E-05	0.194E+06	0.132E-05
11	64	-5.953	0.273E-07	0.273E-07	0.424E-05	-0.424E-05	0.194E+06	0.359E-07
12	64	-5.950	0.276E-07	0.276E-07	0.426E-05	-0.426E-05	0.194E+06	0.606E-06
13	64	-5.951	0.274E-07	0.274E-07	0.425E-05	-0.425E-05	0.194E+06	0.285E-06
14	64	-5.952	0.274E-07	0.274E-07	0.425E-05	-0.425E-05	0.194E+06	0.124E-06
15	64	-5.952	0.273E-07	0.273E-07	0.424E-05	-0.424E-05	0.194E+06	0.442E-07
16	64	-5.953	0.273E-07	0.273E-07	0.424E-05	-0.424E-05	0.194E+06	0.415E-08

Factor of Safety [FS]:

$$\text{Since } F_{ic} [64.82] < 0.55 \cdot 234.4, FS = 2$$

Allowable Circumferential Compressive Stress [Fha]:

$$= F_{ic}/FS$$

$$= 64.816/2.0$$

$$= 32.408 \text{ MPa}$$

Allowable External Pressure at the given Thickness [Pa]:

$$= 2 \cdot F_{ha} (t / D_o)$$

$$= 2 \cdot 32.408 (16.0/7144.0)$$

$$= 0.145 \text{ MPa}$$

Cylindrical Shell From Ring R3 to 210 Ext. Chart: CS-2 at 130 °C

Maximum External Pressure per ASME Division 2 Section 4.4:

Determine Parameter [Mx]:

$$= L / \sqrt{(R_o \cdot t)}$$

$$= 2499.899 / \sqrt{(3572.0 \cdot 16.0)}$$

$$= 10.457$$

Compute the Factor [Ch]:

$$\text{Since } M_x > 1.5 \text{ and } M_x < 13, Ch = 0.92 / (M_x - 0.579)$$

$$= 0.92 / (10.457 - 0.579)$$

$$= 0.0931$$

Elastic Hoop Compressive Membrane Failure Stress [Fhe]:

$$= 1.6 \cdot Ch \cdot E_y (t / D_o)$$

$$= 1.6 \cdot 0.093 \cdot 196301 (16.0/7144.0)$$

$$= 65.515 \text{ MPa}$$

Ratio of Elastic Hoop Compressive Membrane Failure Stress to Yield [Fr]:

$$= F_{he} / F_y$$

$$= 65.5/234.4$$

$$= 0.280$$

Predicted Buckling Stress [Fic]:

Engineering Yield to Tensile Stress ratio [R]:

$$= \sigma_{ys} / \sigma_{uts}$$

$$= 234.4/482.6$$

$$= 0.486$$

Ratio [AE]:

$$= F_{he} / E$$

$$= 65.5/196301.4$$

$$= 0.000334$$

Factor [m1]:

$$= (\ln(r) + (\epsilon_{psp} - \epsilon_{psys})) / \ln(\ln(1 + \epsilon_{psp}) / \ln(1 + \epsilon_{psys}))$$

$$= (\ln(0.486) + (0.00002 - 0.002)) / \ln(\ln(1 + 0.00002) / \ln(1 + 0.002))$$

$$= 0.157$$

Curve Fitting Constant for the Elastic Region of the Stress-Strain Curve [A1]:

$$= \sigma_{ys} (1 + \epsilon_{psys}) / \ln(1 + \epsilon_{psys}) m1$$

$$= 234.4 (1 + 0.002) / \ln(1 + 0.002) 0.157301$$

$$= 624.374 \text{ MPa}$$

Curve Fitting Exponent for, Ferritic Steel [m2]:

$$= m2f(m2f2 - R)$$

$$= 0.6(1.0 - 0.486)$$

$$= 0.309$$

Curve Fitting Constant [A2]:

$$= (\sigma_{uts} \cdot \exp(m2)) / (m2 m2)$$

$$= (482.6 \cdot \exp(0.309)) / (0.309 0.309)$$

$$= 944.566 \text{ MPa}$$

Material Parameter for Stress-Strain Curve Model [K]:

$$= 1.5 \cdot R1.5 - 0.5 \cdot R2.5 - R3.5$$

$$= 1.5 \cdot 0.4861.5 - 0.5 \cdot 0.4862.5 - 0.4863.5$$

$$= 0.346$$

Compute the Allowable Stress per 4.4.3

	Fic	H	D1	D2	D3	D4	Tangent Modulus	Iteration Diff
1	241	-1.839	0.313E-04	0.271E-04	0.807E-04	-0.632E-04	0.124E+05	0.192E-01
2	120	-4.651	0.763E-06	0.763E-06	0.171E-04	-0.171E-04	0.151E+06	0.466E-03
3	60	-6.057	0.186E-07	0.186E-07	0.361E-05	-0.361E-05	0.195E+06	0.242E-04
4	90	-5.354	0.163E-06	0.163E-06	0.896E-05	-0.896E-05	0.184E+06	0.157E-03
5	75	-5.706	0.615E-07	0.615E-07	0.596E-05	-0.596E-05	0.192E+06	0.597E-04
6	67	-5.881	0.350E-07	0.350E-07	0.470E-05	-0.470E-05	0.194E+06	0.168E-04
7	64	-5.969	0.257E-07	0.257E-07	0.414E-05	-0.414E-05	0.194E+06	0.389E-05
8	65	-5.925	0.301E-07	0.301E-07	0.442E-05	-0.442E-05	0.194E+06	0.638E-05
9	65	-5.947	0.278E-07	0.278E-07	0.428E-05	-0.428E-05	0.194E+06	0.123E-05
10	64	-5.958	0.268E-07	0.268E-07	0.421E-05	-0.421E-05	0.194E+06	0.133E-05
11	64	-5.953	0.273E-07	0.273E-07	0.424E-05	-0.424E-05	0.194E+06	0.502E-07
12	64	-5.950	0.276E-07	0.276E-07	0.426E-05	-0.426E-05	0.194E+06	0.591E-06
13	64	-5.951	0.274E-07	0.274E-07	0.425E-05	-0.425E-05	0.194E+06	0.270E-06
14	64	-5.952	0.274E-07	0.274E-07	0.425E-05	-0.425E-05	0.194E+06	0.110E-06
15	64	-5.952	0.273E-07	0.273E-07	0.424E-05	-0.424E-05	0.194E+06	0.299E-07
16	64	-5.953	0.273E-07	0.273E-07	0.424E-05	-0.424E-05	0.194E+06	0.101E-07
17	64	-5.952	0.273E-07	0.273E-07	0.424E-05	-0.424E-05	0.194E+06	0.990E-08

Factor of Safety [FS]:

Since $Fic [64.82] < 0.55 \cdot 234.4$, $FS = 2$

Allowable Circumferential Compressive Stress [Fha]:

$$= Fic/FS$$

$$= 64.82/2.0$$

$$= 32.410 \text{ MPa}$$

Allowable External Pressure at the given Thickness [Pa]:

$$= 2 \cdot Fha(t / Do)$$

$$= 2 \cdot 32.41(16.0/7144.0)$$

$$= 0.145 \text{ MPa}$$

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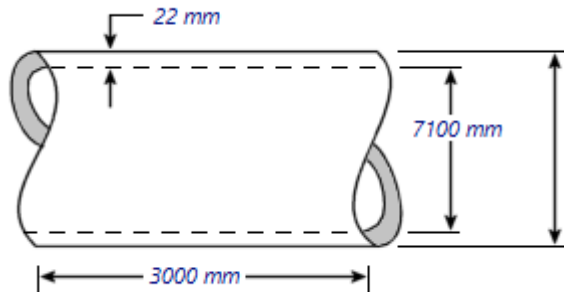


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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 2, 2025



Internal Pressure Results for: 122-D1-Shell-4

Cylindrical Shell Design Information:					
Design Pressure	P	0.388 MPa	Int. Design Temperature		130 °C
Material		SA-516 70	Ext. Design Temperature		130 °C
Length		3000.000 mm	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	6.0000 mm	Allowable Stress (ope)	S	156.4 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	175.0 MPa
Inside Diameter	Di	7100.000 mm	Actual Stress		86.2 MPa
Spec. Min. Thk t,e, or sa		22.0000 mm	Spec. Nominal Thk.		22.000 mm
Surface Area		67330608.0 mm²	SG of Contents		0.000
Element Volume		118797.0 ltr	Weight of Contents		0.0 N
Empty Weight		112225.8 N	Operating Weight		112225.8 N
			Comp. All. Ope. Stress		56.0 MPa

Radiography Information:			
...		...	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Material treatment options: +IT

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 7112.0 \cdot (\exp(0.388/(156.4 \cdot 1.0)) - 1) + 6.0 + 0.0$$

$$= 8.8326 + 6.0 + 0.0 = 14.8326 \text{ mm Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 156.4 \cdot 1.0 \cdot \ln((2 \cdot 16.0 + 7112.0) / 7112.0) - 0.004$$

$$= 0.702 - 0.004 = 0.6981 \text{ MPa}$$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D)$$

$$= 175.0 \cdot 1.0 \cdot \ln((2 \cdot 22.0 + 7100.0) / 7100.0)$$

$$= 1.0811 \text{ MPa}$$

Computed Stress Intensity at Design Pressure [Si]:

$$= P \cdot D / (E (D_o - D))$$

$$= 0.388 \cdot 7112.0 / (1.0 (7144.0 - 7112.0))$$

$$= 86.2341 \text{ MPa}$$

Computed Membrane Stress at Design Pressure [Sm]:

$$= P / (E \cdot \ln((2 \cdot t + D) / D))$$



$$= 0.388 / (1.0 \cdot \ln((2 \cdot 16.0 + 7112.0) / 7112.0))$$
$$= 86.4270 \text{ MPa}$$

Percent Fiber Elongation per Table 6.1

$$\text{Cylinders formed from Plate} \quad (50 \cdot t_{\text{nom}} / R_f) \cdot (1 - R_f / R_o) \quad 0.309 \%$$

Minimum Design Metal Temperature Results:

Note: This Element/Detail was specified as being Impact Tested.

Material treatment options: +IT

EXTERNAL PRESSURE RESULTS for: 122-D1-Shell-4

Cylindrical Shell From 210 to Ring R4 Ext. Chart: CS-2 at 130 °C

Maximum External Pressure per ASME Division 2 Section 4.4:

Determine Parameter [Mx]:

$$= L / \sqrt{(R_o \cdot t)}$$
$$= 2499.899 / \sqrt{(3572.0 \cdot 16.0)}$$
$$= 10.457$$

Compute the Factor [Ch]:

Since $M_x > 1.5$ and $M_x < 13$, $Ch = 0.92 / (M_x - 0.579)$

$$= 0.92 / (10.457 - 0.579)$$
$$= 0.0931$$

Elastic Hoop Compressive Membrane Failure Stress [Fhe]:

$$= 1.6 \cdot Ch \cdot E_y (t / D_o)$$
$$= 1.6 \cdot 0.093 \cdot 196301 (16.0 / 7144.0)$$
$$= 65.515 \text{ MPa}$$

Ratio of Elastic Hoop Compressive Membrane Failure Stress to Yield [Fr]:

$$= F_{he} / F_y$$
$$= 65.5 / 234.4$$
$$= 0.280$$

Predicted Buckling Stress [Fic]:

Engineering Yield to Tensile Stress ratio [R]:

$$= \sigma_{ys} / \sigma_{uts}$$
$$= 234.4 / 482.6$$
$$= 0.486$$

Ratio [AE]:

$$= F_{he} / E$$
$$= 65.5 / 196301.4$$
$$= 0.000334$$

Factor [m1]:

$$= (\ln(r) + (\epsilon_{ps} - \epsilon_{sys})) / \ln(\ln(1 + \epsilon_{ps}) / \ln(1 + \epsilon_{sys}))$$
$$= (\ln(0.486) + (0.00002 - 0.002)) / \ln(\ln(1 + 0.00002) / \ln(1 + 0.002))$$
$$= 0.157$$

Curve Fitting Constant for the Elastic Region of the Stress-Strain Curve [A1]:

$$= \sigma_{ys} (1 + \epsilon_{sys}) / \ln(1 + \epsilon_{sys})^{m1}$$
$$= 234.4 (1 + 0.002) / \ln(1 + 0.002)^{0.157301}$$
$$= 624.374 \text{ MPa}$$

Curve Fitting Exponent for, Ferritic Steel [m2]:

$$= m2f(m2f2 - R)$$
$$= 0.6(1.0 - 0.486)$$
$$= 0.309$$

Curve Fitting Constant [A2]:

$$= (\sigma_{uts} \cdot \exp(m2)) / (m2^{m2})$$
$$= (482.6 \cdot \exp(0.309)) / (0.309^{0.309})$$



= 944.566 MPa

Material Parameter for Stress-Strain Curve Model [K]:

= $1.5 \cdot R1.5 - 0.5 \cdot R2.5 - R3.5$

= $1.5 \cdot 0.4861.5 - 0.5 \cdot 0.4862.5 - 0.4863.5$

= 0.346

Compute the Allowable Stress per 4.4.3

	Fic	H	D1	D2	D3	D4	Tangent Modulus	Iteration Diff
1	241	-1.839	0.313E-04	0.271E-04	0.807E-04	-0.632E-04	0.124E+05	0.192E-01
2	120	-4.651	0.763E-06	0.763E-06	0.171E-04	-0.171E-04	0.151E+06	0.466E-03
3	60	-6.057	0.186E-07	0.186E-07	0.361E-05	-0.361E-05	0.195E+06	0.242E-04
4	90	-5.354	0.163E-06	0.163E-06	0.896E-05	-0.896E-05	0.184E+06	0.157E-03
5	75	-5.706	0.615E-07	0.615E-07	0.596E-05	-0.596E-05	0.192E+06	0.597E-04
6	67	-5.881	0.350E-07	0.350E-07	0.470E-05	-0.470E-05	0.194E+06	0.168E-04
7	64	-5.969	0.257E-07	0.257E-07	0.414E-05	-0.414E-05	0.194E+06	0.389E-05
8	65	-5.925	0.301E-07	0.301E-07	0.442E-05	-0.442E-05	0.194E+06	0.638E-05
9	65	-5.947	0.278E-07	0.278E-07	0.428E-05	-0.428E-05	0.194E+06	0.123E-05
10	64	-5.958	0.268E-07	0.268E-07	0.421E-05	-0.421E-05	0.194E+06	0.133E-05
11	64	-5.953	0.273E-07	0.273E-07	0.424E-05	-0.424E-05	0.194E+06	0.502E-07
12	64	-5.950	0.276E-07	0.276E-07	0.426E-05	-0.426E-05	0.194E+06	0.591E-06
13	64	-5.951	0.274E-07	0.274E-07	0.425E-05	-0.425E-05	0.194E+06	0.270E-06
14	64	-5.952	0.274E-07	0.274E-07	0.425E-05	-0.425E-05	0.194E+06	0.110E-06
15	64	-5.952	0.273E-07	0.273E-07	0.424E-05	-0.424E-05	0.194E+06	0.299E-07
16	64	-5.953	0.273E-07	0.273E-07	0.424E-05	-0.424E-05	0.194E+06	0.101E-07
17	64	-5.952	0.273E-07	0.273E-07	0.424E-05	-0.424E-05	0.194E+06	0.990E-08

Factor of Safety [FS]:

Since Fic [64.82] < 0.55 * 234.4, FS = 2

Allowable Circumferential Compressive Stress [Fha]:

= Fic/FS

= 64.82/2.0

= 32.410 MPa

Allowable External Pressure at the given Thickness [Pa]:

= $2 \cdot Fha(t / Do)$

= $2 \cdot 32.41(16.0/7144.0)$

= 0.145 MPa

Cylindrical Shell From Ring R4 to 220 Ext. Chart: CS-2 at 130 °C

Maximum External Pressure per ASME Division 2 Section 4.4:

Determine Parameter [Mx]:

= $L / \sqrt{(Ro \cdot t)}$

= $2500.005 / \sqrt{(3572.0 \cdot 16.0)}$

= 10.457

Compute the Factor [Ch]:

Since Mx > 1.5 and Mx < 13, Ch = $0.92 / (Mx - 0.579)$

= $0.92 / (10.457 - 0.579)$

= 0.0931

Elastic Hoop Compressive Membrane Failure Stress [Fhe]:

= $1.6 \cdot Ch \cdot Ey(t / Do)$

= $1.6 \cdot 0.093 \cdot 196301(16.0/7144.0)$

= 65.512 MPa

Ratio of Elastic Hoop Compressive Membrane Failure Stress to Yield [Fr]:

= Fhe / Fy

= 65.5/234.4

= 0.279

Predicted Buckling Stress [Fic]:



Engineering Yield to Tensile Stress ratio [R]:

$$= \sigma_{ys} / \sigma_{uts}$$

$$= 234.4/482.6$$

$$= 0.486$$

Ratio [AE]:

$$= F_{he} / E$$

$$= 65.5/196301.4$$

$$= 0.000334$$

Factor [m1]:

$$= (\ln(r) + (\epsilon_{psp} - \epsilon_{sys})) / \ln(\ln(1 + \epsilon_{psp}) / \ln(1 + \epsilon_{sys}))$$

$$= (\ln(0.486) + (0.00002 - 0.002)) / \ln(\ln(1 + 0.00002) / \ln(1 + 0.002))$$

$$= 0.157$$

Curve Fitting Constant for the Elastic Region of the Stress-Strain Curve [A1]:

$$= \sigma_{ys} (1 + \epsilon_{sys}) / \ln(1 + \epsilon_{sys}) m_1$$

$$= 234.4 (1 + 0.002) / \ln(1 + 0.002) 0.157301$$

$$= 624.374 \text{ MPa}$$

Curve Fitting Exponent for, Ferritic Steel [m2]:

$$= m_2 f(m_2 f_2 - R)$$

$$= 0.6 (1.0 - 0.486)$$

$$= 0.309$$

Curve Fitting Constant [A2]:

$$= (\sigma_{uts} \cdot \exp(m_2)) / (m_2 m_2)$$

$$= (482.6 \cdot \exp(0.309)) / (0.309 0.309)$$

$$= 944.566 \text{ MPa}$$

Material Parameter for Stress-Strain Curve Model [K]:

$$= 1.5 \cdot R_{1.5} - 0.5 \cdot R_{2.5} - R_{3.5}$$

$$= 1.5 \cdot 0.4861.5 - 0.5 \cdot 0.4862.5 - 0.4863.5$$

$$= 0.346$$

Compute the Allowable Stress per 4.4.3

	Fic	H	D1	D2	D3	D4	Tangent Modulus	Iteration Diff
1	241	-1.839	0.313E-04	0.271E-04	0.807E-04	-0.632E-04	0.124E+05	0.192E-01
2	120	-4.651	0.763E-06	0.763E-06	0.171E-04	-0.171E-04	0.151E+06	0.466E-03
3	60	-6.057	0.186E-07	0.186E-07	0.361E-05	-0.361E-05	0.195E+06	0.241E-04
4	90	-5.354	0.163E-06	0.163E-06	0.896E-05	-0.896E-05	0.184E+06	0.157E-03
5	75	-5.706	0.615E-07	0.615E-07	0.596E-05	-0.596E-05	0.192E+06	0.597E-04
6	67	-5.881	0.350E-07	0.350E-07	0.470E-05	-0.470E-05	0.194E+06	0.168E-04
7	64	-5.969	0.257E-07	0.257E-07	0.414E-05	-0.414E-05	0.194E+06	0.388E-05
8	65	-5.925	0.301E-07	0.301E-07	0.442E-05	-0.442E-05	0.194E+06	0.640E-05
9	65	-5.947	0.278E-07	0.278E-07	0.428E-05	-0.428E-05	0.194E+06	0.125E-05
10	64	-5.958	0.268E-07	0.268E-07	0.421E-05	-0.421E-05	0.194E+06	0.132E-05
11	64	-5.953	0.273E-07	0.273E-07	0.424E-05	-0.424E-05	0.194E+06	0.352E-07
12	64	-5.950	0.276E-07	0.276E-07	0.426E-05	-0.426E-05	0.194E+06	0.606E-06
13	64	-5.951	0.274E-07	0.274E-07	0.425E-05	-0.425E-05	0.194E+06	0.285E-06
14	64	-5.952	0.274E-07	0.274E-07	0.425E-05	-0.425E-05	0.194E+06	0.125E-06
15	64	-5.952	0.273E-07	0.273E-07	0.424E-05	-0.424E-05	0.194E+06	0.450E-07
16	64	-5.953	0.273E-07	0.273E-07	0.424E-05	-0.424E-05	0.194E+06	0.488E-08

Factor of Safety [FS]:

$$\text{Since } F_{ic} [64.82] < 0.55 \cdot 234.4, FS = 2$$

Allowable Circumferential Compressive Stress [Fha]:

$$= F_{ic} / FS$$

$$= 64.816/2.0$$

$$= 32.408 \text{ MPa}$$

Allowable External Pressure at the given Thickness [Pa]:

$$= 2 \cdot F_{ha} (t / D_o)$$

$$= 2 \cdot 32.408 (16.0/7144.0)$$

$$= 0.145 \text{ MPa}$$

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FileName : 122D1 & 163-D_Ope_Rev-06
Element: 122-D1-Shell-4

Step: 17 5:17pm Aug 25,2026

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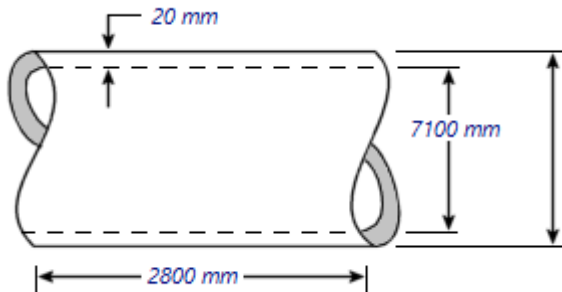


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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 2, 2025



Internal Pressure Results for: 122-D1-Shell-5

Cylindrical Shell Design Information:					
Design Pressure	P	0.388 MPa	Int. Design Temperature		130 °C
Material		SA-516 70	Ext. Design Temperature		130 °C
Length		2800.000 mm	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	6.0000 mm	Allowable Stress (ope)	S	156.4 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	175.0 MPa
Inside Diameter	Di	7100.000 mm	Actual Stress		98.6 MPa
Spec. Min. Thk t _e , or s _a		20.0000 mm	Spec. Nominal Thk.		20.000 mm
Surface Area		62806720.0 mm ²	SG of Contents		0.000
Element Volume		110877.2 ltr	Weight of Contents		0.0 N
Empty Weight		95195.3 N	Operating Weight		95195.3 N
			Comp. All. Ope. Stress		49.0 MPa

Radiography Information:			
...		...	
Circ. Joint Efficiency		Long. Joint Eff.	E or v
	100 %		100 %

Material treatment options: +IT

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 7112.0 \cdot (\exp(0.388/(156.4 \cdot 1.0)) - 1) + 6.0 + 0.0$$

$$= 8.8326 + 6.0 + 0.0 = 14.8326 \text{ mm Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 156.4 \cdot 1.0 \cdot \ln((2 \cdot 14.0 + 7112.0) / 7112.0) - 0.004$$

$$= 0.615 - 0.004 = 0.6105 \text{ MPa}$$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D)$$

$$= 175.0 \cdot 1.0 \cdot \ln((2 \cdot 20.0 + 7100.0) / 7100.0)$$

$$= 0.9832 \text{ MPa}$$

Computed Stress Intensity at Design Pressure [Si]:

$$= P \cdot D / (E (D_o - D))$$

$$= 0.388 \cdot 7112.0 / (1.0 (7140.0 - 7112.0))$$

$$= 98.5526 \text{ MPa}$$

Computed Membrane Stress at Design Pressure [Sm]:

$$= P / (E \cdot \ln((2 \cdot t + D) / D))$$



$$= 0.388 / (1.0 \cdot \ln((2 \cdot 14.0 + 7112.0) / 7112.0))$$
$$= 98.7460 \text{ MPa}$$

Percent Fiber Elongation per Table 6.1

$$\text{Cylinders formed from Plate} \quad (50 \cdot t_{\text{nom}} / R_f) \cdot (1 - R_f / R_o) \quad 0.281 \%$$

Minimum Design Metal Temperature Results:

Note: This Element/Detail was specified as being Impact Tested.

Material treatment options: +IT

EXTERNAL PRESSURE RESULTS for: 122-D1-Shell-5

Cylindrical Shell From 220 to Ring R5 Ext. Chart: CS-2 at 130 °C

Maximum External Pressure per ASME Division 2 Section 4.4:

Determine Parameter [Mx]:

$$= L / \sqrt{(R_o \cdot t)}$$
$$= 2500.005 / \sqrt{(3570.0 \cdot 14.0)}$$
$$= 11.183$$

Compute the Factor [Ch]:

Since $M_x > 1.5$ and $M_x < 13$, $Ch = 0.92 / (M_x - 0.579)$

$$= 0.92 / (11.183 - 0.579)$$
$$= 0.0868$$

Elastic Hoop Compressive Membrane Failure Stress [Fhe]:

$$= 1.6 \cdot Ch \cdot E_y (t / D_o)$$
$$= 1.6 \cdot 0.087 \cdot 196301 (14.0 / 7140.0)$$
$$= 53.433 \text{ MPa}$$

Ratio of Elastic Hoop Compressive Membrane Failure Stress to Yield [Fr]:

$$= F_{he} / F_y$$
$$= 53.4 / 234.4$$
$$= 0.228$$

Predicted Buckling Stress [Fic]:

Engineering Yield to Tensile Stress ratio [R]:

$$= \sigma_{ys} / \sigma_{uts}$$
$$= 234.4 / 482.6$$
$$= 0.486$$

Ratio [AE]:

$$= F_{he} / E$$
$$= 53.4 / 196301.4$$
$$= 0.000272$$

Factor [m1]:

$$= (\ln(r) + (\epsilon_{ps} - \epsilon_{sys})) / \ln(\ln(1 + \epsilon_{ps}) / \ln(1 + \epsilon_{sys}))$$
$$= (\ln(0.486) + (0.00002 - 0.002)) / \ln(\ln(1 + 0.00002) / \ln(1 + 0.002))$$
$$= 0.157$$

Curve Fitting Constant for the Elastic Region of the Stress-Strain Curve [A1]:

$$= \sigma_{ys} (1 + \epsilon_{sys}) / \ln(1 + \epsilon_{sys})^{m1}$$
$$= 234.4 (1 + 0.002) / \ln(1 + 0.002)^{0.157301}$$
$$= 624.374 \text{ MPa}$$

Curve Fitting Exponent for, Ferritic Steel [m2]:

$$= m2f(m2f2 - R)$$
$$= 0.6(1.0 - 0.486)$$
$$= 0.309$$

Curve Fitting Constant [A2]:

$$= (\sigma_{uts} \cdot \exp(m2)) / (m2^{m2})$$
$$= (482.6 \cdot \exp(0.309)) / (0.309^{0.309})$$



= 944.566 MPa

Material Parameter for Stress-Strain Curve Model [K]:

$$= 1.5 \cdot R1.5 - 0.5 \cdot R2.5 - R3.5$$

$$= 1.5 \cdot 0.4861.5 - 0.5 \cdot 0.4862.5 - 0.4863.5$$

$$= 0.346$$

Compute the Allowable Stress per 4.4.3

	Fic	H	D1	D2	D3	D4	Tangent Modulus	Iteration Diff
1	241	-1.839	0.313E-04	0.271E-04	0.807E-04	-0.632E-04	0.124E+05	0.192E-01
2	120	-4.651	0.763E-06	0.763E-06	0.171E-04	-0.171E-04	0.151E+06	0.528E-03
3	60	-6.057	0.186E-07	0.186E-07	0.361E-05	-0.361E-05	0.195E+06	0.374E-04
4	30	-6.760	0.454E-09	0.454E-09	0.765E-06	-0.765E-06	0.196E+06	0.119E-03
5	45	-6.409	0.398E-08	0.398E-08	0.190E-05	-0.190E-05	0.196E+06	0.413E-04
6	52	-6.233	0.910E-08	0.910E-08	0.268E-05	-0.268E-05	0.196E+06	0.231E-05
7	56	-6.145	0.132E-07	0.132E-07	0.313E-05	-0.313E-05	0.195E+06	0.174E-04
8	54	-6.189	0.110E-07	0.110E-07	0.290E-05	-0.290E-05	0.195E+06	0.753E-05
9	53	-6.211	0.100E-07	0.100E-07	0.279E-05	-0.279E-05	0.196E+06	0.260E-05
10	53	-6.222	0.954E-08	0.954E-08	0.273E-05	-0.273E-05	0.196E+06	0.145E-06
11	53	-6.227	0.932E-08	0.932E-08	0.271E-05	-0.271E-05	0.196E+06	0.108E-05
12	53	-6.225	0.943E-08	0.943E-08	0.272E-05	-0.272E-05	0.196E+06	0.470E-06
13	53	-6.223	0.949E-08	0.949E-08	0.273E-05	-0.273E-05	0.196E+06	0.163E-06
14	53	-6.223	0.952E-08	0.952E-08	0.273E-05	-0.273E-05	0.196E+06	0.911E-08

Factor of Safety [FS]:

$$\text{Since } F_{ic} [53.23] < 0.55 \cdot 234.4, FS = 2$$

Allowable Circumferential Compressive Stress [Fha]:

$$= F_{ic}/FS$$

$$= 53.232/2.0$$

$$= 26.616 \text{ MPa}$$

Allowable External Pressure at the given Thickness [Pa]:

$$= 2 \cdot F_{ha} (t / D_o)$$

$$= 2 \cdot 26.616 (14.0/7140.0)$$

$$= 0.104 \text{ MPa}$$

Cylindrical Shell From Ring R5 to 230 Ext. Chart: CS-2 at 130 °C

Maximum External Pressure per ASME Division 2 Section 4.4:

Determine Parameter [Mx]:

$$= L / \sqrt{(R_o \cdot t)}$$

$$= 2499.995 / \sqrt{(3570.0 \cdot 14.0)}$$

$$= 11.183$$

Compute the Factor [Ch]:

$$\text{Since } M_x > 1.5 \text{ and } M_x < 13, Ch = 0.92 / (M_x - 0.579)$$

$$= 0.92 / (11.183 - 0.579)$$

$$= 0.0868$$

Elastic Hoop Compressive Membrane Failure Stress [Fhe]:

$$= 1.6 \cdot Ch \cdot E_y (t / D_o)$$

$$= 1.6 \cdot 0.087 \cdot 196301 (14.0/7140.0)$$

$$= 53.433 \text{ MPa}$$

Ratio of Elastic Hoop Compressive Membrane Failure Stress to Yield [Fr]:

$$= F_{he} / F_y$$

$$= 53.4/234.4$$

$$= 0.228$$

Predicted Buckling Stress [Fic]:

Engineering Yield to Tensile Stress ratio [R]:

$$= \sigma_{ys} / \sigma_{uts}$$

$$= 234.4/482.6$$



= 0.486

Ratio [AE]:

= Fhe / E
 = 53.4/196301.4
 = 0.000272

Factor [m1]:

= (ln(r) + (epsp - epsys)) / ln(ln(1 + epsp) / ln(1 + epsys))
 = (ln(0.486) + (0.00002 - 0.002)) / ln(ln(1 + 0.00002) / ln(1 + 0.002))
 = 0.157

Curve Fitting Constant for the Elastic Region of the Stress-Strain Curve [A1]:

= $\sigma_{ys}(1 + \epsilon_{sys}) / \ln(1 + \epsilon_{sys})^{m1}$
 = 234.4(1 + 0.002) / ln(1 + 0.002) 0.157301
 = 624.374 MPa

Curve Fitting Exponent for, Ferritic Steel [m2]:

= $m2f(m2f^2 - R)$
 = 0.6(1.0 - 0.486)
 = 0.309

Curve Fitting Constant [A2]:

= ($\sigma_{uts} \cdot \exp(m2)$) / ($m2 m2$)
 = (482.6 • exp(0.309)) / (0.309 0.309)
 = 944.566 MPa

Material Parameter for Stress-Strain Curve Model [K]:

= 1.5 • R1.5 - 0.5 • R2.5 - R3.5
 = 1.5 • 0.4861.5 - 0.5 • 0.4862.5 - 0.4863.5
 = 0.346

Compute the Allowable Stress per 4.4.3

	Fic	H	D1	D2	D3	D4	Tangent Modulus	Iteration Diff
1	241	-1.839	0.313E-04	0.271E-04	0.807E-04	-0.632E-04	0.124E+05	0.192E-01
2	120	-4.651	0.763E-06	0.763E-06	0.171E-04	-0.171E-04	0.151E+06	0.528E-03
3	60	-6.057	0.186E-07	0.186E-07	0.361E-05	-0.361E-05	0.195E+06	0.374E-04
4	30	-6.760	0.454E-09	0.454E-09	0.765E-06	-0.765E-06	0.196E+06	0.119E-03
5	45	-6.409	0.398E-08	0.398E-08	0.190E-05	-0.190E-05	0.196E+06	0.413E-04
6	52	-6.233	0.910E-08	0.910E-08	0.268E-05	-0.268E-05	0.196E+06	0.231E-05
7	56	-6.145	0.132E-07	0.132E-07	0.313E-05	-0.313E-05	0.195E+06	0.174E-04
8	54	-6.189	0.110E-07	0.110E-07	0.290E-05	-0.290E-05	0.195E+06	0.753E-05
9	53	-6.211	0.100E-07	0.100E-07	0.279E-05	-0.279E-05	0.196E+06	0.260E-05
10	53	-6.222	0.954E-08	0.954E-08	0.273E-05	-0.273E-05	0.196E+06	0.143E-06
11	53	-6.227	0.932E-08	0.932E-08	0.271E-05	-0.271E-05	0.196E+06	0.109E-05
12	53	-6.225	0.943E-08	0.943E-08	0.272E-05	-0.272E-05	0.196E+06	0.471E-06
13	53	-6.223	0.949E-08	0.949E-08	0.273E-05	-0.273E-05	0.196E+06	0.164E-06
14	53	-6.223	0.952E-08	0.952E-08	0.273E-05	-0.273E-05	0.196E+06	0.102E-07
15	53	-6.222	0.953E-08	0.953E-08	0.273E-05	-0.273E-05	0.196E+06	0.666E-07
16	53	-6.222	0.952E-08	0.952E-08	0.273E-05	-0.273E-05	0.196E+06	0.282E-07
17	53	-6.222	0.952E-08	0.952E-08	0.273E-05	-0.273E-05	0.196E+06	0.899E-08

Factor of Safety [FS]:

Since Fic [53.24] < 0.55 * 234.4, FS = 2

Allowable Circumferential Compressive Stress [Fha]:

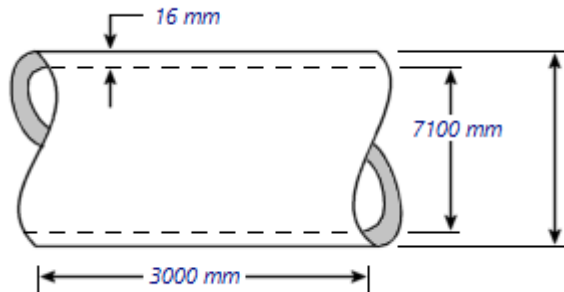
= Fic/FS
 = 53.235/2.0
 = 26.618 MPa

Allowable External Pressure at the given Thickness [Pa]:

= 2 • Fha(t / Do)
 = 2 • 26.618(14.0/7140.0)
 = 0.104 MPa

Internal Pressure Calculation Results:

ASME Code, Section VIII Division 2, 2025



Internal Pressure Results for: 122-D1-Shell-6

Cylindrical Shell Design Information:					
Design Pressure	P	0.388 MPa	Int. Design Temperature		130 °C
Material		SA-516 70	Ext. Design Temperature		130 °C
Length		3000.000 mm	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	1.0000 mm	Allowable Stress (ope)	S	156.4 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	175.0 MPa
Inside Diameter	Di	7100.000 mm	Actual Stress		108.1 MPa
Spec. Min. Thk t,e, or sa		16.0000 mm	Spec. Nominal Thk.		16.000 mm
Surface Area		67217520.0 mm²	SG of Contents		0.000
Element Volume		118797.0 ltr	Weight of Contents		0.0 N
Empty Weight		81551.6 N	Operating Weight		81551.6 N
			Comp. All. Ope. Stress		52.6 MPa

Radiography Information:			
...		...	
Circ. Joint Efficiency	85 %	Long. Joint Eff.	E or v 85 %

Material treatment options: +IT

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:
 $= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$
 $= 0.5 \cdot 7102.0 \cdot (\exp(0.388/(156.4 \cdot 0.85)) - 1) + 1.0 + 0.0$
 $= 10.3792 + 1.0 + 0.0 = 11.3792 \text{ mm Equation 4.3.1}$

Computed Maximum Allowable Working Pressure [MAWP]:
 $= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$
 $= 156.4 \cdot 0.85 \cdot \ln((2 \cdot 15.0 + 7102.0) / 7102.0) - 0.004$
 $= 0.56 - 0.004 = 0.5564 \text{ MPa}$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:
 $= S \cdot E \cdot \ln((2 \cdot t + D) / D)$
 $= 175.0 \cdot 0.85 \cdot \ln((2 \cdot 16.0 + 7100.0) / 7100.0)$
 $= 0.6689 \text{ MPa}$

Computed Stress Intensity at Design Pressure [Si]:
 $= P \cdot D / (E (D_o - D))$
 $= 0.388 \cdot 7102.0 / (0.85 (7132.0 - 7102.0))$
 $= 108.0601 \text{ MPa}$

Computed Membrane Stress at Design Pressure [Sm]:
 $= P / (E \cdot \ln((2 \cdot t + D) / D))$



$$= 0.388 / (0.85 \cdot \ln((2 \cdot 15.0 + 7102.0) / 7102.0))$$

$$= 108.2895 \text{ MPa}$$

Percent Fiber Elongation per Table 6.1

$$\text{Cylinders formed from Plate} \quad (50 \cdot t_{\text{nom}} / R_f) \cdot (1 - R_f / R_o) \quad 0.225 \%$$

Minimum Design Metal Temperature Results:

Note: This Element/Detail was specified as being Impact Tested.

Material treatment options: +IT

EXTERNAL PRESSURE RESULTS for: 122-D1-Shell-6

Cylindrical Shell From 230 to Ring R6 Ext. Chart: CS-2 at 130 °C

Maximum External Pressure per ASME Division 2 Section 4.4:

Determine Parameter [Mx]:

$$= L / \sqrt{(R_o \cdot t)}$$

$$= 2499.995 / \sqrt{(3566.0 \cdot 15.0)}$$

$$= 10.809$$

Compute the Factor [Ch]:

Since $M_x > 1.5$ and $M_x < 13$, $Ch = 0.92 / (M_x - 0.579)$

$$= 0.92 / (10.809 - 0.579)$$

$$= 0.0899$$

Elastic Hoop Compressive Membrane Failure Stress [Fhe]:

$$= 1.6 \cdot Ch \cdot E_y (t / D_o)$$

$$= 1.6 \cdot 0.09 \cdot 196301 (15.0 / 7132.0)$$

$$= 59.404 \text{ MPa}$$

Ratio of Elastic Hoop Compressive Membrane Failure Stress to Yield [Fr]:

$$= Fhe / F_y$$

$$= 59.4 / 234.4$$

$$= 0.253$$

Predicted Buckling Stress [Fic]:

Engineering Yield to Tensile Stress ratio [R]:

$$= \sigma_{ys} / \sigma_{uts}$$

$$= 234.4 / 482.6$$

$$= 0.486$$

Ratio [AE]:

$$= Fhe / E$$

$$= 59.4 / 196301.4$$

$$= 0.000303$$

Factor [m1]:

$$= (\ln(r) + (\epsilon_{ps} - \epsilon_{sys})) / \ln(\ln(1 + \epsilon_{ps}) / \ln(1 + \epsilon_{sys}))$$

$$= (\ln(0.486) + (0.00002 - 0.002)) / \ln(\ln(1 + 0.00002) / \ln(1 + 0.002))$$

$$= 0.157$$

Curve Fitting Constant for the Elastic Region of the Stress-Strain Curve [A1]:

$$= \sigma_{ys} (1 + \epsilon_{sys}) / \ln(1 + \epsilon_{sys})^{m1}$$

$$= 234.4 (1 + 0.002) / \ln(1 + 0.002)^{0.157301}$$

$$= 624.374 \text{ MPa}$$

Curve Fitting Exponent for, Ferritic Steel [m2]:

$$= m2f(m2f2 - R)$$

$$= 0.6(1.0 - 0.486)$$

$$= 0.309$$

Curve Fitting Constant [A2]:

$$= (\sigma_{uts} \cdot \exp(m2)) / (m2^{m2})$$

$$= (482.6 \cdot \exp(0.309)) / (0.309^{0.309})$$



= 944.566 MPa

Material Parameter for Stress-Strain Curve Model [K]:

$$= 1.5 \cdot R1.5 - 0.5 \cdot R2.5 - R3.5$$

$$= 1.5 \cdot 0.4861.5 - 0.5 \cdot 0.4862.5 - 0.4863.5$$

$$= 0.346$$

Compute the Allowable Stress per 4.4.3

	Fic	H	D1	D2	D3	D4	Tangent Modulus	Iteration Diff
1	241	-1.839	0.313E-04	0.271E-04	0.807E-04	-0.632E-04	0.124E+05	0.192E-01
2	120	-4.651	0.763E-06	0.763E-06	0.171E-04	-0.171E-04	0.151E+06	0.497E-03
3	60	-6.057	0.186E-07	0.186E-07	0.361E-05	-0.361E-05	0.195E+06	0.697E-05
4	30	-6.760	0.454E-09	0.454E-09	0.765E-06	-0.765E-06	0.196E+06	0.149E-03
5	45	-6.409	0.398E-08	0.398E-08	0.190E-05	-0.190E-05	0.196E+06	0.718E-04
6	52	-6.233	0.910E-08	0.910E-08	0.268E-05	-0.268E-05	0.196E+06	0.327E-04
7	56	-6.145	0.132E-07	0.132E-07	0.313E-05	-0.313E-05	0.195E+06	0.130E-04
8	58	-6.101	0.157E-07	0.157E-07	0.336E-05	-0.336E-05	0.195E+06	0.304E-05
9	59	-6.079	0.171E-07	0.171E-07	0.349E-05	-0.349E-05	0.195E+06	0.196E-05
10	58	-6.090	0.164E-07	0.164E-07	0.343E-05	-0.343E-05	0.195E+06	0.546E-06
11	59	-6.085	0.167E-07	0.167E-07	0.346E-05	-0.346E-05	0.195E+06	0.705E-06
12	59	-6.087	0.166E-07	0.166E-07	0.344E-05	-0.344E-05	0.195E+06	0.793E-07
13	58	-6.089	0.165E-07	0.165E-07	0.343E-05	-0.343E-05	0.195E+06	0.233E-06
14	59	-6.088	0.165E-07	0.165E-07	0.344E-05	-0.344E-05	0.195E+06	0.770E-07
15	59	-6.088	0.165E-07	0.165E-07	0.344E-05	-0.344E-05	0.195E+06	0.112E-08

Factor of Safety [FS]:

$$\text{Since Fic [59.02]} < 0.55 \cdot 234.4, \text{FS} = 2$$

Allowable Circumferential Compressive Stress [Fha]:

$$= \text{Fic} / \text{FS}$$

$$= 59.02 / 2.0$$

$$= 29.510 \text{ MPa}$$

Allowable External Pressure at the given Thickness [Pa]:

$$= 2 \cdot \text{Fha} (t / D_o)$$

$$= 2 \cdot 29.51 (15.0 / 7132.0)$$

$$= 0.124 \text{ MPa}$$

Cylindrical Shell From Ring R6 to Ring R7 Ext. Chart: CS-2 at 130 °C

Maximum External Pressure per ASME Division 2 Section 4.4:

Determine Parameter [Mx]:

$$= L / \sqrt{(R_o \cdot t)}$$

$$= 2500.001 / \sqrt{(3566.0 \cdot 15.0)}$$

$$= 10.809$$

Compute the Factor [Ch]:

$$\text{Since } Mx > 1.5 \text{ and } Mx < 13, Ch = 0.92 / (Mx - 0.579)$$

$$= 0.92 / (10.809 - 0.579)$$

$$= 0.0899$$

Elastic Hoop Compressive Membrane Failure Stress [Fhe]:

$$= 1.6 \cdot Ch \cdot E_y (t / D_o)$$

$$= 1.6 \cdot 0.09 \cdot 196301 (15.0 / 7132.0)$$

$$= 59.404 \text{ MPa}$$

Ratio of Elastic Hoop Compressive Membrane Failure Stress to Yield [Fr]:

$$= \text{Fhe} / \text{Fy}$$

$$= 59.4 / 234.4$$

$$= 0.253$$

Predicted Buckling Stress [Fic]:

Engineering Yield to Tensile Stress ratio [R]:

$$= \sigma_{ys} / \sigma_{uts}$$

$$= 234.4/482.6$$

$$= 0.486$$

Ratio [AE]:

$$= F_{he} / E$$

$$= 59.4/196301.4$$

$$= 0.000303$$

Factor [m1]:

$$= (\ln(r) + (\epsilon_{ps} - \epsilon_{sys})) / \ln(\ln(1 + \epsilon_{ps}) / \ln(1 + \epsilon_{sys}))$$

$$= (\ln(0.486) + (0.00002 - 0.002)) / \ln(\ln(1 + 0.00002) / \ln(1 + 0.002))$$

$$= 0.157$$

Curve Fitting Constant for the Elastic Region of the Stress-Strain Curve [A1]:

$$= \sigma_{ys} (1 + \epsilon_{sys}) / \ln(1 + \epsilon_{sys})^{m1}$$

$$= 234.4 (1 + 0.002) / \ln(1 + 0.002)^{0.157301}$$

$$= 624.374 \text{ MPa}$$

Curve Fitting Exponent for, Ferritic Steel [m2]:

$$= m2f(m2f2 - R)$$

$$= 0.6(1.0 - 0.486)$$

$$= 0.309$$

Curve Fitting Constant [A2]:

$$= (\sigma_{uts} \cdot \exp(m2)) / (m2^{m2})$$

$$= (482.6 \cdot \exp(0.309)) / (0.309^{0.309})$$

$$= 944.566 \text{ MPa}$$

Material Parameter for Stress-Strain Curve Model [K]:

$$= 1.5 \cdot R1.5 - 0.5 \cdot R2.5 - R3.5$$

$$= 1.5 \cdot 0.486^{1.5} - 0.5 \cdot 0.486^{2.5} - 0.486^{3.5}$$

$$= 0.346$$

Compute the Allowable Stress per 4.4.3

	Fic	H	D1	D2	D3	D4	Tangent Modulus	Iteration Diff
1	241	-1.839	0.313E-04	0.271E-04	0.807E-04	-0.632E-04	0.124E+05	0.192E-01
2	120	-4.651	0.763E-06	0.763E-06	0.171E-04	-0.171E-04	0.151E+06	0.497E-03
3	60	-6.057	0.186E-07	0.186E-07	0.361E-05	-0.361E-05	0.195E+06	0.697E-05
4	30	-6.760	0.454E-09	0.454E-09	0.765E-06	-0.765E-06	0.196E+06	0.149E-03
5	45	-6.409	0.398E-08	0.398E-08	0.190E-05	-0.190E-05	0.196E+06	0.718E-04
6	52	-6.233	0.910E-08	0.910E-08	0.268E-05	-0.268E-05	0.196E+06	0.327E-04
7	56	-6.145	0.132E-07	0.132E-07	0.313E-05	-0.313E-05	0.195E+06	0.130E-04
8	58	-6.101	0.157E-07	0.157E-07	0.336E-05	-0.336E-05	0.195E+06	0.304E-05
9	59	-6.079	0.171E-07	0.171E-07	0.349E-05	-0.349E-05	0.195E+06	0.196E-05
10	58	-6.090	0.164E-07	0.164E-07	0.343E-05	-0.343E-05	0.195E+06	0.545E-06
11	59	-6.085	0.167E-07	0.167E-07	0.346E-05	-0.346E-05	0.195E+06	0.705E-06
12	59	-6.087	0.166E-07	0.166E-07	0.344E-05	-0.344E-05	0.195E+06	0.800E-07
13	58	-6.089	0.165E-07	0.165E-07	0.343E-05	-0.343E-05	0.195E+06	0.232E-06
14	59	-6.088	0.165E-07	0.165E-07	0.344E-05	-0.344E-05	0.195E+06	0.762E-07
15	59	-6.088	0.165E-07	0.165E-07	0.344E-05	-0.344E-05	0.195E+06	0.190E-08

Factor of Safety [FS]:

Since $Fic [59.02] < 0.55 \cdot 234.4$, $FS = 2$

Allowable Circumferential Compressive Stress [Fha]:

$$= Fic / FS$$

$$= 59.02 / 2.0$$

$$= 29.510 \text{ MPa}$$

Allowable External Pressure at the given Thickness [Pa]:

$$= 2 \cdot F_{ha} (t / D_o)$$

$$= 2 \cdot 29.51 (15.0 / 7132.0)$$

$$= 0.124 \text{ MPa}$$

Cylindrical Shell From Ring R7 to 240 Ext. Chart: CS-2 at 130 °C



Maximum External Pressure per ASME Division 2 Section 4.4:

Determine Parameter [Mx]:

$$= L / \sqrt{(R_o \cdot t)}$$

$$= 2500.01 / \sqrt{(3566.0 \cdot 15.0)}$$

$$= 10.809$$

Compute the Factor [Ch]:

$$\text{Since } M_x > 1.5 \text{ and } M_x < 13, Ch = 0.92 / (M_x - 0.579)$$

$$= 0.92 / (10.809 - 0.579)$$

$$= 0.0899$$

Elastic Hoop Compressive Membrane Failure Stress [Fhe]:

$$= 1.6 \cdot Ch \cdot E_y (t / D_o)$$

$$= 1.6 \cdot 0.09 \cdot 196301 (15.0 / 7132.0)$$

$$= 59.404 \text{ MPa}$$

Ratio of Elastic Hoop Compressive Membrane Failure Stress to Yield [Fr]:

$$= Fhe / F_y$$

$$= 59.4 / 234.4$$

$$= 0.253$$

Predicted Buckling Stress [Fic]:

Engineering Yield to Tensile Stress ratio [R]:

$$= \sigma_{ys} / \sigma_{uts}$$

$$= 234.4 / 482.6$$

$$= 0.486$$

Ratio [AE]:

$$= Fhe / E$$

$$= 59.4 / 196301.4$$

$$= 0.000303$$

Factor [m1]:

$$= (\ln(r) + (\epsilon_{psp} - \epsilon_{sys})) / \ln(\ln(1 + \epsilon_{psp}) / \ln(1 + \epsilon_{sys}))$$

$$= (\ln(0.486) + (0.00002 - 0.002)) / \ln(\ln(1 + 0.00002) / \ln(1 + 0.002))$$

$$= 0.157$$

Curve Fitting Constant for the Elastic Region of the Stress-Strain Curve [A1]:

$$= \sigma_{ys} (1 + \epsilon_{sys}) / \ln(1 + \epsilon_{sys}) m1$$

$$= 234.4 (1 + 0.002) / \ln(1 + 0.002) 0.157301$$

$$= 624.374 \text{ MPa}$$

Curve Fitting Exponent for, Ferritic Steel [m2]:

$$= m2f(m2f2 - R)$$

$$= 0.6 (1.0 - 0.486)$$

$$= 0.309$$

Curve Fitting Constant [A2]:

$$= (\sigma_{uts} \cdot \exp(m2)) / (m2 m2)$$

$$= (482.6 \cdot \exp(0.309)) / (0.309 0.309)$$

$$= 944.566 \text{ MPa}$$

Material Parameter for Stress-Strain Curve Model [K]:

$$= 1.5 \cdot R1.5 - 0.5 \cdot R2.5 - R3.5$$

$$= 1.5 \cdot 0.4861.5 - 0.5 \cdot 0.4862.5 - 0.4863.5$$

$$= 0.346$$

Compute the Allowable Stress per 4.4.3

	Fic	H	D1	D2	D3	D4	Tangent Modulus	Iteration Diff
1	241	-1.839	0.313E-04	0.271E-04	0.807E-04	-0.632E-04	0.124E+05	0.192E-01
2	120	-4.651	0.763E-06	0.763E-06	0.171E-04	-0.171E-04	0.151E+06	0.497E-03
3	60	-6.057	0.186E-07	0.186E-07	0.361E-05	-0.361E-05	0.195E+06	0.697E-05
4	30	-6.760	0.454E-09	0.454E-09	0.765E-06	-0.765E-06	0.196E+06	0.149E-03
5	45	-6.409	0.398E-08	0.398E-08	0.190E-05	-0.190E-05	0.196E+06	0.717E-04
6	52	-6.233	0.910E-08	0.910E-08	0.268E-05	-0.268E-05	0.196E+06	0.327E-04



7	56	-6.145	0.132E-07	0.132E-07	0.313E-05	-0.313E-05	0.195E+06	0.130E-04
8	58	-6.101	0.157E-07	0.157E-07	0.336E-05	-0.336E-05	0.195E+06	0.304E-05
9	59	-6.079	0.171E-07	0.171E-07	0.349E-05	-0.349E-05	0.195E+06	0.196E-05
10	58	-6.090	0.164E-07	0.164E-07	0.343E-05	-0.343E-05	0.195E+06	0.544E-06
11	59	-6.085	0.167E-07	0.167E-07	0.346E-05	-0.346E-05	0.195E+06	0.706E-06
12	59	-6.087	0.166E-07	0.166E-07	0.344E-05	-0.344E-05	0.195E+06	0.811E-07
13	58	-6.089	0.165E-07	0.165E-07	0.343E-05	-0.343E-05	0.195E+06	0.231E-06
14	59	-6.088	0.165E-07	0.165E-07	0.344E-05	-0.344E-05	0.195E+06	0.751E-07
15	59	-6.088	0.165E-07	0.165E-07	0.344E-05	-0.344E-05	0.195E+06	0.301E-08

Factor of Safety [FS]:

Since $Fic [59.02] < 0.55 * 234.4$, $FS = 2$

Allowable Circumferential Compressive Stress [Fha]:

= Fic/FS

= $59.02/2.0$

= 29.510 MPa

Allowable External Pressure at the given Thickness [Pa]:

= $2 * Fha(t / Do)$

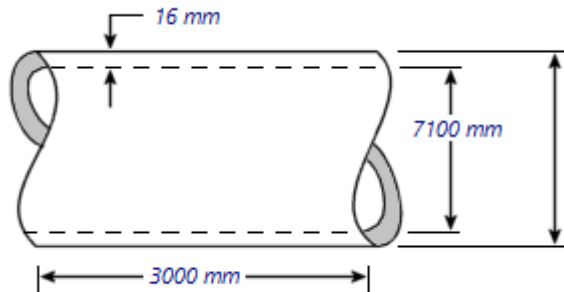
= $2 * 29.51(15.0/7132.0)$

= 0.124 MPa

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 2, 2025



Internal Pressure Results for: 122-D1-Shell-7

Cylindrical Shell Design Information:					
Design Pressure	P	0.388 MPa	Int. Design Temperature		130 °C
Material		SA-516 70	Ext. Design Temperature		130 °C
Length		3000.000 mm	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	1.0000 mm	Allowable Stress (ope)	S	156.4 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	175.0 MPa
Inside Diameter	Di	7100.000 mm	Actual Stress		108.1 MPa
Spec. Min. Thk t _e , or s _a		16.0000 mm	Spec. Nominal Thk.		16.000 mm
Surface Area		67217520.0 mm ²	SG of Contents		0.000
Element Volume		118797.0 ltr	Weight of Contents		0.0 N
Empty Weight		81551.6 N	Operating Weight		81551.6 N
			Comp. All. Ope. Stress		52.6 MPa

Radiography Information:			
...		...	
Circ. Joint Efficiency	85 %	Long. Joint Eff.	E or v 85 %

Material treatment options: +IT

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:
 $= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$
 $= 0.5 \cdot 7102.0 \cdot (\exp(0.388/(156.4 \cdot 0.85)) - 1) + 1.0 + 0.0$
 $= 10.3792 + 1.0 + 0.0 = 11.3792 \text{ mm Equation 4.3.1}$

Computed Maximum Allowable Working Pressure [MAWP]:
 $= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$
 $= 156.4 \cdot 0.85 \cdot \ln((2 \cdot 15.0 + 7102.0) / 7102.0) - 0.004$
 $= 0.56 - 0.004 = 0.5564 \text{ MPa}$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:
 $= S \cdot E \cdot \ln((2 \cdot t + D) / D)$
 $= 175.0 \cdot 0.85 \cdot \ln((2 \cdot 16.0 + 7100.0) / 7100.0)$
 $= 0.6689 \text{ MPa}$

Computed Stress Intensity at Design Pressure [Si]:
 $= P \cdot D / (E (D_o - D))$
 $= 0.388 \cdot 7102.0 / (0.85 (7132.0 - 7102.0))$
 $= 108.0601 \text{ MPa}$

Computed Membrane Stress at Design Pressure [Sm]:
 $= P / (E \cdot \ln((2 \cdot t + D) / D))$



$$= 0.388 / (0.85 \cdot \ln((2 \cdot 15.0 + 7102.0) / 7102.0))$$

$$= 108.2895 \text{ MPa}$$

Percent Fiber Elongation per Table 6.1

$$\text{Cylinders formed from Plate} \quad (50 \cdot t_{\text{nom}} / R_f) \cdot (1 - R_f / R_o) \quad 0.225 \%$$

Minimum Design Metal Temperature Results:

Note: This Element/Detail was specified as being Impact Tested.

Material treatment options: +IT

EXTERNAL PRESSURE RESULTS for: 122-D1-Shell-7

Cylindrical Shell From 240 to Ring R8 Ext. Chart: CS-2 at 130 °C

Maximum External Pressure per ASME Division 2 Section 4.4:

Determine Parameter [Mx]:

$$= L / \sqrt{(R_o \cdot t)}$$

$$= 2500.01 / \sqrt{(3566.0 \cdot 15.0)}$$

$$= 10.809$$

Compute the Factor [Ch]:

Since $M_x > 1.5$ and $M_x < 13$, $Ch = 0.92 / (M_x - 0.579)$

$$= 0.92 / (10.809 - 0.579)$$

$$= 0.0899$$

Elastic Hoop Compressive Membrane Failure Stress [Fhe]:

$$= 1.6 \cdot Ch \cdot E_y (t / D_o)$$

$$= 1.6 \cdot 0.09 \cdot 196301 (15.0 / 7132.0)$$

$$= 59.404 \text{ MPa}$$

Ratio of Elastic Hoop Compressive Membrane Failure Stress to Yield [Fr]:

$$= Fhe / F_y$$

$$= 59.4 / 234.4$$

$$= 0.253$$

Predicted Buckling Stress [Fic]:

Engineering Yield to Tensile Stress ratio [R]:

$$= \sigma_{ys} / \sigma_{uts}$$

$$= 234.4 / 482.6$$

$$= 0.486$$

Ratio [AE]:

$$= Fhe / E$$

$$= 59.4 / 196301.4$$

$$= 0.000303$$

Factor [m1]:

$$= (\ln(r) + (\epsilon_{psp} - \epsilon_{sys})) / \ln(\ln(1 + \epsilon_{psp}) / \ln(1 + \epsilon_{sys}))$$

$$= (\ln(0.486) + (0.00002 - 0.002)) / \ln(\ln(1 + 0.00002) / \ln(1 + 0.002))$$

$$= 0.157$$

Curve Fitting Constant for the Elastic Region of the Stress-Strain Curve [A1]:

$$= \sigma_{ys} (1 + \epsilon_{sys}) / \ln(1 + \epsilon_{sys})^{m1}$$

$$= 234.4 (1 + 0.002) / \ln(1 + 0.002)^{0.157301}$$

$$= 624.374 \text{ MPa}$$

Curve Fitting Exponent for, Ferritic Steel [m2]:

$$= m2f(m2f2 - R)$$

$$= 0.6 (1.0 - 0.486)$$

$$= 0.309$$

Curve Fitting Constant [A2]:

$$= (\sigma_{uts} \cdot \exp(m2)) / (m2^{m2})$$

$$= (482.6 \cdot \exp(0.309)) / (0.309^{0.309})$$



= 944.566 MPa

Material Parameter for Stress-Strain Curve Model [K]:

= $1.5 \cdot R1.5 - 0.5 \cdot R2.5 - R3.5$

= $1.5 \cdot 0.4861.5 - 0.5 \cdot 0.4862.5 - 0.4863.5$

= 0.346

Compute the Allowable Stress per 4.4.3

	Fic	H	D1	D2	D3	D4	Tangent Modulus	Iteration Diff
1	241	-1.839	0.313E-04	0.271E-04	0.807E-04	-0.632E-04	0.124E+05	0.192E-01
2	120	-4.651	0.763E-06	0.763E-06	0.171E-04	-0.171E-04	0.151E+06	0.497E-03
3	60	-6.057	0.186E-07	0.186E-07	0.361E-05	-0.361E-05	0.195E+06	0.697E-05
4	30	-6.760	0.454E-09	0.454E-09	0.765E-06	-0.765E-06	0.196E+06	0.149E-03
5	45	-6.409	0.398E-08	0.398E-08	0.190E-05	-0.190E-05	0.196E+06	0.717E-04
6	52	-6.233	0.910E-08	0.910E-08	0.268E-05	-0.268E-05	0.196E+06	0.327E-04
7	56	-6.145	0.132E-07	0.132E-07	0.313E-05	-0.313E-05	0.195E+06	0.130E-04
8	58	-6.101	0.157E-07	0.157E-07	0.336E-05	-0.336E-05	0.195E+06	0.304E-05
9	59	-6.079	0.171E-07	0.171E-07	0.349E-05	-0.349E-05	0.195E+06	0.196E-05
10	58	-6.090	0.164E-07	0.164E-07	0.343E-05	-0.343E-05	0.195E+06	0.544E-06
11	59	-6.085	0.167E-07	0.167E-07	0.346E-05	-0.346E-05	0.195E+06	0.706E-06
12	59	-6.087	0.166E-07	0.166E-07	0.344E-05	-0.344E-05	0.195E+06	0.811E-07
13	58	-6.089	0.165E-07	0.165E-07	0.343E-05	-0.343E-05	0.195E+06	0.231E-06
14	59	-6.088	0.165E-07	0.165E-07	0.344E-05	-0.344E-05	0.195E+06	0.751E-07
15	59	-6.088	0.165E-07	0.165E-07	0.344E-05	-0.344E-05	0.195E+06	0.301E-08

Factor of Safety [FS]:

Since Fic [59.02] < 0.55 * 234.4, FS = 2

Allowable Circumferential Compressive Stress [Fha]:

= Fic/FS

= 59.02/2.0

= 29.510 MPa

Allowable External Pressure at the given Thickness [Pa]:

= $2 \cdot Fha(t / Do)$

= $2 \cdot 29.51(15.0/7132.0)$

= 0.124 MPa

Cylindrical Shell From Ring R8 to 250 Ext. Chart: CS-2 at 130 °C

Maximum External Pressure per ASME Division 2 Section 4.4:

Determine Parameter [Mx]:

= $L / \sqrt{(Ro \cdot t)}$

= $2500.0 / \sqrt{(3566.0 \cdot 15.0)}$

= 10.809

Compute the Factor [Ch]:

Since Mx > 1.5 and Mx < 13, Ch = 0.92 / (Mx - 0.579)

= 0.92 / (10.809 - 0.579)

= 0.0899

Elastic Hoop Compressive Membrane Failure Stress [Fhe]:

= $1.6 \cdot Ch \cdot Ey(t / Do)$

= $1.6 \cdot 0.09 \cdot 196301(15.0/7132.0)$

= 59.404 MPa

Ratio of Elastic Hoop Compressive Membrane Failure Stress to Yield [Fr]:

= Fhe / Fy

= 59.4/234.4

= 0.253

Predicted Buckling Stress [Fic]:

Engineering Yield to Tensile Stress ratio [R]:

= $\sigma_{ys} / \sigma_{uts}$



$$= 234.4/482.6$$

$$= 0.486$$

Ratio [AE]:

$$= F_{he} / E$$

$$= 59.4/196301.4$$

$$= 0.000303$$

Factor [m1]:

$$= (\ln(r) + (\epsilon_{ps} - \epsilon_{sys})) / \ln(\ln(1 + \epsilon_{ps}) / \ln(1 + \epsilon_{sys}))$$

$$= (\ln(0.486) + (0.00002 - 0.002)) / \ln(\ln(1 + 0.00002) / \ln(1 + 0.002))$$

$$= 0.157$$

Curve Fitting Constant for the Elastic Region of the Stress-Strain Curve [A1]:

$$= \sigma_{ys} (1 + \epsilon_{sys}) / \ln(1 + \epsilon_{sys})^{m1}$$

$$= 234.4 (1 + 0.002) / \ln(1 + 0.002)^{0.157301}$$

$$= 624.374 \text{ MPa}$$

Curve Fitting Exponent for, Ferritic Steel [m2]:

$$= m2f(m2f2 - R)$$

$$= 0.6(1.0 - 0.486)$$

$$= 0.309$$

Curve Fitting Constant [A2]:

$$= (\sigma_{uts} \cdot \exp(m2)) / (m2^{m2})$$

$$= (482.6 \cdot \exp(0.309)) / (0.309^{0.309})$$

$$= 944.566 \text{ MPa}$$

Material Parameter for Stress-Strain Curve Model [K]:

$$= 1.5 \cdot R1.5 - 0.5 \cdot R2.5 - R3.5$$

$$= 1.5 \cdot 0.486^{1.5} - 0.5 \cdot 0.486^{2.5} - 0.486^{3.5}$$

$$= 0.346$$

Compute the Allowable Stress per 4.4.3

	Fic	H	D1	D2	D3	D4	Tangent Modulus	Iteration Diff
1	241	-1.839	0.313E-04	0.271E-04	0.807E-04	-0.632E-04	0.124E+05	0.192E-01
2	120	-4.651	0.763E-06	0.763E-06	0.171E-04	-0.171E-04	0.151E+06	0.497E-03
3	60	-6.057	0.186E-07	0.186E-07	0.361E-05	-0.361E-05	0.195E+06	0.697E-05
4	30	-6.760	0.454E-09	0.454E-09	0.765E-06	-0.765E-06	0.196E+06	0.149E-03
5	45	-6.409	0.398E-08	0.398E-08	0.190E-05	-0.190E-05	0.196E+06	0.718E-04
6	52	-6.233	0.910E-08	0.910E-08	0.268E-05	-0.268E-05	0.196E+06	0.327E-04
7	56	-6.145	0.132E-07	0.132E-07	0.313E-05	-0.313E-05	0.195E+06	0.130E-04
8	58	-6.101	0.157E-07	0.157E-07	0.336E-05	-0.336E-05	0.195E+06	0.304E-05
9	59	-6.079	0.171E-07	0.171E-07	0.349E-05	-0.349E-05	0.195E+06	0.196E-05
10	58	-6.090	0.164E-07	0.164E-07	0.343E-05	-0.343E-05	0.195E+06	0.545E-06
11	59	-6.085	0.167E-07	0.167E-07	0.346E-05	-0.346E-05	0.195E+06	0.705E-06
12	59	-6.087	0.166E-07	0.166E-07	0.344E-05	-0.344E-05	0.195E+06	0.799E-07
13	58	-6.089	0.165E-07	0.165E-07	0.343E-05	-0.343E-05	0.195E+06	0.233E-06
14	59	-6.088	0.165E-07	0.165E-07	0.344E-05	-0.344E-05	0.195E+06	0.764E-07
15	59	-6.088	0.165E-07	0.165E-07	0.344E-05	-0.344E-05	0.195E+06	0.173E-08

Factor of Safety [FS]:

Since $Fic [59.02] < 0.55 \cdot 234.4$, $FS = 2$

Allowable Circumferential Compressive Stress [Fha]:

$$= Fic / FS$$

$$= 59.02 / 2.0$$

$$= 29.510 \text{ MPa}$$

Allowable External Pressure at the given Thickness [Pa]:

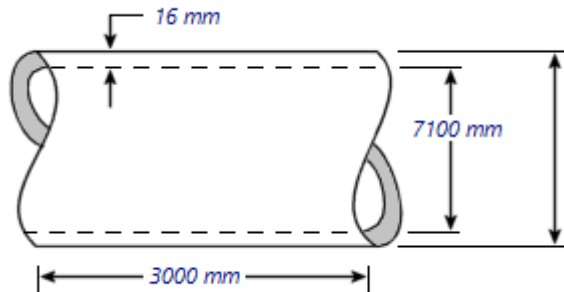
$$= 2 \cdot Fha(t / D_o)$$

$$= 2 \cdot 29.51(15.0/7132.0)$$

$$= 0.124 \text{ MPa}$$

Internal Pressure Calculation Results:

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Internal Pressure Results for: 122-D1-Shell-8

Cylindrical Shell Design Information:					
Design Pressure	P	0.388 MPa	Int. Design Temperature		130 °C
Material		SA-516 70	Ext. Design Temperature		130 °C
Length		3000.000 mm	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	1.0000 mm	Allowable Stress (ope)	S	138.0 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	138.0 MPa
Inside Diameter	Di	7100.000 mm	Actual Stress		108.1 MPa
Spec. Min. Thk t,e, or sa		16.0000 mm	Spec. Nominal Thk.		16.000 mm
Surface Area		67217520.0 mm²	SG of Contents		0.000
Element Volume		118797.0 ltr	Weight of Contents		0.0 N
Empty Weight		81551.6 N	Operating Weight		81551.6 N
			Comp. All. Ope. Stress		52.6 MPa
Radiography Information:					
...			...		
Circ. Joint Efficiency		85 %	Long. Joint Eff.	E or v	85 %

Material treatment options: +IT

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 7102.0 \cdot (\exp(0.388/(138.0 \cdot 0.85)) - 1) + 1.0 + 0.0$$

$$= 11.7651 + 1.0 + 0.0 = 12.7651 \text{ mm Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 138.0 \cdot 0.85 \cdot \ln((2 \cdot 15.0 + 7102.0) / 7102.0) - 0.004$$

$$= 0.494 - 0.004 = 0.4905 \text{ MPa}$$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D)$$

$$= 138.0 \cdot 0.85 \cdot \ln((2 \cdot 16.0 + 7100.0) / 7100.0)$$

$$= 0.5275 \text{ MPa}$$

Computed Stress Intensity at Design Pressure [Si]:

$$= P \cdot D / (E (D_o - D))$$

$$= 0.388 \cdot 7102.0 / (0.85 (7132.0 - 7102.0))$$

$$= 108.0601 \text{ MPa}$$

Computed Membrane Stress at Design Pressure [Sm]:

$$= P / (E \cdot \ln((2 \cdot t + D) / D))$$



$$= 0.388 / (0.85 \cdot \ln((2 \cdot 15.0 + 7102.0) / 7102.0))$$
$$= 108.2895 \text{ MPa}$$

Percent Fiber Elongation per Table 6.1

$$\text{Cylinders formed from Plate} \quad (50 \cdot t_{\text{nom}} / R_f) \cdot (1 - R_f / R_o) \quad 0.225 \%$$

Minimum Design Metal Temperature Results:

Note: This Element/Detail was specified as being Impact Tested.

Material treatment options: +IT

EXTERNAL PRESSURE RESULTS for: 122-D1-Shell-8

Cylindrical Shell From 250 to Ring R9 Ext. Chart: CS-2 at 130 °C

Maximum External Pressure per ASME Division 2 Section 4.4:

Determine Parameter [Mx]:

$$= L / \sqrt{(R_o \cdot t)}$$
$$= 2500.0 / \sqrt{(3566.0 \cdot 15.0)}$$
$$= 10.809$$

Compute the Factor [Ch]:

$$\text{Since } M_x > 1.5 \text{ and } M_x < 13, \text{ Ch} = 0.92 / (M_x - 0.579)$$
$$= 0.92 / (10.809 - 0.579)$$
$$= 0.0899$$

Elastic Hoop Compressive Membrane Failure Stress [Fhe]:

$$= 1.6 \cdot \text{Ch} \cdot E_y (t / D_o)$$
$$= 1.6 \cdot 0.09 \cdot 196301 (15.0 / 7132.0)$$
$$= 59.404 \text{ MPa}$$

Ratio of Elastic Hoop Compressive Membrane Failure Stress to Yield [Fr]:

$$= F_{he} / F_y$$
$$= 59.4 / 234.4$$
$$= 0.253$$

Predicted Buckling Stress [Fic]:

Engineering Yield to Tensile Stress ratio [R]:

$$= \sigma_{ys} / \sigma_{uts}$$
$$= 234.4 / 482.6$$
$$= 0.486$$

Ratio [AE]:

$$= F_{he} / E$$
$$= 59.4 / 196301.4$$
$$= 0.000303$$

Factor [m1]:

$$= (\ln(r) + (\epsilon_{psp} - \epsilon_{sys})) / \ln(\ln(1 + \epsilon_{psp}) / \ln(1 + \epsilon_{sys}))$$
$$= (\ln(0.486) + (0.00002 - 0.002)) / \ln(\ln(1 + 0.00002) / \ln(1 + 0.002))$$
$$= 0.157$$

Curve Fitting Constant for the Elastic Region of the Stress-Strain Curve [A1]:

$$= \sigma_{ys} (1 + \epsilon_{sys}) / \ln(1 + \epsilon_{sys})^{m1}$$
$$= 234.4 (1 + 0.002) / \ln(1 + 0.002)^{0.157301}$$
$$= 624.374 \text{ MPa}$$

Curve Fitting Exponent for, Ferritic Steel [m2]:

$$= m2f(m2f2 - R)$$
$$= 0.6(1.0 - 0.486)$$
$$= 0.309$$

Curve Fitting Constant [A2]:

$$= (\sigma_{uts} \cdot \exp(m2)) / (m2^{m2})$$
$$= (482.6 \cdot \exp(0.309)) / (0.309^{0.309})$$



= 944.566 MPa

Material Parameter for Stress-Strain Curve Model [K]:

$$= 1.5 \cdot R1.5 - 0.5 \cdot R2.5 - R3.5$$

$$= 1.5 \cdot 0.4861.5 - 0.5 \cdot 0.4862.5 - 0.4863.5$$

$$= 0.346$$

Compute the Allowable Stress per 4.4.3

	Fic	H	D1	D2	D3	D4	Tangent Modulus	Iteration Diff
1	241	-1.839	0.313E-04	0.271E-04	0.807E-04	-0.632E-04	0.124E+05	0.192E-01
2	120	-4.651	0.763E-06	0.763E-06	0.171E-04	-0.171E-04	0.151E+06	0.497E-03
3	60	-6.057	0.186E-07	0.186E-07	0.361E-05	-0.361E-05	0.195E+06	0.697E-05
4	30	-6.760	0.454E-09	0.454E-09	0.765E-06	-0.765E-06	0.196E+06	0.149E-03
5	45	-6.409	0.398E-08	0.398E-08	0.190E-05	-0.190E-05	0.196E+06	0.718E-04
6	52	-6.233	0.910E-08	0.910E-08	0.268E-05	-0.268E-05	0.196E+06	0.327E-04
7	56	-6.145	0.132E-07	0.132E-07	0.313E-05	-0.313E-05	0.195E+06	0.130E-04
8	58	-6.101	0.157E-07	0.157E-07	0.336E-05	-0.336E-05	0.195E+06	0.304E-05
9	59	-6.079	0.171E-07	0.171E-07	0.349E-05	-0.349E-05	0.195E+06	0.196E-05
10	58	-6.090	0.164E-07	0.164E-07	0.343E-05	-0.343E-05	0.195E+06	0.545E-06
11	59	-6.085	0.167E-07	0.167E-07	0.346E-05	-0.346E-05	0.195E+06	0.705E-06
12	59	-6.087	0.166E-07	0.166E-07	0.344E-05	-0.344E-05	0.195E+06	0.799E-07
13	58	-6.089	0.165E-07	0.165E-07	0.343E-05	-0.343E-05	0.195E+06	0.233E-06
14	59	-6.088	0.165E-07	0.165E-07	0.344E-05	-0.344E-05	0.195E+06	0.764E-07
15	59	-6.088	0.165E-07	0.165E-07	0.344E-05	-0.344E-05	0.195E+06	0.173E-08

Factor of Safety [FS]:

$$\text{Since Fic [59.02]} < 0.55 \cdot 234.4, \text{FS} = 2$$

Allowable Circumferential Compressive Stress [Fha]:

$$= \text{Fic} / \text{FS}$$

$$= 59.02 / 2.0$$

$$= 29.510 \text{ MPa}$$

Allowable External Pressure at the given Thickness [Pa]:

$$= 2 \cdot \text{Fha} (t / D_o)$$

$$= 2 \cdot 29.51 (15.0 / 7132.0)$$

$$= 0.124 \text{ MPa}$$

Cylindrical Shell From Ring R9 to 260 Ext. Chart: CS-2 at 130 °C

Maximum External Pressure per ASME Division 2 Section 4.4:

Determine Parameter [Mx]:

$$= L / \sqrt{(R_o \cdot t)}$$

$$= 2500.04 / \sqrt{(3566.0 \cdot 15.0)}$$

$$= 10.810$$

Compute the Factor [Ch]:

$$\text{Since } Mx > 1.5 \text{ and } Mx < 13, \text{Ch} = 0.92 / (Mx - 0.579)$$

$$= 0.92 / (10.81 - 0.579)$$

$$= 0.0899$$

Elastic Hoop Compressive Membrane Failure Stress [Fhe]:

$$= 1.6 \cdot \text{Ch} \cdot E_y (t / D_o)$$

$$= 1.6 \cdot 0.09 \cdot 196301 (15.0 / 7132.0)$$

$$= 59.403 \text{ MPa}$$

Ratio of Elastic Hoop Compressive Membrane Failure Stress to Yield [Fr]:

$$= \text{Fhe} / F_y$$

$$= 59.4 / 234.4$$

$$= 0.253$$

Predicted Buckling Stress [Fic]:

Engineering Yield to Tensile Stress ratio [R]:

$$= \sigma_y / \sigma_{uts}$$



$$= 234.4/482.6$$

$$= 0.486$$

Ratio [AE]:

$$= F_{he} / E$$

$$= 59.4/196301.4$$

$$= 0.000303$$

Factor [m1]:

$$= (\ln(r) + (\epsilon_{ps} - \epsilon_{sys})) / \ln(\ln(1 + \epsilon_{ps}) / \ln(1 + \epsilon_{sys}))$$

$$= (\ln(0.486) + (0.00002 - 0.002)) / \ln(\ln(1 + 0.00002) / \ln(1 + 0.002))$$

$$= 0.157$$

Curve Fitting Constant for the Elastic Region of the Stress-Strain Curve [A1]:

$$= \sigma_{ys} (1 + \epsilon_{sys}) / \ln(1 + \epsilon_{sys})^{m1}$$

$$= 234.4 (1 + 0.002) / \ln(1 + 0.002)^{0.157301}$$

$$= 624.374 \text{ MPa}$$

Curve Fitting Exponent for, Ferritic Steel [m2]:

$$= m2f(m2f2 - R)$$

$$= 0.6(1.0 - 0.486)$$

$$= 0.309$$

Curve Fitting Constant [A2]:

$$= (\sigma_{uts} \cdot \exp(m2)) / (m2^{m2})$$

$$= (482.6 \cdot \exp(0.309)) / (0.309^{0.309})$$

$$= 944.566 \text{ MPa}$$

Material Parameter for Stress-Strain Curve Model [K]:

$$= 1.5 \cdot R1.5 - 0.5 \cdot R2.5 - R3.5$$

$$= 1.5 \cdot 0.486^{1.5} - 0.5 \cdot 0.486^{2.5} - 0.486^{3.5}$$

$$= 0.346$$

Compute the Allowable Stress per 4.4.3

	Fic	H	D1	D2	D3	D4	Tangent Modulus	Iteration Diff
1	241	-1.839	0.313E-04	0.271E-04	0.807E-04	-0.632E-04	0.124E+05	0.192E-01
2	120	-4.651	0.763E-06	0.763E-06	0.171E-04	-0.171E-04	0.151E+06	0.497E-03
3	60	-6.057	0.186E-07	0.186E-07	0.361E-05	-0.361E-05	0.195E+06	0.698E-05
4	30	-6.760	0.454E-09	0.454E-09	0.765E-06	-0.765E-06	0.196E+06	0.149E-03
5	45	-6.409	0.398E-08	0.398E-08	0.190E-05	-0.190E-05	0.196E+06	0.717E-04
6	52	-6.233	0.910E-08	0.910E-08	0.268E-05	-0.268E-05	0.196E+06	0.327E-04
7	56	-6.145	0.132E-07	0.132E-07	0.313E-05	-0.313E-05	0.195E+06	0.130E-04
8	58	-6.101	0.157E-07	0.157E-07	0.336E-05	-0.336E-05	0.195E+06	0.304E-05
9	59	-6.079	0.171E-07	0.171E-07	0.349E-05	-0.349E-05	0.195E+06	0.196E-05
10	58	-6.090	0.164E-07	0.164E-07	0.343E-05	-0.343E-05	0.195E+06	0.540E-06
11	59	-6.085	0.167E-07	0.167E-07	0.346E-05	-0.346E-05	0.195E+06	0.710E-06
12	59	-6.087	0.166E-07	0.166E-07	0.344E-05	-0.344E-05	0.195E+06	0.850E-07
13	58	-6.089	0.165E-07	0.165E-07	0.343E-05	-0.343E-05	0.195E+06	0.228E-06
14	59	-6.088	0.165E-07	0.165E-07	0.344E-05	-0.344E-05	0.195E+06	0.713E-07
15	59	-6.088	0.165E-07	0.165E-07	0.344E-05	-0.344E-05	0.195E+06	0.688E-08

Factor of Safety [FS]:

Since $Fic [59.02] < 0.55 \cdot 234.4$, $FS = 2$

Allowable Circumferential Compressive Stress [Fha]:

$$= Fic / FS$$

$$= 59.02 / 2.0$$

$$= 29.510 \text{ MPa}$$

Allowable External Pressure at the given Thickness [Pa]:

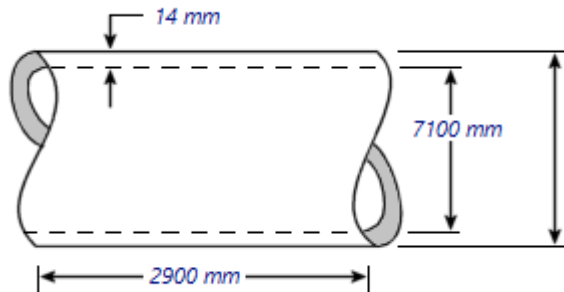
$$= 2 \cdot Fha(t / D_o)$$

$$= 2 \cdot 29.51(15.0/7132.0)$$

$$= 0.124 \text{ MPa}$$

Internal Pressure Calculation Results:

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Internal Pressure Results for: 122-D1-Shell-9

Cylindrical Shell Design Information:					
Design Pressure	P	0.388 MPa	Int. Design Temperature		130 °C
Material		SA-516 70	Ext. Design Temperature		130 °C
Length		2900.000 mm	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	1.0000 mm	Allowable Stress (ope)	S	156.4 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	175.0 MPa
Inside Diameter	Di	7100.000 mm	Actual Stress		124.7 MPa
Spec. Min. Thk t _e , or s _a		14.0000 mm	Spec. Nominal Thk.		14.000 mm
Surface Area		64940488.0 mm ²	SG of Contents		0.000
Element Volume		114837.1 ltr	Weight of Contents		0.0 N
Empty Weight		68958.3 N	Operating Weight		68958.3 N
			Comp. All. Ope. Stress		45.6 MPa

Radiography Information:			
...		...	
Circ. Joint Efficiency	85 %	Long. Joint Eff.	E or v 85 %

Material treatment options: +IT

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:
 $= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$
 $= 0.5 \cdot 7102.0 \cdot (\exp(0.388/(156.4 \cdot 0.85)) - 1) + 1.0 + 0.0$
 $= 10.3792 + 1.0 + 0.0 = 11.3792 \text{ mm Equation 4.3.1}$

Computed Maximum Allowable Working Pressure [MAWP]:
 $= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$
 $= 156.4 \cdot 0.85 \cdot \ln((2 \cdot 13.0 + 7102.0) / 7102.0) - 0.004$
 $= 0.486 - 0.004 = 0.4818 \text{ MPa}$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:
 $= S \cdot E \cdot \ln((2 \cdot t + D) / D)$
 $= 175.0 \cdot 0.85 \cdot \ln((2 \cdot 14.0 + 7100.0) / 7100.0)$
 $= 0.5855 \text{ MPa}$

Computed Stress Intensity at Design Pressure [Si]:
 $= P \cdot D / (E (D_o - D))$
 $= 0.388 \cdot 7102.0 / (0.85 (7128.0 - 7102.0))$
 $= 124.6869 \text{ MPa}$

Computed Membrane Stress at Design Pressure [Sm]:
 $= P / (E \cdot \ln((2 \cdot t + D) / D))$



$$= 0.388 / (0.85 \cdot \ln((2 \cdot 13.0 + 7102.0) / 7102.0))$$

$$= 124.9157 \text{ MPa}$$

Percent Fiber Elongation per Table 6.1

$$\text{Cylinders formed from Plate} \quad (50 \cdot t_{\text{nom}} / R_f) \cdot (1 - R_f / R_o) \quad 0.197 \%$$

Minimum Design Metal Temperature Results:

Note: This Element/Detail was specified as being Impact Tested.

Material treatment options: +IT

EXTERNAL PRESSURE RESULTS for: 122-D1-Shell-9

Cylindrical Shell From 260 to Ring R10 Ext. Chart: CS-2 at 130 °C

Maximum External Pressure per ASME Division 2 Section 4.4:

Determine Parameter [Mx]:

$$= L / \sqrt{(R_o \cdot t)}$$

$$= 2500.04 / \sqrt{(3564.0 \cdot 13.0)}$$

$$= 11.615$$

Compute the Factor [Ch]:

Since $M_x > 1.5$ and $M_x < 13$, $Ch = 0.92 / (M_x - 0.579)$

$$= 0.92 / (11.615 - 0.579)$$

$$= 0.0834$$

Elastic Hoop Compressive Membrane Failure Stress [Fhe]:

$$= 1.6 \cdot Ch \cdot E_y (t / D_o)$$

$$= 1.6 \cdot 0.083 \cdot 196301 (13.0 / 7128.0)$$

$$= 47.754 \text{ MPa}$$

Ratio of Elastic Hoop Compressive Membrane Failure Stress to Yield [Fr]:

$$= F_{he} / F_y$$

$$= 47.8 / 234.4$$

$$= 0.204$$

Predicted Buckling Stress [Fic]:

Engineering Yield to Tensile Stress ratio [R]:

$$= \sigma_{ys} / \sigma_{uts}$$

$$= 234.4 / 482.6$$

$$= 0.486$$

Ratio [AE]:

$$= F_{he} / E$$

$$= 47.8 / 196301.4$$

$$= 0.000243$$

Factor [m1]:

$$= (\ln(r) + (\epsilon_{psp} - \epsilon_{sys})) / \ln(\ln(1 + \epsilon_{psp}) / \ln(1 + \epsilon_{sys}))$$

$$= (\ln(0.486) + (0.00002 - 0.002)) / \ln(\ln(1 + 0.00002) / \ln(1 + 0.002))$$

$$= 0.157$$

Curve Fitting Constant for the Elastic Region of the Stress-Strain Curve [A1]:

$$= \sigma_{ys} (1 + \epsilon_{sys}) / \ln(1 + \epsilon_{sys})^{m1}$$

$$= 234.4 (1 + 0.002) / \ln(1 + 0.002)^{0.157301}$$

$$= 624.374 \text{ MPa}$$

Curve Fitting Exponent for, Ferritic Steel [m2]:

$$= m2f(m2f2 - R)$$

$$= 0.6(1.0 - 0.486)$$

$$= 0.309$$

Curve Fitting Constant [A2]:

$$= (\sigma_{uts} \cdot \exp(m2)) / (m2^{m2})$$

$$= (482.6 \cdot \exp(0.309)) / (0.309^{0.309})$$



= 944.566 MPa

Material Parameter for Stress-Strain Curve Model [K]:

$$= 1.5 \cdot R1.5 - 0.5 \cdot R2.5 - R3.5$$

$$= 1.5 \cdot 0.4861.5 - 0.5 \cdot 0.4862.5 - 0.4863.5$$

$$= 0.346$$

Compute the Allowable Stress per 4.4.3

	Fic	H	D1	D2	D3	D4	Tangent Modulus	Iteration Diff
1	241	-1.839	0.313E-04	0.271E-04	0.807E-04	-0.632E-04	0.124E+05	0.193E-01
2	120	-4.651	0.763E-06	0.763E-06	0.171E-04	-0.171E-04	0.151E+06	0.556E-03
3	60	-6.057	0.186E-07	0.186E-07	0.361E-05	-0.361E-05	0.195E+06	0.663E-04
4	30	-6.760	0.454E-09	0.454E-09	0.765E-06	-0.765E-06	0.196E+06	0.896E-04
5	45	-6.409	0.398E-08	0.398E-08	0.190E-05	-0.190E-05	0.196E+06	0.124E-04
6	52	-6.233	0.910E-08	0.910E-08	0.268E-05	-0.268E-05	0.196E+06	0.266E-04
7	49	-6.321	0.612E-08	0.612E-08	0.227E-05	-0.227E-05	0.196E+06	0.705E-05
8	47	-6.365	0.496E-08	0.496E-08	0.208E-05	-0.208E-05	0.196E+06	0.269E-05
9	48	-6.343	0.551E-08	0.551E-08	0.217E-05	-0.217E-05	0.196E+06	0.217E-05
10	47	-6.354	0.523E-08	0.523E-08	0.212E-05	-0.212E-05	0.196E+06	0.259E-06
11	47	-6.348	0.537E-08	0.537E-08	0.215E-05	-0.215E-05	0.196E+06	0.958E-06
12	47	-6.351	0.530E-08	0.530E-08	0.214E-05	-0.214E-05	0.196E+06	0.349E-06
13	47	-6.352	0.527E-08	0.527E-08	0.213E-05	-0.213E-05	0.196E+06	0.453E-07
14	47	-6.353	0.525E-08	0.525E-08	0.213E-05	-0.213E-05	0.196E+06	0.107E-06
15	47	-6.353	0.526E-08	0.526E-08	0.213E-05	-0.213E-05	0.196E+06	0.307E-07
16	47	-6.352	0.526E-08	0.526E-08	0.213E-05	-0.213E-05	0.196E+06	0.732E-08

Factor of Safety [FS]:

$$\text{Since } F_{ic} [47.66] < 0.55 \cdot 234.4, FS = 2$$

Allowable Circumferential Compressive Stress [Fha]:

$$= F_{ic}/FS$$

$$= 47.657/2.0$$

$$= 23.828 \text{ MPa}$$

Allowable External Pressure at the given Thickness [Pa]:

$$= 2 \cdot F_{ha} (t / D_o)$$

$$= 2 \cdot 23.828 (13.0/7128.0)$$

$$= 0.087 \text{ MPa}$$

Cylindrical Shell From Ring R10 to 270 Ext. Chart: CS-2 at 130 °C

Maximum External Pressure per ASME Division 2 Section 4.4:

Determine Parameter [Mx]:

$$= L / \sqrt{(R_o \cdot t)}$$

$$= 2391.667 / \sqrt{(3564.0 \cdot 13.0)}$$

$$= 11.111$$

Compute the Factor [Ch]:

$$\text{Since } M_x > 1.5 \text{ and } M_x < 13, Ch = 0.92 / (M_x - 0.579)$$

$$= 0.92 / (11.111 - 0.579)$$

$$= 0.0874$$

Elastic Hoop Compressive Membrane Failure Stress [Fhe]:

$$= 1.6 \cdot Ch \cdot E_y (t / D_o)$$

$$= 1.6 \cdot 0.087 \cdot 196301 (13.0/7128.0)$$

$$= 50.037 \text{ MPa}$$

Ratio of Elastic Hoop Compressive Membrane Failure Stress to Yield [Fr]:

$$= F_{he} / F_y$$

$$= 50.0/234.4$$

$$= 0.213$$

Predicted Buckling Stress [Fic]:

Engineering Yield to Tensile Stress ratio [R]:

$$= \sigma_{ys} / \sigma_{uts}$$

$$= 234.4/482.6$$

$$= 0.486$$

Ratio [AE]:

$$= F_{he} / E$$

$$= 50.0/196301.4$$

$$= 0.000255$$

Factor [m1]:

$$= (\ln(r) + (\epsilon_{psp} - \epsilon_{sys})) / \ln(\ln(1 + \epsilon_{psp}) / \ln(1 + \epsilon_{sys}))$$

$$= (\ln(0.486) + (0.00002 - 0.002)) / \ln(\ln(1 + 0.00002) / \ln(1 + 0.002))$$

$$= 0.157$$

Curve Fitting Constant for the Elastic Region of the Stress-Strain Curve [A1]:

$$= \sigma_{ys} (1 + \epsilon_{sys}) / \ln(1 + \epsilon_{sys})^{m1}$$

$$= 234.4 (1 + 0.002) / \ln(1 + 0.002)^{0.157301}$$

$$= 624.374 \text{ MPa}$$

Curve Fitting Exponent for, Ferritic Steel [m2]:

$$= m2f(m2f2 - R)$$

$$= 0.6(1.0 - 0.486)$$

$$= 0.309$$

Curve Fitting Constant [A2]:

$$= (\sigma_{uts} \cdot \exp(m2)) / (m2^{m2})$$

$$= (482.6 \cdot \exp(0.309)) / (0.309^{0.309})$$

$$= 944.566 \text{ MPa}$$

Material Parameter for Stress-Strain Curve Model [K]:

$$= 1.5 \cdot R1.5 - 0.5 \cdot R2.5 - R3.5$$

$$= 1.5 \cdot 0.4861.5 - 0.5 \cdot 0.4862.5 - 0.4863.5$$

$$= 0.346$$

Compute the Allowable Stress per 4.4.3

	Fic	H	D1	D2	D3	D4	Tangent Modulus	Iteration Diff
1	241	-1.839	0.313E-04	0.271E-04	0.807E-04	-0.632E-04	0.124E+05	0.193E-01
2	120	-4.651	0.763E-06	0.763E-06	0.171E-04	-0.171E-04	0.151E+06	0.545E-03
3	60	-6.057	0.186E-07	0.186E-07	0.361E-05	-0.361E-05	0.195E+06	0.547E-04
4	30	-6.760	0.454E-09	0.454E-09	0.765E-06	-0.765E-06	0.196E+06	0.101E-03
5	45	-6.409	0.398E-08	0.398E-08	0.190E-05	-0.190E-05	0.196E+06	0.240E-04
6	52	-6.233	0.910E-08	0.910E-08	0.268E-05	-0.268E-05	0.196E+06	0.150E-04
7	49	-6.321	0.612E-08	0.612E-08	0.227E-05	-0.227E-05	0.196E+06	0.458E-05
8	50	-6.277	0.749E-08	0.749E-08	0.247E-05	-0.247E-05	0.196E+06	0.518E-05
9	49	-6.299	0.678E-08	0.678E-08	0.237E-05	-0.237E-05	0.196E+06	0.297E-06
10	49	-6.310	0.644E-08	0.644E-08	0.232E-05	-0.232E-05	0.196E+06	0.214E-05
11	49	-6.304	0.661E-08	0.661E-08	0.234E-05	-0.234E-05	0.196E+06	0.924E-06
12	49	-6.302	0.669E-08	0.669E-08	0.236E-05	-0.236E-05	0.196E+06	0.314E-06
13	49	-6.300	0.673E-08	0.673E-08	0.236E-05	-0.236E-05	0.196E+06	0.836E-08

Factor of Safety [FS]:

Since Fic [49.9] < 0.55 * 234.4, FS = 2

Allowable Circumferential Compressive Stress [Fha]:

$$= F_{ic}/FS$$

$$= 49.903/2.0$$

$$= 24.951 \text{ MPa}$$

Allowable External Pressure at the given Thickness [Pa]:

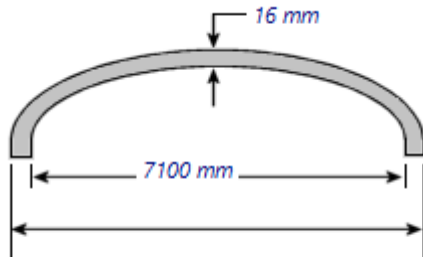
$$= 2 \cdot F_{ha} (t / D_o)$$

$$= 2 \cdot 24.951 (13.0/7128.0)$$

$$= 0.091 \text{ MPa}$$

Internal Pressure Calculation Results:

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Internal Pressure Results for: 122-D1-Top head

Elliptical Head Design Information:					
Design Pressure	P	0.388 MPa	Int. Design Temperature		130 °C
Material		SA-516 70	Ext. Design Temperature		130 °C
Length		50.000 mm	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	1.0000 mm	Allowable Stress (ope)	S	156.4 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	175.0 MPa
Inside Diameter	Di	7100.000 mm	Actual Stress		97.4 MPa
Spec. Min. Thk t _e , or s _a		16.0000 mm	Spec. Nominal Thk.		20.000 mm
Surface Area		56382684.0 mm ²	SG of Contents		0.000
Element Volume		48838.8 ltr	Weight of Contents		0.0 N
Empty Weight		93920.9 N	Operating Weight		93920.9 N
			Comp. All. Ope. Stress		52.6 MPa
			Aspect Ratio		2.00

Radiography Information:			
...		...	
Circ. Joint Efficiency	85 %	Long. Joint Eff.	E or v 85 %

Tolerance for Formed Heads per 4.3.2.2:

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 88.750 mm

Head inner surface maximum deviation inside the specified shape, 0.625% of D: 44.375 mm

Material treatment options: +IT

Head Calculation - Section 4.3.6 (and Section 4.3.7 for Ellipsoidal Head)

Computed Minimum Required Thickness [t]:

tr = 9.8361 mm

t = tr + ci + co
 = 9.8361 + 1.0 + 0.0
 = 10.8361 mm - see below for the derivation

For Elliptical heads: Compute [h, k, r,] per 4.3.7.1

h = D/(2 • Aspect Ratio)
 = 7100.0/(2 • 2.0)
 = 1775.000 mm

k = D/(2 • h)
 = 7100.0/(2 • 1775.0)
 = 2.000

r = D(0.5/k - 0.08) + ca



$$= 7100.0(0.5/2.0 - 0.08) + 1.0$$

$$= 1208.000 \text{ mm}$$

$$L = D(0.44 \cdot k + 0.02) + ca$$

$$= 7100.0(0.44 \cdot 2.0 + 0.02) + 1.0$$

$$= 6391.000 \text{ mm}$$

Step 3: Geometric Constants [βTh , ϕTh , Rth]:

Equations 4.3.8 to 4.3.11 as appropriate

t (corroded) and P for the calc.: $t = 9.8361 \text{ mm}$, $P = 0.388 \text{ MPa}$

Angle [βTh]:

$$= \arccos[(0.5 \cdot D - r)/(L - r)]$$

$$= \arccos[(0.5 \cdot 7102.0 - 1208.0)/(6391.0 - 1208.0)]$$

$$= 1.1017 \text{ radians - Equation 4.3.8}$$

$$\phi Th = \sqrt{(L \cdot t) / r}$$

$$= \sqrt{(6391.0 \cdot 9.836) / 1208.0}$$

$$= 0.2076 \text{ radians - Equation 4.3.9}$$

$\phi Th < \beta Th$, Thus:

$$Rth = (0.5 \cdot D - r) / \cos(\beta Th - \phi Th) + r$$

$$= (0.5 \cdot 7102.0 - 1208.0) / \cos(1.102 - 0.208) + 1208.0$$

$$= 4949.8438 \text{ mm - Equation 4.3.10}$$

Step 4: Coefficients: Equations 4.3.12 to 4.3.15 [$C1$ and $C2$]:

$r/D > 0.08$, Thus:

$$C1 = 0.692(r/D) + 0.605$$

$$= 0.692(1208.0/7102.0) + 0.605$$

$$= 0.7227 \text{ - Equation 4.3.13}$$

$$C2 = 1.46 - 2.6(r/D) = 1.46 - 2.6(1208.0/7102.0)$$

$$= 1.0178, \text{ Equation 4.3.15}$$

Step 5: Pressure to produce knuckle elastic buckling [$Peth$]:

$$= C1 \cdot Et \cdot t^2 / (C2 \cdot Rth(0.5 \cdot Rth - r))$$

$$= 0.723 \cdot 196301.391 \cdot 9.836^2 / (1.018 \cdot 4949.844(0.5 \cdot 4949.844 - 1208.0))$$

$$= 2.15 \text{ MPa, Equation 4.3.16}$$

Step 6: Internal Pressure for max stress in knuckle equals yield [Py]:

$C3 = Sy = 234.4 \text{ MPa}$, Not in the creep range - 4.3.6.1(f)

$$Py = C3 \cdot t / (C2 \cdot Rth(0.5 \cdot Rth / r - 1))$$

$$= 234.4 \cdot 9.836 / (1.018 \cdot 4949.844(0.5 \cdot 4949.844 / 1208.0 - 1))$$

$$= 0.436 \text{ MPa, Equation 4.3.17}$$

Step 7: Value of Internal Press. resulting in buckling of knuckle [G and Pck]:

$$G = Peth/Py = 2.151/0.436 = 4.928$$

$G > 1$, Thus:

Buckling Pressure [Pck]:

$$= ((0.77508 \cdot G - 0.20354 \cdot G^2 + 0.019274 \cdot G^3) /$$

$$(1 + 0.19014 \cdot G - 0.089534 \cdot G^2 + 0.0093965 \cdot G^3)) Py$$

$$= ((0.77508 \cdot 4.928 - 0.20354 \cdot 4.928^2 + 0.019274 \cdot 4.928^3) /$$

$$(1 + 0.19014 \cdot 4.928 - 0.089534 \cdot 4.928^2 + 0.0093965 \cdot 4.928^3)) \cdot 0.436$$

$$= 0.58 \text{ MPa, Equation 4.3.19}$$

Step 8: Compute Allowable Pressure base on buckling of knuckle [Pak]:

$$= Pck/1.5 = 0.58/1.5 = 0.39 \text{ MPa - Equation 4.3.21}$$

Step 9: Compute allowable pressure based on rupture of crown [Pac]:

$$= 2 \cdot S \cdot E / (L/t + 0.5)$$

$$= 2 \cdot 156.4 \cdot 0.85 / (6391.0/9.836 + 0.5)$$

$$= 0.41 \text{ MPa, Equation 4.3.22}$$

Step 10: Compute final pressure [Pa]:

$$= \min(Pak, Pac) = \min(0.388, 0.409)$$

$$= 0.39 \text{ MPa, Equation 4.3.23: (same as original design pressure)}$$



Allowable pressure in Crown at given thickness [Pac]:

$$\begin{aligned} &= 2 \cdot S \cdot E / (L / t + 0.5) \\ &= 2 \cdot 156.4 \cdot 0.85 / (6391.0 / 15.0 + 0.5) \\ &= 0.623 \text{ MPa} \end{aligned}$$

Computed Maximum Allowable Pressure [MAWP]:

$$= \text{MaxPressure} - \text{Pliq} = 0.6233 - 0.004 = 0.6193 \text{ MPa}$$

Computed Maximum Allowable Pressure New and Cold [MAPnc]:

$$= 0.744 \text{ MPa}$$

Computed Membrane Stress Intensity at Design Pressure [Sm]:

$$= 97.4 \text{ MPa}$$

Required Thickness of Straight Flange = 11.379 mm

Percent Fiber Elongation per Table 6.1

$$\begin{aligned} \text{For Heads formed from Segments } (75 \cdot t_{\text{nom}} / R_f) \cdot (1 - R_f / R_o) & \quad 1.233 \% \\ \text{For one Piece doubly Curved Heads } 100 \cdot \ln(D_b / (D_f - 2t)) & \quad 17.677 \% \end{aligned}$$

Where D_b is the blank diameter for one piece doubly curved surfaces [D_b]:

$$\begin{aligned} &= \text{square root}(4 \cdot \text{Head Surface Area} / 3.14159) \\ &= \text{square root}(4 \cdot 56382684 / 3.14159) \\ &= 8472.819 \text{ mm} \end{aligned}$$

MDMT Calculations in the Knuckle Portion:

Note: This Element/Detail was specified as being Impact Tested.

MDMT Calculations in the Head Straight Flange:

Note: This Element/Detail was specified as being Impact Tested.

Tolerance for Formed Heads per 4.3.2.2:

$$\begin{aligned} \text{Head inner surface maximum deviation outside the specified shape, } & 1.25\% \text{ of } D: 88.750 \text{ mm} \\ \text{Head inner surface maximum deviation inside the specified shape, } & 0.625\% \text{ of } D: 44.375 \text{ mm} \end{aligned}$$

Material treatment options: +IT

EXTERNAL PRESSURE RESULTS for: 122-D1-Top head

Elliptical Head From 270 to 280 Ext. Chart: CS-2 at 130 °C

Maximum External Pressure per ASME Division 2 Section 4.4:

Elastic Hoop Compressive Membrane Failure Stress [Fhe]:

$$\begin{aligned} &= 0.075 \cdot E_y(t / R_o) \\ &= 0.075 \cdot 196301(15.0 / 6394.002) \\ &= 34.538 \text{ MPa} \end{aligned}$$

Ratio of Elastic Hoop Compressive Membrane Failure Stress to Yield [Fr]:

$$\begin{aligned} &= F_{he} / S_y \\ &= 34.5 / 234.4 \\ &= 0.147 \end{aligned}$$

Engineering Yield to Tensile Stress ratio [R]:

$$\begin{aligned} &= \sigma_y / \sigma_{uts} \\ &= 234.4 / 482.6 \\ &= 0.486 \end{aligned}$$

Ratio [AE]:

$$\begin{aligned} &= F_{he} / E \\ &= 34.5 / 196301.4 \\ &= 0.000176 \end{aligned}$$

Factor [m1]:

$$= (\ln(r) + (\epsilon_{psp} - \epsilon_{sys})) / (\ln(1 + \epsilon_{psp}) / \ln(1 + \epsilon_{sys}))$$



$$= (\ln(0.486) + (0.00002 - 0.002)) / (\ln(1 + 0.00002) / \ln(1 + 0.002))$$

$$= 0.157$$

Curve Fitting Constant for the Elastic Region of the Stress-Strain Curve [A1]:

$$= \sigma_{ys} (1 + \epsilon_{sys}) / \ln(1 + \epsilon_{sys})^{m1}$$

$$= 234.4 (1 + 0.002) / \ln(1 + 0.002) \cdot 0.157301$$

$$= 624.374 \text{ MPa}$$

Curve Fitting Exponent for, Ferritic Steel [m2]:

$$= m2f / (m2f^2 - R)$$

$$= 0.6 (1.0 - 0.486)$$

$$= 0.309$$

Curve Fitting Constant [A2]:

$$= (\sigma_{uts} \cdot \exp(m2)) / (m2^{m2})$$

$$= (482.6 \cdot \exp(0.309)) / (0.309^{0.309})$$

$$= 944.566 \text{ MPa}$$

Material Parameter for Stress-Strain Curve Model [K]:

$$= 1.5 \cdot R1.5 - 0.5 \cdot R2.5 - R3.5$$

$$= 1.5 \cdot 0.4861.5 - 0.5 \cdot 0.4862.5 - 0.4863.5$$

$$= 0.346$$

Compute the Allowable Stress per 4.4.3

	Fic	H	D1	D2	D3	D4	Tangent Modulus	Iteration Diff
1	241	-1.839	0.313E-04	0.271E-04	0.807E-04	-0.632E-04	0.124E+05	0.193E-01
2	120	-4.651	0.763E-06	0.763E-06	0.171E-04	-0.171E-04	0.151E+06	0.624E-03
3	60	-6.057	0.186E-07	0.186E-07	0.361E-05	-0.361E-05	0.195E+06	0.134E-03
4	30	-6.760	0.454E-09	0.454E-09	0.765E-06	-0.765E-06	0.196E+06	0.222E-04
5	45	-6.409	0.398E-08	0.398E-08	0.190E-05	-0.190E-05	0.196E+06	0.549E-04
6	37	-6.584	0.150E-08	0.150E-08	0.126E-05	-0.126E-05	0.196E+06	0.163E-04
7	33	-6.672	0.853E-09	0.853E-09	0.995E-06	-0.995E-06	0.196E+06	0.301E-05
8	35	-6.628	0.114E-08	0.114E-08	0.112E-05	-0.112E-05	0.196E+06	0.662E-05
9	34	-6.650	0.988E-09	0.988E-09	0.106E-05	-0.106E-05	0.196E+06	0.180E-05
10	34	-6.661	0.919E-09	0.919E-09	0.103E-05	-0.103E-05	0.196E+06	0.603E-06
11	34	-6.656	0.953E-09	0.953E-09	0.104E-05	-0.104E-05	0.196E+06	0.600E-06
12	34	-6.659	0.936E-09	0.936E-09	0.103E-05	-0.103E-05	0.196E+06	0.130E-08

Factor of Safety [FS]:

$$\text{Since } Fic [34.53] < 0.55 \cdot 234.4, FS = 2$$

Allowable Circumferential Compressive Stress [Fha]:

$$= Fic / FS$$

$$= 34.526 / 2.0$$

$$= 17.263 \text{ MPa}$$

Allowable External Pressure at the given Thickness [Pa]:

$$= 2 \cdot Fha [t / Ro]$$

$$= 2 \cdot 17.263 [15.0 / 6394.002]$$

$$= 0.081 \text{ MPa}$$



Hydrostatic Test Pressure Results:

Test Pressure at high point in vessel per Equation 8.1
 $= \gamma_{St/s} \cdot MAWP(St/S)$
 $= 1.25 \cdot 0.4818(1.0)$
 $= 0.602 \text{ MPa}$

Horizontal Test performed per: $\gamma_{St/s} \cdot MAWP(St/S)$

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (MPa):

From To	Stress	Allowable	Ratio	Pressure
163-D- BTM dish	18.9	247.0	0.077	0.67
163-D-Shell-1	23.2	247.0	0.094	0.67
163-D-Shell-2	23.2	247.0	0.094	0.67
163-D-Shell-3	23.6	247.0	0.096	0.67
163-D-Shell-4	24.7	247.0	0.100	0.67
163-D-Shell-5	24.7	247.0	0.100	0.67
163-D-Shell-6	24.7	247.0	0.100	0.67
163-D-Shell-7	24.7	247.0	0.100	0.67
163-D-Top head	21.8	247.0	0.088	0.67
122-D1-Btm head	59.8	247.0	0.242	0.67
122-D1-Shell-1	99.7	247.0	0.404	0.67
122-D1-Shell-2	99.7	247.0	0.404	0.67
122-D1-Shell-3	108.7	247.0	0.440	0.67
122-D1-Shell-4	108.7	247.0	0.440	0.67
122-D1-Shell-5	119.6	247.0	0.484	0.67
122-D1-Shell-6	175.8	247.0	0.712	0.67
122-D1-Shell-7	175.8	247.0	0.712	0.67
122-D1-Shell-8	175.8	234.0	0.751	0.67
122-D1-Shell-9	200.8	247.0	0.813	0.67
122-D1-Top head	158.0	247.0	0.640	0.67

Stress ratios for Nozzle and Pad Materials (MPa):

Description	Pad/Nozzle	Ambient	Operating	Ratio
D 36"	Nozzle	165.00	147.60	1.118
L12 2"	Nozzle	165.00	147.60	1.118
L14 1.5"	Nozzle	165.00	147.60	1.118
MH-6 24"	Nozzle	165.00	147.60	1.118
L11 2"	Nozzle	165.00	147.60	1.118
L13 1.5"	Nozzle	165.00	147.60	1.118
UC2 24"	Nozzle	175.00	156.40	1.119
MH-5 24"	Nozzle	165.00	147.60	1.118
MH-4 24"	Nozzle	165.00	147.60	1.118
A 36"	Nozzle	165.00	147.60	1.118
E 8"	Nozzle	138.00	138.00	1.000
C 48"	Nozzle	175.00	156.40	1.119
C 48"	Pad	175.00	156.40	1.119
MH-3 24"	Nozzle	175.00	156.40	1.119
MH-3 24"	Pad	175.00	156.40	1.119
L2 2"	Nozzle	138.00	138.00	1.000
L4 2"	Nozzle	138.00	138.00	1.000
L6 2"	Nozzle	138.00	138.00	1.000
L8 2"	Nozzle	138.00	138.00	1.000
L10 2"	Nozzle	138.00	138.00	1.000
B 36"	Nozzle	175.00	156.40	1.119
B 36"	Pad	175.00	156.40	1.119
K 6"	Nozzle	161.00	143.60	1.121
K 6"	Pad	175.00	156.40	1.119
I 28"	Nozzle	175.00	156.40	1.119
I 28"	Pad	175.00	156.40	1.119

L1 2"	Nozzle	138.00	138.00	1.000
L3 2"	Nozzle	138.00	138.00	1.000
L5 2"	Nozzle	138.00	138.00	1.000
L7 2"	Nozzle	138.00	138.00	1.000
L9 2"	Nozzle	138.00	138.00	1.000
MH-2 24"	Nozzle	175.00	156.40	1.119
MH-2 24"	Pad	175.00	156.40	1.119
UC1 24"	Nozzle	175.00	156.40	1.119
UC1 24"	Pad	175.00	156.40	1.119
MH-1 24"	Nozzle	175.00	156.40	1.119
MH-1 24"	Pad	175.00	156.40	1.119
S1 1"	Nozzle	138.00	138.00	1.000
H1 42"	Nozzle	175.00	156.40	1.119
H1 42"	Pad	175.00	156.40	1.119
H2 42"	Nozzle	175.00	156.40	1.119
H2 42"	Pad	175.00	156.40	1.119
F 42"	Nozzle	175.00	156.40	1.119
F 42"	Pad	175.00	156.40	1.119

Minimum 1.000

Stress ratios for Stiffening Ring Materials (MPa):

Description	Ambient	Operating	Ratio
3	175.00	156.40	1.119
Ring R2	175.00	156.40	1.119
Ring R3	175.00	156.40	1.119
Ring R4	175.00	156.40	1.119
Ring R5	175.00	156.40	1.119
Ring R6	175.00	156.40	1.119
Ring R7	175.00	156.40	1.119
Ring R8	175.00	156.40	1.119
Ring R9	175.00	156.40	1.119
Ring R10	175.00	156.40	1.119

Minimum 1.119

Stress ratios for Pressurized Vessel Elements (MPa):

Description	Ambient	Operating	Ratio
163-D- BTM dish	175.00	156.40	1.119
163-D-Shell-1	175.00	156.40	1.119
163-D-Shell-2	175.00	156.40	1.119
163-D-Shell-3	175.00	156.40	1.119
163-D-Shell-4	175.00	156.40	1.119
163-D-Shell-5	175.00	156.40	1.119
163-D-Shell-6	175.00	156.40	1.119
163-D-Shell-7	175.00	156.40	1.119
163-D-Top head	175.00	156.40	1.119
122-D1-Btm head	175.00	156.40	1.119
122-D1-Shell-1	175.00	156.40	1.119
122-D1-Shell-2	175.00	156.40	1.119
122-D1-Shell-3	175.00	156.40	1.119
122-D1-Shell-4	175.00	156.40	1.119
122-D1-Shell-5	175.00	156.40	1.119
122-D1-Shell-6	175.00	156.40	1.119
122-D1-Shell-7	175.00	156.40	1.119
122-D1-Shell-8	138.00	138.00	1.000
122-D1-Shell-9	175.00	156.40	1.119
122-D1-Top head	175.00	156.40	1.119

Minimum 1.000

Hoop Stress in Nozzle Wall during Pressure Test (MPa):

Description	Ambient	Operating	Ratio
D 36"	15.61	237.50	0.066



L12 2"	1.73	237.50	0.007
L14 1.5"	1.58	237.50	0.007
MH-6 24"	13.05	237.50	0.055
L11 2"	1.73	237.50	0.007
L13 1.5"	1.58	237.50	0.007
UC2 24"	13.05	247.00	0.053
MH-5 24"	13.06	237.50	0.055
MH-4 24"	13.06	237.50	0.055
A 36"	15.60	237.50	0.066
E 8"	3.90	225.00	0.017
C 48"	25.26	247.00	0.102
MH-3 24"	12.46	247.00	0.050
L2 2"	1.33	225.00	0.006
L4 2"	1.33	225.00	0.006
L6 2"	1.33	225.00	0.006
L8 2"	1.33	225.00	0.006
L10 2"	1.33	225.00	0.006
B 36"	12.46	247.00	0.050
K 6"	3.19	228.00	0.014
I 28"	11.19	247.00	0.045
L1 2"	1.33	225.00	0.006
L3 2"	1.33	225.00	0.006
L5 2"	1.33	225.00	0.006
L7 2"	1.33	225.00	0.006
L9 2"	1.33	225.00	0.006
MH-2 24"	12.46	247.00	0.050
UC1 24"	13.13	247.00	0.053
MH-1 24"	12.46	247.00	0.050
S1 1"	0.97	225.00	0.004
H1 42"	13.44	247.00	0.054
H2 42"	13.44	247.00	0.054
F 42"	14.59	247.00	0.059

Elements Suitable for Test Pressure.

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External Pressure Calculation Results Summary:

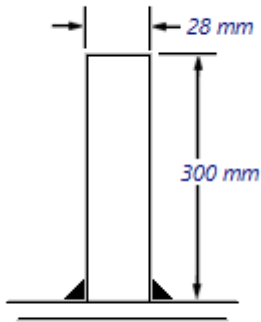
ASME Code, Section VIII Division 2, 2025

MDMT Calculations for: 3, Shell to Ring at Ring (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Stiffening Ring Calculations for: 3, SA-516 70 , Bar Ring: 300.0 x 28.0 mm

Material treatment options:



Local Stiffener Geometry Requirements per [6.4] or [Div. 2 4.4.37]:

$$h/t \leq 0.375 \cdot \sqrt{E_y/F_y} \quad [\text{eq 6-6}] \text{ or } [\text{eq 4.4.32}]$$

$$300.0/28.0 \leq 0.375 \cdot \sqrt{196301/234}$$

$$10.714 \leq 10.852 \quad \text{Ok}$$

Effective Length of Shell: 395 mm

	Area (mm ²)	Distance (mm)	Area*Dist
Shell:	7102.213	9.0000	63919.922
Ring :	8399.997	168.0000	1411199.125
Total:	15502.211		1475119.000

Centroid of Ring plus Shell: 95 mm

	Inertia	Distance	A*Dist ²
Shell:	191759.609	86.1554	52717972.000
Ring :	62999880.000	-72.8446	44573152.000
Total:	63191640.000		97291120.000

Available Moment of Inertia, Ring + Shell: 0.160E+09 mm⁴

Determine Parameter [Mx]:

$$= L / \sqrt{R_o \cdot t}$$

$$= 2795.858 / \sqrt{3574.0 \cdot 18.0}$$

$$= 11.023$$

Compute the Factor [Ch]:

$$\text{Since } M_x > 1.5 \text{ and } M_x < 13, \text{ Ch} = 0.92 / (M_x - 0.579)$$

$$= 0.92 / (11.023 - 0.579)$$

$$= 0.0881$$

Elastic Hoop Compressive Membrane Failure Stress [Fhe]:

$$= 1.6 \cdot \text{Ch} \cdot E_y (t / D_o)$$

$$= 1.6 \cdot 0.088 \cdot 196301 (18.0 / 7148.0)$$



= 69.671 MPa

Ratio of Elastic Hoop Compressive Membrane Failure Stress to Yield [Fr]:

= Fhe / Fy
 = 69.7/234.4
 = 0.297

Predicted Buckling Stress [Fic]:

Engineering Yield to Tensile Stress ratio [R]:

= $\sigma_{ys} / \sigma_{uts}$
 = 234.4/482.6
 = 0.486

Ratio [AE]:

= Fhe / E
 = 69.7/196301.4
 = 0.000355

Factor [m1]:

= $(\ln(r) + (\text{epsp} - \text{epsys})) / \ln(\ln(1 + \text{epsp}) / \ln(1 + \text{epsys}))$
 = $(\ln(0.486) + (0.00002 - 0.002)) / \ln(\ln(1 + 0.00002) / \ln(1 + 0.002))$
 = 0.157

Curve Fitting Constant for the Elastic Region of the Stress-Strain Curve [A1]:

= $\sigma_{ys}(1 + \text{epsys}) / \ln(1 + \text{epsys})^{m1}$
 = $234.4(1 + 0.002) / \ln(1 + 0.002)^{0.157301}$
 = 624.374 MPa

Curve Fitting Exponent for, Ferritic Steel [m2]:

= $m2f(m2f^2 - R)$
 = $0.6(1.0 - 0.486)$
 = 0.309

Curve Fitting Constant [A2]:

= $(\sigma_{uts} \cdot \exp(m2)) / (m2^{m2})$
 = $(482.6 \cdot \exp(0.309)) / (0.309^{0.309})$
 = 944.566 MPa

Material Parameter for Stress-Strain Curve Model [K]:

= $1.5 \cdot R1.5 - 0.5 \cdot R2.5 - R3.5$
 = $1.5 \cdot 0.486^{1.5} - 0.5 \cdot 0.486^{2.5} - 0.486^{3.5}$
 = 0.346

Compute the Allowable Stress per 4.4.3

	Fic	H	D1	D2	D3	D4	Tangent Modulus	Iteration Diff
1	241	-1.839	0.313E-04	0.271E-04	0.807E-04	-0.632E-04	0.124E+05	0.192E-01
2	120	-4.651	0.763E-06	0.763E-06	0.171E-04	-0.171E-04	0.151E+06	0.445E-03
3	60	-6.057	0.186E-07	0.186E-07	0.361E-05	-0.361E-05	0.195E+06	0.453E-04
4	90	-5.354	0.163E-06	0.163E-06	0.896E-05	-0.896E-05	0.184E+06	0.136E-03
5	75	-5.706	0.615E-07	0.615E-07	0.596E-05	-0.596E-05	0.192E+06	0.386E-04
6	67	-5.881	0.350E-07	0.350E-07	0.470E-05	-0.470E-05	0.194E+06	0.440E-05
7	71	-5.793	0.467E-07	0.467E-07	0.531E-05	-0.531E-05	0.193E+06	0.168E-04
8	69	-5.837	0.405E-07	0.405E-07	0.500E-05	-0.500E-05	0.193E+06	0.611E-05
9	68	-5.859	0.377E-07	0.377E-07	0.485E-05	-0.485E-05	0.193E+06	0.836E-06
10	68	-5.870	0.363E-07	0.363E-07	0.478E-05	-0.478E-05	0.194E+06	0.179E-05
11	68	-5.865	0.370E-07	0.370E-07	0.481E-05	-0.481E-05	0.193E+06	0.477E-06
12	68	-5.862	0.373E-07	0.373E-07	0.483E-05	-0.483E-05	0.193E+06	0.179E-06
13	68	-5.863	0.371E-07	0.371E-07	0.482E-05	-0.482E-05	0.193E+06	0.149E-06
14	68	-5.863	0.372E-07	0.372E-07	0.483E-05	-0.483E-05	0.193E+06	0.154E-07
15	68	-5.863	0.372E-07	0.372E-07	0.483E-05	-0.483E-05	0.193E+06	0.666E-07
16	68	-5.863	0.372E-07	0.372E-07	0.483E-05	-0.483E-05	0.193E+06	0.256E-07
17	68	-5.863	0.372E-07	0.372E-07	0.483E-05	-0.483E-05	0.193E+06	0.512E-08

Factor of Safety [FS]:

Since Fic [68.66] < 0.55 * 234.4, FS = 2

Allowable Circumferential Compressive Stress [Fha]:

$$= F_{ic}/FS$$

$$= 68.664/2.0$$

$$= 34.332 \text{ MPa}$$

Allowable External Pressure at the given Thickness [Pa]:

$$= 2 \cdot F_{ha} (t / D_o)$$

$$= 2 \cdot 34.332 (18.0/7148.0)$$

$$= 0.173 \text{ MPa}$$

Determine the value of [n]:

$$= \sqrt[2]{(2 \cdot D_o^{1.5} / (3 \cdot L_B \cdot (t)))}$$

$$= \sqrt[2]{(2 \cdot 7148.01^{1.5} / (3 \cdot 27983.334 \cdot (24.0)))}$$

$$= 2.000 \text{ Note: Rounded to Nearest Integer Value}$$

Because $n < 2$, set $n = 2$

Required Moment of Inertia for the Ring/Shell Junction [Rir]:

$$= 1.5 \cdot F_{he} \cdot L_s \cdot R_c^2 \cdot t / (E \cdot (n^2 - 1)) \text{ [eq 6-1] or [4.4.30]}$$

$$= 1.5 \cdot 70 \cdot 2795.858 \cdot 3724.0^2 \cdot 24.0 / (196301 \cdot (2.0^2 - 1))$$

$$= 123852144.000 \text{ mm}^4$$

Factor of Safety [FS]:

$$\text{Since } F_{ic} [118.82] < 0.55 \cdot 234.4, FS = 2$$

Engineering Yield to Tensile Stress ratio [R]:

$$= \sigma_{ys} / \sigma_{uts}$$

$$= 234.4/482.6$$

$$= 0.486$$

Ratio [AE]:

$$= F_{he} / E$$

$$= 180.3/196301.4$$

$$= 0.000918$$

Factor [m1]:

$$= (\ln(r) + (\epsilon_{psp} - \epsilon_{sys})) / \ln(\ln(1 + \epsilon_{psp}) / \ln(1 + \epsilon_{sys}))$$

$$= (\ln(0.486) + (0.00002 - 0.002)) / \ln(\ln(1 + 0.00002) / \ln(1 + 0.002))$$

$$= 0.157$$

Curve Fitting Constant for the Elastic Region of the Stress-Strain Curve [A1]:

$$= \sigma_{ys} (1 + \epsilon_{sys}) / \ln(1 + \epsilon_{sys})^{m1}$$

$$= 234.4 (1 + 0.002) / \ln(1 + 0.002)^{0.157301}$$

$$= 624.374 \text{ MPa}$$

Curve Fitting Exponent for, Ferritic Steel [m2]:

$$= m2f(m2f2 - R)$$

$$= 0.6 (1.0 - 0.486)$$

$$= 0.309$$

Curve Fitting Constant [A2]:

$$= (\sigma_{uts} \cdot \exp(m2)) / (m2^{m2})$$

$$= (482.6 \cdot \exp(0.309)) / (0.309^{0.309})$$

$$= 944.566 \text{ MPa}$$

Material Parameter for Stress-Strain Curve Model [K]:

$$= 1.5 \cdot R1.5 - 0.5 \cdot R2.5 - R3.5$$

$$= 1.5 \cdot 0.4861.5 - 0.5 \cdot 0.4862.5 - 0.4863.5$$

$$= 0.346$$

Compute the Allowable Stress per 4.4.3

	Fic	H	D1	D2	D3	D4	Tangent Modulus	Iteration Diff
1	241	-1.839	0.313E-04	0.271E-04	0.807E-04	-0.632E-04	0.124E+05	0.186E-01
2	120	-4.651	0.763E-06	0.763E-06	0.171E-04	-0.171E-04	0.151E+06	0.119E-03
3	180	-3.245	0.670E-05	0.665E-05	0.423E-04	-0.419E-04	0.529E+05	0.250E-02
4	150	-3.948	0.252E-05	0.252E-05	0.281E-04	-0.281E-04	0.981E+05	0.620E-03
5	135	-4.299	0.143E-05	0.143E-05	0.222E-04	-0.222E-04	0.125E+06	0.166E-03



6	128	-4.475	0.106E-05	0.105E-05	0.196E-04	-0.195E-04	0.139E+06	0.722E-05
7	124	-4.563	0.900E-06	0.899E-06	0.183E-04	-0.183E-04	0.145E+06	0.592E-04
8	126	-4.519	0.975E-06	0.975E-06	0.189E-04	-0.189E-04	0.142E+06	0.269E-04
9	127	-4.497	0.101E-05	0.101E-05	0.192E-04	-0.192E-04	0.140E+06	0.101E-04
10	127	-4.486	0.103E-05	0.103E-05	0.194E-04	-0.194E-04	0.139E+06	0.150E-05
11	127	-4.481	0.105E-05	0.104E-05	0.195E-04	-0.195E-04	0.139E+06	0.284E-05
12	127	-4.483	0.104E-05	0.104E-05	0.194E-04	-0.194E-04	0.139E+06	0.666E-06
13	127	-4.485	0.104E-05	0.104E-05	0.194E-04	-0.194E-04	0.139E+06	0.420E-06
14	127	-4.484	0.104E-05	0.104E-05	0.194E-04	-0.194E-04	0.139E+06	0.123E-06
15	127	-4.484	0.104E-05	0.104E-05	0.194E-04	-0.194E-04	0.139E+06	0.149E-06
16	127	-4.484	0.104E-05	0.104E-05	0.194E-04	-0.194E-04	0.139E+06	0.128E-07
17	127	-4.484	0.104E-05	0.104E-05	0.194E-04	-0.194E-04	0.139E+06	0.550E-07

Factor of Safety [FS]:

Since $Fic [127.82] < 0.55 * 234.4$, $FS = 2$

Required Moment of Inertia for the Ring/Shell Junction, Shear [Irs]:

$= 0.184 \cdot Cv \cdot Ms0.8 \cdot t^3 \cdot Ls$ [eq 6-5] or [eq 4.4.43]

$= 0.184 \cdot 0.4558 \cdot 11.0230.8 \cdot 18.0^3 \cdot 2795.8584$

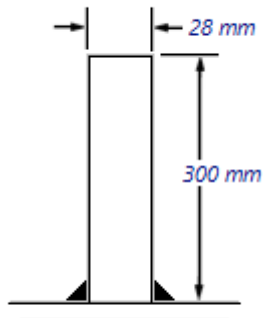
$= 9328167.000 \text{ mm}^4$

MDMT Calculations for: Ring R2, Shell to Ring at Ring (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Stiffening Ring Calculations for: Ring R2, SA-516 70 , Bar Ring: 300.0 x 28.0 mm

Material treatment options:



Local Stiffener Geometry Requirements per [6.4] or [Div. 2 4.4.37]:

$h/t \leq 0.375 \cdot \sqrt{E_y/F_y}$ [eq 6-6] or [eq 4.4.32]

$300.0/28.0 \leq 0.375 \cdot \sqrt{196301/234}$

$10.714 \leq 10.852$ Ok

Effective Length of Shell: 395 mm

	Area (mm ²)	Distance (mm)	Area*Dist
Shell:	7102.213	9.0000	63919.922
Ring :	8399.997	168.0000	1411199.125
Total:	15502.211		1475119.000

Centroid of Ring plus Shell: 95 mm

	Inertia	Distance	A*Dist ²
Shell:	191759.609	86.1554	52717972.000
Ring :	62999880.000	-72.8446	44573152.000
Total:	63191640.000		97291120.000



Available Moment of Inertia, Ring + Shell: 0.160E+09 mm⁴

Determine Parameter [Mx]:

$$= L / \sqrt{(Ro \cdot t)}$$

$$= 2625.0 / \sqrt{(3574.0 \cdot 18.0)}$$

$$= 10.349$$

Compute the Factor [Ch]:

$$\text{Since } Mx > 1.5 \text{ and } Mx < 13, Ch = 0.92 / (Mx - 0.579)$$

$$= 0.92 / (10.349 - 0.579)$$

$$= 0.0942$$

Elastic Hoop Compressive Membrane Failure Stress [Fhe]:

$$= 1.6 \cdot Ch \cdot Ey(t / Do)$$

$$= 1.6 \cdot 0.094 \cdot 196301(18.0 / 7148.0)$$

$$= 74.474 \text{ MPa}$$

Ratio of Elastic Hoop Compressive Membrane Failure Stress to Yield [Fr]:

$$= Fhe / Fy$$

$$= 74.5 / 234.4$$

$$= 0.318$$

Predicted Buckling Stress [Fic]:

Engineering Yield to Tensile Stress ratio [R]:

$$= \sigma_{ys} / \sigma_{uts}$$

$$= 234.4 / 482.6$$

$$= 0.486$$

Ratio [AE]:

$$= Fhe / E$$

$$= 74.5 / 196301.4$$

$$= 0.000379$$

Factor [m1]:

$$= (\ln(r) + (\epsilon_{psp} - \epsilon_{sys})) / \ln(\ln(1 + \epsilon_{psp}) / \ln(1 + \epsilon_{sys}))$$

$$= (\ln(0.486) + (0.00002 - 0.002)) / \ln(\ln(1 + 0.00002) / \ln(1 + 0.002))$$

$$= 0.157$$

Curve Fitting Constant for the Elastic Region of the Stress-Strain Curve [A1]:

$$= \sigma_{ys}(1 + \epsilon_{sys}) / \ln(1 + \epsilon_{sys})^{m1}$$

$$= 234.4(1 + 0.002) / \ln(1 + 0.002)^{0.157301}$$

$$= 624.374 \text{ MPa}$$

Curve Fitting Exponent for, Ferritic Steel [m2]:

$$= m2f(m2f2 - R)$$

$$= 0.6(1.0 - 0.486)$$

$$= 0.309$$

Curve Fitting Constant [A2]:

$$= (\sigma_{uts} \cdot \exp(m2)) / (m2^{m2})$$

$$= (482.6 \cdot \exp(0.309)) / (0.309^{0.309})$$

$$= 944.566 \text{ MPa}$$

Material Parameter for Stress-Strain Curve Model [K]:

$$= 1.5 \cdot R1.5 - 0.5 \cdot R2.5 - R3.5$$

$$= 1.5 \cdot 0.4861.5 - 0.5 \cdot 0.4862.5 - 0.4863.5$$

$$= 0.346$$

Compute the Allowable Stress per 4.4.3

	Fic	H	D1	D2	D3	D4	Tangent Modulus	Iteration Diff
1	241	-1.839	0.313E-04	0.271E-04	0.807E-04	-0.632E-04	0.124E+05	0.191E-01
2	120	-4.651	0.763E-06	0.763E-06	0.171E-04	-0.171E-04	0.151E+06	0.420E-03
3	60	-6.057	0.186E-07	0.186E-07	0.361E-05	-0.361E-05	0.195E+06	0.698E-04
4	90	-5.354	0.163E-06	0.163E-06	0.896E-05	-0.896E-05	0.184E+06	0.111E-03
5	75	-5.706	0.615E-07	0.615E-07	0.596E-05	-0.596E-05	0.192E+06	0.141E-04



6	67	-5.881	0.350E-07	0.350E-07	0.470E-05	-0.470E-05	0.194E+06	0.289E-04
7	71	-5.793	0.467E-07	0.467E-07	0.531E-05	-0.531E-05	0.193E+06	0.771E-05
8	73	-5.749	0.537E-07	0.537E-07	0.563E-05	-0.563E-05	0.192E+06	0.310E-05
9	72	-5.771	0.501E-07	0.501E-07	0.547E-05	-0.547E-05	0.193E+06	0.233E-05
10	73	-5.760	0.519E-07	0.519E-07	0.555E-05	-0.555E-05	0.192E+06	0.379E-06
11	72	-5.766	0.510E-07	0.510E-07	0.551E-05	-0.551E-05	0.192E+06	0.976E-06
12	72	-5.763	0.514E-07	0.514E-07	0.553E-05	-0.553E-05	0.192E+06	0.298E-06
13	72	-5.762	0.517E-07	0.517E-07	0.554E-05	-0.554E-05	0.192E+06	0.404E-07
14	72	-5.763	0.515E-07	0.515E-07	0.553E-05	-0.553E-05	0.192E+06	0.129E-06
15	72	-5.762	0.516E-07	0.516E-07	0.553E-05	-0.553E-05	0.192E+06	0.444E-07
16	72	-5.762	0.516E-07	0.516E-07	0.554E-05	-0.554E-05	0.192E+06	0.200E-08

Factor of Safety [FS]:

Since $Fic [72.99] < 0.55 * 234.4$, FS = 2

Allowable Circumferential Compressive Stress [Fha]:

= Fic/FS

= $72.991/2.0$

= 36.496 MPa

Allowable External Pressure at the given Thickness [Pa]:

= $2 * Fha (t / Do)$

= $2 * 36.496 (18.0/7148.0)$

= 0.184 MPa

Determine the value of [n]:

= $\sqrt[3]{(2 * Do^{1.5} / (3 * LB * (t)))}$

= $\sqrt[3]{(2 * 7148.01.5 / (3 * 27983.334 * (24.0)))}$

= 2.000 Note: Rounded to Nearest Integer Value

Because $n < 2$, set $n = 2$

Required Moment of Inertia for the Ring/Shell Junction [Rir]:

= $1.5 * Fhe * Ls * Rc^2 * t / (E * (n^2 - 1))$ [eq 6-1] or [4.4.30]

= $1.5 * 74 * 2625.0 * 3724.0^2 * 24.0 / (196301 * (2.0^2 - 1))$

= 124300688.000 mm⁴

Factor of Safety [FS]:

Since $Fic [119.45] < 0.55 * 234.4$, FS = 2

Engineering Yield to Tensile Stress ratio [R]:

= $\sigma_{ys} / \sigma_{ts}$

= $234.4/482.6$

= 0.486

Ratio [AE]:

= Fhe / E

= $186.6/196301.4$

= 0.000951

Factor [m1]:

= $(\ln(r) + (\epsilon_{psp} - \epsilon_{sys})) / \ln(\ln(1 + \epsilon_{psp}) / \ln(1 + \epsilon_{sys}))$

= $(\ln(0.486) + (0.00002 - 0.002)) / \ln(\ln(1 + 0.00002) / \ln(1 + 0.002))$

= 0.157

Curve Fitting Constant for the Elastic Region of the Stress-Strain Curve [A1]:

= $\sigma_{ys} (1 + \epsilon_{sys}) / \ln(1 + \epsilon_{sys}) m1$

= $234.4 (1 + 0.002) / \ln(1 + 0.002) 0.157301$

= 624.374 MPa

Curve Fitting Exponent for, Ferritic Steel [m2]:

= $m2f(m2f2 - R)$

= $0.6 (1.0 - 0.486)$

= 0.309

Curve Fitting Constant [A2]:

= $(\sigma_{ts} * \exp(m2)) / (m2 m2)$

= $(482.6 * \exp(0.309)) / (0.309 0.309)$

= 944.566 MPa

Material Parameter for Stress-Strain Curve Model [K]:

$$= 1.5 \cdot R1.5 - 0.5 \cdot R2.5 - R3.5$$

$$= 1.5 \cdot 0.4861.5 - 0.5 \cdot 0.4862.5 - 0.4863.5$$

$$= 0.346$$

Compute the Allowable Stress per 4.4.3

	Fic	H	D1	D2	D3	D4	Tangent Modulus	Iteration Diff
1	241	-1.839	0.313E-04	0.271E-04	0.807E-04	-0.632E-04	0.124E+05	0.186E-01
2	120	-4.651	0.763E-06	0.763E-06	0.171E-04	-0.171E-04	0.151E+06	0.151E-03
3	180	-3.245	0.670E-05	0.665E-05	0.423E-04	-0.419E-04	0.529E+05	0.247E-02
4	150	-3.948	0.252E-05	0.252E-05	0.281E-04	-0.281E-04	0.981E+05	0.588E-03
5	135	-4.299	0.143E-05	0.143E-05	0.222E-04	-0.222E-04	0.125E+06	0.133E-03
6	128	-4.475	0.106E-05	0.105E-05	0.196E-04	-0.195E-04	0.139E+06	0.252E-04
7	131	-4.387	0.123E-05	0.123E-05	0.209E-04	-0.208E-04	0.132E+06	0.494E-04
8	130	-4.431	0.114E-05	0.114E-05	0.202E-04	-0.202E-04	0.135E+06	0.110E-04
9	129	-4.453	0.110E-05	0.110E-05	0.199E-04	-0.199E-04	0.137E+06	0.734E-05
10	129	-4.442	0.112E-05	0.112E-05	0.200E-04	-0.200E-04	0.136E+06	0.178E-05
11	129	-4.448	0.111E-05	0.111E-05	0.200E-04	-0.199E-04	0.136E+06	0.280E-05
12	129	-4.445	0.111E-05	0.111E-05	0.200E-04	-0.200E-04	0.136E+06	0.515E-06
13	129	-4.444	0.112E-05	0.112E-05	0.200E-04	-0.200E-04	0.136E+06	0.629E-06
14	129	-4.444	0.112E-05	0.111E-05	0.200E-04	-0.200E-04	0.136E+06	0.568E-07
15	129	-4.445	0.111E-05	0.111E-05	0.200E-04	-0.200E-04	0.136E+06	0.229E-06
16	129	-4.444	0.111E-05	0.111E-05	0.200E-04	-0.200E-04	0.136E+06	0.863E-07
17	129	-4.444	0.112E-05	0.111E-05	0.200E-04	-0.200E-04	0.136E+06	0.147E-07

Factor of Safety [FS]:

Since $Fic > 0.55 F_y$ and $Fic < F_y$

$$= 2.407 - 0.741 \cdot Fic / F_y$$

$$= 2.407 - 0.741 \cdot 129.53/234.4$$

$$= 1.998$$

Required Moment of Inertia for the Ring/Shell Junction, Shear [Irs]:

$$= 0.184 \cdot C_v \cdot M_s 0.8 \cdot t^3 \cdot L_s \quad [\text{eq 6-5}] \text{ or } [\text{eq 4.4.43}]$$

$$= 0.184 \cdot 0.4719 \cdot 10.3490.8 \cdot 18.0^3 \cdot 2625.0002$$

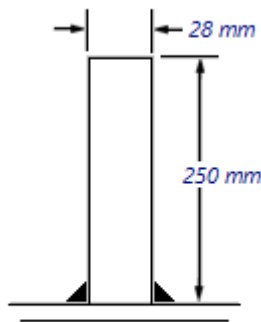
$$= 8621052.000 \text{ mm}^4$$

MDMT Calculations for: Ring R3, Shell to Ring at Ring (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Stiffening Ring Calculations for: Ring R3, SA-516 70 , Bar Ring: 250.0 x 28.0 mm

Material treatment options:



Local Stiffener Geometry Requirements per [6.4] or [Div. 2 4.4.37]:

$$h/t \leq 0.375 \cdot \sqrt{E_y/F_y} \quad [\text{eq 6-6}] \text{ or } [\text{eq 4.4.32}]$$

$$250.0/28.0 \leq 0.375 \cdot \sqrt{196301/234}$$



8.929 <= 10.852 Ok

Effective Length of Shell: 372 mm

	Area (mm ²)	Distance (mm)	Area*Dist
Shell:	5950.362	8.0000	47602.891
Ring :	6999.997	140.9999	986999.062
Total:	12950.358		1034601.938

Centroid of Ring plus Shell: 80 mm

	Inertia	Distance	A*Dist ²
Shell:	126940.922	71.8898	30752338.000
Ring :	36458232.000	-61.1101	26141104.000
Total:	36585176.000		56893440.000

Available Moment of Inertia, Ring + Shell: 0.935E+08 mm⁴

Determine Parameter [Mx]:

$$= L / \sqrt{(Ro \cdot t)}$$

$$= 2499.949 / \sqrt{(3572.0 \cdot 16.0)}$$

$$= 10.457$$

Compute the Factor [Ch]:

$$\text{Since } Mx > 1.5 \text{ and } Mx < 13, Ch = 0.92 / (Mx - 0.579)$$

$$= 0.92 / (10.457 - 0.579)$$

$$= 0.0931$$

Elastic Hoop Compressive Membrane Failure Stress [Fhe]:

$$= 1.6 \cdot Ch \cdot Ey(t / Do)$$

$$= 1.6 \cdot 0.093 \cdot 196301(16.0 / 7144.0)$$

$$= 65.514 \text{ MPa}$$

Ratio of Elastic Hoop Compressive Membrane Failure Stress to Yield [Fr]:

$$= Fhe / Fy$$

$$= 65.5 / 234.4$$

$$= 0.279$$

Predicted Buckling Stress [Fic]:

Engineering Yield to Tensile Stress ratio [R]:

$$= \sigma_{ys} / \sigma_{uts}$$

$$= 234.4 / 482.6$$

$$= 0.486$$

Ratio [AE]:

$$= Fhe / E$$

$$= 65.5 / 196301.4$$

$$= 0.000334$$

Factor [m1]:

$$= (\ln(r) + (\epsilon_{psp} - \epsilon_{sys})) / \ln(\ln(1 + \epsilon_{psp}) / \ln(1 + \epsilon_{sys}))$$

$$= (\ln(0.486) + (0.00002 - 0.002)) / \ln(\ln(1 + 0.00002) / \ln(1 + 0.002))$$

$$= 0.157$$

Curve Fitting Constant for the Elastic Region of the Stress-Strain Curve [A1]:

$$= \sigma_{ys}(1 + \epsilon_{sys}) / \ln(1 + \epsilon_{sys})^{m1}$$

$$= 234.4(1 + 0.002) / \ln(1 + 0.002)^{0.157301}$$

$$= 624.374 \text{ MPa}$$

Curve Fitting Exponent for, Ferritic Steel [m2]:

$$= m2f(m2f2 - R)$$

$$= 0.6(1.0 - 0.486)$$

$$= 0.309$$

Curve Fitting Constant [A2]:

$$= (\sigma_{uts} \cdot \exp(m2)) / (m2^{m2})$$



$$= (482.6 \cdot \exp(0.309)) / (0.309 \cdot 0.309)$$

$$= 944.566 \text{ MPa}$$

Material Parameter for Stress-Strain Curve Model [K]:

$$= 1.5 \cdot R1.5 - 0.5 \cdot R2.5 - R3.5$$

$$= 1.5 \cdot 0.4861.5 - 0.5 \cdot 0.4862.5 - 0.4863.5$$

$$= 0.346$$

Compute the Allowable Stress per 4.4.3

	Fic	H	D1	D2	D3	D4	Tangent Modulus	Iteration Diff
1	241	-1.839	0.313E-04	0.271E-04	0.807E-04	-0.632E-04	0.124E+05	0.192E-01
2	120	-4.651	0.763E-06	0.763E-06	0.171E-04	-0.171E-04	0.151E+06	0.466E-03
3	60	-6.057	0.186E-07	0.186E-07	0.361E-05	-0.361E-05	0.195E+06	0.242E-04
4	90	-5.354	0.163E-06	0.163E-06	0.896E-05	-0.896E-05	0.184E+06	0.157E-03
5	75	-5.706	0.615E-07	0.615E-07	0.596E-05	-0.596E-05	0.192E+06	0.597E-04
6	67	-5.881	0.350E-07	0.350E-07	0.470E-05	-0.470E-05	0.194E+06	0.168E-04
7	64	-5.969	0.257E-07	0.257E-07	0.414E-05	-0.414E-05	0.194E+06	0.388E-05
8	65	-5.925	0.301E-07	0.301E-07	0.442E-05	-0.442E-05	0.194E+06	0.639E-05
9	65	-5.947	0.278E-07	0.278E-07	0.428E-05	-0.428E-05	0.194E+06	0.124E-05
10	64	-5.958	0.268E-07	0.268E-07	0.421E-05	-0.421E-05	0.194E+06	0.132E-05
11	64	-5.953	0.273E-07	0.273E-07	0.424E-05	-0.424E-05	0.194E+06	0.431E-07
12	64	-5.950	0.276E-07	0.276E-07	0.426E-05	-0.426E-05	0.194E+06	0.598E-06
13	64	-5.951	0.274E-07	0.274E-07	0.425E-05	-0.425E-05	0.194E+06	0.278E-06
14	64	-5.952	0.274E-07	0.274E-07	0.425E-05	-0.425E-05	0.194E+06	0.117E-06
15	64	-5.952	0.273E-07	0.273E-07	0.424E-05	-0.424E-05	0.194E+06	0.370E-07
16	64	-5.953	0.273E-07	0.273E-07	0.424E-05	-0.424E-05	0.194E+06	0.304E-08

Factor of Safety [FS]:

$$\text{Since } F_{ic} [64.82] < 0.55 \cdot 234.4, FS = 2$$

Allowable Circumferential Compressive Stress [Fha]:

$$= F_{ic} / FS$$

$$= 64.816 / 2.0$$

$$= 32.408 \text{ MPa}$$

Allowable External Pressure at the given Thickness [Pa]:

$$= 2 \cdot F_{ha} (t / D_o)$$

$$= 2 \cdot 32.408 (16.0 / 7144.0)$$

$$= 0.145 \text{ MPa}$$

Determine the value of [n]:

$$= \sqrt[3]{ (2 \cdot D_o^{1.5} / (3 \cdot L_B \cdot (t))) }$$

$$= \sqrt[3]{ (2 \cdot 7144.01.5 / (3 \cdot 27983.334 \cdot (22.0))) }$$

$$= 2.000 \text{ Note: Rounded to Nearest Integer Value}$$

Because $n < 2$, set $n = 2$

Required Moment of Inertia for the Ring/Shell Junction [Rir]:

$$= 1.5 \cdot F_{he} \cdot L_s \cdot R_c^2 \cdot t / (E \cdot (n^2 - 1)) \text{ [eq 6-1] or [4.4.30]}$$

$$= 1.5 \cdot 66 \cdot 2499.949 \cdot 3697.0^2 \cdot 22.0 / (196301 \cdot (2.0^2 - 1))$$

$$= 91227984.000 \text{ mm}^4$$

Factor of Safety [FS]:

$$\text{Since } F_{ic} [117.3] < 0.55 \cdot 234.4, FS = 2$$

Engineering Yield to Tensile Stress ratio [R]:

$$= \sigma_{ys} / \sigma_{uts}$$

$$= 234.4 / 482.6$$

$$= 0.486$$

Ratio [AE]:

$$= F_{he} / E$$

$$= 165.0 / 196301.4$$

$$= 0.000841$$

Factor [m1]:

$$= (\ln(r) + (\epsilon_{psp} - \epsilon_{sys})) / \ln(\ln(1 + \epsilon_{psp}) / \ln(1 + \epsilon_{sys}))$$



$$= (\ln(0.486) + (0.00002 - 0.002)) / (\ln(1 + 0.00002) / \ln(1 + 0.002))$$

$$= 0.157$$

Curve Fitting Constant for the Elastic Region of the Stress-Strain Curve [A1]:

$$= \sigma_{ys} (1 + \epsilon_{sys}) / \ln(1 + \epsilon_{sys})^{m1}$$

$$= 234.4 (1 + 0.002) / \ln(1 + 0.002)^{0.157301}$$

$$= 624.374 \text{ MPa}$$

Curve Fitting Exponent for, Ferritic Steel [m2]:

$$= m2f(m2f^2 - R)$$

$$= 0.6(1.0 - 0.486)$$

$$= 0.309$$

Curve Fitting Constant [A2]:

$$= (\sigma_{uts} \cdot \exp(m2)) / (m2^{m2})$$

$$= (482.6 \cdot \exp(0.309)) / (0.309^{0.309})$$

$$= 944.566 \text{ MPa}$$

Material Parameter for Stress-Strain Curve Model [K]:

$$= 1.5 \cdot R1.5 - 0.5 \cdot R2.5 - R3.5$$

$$= 1.5 \cdot 0.4861.5 - 0.5 \cdot 0.4862.5 - 0.4863.5$$

$$= 0.346$$

Compute the Allowable Stress per 4.4.3

	Fic	H	D1	D2	D3	D4	Tangent Modulus	Iteration Diff
1	241	-1.839	0.313E-04	0.271E-04	0.807E-04	-0.632E-04	0.124E+05	0.187E-01
2	120	-4.651	0.763E-06	0.763E-06	0.171E-04	-0.171E-04	0.151E+06	0.409E-04
3	180	-3.245	0.670E-05	0.665E-05	0.423E-04	-0.419E-04	0.529E+05	0.258E-02
4	150	-3.948	0.252E-05	0.252E-05	0.281E-04	-0.281E-04	0.981E+05	0.698E-03
5	135	-4.299	0.143E-05	0.143E-05	0.222E-04	-0.222E-04	0.125E+06	0.243E-03
6	128	-4.475	0.106E-05	0.105E-05	0.196E-04	-0.195E-04	0.139E+06	0.848E-04
7	124	-4.563	0.900E-06	0.899E-06	0.183E-04	-0.183E-04	0.145E+06	0.184E-04
8	122	-4.607	0.829E-06	0.829E-06	0.177E-04	-0.177E-04	0.148E+06	0.121E-04
9	123	-4.585	0.864E-06	0.863E-06	0.180E-04	-0.180E-04	0.146E+06	0.292E-05
10	123	-4.596	0.846E-06	0.846E-06	0.178E-04	-0.178E-04	0.147E+06	0.465E-05
11	123	-4.590	0.855E-06	0.854E-06	0.179E-04	-0.179E-04	0.147E+06	0.875E-06
12	123	-4.588	0.859E-06	0.859E-06	0.179E-04	-0.179E-04	0.147E+06	0.102E-05
13	123	-4.589	0.857E-06	0.857E-06	0.179E-04	-0.179E-04	0.147E+06	0.719E-07
14	123	-4.590	0.856E-06	0.856E-06	0.179E-04	-0.179E-04	0.147E+06	0.402E-06
15	123	-4.589	0.856E-06	0.856E-06	0.179E-04	-0.179E-04	0.147E+06	0.165E-06
16	123	-4.589	0.857E-06	0.856E-06	0.179E-04	-0.179E-04	0.147E+06	0.466E-07
17	123	-4.589	0.857E-06	0.857E-06	0.179E-04	-0.179E-04	0.147E+06	0.127E-07

Factor of Safety [FS]:

$$\text{Since } Fic [123.31] < 0.55 \cdot 234.4, FS = 2$$

Required Moment of Inertia for the Ring/Shell Junction, Shear [Irs]:

$$= 0.184 \cdot C_v \cdot M_s 0.8 \cdot t^3 \cdot L_s \text{ [eq 6-5] or [eq 4.4.43]}$$

$$= 0.184 \cdot 0.4692 \cdot 10.4570.8 \cdot 16.0^3 \cdot 2499.9492$$

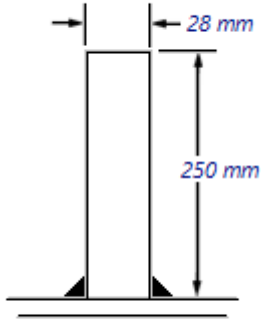
$$= 5781127.500 \text{ mm}^4$$

MDMT Calculations for: Ring R4, Shell to Ring at Ring (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Stiffening Ring Calculations for: Ring R4, SA-516 70 , Bar Ring: 250.0 x 28.0 mm

Material treatment options:



Local Stiffener Geometry Requirements per [6.4] or [Div. 2 4.4.37]:

$$h/t \leq 0.375 \cdot \sqrt{E_y/F_y} \quad [\text{eq 6-6}] \text{ or } [\text{eq 4.4.32}]$$

$$250.0/28.0 \leq 0.375 \cdot \sqrt{196301/234} \quad [\text{eq 6-6}] \text{ or } [\text{eq 4.4.32}]$$

$$8.929 \leq 10.852 \quad \text{Ok}$$

Effective Length of Shell: 372 mm

	Area (mm ²)	Distance (mm)	Area*Dist
Shell:	5950.362	8.0000	47602.891
Ring :	6999.997	140.9999	986999.062
Total:	12950.358		1034601.938

Centroid of Ring plus Shell: 80 mm

	Inertia	Distance	A*Dist ²
Shell:	126940.922	71.8898	30752338.000
Ring :	36458232.000	-61.1101	26141104.000
Total:	36585176.000		56893440.000

Available Moment of Inertia, Ring + Shell: 0.935E+08 mm⁴

Determine Parameter [Mx]:

$$= L / \sqrt{R_o \cdot t}$$

$$= 2499.952 / \sqrt{3572.0 \cdot 16.0}$$

$$= 10.457$$

Compute the Factor [Ch]:

$$\text{Since } M_x > 1.5 \text{ and } M_x < 13, \text{ Ch} = 0.92 / (M_x - 0.579)$$

$$= 0.92 / (10.457 - 0.579)$$

$$= 0.0931$$

Elastic Hoop Compressive Membrane Failure Stress [Fhe]:

$$= 1.6 \cdot \text{Ch} \cdot E_y (t / D_o)$$

$$= 1.6 \cdot 0.093 \cdot 196301 (16.0 / 7144.0)$$

$$= 65.514 \text{ MPa}$$

Ratio of Elastic Hoop Compressive Membrane Failure Stress to Yield [Fr]:

$$= F_{he} / F_y$$

$$= 65.5 / 234.4$$

$$= 0.279$$

Predicted Buckling Stress [Fic]:

Engineering Yield to Tensile Stress ratio [R]:

$$= \sigma_{ys} / \sigma_{uts}$$

$$= 234.4 / 482.6$$



$$= 0.486$$

Ratio [AE]:

$$= F_{he} / E$$

$$= 65.5 / 196301.4$$

$$= 0.000334$$

Factor [m1]:

$$= (\ln(r) + (\epsilon_{ps} - \epsilon_{sys})) / \ln(\ln(1 + \epsilon_{ps}) / \ln(1 + \epsilon_{sys}))$$

$$= (\ln(0.486) + (0.00002 - 0.002)) / \ln(\ln(1 + 0.00002) / \ln(1 + 0.002))$$

$$= 0.157$$

Curve Fitting Constant for the Elastic Region of the Stress-Strain Curve [A1]:

$$= \sigma_{ys} (1 + \epsilon_{sys}) / \ln(1 + \epsilon_{sys})^{m1}$$

$$= 234.4 (1 + 0.002) / \ln(1 + 0.002)^{0.157301}$$

$$= 624.374 \text{ MPa}$$

Curve Fitting Exponent for, Ferritic Steel [m2]:

$$= m2f / (m2f^2 - R)$$

$$= 0.6 (1.0 - 0.486)$$

$$= 0.309$$

Curve Fitting Constant [A2]:

$$= (\sigma_{uts} \cdot \exp(m2)) / (m2^{m2})$$

$$= (482.6 \cdot \exp(0.309)) / (0.309^{0.309})$$

$$= 944.566 \text{ MPa}$$

Material Parameter for Stress-Strain Curve Model [K]:

$$= 1.5 \cdot R1.5 - 0.5 \cdot R2.5 - R3.5$$

$$= 1.5 \cdot 0.486^{1.5} - 0.5 \cdot 0.486^{2.5} - 0.486^{3.5}$$

$$= 0.346$$

Compute the Allowable Stress per 4.4.3

	Fic	H	D1	D2	D3	D4	Tangent Modulus	Iteration Diff
1	241	-1.839	0.313E-04	0.271E-04	0.807E-04	-0.632E-04	0.124E+05	0.192E-01
2	120	-4.651	0.763E-06	0.763E-06	0.171E-04	-0.171E-04	0.151E+06	0.466E-03
3	60	-6.057	0.186E-07	0.186E-07	0.361E-05	-0.361E-05	0.195E+06	0.241E-04
4	90	-5.354	0.163E-06	0.163E-06	0.896E-05	-0.896E-05	0.184E+06	0.157E-03
5	75	-5.706	0.615E-07	0.615E-07	0.596E-05	-0.596E-05	0.192E+06	0.597E-04
6	67	-5.881	0.350E-07	0.350E-07	0.470E-05	-0.470E-05	0.194E+06	0.168E-04
7	64	-5.969	0.257E-07	0.257E-07	0.414E-05	-0.414E-05	0.194E+06	0.388E-05
8	65	-5.925	0.301E-07	0.301E-07	0.442E-05	-0.442E-05	0.194E+06	0.639E-05
9	65	-5.947	0.278E-07	0.278E-07	0.428E-05	-0.428E-05	0.194E+06	0.124E-05
10	64	-5.958	0.268E-07	0.268E-07	0.421E-05	-0.421E-05	0.194E+06	0.132E-05
11	64	-5.953	0.273E-07	0.273E-07	0.424E-05	-0.424E-05	0.194E+06	0.427E-07
12	64	-5.950	0.276E-07	0.276E-07	0.426E-05	-0.426E-05	0.194E+06	0.599E-06
13	64	-5.951	0.274E-07	0.274E-07	0.425E-05	-0.425E-05	0.194E+06	0.278E-06
14	64	-5.952	0.274E-07	0.274E-07	0.425E-05	-0.425E-05	0.194E+06	0.118E-06
15	64	-5.952	0.273E-07	0.273E-07	0.424E-05	-0.424E-05	0.194E+06	0.375E-07
16	64	-5.953	0.273E-07	0.273E-07	0.424E-05	-0.424E-05	0.194E+06	0.260E-08

Factor of Safety [FS]:

$$\text{Since } F_{ic} [64.82] < 0.55 \cdot 234.4, FS = 2$$

Allowable Circumferential Compressive Stress [Fha]:

$$= F_{ic} / FS$$

$$= 64.816 / 2.0$$

$$= 32.408 \text{ MPa}$$

Allowable External Pressure at the given Thickness [Pa]:

$$= 2 \cdot F_{ha} (t / D_o)$$

$$= 2 \cdot 32.408 (16.0 / 7144.0)$$

$$= 0.145 \text{ MPa}$$

Determine the value of [n]:

$$= \sqrt{(2 \cdot D_o^{1.5} / (3 \cdot LB \cdot (t)))}$$

$$= \sqrt{(2 \cdot 7144.0^{1.5} / (3 \cdot 27983.334 \cdot (22.0)))}$$



= 2.000 Note: Rounded to Nearest Integer Value

Because $n < 2$, set $n = 2$

Required Moment of Inertia for the Ring/Shell Junction [Rir]:

$$= 1.5 \cdot F_{he} \cdot L_s \cdot R_c^2 \cdot t / (E \cdot (n^2 - 1)) \text{ [eq 6-1] or [4.4.30]}$$

$$= 1.5 \cdot 66 \cdot 2499.952 \cdot 3697.0^2 \cdot 22.0 / (196301 \cdot (2.0^2 - 1))$$

$$= 91227968.000 \text{ mm}^4$$

Factor of Safety [FS]:

$$\text{Since } F_{ic} [117.3] < 0.55 \cdot 234.4, FS = 2$$

Engineering Yield to Tensile Stress ratio [R]:

$$= \sigma_{ys} / \sigma_{uts}$$

$$= 234.4 / 482.6$$

$$= 0.486$$

Ratio [AE]:

$$= F_{he} / E$$

$$= 165.0 / 196301.4$$

$$= 0.000841$$

Factor [m1]:

$$= (\ln(r) + (\epsilon_{psp} - \epsilon_{sys})) / \ln(\ln(1 + \epsilon_{psp}) / \ln(1 + \epsilon_{sys}))$$

$$= (\ln(0.486) + (0.00002 - 0.002)) / \ln(\ln(1 + 0.00002) / \ln(1 + 0.002))$$

$$= 0.157$$

Curve Fitting Constant for the Elastic Region of the Stress-Strain Curve [A1]:

$$= \sigma_{ys} (1 + \epsilon_{sys}) / \ln(1 + \epsilon_{sys}) m_1$$

$$= 234.4 (1 + 0.002) / \ln(1 + 0.002) \cdot 0.157301$$

$$= 624.374 \text{ MPa}$$

Curve Fitting Exponent for, Ferritic Steel [m2]:

$$= m_2 f(m_2^2 - R)$$

$$= 0.6 (1.0 - 0.486)$$

$$= 0.309$$

Curve Fitting Constant [A2]:

$$= (\sigma_{uts} \cdot \exp(m_2)) / (m_2 m_2)$$

$$= (482.6 \cdot \exp(0.309)) / (0.309 \cdot 0.309)$$

$$= 944.566 \text{ MPa}$$

Material Parameter for Stress-Strain Curve Model [K]:

$$= 1.5 \cdot R_{1.5} - 0.5 \cdot R_{2.5} - R_{3.5}$$

$$= 1.5 \cdot 0.486^{1.5} - 0.5 \cdot 0.486^{2.5} - 0.486^{3.5}$$

$$= 0.346$$

Compute the Allowable Stress per 4.4.3

	Fic	H	D1	D2	D3	D4	Tangent Modulus	Iteration Diff
1	241	-1.839	0.313E-04	0.271E-04	0.807E-04	-0.632E-04	0.124E+05	0.187E-01
2	120	-4.651	0.763E-06	0.763E-06	0.171E-04	-0.171E-04	0.151E+06	0.409E-04
3	180	-3.245	0.670E-05	0.665E-05	0.423E-04	-0.419E-04	0.529E+05	0.258E-02
4	150	-3.948	0.252E-05	0.252E-05	0.281E-04	-0.281E-04	0.981E+05	0.698E-03
5	135	-4.299	0.143E-05	0.143E-05	0.222E-04	-0.222E-04	0.125E+06	0.243E-03
6	128	-4.475	0.106E-05	0.105E-05	0.196E-04	-0.195E-04	0.139E+06	0.848E-04
7	124	-4.563	0.900E-06	0.899E-06	0.183E-04	-0.183E-04	0.145E+06	0.184E-04
8	122	-4.607	0.829E-06	0.829E-06	0.177E-04	-0.177E-04	0.148E+06	0.121E-04
9	123	-4.585	0.864E-06	0.863E-06	0.180E-04	-0.180E-04	0.146E+06	0.292E-05
10	123	-4.596	0.846E-06	0.846E-06	0.178E-04	-0.178E-04	0.147E+06	0.465E-05
11	123	-4.590	0.855E-06	0.854E-06	0.179E-04	-0.179E-04	0.147E+06	0.875E-06
12	123	-4.588	0.859E-06	0.859E-06	0.179E-04	-0.179E-04	0.147E+06	0.102E-05
13	123	-4.589	0.857E-06	0.857E-06	0.179E-04	-0.179E-04	0.147E+06	0.723E-07
14	123	-4.590	0.856E-06	0.856E-06	0.179E-04	-0.179E-04	0.147E+06	0.401E-06
15	123	-4.589	0.856E-06	0.856E-06	0.179E-04	-0.179E-04	0.147E+06	0.165E-06
16	123	-4.589	0.857E-06	0.856E-06	0.179E-04	-0.179E-04	0.147E+06	0.461E-07
17	123	-4.589	0.857E-06	0.857E-06	0.179E-04	-0.179E-04	0.147E+06	0.131E-07



Factor of Safety [FS]:

Since $Fic [123.31] < 0.55 * 234.4$, $FS = 2$

Required Moment of Inertia for the Ring/Shell Junction, Shear [Irs]:

$= 0.184 \cdot Cv \cdot Ms0.8 \cdot t^3 \cdot Ls$ [eq 6-5] or [eq 4.4.43]

$= 0.184 \cdot 0.4692 \cdot 10.4570.8 \cdot 16.0^3 \cdot 2499.9519$

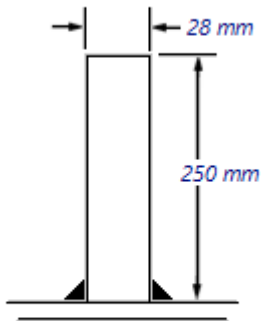
$= 5781136.500 \text{ mm}^4$

MDMT Calculations for: Ring R5, Shell to Ring at Ring (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Stiffening Ring Calculations for: Ring R5, SA-516 70 , Bar Ring: 250.0 x 28.0 mm

Material treatment options:



Local Stiffener Geometry Requirements per [6.4] or [Div. 2 4.4.37]:

$h/t \leq 0.375 \cdot \sqrt{E_y/F_y}$ [eq 6-6] or [eq 4.4.32]

$250.0/28.0 \leq 0.375 \cdot \sqrt{196301/234}$

$8.929 \leq 10.852$ Ok

Effective Length of Shell: 348 mm

	Area (mm ²)	Distance (mm)	Area*Dist
Shell:	4868.933	7.0000	34082.531
Ring :	7000.007	139.0001	973001.750
Total:	11868.939		1007084.250

Centroid of Ring plus Shell: 85 mm

	Inertia	Distance	A*Dist ²
Shell:	79525.820	77.8504	29509072.000
Ring :	36458404.000	-54.1497	20525358.000
Total:	36537932.000		50034432.000

Available Moment of Inertia, Ring + Shell: 0.866E+08 mm⁴

Determine Parameter [Mx]:

$= L / \sqrt{Ro \cdot t}$

$= 2500.0 / \sqrt{3570.0 \cdot 14.0}$

$= 11.183$

Compute the Factor [Ch]:

Since $Mx > 1.5$ and $Mx < 13$, $Ch = 0.92 / (Mx - 0.579)$

$= 0.92 / (11.183 - 0.579)$

$= 0.0868$



Elastic Hoop Compressive Membrane Failure Stress [Fhe]:

$$= 1.6 \cdot Ch \cdot Ey(t / Do)$$

$$= 1.6 \cdot 0.087 \cdot 196301(14.0/7140.0)$$

$$= 53.433 \text{ MPa}$$

Ratio of Elastic Hoop Compressive Membrane Failure Stress to Yield [Fr]:

$$= Fhe / Fy$$

$$= 53.4/234.4$$

$$= 0.228$$

Predicted Buckling Stress [Fic]:

Engineering Yield to Tensile Stress ratio [R]:

$$= \sigma_{ys} / \sigma_{uts}$$

$$= 234.4/482.6$$

$$= 0.486$$

Ratio [AE]:

$$= Fhe / E$$

$$= 53.4/196301.4$$

$$= 0.000272$$

Factor [m1]:

$$= (\ln(r) + (\epsilon_{psp} - \epsilon_{sys})) / \ln(\ln(1 + \epsilon_{psp}) / \ln(1 + \epsilon_{sys}))$$

$$= (\ln(0.486) + (0.00002 - 0.002)) / \ln(\ln(1 + 0.00002) / \ln(1 + 0.002))$$

$$= 0.157$$

Curve Fitting Constant for the Elastic Region of the Stress-Strain Curve [A1]:

$$= \sigma_{ys}(1 + \epsilon_{sys}) / \ln(1 + \epsilon_{sys}) m1$$

$$= 234.4(1 + 0.002) / \ln(1 + 0.002) 0.157301$$

$$= 624.374 \text{ MPa}$$

Curve Fitting Exponent for, Ferritic Steel [m2]:

$$= m2f(m2f2 - R)$$

$$= 0.6(1.0 - 0.486)$$

$$= 0.309$$

Curve Fitting Constant [A2]:

$$= (\sigma_{uts} \cdot \exp(m2)) / (m2 m2)$$

$$= (482.6 \cdot \exp(0.309)) / (0.309 0.309)$$

$$= 944.566 \text{ MPa}$$

Material Parameter for Stress-Strain Curve Model [K]:

$$= 1.5 \cdot R1.5 - 0.5 \cdot R2.5 - R3.5$$

$$= 1.5 \cdot 0.4861.5 - 0.5 \cdot 0.4862.5 - 0.4863.5$$

$$= 0.346$$

Compute the Allowable Stress per 4.4.3

	Fic	H	D1	D2	D3	D4	Tangent Modulus	Iteration Diff
1	241	-1.839	0.313E-04	0.271E-04	0.807E-04	-0.632E-04	0.124E+05	0.192E-01
2	120	-4.651	0.763E-06	0.763E-06	0.171E-04	-0.171E-04	0.151E+06	0.528E-03
3	60	-6.057	0.186E-07	0.186E-07	0.361E-05	-0.361E-05	0.195E+06	0.374E-04
4	30	-6.760	0.454E-09	0.454E-09	0.765E-06	-0.765E-06	0.196E+06	0.119E-03
5	45	-6.409	0.398E-08	0.398E-08	0.190E-05	-0.190E-05	0.196E+06	0.413E-04
6	52	-6.233	0.910E-08	0.910E-08	0.268E-05	-0.268E-05	0.196E+06	0.231E-05
7	56	-6.145	0.132E-07	0.132E-07	0.313E-05	-0.313E-05	0.195E+06	0.174E-04
8	54	-6.189	0.110E-07	0.110E-07	0.290E-05	-0.290E-05	0.195E+06	0.753E-05
9	53	-6.211	0.100E-07	0.100E-07	0.279E-05	-0.279E-05	0.196E+06	0.260E-05
10	53	-6.222	0.954E-08	0.954E-08	0.273E-05	-0.273E-05	0.196E+06	0.144E-06
11	53	-6.227	0.932E-08	0.932E-08	0.271E-05	-0.271E-05	0.196E+06	0.108E-05
12	53	-6.225	0.943E-08	0.943E-08	0.272E-05	-0.272E-05	0.196E+06	0.471E-06
13	53	-6.223	0.949E-08	0.949E-08	0.273E-05	-0.273E-05	0.196E+06	0.163E-06
14	53	-6.223	0.952E-08	0.952E-08	0.273E-05	-0.273E-05	0.196E+06	0.964E-08

Factor of Safety [FS]:

$$\text{Since } Fic [53.23] < 0.55 \cdot 234.4, FS = 2$$



Allowable Circumferential Compressive Stress [Fha]:

$$= F_{ic}/F_S$$

$$= 53.232/2.0$$

$$= 26.616 \text{ MPa}$$

Allowable External Pressure at the given Thickness [Pa]:

$$= 2 \cdot F_{ha}(t / D_o)$$

$$= 2 \cdot 26.616(14.0/7140.0)$$

$$= 0.104 \text{ MPa}$$

Determine the value of [n]:

$$= \sqrt[2]{(2 \cdot D_o^{1.5} / (3 \cdot L_B \cdot (t)))}$$

$$= \sqrt[2]{(2 \cdot 7140.01.5 / (3 \cdot 27983.334 \cdot (20.0)))}$$

$$= 2.000 \text{ Note: Rounded to Nearest Integer Value}$$

Because $n < 2$, set $n = 2$

Required Moment of Inertia for the Ring/Shell Junction [Rir]:

$$= 1.5 \cdot F_{he} \cdot L_s \cdot R_c^2 \cdot t / (E \cdot (n^2 - 1)) \text{ [eq 6-1] or [4.4.30]}$$

$$= 1.5 \cdot 53 \cdot 2500.0 \cdot 3695.0^2 \cdot 20.0 / (196301 \cdot (2.0^2 - 1))$$

$$= 65035676.000 \text{ mm}^4$$

Factor of Safety [FS]:

$$\text{Since } F_{ic} [114.7] < 0.55 \cdot 234.4, FS = 2$$

Engineering Yield to Tensile Stress ratio [R]:

$$= \sigma_{ys} / \sigma_{uts}$$

$$= 234.4/482.6$$

$$= 0.486$$

Ratio [AE]:

$$= F_{he} / E$$

$$= 139.0/196301.4$$

$$= 0.000708$$

Factor [m1]:

$$= (\ln(r) + (\epsilon_{psp} - \epsilon_{sys})) / \ln(\ln(1 + \epsilon_{psp}) / \ln(1 + \epsilon_{sys}))$$

$$= (\ln(0.486) + (0.00002 - 0.002)) / \ln(\ln(1 + 0.00002) / \ln(1 + 0.002))$$

$$= 0.157$$

Curve Fitting Constant for the Elastic Region of the Stress-Strain Curve [A1]:

$$= \sigma_{ys}(1 + \epsilon_{sys}) / \ln(1 + \epsilon_{sys})^{m1}$$

$$= 234.4(1 + 0.002) / \ln(1 + 0.002)^{0.157301}$$

$$= 624.374 \text{ MPa}$$

Curve Fitting Exponent for, Ferritic Steel [m2]:

$$= m2f(m2f2 - R)$$

$$= 0.6(1.0 - 0.486)$$

$$= 0.309$$

Curve Fitting Constant [A2]:

$$= (\sigma_{uts} \cdot \exp(m2)) / (m2^{m2})$$

$$= (482.6 \cdot \exp(0.309)) / (0.309^{0.309})$$

$$= 944.566 \text{ MPa}$$

Material Parameter for Stress-Strain Curve Model [K]:

$$= 1.5 \cdot R1.5 - 0.5 \cdot R2.5 - R3.5$$

$$= 1.5 \cdot 0.4861.5 - 0.5 \cdot 0.4862.5 - 0.4863.5$$

$$= 0.346$$

Compute the Allowable Stress per 4.4.3

	Fic	H	D1	D2	D3	D4	Tangent Modulus	Iteration Diff
1	241	-1.839	0.313E-04	0.271E-04	0.807E-04	-0.632E-04	0.124E+05	0.188E-01
2	120	-4.651	0.763E-06	0.763E-06	0.171E-04	-0.171E-04	0.151E+06	0.914E-04
3	60	-6.057	0.186E-07	0.186E-07	0.361E-05	-0.361E-05	0.195E+06	0.399E-03
4	90	-5.354	0.163E-06	0.163E-06	0.896E-05	-0.896E-05	0.184E+06	0.218E-03
5	105	-5.002	0.373E-06	0.373E-06	0.127E-04	-0.127E-04	0.171E+06	0.914E-04



6	113	-4.827	0.540E-06	0.540E-06	0.148E-04	-0.148E-04	0.162E+06	0.941E-05
7	116	-4.739	0.643E-06	0.643E-06	0.159E-04	-0.159E-04	0.157E+06	0.383E-04
8	115	-4.783	0.590E-06	0.590E-06	0.153E-04	-0.153E-04	0.159E+06	0.138E-04
9	114	-4.805	0.564E-06	0.564E-06	0.150E-04	-0.150E-04	0.161E+06	0.206E-05
10	113	-4.816	0.552E-06	0.552E-06	0.149E-04	-0.149E-04	0.161E+06	0.371E-05
11	113	-4.810	0.558E-06	0.558E-06	0.150E-04	-0.150E-04	0.161E+06	0.837E-06
12	113	-4.807	0.561E-06	0.561E-06	0.150E-04	-0.150E-04	0.161E+06	0.607E-06
13	113	-4.809	0.560E-06	0.560E-06	0.150E-04	-0.150E-04	0.161E+06	0.116E-06
14	113	-4.808	0.560E-06	0.560E-06	0.150E-04	-0.150E-04	0.161E+06	0.245E-06
15	113	-4.808	0.560E-06	0.560E-06	0.150E-04	-0.150E-04	0.161E+06	0.647E-07
16	113	-4.809	0.560E-06	0.560E-06	0.150E-04	-0.150E-04	0.161E+06	0.256E-07
17	113	-4.809	0.560E-06	0.560E-06	0.150E-04	-0.150E-04	0.161E+06	0.195E-07

Factor of Safety [FS]:

Since $Fic [113.9] < 0.55 * 234.4$, $FS = 2$

Required Moment of Inertia for the Ring/Shell Junction, Shear [Irs]:

$= 0.184 \cdot Cv \cdot Ms0.8 \cdot t^3 \cdot Ls$ [eq 6-5] or [eq 4.4.3]

$= 0.184 \cdot 0.4523 \cdot 11.1830.8 \cdot 14.0^3 \cdot 2500.0$

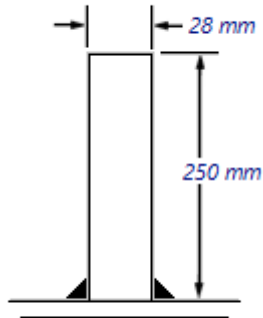
$= 3938964.000 \text{ mm}^4$

MDMT Calculations for: Ring R6, Shell to Ring at Ring (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Stiffening Ring Calculations for: Ring R6, SA-516 70 , Bar Ring: 250.0 x 28.0 mm

Material treatment options:



Local Stiffener Geometry Requirements per [6.4] or [Div. 2 4.4.37]:

$h/t \leq 0.375 \cdot \sqrt{E_y/F_y}$ [eq 6-6] or [eq 4.4.32]

$250.0/28.0 \leq 0.375 \cdot \sqrt{196301/234}$

$8.929 \leq 10.852$ Ok

Effective Length of Shell: 360 mm

	Area (mm ²)	Distance (mm)	Area*Dist
Shell:	5396.787	7.5000	40475.902
Ring :	6999.997	139.9999	979999.062
Total:	12396.783		1020474.938

Centroid of Ring plus Shell: 82 mm

	Inertia	Distance	A*Dist ²
Shell:	101189.664	74.8177	30209544.000
Ring :	36458232.000	-57.6822	23290656.000
Total:	36559424.000		53500200.000



Available Moment of Inertia, Ring + Shell: 0.901E+08 mm⁴

Determine Parameter [Mx]:

$$= L / \sqrt{(Ro \cdot t)}$$

$$= 2499.998 / \sqrt{(3566.0 \cdot 15.0)}$$

$$= 10.809$$

Compute the Factor [Ch]:

$$\text{Since } Mx > 1.5 \text{ and } Mx < 13, Ch = 0.92 / (Mx - 0.579)$$

$$= 0.92 / (10.809 - 0.579)$$

$$= 0.0899$$

Elastic Hoop Compressive Membrane Failure Stress [Fhe]:

$$= 1.6 \cdot Ch \cdot Ey(t / Do)$$

$$= 1.6 \cdot 0.09 \cdot 196301(15.0 / 7132.0)$$

$$= 59.404 \text{ MPa}$$

Ratio of Elastic Hoop Compressive Membrane Failure Stress to Yield [Fr]:

$$= Fhe / Fy$$

$$= 59.4 / 234.4$$

$$= 0.253$$

Predicted Buckling Stress [Fic]:

Engineering Yield to Tensile Stress ratio [R]:

$$= \sigma_{ys} / \sigma_{uts}$$

$$= 234.4 / 482.6$$

$$= 0.486$$

Ratio [AE]:

$$= Fhe / E$$

$$= 59.4 / 196301.4$$

$$= 0.000303$$

Factor [m1]:

$$= (\ln(r) + (\epsilon_{ps} - \epsilon_{sys})) / \ln(\ln(1 + \epsilon_{ps}) / \ln(1 + \epsilon_{sys}))$$

$$= (\ln(0.486) + (0.00002 - 0.002)) / \ln(\ln(1 + 0.00002) / \ln(1 + 0.002))$$

$$= 0.157$$

Curve Fitting Constant for the Elastic Region of the Stress-Strain Curve [A1]:

$$= \sigma_{ys}(1 + \epsilon_{sys}) / \ln(1 + \epsilon_{sys})^{m1}$$

$$= 234.4(1 + 0.002) / \ln(1 + 0.002)^{0.157301}$$

$$= 624.374 \text{ MPa}$$

Curve Fitting Exponent for, Ferritic Steel [m2]:

$$= m2f(m2f2 - R)$$

$$= 0.6(1.0 - 0.486)$$

$$= 0.309$$

Curve Fitting Constant [A2]:

$$= (\sigma_{uts} \cdot \exp(m2)) / (m2^{m2})$$

$$= (482.6 \cdot \exp(0.309)) / (0.309^{0.309})$$

$$= 944.566 \text{ MPa}$$

Material Parameter for Stress-Strain Curve Model [K]:

$$= 1.5 \cdot R1.5 - 0.5 \cdot R2.5 - R3.5$$

$$= 1.5 \cdot 0.4861.5 - 0.5 \cdot 0.4862.5 - 0.4863.5$$

$$= 0.346$$

Compute the Allowable Stress per 4.4.3

	Fic	H	D1	D2	D3	D4	Tangent Modulus	Iteration Diff
1	241	-1.839	0.313E-04	0.271E-04	0.807E-04	-0.632E-04	0.124E+05	0.192E-01
2	120	-4.651	0.763E-06	0.763E-06	0.171E-04	-0.171E-04	0.151E+06	0.497E-03
3	60	-6.057	0.186E-07	0.186E-07	0.361E-05	-0.361E-05	0.195E+06	0.697E-05
4	30	-6.760	0.454E-09	0.454E-09	0.765E-06	-0.765E-06	0.196E+06	0.149E-03
5	45	-6.409	0.398E-08	0.398E-08	0.190E-05	-0.190E-05	0.196E+06	0.718E-04



6	52	-6.233	0.910E-08	0.910E-08	0.268E-05	-0.268E-05	0.196E+06	0.327E-04
7	56	-6.145	0.132E-07	0.132E-07	0.313E-05	-0.313E-05	0.195E+06	0.130E-04
8	58	-6.101	0.157E-07	0.157E-07	0.336E-05	-0.336E-05	0.195E+06	0.304E-05
9	59	-6.079	0.171E-07	0.171E-07	0.349E-05	-0.349E-05	0.195E+06	0.196E-05
10	58	-6.090	0.164E-07	0.164E-07	0.343E-05	-0.343E-05	0.195E+06	0.545E-06
11	59	-6.085	0.167E-07	0.167E-07	0.346E-05	-0.346E-05	0.195E+06	0.705E-06
12	59	-6.087	0.166E-07	0.166E-07	0.344E-05	-0.344E-05	0.195E+06	0.796E-07
13	58	-6.089	0.165E-07	0.165E-07	0.343E-05	-0.343E-05	0.195E+06	0.233E-06
14	59	-6.088	0.165E-07	0.165E-07	0.344E-05	-0.344E-05	0.195E+06	0.766E-07
15	59	-6.088	0.165E-07	0.165E-07	0.344E-05	-0.344E-05	0.195E+06	0.149E-08

Factor of Safety [FS]:

Since $Fic [59.02] < 0.55 * 234.4$, $FS = 2$

Allowable Circumferential Compressive Stress [Fha]:

= Fic/FS

= $59.02/2.0$

= 29.510 MPa

Allowable External Pressure at the given Thickness [Pa]:

= $2 * Fha (t / Do)$

= $2 * 29.51 (15.0/7132.0)$

= 0.124 MPa

Determine the value of [n]:

= $\sqrt[3]{(2 * Do^{1.5} / (3 * LB * (t)))}$

= $\sqrt[3]{(2 * 7132.01.5 / (3 * 27983.334 * (16.0)))}$

= 2.000 Note: Rounded to Nearest Integer Value

Because $n < 2$, set $n = 2$

Required Moment of Inertia for the Ring/Shell Junction [Rir]:

= $1.5 * Fhe * Ls * Rc^2 * t / (E * (n^2 - 1))$ [eq 6-1] or [4.4.30]

= $1.5 * 59 * 2499.998 * 3691.0^2 * 16.0 / (196301 * (2.0^2 - 1))$

= 77300416.000 mm⁴

Factor of Safety [FS]:

Since $Fic [116.01] < 0.55 * 234.4$, $FS = 2$

Engineering Yield to Tensile Stress ratio [R]:

= $\sigma_{ys} / \sigma_{uts}$

= $234.4/482.6$

= 0.486

Ratio [AE]:

= Fhe / E

= $152.2/196301.4$

= 0.000775

Factor [m1]:

= $(\ln(r) + (\epsilon_{psp} - \epsilon_{sys})) / \ln(\ln(1 + \epsilon_{psp}) / \ln(1 + \epsilon_{sys}))$

= $(\ln(0.486) + (0.00002 - 0.002)) / \ln(\ln(1 + 0.00002) / \ln(1 + 0.002))$

= 0.157

Curve Fitting Constant for the Elastic Region of the Stress-Strain Curve [A1]:

= $\sigma_{ys} (1 + \epsilon_{sys}) / \ln(1 + \epsilon_{sys})^{m1}$

= $234.4 (1 + 0.002) / \ln(1 + 0.002)^{0.157301}$

= 624.374 MPa

Curve Fitting Exponent for, Ferritic Steel [m2]:

= $m2f(m2f2 - R)$

= $0.6 (1.0 - 0.486)$

= 0.309

Curve Fitting Constant [A2]:

= $(\sigma_{uts} * \exp(m2)) / (m2^{m2})$

= $(482.6 * \exp(0.309)) / (0.309^{0.309})$

= 944.566 MPa

Material Parameter for Stress-Strain Curve Model [K]:



$$\begin{aligned}
 &= 1.5 \cdot R1.5 - 0.5 \cdot R2.5 - R3.5 \\
 &= 1.5 \cdot 0.4861.5 - 0.5 \cdot 0.4862.5 - 0.4863.5 \\
 &= 0.346
 \end{aligned}$$

Compute the Allowable Stress per 4.4.3

	Fic	H	D1	D2	D3	D4	Tangent Modulus	Iteration Diff
1	241	-1.839	0.313E-04	0.271E-04	0.807E-04	-0.632E-04	0.124E+05	0.187E-01
2	120	-4.651	0.763E-06	0.763E-06	0.171E-04	-0.171E-04	0.151E+06	0.245E-04
3	60	-6.057	0.186E-07	0.186E-07	0.361E-05	-0.361E-05	0.195E+06	0.466E-03
4	90	-5.354	0.163E-06	0.163E-06	0.896E-05	-0.896E-05	0.184E+06	0.285E-03
5	105	-5.002	0.373E-06	0.373E-06	0.127E-04	-0.127E-04	0.171E+06	0.158E-03
6	113	-4.827	0.540E-06	0.540E-06	0.148E-04	-0.148E-04	0.162E+06	0.763E-04
7	116	-4.739	0.643E-06	0.643E-06	0.159E-04	-0.159E-04	0.157E+06	0.286E-04
8	118	-4.695	0.701E-06	0.701E-06	0.165E-04	-0.165E-04	0.154E+06	0.273E-05
9	119	-4.673	0.731E-06	0.731E-06	0.168E-04	-0.168E-04	0.152E+06	0.107E-04
10	119	-4.684	0.716E-06	0.716E-06	0.166E-04	-0.166E-04	0.153E+06	0.395E-05
11	119	-4.689	0.709E-06	0.708E-06	0.166E-04	-0.165E-04	0.153E+06	0.597E-06
12	118	-4.692	0.705E-06	0.705E-06	0.165E-04	-0.165E-04	0.154E+06	0.107E-05
13	118	-4.691	0.707E-06	0.706E-06	0.165E-04	-0.165E-04	0.153E+06	0.237E-06
14	118	-4.690	0.708E-06	0.707E-06	0.165E-04	-0.165E-04	0.153E+06	0.180E-06
15	118	-4.690	0.707E-06	0.707E-06	0.165E-04	-0.165E-04	0.153E+06	0.288E-07
16	118	-4.690	0.707E-06	0.707E-06	0.165E-04	-0.165E-04	0.153E+06	0.754E-07
17	118	-4.690	0.707E-06	0.707E-06	0.165E-04	-0.165E-04	0.153E+06	0.233E-07

Factor of Safety [FS]:

Since Fic [118.97] < 0.55 * 234.4, FS = 2

Required Moment of Inertia for the Ring/Shell Junction, Shear [Irs]:

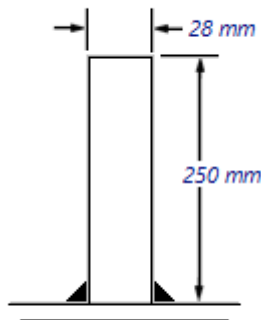
$$\begin{aligned}
 &= 0.184 \cdot Cv \cdot Ms0.8 \cdot t^3 \cdot Ls \quad [\text{eq 6-5}] \text{ or } [\text{eq 4.4.43}] \\
 &= 0.184 \cdot 0.4607 \cdot 10.8090.8 \cdot 15.0^3 \cdot 2499.998 \\
 &= 4803135.500 \text{ mm}^4
 \end{aligned}$$

MDMT Calculations for: Ring R7, Shell to Ring at Ring (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Stiffening Ring Calculations for: Ring R7, SA-516 70 , Bar Ring: 250.0 x 28.0 mm

Material treatment options:



Local Stiffener Geometry Requirements per [6.4] or [Div. 2 4.4.37]:

$$\begin{aligned}
 h/t &\leq 0.375 \cdot \sqrt{E_y/F_y} \quad [\text{eq 6-6}] \text{ or } [\text{eq 4.4.32}] \\
 250.0/28.0 &\leq 0.375 \cdot \sqrt{196301/234} \\
 8.929 &\leq 10.852 \quad \text{Ok}
 \end{aligned}$$

Effective Length of Shell:

360 mm



	Area (mm ²)	Distance (mm)	Area*Dist
Shell:	5396.787	7.5000	40475.902
Ring :	6999.997	139.9999	979999.062
Total:	12396.783		1020474.938

Centroid of Ring plus Shell: 82 mm

	Inertia	Distance	A*Dist ²
Shell:	101189.664	74.8177	30209544.000
Ring :	36458232.000	-57.6822	23290656.000
Total:	36559424.000		53500200.000

Available Moment of Inertia, Ring + Shell: 0.901E+08 mm⁴

Determine Parameter [Mx]:

$$= L / \sqrt{(Ro \cdot t)}$$

$$= 2500.006 / \sqrt{(3566.0 \cdot 15.0)}$$

$$= 10.809$$

Compute the Factor [Ch]:

$$\text{Since } Mx > 1.5 \text{ and } Mx < 13, Ch = 0.92 / (Mx - 0.579)$$

$$= 0.92 / (10.809 - 0.579)$$

$$= 0.0899$$

Elastic Hoop Compressive Membrane Failure Stress [Fhe]:

$$= 1.6 \cdot Ch \cdot Ey(t / Do)$$

$$= 1.6 \cdot 0.09 \cdot 196301(15.0 / 7132.0)$$

$$= 59.404 \text{ MPa}$$

Ratio of Elastic Hoop Compressive Membrane Failure Stress to Yield [Fr]:

$$= Fhe / Fy$$

$$= 59.4 / 234.4$$

$$= 0.253$$

Predicted Buckling Stress [Fic]:

Engineering Yield to Tensile Stress ratio [R]:

$$= \sigma_{ys} / \sigma_{uts}$$

$$= 234.4 / 482.6$$

$$= 0.486$$

Ratio [AE]:

$$= Fhe / E$$

$$= 59.4 / 196301.4$$

$$= 0.000303$$

Factor [m1]:

$$= (\ln(r) + (\epsilon_{psp} - \epsilon_{sys})) / \ln(\ln(1 + \epsilon_{psp}) / \ln(1 + \epsilon_{sys}))$$

$$= (\ln(0.486) + (0.00002 - 0.002)) / \ln(\ln(1 + 0.00002) / \ln(1 + 0.002))$$

$$= 0.157$$

Curve Fitting Constant for the Elastic Region of the Stress-Strain Curve [A1]:

$$= \sigma_{ys}(1 + \epsilon_{sys}) / \ln(1 + \epsilon_{sys})^{m1}$$

$$= 234.4(1 + 0.002) / \ln(1 + 0.002)^{0.157301}$$

$$= 624.374 \text{ MPa}$$

Curve Fitting Exponent for, Ferritic Steel [m2]:

$$= m2f(m2f2 - R)$$

$$= 0.6(1.0 - 0.486)$$

$$= 0.309$$

Curve Fitting Constant [A2]:

$$= (\sigma_{uts} \cdot \exp(m2)) / (m2^{m2})$$

$$= (482.6 \cdot \exp(0.309)) / (0.309^{0.309})$$

$$= 944.566 \text{ MPa}$$

Material Parameter for Stress-Strain Curve Model [K]:



$$= 1.5 \cdot R1.5 - 0.5 \cdot R2.5 - R3.5$$

$$= 1.5 \cdot 0.4861.5 - 0.5 \cdot 0.4862.5 - 0.4863.5$$

$$= 0.346$$

Compute the Allowable Stress per 4.4.3

	Fic	H	D1	D2	D3	D4	Tangent Modulus	Iteration Diff
1	241	-1.839	0.313E-04	0.271E-04	0.807E-04	-0.632E-04	0.124E+05	0.192E-01
2	120	-4.651	0.763E-06	0.763E-06	0.171E-04	-0.171E-04	0.151E+06	0.497E-03
3	60	-6.057	0.186E-07	0.186E-07	0.361E-05	-0.361E-05	0.195E+06	0.697E-05
4	30	-6.760	0.454E-09	0.454E-09	0.765E-06	-0.765E-06	0.196E+06	0.149E-03
5	45	-6.409	0.398E-08	0.398E-08	0.190E-05	-0.190E-05	0.196E+06	0.717E-04
6	52	-6.233	0.910E-08	0.910E-08	0.268E-05	-0.268E-05	0.196E+06	0.327E-04
7	56	-6.145	0.132E-07	0.132E-07	0.313E-05	-0.313E-05	0.195E+06	0.130E-04
8	58	-6.101	0.157E-07	0.157E-07	0.336E-05	-0.336E-05	0.195E+06	0.304E-05
9	59	-6.079	0.171E-07	0.171E-07	0.349E-05	-0.349E-05	0.195E+06	0.196E-05
10	58	-6.090	0.164E-07	0.164E-07	0.343E-05	-0.343E-05	0.195E+06	0.544E-06
11	59	-6.085	0.167E-07	0.167E-07	0.346E-05	-0.346E-05	0.195E+06	0.706E-06
12	59	-6.087	0.166E-07	0.166E-07	0.344E-05	-0.344E-05	0.195E+06	0.806E-07
13	58	-6.089	0.165E-07	0.165E-07	0.343E-05	-0.343E-05	0.195E+06	0.232E-06
14	59	-6.088	0.165E-07	0.165E-07	0.344E-05	-0.344E-05	0.195E+06	0.757E-07
15	59	-6.088	0.165E-07	0.165E-07	0.344E-05	-0.344E-05	0.195E+06	0.245E-08

Factor of Safety [FS]:

$$\text{Since } F_{ic} [59.02] < 0.55 \cdot 234.4, FS = 2$$

Allowable Circumferential Compressive Stress [Fha]:

$$= F_{ic}/FS$$

$$= 59.02/2.0$$

$$= 29.510 \text{ MPa}$$

Allowable External Pressure at the given Thickness [Pa]:

$$= 2 \cdot F_{ha}(t / D_o)$$

$$= 2 \cdot 29.51(15.0/7132.0)$$

$$= 0.124 \text{ MPa}$$

Determine the value of [n]:

$$= \sqrt[3]{(2 \cdot D_o^{1.5} / (3 \cdot L_B \cdot (t)))}$$

$$= \sqrt[3]{(2 \cdot 7132.01.5 / (3 \cdot 27983.334 \cdot (16.0)))}$$

$$= 2.000 \text{ Note: Rounded to Nearest Integer Value}$$

Because $n < 2$, set $n = 2$

Required Moment of Inertia for the Ring/Shell Junction [Rir]:

$$= 1.5 \cdot F_{he} \cdot L_s \cdot R_c^2 \cdot t / (E \cdot (n^2 - 1)) \text{ [eq 6-1] or [4.4.30]}$$

$$= 1.5 \cdot 59 \cdot 2500.006 \cdot 3691.0^2 \cdot 16.0 / (196301 \cdot (2.0^2 - 1))$$

$$= 77300392.000 \text{ mm}^4$$

Factor of Safety [FS]:

$$\text{Since } F_{ic} [116.01] < 0.55 \cdot 234.4, FS = 2$$

Engineering Yield to Tensile Stress ratio [R]:

$$= \sigma_{ys} / \sigma_{uts}$$

$$= 234.4/482.6$$

$$= 0.486$$

Ratio [AE]:

$$= F_{he} / E$$

$$= 152.2/196301.4$$

$$= 0.000775$$

Factor [m1]:

$$= (\ln(r) + (\epsilon_{psp} - \epsilon_{sys})) / \ln(\ln(1 + \epsilon_{psp}) / \ln(1 + \epsilon_{sys}))$$

$$= (\ln(0.486) + (0.00002 - 0.002)) / \ln(\ln(1 + 0.00002) / \ln(1 + 0.002))$$

$$= 0.157$$

Curve Fitting Constant for the Elastic Region of the Stress-Strain Curve [A1]:

$$= \sigma_{ys}(1 + \epsilon_{psys}) / \ln(1 + \epsilon_{psys}) m1$$



$$= 234.4(1 + 0.002) / \ln(1 + 0.002) \cdot 0.157301$$

$$= 624.374 \text{ MPa}$$

Curve Fitting Exponent for, Ferritic Steel [m2]:

$$= m2f(m2f2 - R)$$

$$= 0.6(1.0 - 0.486)$$

$$= 0.309$$

Curve Fitting Constant [A2]:

$$= (\sigma_{uts} \cdot \exp(m2)) / (m2 \cdot m2)$$

$$= (482.6 \cdot \exp(0.309)) / (0.309 \cdot 0.309)$$

$$= 944.566 \text{ MPa}$$

Material Parameter for Stress-Strain Curve Model [K]:

$$= 1.5 \cdot R1.5 - 0.5 \cdot R2.5 - R3.5$$

$$= 1.5 \cdot 0.4861.5 - 0.5 \cdot 0.4862.5 - 0.4863.5$$

$$= 0.346$$

Compute the Allowable Stress per 4.4.3

	Fic	H	D1	D2	D3	D4	Tangent Modulus	Iteration Diff
1	241	-1.839	0.313E-04	0.271E-04	0.807E-04	-0.632E-04	0.124E+05	0.187E-01
2	120	-4.651	0.763E-06	0.763E-06	0.171E-04	-0.171E-04	0.151E+06	0.245E-04
3	60	-6.057	0.186E-07	0.186E-07	0.361E-05	-0.361E-05	0.195E+06	0.466E-03
4	90	-5.354	0.163E-06	0.163E-06	0.896E-05	-0.896E-05	0.184E+06	0.285E-03
5	105	-5.002	0.373E-06	0.373E-06	0.127E-04	-0.127E-04	0.171E+06	0.158E-03
6	113	-4.827	0.540E-06	0.540E-06	0.148E-04	-0.148E-04	0.162E+06	0.763E-04
7	116	-4.739	0.643E-06	0.643E-06	0.159E-04	-0.159E-04	0.157E+06	0.286E-04
8	118	-4.695	0.701E-06	0.701E-06	0.165E-04	-0.165E-04	0.154E+06	0.273E-05
9	119	-4.673	0.731E-06	0.731E-06	0.168E-04	-0.168E-04	0.152E+06	0.107E-04
10	119	-4.684	0.716E-06	0.716E-06	0.166E-04	-0.166E-04	0.153E+06	0.395E-05
11	119	-4.689	0.709E-06	0.708E-06	0.166E-04	-0.165E-04	0.153E+06	0.598E-06
12	118	-4.692	0.705E-06	0.705E-06	0.165E-04	-0.165E-04	0.154E+06	0.107E-05
13	118	-4.691	0.707E-06	0.706E-06	0.165E-04	-0.165E-04	0.153E+06	0.236E-06
14	118	-4.690	0.708E-06	0.707E-06	0.165E-04	-0.165E-04	0.153E+06	0.181E-06
15	118	-4.690	0.707E-06	0.707E-06	0.165E-04	-0.165E-04	0.153E+06	0.276E-07
16	118	-4.690	0.707E-06	0.707E-06	0.165E-04	-0.165E-04	0.153E+06	0.767E-07
17	118	-4.690	0.707E-06	0.707E-06	0.165E-04	-0.165E-04	0.153E+06	0.245E-07

Factor of Safety [FS]:

$$\text{Since } Fic[118.97] < 0.55 \cdot 234.4, FS = 2$$

Required Moment of Inertia for the Ring/Shell Junction, Shear [Irs]:

$$= 0.184 \cdot Cv \cdot Ms0.8 \cdot t^3 \cdot Ls \quad [\text{eq 6-5}] \text{ or } [\text{eq 4.4.43}]$$

$$= 0.184 \cdot 0.4607 \cdot 10.8090.8 \cdot 15.0^3 \cdot 2500.0056$$

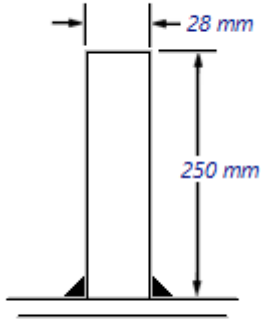
$$= 4803153.500 \text{ mm}^4$$

MDMT Calculations for: Ring R8, Shell to Ring at Ring (Impact tested) :

[Note: This Element/Detail was specified as being Impact Tested.](#)

Stiffening Ring Calculations for: Ring R8, SA-516 70 , Bar Ring: 250.0 x 28.0 mm

Material treatment options:



Local Stiffener Geometry Requirements per [6.4] or [Div. 2 4.4.37]:

$$h/t \leq 0.375 \cdot \sqrt{E_y/F_y} \quad [\text{eq 6-6}] \text{ or } [\text{eq 4.4.32}]$$

$$250.0/28.0 \leq 0.375 \cdot \sqrt{196301/234}$$

$$8.929 \leq 10.852 \quad \text{Ok}$$

Effective Length of Shell: 360 mm

	Area (mm ²)	Distance (mm)	Area*Dist
Shell:	5396.787	7.5000	40475.902
Ring :	6999.997	139.9999	979999.062
Total:	12396.783		1020474.938

Centroid of Ring plus Shell: 82 mm

	Inertia	Distance	A*Dist ²
Shell:	101189.664	74.8177	30209544.000
Ring :	36458232.000	-57.6822	23290656.000
Total:	36559424.000		53500200.000

Available Moment of Inertia, Ring + Shell: 0.901E+08 mm⁴

Determine Parameter [Mx]:

$$= L / \sqrt{R_o \cdot t}$$

$$= 2500.005 / \sqrt{3566.0 \cdot 15.0}$$

$$= 10.809$$

Compute the Factor [Ch]:

$$\text{Since } M_x > 1.5 \text{ and } M_x < 13, Ch = 0.92 / (M_x - 0.579)$$

$$= 0.92 / (10.809 - 0.579)$$

$$= 0.0899$$

Elastic Hoop Compressive Membrane Failure Stress [Fhe]:

$$= 1.6 \cdot Ch \cdot E_y (t / D_o)$$

$$= 1.6 \cdot 0.09 \cdot 196301 (15.0 / 7132.0)$$

$$= 59.404 \text{ MPa}$$

Ratio of Elastic Hoop Compressive Membrane Failure Stress to Yield [Fr]:

$$= Fhe / F_y$$

$$= 59.4/234.4$$

$$= 0.253$$

Predicted Buckling Stress [Fic]:

Engineering Yield to Tensile Stress ratio [R]:

$$= \sigma_{ys} / \sigma_{uts}$$

$$= 234.4/482.6$$



$$= 0.486$$

Ratio [AE]:

$$= F_{he} / E$$

$$= 59.4 / 196301.4$$

$$= 0.000303$$

Factor [m1]:

$$= (\ln(r) + (\epsilon_{ps} - \epsilon_{sys})) / \ln(\ln(1 + \epsilon_{ps}) / \ln(1 + \epsilon_{sys}))$$

$$= (\ln(0.486) + (0.00002 - 0.002)) / \ln(\ln(1 + 0.00002) / \ln(1 + 0.002))$$

$$= 0.157$$

Curve Fitting Constant for the Elastic Region of the Stress-Strain Curve [A1]:

$$= \sigma_{ys} (1 + \epsilon_{sys}) / \ln(1 + \epsilon_{sys})^{m1}$$

$$= 234.4 (1 + 0.002) / \ln(1 + 0.002)^{0.157301}$$

$$= 624.374 \text{ MPa}$$

Curve Fitting Exponent for, Ferritic Steel [m2]:

$$= m2f / (m2f^2 - R)$$

$$= 0.6 (1.0 - 0.486)$$

$$= 0.309$$

Curve Fitting Constant [A2]:

$$= (\sigma_{uts} \cdot \exp(m2)) / (m2^{m2})$$

$$= (482.6 \cdot \exp(0.309)) / (0.309^{0.309})$$

$$= 944.566 \text{ MPa}$$

Material Parameter for Stress-Strain Curve Model [K]:

$$= 1.5 \cdot R1.5 - 0.5 \cdot R2.5 - R3.5$$

$$= 1.5 \cdot 0.486^{1.5} - 0.5 \cdot 0.486^{2.5} - 0.486^{3.5}$$

$$= 0.346$$

Compute the Allowable Stress per 4.4.3

	Fic	H	D1	D2	D3	D4	Tangent Modulus	Iteration Diff
1	241	-1.839	0.313E-04	0.271E-04	0.807E-04	-0.632E-04	0.124E+05	0.192E-01
2	120	-4.651	0.763E-06	0.763E-06	0.171E-04	-0.171E-04	0.151E+06	0.497E-03
3	60	-6.057	0.186E-07	0.186E-07	0.361E-05	-0.361E-05	0.195E+06	0.697E-05
4	30	-6.760	0.454E-09	0.454E-09	0.765E-06	-0.765E-06	0.196E+06	0.149E-03
5	45	-6.409	0.398E-08	0.398E-08	0.190E-05	-0.190E-05	0.196E+06	0.717E-04
6	52	-6.233	0.910E-08	0.910E-08	0.268E-05	-0.268E-05	0.196E+06	0.327E-04
7	56	-6.145	0.132E-07	0.132E-07	0.313E-05	-0.313E-05	0.195E+06	0.130E-04
8	58	-6.101	0.157E-07	0.157E-07	0.336E-05	-0.336E-05	0.195E+06	0.304E-05
9	59	-6.079	0.171E-07	0.171E-07	0.349E-05	-0.349E-05	0.195E+06	0.196E-05
10	58	-6.090	0.164E-07	0.164E-07	0.343E-05	-0.343E-05	0.195E+06	0.545E-06
11	59	-6.085	0.167E-07	0.167E-07	0.346E-05	-0.346E-05	0.195E+06	0.706E-06
12	59	-6.087	0.166E-07	0.166E-07	0.344E-05	-0.344E-05	0.195E+06	0.805E-07
13	58	-6.089	0.165E-07	0.165E-07	0.343E-05	-0.343E-05	0.195E+06	0.232E-06
14	59	-6.088	0.165E-07	0.165E-07	0.344E-05	-0.344E-05	0.195E+06	0.758E-07
15	59	-6.088	0.165E-07	0.165E-07	0.344E-05	-0.344E-05	0.195E+06	0.237E-08

Factor of Safety [FS]:

$$\text{Since } F_{ic} [59.02] < 0.55 \cdot 234.4, FS = 2$$

Allowable Circumferential Compressive Stress [Fha]:

$$= F_{ic} / FS$$

$$= 59.02 / 2.0$$

$$= 29.510 \text{ MPa}$$

Allowable External Pressure at the given Thickness [Pa]:

$$= 2 \cdot F_{ha} (t / D_o)$$

$$= 2 \cdot 29.51 (15.0 / 7132.0)$$

$$= 0.124 \text{ MPa}$$

Determine the value of [n]:

$$= \sqrt[3]{2 \cdot D_o^{1.5} / (3 \cdot L_B \cdot (t))}$$

$$= \sqrt[3]{2 \cdot 7132.0^{1.5} / (3 \cdot 27983.334 \cdot (16.0))}$$

$$= 2.000 \text{ Note: Rounded to Nearest Integer Value}$$



Because $n < 2$, set $n = 2$

Required Moment of Inertia for the Ring/Shell Junction [Rir]:

$$= 1.5 \cdot F_{he} \cdot L_s \cdot R_c^2 \cdot t / (E \cdot (n^2 - 1)) \text{ [eq 6-1] or [4.4.30]}$$

$$= 1.5 \cdot 59 \cdot 2500.005 \cdot 3691.0^2 \cdot 16.0 / (196301 \cdot (2.0^2 - 1))$$

$$= 77300416.000 \text{ mm}^4$$

Factor of Safety [FS]:

$$\text{Since } F_{ic} [116.01] < 0.55 \cdot 234.4, \text{ FS} = 2$$

Engineering Yield to Tensile Stress ratio [R]:

$$= \sigma_{ys} / \sigma_{uts}$$

$$= 234.4 / 482.6$$

$$= 0.486$$

Ratio [AE]:

$$= F_{he} / E$$

$$= 152.2 / 196301.4$$

$$= 0.000775$$

Factor [m1]:

$$= (\ln(r) + (\epsilon_{psp} - \epsilon_{sys})) / (\ln(1 + \epsilon_{psp}) / \ln(1 + \epsilon_{sys}))$$

$$= (\ln(0.486) + (0.00002 - 0.002)) / (\ln(1 + 0.00002) / \ln(1 + 0.002))$$

$$= 0.157$$

Curve Fitting Constant for the Elastic Region of the Stress-Strain Curve [A1]:

$$= \sigma_{ys} (1 + \epsilon_{sys}) / \ln(1 + \epsilon_{sys}) m_1$$

$$= 234.4 (1 + 0.002) / \ln(1 + 0.002) \cdot 0.157301$$

$$= 624.374 \text{ MPa}$$

Curve Fitting Exponent for, Ferritic Steel [m2]:

$$= m_2 f(m_2^2 - R)$$

$$= 0.6 (1.0 - 0.486)$$

$$= 0.309$$

Curve Fitting Constant [A2]:

$$= (\sigma_{uts} \cdot \exp(m_2)) / (m_2 m_2)$$

$$= (482.6 \cdot \exp(0.309)) / (0.309 \cdot 0.309)$$

$$= 944.566 \text{ MPa}$$

Material Parameter for Stress-Strain Curve Model [K]:

$$= 1.5 \cdot R_{1.5} - 0.5 \cdot R_{2.5} - R_{3.5}$$

$$= 1.5 \cdot 0.4861.5 - 0.5 \cdot 0.4862.5 - 0.4863.5$$

$$= 0.346$$

Compute the Allowable Stress per 4.4.3

	Fic	H	D1	D2	D3	D4	Tangent Modulus	Iteration Diff
1	241	-1.839	0.313E-04	0.271E-04	0.807E-04	-0.632E-04	0.124E+05	0.187E-01
2	120	-4.651	0.763E-06	0.763E-06	0.171E-04	-0.171E-04	0.151E+06	0.245E-04
3	60	-6.057	0.186E-07	0.186E-07	0.361E-05	-0.361E-05	0.195E+06	0.466E-03
4	90	-5.354	0.163E-06	0.163E-06	0.896E-05	-0.896E-05	0.184E+06	0.285E-03
5	105	-5.002	0.373E-06	0.373E-06	0.127E-04	-0.127E-04	0.171E+06	0.158E-03
6	113	-4.827	0.540E-06	0.540E-06	0.148E-04	-0.148E-04	0.162E+06	0.763E-04
7	116	-4.739	0.643E-06	0.643E-06	0.159E-04	-0.159E-04	0.157E+06	0.286E-04
8	118	-4.695	0.701E-06	0.701E-06	0.165E-04	-0.165E-04	0.154E+06	0.273E-05
9	119	-4.673	0.731E-06	0.731E-06	0.168E-04	-0.168E-04	0.152E+06	0.107E-04
10	119	-4.684	0.716E-06	0.716E-06	0.166E-04	-0.166E-04	0.153E+06	0.395E-05
11	119	-4.689	0.709E-06	0.708E-06	0.166E-04	-0.165E-04	0.153E+06	0.598E-06
12	118	-4.692	0.705E-06	0.705E-06	0.165E-04	-0.165E-04	0.154E+06	0.107E-05
13	118	-4.691	0.707E-06	0.706E-06	0.165E-04	-0.165E-04	0.153E+06	0.236E-06
14	118	-4.690	0.708E-06	0.707E-06	0.165E-04	-0.165E-04	0.153E+06	0.181E-06
15	118	-4.690	0.707E-06	0.707E-06	0.165E-04	-0.165E-04	0.153E+06	0.277E-07
16	118	-4.690	0.707E-06	0.707E-06	0.165E-04	-0.165E-04	0.153E+06	0.765E-07
17	118	-4.690	0.707E-06	0.707E-06	0.165E-04	-0.165E-04	0.153E+06	0.244E-07

Factor of Safety [FS]:



Since $Fic [118.97] < 0.55 \cdot 234.4$, $FS = 2$

Required Moment of Inertia for the Ring/Shell Junction, Shear [Irs]:

$$= 0.184 \cdot Cv \cdot Ms0.8 \cdot t^3 \cdot Ls \quad [\text{eq 6-5}] \text{ or } [\text{eq 4.4.43}]$$

$$= 0.184 \cdot 0.4607 \cdot 10.8090.8 \cdot 15.0^3 \cdot 2500.0049$$

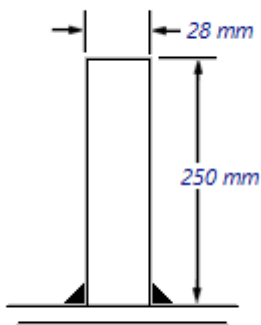
$$= 4803153.000 \text{ mm}^4$$

MDMT Calculations for: Ring R9, Shell to Ring at Ring (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Stiffening Ring Calculations for: Ring R9, SA-516 70 , Bar Ring: 250.0 x 28.0 mm

Material treatment options:



Local Stiffener Geometry Requirements per [6.4] or [Div. 2 4.4.37]:

$$h/t \leq 0.375 \cdot \sqrt{E_y/F_y} \quad [\text{eq 6-6}] \text{ or } [\text{eq 4.4.32}]$$

$$250.0/28.0 \leq 0.375 \cdot \sqrt{196301/234}$$

$$8.929 \leq 10.852 \quad \text{Ok}$$

Effective Length of Shell: 360 mm

	Area (mm ²)	Distance (mm)	Area*Dist
Shell:	5396.787	7.5000	40475.902
Ring :	6999.997	139.9999	979999.062
Total:	12396.783		1020474.938

Centroid of Ring plus Shell: 82 mm

	Inertia	Distance	A*Dist ²
Shell:	101189.664	74.8177	30209544.000
Ring :	36458232.000	-57.6822	23290656.000
Total:	36559424.000		53500200.000

Available Moment of Inertia, Ring + Shell: 0.901E+08 mm⁴

Determine Parameter [Mx]:

$$= L / \sqrt{Ro \cdot t}$$

$$= 2500.02 / \sqrt{3566.0 \cdot 15.0}$$

$$= 10.810$$

Compute the Factor [Ch]:

$$\text{Since } Mx > 1.5 \text{ and } Mx < 13, Ch = 0.92 / (Mx - 0.579)$$

$$= 0.92 / (10.81 - 0.579)$$

$$= 0.0899$$

Elastic Hoop Compressive Membrane Failure Stress [Fhe]:



$$= 1.6 \cdot Ch \cdot Ey(t / Do)$$

$$= 1.6 \cdot 0.09 \cdot 196301(15.0/7132.0)$$

$$= 59.404 \text{ MPa}$$

Ratio of Elastic Hoop Compressive Membrane Failure Stress to Yield [Fr]:

$$= Fhe / Fy$$

$$= 59.4/234.4$$

$$= 0.253$$

Predicted Buckling Stress [Fic]:

Engineering Yield to Tensile Stress ratio [R]:

$$= \sigma_{ys} / \sigma_{uts}$$

$$= 234.4/482.6$$

$$= 0.486$$

Ratio [AE]:

$$= Fhe / E$$

$$= 59.4/196301.4$$

$$= 0.000303$$

Factor [m1]:

$$= (\ln(r) + (\epsilon_{psp} - \epsilon_{sys})) / \ln(\ln(1 + \epsilon_{psp}) / \ln(1 + \epsilon_{sys}))$$

$$= (\ln(0.486) + (0.00002 - 0.002)) / \ln(\ln(1 + 0.00002) / \ln(1 + 0.002))$$

$$= 0.157$$

Curve Fitting Constant for the Elastic Region of the Stress-Strain Curve [A1]:

$$= \sigma_{ys}(1 + \epsilon_{sys}) / \ln(1 + \epsilon_{sys}) m1$$

$$= 234.4(1 + 0.002) / \ln(1 + 0.002) 0.157301$$

$$= 624.374 \text{ MPa}$$

Curve Fitting Exponent for, Ferritic Steel [m2]:

$$= m2f(m2f2 - R)$$

$$= 0.6(1.0 - 0.486)$$

$$= 0.309$$

Curve Fitting Constant [A2]:

$$= (\sigma_{uts} \cdot \exp(m2)) / (m2 m2)$$

$$= (482.6 \cdot \exp(0.309)) / (0.309 0.309)$$

$$= 944.566 \text{ MPa}$$

Material Parameter for Stress-Strain Curve Model [K]:

$$= 1.5 \cdot R1.5 - 0.5 \cdot R2.5 - R3.5$$

$$= 1.5 \cdot 0.4861.5 - 0.5 \cdot 0.4862.5 - 0.4863.5$$

$$= 0.346$$

Compute the Allowable Stress per 4.4.3

	Fic	H	D1	D2	D3	D4	Tangent Modulus	Iteration Diff
1	241	-1.839	0.313E-04	0.271E-04	0.807E-04	-0.632E-04	0.124E+05	0.192E-01
2	120	-4.651	0.763E-06	0.763E-06	0.171E-04	-0.171E-04	0.151E+06	0.497E-03
3	60	-6.057	0.186E-07	0.186E-07	0.361E-05	-0.361E-05	0.195E+06	0.698E-05
4	30	-6.760	0.454E-09	0.454E-09	0.765E-06	-0.765E-06	0.196E+06	0.149E-03
5	45	-6.409	0.398E-08	0.398E-08	0.190E-05	-0.190E-05	0.196E+06	0.717E-04
6	52	-6.233	0.910E-08	0.910E-08	0.268E-05	-0.268E-05	0.196E+06	0.327E-04
7	56	-6.145	0.132E-07	0.132E-07	0.313E-05	-0.313E-05	0.195E+06	0.130E-04
8	58	-6.101	0.157E-07	0.157E-07	0.336E-05	-0.336E-05	0.195E+06	0.304E-05
9	59	-6.079	0.171E-07	0.171E-07	0.349E-05	-0.349E-05	0.195E+06	0.196E-05
10	58	-6.090	0.164E-07	0.164E-07	0.343E-05	-0.343E-05	0.195E+06	0.543E-06
11	59	-6.085	0.167E-07	0.167E-07	0.346E-05	-0.346E-05	0.195E+06	0.708E-06
12	59	-6.087	0.166E-07	0.166E-07	0.344E-05	-0.344E-05	0.195E+06	0.825E-07
13	58	-6.089	0.165E-07	0.165E-07	0.343E-05	-0.343E-05	0.195E+06	0.230E-06
14	59	-6.088	0.165E-07	0.165E-07	0.344E-05	-0.344E-05	0.195E+06	0.738E-07
15	59	-6.088	0.165E-07	0.165E-07	0.344E-05	-0.344E-05	0.195E+06	0.435E-08

Factor of Safety [FS]:

$$\text{Since } Fic [59.02] < 0.55 \cdot 234.4, FS = 2$$



Allowable Circumferential Compressive Stress [Fha]:

$$= F_{ic}/FS$$

$$= 59.02/2.0$$

$$= 29.510 \text{ MPa}$$

Allowable External Pressure at the given Thickness [Pa]:

$$= 2 \cdot F_{ha}(t / D_o)$$

$$= 2 \cdot 29.51(15.0/7132.0)$$

$$= 0.124 \text{ MPa}$$

Determine the value of [n]:

$$= \sqrt[3]{(2 \cdot D_o^{1.5} / (3 \cdot L_B \cdot (t)))}$$

$$= \sqrt[3]{(2 \cdot 7132.01^{1.5} / (3 \cdot 27983.334 \cdot (16.0)))}$$

$$= 2.000 \text{ Note: Rounded to Nearest Integer Value}$$

Because $n < 2$, set $n = 2$

Required Moment of Inertia for the Ring/Shell Junction [Rir]:

$$= 1.5 \cdot F_{he} \cdot L_s \cdot R_c^2 \cdot t / (E \cdot (n^2 - 1)) \text{ [eq 6-1] or [4.4.30]}$$

$$= 1.5 \cdot 59 \cdot 2500.02 \cdot 3691.0^2 \cdot 16.0 / (196301 \cdot (2.0^2 - 1))$$

$$= 77300368.000 \text{ mm}^4$$

Factor of Safety [FS]:

$$\text{Since } F_{ic} [116.01] < 0.55 \cdot 234.4, FS = 2$$

Engineering Yield to Tensile Stress ratio [R]:

$$= \sigma_{ys} / \sigma_{uts}$$

$$= 234.4/482.6$$

$$= 0.486$$

Ratio [AE]:

$$= F_{he} / E$$

$$= 152.2/196301.4$$

$$= 0.000775$$

Factor [m1]:

$$= (\ln(r) + (\epsilon_{psp} - \epsilon_{sys})) / \ln(\ln(1 + \epsilon_{psp}) / \ln(1 + \epsilon_{sys}))$$

$$= (\ln(0.486) + (0.00002 - 0.002)) / \ln(\ln(1 + 0.00002) / \ln(1 + 0.002))$$

$$= 0.157$$

Curve Fitting Constant for the Elastic Region of the Stress-Strain Curve [A1]:

$$= \sigma_{ys}(1 + \epsilon_{sys}) / \ln(1 + \epsilon_{sys})^{m1}$$

$$= 234.4(1 + 0.002) / \ln(1 + 0.002)^{0.157301}$$

$$= 624.374 \text{ MPa}$$

Curve Fitting Exponent for, Ferritic Steel [m2]:

$$= m2f(m2f2 - R)$$

$$= 0.6(1.0 - 0.486)$$

$$= 0.309$$

Curve Fitting Constant [A2]:

$$= (\sigma_{uts} \cdot \exp(m2)) / (m2^{m2})$$

$$= (482.6 \cdot \exp(0.309)) / (0.309^{0.309})$$

$$= 944.566 \text{ MPa}$$

Material Parameter for Stress-Strain Curve Model [K]:

$$= 1.5 \cdot R1.5 - 0.5 \cdot R2.5 - R3.5$$

$$= 1.5 \cdot 0.4861.5 - 0.5 \cdot 0.4862.5 - 0.4863.5$$

$$= 0.346$$

Compute the Allowable Stress per 4.4.3

	Fic	H	D1	D2	D3	D4	Tangent Modulus	Iteration Diff
1	241	-1.839	0.313E-04	0.271E-04	0.807E-04	-0.632E-04	0.124E+05	0.187E-01
2	120	-4.651	0.763E-06	0.763E-06	0.171E-04	-0.171E-04	0.151E+06	0.246E-04
3	60	-6.057	0.186E-07	0.186E-07	0.361E-05	-0.361E-05	0.195E+06	0.466E-03
4	90	-5.354	0.163E-06	0.163E-06	0.896E-05	-0.896E-05	0.184E+06	0.285E-03
5	105	-5.002	0.373E-06	0.373E-06	0.127E-04	-0.127E-04	0.171E+06	0.158E-03



6	113	-4.827	0.540E-06	0.540E-06	0.148E-04	-0.148E-04	0.162E+06	0.763E-04
7	116	-4.739	0.643E-06	0.643E-06	0.159E-04	-0.159E-04	0.157E+06	0.286E-04
8	118	-4.695	0.701E-06	0.701E-06	0.165E-04	-0.165E-04	0.154E+06	0.273E-05
9	119	-4.673	0.731E-06	0.731E-06	0.168E-04	-0.168E-04	0.152E+06	0.107E-04
10	119	-4.684	0.716E-06	0.716E-06	0.166E-04	-0.166E-04	0.153E+06	0.395E-05
11	119	-4.689	0.709E-06	0.708E-06	0.166E-04	-0.165E-04	0.153E+06	0.601E-06
12	118	-4.692	0.705E-06	0.705E-06	0.165E-04	-0.165E-04	0.154E+06	0.107E-05
13	118	-4.691	0.707E-06	0.706E-06	0.165E-04	-0.165E-04	0.153E+06	0.233E-06
14	118	-4.690	0.708E-06	0.707E-06	0.165E-04	-0.165E-04	0.153E+06	0.183E-06
15	118	-4.690	0.707E-06	0.707E-06	0.165E-04	-0.165E-04	0.153E+06	0.250E-07
16	118	-4.690	0.707E-06	0.707E-06	0.165E-04	-0.165E-04	0.153E+06	0.792E-07
17	118	-4.690	0.707E-06	0.707E-06	0.165E-04	-0.165E-04	0.153E+06	0.271E-07

Factor of Safety [FS]:

Since $Fic [118.97] < 0.55 * 234.4$, $FS = 2$

Required Moment of Inertia for the Ring/Shell Junction, Shear [Irs]:

$= 0.184 \cdot Cv \cdot Ms0.8 \cdot t^3 \cdot Ls$ [eq 6-5] or [eq 4.4.3]

$= 0.184 \cdot 0.4607 \cdot 10.810.8 \cdot 15.0^3 \cdot 2500.02$

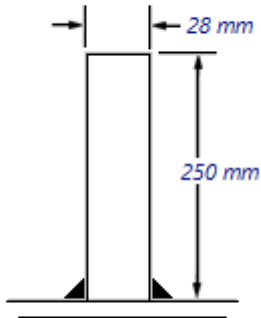
$= 4803188.500 \text{ mm}^4$

MDMT Calculations for: Ring R10, Shell to Ring at Ring (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Stiffening Ring Calculations for: Ring R10, SA-516 70 , Bar Ring: 250.0 x 28.0 mm

Material treatment options:



Local Stiffener Geometry Requirements per [6.4] or [Div. 2 4.4.37]:

$h/t \leq 0.375 \cdot \sqrt{E_y/F_y}$ [eq 6-6] or [eq 4.4.32]

$250.0/28.0 \leq 0.375 \cdot \sqrt{196301/234}$

$8.929 \leq 10.852$ Ok

Effective Length of Shell: 335 mm

	Area (mm ²)	Distance (mm)	Area*Dist
Shell:	4353.029	6.5000	28294.693
Ring :	7000.007	138.0001	966001.875
Total:	11353.036		994296.562

Centroid of Ring plus Shell: 88 mm

	Inertia	Distance	A*Dist ²
Shell:	61305.117	81.0798	28616516.000
Ring :	36458404.000	-50.4203	17795494.000
Total:	36519712.000		46412008.000



Available Moment of Inertia, Ring + Shell: 0.829E+08 mm⁴

Determine Parameter [Mx]:

$$= L / \sqrt{(Ro \cdot t)}$$

$$= 2445.854 / \sqrt{(3564.0 \cdot 13.0)}$$

$$= 11.363$$

Compute the Factor [Ch]:

$$\text{Since } Mx > 1.5 \text{ and } Mx < 13, Ch = 0.92 / (Mx - 0.579)$$

$$= 0.92 / (11.363 - 0.579)$$

$$= 0.0853$$

Elastic Hoop Compressive Membrane Failure Stress [Fhe]:

$$= 1.6 \cdot Ch \cdot Ey(t / Do)$$

$$= 1.6 \cdot 0.085 \cdot 196301(13.0 / 7128.0)$$

$$= 48.869 \text{ MPa}$$

Ratio of Elastic Hoop Compressive Membrane Failure Stress to Yield [Fr]:

$$= Fhe / Fy$$

$$= 48.9 / 234.4$$

$$= 0.208$$

Predicted Buckling Stress [Fic]:

Engineering Yield to Tensile Stress ratio [R]:

$$= \sigma_{ys} / \sigma_{uts}$$

$$= 234.4 / 482.6$$

$$= 0.486$$

Ratio [AE]:

$$= Fhe / E$$

$$= 48.9 / 196301.4$$

$$= 0.000249$$

Factor [m1]:

$$= (\ln(r) + (\epsilon_{ps} - \epsilon_{sys})) / \ln(\ln(1 + \epsilon_{ps}) / \ln(1 + \epsilon_{sys}))$$

$$= (\ln(0.486) + (0.00002 - 0.002)) / \ln(\ln(1 + 0.00002) / \ln(1 + 0.002))$$

$$= 0.157$$

Curve Fitting Constant for the Elastic Region of the Stress-Strain Curve [A1]:

$$= \sigma_{ys}(1 + \epsilon_{sys}) / \ln(1 + \epsilon_{sys})^{m1}$$

$$= 234.4(1 + 0.002) / \ln(1 + 0.002)^{0.157304}$$

$$= 624.376 \text{ MPa}$$

Curve Fitting Exponent for, Ferritic Steel [m2]:

$$= m2f(m2f2 - R)$$

$$= 0.6(1.0 - 0.486)$$

$$= 0.309$$

Curve Fitting Constant [A2]:

$$= (\sigma_{uts} \cdot \exp(m2)) / (m2^{m2})$$

$$= (482.6 \cdot \exp(0.309)) / (0.309^{0.309})$$

$$= 944.570 \text{ MPa}$$

Material Parameter for Stress-Strain Curve Model [K]:

$$= 1.5 \cdot R1.5 - 0.5 \cdot R2.5 - R3.5$$

$$= 1.5 \cdot 0.4861.5 - 0.5 \cdot 0.4862.5 - 0.4863.5$$

$$= 0.346$$

Compute the Allowable Stress per 4.4.3

	Fic	H	D1	D2	D3	D4	Tangent Modulus	Iteration Diff
1	241	-1.839	0.313E-04	0.271E-04	0.807E-04	-0.632E-04	0.124E+05	0.193E-01
2	120	-4.651	0.763E-06	0.763E-06	0.171E-04	-0.171E-04	0.151E+06	0.551E-03
3	60	-6.057	0.186E-07	0.186E-07	0.361E-05	-0.361E-05	0.195E+06	0.606E-04
4	30	-6.760	0.454E-09	0.454E-09	0.765E-06	-0.765E-06	0.196E+06	0.952E-04
5	45	-6.409	0.399E-08	0.399E-08	0.190E-05	-0.190E-05	0.196E+06	0.181E-04



6	52	-6.233	0.910E-08	0.910E-08	0.268E-05	-0.268E-05	0.196E+06	0.209E-04
7	49	-6.321	0.612E-08	0.612E-08	0.227E-05	-0.227E-05	0.196E+06	0.137E-05
8	47	-6.365	0.496E-08	0.496E-08	0.208E-05	-0.208E-05	0.196E+06	0.837E-05
9	48	-6.343	0.552E-08	0.552E-08	0.217E-05	-0.217E-05	0.196E+06	0.350E-05
10	48	-6.332	0.581E-08	0.581E-08	0.222E-05	-0.222E-05	0.196E+06	0.107E-05
11	48	-6.326	0.596E-08	0.596E-08	0.224E-05	-0.224E-05	0.196E+06	0.149E-06
12	48	-6.329	0.589E-08	0.589E-08	0.223E-05	-0.223E-05	0.196E+06	0.460E-06
13	48	-6.328	0.593E-08	0.593E-08	0.224E-05	-0.224E-05	0.196E+06	0.155E-06
14	48	-6.327	0.594E-08	0.594E-08	0.224E-05	-0.224E-05	0.196E+06	0.310E-08

Factor of Safety [FS]:

Since $Fic [48.75] < 0.55 * 234.4$, $FS = 2$

Allowable Circumferential Compressive Stress [Fha]:

= Fic/FS

= $48.754/2.0$

= 24.377 MPa

Allowable External Pressure at the given Thickness [Pa]:

= $2 * Fha (t / Do)$

= $2 * 24.377 (13.0/7128.0)$

= 0.089 MPa

Determine the value of [n]:

= $\sqrt[3]{(2 * Do^{1.5} / (3 * LB * (t)))}$

= $\sqrt[3]{(2 * 7128.0^{1.5} / (3 * 27983.334 * (14.0)))}$

= 2.000 Note: Rounded to Nearest Integer Value

Because $n < 2$, set $n = 2$

Required Moment of Inertia for the Ring/Shell Junction [Rir]:

= $1.5 * Fhe * Ls * Rc^2 * t / (E * (n^2 - 1))$ [eq 6-1] or [4.4.30]

= $1.5 * 49 * 2445.854 * 3689.0^2 * 14.0 / (196301 * (2.0^2 - 1))$

= 53860156.000 mm⁴

Factor of Safety [FS]:

Since $Fic [113.5] < 0.55 * 234.4$, $FS = 2$

Engineering Yield to Tensile Stress ratio [R]:

= $\sigma_{ys} / \sigma_{uts}$

= $234.4/482.6$

= 0.486

Ratio [AE]:

= Fhe / E

= $127.1/196301.4$

= 0.000648

Factor [m1]:

= $(\ln(r) + (\epsilon_{psp} - \epsilon_{sys})) / \ln(\ln(1 + \epsilon_{psp}) / \ln(1 + \epsilon_{sys}))$

= $(\ln(0.486) + (0.00002 - 0.002)) / \ln(\ln(1 + 0.00002) / \ln(1 + 0.002))$

= 0.157

Curve Fitting Constant for the Elastic Region of the Stress-Strain Curve [A1]:

= $\sigma_{ys} (1 + \epsilon_{sys}) / \ln(1 + \epsilon_{sys})^{m1}$

= $234.4 (1 + 0.002) / \ln(1 + 0.002)^{0.157304}$

= 624.376 MPa

Curve Fitting Exponent for, Ferritic Steel [m2]:

= $m2f / (m2f^2 - R)$

= $0.6 (1.0 - 0.486)$

= 0.309

Curve Fitting Constant [A2]:

= $(\sigma_{uts} * \exp(m2)) / (m2^{m2})$

= $(482.6 * \exp(0.309)) / (0.309^{0.309})$

= 944.570 MPa

Material Parameter for Stress-Strain Curve Model [K]:

= $1.5 * R1.5 - 0.5 * R2.5 - R3.5$

$$= 1.5 \cdot 0.4861.5 - 0.5 \cdot 0.4862.5 - 0.4863.5$$

$$= 0.346$$

Compute the Allowable Stress per 4.4.3

	Fic	H	D1	D2	D3	D4	Tangent Modulus	Iteration Diff
1	241	-1.839	0.313E-04	0.271E-04	0.807E-04	-0.632E-04	0.124E+05	0.189E-01
2	120	-4.651	0.763E-06	0.763E-06	0.171E-04	-0.171E-04	0.151E+06	0.152E-03
3	60	-6.057	0.186E-07	0.186E-07	0.361E-05	-0.361E-05	0.195E+06	0.338E-03
4	90	-5.354	0.163E-06	0.163E-06	0.896E-05	-0.896E-05	0.184E+06	0.157E-03
5	105	-5.002	0.373E-06	0.373E-06	0.127E-04	-0.127E-04	0.171E+06	0.306E-04
6	113	-4.827	0.540E-06	0.540E-06	0.148E-04	-0.148E-04	0.162E+06	0.514E-04
7	109	-4.914	0.450E-06	0.450E-06	0.137E-04	-0.137E-04	0.167E+06	0.837E-05
8	107	-4.958	0.410E-06	0.410E-06	0.132E-04	-0.132E-04	0.169E+06	0.116E-04
9	108	-4.936	0.430E-06	0.430E-06	0.134E-04	-0.134E-04	0.168E+06	0.174E-05
10	108	-4.925	0.440E-06	0.440E-06	0.136E-04	-0.136E-04	0.167E+06	0.328E-05
11	108	-4.931	0.435E-06	0.435E-06	0.135E-04	-0.135E-04	0.168E+06	0.766E-06
12	108	-4.934	0.432E-06	0.432E-06	0.135E-04	-0.135E-04	0.168E+06	0.487E-06
13	108	-4.932	0.434E-06	0.434E-06	0.135E-04	-0.135E-04	0.168E+06	0.139E-06
14	108	-4.933	0.433E-06	0.433E-06	0.135E-04	-0.135E-04	0.168E+06	0.174E-06
15	108	-4.933	0.433E-06	0.433E-06	0.135E-04	-0.135E-04	0.168E+06	0.178E-07
16	108	-4.933	0.433E-06	0.433E-06	0.135E-04	-0.135E-04	0.168E+06	0.605E-07
17	108	-4.933	0.433E-06	0.433E-06	0.135E-04	-0.135E-04	0.168E+06	0.213E-07

Factor of Safety [FS]:

Since Fic [108.57] < 0.55 * 234.4, FS = 2

Required Moment of Inertia for the Ring/Shell Junction, Shear [Irs]:

$$= 0.184 \cdot Cv \cdot Ms0.8 \cdot t^3 \cdot Ls \quad [\text{eq 6-5}] \text{ or } [\text{eq 4.4.43}]$$

$$= 0.184 \cdot 0.4484 \cdot 11.3630.8 \cdot 13.0^3 \cdot 2445.8535$$

$$= 3098184.000 \text{ mm}^4$$

External Pressure Calculations:

From	To	Section Length mm	Outside Diameter mm	Corroded Thickness mm	Factor A	Factor B MPa
10	20	15700	...	...	No Calc	No Calc
20	30	15700	...	...	No Calc	No Calc
30	40	15700	...	...	No Calc	No Calc
40	50	No Calc	6858	54	No Calc	No Calc
50	60	19646.3	6896	92	No Calc	No Calc
60	70	19646.3	6896	92	No Calc	No Calc
70	80	19646.3	6892	90	No Calc	No Calc
80	90	19646.3	6890	91	No Calc	No Calc
90	100	19646.3	6890	91	No Calc	No Calc
100	110	19646.3	6890	91	No Calc	No Calc
110	120	19646.3	6890	91	No Calc	No Calc
120	130	No Calc	6844	51	No Calc	No Calc
130	140	15700	...	...	No Calc	No Calc
140	150	15700	...	...	No Calc	No Calc
150	160	15700	...	...	No Calc	No Calc
160	170	15700	...	...	No Calc	No Calc
170	180	No Calc	7172	30	No Calc	No Calc
180	Ring	2841.72	7148	18	No Calc	No Calc
Ring	190	2750	7148	18	No Calc	No Calc
190	Ring	2750	7148	18	No Calc	No Calc
Ring	200	2500	7148	18	No Calc	No Calc
200	Ring	2500	7144	16	No Calc	No Calc
Ring	210	2499.9	7144	16	No Calc	No Calc
210	Ring	2499.9	7144	16	No Calc	No Calc
Ring	220	2500	7144	16	No Calc	No Calc
220	Ring	2500	7140	14	No Calc	No Calc
Ring	230	2499.99	7140	14	No Calc	No Calc
230	Ring	2499.99	7132	15	No Calc	No Calc
Ring	Ring	2500	7132	15	No Calc	No Calc
Ring	240	2500.01	7132	15	No Calc	No Calc



240	Ring	2500.01	7132	15	No Calc	No Calc
Ring	250	2500	7132	15	No Calc	No Calc
250	Ring	2500	7132	15	No Calc	No Calc
Ring	260	2500.04	7132	15	No Calc	No Calc
260	Ring	2500.04	7128	13	No Calc	No Calc
Ring	270	2391.67	7128	13	No Calc	No Calc
270	280	No Calc	7132	15	No Calc	No Calc

External Pressure Calculations:

From	To	External Actual T. mm	External Required T. mm	External Design Pressure MPa	External M.A.W.P. MPa
10	20	...	No Calc	...	No Calc
20	30	...	No Calc	...	No Calc
30	40	...	No Calc	...	No Calc
40	50	60	No Calc	...	2.60545
50	60	98	No Calc	...	1.45355
60	70	98	No Calc	...	1.45355
70	80	96	No Calc	...	1.38435
80	90	92	No Calc	...	1.42762
90	100	92	No Calc	...	1.42762
100	110	92	No Calc	...	1.42762
110	120	92	No Calc	...	1.42762
120	130	52	No Calc	...	2.41201
130	140	...	No Calc	...	No Calc
140	150	...	No Calc	...	No Calc
150	160	...	No Calc	...	No Calc
160	170	...	No Calc	...	No Calc
170	180	36	17.9883	0.051733	0.31953
180	Ring	24	17.1675	0.051733	0.17017
Ring	190	24	17.0156	0.051733	0.17573
190	Ring	24	17.0156	0.051733	0.17573
Ring	200	24	16.5959	0.051733	0.19249
200	Ring	22	16.5921	0.051733	0.14517
Ring	210	22	16.5922	0.051733	0.14517
210	Ring	22	16.5922	0.051733	0.14517
Ring	220	22	16.5921	0.051733	0.14517
220	Ring	20	16.5885	0.051733	0.10438
Ring	230	20	16.5885	0.051733	0.10438
230	Ring	16	11.5816	0.051733	0.12413
Ring	Ring	16	11.5816	0.051733	0.12413
Ring	240	16	11.5818	0.051733	0.12413
240	Ring	16	11.5818	0.051733	0.12413
Ring	250	16	11.5816	0.051733	0.12413
250	Ring	16	11.5816	0.051733	0.12413
Ring	260	16	11.5818	0.051733	0.12413
260	Ring	14	11.5785	0.051733	0.086916
Ring	270	14	11.3855	0.051733	0.091013
270	280	16	12.9865	0.051733	0.080995

Minimum

0.1

External Pressure Calculations:

From	To	Actual Length Bet. Stiffeners mm	Allowable Length Bet. Stiffeners mm	Ring Inertia Required mm ⁴	Ring Inertia Available mm ⁴
10	20	15700	No Calc	No Calc	No Calc
20	30	15700	No Calc	No Calc	No Calc
30	40	15700	No Calc	No Calc	No Calc
40	50	No Calc	No Calc	No Calc	No Calc
50	60	19646.3	No Calc	No Calc	No Calc
60	70	19646.3	No Calc	No Calc	No Calc
70	80	19646.3	No Calc	No Calc	No Calc
80	90	19646.3	No Calc	No Calc	No Calc
90	100	19646.3	No Calc	No Calc	No Calc



100	110	19646.3	No Calc	No Calc	No Calc
110	120	19646.3	No Calc	No Calc	No Calc
120	130	No Calc	No Calc	No Calc	No Calc
130	140	15700	No Calc	No Calc	No Calc
140	150	15700	No Calc	No Calc	No Calc
150	160	15700	No Calc	No Calc	No Calc
160	170	15700	No Calc	No Calc	No Calc
170	180	No Calc	No Calc	No Calc	No Calc
180	Ring	2841.72	8896.97	No Calc	No Calc
Ring	190	2750	8896.93	123852144	160482688
190	Ring	2750	8896.93	No Calc	No Calc
Ring	200	2500	8897.1	124300688	160482688
200	Ring	2500	6718.92	No Calc	No Calc
Ring	210	2499.9	6719.17	91227984	93478560
210	Ring	2499.9	6719.17	No Calc	No Calc
Ring	220	2500	6718.94	91227968	93478560
220	Ring	2500	4886.65	No Calc	No Calc
Ring	230	2499.99	4886.55	65035676	86572312
230	Ring	2499.99	5771.89	No Calc	No Calc
Ring	Ring	2500	5771.9	77300416	90059568
Ring	240	2500.01	5771.92	77300392	90059568
240	Ring	2500.01	5771.92	No Calc	No Calc
Ring	250	2500	5771.9	77300416	90059568
250	Ring	2500	5771.9	No Calc	No Calc
Ring	260	2500.04	5771.99	77300368	90059568
260	Ring	2500.04	4102.43	No Calc	No Calc
Ring	270	2391.67	4102.41	53860156	82931680
270	280	No Calc	No Calc	No Calc	No Calc

Elements Suitable for External Pressure.

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Element and Detail Weights:

From	To	Element Metal Wgt. kgm	Element ID Volume ltr	Corroded Metal Wgt. kgm	Corroded ID Volume ltr	Extra due Misc % kgm
10	20	27044.8	...	26254.5	...	...
20	30	16103.2	...	15391.1	...	...
30	40	10436.7	...	10041.5	...	...
40	50	38347.4	80101.3	35025.3	80530	...
50	60	48661.4	105789	45722.5	106168	...
60	70	48661.4	105789	45722.5	106168	...
70	80	30181.1	66999.4	28319.8	67239.6	...
80	90	45682.2	105978	45192.3	106041	...
90	100	45682.2	105978	45192.3	106041	...
100	110	30454.8	70652.1	30128.2	70694.2	...
110	120	22841.1	52989	22596.1	53020.6	...
120	130	32630.9	80172.6	32077.7	80244	...
130	140	5401.87	...	5236.48	...	...
140	150	12109.2	...	11708	...	...
150	160	12309.9	...	11902.1	...	...
160	170	5188.45	...	5016.24	...	...
170	180	21212.4	48838.8	18319.8	49083.5	...
180	190	12488.6	118797	9374.36	119199	...
190	200	12488.6	118797	9374.36	119199	...
200	210	11444.6	118797	8330.34	119199	...
210	220	11444.6	118797	8330.34	119199	...
220	230	9707.86	110877	6801.18	111252	...
230	240	8316.5	118797	7797.83	118864	...
240	250	8316.5	118797	7797.83	118864	...
250	260	8316.5	118797	7797.83	118864	...
260	270	7032.26	114837	6530.86	114902	...
270	280	9577.91	48838.8	9099.01	48879.5	...
Total		552082	1929419.25	515080	1933651.62	0

Weight of Details:

From	Type	Weight of Detail kgm	X Offset, Dtl. Cent. mm	Y Offset, Dtl. Cent. mm	Z Offset Dtl. Cent. mm	Description
10	Insl	4182.56	...	1500	...	FP
20	Plat	1731.68	-2733.09	300	2733.09	SKRT PLAT-1
20	Insl	3761.04	...	1350	...	FP
30	Insl	2091.28	...	750	...	FP
40	Pack	13309.4	...	-1675	...	HP_Pack: 1_1
40	Liqd	84855.2	...	-1684.5	...	Liquid 40
40	Insl	2372.86	...	-1675	...	Ins: 70
40	Nozl	4037.73	...	-3439	...	D 36"
40	Wght	170	...	-3360	...	HP_VB
40	Wght	2000	...	...	...	163-D_Btm SF wt
50	Plat	6913.9	0.37717E-04	...	...	PLAT:1
50	Pack	5859.44	...	500	...	HP_Pack: 1_2
50	Liqd	112067	...	1500	...	Liquid 50
50	Insl	1853.61	...	1500	...	Ins: 70
50	Nozl	30.284	3448.43	150	...	L12 2"
50	Nozl	26.4796	3009.44	150	-1737.5	L14 1.5"
50	Wght	2970	-5048	0.0040691	...	D Fy wt
60	Plat	3456.95	-2711.67	799.975	...	PLAT-2
60	Liqd	112067	...	1500	...	Liquid 60
60	Insl	1853.61	...	1500	...	Ins: 70
60	Nozl	1356.88	-2478.41	1399.97	2478.41	MH-6 24"
60	Wght	50	-3535.53	1399.99	3535.53	MH6 davit
70	Plat	3455.34	2711.67	1799.97	...	PLAT-3
70	Pack	5859.55	...	1399.98	...	HP_Pack: 2_1
70	Liqd	22413.4	...	300	...	Liquid: 40
70	Insl	1173.28	...	950	...	Ins: 70
70	Nozl	30.284	3448.43	750.004	...	L11 2"



70	Nozl	26.2651	3009.44	750.004	-1737.5
70	Nozl	1395.61	...	899.994	-3600
70	Wght	3100	...	899.975	...
70	Wght	300	...	899.975	...
80	Pack	5859.3	...	499.993	...
80	Insl	1852.02	...	1500	...
80	Lini	1518.03	...	1500	...
80	Wght	960	...	999.974	...
80	Wght	5900	...	1599.97	...
80	Wght	300	...	1599.97	...
80	Wght	6000	...	2799.97	...
90	Plat	3454.54	-2713.56	899.975	...
90	Insl	1852.02	...	1500	...
90	Lini	1518.03	...	1500	...
90	Nozl	1350.43	-2480.53	1499.97	2480.53
90	Wght	50	-3535.53	1499.97	3535.53
100	Plat	3454.54	2713.56	1899.97	...
100	Insl	1234.68	...	1000	...
100	Lini	1012.02	...	1000	...
100	Nozl	1350.43	-2480.53	999.975	2480.53
100	Wght	50	-3535.53	999.975	3535.53
110	Insl	926.008	...	750	...
110	Lini	759.013	...	750	...
110	Nozl	1730.71	-3533	749.976	...
110	Wght	6800	...	1099.98	...
110	Wght	11746	-5059	749.976	...
120	Plat	3436.11	2724.27	2399.97	...
120	Insl	2327.12	...	1676.5	...
120	Lini	1713.06	...	1685	...
120	Nozl	104.131	...	3474	...
120	Wght	2000	...	...	...
120	Wght	1789	4686	-0.025579	...
120	Forc	...	...	3403	...
130	Insl	770.013	...	625	...
140	Wght	10000	...	1649.95	...
160	Plat	3577.98	-2838.34	899.95	...
160	Insl	800.459	...	625	...
170	Liqd	51737.3	...	-887.5	...
170	Insl	2178.33	...	-862.5	...
170	Nozl	3919.83	...	-1962	...
170	Wght	250	...	-1800	...
170	Wght	50950	-5350	49.9504	...
180	Plat	3557.94	-2837.71	1849.95	...
180	Liqd	125848	...	1500	...
180	Insl	1920.57	...	1500	...
180	Ring	1523.27	...	2200.05	...
180	Wght	18300	...	249.95	...
180	Wght	300	...	49.9504	...
190	Plat	3557.94	2837.71	2849.95	...
190	Liqd	125848	...	1500	...
190	Insl	1920.57	...	1500	...
190	Ring	1523.27	...	1950.05	...
190	Nozl	521.303	-2619.83	600.049	2619.83
190	Wght	50	-3535.53	600.049	3535.53
200	Liqd	4194.92	...	50	...
200	Insl	1919.51	...	1500	...
200	Ring	1260.19	...	1450.05	...
200	Nozl	10.5771	3653.19	1299.95	...
200	Nozl	10.5771	3163.75	1299.95	-1826.59
200	Nozl	10.5771	1826.59	1299.95	-3163.75
200	Nozl	10.5771	...	1299.95	-3653.19
200	Nozl	10.5771	-1826.59	1299.95	-3163.75
200	Nozl	501.501	-3705	2300.05	...
200	Nozl	42.4199	3655	2500.05	...
200	Wght	5295	4808	2500.05	...
200	Wght	75755	-5189	2299.95	...
210	Plat	3556.34	-2837.71	849.955	...
210	Pack	7895.61	...	2400.03	...
210	Insl	1919.51	...	1500	...
210	Ring	1260.19	...	949.949	...

L13 1.5"
 UC2 24"
 HP_B#1_btm grd
 HP_B#1_btm grd
 HP_Pack: 2_2
 Ins: 70
 SS304L_clad
 HP_B#1_top HDG
 HP_B#1_dis. dea
 HP_B#1_dis. TA
 HP_Shroud
 PLAT-4
 Ins: 70
 SS304L_clad
 MH-5 24"
 MH5 davit
 PLAT-5
 Ins: 70
 SS304L_clad
 MH-4 24"
 MH4 davit
 Ins: 70
 SS304L_clad
 A 36"
 HP_Gallery
 A Fy wt
 PLAT-6
 Ins: 70
 SS304L_clad
 E 8"
 163-D_Top SF wt
 E Fy wt
 HP_E_FxFz
 Ins: 70
 Misc Margin
 PLAT-7
 Ins: 70
 Liquid: 10
 Ins: 70
 C 48"
 LP_VB
 C Fy wt
 PLAT-8
 Liquid: 20
 Ins: 70
 3
 Cleat wt
 Nozzles clad wt
 PLAT-9
 Liquid: 30
 Ins: 70
 Ring R2
 MH-3 24"
 MH3 davit
 Liquid: 40
 Ins: 70
 Ring R3
 L2 2"
 L4 2"
 L6 2"
 L8 2"
 L10 2"
 B 36"
 K 6"
 K Fy wt
 B Fy wt
 PLAT-10
 LP_Pack_1
 Ins: 70
 Ring R4



210	Nozl	373.512	3700	1500.05	...
210	Wght	4000	...	1800.05	...
210	Wght	350	...	1800.05	...
210	Wght	49798	5080	1500.05	...
220	Plat	3554.73	2837.71	1849.95	...
220	Pack	18423.9	...	1400	...
220	Insl	1790.55	...	1400	...
220	Ring	1259.51	...	449.954	...
220	Nozl	10.5771	3653.19	1399.95	...
220	Nozl	10.5771	3163.75	1399.95	-1826.59
220	Nozl	10.5771	1826.59	1399.95	-3163.75
220	Nozl	10.5771	...	1399.95	-3653.19
220	Nozl	10.5771	-1826.59	1399.95	-3163.75
230	Pack	6568.84	...	500	...
230	Insl	1916.32	...	1500	...
230	Lini	1607.27	...	1500	...
230	Ring	1258.48	...	149.949	...
230	Ring	1258.48	...	2649.95	...
230	Nozl	466.763	-2619.83	2100.05	2619.83
230	Wght	900	...	999.948	...
230	Wght	5750	...	1599.95	...
230	Wght	350	...	1599.95	...
230	Wght	9800	...	2800.05	...
230	Wght	50	-3535.53	2100.05	3535.53
240	Plat	3551.53	-2837.71	49.9551	...
240	Insl	1916.32	...	1500	...
240	Lini	1607.27	...	1500	...
240	Ring	1258.48	...	2149.96	...
240	Nozl	277.974	...	199.955	-3700
240	Nozl	493.339	-2619.83	1399.96	2619.83
240	Wght	80	...	2749.96	...
240	Wght	50	-3535.53	1399.96	3535.53
250	Plat	5579.13	2837.71	1500.05	...
250	Insl	1916.32	...	1500	...
250	Lini	1607.27	...	1500	...
250	Ring	1258.48	...	1649.96	...
250	Nozl	4.96571	-2586	1199.95	2586
250	Nozl	605.11	-3705	800.053	...
250	Nozl	625.255	3705	800.053	...
250	Tray	8592.07	...	1550.05	...
250	Wght	3100	...	650.053	...
250	Wght	150	...	2450.05	...
250	Wght	20	...	2450.05	...
250	Wght	99326	...	799.951	5257
260	Plat	3663.12	-2837.71	1000	...
260	Insl	1851.42	...	1450	...
260	Lini	1553.7	...	1450	...
260	Ring	1259.23	...	1150	...
260	Wght	20400	...	1949.95	...
270	Plat	9733.92	...	2300.05	...
270	Insl	2207.01	...	925.75	...
270	Lini	1475.26	...	912.5	...
270	Nozl	837.337	...	2067	...
270	Wght	960	...	...	...
270	Wght	7500	-3750	1825	...
270	Wght	150	...	...	...
270	Wght	28112	-5259	0.041858	...
270	Wght	900	3600	0.041858	...
270	Forc	...	...	1800	...
70	Pliq	9339.14	...	1399.985	...
80	Pliq	9355.46	...	499.993	...
210	Pliq	12584.28	...	2400.025	...
220	Pliq	29364.54	...	1400.000	...
230	Pliq	10487.33	...	500.000	...

I 28"
 LP_B#1_btm grid
 LP_B#1_TA
 I Fy wt
 PLAT-11
 LP_Pack_2
 Ins: 70
 Ring R5
 L1 2"
 L3 2"
 L5 2"
 L7 2"
 L9 2"
 LP_Pack_3
 Ins: 70
 Clad SS304L
 Ring R6
 Ring R7
 MH-2 24"
 LP_B#1_HDG
 LP_B#1_dist. de
 LP_B#1_dist.TA
 LP_Shroud
 MH2 davit
 PLAT-12
 Ins: 70
 Clad SS304L
 Ring R8
 UC1 24"
 MH-1 24"
 LP_dwncm PP dea
 MH1 davit
 PLAT-13
 Ins: 70
 Clad SS304L
 Ring R9
 S1 1"
 H1 42"
 H2 42"
 Tray#1-3
 LP_Gallery
 LP_FP_G_dead
 LP_FP_G_TA
 H1/H2 Fy wt
 PL-14
 Ins: 70
 Clad SS304L
 Ring R10
 Lifting wt
 TOP PLT
 Ins: 70
 Clad SS304L
 F 42"
 LP_Demister
 PSV
 LP_Demister_TA
 F Fy wt
 LP_Davit
 F_FxFz

Note: The individual tray liquid weights are listed below, but these weights are included in the tray weights above.

From	Tliq	Weight	X Offset	Y Offset
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250 Tliq	0.00	0.000	1550.053
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Total Weight of Each Detail Type:

Platforms	66235.7
Packing	63776.0
Packing Liquid	71130.8
Liquid	639030.1
Insulation	48507.0
Cladding & Overlay	14370.9
Stiffeners	13119.6
Nozzles	20214.3
Trays	7962.1
Tray Bolting Weight	630.0
Weights	436831.0

Sum of the Detail Weights 1381807.2 kgm

Weight Summation Results: (kgm)

	Fabricated	Shop Test	Shipping	Erected	Empty	Operating
Main Elements	552082.8	552082.8	552082.8	552082.8	552082.8	552082.8
Cladding	14370.9	14370.9	14370.9	14370.9	14370.9	14370.9
Stif. Rings	13119.6	13119.6	13119.6	13119.6	13119.6	13119.6
Nozzles	20214.3	20214.3	20214.3	20214.3	20214.3	20214.3
Tray Bar	630.0	630.0	630.0	630.0	630.0	630.0
Wld Weights	81870.0	81870.0	81870.0	81870.0	81870.0	81870.0
Platforms	...	...	66235.7	66235.7	66235.7	66235.7
Packing	...	...	63776.0	63776.0	63776.0	63776.0
Insulation	...	...	48507.0	48507.0	48507.0	48507.0
Trays	...	...	7962.1	7962.1	7962.1	7962.1
Empty Weights	...	...	...	29220.0	29220.0	...
Test Weights	...	29220.0	...	...	...	...
Ope Weights	...	...	...	...	...	354961.0
Ope. Liquid	...	...	...	...	...	639030.1
Packing Liq	...	...	...	...	...	71130.8
Test Liquid	...	1926448.0	...	...	...	...
Totals	682287.5	2637955.8	868768.2	897988.2	897988.2	1933890.0

Field Installation Options:

Weight Summary:

Fabricated Wt.	- Bare weight without removable internals	682287.5 kgm
Shop Test Wt.	- Fabricated weight + water (full)	2637955.8 kgm
Shipping Wt.	- Fab. weight + removable intls.+ shipping app.	868768.2 kgm
Erected Wt.	- Fab. wt + or - loose items (trays,platforms etc.)	897988.2 kgm
Ope. Wt. no Liq	- Fab. weight + internals. + details + weights	897988.2 kgm
Operating Wt.	- Empty weight + operating liq. uncorroded	1933890.0 kgm
Field Test Wt.	- Empty weight + test medium (full)	2760660.0 kgm

Weights used in Vortex Shedding Calculations:

Mass of the Upper 1/3 of the Vertical Vessel (operating)	533866.0 kgm
Mass of the Upper 1/3 of the Vertical Vessel (filled w/water)	1062825.6 kgm
Mass of the Upper 1/3 of the Vertical Vessel (empty)	234002.2 kgm

Surface Areas of Elements:

From	To	Outside Surface Area mm ²	Inside Surface Area mm ²
10	20	64522024	63730344
20	30	58018932	57408208
30	40	32261012	31865172
40	50	74223088	71315168



50	60	64993268	63146016
60	70	64993268	63146016
70	80	41138528	39992472
80	90	64936716	63202556
90	100	64936716	63202556
100	110	43291148	42135040
110	120	32468358	31601278
120	130	73834872	71357504
130	140	26998062	26793860
140	150	65850924	65398532
150	160	66934772	66482380
160	170	28077988	27889490
170	180	57135736	55758964
180	190	67368312	66915920
190	200	67368312	66915920
200	210	67330608	66915920
210	220	67330608	66915920
220	230	62806720	62454864
230	240	67217520	66915920
240	250	67217520	66915920
250	260	67217520	66915920
260	270	64940488	64685392
270	280	56382684	55758964

Total 1579795584.0 1555736192.0 mm²
 Total 1579.8 1555.7 m²

Element and Detail Weights:

From	To	Total Ele. Empty Wgt. kgm	Total. Ele. Oper. Wgt. kgm	Total. Ele. Hydro. Wgt. kgm	Total Dtl. Offset Mom. N-mm	Oper. Wgt. No Liquid kgm
10	20	31227.4	31227.4	31227.4	...	31227.4
20	30	21595.9	21595.9	21595.9	46429000	21595.9
30	40	12528	12528	12528	...	12528
40	50	60237.4	145093	126980	...	60237.4
50	60	63345.1	178382	163210	145270368	66315.1
60	70	55378.8	167446	161103	126683936	55378.8
70	80	45521.4	77274	106620	93716960	45521.4
80	90	68071.5	77427	168126	...	68071.5
90	100	53907.2	53907.2	159821	126555280	53907.2
100	110	37556.4	37556.4	108165	57364292	37556.4
110	120	33056.8	44802.8	86013.5	642922112	44802.8
120	130	42211.3	44000.3	122335	174069568	44000.3
130	140	6171.89	6171.89	6171.89	...	6171.89
140	150	22109.2	22109.2	22109.2	...	22109.2
150	160	12309.9	12309.9	12309.9	...	12309.9
160	170	9566.89	9566.89	9566.89	99625400	9566.89
170	180	27560.6	130248	76369.5	2.674E+09	78510.6
180	190	38090.4	163938	156815	99045456	38090.4
190	200	20061.7	145909	138786	83913536	20061.7
200	210	15221.1	100466	133946	3.622E+09	96271.1
210	220	30799.8	93182	141629	2.396E+09	80597.8
220	230	34789.4	64153.9	127175	99663552	34789.4
230	240	38242.7	48730	150398	13730212	38242.7
240	250	17551.4	17551.4	136276	113280200	17551.4
250	260	31775.1	131101	150500	155917264	131101
260	270	35759.7	35759.7	150527	101973368	35759.7
270	280	33341.4	61453.4	82150.4	1694437120	61453.4

Empty Support Force + the Sum of the Y forces 8214547.00 N
 Operating Support Force + the Sum of the Y forces 19070104.00 N
 Hydro Support Force + the Sum of the Y forces 27194994.00 N

Cumulative Vessel Weight

From	To	Cumulative Ope Wgt. No Liquid	Cumulative Oper. Wgt.	Cumulative Hydro. Wgt.
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		kgm	kgm	kgm
10	20	1223729	1933890	2762453
20	30	1192501	1902662	2731226
30	40	1170906	1881066	2709630
40	50	1158378	1868538	2697102
50	60	1098140	1723446	2570121
60	70	1031825	1545064	2406912
70	80	976447	1377618	2245809
80	90	930925	1300344	2139189
90	100	862854	1222917	1971063
100	110	808947	1169010	1811242
110	120	771390	1131453	1703077
120	130	726588	1086650	1617064
130	140	682587	1042650	1494729
140	150	676415	1036478	1488557
150	160	654306	1014369	1466448
160	170	641996	1002059	1454137
170	180	632429	992493	1444571
180	190	553919	862245	1368201
190	200	515828	698307	1211386
200	210	495767	552398	1072600
210	220	399496	451932	938655
220	230	318898	358750	797026
230	240	284108	294596	669851
240	250	245866	245866	519453
250	260	228314	228314	383177
260	270	97213.1	97213.1	232677
270	280	61453.4	61453.4	82150.4

The cumulative operating weights no liquid in the column above are the cumulative operating weights minus the operating liquid weight minus any weights absent in the empty condition.

Cumulative Vessel Moment

From	To	Cumulative Empty Mom. N-mm	Cumulative Oper. Mom. N-mm	Cumulative Hydro. Mom. N-mm
10	20	12.57E+09	12.57E+09	12.57E+09
20	30	12.57E+09	12.57E+09	12.57E+09
30	40	12.52E+09	12.52E+09	12.52E+09
40	50	12.52E+09	12.52E+09	12.52E+09
50	60	12.52E+09	12.52E+09	12.52E+09
60	70	12.38E+09	12.38E+09	12.38E+09
70	80	12.25E+09	12.25E+09	12.25E+09
80	90	12.16E+09	12.16E+09	12.16E+09
90	100	12.16E+09	12.16E+09	12.16E+09
100	110	12.03E+09	12.03E+09	12.03E+09
110	120	11.97E+09	11.97E+09	11.97E+09
120	130	11.33E+09	11.33E+09	11.33E+09
130	140	11.15E+09	11.15E+09	11.15E+09
140	150	11.15E+09	11.15E+09	11.15E+09
150	160	11.15E+09	11.15E+09	11.15E+09
160	170	11.15E+09	11.15E+09	11.15E+09
170	180	11.05E+09	11.05E+09	11.05E+09
180	190	8.381E+09	8.381E+09	8.381E+09
190	200	8.282E+09	8.282E+09	8.282E+09
200	210	8.198E+09	8.198E+09	8.198E+09
210	220	4.575E+09	4.575E+09	4.575E+09
220	230	2.179E+09	2.179E+09	2.179E+09
230	240	2079338240	2079338240	2079338240
240	250	2065608064	2065608064	2065608064
250	260	1952327808	1952327808	1952327808
260	270	1796410624	1796410624	1796410624
270	280	1694437120	1694437120	1694437120



Nozzle Flange MAWP Results: (MPa & °C)

Nozzle Description	Flange Ope.	Rating Ambient	Des. Temp	Class	Grade/Group	Equiv. Press	- - - UG-44	Max Pressure 50%	DNV
D 36"	9.14	10.20	130	600	GR 1.1	...	...	...	...
L12 2"	9.14	10.20	130	600	GR 1.1	...	...	...	...
L14 1.5"	9.14	10.20	130	600	GR 1.1	...	...	...	...
MH-6 24"	9.14	10.20	130	600	GR 1.1	...	...	...	...
L11 2"	9.14	10.20	130	600	GR 1.1	...	...	...	...
L13 1.5"	9.14	10.20	130	600	GR 1.1	...	...	...	...
UC2 24"	9.14	10.20	130	600	GR 1.1	...	...	...	...
MH-5 24"	9.14	10.20	130	600	GR 1.1	...	...	...	...
MH-4 24"	9.14	10.20	130	600	GR 1.1	...	...	...	...
A 36"	9.14	10.20	130	600	GR 1.1	...	...	...	...
E 8"	9.14	10.20	130	600	GR 1.1	3.936	9.143	9.143	9.143
C 48"	1.66	1.96	130	150	GR 1.1	1.460	1.190	1.656	1.024
MH-3 24"	1.66	1.96	130	150	GR 1.1	...	...	...	...
L2 2"	4.57	5.10	130	300	GR 1.1	...	...	...	...
L4 2"	4.57	5.10	130	300	GR 1.1	...	...	...	...
L6 2"	4.57	5.10	130	300	GR 1.1	...	...	...	...
L8 2"	4.57	5.10	130	300	GR 1.1	...	...	...	...
L10 2"	4.57	5.10	130	300	GR 1.1	...	...	...	...
B 36"	1.66	1.96	130	150	GR 1.1	1.055	1.595	1.656	1.429
K 6"	1.66	1.96	130	150	GR 1.1	3.008	0.635	1.656	-0.524
I 28"	1.66	1.96	130	150	GR 1.1	...	...	...	...
L1 2"	4.57	5.10	130	300	GR 1.1	...	...	...	...
L3 2"	4.57	5.10	130	300	GR 1.1	...	...	...	...
L5 2"	4.57	5.10	130	300	GR 1.1	...	...	...	...
L7 2"	4.57	5.10	130	300	GR 1.1	...	...	...	...
L9 2"	4.57	5.10	130	300	GR 1.1	...	...	...	...
MH-2 24"	1.66	1.96	130	150	GR 1.1	...	...	...	...
UC1 24"	1.66	1.96	130	150	GR 1.1	...	...	...	...
MH-1 24"	1.66	1.96	130	150	GR 1.1	...	...	...	...
S1 1"	1.66	1.96	130	150	GR 1.1	...	...	...	...
H1 42"	1.66	1.96	130	150	GR 1.1	0.925	1.074	1.656	1.559
H2 42"	1.66	1.96	130	150	GR 1.1	0.914	1.656	1.656	1.571
F 42"	1.66	1.96	130	150	GR 1.1	...	...	...	...
Min Rating	0.635	1.960, [for Core Elements]					...	...	-0.524

Selected Method for Derating ASME B16.5 Flange MAWP: ASME 4.16.12

The 4.16.12 Method is based on the paper PVP 2013-97814. PV Elite uses the maximum loads from each load category to determine ME and FE. In many cases, the computed maximum allowable pressure will be greater than the flange rating. In these cases, the minimum of the rating from the table and the 4.16.12 method will be used. SA-193 B8 Cl. 2 bolts or ones with higher allowable stresses at the specified bolt size shall be used.

The axial force FE must put the nozzle in tension (-P for 107/297; +Fr for PD 5500 AnnexG) to be considered.

Nozzle Description	Gasket Dimension G mm	Fm Value	Resolved Moment ME N-mm	Axial Force FE N	MAWP PD MPa	Maximum Pressure Pmax MPa	Pass Fail
E 8"	244.602	0.50	11314305	0	0.482	9.143	Pass
C 48"	1255.776	0.60	285106080	899999	0.482	1.190	Pass
B 36"	947.801	0.60	160434128	67500	0.482	1.595	Pass
K 6"	196.088	1.20	4455013	0	0.482	0.635	Pass
H1 42"	1097.026	0.21	218283392	78750	0.482	1.074	Pass
H2 42"	1101.725	0.60	218283392	78750	0.482	1.656	Pass

Evaluating Flange on Nozzle [E 8"]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 \cdot [(PR - PD) + FM \cdot PR]$$

$$= 16(11314305) + 4 \cdot \text{abs}(0) \cdot 244.602 \leq \pi \cdot 244.602^3 \cdot [(9.143 - 0.482) + 0.500 \cdot 9.143]$$



= 0.18103E+09 ≤ 0.60855E+09 Check Passes

Maximum Pressure for Flange on Nozzle E 8" [Pmax]:

$$\begin{aligned} &= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3)) \\ &= \min(9.143, 0.500 \cdot 9.143 + 9.143 - 181028880.0 / (\pi \cdot 244.602^3)) \\ &= \min(9.143, 9.778) \\ &= 9.143 \text{ MPa} \end{aligned}$$

Evaluating Flange on Nozzle [C 48"]:

$$\begin{aligned} &= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 [(PR - PD) + FM \cdot PR] \\ &= 16(285106080) + 4 \cdot \text{abs}(899999) 1255.776 \leq \pi \cdot 1255.776^3 [(1.656 - 0.482) + .600 \cdot 1.656] \\ &= 0.90843E+10 \leq 0.13492E+11 \text{ Check Passes} \end{aligned}$$

Maximum Pressure for Flange on Nozzle C 48" [Pmax]:

$$\begin{aligned} &= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3)) \\ &= \min(1.656, 0.600 \cdot 1.656 + 1.656 - 9084320768.0 / (\pi \cdot 1255.776^3)) \\ &= \min(1.656, 1.190) \\ &= 1.190 \text{ MPa} \end{aligned}$$

Evaluating Flange on Nozzle [B 36"]:

$$\begin{aligned} &= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 [(PR - PD) + FM \cdot PR] \\ &= 16(160434128) + 4 \cdot \text{abs}(67500) 947.801 \leq \pi \cdot 947.801^3 [(1.656 - 0.482) + .600 \cdot 1.656] \\ &= 0.28230E+10 \leq 0.58010E+10 \text{ Check Passes} \end{aligned}$$

Maximum Pressure for Flange on Nozzle B 36" [Pmax]:

$$\begin{aligned} &= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3)) \\ &= \min(1.656, 0.600 \cdot 1.656 + 1.656 - 2822956544.0 / (\pi \cdot 947.801^3)) \\ &= \min(1.656, 1.595) \\ &= 1.595 \text{ MPa} \end{aligned}$$

Evaluating Flange on Nozzle [K 6"]:

$$\begin{aligned} &= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 [(PR - PD) + FM \cdot PR] \\ &= 16(4455013) + 4 \cdot \text{abs}(0) 196.088 \leq \pi \cdot 196.088^3 [(1.656 - 0.482) + 1.200 \cdot 1.656] \\ &= 0.71280E+08 \leq 0.74914E+08 \text{ Check Passes} \end{aligned}$$

Maximum Pressure for Flange on Nozzle K 6" [Pmax]:

$$\begin{aligned} &= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3)) \\ &= \min(1.656, 1.200 \cdot 1.656 + 1.656 - 71280240.0 / (\pi \cdot 196.088^3)) \\ &= \min(1.656, 0.635) \\ &= 0.635 \text{ MPa} \end{aligned}$$

Evaluating Flange on Nozzle [H1 42"]:

$$\begin{aligned} &= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 [(PR - PD) + FM \cdot PR] \\ &= 16(218283392) + 4 \cdot \text{abs}(78750) 1097.026 \leq \pi \cdot 1097.026^3 [(1.656 - 0.482) + .207 \cdot 1.656] \\ &= 0.38382E+10 \leq 0.62954E+10 \text{ Check Passes} \end{aligned}$$

Maximum Pressure for Flange on Nozzle H1 42" [Pmax]:

$$\begin{aligned} &= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3)) \\ &= \min(1.656, 0.207 \cdot 1.656 + 1.656 - 3838237952.0 / (\pi \cdot 1097.026^3)) \\ &= \min(1.656, 1.074) \\ &= 1.074 \text{ MPa} \end{aligned}$$

Evaluating Flange on Nozzle [H2 42"]:

$$\begin{aligned} &= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 [(PR - PD) + FM \cdot PR] \\ &= 16(218283392) + 4 \cdot \text{abs}(78750) 1101.725 \leq \pi \cdot 1101.725^3 [(1.656 - 0.482) + .600 \cdot 1.656] \\ &= 0.38397E+10 \leq 0.91112E+10 \text{ Check Passes} \end{aligned}$$

Maximum Pressure for Flange on Nozzle H2 42" [Pmax]:

$$\begin{aligned} &= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3)) \\ &= \min(1.656, 0.600 \cdot 1.656 + 1.656 - 3839718656.0 / (\pi \cdot 1101.725^3)) \\ &= \min(1.656, 1.736) \\ &= 1.656 \text{ MPa} \end{aligned}$$

Flange Pressure Ratings are per ASME B16.5 2025 Metric Edition

Large Diameter Flange Pressure Ratings are per ASME B16.47 2025 Edition

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The Natural Frequencies for the vessel have been computed iteratively by solving a system of matrices. These matrices describe the mass and the stiffness of the vessel. This is the generalized eigenvalue/eigenvector problem and is referenced in some mathematical texts.

The Natural Frequency for the Vessel (Empty.) is 1.05893 Hz.

The Natural Frequency for the Vessel (Ope...) is 0.95315 Hz.

The Natural Frequency for the Vessel (Filled) is 0.71685 Hz.

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Forces/Moments Applied to Vessel (Combined w/Wind Loads)

From	To	X and Z Dir Force Res. N	X,Z Moment and For Res N-mm
10	20	...	...
20	30	...	...
30	40	...	...
40	50	...	...
50	60	...	...
60	70	...	...
70	80	...	...
80	90	...	...
90	100	...	...
100	110	...	...
110	120	...	...
120	130	20215.5	81266880
130	140	...	...
140	150	...	...
150	160	...	...
160	170	...	...
170	180	...	...
180	190	...	...
190	200	...	...
200	210	...	...
210	220	...	...
220	230	...	...
230	240	...	...
240	250	...	...
250	260	...	...
260	270	...	...
270	280	81642.4	387138784

Forces/Moments Applied to Vessel (Combined w/Seismic Loads)

From	To	X and Z Dir Force Res. N	X,Z Moment and For Res N-mm
10	20	...	...
20	30	...	...
30	40	...	...
40	50	...	...
50	60	...	...
60	70	...	...
70	80	...	...
80	90	...	...
90	100	...	...
100	110	...	...
110	120	...	...
120	130	20215.5	81266880
130	140	...	...
140	150	...	...
150	160	...	...
160	170	...	...
170	180	...	...
180	190	...	...
190	200	...	...
200	210	...	...
210	220	...	...
220	230	...	...
230	240	...	...
240	250	...	...
250	260	...	...
260	270	...	...
270	280	81642.4	387138784



User Input Forces and Moments:

From Node	Distance From	- - - - Fx	Forces - - - - Fy	- - - - Fz	- - - - Mx	Moments - - - - My	- - - - Mz
120	3403.00	14294.	-19792.	14294.	0.880E+07	0.880E+07	
270	1800.00	57730.	-86592.	57730.	0.170E+09	0.170E+09	

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Wind Load Calculation Results:

Input Values:

Wind Design Code	ASCE/SEI 7-16
Wind Load Reduction Scale Factor	1.100
Basic Wind Speed	[V] 97.2 km/hr
Surface Roughness Category	C: Open Terrain
Type of Surface	Moderately Smooth
Base Elevation	300 mm
Percent Wind for Hydrotest	50.0
Using User defined Wind Press. Vs Elev.	N
Height of Hill or Escarpment H or Hh	0 mm
Distance Upwind of Crest Lh	0 mm
Distance from Crest to the Vessel x	0 mm
Type of Terrain (Hill, Escarpment)	Flat
Damping Factor (Beta) for Wind (Ope)	0.0100
Damping Factor (Beta) for Wind (Empty)	0.0000
Damping Factor (Beta) for Wind (Filled)	0.0000
Site Elevation above Sea Level	0 mm

Wind Analysis Results

The importance factor is not used in the calculation of wind pressure vs. elevation for the selected wind code.

Dynamic Gust-Effect Factor, Operating Case [Gf]:

$$= 0.925[(1+1.7 \cdot I_{zbar} \cdot \sqrt{(gQ^2 + gR^2 \cdot R^2)}) / (1+1.7 \cdot gV \cdot I_{zbar})]$$

$$= 0.925[(1+1.7 \cdot 0.162 \cdot \sqrt{(3.4^2 + 0.871^2 + 4.178^2 \cdot 0.283^2)}) / (1+1.7 \cdot 3.4 \cdot 0.162)]$$

$$= 0.897$$

Natural Frequency of Vessel (Operating)	0.953 Hz
Natural Frequency of Vessel (Empty)	1.059 Hz
Natural Frequency of Vessel (Test)	0.717 Hz

User Entered Importance Factor is	1.000
Force Coefficient	[Cf] 0.609
Structure Height to Diameter ratio	8.694
Height to top of Structure	61770.996 mm

This is classified as a flexible structure. Dynamic analysis performed.

Sample Calculation for the First Element:

The ASCE code performs calculations in Imperial Units. The wind pressure is therefore computed in these units.

Value of $[\alpha]$ and $[Zg]$:
 Exposure Category: C from Table 26.11-1
 $\alpha = 9.5$: $Zg = 274320.0$ mm

Effective Height $[z]$:
 $=$ Centroid Height + Vessel Base Elevation
 $= 1500.0 + 300.0 = 1800.0$ mm
 $= 5.906$ ft. Imperial Units

Velocity Pressure coefficient evaluated at height z $[Kz]$:
 Because z (5.906 ft.) $<$ 15 ft.
 $= 2.01(15 / Zg)^{2/\alpha}$
 $= 2.01(15/900.0)^{2/9.5}$
 $= 0.849$

Type of Hill: No Hill

Wind Directionality Factor $[Kd]$:
 $= 1$, per Table 26.6-1

As there is No Hill Present: $[Kzt]$:



K1 = 0, K2 = 0, K3 = 0

Topographical Factor [Kzt]:

$$= (1 + K1 \cdot K2 \cdot K3)^2$$

$$= (1 + 0.0 \cdot 0.0 \cdot 0.0)^2$$

$$= 1.0$$

Velocity Pressure evaluated at height z, Imperial Units [qz]:

$$= \max(16, 0.00256 \cdot Kz \cdot Kzt \cdot Kd \cdot Ke \cdot V(\text{mph})^2)$$

$$= \max(16, 0.00256 \cdot 0.849 \cdot 1.0 \cdot 1.0 \cdot 1.0 \cdot 60.399^2)$$

$$= 16.0 \text{ psf } [0.766] \text{ kPa}$$

Force on the first element [F]:

$$= qz \cdot Gf \cdot Cf \cdot \text{WindArea}$$

$$= 16.0 \cdot 0.897 \cdot 0.609 \cdot 271.483$$

$$= 2374.6 \text{ lbs. } [10562.3] \text{ N}$$

Element	Hgt (z) mm	K1	K2	K3	Kz	Kzt	qz kPa
Skirt-3	1800.0	0.000	0.000	0.000	0.849	1.000	0.766
Skirt-2	4650.0	0.000	0.000	0.000	0.852	1.000	0.766
Skirt-1	6750.0	0.000	0.000	0.000	0.921	1.000	0.766
163-D- BTM dish	7500.0	0.000	0.000	0.000	0.942	1.000	0.766
163-D-Shell-1	9000.0	0.000	0.000	0.000	0.979	1.000	0.766
163-D-Shell-2	12000.0	0.000	0.000	0.000	1.040	1.000	0.766
163-D-Shell-3	14450.0	0.000	0.000	0.000	1.082	1.000	0.766
163-D-Shell-4	16900.0	0.000	0.000	0.000	1.118	1.000	0.766
163-D-Shell-5	19900.0	0.000	0.000	0.000	1.157	1.000	0.766
163-D-Shell-6	22400.0	0.000	0.000	0.000	1.186	1.000	0.766
163-D-Shell-7	24150.0	0.000	0.000	0.000	1.205	1.000	0.766
163-D-Top head	24900.0	0.000	0.000	0.000	1.213	1.000	0.766
Int skirt 1	25525.7	0.000	0.000	0.000	1.219	1.000	0.766
Int skirt 2	27654.1	0.000	0.000	0.000	1.240	1.000	0.766
Int skirt 3	30654.1	0.000	0.000	0.000	1.267	1.000	0.766
Int skirt 4	32775.7	0.000	0.000	0.000	1.285	1.000	0.766
122-D1-Btm head	33425.0	0.000	0.000	0.000	1.290	1.000	0.766
122-D1-Shell-1	34950.0	0.000	0.000	0.000	1.303	1.000	0.766
122-D1-Shell-2	37950.0	0.000	0.000	0.000	1.325	1.000	0.766
122-D1-Shell-3	40950.0	0.000	0.000	0.000	1.347	1.000	0.766
122-D1-Shell-4	43950.0	0.000	0.000	0.000	1.367	1.000	0.766
122-D1-Shell-5	46850.0	0.000	0.000	0.000	1.386	1.000	0.766
122-D1-Shell-6	49750.0	0.000	0.000	0.000	1.403	1.000	0.766
122-D1-Shell-7	52750.0	0.000	0.000	0.000	1.421	1.000	0.766
122-D1-Shell-8	55750.0	0.000	0.000	0.000	1.437	1.000	0.766
122-D1-Shell-9	58700.0	0.000	0.000	0.000	1.453	1.000	0.766
122-D1-Top head	60967.1	0.000	0.000	0.000	1.464	1.000	0.766

Note:

The height/(avg. diameter) ratio [8.4], was less than or equal to 15,
 therefore no wind vibration calculations have been performed.

Platform Load Calculations

ID	Wind Area mm ²	Elevation mm	Pressure kPa	Force N	Cf
SKRT PLAT-1	1499999.88	3600.00	0.77	1149.07	1.00
PLAT:1	2999999.75	7500.00	0.77	2298.14	1.00
PLAT-2	2999999.75	11299.97	0.77	2298.14	1.00
PLAT-3	2999999.75	15299.97	0.77	2298.14	1.00
PLAT-4	2999999.75	19299.98	0.77	2298.14	1.00
PLAT-5	2999999.75	23299.97	0.77	2298.14	1.00
PLAT-6	2999999.75	27299.94	0.77	2298.14	1.00
PLAT-7	2999999.75	33049.95	0.77	2298.14	1.00
PLAT-8	2999999.75	35299.95	0.77	2298.14	1.00
PLAT-9	2999999.75	39299.95	0.77	2298.14	1.00
PLAT-10	2999999.75	43299.96	0.77	2298.14	1.00
PLAT-11	2999999.75	47299.95	0.77	2298.14	1.00



PLAT-12	2999999.75	51299.95	0.77	2298.14	1.00
PLAT-13	25270700.00	55750.05	0.77	19358.52	1.00
PL-14	5063040.00	58250.00	0.77	3878.52	1.20
TOP PLT	9328800.00	62450.05	0.77	7146.29	1.00

Wind Loads on Masses/Equipment/Piping

ID	Wind Area mm ²	Elevation mm	Pressure kPa	Force N
HPVB	0.00	4139.97	0.77	0.00
163-DBtm SF	0.00	7499.97	0.77	0.00
D Fy wt	0.00	7500.00	0.77	0.00
MH6 davit	0.00	11900.00	0.77	0.00
HPB#1btm gr	0.00	14399.98	0.77	0.00
HPB#1btm gr	0.00	14399.98	0.77	0.00
HPB#1top HD	0.00	16399.97	0.77	0.00
HPB#1dis. d	0.00	16999.98	0.77	0.00
HPB#1dis. T	0.00	16999.98	0.77	0.00
HPShroud	0.00	18199.97	0.77	0.00
MH5 davit	0.00	19899.97	0.77	0.00
MH4 davit	0.00	22399.98	0.77	0.00
HPGallary	0.00	24499.98	0.77	0.00
A Fy wt	0.00	24149.98	0.77	0.00
163-DTop SF	0.00	24899.97	0.77	0.00
E Fy wt	0.00	24899.95	0.77	0.00
Misc Margin	0.00	27799.95	0.77	0.00
LPVB	0.00	31600.00	0.77	0.00
C Fy wt	0.00	33449.95	0.77	0.00
Cleat wt	0.00	33699.95	0.77	0.00
Nozzles clad	0.00	33499.95	0.77	0.00
MH3 davit	0.00	37050.05	0.77	0.00
K Fy wt	0.00	41950.05	0.77	0.00
B Fy wt	0.00	41749.95	0.77	0.00
LPB#1btm gr	0.00	44250.05	0.77	0.00
LPB#1TA	0.00	44250.05	0.77	0.00
I Fy wt	0.00	43950.05	0.77	0.00
LPB#1HDG	0.00	49249.95	0.77	0.00
LPB#1dist.	0.00	49849.95	0.77	0.00
LPB#1dist.T	0.00	49849.95	0.77	0.00
LPShroud	0.00	51050.05	0.77	0.00
MH2 davit	0.00	50350.05	0.77	0.00
LPdwncm PP d	0.00	53999.96	0.77	0.00
MH1 davit	0.00	52649.96	0.77	0.00
LPGallary	0.00	54900.05	0.77	0.00
LPFPGdead	0.00	56700.05	0.77	0.00
LPFPGTA	0.00	56700.05	0.77	0.00
H1/H2 Fy wt	0.00	55049.95	0.77	0.00
Lifting wt	0.00	59199.95	0.77	0.00
LPdemister	0.00	60150.00	0.77	0.00
PSV	0.00	61975.00	0.77	0.00
LPDemisterT	0.00	60150.00	0.77	0.00
F Fy wt	0.00	60150.04	0.77	0.00
LPDavit	0.00	60150.04	0.77	0.00

The Natural Frequency for the Vessel (Ope...) is 0.95315 Hz.

Wind Load Calculation:

From	To	Wind Height mm	Wind Diameter mm	Wind Area mm ²	Wind Pressure kPa	Element Wind Load N
10	20	1800	8407.2	25221602	0.76606	11618.5
20	30	4650	8400	22680002	0.76606	11426.3
30	40	6750	8407.2	12610801	0.76606	5809.27
40	50	7499.98	8421.6	256.69	0.76606	0.11825
50	60	9000	8467.2	25401604	0.76606	13808.1
60	70	12000	8467.2	25401604	0.76606	14081.9



70	80	14450	8462.4	16078562	0.76606	10083.4
80	90	16900	8460	25379998	0.76606	11691.5
90	100	19900	8460	25379998	0.76606	14143.2
100	110	22400	8460	16920000	0.76606	10423.9
110	120	24150	8460	12689999	0.76606	5845.76
120	130	24900	8404.8	256.178	0.76606	2771.72
130	140	25525.7	8413.2	10516501	0.76606	4844.51
140	150	27654.1	8316	24948004	0.76606	11492.5
150	160	30654.1	8453.4	25360200	0.76606	11682.4
160	170	32775.7	8743.2	10928999	0.76606	7583.64
170	180	33425	8798.4	439920	0.76606	202.653
180	190	34950	8769.6	26308798	0.76606	14672.6
190	200	37950	8769.6	26308798	0.76606	14737.2
200	210	40950	8764.8	26294400	0.76606	12112.7
210	220	43950	8764.8	26294400	0.76606	14603.3
220	230	46850	8760	24528000	0.76606	13851.3
230	240	49750	8750.4	26251202	0.76606	12092.8
240	250	52750	8750.4	26251202	0.76606	14551.3
250	260	55750	8750.4	26251202	0.76606	33387.2
260	270	58700	8745.6	25362240	0.76606	15917
270	280	60967.1	8750.4	13296059	0.76606	14177.1

The Wind Loads calculated and printed in the Wind Load calculation report have been factored by the input scalar/load reduction factor of: 1.100.

Be sure the wind speed is in accordance with the specified wind design code.

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Earthquake Load Calculation Results:

Earthquake Load Calculation:

Input Values:

Seismic Design Code	ASCE/SEI 7-16
Seismic Load Reduction Scale Factor	1.000
Importance Factor	1.250
Table Value Fa	1.600
Table Value Fv	2.400
Max. Mapped Res. Acceleration [Ss]	0.147
Max. Eff. Ground Acceleration [S]	0.065
Force Modification Factor R	1.250
Site Class	D
Component Elevation Ratio z/h	0.000
Amplification Factor Ap	0.000
Force Factor	0.000
Consider Vertical Acceleration	Yes
Minimum Acceleration Multiplier	0.000
User Value of Sds (used if > 0)	0.000
User Value of Sd1 (used if > 0)	0.000
Moment Reduction Factor Tau	1.000

Seismic Analysis Results:

$$\begin{aligned} SMS &= Fa * Ss = 1.6 * 0.147 = 0.235 \\ SM1 &= Fv * S1 = 2.4 * 0.065 = 0.156 \\ SDS &= 2/3 * Sms = 2/3 * 0.235 = 0.157 \\ SD1 &= 2/3 * Sm1 = 2/3 * 0.156 = 0.104 \end{aligned}$$

Check Approximate Fundamental Period from 12.8-7 [Ta]:
 $= Ct * hnx$ where $Ct = 0.020$, $x = 0.75$ and $hn =$ Structural Height (ft.)
 $= 0.020(202.34580.75)$
 $= 1.073$ seconds

The Coefficient Cu from Table 12.8-1 is : 1.700

Fundamental Period (1/Frequency) [T]:
 $= (1/\text{Natural Frequency}) = (1/0.953)$
 $= 1.049$

Check the Value of T which is the smaller of $Cu \cdot Ta$ and T:
 $=$ Minimum Value of ($1.7 * 1.073$, 1.049) per 12.8.2
 $= 1.049$

Seismic Response Coefficient per equation 12.8-2 [Cs]:
 $= Sds / (R / I)$
 $= 0.157 / (1.25 / 1.25)$
 $= 0.157$

Maximum value of Cs per equation 12.8-3 [Cs]:
 $= Sd1 / (T(R / Ie))$
 $= 0.104 / (1.049(1.25 / 1.25))$
 $= 0.099$

Minimum value of Cs per equation 15.4-1 [Cs]:
 $= \max((0.044 * SDS * Ie), 0.030)$
 $= \max((0.044 * 0.1568 * 1.25), 0.030)$
 $= 0.030$

Total Base Shear [V]:
 $= Cs * W$ (Equation 12.8-1):
 $= 0.0991 * 18963718$
 $= 1879828.625$ N

Vertical load per 12.4-4, [Ev]:
 $= 0.2 * Sds * W$
 $= 0.2 * 0.157 * 18963718$



= 594702.2 N

Distribute the Base shear force to each element according to the equations

$F_x = C_{vx} * V$ (eqn. 12.8-11) and the vertical distribution factor

$C_{vx} = W_x * h_x k / (\text{Sum of } W_i * h_i k)$) and k is an exponent which is related to the period of Vibration.

In this case, the value of k was 1.2746

The Natural Frequency for the Vessel (Ope...) is 0.95315 Hz.

Earthquake Load Calculation:

From	To	Earthquake Height mm	Earthquake Weight N	Element Ope Load N	Element Emp Load N	Element Vertical Load N
10	20	1500	306216	661.218	942.397	574910
20	30	4350	211769	1776.4	2394.42	565307
30	40	6450	122849	1702.52	2247.96	558666
40	50	7199.98	1422777	22685.4	12364	554813
50	60	8700	1749214	35498.3	16385.2	510195
60	70	11700	1641973	48610	20574.8	455340
70	80	14150	757748	28584.3	21336.6	403847
80	90	16600	759248	35105.9	38782.3	380084
90	100	19600	528614	30206	37626.4	356274
100	110	22100	368278	24523.5	30356.4	339697
110	120	23850	439336	32239.3	29327.5	328148
120	130	24600	431467	32936.3	38893.5	314370
130	140	25225	60521.5	4770.08	5863.86	320632
140	150	27350	216803	18943.1	23188.3	318734
150	160	30350	120711	12043.3	14662.1	311935
160	170	32475	93812.9	10202.9	12377.4	308149
170	180	33125	1277210	142460	36531.5	305207
180	190	34650	1607575	189896	53343.9	265154
190	200	37650	1430785	187881	31096.5	214741
200	210	40650	985169	142645	25910.9	169871
210	220	43650	913743	144872	57197.6	138976
220	230	46550	629093	108263	69890.7	110321
230	240	49450	477846	88819.3	82717.6	90592.9
240	250	52450	172109	34484.6	40796.6	75607.6
250	260	55450	1285576	276509	79053.7	70210.3
260	270	58400	350660	80572.9	94785.5	29894.6
270	280	59875	602612	142938	91111.1	18897.9



Bending Moments due to user defined forces and moments.

User Force/Moment Shear and Bending

From	To	Distance to Support mm	Cumulative Shr Wind Cas N	Cumulative Shr Eqk Cas N	Wind Bending N-mm	Earthquake Bending N-mm
10	20	1500	101858	101858	5.854E+09	5.854E+09
20	30	4350	101858	101858	5.548E+09	5.548E+09
30	40	6450	101858	101858	5.273E+09	5.273E+09
40	50	7199.98	101858	101858	5.121E+09	5.121E+09
50	60	8700	101858	101858	5.121E+09	5.121E+09
60	70	11700	101858	101858	4.815E+09	4.815E+09
70	80	14150	101858	101858	4.509E+09	4.509E+09
80	90	16600	101858	101858	4.316E+09	4.316E+09
90	100	19600	101858	101858	4.010E+09	4.010E+09
100	110	22100	101858	101858	3.704E+09	3.704E+09
110	120	23850	101858	101858	3.500E+09	3.500E+09
120	130	24600	101858	101858	3.347E+09	3.347E+09
130	140	25225	81642.4	81642.4	3.266E+09	3.266E+09
140	150	27350	81642.4	81642.4	3.164E+09	3.164E+09
150	160	30350	81642.4	81642.4	2.919E+09	2.919E+09
160	170	32475	81642.4	81642.4	2.674E+09	2.674E+09
170	180	33125	81642.4	81642.4	2.572E+09	2.572E+09
180	190	34650	81642.4	81642.4	2.568E+09	2.568E+09
190	200	37650	81642.4	81642.4	2.323E+09	2.323E+09
200	210	40650	81642.4	81642.4	2077820160	2077820160
210	220	43650	81642.4	81642.4	1832793856	1832793856
220	230	46550	81642.4	81642.4	1587767552	1587767552
230	240	49450	81642.4	81642.4	1359076224	1359076224
240	250	52450	81642.4	81642.4	1114050048	1114050048
250	260	55450	81642.4	81642.4	869023808	869023808
260	270	58400	81642.4	81642.4	623997568	623997568
270	280	59875	81642.4	81642.4	387138784	387138784

Note:

The Wind Shears/Moments and the Earthquake Shears/Moments calculated and printed in the Wind/Earthquake Shear and Bending report have been factored by the input Scalar/Load reductions factors of;
 Wind: 1.100; Earthquake: 1.000.

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The following table is for the Operating Case.

Wind/Earthquake Shear, Bending:

From	To	Distance to Support mm	Cumulative Wind Shear N	Earthquake Shear N	Wind Bending N-mm	Earthquake Bending N-mm
10	20	1500	307612	1879828	10.33E+09	76.57E+09
20	30	4350	295994	1879167	9.424E+09	70.93E+09
30	40	6450	284567	1877391	8.640E+09	65.86E+09
40	50	7199.98	278758	1875688	8.217E+09	63.04E+09
50	60	8700	278758	1853003	8.217E+09	63.04E+09
60	70	11700	264950	1817505	7.401E+09	57.53E+09
70	80	14150	250868	1768895	6.627E+09	52.15E+09
80	90	16600	240784	1740310	6.160E+09	48.82E+09
90	100	19600	229093	1705204	5.455E+09	43.65E+09
100	110	22100	214950	1674998	4.789E+09	38.57E+09
110	120	23850	204526	1650475	4.369E+09	35.25E+09
120	130	24600	198680	1618236	4.066E+09	32.79E+09
130	140	25225	195908	1585299	4.066E+09	32.79E+09
140	150	27350	191064	1580529	3.824E+09	30.82E+09
150	160	30350	179571	1561586	3.268E+09	26.10E+09
160	170	32475	167889	1549543	2.747E+09	21.43E+09
170	180	33125	160305	1539340	2.542E+09	19.50E+09
180	190	34650	160103	1396880	2.534E+09	19.43E+09
190	200	37650	145430	1206984	2075159808	15.52E+09
200	210	40650	130693	1019103	1660807680	12.18E+09
210	220	43650	118580	876459	1286747264	9.335E+09
220	230	46550	103977	731587	952777152	6.922E+09
230	240	49450	90125.4	623324	680923904	5.024E+09
240	250	52450	78032.6	534504	428584800	3.287E+09
250	260	55450	63481.3	500020	216228064	1734147072
260	270	58400	30094.1	223511	75808312	648412480
270	280	59875	14177.1	142938	11589157	116845904

Note:

The Wind Shears/Moments and the Earthquake Shears/Moments calculated and printed in the Wind/Earthquake Shear and Bending report have been factored by the input Scalar/Load reductions factors of;
 Wind: 1.100; Earthquake: 1.000.

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Wind Deflection Calculations:

The following table is for Wind Deflection; Applied Forces Case.

Wind Load Deflection:

From	To	Cumulative Shear N	Centroid Deflection mm	Elem. End Deflection mm	Elem. Ang. Rotation
10	20	101858	0.006345	0.025157	0.16623E-04
20	30	101858	0.053302	0.092665	0.33242E-04
30	40	101858	0.11903	0.14824	0.40818E-04
40	50	101858	0.14824	0.14824	0.40818E-04
50	60	101858	0.21197	0.28059	0.47350E-04
60	70	101858	0.35397	0.43194	0.53481E-04
70	80	101858	0.48365	0.53715	0.57240E-04
80	90	101858	0.62512	0.71723	0.62748E-04
90	100	101858	0.81332	0.91323	0.67852E-04
100	110	101858	0.98189	1.05215	0.71030E-04
110	120	101858	1.10585	1.1604	0.73295E-04
120	130	101858	1.1604	1.1604	0.73295E-04
130	140	81642.4	1.20721	1.256	0.79622E-04
140	150	81642.4	1.38123	1.51776	0.94684E-04
150	160	81642.4	1.66487	1.82187	0.00010787
160	170	81642.4	1.89008	1.95983	0.00011284
170	180	81642.4	1.96266	1.96548	0.00011299
180	190	81642.4	2.14046	2.32609	0.00012718
190	200	81642.4	2.52181	2.72712	0.00013994
200	210	81642.4	2.94202	3.1665	0.00015271
210	220	81642.4	3.39997	3.64181	0.00016389
220	230	81642.4	3.87503	4.11545	0.00017416
230	240	81642.4	4.38015	4.65135	0.00018282
240	250	81642.4	4.92839	5.21064	0.00018976
250	260	81642.4	5.49746	5.78819	0.00019498
260	270	81642.4	6.07257	6.35982	0.00019894
270	280	81642.4	6.36479	6.36976	0.00019894

Allowable deflection at the Tower Top (For)(6.000"/100ft. Criteria)
 Allowable deflection : 299.500 Actual deflection : 6.370 mm

The following table is for Wind Deflection; Operating Case.

Wind Load Deflection:

From	To	Cumulative Shear N	Centroid Deflection mm	Elem. End Deflection mm	Elem. Ang. Rotation
10	20	307612	0.011268	0.0444	0.29154E-04
20	30	295994	0.093513	0.16158	0.57239E-04
30	40	284567	0.20692	0.25699	0.69867E-04
40	50	278758	0.25699	0.25699	0.69867E-04
50	60	278758	0.36589	0.4827	0.80419E-04
60	70	264950	0.60701	0.73843	0.89896E-04
70	80	250868	0.82519	0.91462	0.95495E-04
80	90	240784	1.06094	1.21318	0.00010339
90	100	229093	1.37099	1.53401	0.00011035
100	110	214950	1.64543	1.75893	0.0001145
110	120	204526	1.84535	1.93285	0.00011737
120	130	198680	1.93286	1.93286	0.00011737
130	140	195908	2.00748	2.0846	0.00012535
140	150	191064	2.27974	2.4884	0.00014338
150	160	179571	2.70925	2.94101	0.00015794
160	170	167889	3.04055	3.1417	0.00016309
170	180	160305	3.14577	3.14985	0.00016324
180	190	160103	3.4002	3.66084	0.00017696
190	200	145430	3.93076	4.20902	0.00018808



200	210	130693	4.49515	4.78869	0.00019795
210	220	118580	5.08871	5.39434	0.00020545
220	230	103977	5.68424	5.97821	0.00021128
230	240	90125.4	6.29684	6.61845	0.00021525
240	250	78032.6	6.94237	7.26801	0.00021754
250	260	63481.3	7.59483	7.92237	0.00021853
260	270	30094.1	8.23942	8.5567	0.00021885
270	280	14177.1	8.56217	8.56764	0.00021885

Critical Wind Velocity for Tower Vibration:

From	To	1st Crit. Wind Speed km/hr	2nd Crit. Wind Speed km/hr
10	20	143.851	899.071
20	30	143.728	898.301
30	40	143.851	899.071
40	50	144.098	900.611
50	60	144.878	905.487
60	70	144.878	905.487
70	80	144.796	904.974
80	90	144.755	904.717
90	100	144.755	904.717
100	110	144.755	904.717
110	120	144.755	904.717
120	130	143.81	898.814
130	140	143.954	899.712
140	150	142.291	889.318
150	160	144.642	904.011
160	170	149.6	935.002
170	180	150.545	940.906
180	190	150.052	937.826
190	200	150.052	937.826
200	210	149.97	937.312
210	220	149.97	937.312
220	230	149.888	936.799
230	240	149.724	935.773
240	250	149.724	935.773
250	260	149.724	935.773
260	270	149.641	935.259
270	280	149.724	935.773

Allowable deflection at the Tower Top (Ope)(6.000"/100ft. Criteria)

Allowable deflection : 299.500 Actual deflection : 8.568 mm

Total Deflection in the Operating Condition + Applied Forces:

Allowable deflection : 299.500 Actual deflection : 14.937 mm

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Longitudinal Stress Constants:

From	To	Metal Area New mm ²	Metal Area Corroded mm ²	Section Modulus New mm ³	Section Modulus Corroded mm ³
10	20	897767	863774	1517794560	1461003008
20	30	769514	735490	1302096128	1245104896
30	40	897767	863774	1517794560	1461003008
40	50	1281388	1154267	2.159E+09	1948071296
50	60	2092935	1966531	3.507E+09	3.301E+09
60	70	2092935	1966531	3.507E+09	3.301E+09
70	80	2049623	1923220	3.434E+09	3.228E+09
80	90	1964797	1943725	3.295E+09	3.261E+09
90	100	1964797	1943725	3.295E+09	3.261E+09
100	110	1964797	1943725	3.295E+09	3.261E+09
110	120	1964797	1943725	3.295E+09	3.261E+09
120	130	1109558	1088379	1869826048	1834676864
130	140	555518	521457	940932416	883654848
140	150	516402	482093	881137920	822974400
150	160	525000	490117	910767488	850636352
160	170	533670	498208	941148608	879004032
170	180	807066	673121	1432613504	1196849280
180	190	537138	403193	953444096	716888256
190	200	537138	403193	953444096	716888256
200	210	492235	358290	873730816	637042560
210	220	492235	358290	873730816	637042560
220	230	447359	313414	794077632	557256768
230	240	357692	335385	634910912	595484032
240	250	357692	335385	634910912	595484032
250	260	357692	335385	634910912	595484032
260	270	312888	290581	555380032	515931008
270	280	357692	335385	634910912	595484032



Longitudinal Allowable Stresses:

From	To	Tensile MPa	Hydrotest Tensile MPa	Compressive MPa	Hydrotest Compressive MPa
10	20	113.099	246.996	-96.3574	-96.3574
20	30	113.099	246.996	-92.0283	-92.0283
30	40	86.02	247	-96.3574	-96.3574
40	50	156.4	247	-103.397	-103.397
50	60	156.4	247	-113.713	-113.713
60	70	156.4	247	-113.713	-113.713
70	80	156.4	247	-113.359	-113.359
80	90	156.4	247	-113.547	-113.547
90	100	156.4	247	-113.547	-113.547
100	110	156.4	247	-113.547	-113.547
110	120	156.4	247	-113.547	-113.547
120	130	156.4	247	-102.128	-102.128
130	140	156.399	246.996	-81.7628	-81.7628
140	150	156.399	246.996	-78.753	-78.753
150	160	156.399	246.996	-78.2056	-78.2056
160	170	156.4	247	-77.6592	-77.6592
170	180	156.4	247	-86.6629	-86.6629
180	190	156.4	247	-62.9405	-62.9405
190	200	156.4	247	-62.9405	-62.9405
200	210	156.4	247	-55.9784	-55.9784
210	220	156.4	247	-55.9784	-55.9784
220	230	156.4	247	-49.0086	-49.0086
230	240	156.4	247	-52.5681	-52.5681
240	250	156.4	247	-52.5681	-52.5681
250	260	138	234	-52.5681	-52.5681
260	270	156.4	247	-45.5846	-45.5846
270	280	156.4	247	-52.5681	-52.5681



Longitudinal Stress Report



Longitudinal Operating and Empty Stresses are computed in the corroded condition. Stresses due to loads in the hydrostatic test cases have also been computed in the corroded condition.

Longitudinal Pressure Stresses due to:

From	To	Longitudinal Stress Internal Pressure MPa	Longitudinal Stress External Pressure MPa	Longitudinal Stress Hydrotest Pressure MPa
10	20	...	...	...
20	30	...	...	...
30	40	...	...	...
40	50	127.254	...	18.6708
50	60	73.854	...	10.8359
60	70	73.854	...	10.8359
70	80	75.5173	...	11.0799
80	90	74.6315	...	10.9499
90	100	74.6315	...	10.9499
100	110	74.6315	...	10.9499
110	120	74.6315	...	10.9499
120	130	134.638	...	19.7541
130	140	...	...	...
140	150	...	...	...
150	160	...	...	...
160	170	...	...	...
170	180	22.8988	-3.10488	35.5427
180	190	38.229	-5.14889	59.3378
190	200	38.229	-5.14889	59.3378
200	210	43.0201	-5.7877	66.7744
210	220	43.0201	-5.7877	66.7744
220	230	49.1798	-6.60899	76.3353
230	240	45.8289	-6.16221	71.1342
240	250	45.8289	-6.16221	71.1342
250	260	45.8289	-6.16221	71.1342
260	270	52.8952	-7.10438	82.1022
270	280	45.8289	-6.16221	71.1342

Longitudinal Stresses due to Weight Loads for these Conditions:

From	To	Wght. Str. Empty MPa	Wght. Str. Operating MPa	Wght. Str. Hydrotest MPa	Wght. Str. Emp. Mom. MPa	Wght. Str. Opr. Mom. MPa
10	20	-13.8936	-21.9564	-31.3634	8.5991	8.5991
20	30	-15.9005	-25.3696	-36.4175	10.0902	10.0902
30	40	-13.2938	-21.3566	-30.7637	8.56733	8.56733
40	50	-9.84175	-13.0597	-9.84175	6.42527	6.42527
50	60	-5.47628	-7.36509	-5.47628	3.79179	3.79179
60	70	-5.14558	-7.03439	-5.14558	3.7478	3.7478
70	80	-4.97907	-6.91042	-4.97907	3.79303	3.79303
80	90	-4.69687	-6.56073	-4.69687	3.72649	3.72649
90	100	-4.35342	-6.17008	-4.35342	3.72649	3.72649
100	110	-4.08144	-5.8981	-4.08144	3.68769	3.68769
110	120	-3.89196	-5.70861	-3.89196	3.6701	3.6701
120	130	-6.54691	-9.79125	-6.54691	6.17262	6.17262
130	140	-12.8371	-19.6087	-28.1107	12.6189	12.6189
140	150	-13.7598	-21.0843	-30.2805	13.5493	13.5493
150	160	-13.0921	-20.2967	-29.3424	13.1087	13.1087
160	170	-12.6372	-19.7247	-28.6235	12.6857	12.6857
170	180	-9.21398	-9.97793	-9.21398	9.23354	9.23354
180	190	-13.4729	-14.7483	-13.4729	11.6866	11.6866
190	200	-12.5465	-13.8219	-12.5465	11.5485	11.5485
200	210	-13.5697	-15.005	-13.5697	12.8643	12.8643



210	220	-10.9347	-12.3699	-10.9347	7.17968	7.17968
220	230	-9.9784	-11.2254	-9.9784	3.90898	3.90898
230	240	-8.30748	-8.61413	-8.30748	3.49073	3.49073
240	250	-7.18924	-7.18924	-7.18924	3.46768	3.46768
250	260	-6.67603	-6.67603	-6.67603	3.27751	3.27751
260	270	-3.28085	-3.28085	-3.28085	3.48077	3.48077
270	280	-1.79693	-1.79693	-1.79693	2.84457	2.84457

Longitudinal Stresses due to Weight Loads and Bending for these Conditions:

From	To	Wght. Str. Hyd. Mom. MPa	Bend. Str. Oper. Wind MPa	Bend. Str. Oper. Equ. MPa	Bend. Str. Hyd. Wind MPa	Bend. Str. Hyd. Equ. MPa
10	20	8.5991	7.06802	52.3938	3.53401	...
20	30	10.0902	7.56637	56.9499	3.78318	...
30	40	8.56733	5.91176	45.0627	2.95588	...
40	50	6.42527	4.21677	32.3508	2.10839	...
50	60	3.79179	2.48847	19.0914	1.24424	...
60	70	3.7478	2.24139	17.4234	1.12069	...
70	80	3.79303	2.05221	16.1495	1.02611	...
80	90	3.72649	1.8885	14.9661	0.94425	...
90	100	3.72649	1.67234	13.381	0.83617	...
100	110	3.68769	1.46806	11.8259	0.73403	...
110	120	3.6701	1.3394	10.806	0.6697	...
120	130	6.17262	2.2157	17.8694	1.10785	...
130	140	12.6189	4.60032	37.101	2.30016	...
140	150	13.5493	4.64561	37.4321	2.3228	...
150	160	13.1087	3.84091	30.6736	1.92045	...
160	170	12.6857	3.12397	24.3741	1.56199	...
170	180	9.23354	2.12294	16.288	1.06147	...
180	190	11.6866	3.53309	27.0905	1.76655	...
190	200	11.5485	2.89375	21.6417	1.44687	...
200	210	12.8643	2.60622	19.1122	1.30311	...
210	220	7.17968	2.01923	14.6485	1.00961	...
220	230	3.90898	1.70922	12.417	0.85461	...
230	240	3.49073	1.14311	8.43414	0.57156	...
240	250	3.46768	0.71949	5.51737	0.35975	...
250	260	3.27751	0.363	2.91123	0.1815	...
260	270	3.48077	0.14689	1.25638	0.073444	...
270	280	2.84457	0.019456	0.19616	0.0097278	...

Longitudinal Stresses due to these Conditions:

From	To	Vortex Shedding Operating Case MPa	Vortex Shedding Empty Case MPa	Vortex Shedding Test Case MPa	Earthquake Empty Case MPa
10	20	...	...	...	26.5917
20	30	...	...	...	28.867
30	40	...	...	...	22.8128
40	50	...	...	...	16.3657
50	60	...	...	...	9.65801
60	70	...	...	...	8.80037
70	80	...	...	...	8.13892
80	90	...	...	...	7.53089
90	100	...	...	...	6.72661
100	110	...	...	...	5.95748
110	120	...	...	...	5.46558
120	130	...	...	...	9.08278
130	140	...	...	...	18.858
140	150	...	...	...	19.1614
150	160	...	...	...	16.0656
160	170	...	...	...	13.2188
170	180	...	...	...	9.00993
180	190	...	...	...	14.9972
190	200	...	...	...	12.4894
200	210	...	...	...	11.4314



210	220	...	...	...	8.94232
220	230	...	...	...	7.6009
230	240	...	...	...	5.12189
240	250	...	...	...	3.37305
250	260	...	...	...	1.93536
260	270	...	...	...	0.9229
270	280	...	...	...	0.12503

Longitudinal Stresses due to Applied Axial Forces:

From	To	Longitudinal Stress Y Forces Wind MPa	Longitudinal Stress Y Forces Seismic MPa
10	20	-0.022916	0.66564
20	30	-0.026913	0.76868
30	40	-0.022916	0.64683
40	50	-0.017149	0.4807
50	60	-0.010065	0.25946
60	70	-0.010065	0.23156
70	80	-0.010292	0.21
80	90	-0.010184	0.19556
90	100	-0.010184	0.18331
100	110	-0.010184	0.17478
110	120	-0.010184	0.16884
120	130	-0.018187	0.28887
130	140	...	0.61493
140	150	...	0.6612
150	160	...	0.6365
160	170	...	0.61857
170	180	...	0.45346
180	190	...	0.65769
190	200	...	0.53265
200	210	...	0.47416
210	220	...	0.38792
220	230	...	0.35203
230	240	...	0.27014
240	250	...	0.22545
250	260	...	0.20936
260	270	...	0.10289
270	280	...	0.056352

Longitudinal Stresses due to User Forces and Moments:

From	To	Wind For/Mom Corroded MPa	Earthquake For/Mom Corroded MPa	Wind For/Mom No Corrosion MPa	Earthquake For/Mom No Corrosion MPa
10	20	4.00568	4.00568	3.8558	3.8558
20	30	4.45481	4.45481	4.25983	4.25983
30	40	3.60825	3.60825	3.47324	3.47324
40	50	2.62766	2.62766	2.37113	2.37113
50	60	1.55068	1.55068	1.45956	1.45956
60	70	1.4581	1.4581	1.37242	1.37242
70	80	1.3963	1.3963	1.31247	1.31247
80	90	1.32303	1.32303	1.30922	1.30922
90	100	1.22931	1.22931	1.21648	1.21648
100	110	1.13559	1.13559	1.12374	1.12374
110	120	1.07311	1.07311	1.06191	1.06191
120	130	1.82397	1.82397	1.78968	1.78968
130	140	3.69505	3.69505	3.47012	3.47012
140	150	3.84349	3.84349	3.58978	3.58978
150	160	3.43054	3.43054	3.20405	3.20405
160	170	3.04116	3.04116	2.84035	2.84035
170	180	2.14825	2.14825	1.79472	1.79472
180	190	3.58082	3.58082	2.6924	2.6924
190	200	3.23914	3.23914	2.43549	2.43549
200	210	3.26062	3.26062	2.37734	2.37734



210	220	2.87611	2.87611	2.09699	2.09699
220	230	2.84834	2.84834	1.99887	1.99887
230	240	2.28157	2.28157	2.13989	2.13989
240	250	1.87023	1.87023	1.75409	1.75409
250	260	1.45889	1.45889	1.36829	1.36829
260	270	1.20907	1.20907	1.12319	1.12319
270	280	0.64992	0.64992	0.60956	0.60956

Bending Stresses due to Blast Load:

From	To	Bending Stress MPa
10	20	...
20	30	...
30	40	...
40	50	...
50	60	...
60	70	...
70	80	...
80	90	...
90	100	...
100	110	...
110	120	...
120	130	...
130	140	...
140	150	...
150	160	...
160	170	...
170	180	...
180	190	...
190	200	...
200	210	...
210	220	...
220	230	...
230	240	...
240	250	...
250	260	...
260	270	...
270	280	...



Stress Combination Load Cases for Vertical Vessels:

Load Case Definition Key

IP = Longitudinal Stress due to Internal Pressure
 EP = Longitudinal Stress due to External Pressure
 HP = Longitudinal Stress due to Hydrotest Pressure
 NP = No Pressure
 EW = Longitudinal Stress due to Weight (No Liquid)
 OW = Longitudinal Stress due to Weight (Operating)
 HW = Longitudinal Stress due to Weight (Hydrotest)
 WI = Bending Stress due to Wind Moment (Operating)
 EQ = Bending Stress due to Earthquake Moment (Operating)
 EE = Bending Stress due to Earthquake Moment (Empty)
 HI = Bending Stress due to Wind Moment (Hydrotest)
 HE = Bending Stress due to Earthquake Moment (Hydrotest)
 WE = Bending Stress due to Wind Moment (Empty) (no CA)
 WF = Bending Stress due to Wind Moment (Filled) (no CA)
 CW = Longitudinal Stress due to Weight (Empty) (no CA)
 VO = Bending Stress due to Vortex Shedding Loads (Ope)
 VE = Bending Stress due to Vortex Shedding Loads (Emp)
 VF = Bending Stress due to Vortex Shedding Loads (Test No CA.)
 FW = Axial Stress due to Vertical Forces for the Wind Case
 FS = Axial Stress due to Vertical Forces for the Seismic Case
 BW = Bending Stress due to Lat. Forces for the Wind Case, Corroded
 BS = Bending Stress due to Lat. Forces for the Seismic Case, Corroded
 BN = Bending Stress due to Lat. Forces for the Wind Case, UnCorroded
 BU = Bending Stress due to Lat. Forces for the Seismic Case, UnCorroded
 BL = Bending Stress due to Blast Loads, Corroded

General Notes:

Case types HI and HE are in the Corroded condition.

Case types WE, WF, and CW are in the Un-Corroded condition.

A blank stress and stress ratio indicates that the corresponding stress comprising those components that did not contribute to that type of stress.

For Division 2, the overall stress intensity is computed in accordance with the Section VIII Division II Example problem manual PTB-3, example 4.3.6, stress due to combined loads. The stresses are combined per equation 4.3.44 to determine the overall stress intensity. The unity check is from Section 4.4.12 equations 4.4.85 and 4.4.90.

An asterisk (*) in the final column denotes overstress.

Note: Performing Combined Stress Analysis per VIII-2 paragraph 4.3.10.2.

Analysis of Load Case 1 : NP+EW+WI+FW+BW

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	33.59	113.10	-33.59	108.67	0.2970	0.3091	0.1725
20	38.04	113.10	-38.04	103.20	0.3363	0.3686	0.2012
30	31.40	86.02	-31.40	104.69	0.3651	0.3000	0.1643
40	23.13	156.40	-23.13	135.99	0.1479	0.1701	0.2152
50	13.32	156.40	-13.32	157.15	0.0851	0.0847	0.0498
60	12.60	156.40	-12.60	157.15	0.0806	0.0802	0.0473
70	12.23	156.40	-12.23	156.12	0.0782	0.0783	0.0463
80	11.65	156.40	-11.65	156.93	0.0745	0.0742	0.0441
90	10.99	156.40	-10.99	156.93	0.0703	0.0700	0.0422
100	10.38	156.40	-10.38	156.93	0.0664	0.0662	0.0400
110	9.98	156.40	-9.98	156.93	0.0638	0.0636	0.0387
120	16.78	156.40	-16.78	134.10	0.1073	0.1251	0.0931
130	33.75	156.40	-33.75	86.95	0.2158	0.3882	0.2159
140	35.80	156.40	-35.80	83.22	0.2289	0.4301	0.2337
150	33.47	156.40	-33.47	82.61	0.2140	0.4052	0.2171
160	31.49	156.40	-31.49	82.37	0.2013	0.3823	0.2012
170	22.72	156.40	-22.72	76.01	0.1453	0.2989	0.3689



180	32.27	156.40	-32.27	77.24	0.2064	0.4178	0.2133
190	30.23	156.40	-30.23	77.60	0.1933	0.3895	0.2006
200	32.30	156.40	-32.30	73.25	0.2065	0.4410	0.2200
210	23.01	156.40	-23.01	73.25	0.1471	0.3141	0.1600
220	18.44	156.40	-18.44	66.70	0.1179	0.2766	0.1596
230	15.22	156.40	-15.22	70.20	0.0973	0.2169	0.1266
240	13.25	156.40	-13.25	70.20	0.0847	0.1887	0.1096
250	11.78	138.00	-11.78	70.20	0.0853	0.1677	0.1017
260	8.12	156.40	-8.12	63.10	0.0519	0.1286	0.0606
270	5.31	156.40	-5.31	38.36	0.0340	0.1384	0.1950

Analysis of Load Case 2 : NP+EW+EE+FS+BS

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	52.42	113.10	-52.42	108.67	0.4635	0.4824	0.3436
20	58.54	113.10	-58.54	103.20	0.5176	0.5673	0.3951
30	47.64	86.02	-47.64	104.69	0.5538	0.4550	0.3178
40	34.78	156.40	-34.78	136.40	0.2224	0.2550	0.3335
50	20.22	156.40	-20.22	157.15	0.1293	0.1287	0.0953
60	18.92	156.40	-18.92	157.15	0.1210	0.1204	0.0890
70	18.10	156.40	-18.10	156.12	0.1157	0.1159	0.0852
80	17.08	156.40	-17.08	156.93	0.1092	0.1088	0.0801
90	15.85	156.40	-15.85	156.93	0.1014	0.1010	0.0743
100	14.69	156.40	-14.69	156.93	0.0939	0.0936	0.0686
110	13.93	156.40	-13.93	156.93	0.0891	0.0888	0.0650
120	23.34	156.40	-23.34	134.21	0.1492	0.1739	0.1224
130	47.39	156.40	-47.39	86.95	0.3030	0.5451	0.3630
140	49.65	156.40	-49.65	83.22	0.3175	0.5966	0.3877
150	45.06	156.40	-45.06	82.61	0.2881	0.5454	0.3473
160	40.96	156.40	-40.96	82.37	0.2619	0.4973	0.3089
170	29.15	156.40	-29.15	77.25	0.1864	0.3774	0.4856
180	43.08	156.40	-43.08	77.24	0.2754	0.5578	0.3434
190	39.29	156.40	-39.29	77.60	0.2512	0.5063	0.3095
200	40.65	156.40	-40.65	73.25	0.2599	0.5550	0.3237
210	29.54	156.40	-29.54	73.25	0.1889	0.4034	0.2231
220	23.98	156.40	-23.98	66.70	0.1534	0.3596	0.1757
230	18.93	156.40	-18.93	70.20	0.1210	0.2697	0.1431
240	15.67	156.40	-15.67	70.20	0.1002	0.2233	0.1465
250	13.14	138.00	-13.14	70.20	0.0952	0.1872	0.1742
260	8.79	156.40	-8.79	63.10	0.0562	0.1393	0.2321
270	5.36	156.40	-5.36	38.82	0.0343	0.1381	0.2101

Analysis of Load Case 3 : NP+OW+WI+FW+BW

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	41.65	113.10	-41.65	108.67	0.3683	0.3833	0.2228
20	47.51	113.10	-47.51	103.20	0.4201	0.4603	0.2702
30	39.47	86.02	-39.47	104.69	0.4588	0.3770	0.2246
40	26.35	156.40	-26.35	71.03	0.1685	0.3709	0.5254
50	15.21	156.40	-15.21	157.15	0.0972	0.0968	0.0533
60	14.49	156.40	-14.49	157.15	0.0927	0.0922	0.0509
70	14.16	156.40	-14.16	156.12	0.0906	0.0907	0.0503
80	13.51	156.40	-13.51	156.93	0.0864	0.0861	0.0475
90	12.81	156.40	-12.81	156.93	0.0819	0.0816	0.0447
100	12.20	156.40	-12.20	156.93	0.0780	0.0777	0.0427
110	11.80	156.40	-11.80	156.93	0.0755	0.0752	0.0413
120	20.02	156.40	-20.02	134.05	0.1280	0.1494	0.1067
130	40.52	156.40	-40.52	86.95	0.2591	0.4660	0.2451
140	43.12	156.40	-43.12	83.22	0.2757	0.5182	0.2741
150	40.68	156.40	-40.68	82.61	0.2601	0.4924	0.2652
160	38.58	156.40	-38.58	82.37	0.2466	0.4683	0.2582
170	23.48	156.40	-23.48	34.06	0.1501	0.6895	...
180	33.55	156.40	-33.55	77.24	0.2145	0.4344	0.2133
190	31.50	156.40	-31.50	77.60	0.2014	0.4060	0.2006
200	33.74	156.40	-33.74	73.25	0.2157	0.4606	0.2200
210	24.44	156.40	-24.44	73.25	0.1563	0.3337	0.1810
220	19.69	156.40	-19.69	66.70	0.1259	0.2953	0.1796
230	15.53	156.40	-15.53	70.20	0.0993	0.2212	0.1313



240	13.25	156.40	-13.25	70.20	0.0847	0.1887	0.1096
250	11.78	138.00	-11.78	70.20	0.0853	0.1677	0.1017
260	8.12	156.40	-8.12	63.10	0.0519	0.1286	0.0606
270	5.31	156.40	-5.31	38.36	0.0340	0.1384	0.1950

Analysis of Load Case 4 : NP+OW+EQ+FS+BS

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	86.29	113.10	-86.29	108.67	0.7630	0.7941	0.5698
20	96.10	113.10	-96.10	103.20	0.8497	0.9311	0.6507
30	77.95	86.02	-77.95	104.69	0.9062	0.7445	0.5198
40	53.98	156.40	-53.98	136.71	0.3452	0.3949	0.5545
50	31.54	156.40	-31.54	157.15	0.2017	0.2007	0.1553
60	29.43	156.40	-29.43	157.15	0.1882	0.1873	0.1438
70	28.04	156.40	-28.04	156.12	0.1793	0.1796	0.1365
80	26.38	156.40	-26.38	156.93	0.1687	0.1681	0.1274
90	24.32	156.40	-24.32	156.93	0.1555	0.1550	0.1167
100	22.37	156.40	-22.37	156.93	0.1430	0.1426	0.1059
110	21.09	156.40	-21.09	156.93	0.1348	0.1344	0.0989
120	35.37	156.40	-35.37	134.31	0.2261	0.2633	0.1757
130	72.41	156.40	-72.41	86.95	0.4630	0.8327	0.5513
140	75.25	156.40	-75.25	83.22	0.4811	0.9042	0.5815
150	66.87	156.40	-66.87	82.61	0.4276	0.8095	0.5030
160	59.21	156.40	-59.21	82.37	0.3786	0.7188	0.4280
170	37.19	156.40	-37.19	77.70	0.2378	0.4787	0.7176
180	56.45	156.40	-56.45	77.24	0.3609	0.7308	0.4806
190	49.72	156.40	-49.72	77.60	0.3179	0.6407	0.4133
200	49.77	156.40	-49.77	73.25	0.3182	0.6794	0.4139
210	36.69	156.40	-36.69	73.25	0.2346	0.5009	0.2902
220	30.05	156.40	-30.05	66.70	0.1921	0.4505	0.2346
230	22.55	156.40	-22.55	70.20	0.1442	0.3212	0.1701
240	17.82	156.40	-17.82	70.20	0.1139	0.2538	0.1465
250	14.11	138.00	-14.11	70.20	0.1023	0.2011	0.1742
260	9.12	156.40	-9.12	63.10	0.0583	0.1446	0.2321
270	5.43	156.40	-5.43	38.99	0.0347	0.1393	0.2177

Analysis of Load Case 5 : NP+HW+HI

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	43.50	247.00	-43.50	96.36	0.1761	0.4514	0.3174
20	50.29	247.00	-50.29	92.03	0.2036	0.5465	0.3868
30	42.29	247.00	-42.29	96.36	0.1712	0.4389	0.3220
40	18.38	247.00	-18.38	103.40	0.0744	0.1777	0.5616
50	10.51	247.00	-10.51	113.71	0.0426	0.0924	0.0392
60	10.01	247.00	-10.01	113.71	0.0405	0.0881	0.0369
70	9.80	247.00	-9.80	113.36	0.0397	0.0864	0.0359
80	9.37	247.00	-9.37	113.55	0.0379	0.0825	0.0337
90	8.92	247.00	-8.92	113.55	0.0361	0.0785	0.0312
100	8.50	247.00	-8.50	113.55	0.0344	0.0749	0.0293
110	8.23	247.00	-8.23	113.55	0.0333	0.0725	0.0279
120	13.83	247.00	-13.83	102.13	0.0560	0.1354	0.0796
130	43.03	247.00	-43.03	81.76	0.1742	0.5263	0.3497
140	46.15	247.00	-46.15	78.75	0.1869	0.5860	0.3916
150	44.37	247.00	-44.37	78.21	0.1796	0.5674	0.3813
160	42.87	247.00	-42.87	77.66	0.1736	0.5520	0.3727
170	19.51	247.00	-19.51	86.66	0.0790	0.2251	0.2174
180	26.93	247.00	-26.93	62.94	0.1090	0.4278	0.1863
190	25.54	247.00	-25.54	62.94	0.1034	0.4058	0.1727
200	27.74	247.00	-27.74	55.98	0.1123	0.4955	0.1972
210	19.12	247.00	-19.12	55.98	0.0774	0.3416	0.1589
220	14.74	247.00	-14.74	49.01	0.0597	0.3008	0.1582
230	12.37	247.00	-12.37	52.57	0.0501	0.2353	0.1256
240	11.02	247.00	-11.02	52.57	0.0446	0.2096	0.1087
250	10.14	234.00	-10.14	52.57	0.0433	0.1928	0.1009
260	6.84	247.00	-6.84	45.58	0.0277	0.1499	0.0549
270	4.65	247.00	-4.65	52.57	0.0188	0.0885	...

Analysis of Load Case 6 : NP+HW+HE



From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	39.96	247.00	-39.96	96.36	0.1618	0.4147	0.3174
20	46.51	247.00	-46.51	92.03	0.1883	0.5054	0.3868
30	39.33	247.00	-39.33	96.36	0.1592	0.4082	0.3220
40	16.27	247.00	-16.27	103.40	0.0659	0.1573	0.5203
50	9.27	247.00	-9.27	113.71	0.0375	0.0815	0.0392
60	8.89	247.00	-8.89	113.71	0.0360	0.0782	0.0369
70	8.77	247.00	-8.77	113.36	0.0355	0.0774	0.0359
80	8.42	247.00	-8.42	113.55	0.0341	0.0742	0.0337
90	8.08	247.00	-8.08	113.55	0.0327	0.0712	0.0312
100	7.77	247.00	-7.77	113.55	0.0315	0.0684	0.0293
110	7.56	247.00	-7.56	113.55	0.0306	0.0666	0.0279
120	12.72	247.00	-12.72	102.13	0.0515	0.1245	0.0747
130	40.73	247.00	-40.73	81.76	0.1649	0.4981	0.3497
140	43.83	247.00	-43.83	78.75	0.1775	0.5565	0.3916
150	42.45	247.00	-42.45	78.21	0.1719	0.5428	0.3813
160	41.31	247.00	-41.31	77.66	0.1672	0.5319	0.3727
170	18.45	247.00	-18.45	86.66	0.0747	0.2129	0.3958
180	25.16	247.00	-25.16	62.94	0.1019	0.3997	0.1863
190	24.09	247.00	-24.09	62.94	0.0976	0.3828	0.1727
200	26.43	247.00	-26.43	55.98	0.1070	0.4722	0.1972
210	18.11	247.00	-18.11	55.98	0.0733	0.3236	0.1589
220	13.89	247.00	-13.89	49.01	0.0562	0.2834	0.1582
230	11.80	247.00	-11.80	52.57	0.0478	0.2244	0.1256
240	10.66	247.00	-10.66	52.57	0.0431	0.2027	0.1087
250	9.95	234.00	-9.95	52.57	0.0425	0.1893	0.1009
260	6.76	247.00	-6.76	45.58	0.0274	0.1483	0.0549
270	4.64	247.00	-4.64	52.57	0.0188	0.0883	...

Analysis of Load Case 7 : IP+OW+WI+FW+BW

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	41.65	113.10	-41.65	108.67	0.3683	0.3833	0.2228
20	47.51	113.10	-47.51	103.20	0.4201	0.4603	0.2702
30	39.47	86.02	-39.47	104.69	0.4588	0.3770	0.2246
40	128.98	156.40		71.03	0.8247		0.5254
50	132.99	156.40		157.15	0.8503		0.0533
60	131.85	156.40		157.15	0.8430		0.0509
70	133.63	156.40		156.12	0.8544		0.0503
80	131.82	156.40		156.93	0.8428		0.0475
90	131.75	156.40		156.93	0.8424		0.0447
100	131.68	156.40		156.93	0.8420		0.0427
110	131.64	156.40		156.93	0.8417		0.0413
120	134.84	156.40		102.13	0.8621		...
130	40.52	156.40	-40.52	86.95	0.2591	0.4660	0.2451
140	43.12	156.40	-43.12	83.22	0.2757	0.5182	0.2741
150	40.68	156.40	-40.68	82.61	0.2601	0.4924	0.2652
160	38.58	156.40	-38.58	82.37	0.2466	0.4683	0.2582
170	50.55	156.40	-0.58	34.06	0.3232	0.0171	...
180	86.93	156.40		77.24	0.5558		0.2133
190	79.86	156.40		77.60	0.5106		0.2006
200	82.22	156.40		73.25	0.5257		0.2200
210	78.61	156.40		73.25	0.5026		0.1810
220	87.61	156.40		66.70	0.5602		0.1796
230	96.54	156.40		70.20	0.6173		0.1313
240	96.01	156.40		70.20	0.6139		0.1096
250	95.69	138.00		70.20	0.6934		0.1017
260	109.40	156.40		63.10	0.6995		0.0606
270	84.71	156.40		52.57	0.5416		...

Analysis of Load Case 8 : IP+OW+EQ+FS+BS

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	86.29	113.10	-86.29	108.67	0.7630	0.7941	0.5698
20	96.10	113.10	-96.10	103.20	0.8497	0.9311	0.6507
30	77.95	86.02	-77.95	104.69	0.9062	0.7445	0.5198



40	144.98	156.40		136.71	0.9270	0.5545
50	136.10	156.40		157.15	0.8702	0.1553
60	134.51	156.40		157.15	0.8600	0.1438
70	135.92	156.40		156.12	0.8691	0.1365
80	133.85	156.40		156.93	0.8558	0.1274
90	133.45	156.40		156.93	0.8532	0.1167
100	133.09	156.40		156.93	0.8510	0.1059
110	132.87	156.40		156.93	0.8496	0.0989
120	143.52	156.40		102.13	0.9177	...
130	72.41	156.40	-72.41	86.95	0.4630	0.8327
140	75.25	156.40	-75.25	83.22	0.4811	0.9042
150	66.87	156.40	-66.87	82.61	0.4276	0.8095
160	59.21	156.40	-59.21	82.37	0.3786	0.7188
170	58.72	156.40	-14.30	77.70	0.3755	0.1840
180	99.54	156.40	-18.22	77.24	0.6365	0.2359
190	89.32	156.40	-11.49	77.60	0.5711	0.1481
200	90.03	156.40	-6.75	73.25	0.5756	0.0921
210	83.25	156.40		73.25	0.5323	0.2902
220	90.52	156.40		66.70	0.5787	0.2346
230	98.51	156.40		70.20	0.6298	0.1701
240	97.13	156.40		70.20	0.6211	0.1465
250	96.21	138.00		70.20	0.6971	0.1742
260	109.57	156.40		63.10	0.7006	0.2321
270	84.72	156.40		52.57	0.5417	...

Analysis of Load Case 9 : EP+OW+WI+FW+BW

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	41.65	113.10	-41.65	108.81	0.3683	0.3828	0.2221
20	47.51	113.10	-47.51	103.37	0.4201	0.4596	0.2693
30	39.47	86.02	-39.47	104.92	0.4588	0.3762	0.2234
40	26.35	156.40	-26.35	71.67	0.1685	0.3676	0.5208
50	15.21	156.40	-15.21	158.58	0.0972	0.0959	0.0526
60	14.49	156.40	-14.49	158.58	0.0927	0.0914	0.0503
70	14.16	156.40	-14.16	157.57	0.0906	0.0899	0.0497
80	13.51	156.40	-13.51	158.35	0.0864	0.0853	0.0469
90	12.81	156.40	-12.81	158.35	0.0819	0.0809	0.0441
100	12.20	156.40	-12.20	158.35	0.0780	0.0770	0.0422
110	11.80	156.40	-11.80	158.35	0.0755	0.0745	0.0408
120	20.02	156.40	-20.02	135.16	0.1280	0.1481	0.1058
130	40.52	156.40	-40.52	86.95	0.2591	0.4660	0.2451
140	43.12	156.40	-43.12	83.22	0.2757	0.5182	0.2741
150	40.68	156.40	-40.68	82.61	0.2601	0.4924	0.2652
160	38.58	156.40	-38.58	82.37	0.2466	0.4683	0.2582
170	26.59	156.40	-26.59	34.06	0.1700	0.7807	...
180	38.70	156.40	-38.70	77.24	0.2474	0.5010	0.6563
190	36.65	156.40	-36.65	77.60	0.2343	0.4723	0.6238
200	39.52	156.40	-39.52	73.25	0.2527	0.5396	0.7417
210	30.23	156.40	-30.23	73.25	0.1933	0.4127	0.6802
220	26.30	156.40	-26.30	66.70	0.1682	0.3943	0.8709
230	21.69	156.40	-21.69	70.20	0.1387	0.3090	0.6964
240	19.41	156.40	-19.41	70.20	0.1241	0.2765	0.6644
250	17.94	138.00	-17.94	70.20	0.1300	0.2555	0.6513
260	15.22	156.40	-15.22	63.10	0.0973	0.2412	0.7621
270	11.47	156.40	-11.47	41.90	0.0734	0.2738	0.0090

Analysis of Load Case 10 : EP+OW+EQ+FS+BS

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	86.29	113.10	-86.29	108.81	0.7630	0.7930	0.5692
20	96.10	113.10	-96.10	103.37	0.8497	0.9297	0.6499
30	77.95	86.02	-77.95	104.92	0.9062	0.7430	0.5189
40	53.98	156.40	-53.98	137.74	0.3452	0.3919	0.5504
50	31.54	156.40	-31.54	158.58	0.2017	0.1989	0.1539
60	29.43	156.40	-29.43	158.58	0.1882	0.1856	0.1425
70	28.04	156.40	-28.04	157.57	0.1793	0.1779	0.1352
80	26.38	156.40	-26.38	158.35	0.1687	0.1666	0.1262
90	24.32	156.40	-24.32	158.35	0.1555	0.1536	0.1156



100	22.37	156.40	-22.37	158.35	0.1430	0.1413	0.1050
110	21.09	156.40	-21.09	158.35	0.1348	0.1332	0.0981
120	35.37	156.40	-35.37	135.40	0.2261	0.2612	0.1742
130	72.41	156.40	-72.41	86.95	0.4630	0.8327	0.5513
140	75.25	156.40	-75.25	83.22	0.4811	0.9042	0.5815
150	66.87	156.40	-66.87	82.61	0.4276	0.8095	0.5030
160	59.21	156.40	-59.21	82.37	0.3786	0.7188	0.4280
170	40.30	156.40	-40.30	77.70	0.2577	0.5187	0.7176
180	61.60	156.40	-61.60	77.24	0.3938	0.7975	0.8475
190	54.87	156.40	-54.87	77.60	0.3508	0.7070	0.7668
200	55.56	156.40	-55.56	73.25	0.3552	0.7585	0.8557
210	42.47	156.40	-42.47	73.25	0.2716	0.5799	0.7340
220	36.66	156.40	-36.66	66.70	0.2344	0.5496	0.9066
230	28.71	156.40	-28.71	70.20	0.1836	0.4090	0.7193
240	23.98	156.40	-23.98	70.20	0.1533	0.3416	0.6812
250	20.28	138.00	-20.28	70.20	0.1469	0.2888	0.6693
260	16.23	156.40	-16.23	63.10	0.1038	0.2572	0.7995
270	11.59	156.40	-11.59	42.47	0.0741	0.2730	0.0326

Analysis of Load Case 11 : HP+HW+HI

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	43.50	247.00	-43.50	96.36	0.1761	0.4514	0.3174
20	50.29	247.00	-50.29	92.03	0.2036	0.5465	0.3868
30	42.29	247.00	-42.29	96.36	0.1712	0.4389	0.3220
40	18.78	247.00		103.40	0.0760		0.5616
50	23.00	247.00		113.71	0.0931		0.0392
60	22.76	247.00		113.71	0.0922		0.0369
70	23.03	247.00		113.36	0.0932		0.0359
80	23.93	247.00		113.55	0.0969		0.0337
90	23.73	247.00		113.55	0.0961		0.0312
100	23.55	247.00		113.55	0.0953		0.0293
110	23.44	247.00		113.55	0.0949		0.0279
120	21.19	247.00		102.13	0.0858		...
130	43.03	247.00	-43.03	81.76	0.1742	0.5263	0.3497
140	46.15	247.00	-46.15	78.75	0.1869	0.5860	0.3916
150	44.37	247.00	-44.37	78.21	0.1796	0.5674	0.3813
160	42.87	247.00	-42.87	77.66	0.1736	0.5520	0.3727
170	53.61	247.00		86.66	0.2170		0.2174
180	88.10	247.00		62.94	0.3567		0.1863
190	87.83	247.00		62.94	0.3556		0.1727
200	95.42	247.00		55.98	0.3863		0.1972
210	94.67	247.00		55.98	0.3833		0.1589
220	104.18	247.00		49.01	0.4218		0.1582
230	154.98	247.00		52.57	0.6274		0.1256
240	154.73	247.00		52.57	0.6264		0.1087
250	154.57	234.00		52.57	0.6606		0.1009
260	175.72	247.00		45.58	0.7114		0.0549
270	137.43	247.00		52.57	0.5564		...

Analysis of Load Case 12 : HP+HW+HE

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	39.96	247.00	-39.96	96.36	0.1618	0.4147	0.3174
20	46.51	247.00	-46.51	92.03	0.1883	0.5054	0.3868
30	39.33	247.00	-39.33	96.36	0.1592	0.4082	0.3220
40	17.85	247.00		103.40	0.0723		0.5203
50	22.42	247.00		113.71	0.0908		0.0392
60	22.26	247.00		113.71	0.0901		0.0369
70	22.57	247.00		113.36	0.0914		0.0359
80	23.52	247.00		113.55	0.0952		0.0337
90	23.37	247.00		113.55	0.0946		0.0312
100	23.25	247.00		113.55	0.0941		0.0293
110	23.17	247.00		113.55	0.0938		0.0279
120	20.71	247.00		102.13	0.0838		...
130	40.73	247.00	-40.73	81.76	0.1649	0.4981	0.3497
140	43.83	247.00	-43.83	78.75	0.1775	0.5565	0.3916
150	42.45	247.00	-42.45	78.21	0.1719	0.5428	0.3813



160	41.31	247.00	-41.31	77.66	0.1672	0.5319	0.3727
170	53.34	247.00		86.66	0.2160		0.3958
180	87.76	247.00		62.94	0.3553		0.1863
190	87.58	247.00		62.94	0.3546		0.1727
200	95.22	247.00		55.98	0.3855		0.1972
210	94.57	247.00		55.98	0.3829		0.1589
220	104.09	247.00		49.01	0.4214		0.1582
230	154.87	247.00		52.57	0.6270		0.1256
240	154.66	247.00		52.57	0.6262		0.1087
250	154.54	234.00		52.57	0.6604		0.1009
260	175.71	247.00		45.58	0.7114		0.0549
270	137.43	247.00		52.57	0.5564		...

Analysis of Load Case 13 : IP+WE+EW

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	22.49	113.10	-22.49	108.67	0.1989	0.2070	0.1408
20	25.99	113.10	-25.99	103.20	0.2298	0.2518	0.1692
30	21.86	86.02	-21.86	104.69	0.2541	0.2088	0.1396
40	127.28	156.40		71.03	0.8138		0.2776
50	132.34	156.40		157.15	0.8462		0.0396
60	131.28	156.40		157.15	0.8394		0.0372
70	133.12	156.40		156.12	0.8512		0.0362
80	131.36	156.40		156.93	0.8399		0.0340
90	131.33	156.40		156.93	0.8397		0.0315
100	131.31	156.40		156.93	0.8396		0.0295
110	131.30	156.40		156.93	0.8395		0.0281
120	134.45	156.40		102.13	0.8597		...
130	25.46	156.40	-25.46	86.95	0.1628	0.2928	0.1605
140	27.31	156.40	-27.31	83.22	0.1746	0.3281	0.1789
150	26.20	156.40	-26.20	82.61	0.1675	0.3171	0.1710
160	25.32	156.40	-25.32	82.37	0.1619	0.3074	0.1654
170	48.19	156.40		34.06	0.3081		...
180	83.41	156.40		77.24	0.5333		0.1875
190	76.93	156.40		77.60	0.4919		0.1738
200	79.48	156.40		73.25	0.5082		0.1986
210	76.87	156.40		73.25	0.4915		0.1600
220	86.49	156.40		66.70	0.5530		0.1596
230	95.70	156.40		70.20	0.6119		0.1266
240	95.46	156.40		70.20	0.6104		0.1096
250	95.33	138.00		70.20	0.6908		0.1017
260	109.19	156.40		63.10	0.6982		0.0553
270	84.65	156.40		52.57	0.5412		...

Analysis of Load Case 14 : IP+WF+CW

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	21.13	113.10	-21.13	108.67	0.1868	0.1944	0.2141
20	24.25	113.10	-24.25	103.20	0.2144	0.2350	0.2580
30	20.55	86.02	-20.55	104.69	0.2389	0.1963	0.2158
40	123.66	156.40		71.03	0.7907		0.2893
50	132.16	156.40		157.15	0.8450		0.0500
60	131.12	156.40		157.15	0.8383		0.0477
70	132.97	156.40		156.12	0.8502		0.0471
80	131.23	156.40		156.93	0.8391		0.0469
90	131.21	156.40		156.93	0.8390		0.0441
100	131.20	156.40		156.93	0.8389		0.0422
110	131.19	156.40		156.93	0.8388		0.0408
120	130.10	156.40		102.13	0.8319		...
130	18.41	156.40	-18.41	86.95	0.1177	0.2117	0.2301
140	19.68	156.40	-19.68	83.22	0.1259	0.2365	0.2559
150	18.95	156.40	-18.95	82.61	0.1212	0.2294	0.2475
160	18.41	156.40	-18.41	82.37	0.1177	0.2236	0.2410
170	44.78	156.40		34.06	0.2863		...
180	79.17	156.40		77.24	0.5062		0.1540
190	73.18	156.40		77.60	0.4679		0.1437
200	75.70	156.40		73.25	0.4840		0.1598
210	75.23	156.40		73.25	0.4810		0.1318



220	85.72	156.40	66.70	0.5481	0.1258
230	94.99	156.40	70.20	0.6073	0.1231
240	94.77	156.40	70.20	0.6059	0.1027
250	94.69	138.00	70.20	0.6862	0.0954
260	108.70	156.40	63.10	0.6950	0.0514
270	84.44	156.40	52.57	0.5399	...

Analysis of Load Case 15 : IP+VO+OW

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	30.56	113.10	-30.56	108.67	0.2702	0.2812	0.2226
20	35.46	113.10	-35.46	103.20	0.3135	0.3436	0.2700
30	29.92	86.02	-29.92	104.69	0.3479	0.2858	0.2243
40	125.83	156.40		71.03	0.8046		0.3911
50	132.52	156.40		157.15	0.8473		0.0532
60	131.45	156.40		157.15	0.8405		0.0508
70	133.28	156.40		156.12	0.8522		0.0502
80	131.51	156.40		156.93	0.8408		0.0474
90	131.47	156.40		156.93	0.8406		0.0446
100	131.45	156.40		156.93	0.8405		0.0426
110	131.43	156.40		156.93	0.8403		0.0413
120	132.87	156.40		102.13	0.8495		...
130	32.23	156.40	-32.23	86.95	0.2061	0.3706	0.2451
140	34.63	156.40	-34.63	83.22	0.2214	0.4162	0.2741
150	33.41	156.40	-33.41	82.61	0.2136	0.4044	0.2652
160	32.41	156.40	-32.41	82.37	0.2072	0.3935	0.2582
170	48.52	156.40		34.06	0.3102		...
180	83.90	156.40		77.24	0.5364		0.2052
190	77.39	156.40		77.60	0.4948		0.1914
200	79.97	156.40		73.25	0.5113		0.2196
210	77.22	156.40		73.25	0.4937		0.1810
220	86.70	156.40		66.70	0.5543		0.1796
230	95.76	156.40		70.20	0.6123		0.1313
240	95.46	156.40		70.20	0.6104		0.1096
250	95.33	138.00		70.20	0.6908		0.1017
260	109.19	156.40		63.10	0.6982		0.0553
270	84.65	156.40		52.57	0.5412		...

Analysis of Load Case 16 : IP+VE+EW

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	22.49	113.10	-22.49	108.67	0.1989	0.2070	0.1408
20	25.99	113.10	-25.99	103.20	0.2298	0.2518	0.1692
30	21.86	86.02	-21.86	104.69	0.2541	0.2088	0.1396
40	127.28	156.40		71.03	0.8138		0.2776
50	132.34	156.40		157.15	0.8462		0.0396
60	131.28	156.40		157.15	0.8394		0.0372
70	133.12	156.40		156.12	0.8512		0.0362
80	131.36	156.40		156.93	0.8399		0.0340
90	131.33	156.40		156.93	0.8397		0.0315
100	131.31	156.40		156.93	0.8396		0.0295
110	131.30	156.40		156.93	0.8395		0.0281
120	134.45	156.40		102.13	0.8597		...
130	25.46	156.40	-25.46	86.95	0.1628	0.2928	0.1605
140	27.31	156.40	-27.31	83.22	0.1746	0.3281	0.1789
150	26.20	156.40	-26.20	82.61	0.1675	0.3171	0.1710
160	25.32	156.40	-25.32	82.37	0.1619	0.3074	0.1654
170	48.19	156.40		34.06	0.3081		...
180	83.41	156.40		77.24	0.5333		0.1875
190	76.93	156.40		77.60	0.4919		0.1738
200	79.48	156.40		73.25	0.5082		0.1986
210	76.87	156.40		73.25	0.4915		0.1600
220	86.49	156.40		66.70	0.5530		0.1596
230	95.70	156.40		70.20	0.6119		0.1266
240	95.46	156.40		70.20	0.6104		0.1096
250	95.33	138.00		70.20	0.6908		0.1017
260	109.19	156.40		63.10	0.6982		0.0553
270	84.65	156.40		52.57	0.5412		...



Analysis of Load Case 17 : NP+VO+OW

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	30.56	113.10	-30.56	108.67	0.2702	0.2812	0.2226
20	35.46	113.10	-35.46	103.20	0.3135	0.3436	0.2700
30	29.92	86.02	-29.92	104.69	0.3479	0.2858	0.2243
40	19.49	156.40	-19.49	71.03	0.1246	0.2743	0.3911
50	11.16	156.40	-11.16	157.15	0.0713	0.0710	0.0532
60	10.78	156.40	-10.78	157.15	0.0689	0.0686	0.0508
70	10.70	156.40	-10.70	156.12	0.0684	0.0686	0.0502
80	10.29	156.40	-10.29	156.93	0.0658	0.0656	0.0474
90	9.90	156.40	-9.90	156.93	0.0633	0.0631	0.0446
100	9.59	156.40	-9.59	156.93	0.0613	0.0611	0.0426
110	9.38	156.40	-9.38	156.93	0.0600	0.0598	0.0413
120	15.96	156.40	-15.96	133.99	0.1021	0.1191	0.0887
130	32.23	156.40	-32.23	86.95	0.2061	0.3706	0.2451
140	34.63	156.40	-34.63	83.22	0.2214	0.4162	0.2741
150	33.41	156.40	-33.41	82.61	0.2136	0.4044	0.2652
160	32.41	156.40	-32.41	82.37	0.2072	0.3935	0.2582
170	19.21	156.40	-19.21	34.06	0.1228	0.5641	...
180	26.43	156.40	-26.43	77.24	0.1690	0.3423	0.2052
190	25.37	156.40	-25.37	77.60	0.1622	0.3269	0.1914
200	27.87	156.40	-27.87	73.25	0.1782	0.3805	0.2196
210	19.55	156.40	-19.55	73.25	0.1250	0.2669	0.1810
220	15.13	156.40	-15.13	66.70	0.0968	0.2269	0.1796
230	12.10	156.40	-12.10	70.20	0.0774	0.1724	0.1313
240	10.66	156.40	-10.66	70.20	0.0681	0.1518	0.1096
250	9.95	138.00	-9.95	70.20	0.0721	0.1418	0.1017
260	6.76	156.40	-6.76	63.10	0.0432	0.1072	0.0553
270	4.64	156.40	-4.64	17.26	0.0297	0.2689	...

Analysis of Load Case 18 : FS+BS+IP+OW

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	33.90	113.10	-33.90	108.67	0.2997	0.3119	0.2158
20	39.15	113.10	-39.15	103.20	0.3461	0.3793	0.2618
30	32.89	86.02	-32.89	104.69	0.3823	0.3141	0.2175
40	127.23	156.40		71.03	0.8135		0.4328
50	132.66	156.40		157.15	0.8482		0.0513
60	131.57	156.40		157.15	0.8412		0.0491
70	133.39	156.40		156.12	0.8529		0.0487
80	131.61	156.40		156.93	0.8415		0.0460
90	131.56	156.40		156.93	0.8412		0.0433
100	131.53	156.40		156.93	0.8410		0.0414
110	131.51	156.40		156.93	0.8408		0.0400
120	133.89	156.40		102.13	0.8561		...
130	35.31	156.40	-35.31	86.95	0.2258	0.4061	0.2374
140	37.82	156.40	-37.82	83.22	0.2418	0.4544	0.2655
150	36.20	156.40	-36.20	82.61	0.2315	0.4382	0.2568
160	34.83	156.40	-34.83	82.37	0.2227	0.4229	0.2501
170	49.29	156.40		34.06	0.3152		...
180	85.09	156.40		77.24	0.5440		0.1960
190	78.43	156.40		77.60	0.5015		0.1841
200	80.99	156.40		73.25	0.5179		0.2126
210	77.89	156.40		73.25	0.4980		0.1753
220	87.17	156.40		66.70	0.5574		0.1740
230	96.21	156.40		70.20	0.6151		0.1272
240	95.80	156.40		70.20	0.6125		0.1061
250	95.57	138.00		70.20	0.6926		0.0986
260	109.36	156.40		63.10	0.6992		0.0536
270	84.70	156.40		52.57	0.5416		...

Analysis of Load Case 19 : FS+BS+EP+OW

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	33.90	113.10	-33.90	108.81	0.2997	0.3115	0.2151



20	39.15	113.10	-39.15	103.37	0.3461	0.3787	0.2609
30	32.89	86.02	-32.89	104.92	0.3823	0.3134	0.2164
40	21.63	156.40	-21.63	71.67	0.1383	0.3018	0.4290
50	12.45	156.40	-12.45	158.58	0.0796	0.0785	0.0507
60	12.01	156.40	-12.01	158.58	0.0768	0.0757	0.0485
70	11.89	156.40	-11.89	157.57	0.0760	0.0755	0.0481
80	11.41	156.40	-11.41	158.35	0.0730	0.0721	0.0455
90	10.94	156.40	-10.94	158.35	0.0700	0.0691	0.0428
100	10.55	156.40	-10.55	158.35	0.0674	0.0666	0.0409
110	10.28	156.40	-10.28	158.35	0.0657	0.0649	0.0396
120	17.50	156.40	-17.50	135.13	0.1119	0.1295	0.0946
130	35.31	156.40	-35.31	86.95	0.2258	0.4061	0.2374
140	37.82	156.40	-37.82	83.22	0.2418	0.4544	0.2655
150	36.20	156.40	-36.20	82.61	0.2315	0.4382	0.2568
160	34.83	156.40	-34.83	82.37	0.2227	0.4229	0.2501
170	24.01	156.40	-24.01	34.06	0.1535	0.7050	...
180	34.51	156.40	-34.51	77.24	0.2206	0.4468	0.6062
190	33.23	156.40	-33.23	77.60	0.2124	0.4282	0.5786
200	36.44	156.40	-36.44	73.25	0.2330	0.4975	0.6969
210	27.83	156.40	-27.83	73.25	0.1779	0.3799	0.6576
220	24.24	156.40	-24.24	66.70	0.1550	0.3634	0.8556
230	20.28	156.40	-20.28	70.20	0.1297	0.2889	0.6844
240	18.46	156.40	-18.46	70.20	0.1181	0.2630	0.6536
250	17.37	138.00	-17.37	70.20	0.1258	0.2474	0.6413
260	14.97	156.40	-14.97	63.10	0.0957	0.2373	0.7516
270	11.40	156.40	-11.40	41.88	0.0729	0.2721	0.0030

Analysis of Load Case 20 : BL+IP+OW

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	30.56	113.10	-30.56	108.67	0.2702	0.2812	0.2226
20	35.46	113.10	-35.46	103.20	0.3135	0.3436	0.2700
30	29.92	86.02	-29.92	104.69	0.3479	0.2858	0.2243
40	125.83	156.40		71.03	0.8046		0.3911
50	132.52	156.40		157.15	0.8473		0.0532
60	131.45	156.40		157.15	0.8405		0.0508
70	133.28	156.40		156.12	0.8522		0.0502
80	131.51	156.40		156.93	0.8408		0.0474
90	131.47	156.40		156.93	0.8406		0.0446
100	131.45	156.40		156.93	0.8405		0.0426
110	131.43	156.40		156.93	0.8403		0.0413
120	132.87	156.40		102.13	0.8495		...
130	32.23	156.40	-32.23	86.95	0.2061	0.3706	0.2451
140	34.63	156.40	-34.63	83.22	0.2214	0.4162	0.2741
150	33.41	156.40	-33.41	82.61	0.2136	0.4044	0.2652
160	32.41	156.40	-32.41	82.37	0.2072	0.3935	0.2582
170	48.52	156.40		34.06	0.3102		...
180	83.90	156.40		77.24	0.5364		0.2052
190	77.39	156.40		77.60	0.4948		0.1914
200	79.97	156.40		73.25	0.5113		0.2196
210	77.22	156.40		73.25	0.4937		0.1810
220	86.70	156.40		66.70	0.5543		0.1796
230	95.76	156.40		70.20	0.6123		0.1313
240	95.46	156.40		70.20	0.6104		0.1096
250	95.33	138.00		70.20	0.6908		0.1017
260	109.19	156.40		63.10	0.6982		0.0553
270	84.65	156.40		52.57	0.5412		...

Absolute Maximum of the all of the Stress Ratio's

0.9311

Governing Element: Skirt-2

Governing Load Case 4 : NP+OW+EQ+FS+BS

Element Description	Long Pressure Stress	Weight + Axial Load Stress	Eccentric Moment Stress	Wind or Earthquake Stress	Combined Stress	Which Side
Skirt-3	...	-21.3	8.6	56.4	43.7	+ Tens
Skirt-3	...	-21.3	-8.6	-56.4	-86.3	- Comp

Skirt-2	...	-24.6	10.1	61.4	46.9	+ Tens
Skirt-2	...	-24.6	-10.1	-61.4	-96.1	- Comp
Skirt-1	...	-20.7	8.6	48.7	36.5	+ Tens
Skirt-1	...	-20.7	-8.6	-48.7	-77.9	- Comp
163-D- BTM dish	...	-12.6	6.4	35.0	28.8	+ Tens
163-D- BTM dish	...	-12.6	-6.4	-35.0	-54.0	- Comp
163-D-Shell-1	...	-7.1	3.8	20.6	17.3	+ Tens
163-D-Shell-1	...	-7.1	-3.8	-20.6	-31.5	- Comp
163-D-Shell-2	...	-6.8	3.7	18.9	15.8	+ Tens
163-D-Shell-2	...	-6.8	-3.7	-18.9	-29.4	- Comp
163-D-Shell-3	...	-6.7	3.8	17.5	14.6	+ Tens
163-D-Shell-3	...	-6.7	-3.8	-17.5	-28.0	- Comp
163-D-Shell-4	...	-6.4	3.7	16.3	13.7	+ Tens
163-D-Shell-4	...	-6.4	-3.7	-16.3	-26.4	- Comp
163-D-Shell-5	...	-6.0	3.7	14.6	12.3	+ Tens
163-D-Shell-5	...	-6.0	-3.7	-14.6	-24.3	- Comp
163-D-Shell-6	...	-5.7	3.7	13.0	10.9	+ Tens
163-D-Shell-6	...	-5.7	-3.7	-13.0	-22.4	- Comp
163-D-Shell-7	...	-5.5	3.7	11.9	10.0	+ Tens
163-D-Shell-7	...	-5.5	-3.7	-11.9	-21.1	- Comp
163-D-Top head	...	-9.5	6.2	19.7	16.4	+ Tens
163-D-Top head	...	-9.5	-6.2	-19.7	-35.4	- Comp
Int skirt 1	...	-19.0	12.6	40.8	34.4	+ Tens
Int skirt 1	...	-19.0	-12.6	-40.8	-72.4	- Comp
Int skirt 2	...	-20.4	13.5	41.3	34.4	+ Tens
Int skirt 2	...	-20.4	-13.5	-41.3	-75.2	- Comp
Int skirt 3	...	-19.7	13.1	34.1	27.6	+ Tens
Int skirt 3	...	-19.7	-13.1	-34.1	-66.9	- Comp
Int skirt 4	...	-19.1	12.7	27.4	21.0	+ Tens
Int skirt 4	...	-19.1	-12.7	-27.4	-59.2	- Comp
122-D1-Btm head	...	-9.5	9.2	18.4	18.1	+ Tens
122-D1-Btm head	...	-9.5	-9.2	-18.4	-37.2	- Comp
122-D1-Shell-1	...	-14.1	11.7	30.7	28.3	+ Tens
122-D1-Shell-1	...	-14.1	-11.7	-30.7	-56.4	- Comp
122-D1-Shell-2	...	-13.3	11.5	24.9	23.1	+ Tens
122-D1-Shell-2	...	-13.3	-11.5	-24.9	-49.7	- Comp
122-D1-Shell-3	...	-14.5	12.9	22.4	20.7	+ Tens
122-D1-Shell-3	...	-14.5	-12.9	-22.4	-49.8	- Comp
122-D1-Shell-4	...	-12.0	7.2	17.5	12.7	+ Tens
122-D1-Shell-4	...	-12.0	-7.2	-17.5	-36.7	- Comp
122-D1-Shell-5	...	-10.9	3.9	15.3	8.3	+ Tens
122-D1-Shell-5	...	-10.9	-3.9	-15.3	-30.0	- Comp
122-D1-Shell-6	...	-8.3	3.5	10.7	5.9	+ Tens
122-D1-Shell-6	...	-8.3	-3.5	-10.7	-22.6	- Comp
122-D1-Shell-7	...	-7.0	3.5	7.4	3.9	+ Tens
122-D1-Shell-7	...	-7.0	-3.5	-7.4	-17.8	- Comp



122-D1-Shell-8	...	-6.5	3.3	4.4	1.2	+ Tens
122-D1-Shell-8	...	-6.5	-3.3	-4.4	-14.1	- Comp
122-D1-Shell-9	...	-3.2	3.5	2.5	2.8	+ Tens
122-D1-Shell-9	...	-3.2	-3.5	-2.5	-9.1	- Comp
122-D1-Top head	...	-1.7	2.8	0.8	2.0	+ Tens
122-D1-Top head	...	-1.7	-2.8	-0.8	-5.4	- Comp

Additional Results per Section 4.4, see 4.4.15

Nomenclature :

- L - Section Length mm
- Fxa - Allowable comp mem stress due to compression with lamc <= 0.15, MPa
- Fba - Allowable all comp mem stress due to bending moment, MPa
- Fva - Allowable shear stress when only shear is present, MPa
- Fha - Allowable hoop comp mem stress under ext pressure alone, MPa
- Fxha - Allowable axial comp w/hoop compression and lamc <= 0.15, MPa
- Fhxa - Allowable hoop comp w/axial compression and lamc <= 0.15, MPa
- Faha - Allowable axial comp w/hoop compression and lamc > 0.15, MPa
- Fbha - Allowable axial comp from bending w/hoop compression, MPa
- Fhba - Allowable hoop comp w/long compression due to moment, MPa

Allowable Stress Results for Case 1 : NP+EW+1.1WI+FW+BW

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10	0.522	15700	108	114	71		6895	108	6895	6895	6895
20	0.512	15700	103	109	65		6895	103	6895	6895	6895
30	0.517	15700	104	110	68		6895	104	6895	6895	6895
40						135	6895		6895	6895	6895
50	0.616	19646	157	157	92	49	6895	157	6895	6895	6895
60	0.616	19646	157	157	92	49	6895	157	6895	6895	6895
70	0.614	19646	156	156	91	48	6895	156	6895	6895	6895
80	0.614	19646	156	157	92	49	6895	156	6895	6895	6895
90	0.614	19646	156	157	92	49	6895	156	6895	6895	6895
100	0.614	19646	156	157	92	49	6895	156	6895	6895	6895
110	0.614	19646	156	157	92	49	6895	156	6895	6895	6895
120						134	6895		6895	6895	6895
130	0.478	15700	86	96	50		6895	86	6895	6895	6895
140	0.462	15700	83	94	46		6895	83	6895	6895	6895
150	0.453	15700	82	93	46		6895	82	6895	6895	6895
160	0.450	15700	82	93	46		6895	82	6895	6895	6895
170						76	6895		6895	6895	6895
180	0.438	2841	77	88	63	33	6895	77	6895	6895	6895
190	0.439	2750	77	88	64	34	6895	77	6895	6895	6895
200	0.429	2500	73	85	61	32	6895	73	6895	6895	6895
210	0.429	2499	73	85	61	32	6895	73	6895	6895	6895
220	0.413	2500	66	81	56	26	6895	66	6895	6895	6895
230	0.422	2499	70	83	59	29	6895	70	6895	6895	6895
240	0.422	2500	70	83	59	29	6895	70	6895	6895	6895
250	0.422	2500	70	83	59	29	6895	70	6895	6895	6895
260	0.404	2500	63	79	53	23	6895	63	6895	6895	6895
270						38	6895		6895	6895	6895

Allowable Stress Results for Case 2 : NP+EW+EE+FS+BS

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10	0.522	15700	108	114	71		6895	108	6895	6895	6895
20	0.512	15700	103	109	65		6895	103	6895	6895	6895
30	0.517	15700	104	110	68		6895	104	6895	6895	6895
40						136	6895		6895	6895	6895
50	0.616	19646	157	157	92	49	6895	157	6895	6895	6895
60	0.616	19646	157	157	92	49	6895	157	6895	6895	6895
70	0.614	19646	156	156	91	48	6895	156	6895	6895	6895
80	0.614	19646	156	157	92	49	6895	156	6895	6895	6895
90	0.614	19646	156	157	92	49	6895	156	6895	6895	6895
100	0.614	19646	156	157	92	49	6895	156	6895	6895	6895
110	0.614	19646	156	157	92	49	6895	156	6895	6895	6895
120						134	6895		6895	6895	6895
130	0.478	15700	86	96	50		6895	86	6895	6895	6895
140	0.462	15700	83	94	46		6895	83	6895	6895	6895



150	0.453	15700	82	93	46	6895	82	6895	6895	6895
160	0.450	15700	82	93	46	6895	82	6895	6895	6895
170						77	6895		6895	6895
180	0.438	2841	77	88	63	33	6895	77	6895	6895
190	0.439	2750	77	88	64	34	6895	77	6895	6895
200	0.429	2500	73	85	61	32	6895	73	6895	6895
210	0.429	2499	73	85	61	32	6895	73	6895	6895
220	0.413	2500	66	81	56	26	6895	66	6895	6895
230	0.422	2499	70	83	59	29	6895	70	6895	6895
240	0.422	2500	70	83	59	29	6895	70	6895	6895
250	0.422	2500	70	83	59	29	6895	70	6895	6895
260	0.404	2500	63	79	53	23	6895	63	6895	6895
270						38	6895		6895	6895

Allowable Stress Results for Case 3 : NP+OW+1.1WI+FW+BW

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10	0.522	15700	108	114	71		6895	108	6895	6895	6895
20	0.512	15700	103	109	65		6895	103	6895	6895	6895
30	0.517	15700	104	110	68		6895	104	6895	6895	6895
40						71	6895		6895	6895	6895
50	0.616	19646	157	157	92	49	6895	157	6895	6895	6895
60	0.616	19646	157	157	92	49	6895	157	6895	6895	6895
70	0.614	19646	156	156	91	48	6895	156	6895	6895	6895
80	0.614	19646	156	157	92	49	6895	156	6895	6895	6895
90	0.614	19646	156	157	92	49	6895	156	6895	6895	6895
100	0.614	19646	156	157	92	49	6895	156	6895	6895	6895
110	0.614	19646	156	157	92	49	6895	156	6895	6895	6895
120						134	6895		6895	6895	6895
130	0.478	15700	86	96	50		6895	86	6895	6895	6895
140	0.462	15700	83	94	46		6895	83	6895	6895	6895
150	0.453	15700	82	93	46		6895	82	6895	6895	6895
160	0.450	15700	82	93	46		6895	82	6895	6895	6895
170						34	6895		6895	6895	6895
180	0.438	2841	77	88	63	33	6895	77	6895	6895	6895
190	0.439	2750	77	88	64	34	6895	77	6895	6895	6895
200	0.429	2500	73	85	61	32	6895	73	6895	6895	6895
210	0.429	2499	73	85	61	32	6895	73	6895	6895	6895
220	0.413	2500	66	81	56	26	6895	66	6895	6895	6895
230	0.422	2499	70	83	59	29	6895	70	6895	6895	6895
240	0.422	2500	70	83	59	29	6895	70	6895	6895	6895
250	0.422	2500	70	83	59	29	6895	70	6895	6895	6895
260	0.404	2500	63	79	53	23	6895	63	6895	6895	6895
270						38	6895		6895	6895	6895

Allowable Stress Results for Case 4 : NP+OW+EQ+FS+BS

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10	0.522	15700	108	114	71		6895	108	6895	6895	6895
20	0.512	15700	103	109	65		6895	103	6895	6895	6895
30	0.517	15700	104	110	68		6895	104	6895	6895	6895
40						136	6895		6895	6895	6895
50	0.616	19646	157	157	92	49	6895	157	6895	6895	6895
60	0.616	19646	157	157	92	49	6895	157	6895	6895	6895
70	0.614	19646	156	156	91	48	6895	156	6895	6895	6895
80	0.614	19646	156	157	92	49	6895	156	6895	6895	6895
90	0.614	19646	156	157	92	49	6895	156	6895	6895	6895
100	0.614	19646	156	157	92	49	6895	156	6895	6895	6895
110	0.614	19646	156	157	92	49	6895	156	6895	6895	6895
120						134	6895		6895	6895	6895
130	0.478	15700	86	96	50		6895	86	6895	6895	6895
140	0.462	15700	83	94	46		6895	83	6895	6895	6895
150	0.453	15700	82	93	46		6895	82	6895	6895	6895
160	0.450	15700	82	93	46		6895	82	6895	6895	6895
170						77	6895		6895	6895	6895
180	0.438	2841	77	88	63	33	6895	77	6895	6895	6895
190	0.439	2750	77	88	64	34	6895	77	6895	6895	6895
200	0.429	2500	73	85	61	32	6895	73	6895	6895	6895
210	0.429	2499	73	85	61	32	6895	73	6895	6895	6895
220	0.413	2500	66	81	56	26	6895	66	6895	6895	6895
230	0.422	2499	70	83	59	29	6895	70	6895	6895	6895
240	0.422	2500	70	83	59	29	6895	70	6895	6895	6895



250	0.422	2500	70	83	59	29	6895	70	6895	6895	6895
260	0.404	2500	63	79	53	23	6895	63	6895	6895	6895
270						38	6895		6895	6895	6895

Allowable Stress Results for Case 5 : NP+HW+HI

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10	0.519	15700	108	114	72		6895	108	6895	6895	6895
20	0.509	15700	103	109	66		6895	103	6895	6895	6895
30	0.510	15700	104	110	69		6895	104	6895	6895	6895
40						71	6895		6895	6895	6895
50	0.609	19646	158	158	92	50	6895	158	6895	6895	6895
60	0.609	19646	158	158	92	50	6895	158	6895	6895	6895
70	0.608	19646	157	157	92	49	6895	157	6895	6895	6895
80	0.607	19646	157	158	92	49	6895	157	6895	6895	6895
90	0.607	19646	157	158	92	49	6895	157	6895	6895	6895
100	0.607	19646	157	158	92	49	6895	157	6895	6895	6895
110	0.607	19646	157	158	92	49	6895	157	6895	6895	6895
120						134	6895		6895	6895	6895
130	0.472	15700	87	97	51		6895	87	6895	6895	6895
140	0.457	15700	83	94	47		6895	83	6895	6895	6895
150	0.448	15700	82	94	47		6895	82	6895	6895	6895
160	0.444	15700	82	93	46		6895	82	6895	6895	6895
170						34	6895		6895	6895	6895
180	0.433	2841	77	88	64	34	6895	77	6895	6895	6895
190	0.434	2750	77	88	64	35	6895	77	6895	6895	6895
200	0.424	2500	73	85	62	33	6895	73	6895	6895	6895
210	0.424	2499	73	85	62	33	6895	73	6895	6895	6895
220	0.409	2500	67	82	57	27	6895	67	6895	6895	6895
230	0.418	2499	70	83	60	30	6895	70	6895	6895	6895
240	0.418	2500	70	83	60	30	6895	70	6895	6895	6895
250	0.418	2500	70	83	60	30	6895	70	6895	6895	6895
260	0.400	2500	63	80	54	24	6895	63	6895	6895	6895
270						17	6895		6895	6895	6895

Allowable Stress Results for Case 6 : NP+HW+HE

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10	0.519	15700	108	114	72		6895	108	6895	6895	6895
20	0.509	15700	103	109	66		6895	103	6895	6895	6895
30	0.510	15700	104	110	69		6895	104	6895	6895	6895
40						71	6895		6895	6895	6895
50	0.609	19646	158	158	92	50	6895	158	6895	6895	6895
60	0.609	19646	158	158	92	50	6895	158	6895	6895	6895
70	0.608	19646	157	157	92	49	6895	157	6895	6895	6895
80	0.607	19646	157	158	92	49	6895	157	6895	6895	6895
90	0.607	19646	157	158	92	49	6895	157	6895	6895	6895
100	0.607	19646	157	158	92	49	6895	157	6895	6895	6895
110	0.607	19646	157	158	92	49	6895	157	6895	6895	6895
120						134	6895		6895	6895	6895
130	0.472	15700	87	97	51		6895	87	6895	6895	6895
140	0.457	15700	83	94	47		6895	83	6895	6895	6895
150	0.448	15700	82	94	47		6895	82	6895	6895	6895
160	0.444	15700	82	93	46		6895	82	6895	6895	6895
170						34	6895		6895	6895	6895
180	0.433	2841	77	88	64	34	6895	77	6895	6895	6895
190	0.434	2750	77	88	64	35	6895	77	6895	6895	6895
200	0.424	2500	73	85	62	33	6895	73	6895	6895	6895
210	0.424	2499	73	85	62	33	6895	73	6895	6895	6895
220	0.409	2500	67	82	57	27	6895	67	6895	6895	6895
230	0.418	2499	70	83	60	30	6895	70	6895	6895	6895
240	0.418	2500	70	83	60	30	6895	70	6895	6895	6895
250	0.418	2500	70	83	60	30	6895	70	6895	6895	6895
260	0.400	2500	63	80	54	24	6895	63	6895	6895	6895
270						17	6895		6895	6895	6895

Allowable Stress Results for Case 7 : IP+OW+1.1WI+FW+BW

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10	0.522	15700	108	114	71		6895	108	6895	6895	6895
20	0.512	15700	103	109	65		6895	103	6895	6895	6895
30	0.517	15700	104	110	68		6895	104	6895	6895	6895
40						71	6895		6895	6895	6895



50	0.616	19646	157	157	92	49	6895	157	6895	6895	6895
60	0.616	19646	157	157	92	49	6895	157	6895	6895	6895
70	0.614	19646	156	156	91	48	6895	156	6895	6895	6895
80	0.614	19646	156	157	92	49	6895	156	6895	6895	6895
90	0.614	19646	156	157	92	49	6895	156	6895	6895	6895
100	0.614	19646	156	157	92	49	6895	156	6895	6895	6895
110	0.614	19646	156	157	92	49	6895	156	6895	6895	6895
120							6895		6895	6895	6895
130	0.478	15700	86	96	50		6895	86	6895	6895	6895
140	0.462	15700	83	94	46		6895	83	6895	6895	6895
150	0.453	15700	82	93	46		6895	82	6895	6895	6895
160	0.450	15700	82	93	46		6895	82	6895	6895	6895
170						34	6895		6895	6895	6895
180	0.438	2841	77	88	63	33	6895	77	6895	6895	6895
190	0.439	2750	77	88	64	34	6895	77	6895	6895	6895
200	0.429	2500	73	85	61	32	6895	73	6895	6895	6895
210	0.429	2499	73	85	61	32	6895	73	6895	6895	6895
220	0.413	2500	66	81	56	26	6895	66	6895	6895	6895
230	0.422	2499	70	83	59	29	6895	70	6895	6895	6895
240	0.422	2500	70	83	59	29	6895	70	6895	6895	6895
250	0.422	2500	70	83	59	29	6895	70	6895	6895	6895
260	0.404	2500	63	79	53	23	6895	63	6895	6895	6895
270							6895		6895	6895	6895

Allowable Stress Results for Case 8 : IP+OW+EQ+FS+BS

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10	0.522	15700	108	114	71		6895	108	6895	6895	6895
20	0.512	15700	103	109	65		6895	103	6895	6895	6895
30	0.517	15700	104	110	68		6895	104	6895	6895	6895
40						136	6895		6895	6895	6895
50	0.616	19646	157	157	92	49	6895	157	6895	6895	6895
60	0.616	19646	157	157	92	49	6895	157	6895	6895	6895
70	0.614	19646	156	156	91	48	6895	156	6895	6895	6895
80	0.614	19646	156	157	92	49	6895	156	6895	6895	6895
90	0.614	19646	156	157	92	49	6895	156	6895	6895	6895
100	0.614	19646	156	157	92	49	6895	156	6895	6895	6895
110	0.614	19646	156	157	92	49	6895	156	6895	6895	6895
120							6895		6895	6895	6895
130	0.478	15700	86	96	50		6895	86	6895	6895	6895
140	0.462	15700	83	94	46		6895	83	6895	6895	6895
150	0.453	15700	82	93	46		6895	82	6895	6895	6895
160	0.450	15700	82	93	46		6895	82	6895	6895	6895
170						77	6895		6895	6895	6895
180	0.438	2841	77	88	63	33	6895	77	6895	6895	6895
190	0.439	2750	77	88	64	34	6895	77	6895	6895	6895
200	0.429	2500	73	85	61	32	6895	73	6895	6895	6895
210	0.429	2499	73	85	61	32	6895	73	6895	6895	6895
220	0.413	2500	66	81	56	26	6895	66	6895	6895	6895
230	0.422	2499	70	83	59	29	6895	70	6895	6895	6895
240	0.422	2500	70	83	59	29	6895	70	6895	6895	6895
250	0.422	2500	70	83	59	29	6895	70	6895	6895	6895
260	0.404	2500	63	79	53	23	6895	63	6895	6895	6895
270							6895		6895	6895	6895

Allowable Stress Results for Case 9 : EP+OW+1.1WI+FW+BW

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10	0.516	15700	108	114	72		6895	108	6895	6895	6895
20	0.507	15700	103	110	66		6895	103	6895	6895	6895
30	0.507	15700	104	110	69		6895	104	6895	6895	6895
40						71	6895		6895	6895	6895
50	0.607	19646	158	158	92	50	6895	158	6895	6895	6895
60	0.607	19646	158	158	92	50	6895	158	6895	6895	6895
70	0.605	19646	157	157	92	49	6895	157	6895	6895	6895
80	0.605	19646	158	158	92	50	6895	158	6895	6895	6895
90	0.605	19646	158	158	92	50	6895	158	6895	6895	6895
100	0.605	19646	158	158	92	50	6895	158	6895	6895	6895
110	0.605	19646	158	158	92	50	6895	158	6895	6895	6895
120						135	6895		6895	6895	6895
130	0.478	15700	86	96	50		6895	86	6895	6895	6895
140	0.462	15700	83	94	46		6895	83	6895	6895	6895



150	0.453	15700	82	93	46	6895	82	6895	6895	6895
160	0.450	15700	82	93	46	6895	82	6895	6895	6895
170						34	6895	6895	6895	6895
180	0.438	2841	77	88	63	33	48	25	44	53
190	0.439	2750	77	88	64	34	48	26	44	52
200	0.429	2500	73	85	61	32	44	24	40	46
210	0.429	2499	73	85	61	32	40	25	37	32
220	0.413	2500	66	81	56	26	30	22	27	16
230	0.422	2499	70	83	59	29	30	25	27	16
240	0.422	2500	70	83	59	29	28	26	25	14
250	0.422	2500	70	83	59	29	27	26	24	12
260	0.404	2500	63	79	53	23	16	22	14	8
270						41	6895	6895	6895	6895

Allowable Stress Results for Case 10 : EP+OW+EQ+FS+BS

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10	0.516	15700	108	114	72	6895	108	6895	6895	6895	6895
20	0.507	15700	103	110	66	6895	103	6895	6895	6895	6895
30	0.507	15700	104	110	69	6895	104	6895	6895	6895	6895
40						137	6895	6895	6895	6895	6895
50	0.607	19646	158	158	92	50	6895	158	6895	6895	6895
60	0.607	19646	158	158	92	50	6895	158	6895	6895	6895
70	0.605	19646	157	157	92	49	6895	157	6895	6895	6895
80	0.605	19646	158	158	92	50	6895	158	6895	6895	6895
90	0.605	19646	158	158	92	50	6895	158	6895	6895	6895
100	0.605	19646	158	158	92	50	6895	158	6895	6895	6895
110	0.605	19646	158	158	92	50	6895	158	6895	6895	6895
120						135	6895	6895	6895	6895	6895
130	0.478	15700	86	96	50	6895	86	6895	6895	6895	6895
140	0.462	15700	83	94	46	6895	83	6895	6895	6895	6895
150	0.453	15700	82	93	46	6895	82	6895	6895	6895	6895
160	0.450	15700	82	93	46	6895	82	6895	6895	6895	6895
170						77	6895	6895	6895	6895	6895
180	0.438	2841	77	88	63	33	48	25	43	74	18
190	0.439	2750	77	88	64	34	47	26	43	73	20
200	0.429	2500	73	85	61	32	43	24	39	67	22
210	0.429	2499	73	85	61	32	40	26	36	57	26
220	0.413	2500	66	81	56	26	29	22	27	36	24
230	0.422	2499	70	83	59	29	30	25	27	32	27
240	0.422	2500	70	83	59	29	27	26	25	25	28
250	0.422	2500	70	83	59	29	26	26	24	17	28
260	0.404	2500	63	79	53	23	16	22	14	9	23
270						42	6895	6895	6895	6895	6895

Allowable Stress Results for Case 11 : HP+HW+HI

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10	0.519	15700	108	114	72	6895	108	6895	6895	6895	6895
20	0.509	15700	103	109	66	6895	103	6895	6895	6895	6895
30	0.510	15700	104	110	69	6895	104	6895	6895	6895	6895
40						71	6895	6895	6895	6895	6895
50	0.609	19646	158	158	92	50	6895	158	6895	6895	6895
60	0.609	19646	158	158	92	50	6895	158	6895	6895	6895
70	0.608	19646	157	157	92	49	6895	157	6895	6895	6895
80	0.607	19646	157	158	92	49	6895	157	6895	6895	6895
90	0.607	19646	157	158	92	49	6895	157	6895	6895	6895
100	0.607	19646	157	158	92	49	6895	157	6895	6895	6895
110	0.607	19646	157	158	92	49	6895	157	6895	6895	6895
120						6895	6895	6895	6895	6895	6895
130	0.472	15700	87	97	51	6895	87	6895	6895	6895	6895
140	0.457	15700	83	94	47	6895	83	6895	6895	6895	6895
150	0.448	15700	82	94	47	6895	82	6895	6895	6895	6895
160	0.444	15700	82	93	46	6895	82	6895	6895	6895	6895
170						34	6895	6895	6895	6895	6895
180	0.433	2841	77	88	64	34	6895	77	6895	6895	6895
190	0.434	2750	77	88	64	35	6895	77	6895	6895	6895
200	0.424	2500	73	85	62	33	6895	73	6895	6895	6895
210	0.424	2499	73	85	62	33	6895	73	6895	6895	6895
220	0.409	2500	67	82	57	27	6895	67	6895	6895	6895
230	0.418	2499	70	83	60	30	6895	70	6895	6895	6895
240	0.418	2500	70	83	60	30	6895	70	6895	6895	6895



250	0.418	2500	70	83	60	30	6895	70	6895	6895	6895
260	0.400	2500	63	80	54	24	6895	63	6895	6895	6895
270							6895		6895	6895	6895

Allowable Stress Results for Case 12 : HP+HW+HE

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10	0.519	15700	108	114	72		6895	108	6895	6895	6895
20	0.509	15700	103	109	66		6895	103	6895	6895	6895
30	0.510	15700	104	110	69		6895	104	6895	6895	6895
40						71	6895		6895	6895	6895
50	0.609	19646	158	158	92	50	6895	158	6895	6895	6895
60	0.609	19646	158	158	92	50	6895	158	6895	6895	6895
70	0.608	19646	157	157	92	49	6895	157	6895	6895	6895
80	0.607	19646	157	158	92	49	6895	157	6895	6895	6895
90	0.607	19646	157	158	92	49	6895	157	6895	6895	6895
100	0.607	19646	157	158	92	49	6895	157	6895	6895	6895
110	0.607	19646	157	158	92	49	6895	157	6895	6895	6895
120							6895		6895	6895	6895
130	0.472	15700	87	97	51		6895	87	6895	6895	6895
140	0.457	15700	83	94	47		6895	83	6895	6895	6895
150	0.448	15700	82	94	47		6895	82	6895	6895	6895
160	0.444	15700	82	93	46		6895	82	6895	6895	6895
170						34	6895		6895	6895	6895
180	0.433	2841	77	88	64	34	6895	77	6895	6895	6895
190	0.434	2750	77	88	64	35	6895	77	6895	6895	6895
200	0.424	2500	73	85	62	33	6895	73	6895	6895	6895
210	0.424	2499	73	85	62	33	6895	73	6895	6895	6895
220	0.409	2500	67	82	57	27	6895	67	6895	6895	6895
230	0.418	2499	70	83	60	30	6895	70	6895	6895	6895
240	0.418	2500	70	83	60	30	6895	70	6895	6895	6895
250	0.418	2500	70	83	60	30	6895	70	6895	6895	6895
260	0.400	2500	63	80	54	24	6895	63	6895	6895	6895
270							6895		6895	6895	6895

Allowable Stress Results for Case 13 : IP+WE+EW

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10	0.522	15700	108	114	71		6895	108	6895	6895	6895
20	0.512	15700	103	109	65		6895	103	6895	6895	6895
30	0.517	15700	104	110	68		6895	104	6895	6895	6895
40						71	6895		6895	6895	6895
50	0.616	19646	157	157	92	49	6895	157	6895	6895	6895
60	0.616	19646	157	157	92	49	6895	157	6895	6895	6895
70	0.614	19646	156	156	91	48	6895	156	6895	6895	6895
80	0.614	19646	156	157	92	49	6895	156	6895	6895	6895
90	0.614	19646	156	157	92	49	6895	156	6895	6895	6895
100	0.614	19646	156	157	92	49	6895	156	6895	6895	6895
110	0.614	19646	156	157	92	49	6895	156	6895	6895	6895
120							6895		6895	6895	6895
130	0.478	15700	86	96	50		6895	86	6895	6895	6895
140	0.462	15700	83	94	46		6895	83	6895	6895	6895
150	0.453	15700	82	93	46		6895	82	6895	6895	6895
160	0.450	15700	82	93	46		6895	82	6895	6895	6895
170						34	6895		6895	6895	6895
180	0.438	2841	77	88	63	33	6895	77	6895	6895	6895
190	0.439	2750	77	88	64	34	6895	77	6895	6895	6895
200	0.429	2500	73	85	61	32	6895	73	6895	6895	6895
210	0.429	2499	73	85	61	32	6895	73	6895	6895	6895
220	0.413	2500	66	81	56	26	6895	66	6895	6895	6895
230	0.422	2499	70	83	59	29	6895	70	6895	6895	6895
240	0.422	2500	70	83	59	29	6895	70	6895	6895	6895
250	0.422	2500	70	83	59	29	6895	70	6895	6895	6895
260	0.404	2500	63	79	53	23	6895	63	6895	6895	6895
270							6895		6895	6895	6895

Allowable Stress Results for Case 14 : IP+WF+CW

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10	0.522	15700	108	114	71		6895	108	6895	6895	6895
20	0.512	15700	103	109	65		6895	103	6895	6895	6895
30	0.517	15700	104	110	68		6895	104	6895	6895	6895
40						71	6895		6895	6895	6895



50	0.616	19646	157	157	92	49	6895	157	6895	6895	6895
60	0.616	19646	157	157	92	49	6895	157	6895	6895	6895
70	0.614	19646	156	156	91	48	6895	156	6895	6895	6895
80	0.614	19646	156	157	92	49	6895	156	6895	6895	6895
90	0.614	19646	156	157	92	49	6895	156	6895	6895	6895
100	0.614	19646	156	157	92	49	6895	156	6895	6895	6895
110	0.614	19646	156	157	92	49	6895	156	6895	6895	6895
120							6895		6895	6895	6895
130	0.478	15700	86	96	50		6895	86	6895	6895	6895
140	0.462	15700	83	94	46		6895	83	6895	6895	6895
150	0.453	15700	82	93	46		6895	82	6895	6895	6895
160	0.450	15700	82	93	46		6895	82	6895	6895	6895
170						34	6895		6895	6895	6895
180	0.438	2841	77	88	63	33	6895	77	6895	6895	6895
190	0.439	2750	77	88	64	34	6895	77	6895	6895	6895
200	0.429	2500	73	85	61	32	6895	73	6895	6895	6895
210	0.429	2499	73	85	61	32	6895	73	6895	6895	6895
220	0.413	2500	66	81	56	26	6895	66	6895	6895	6895
230	0.422	2499	70	83	59	29	6895	70	6895	6895	6895
240	0.422	2500	70	83	59	29	6895	70	6895	6895	6895
250	0.422	2500	70	83	59	29	6895	70	6895	6895	6895
260	0.404	2500	63	79	53	23	6895	63	6895	6895	6895
270							6895		6895	6895	6895

Allowable Stress Results for Case 15 : IP+VO+OW

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10	0.522	15700	108	114	71		6895	108	6895	6895	6895
20	0.512	15700	103	109	65		6895	103	6895	6895	6895
30	0.517	15700	104	110	68		6895	104	6895	6895	6895
40						71	6895		6895	6895	6895
50	0.616	19646	157	157	92	49	6895	157	6895	6895	6895
60	0.616	19646	157	157	92	49	6895	157	6895	6895	6895
70	0.614	19646	156	156	91	48	6895	156	6895	6895	6895
80	0.614	19646	156	157	92	49	6895	156	6895	6895	6895
90	0.614	19646	156	157	92	49	6895	156	6895	6895	6895
100	0.614	19646	156	157	92	49	6895	156	6895	6895	6895
110	0.614	19646	156	157	92	49	6895	156	6895	6895	6895
120							6895		6895	6895	6895
130	0.478	15700	86	96	50		6895	86	6895	6895	6895
140	0.462	15700	83	94	46		6895	83	6895	6895	6895
150	0.453	15700	82	93	46		6895	82	6895	6895	6895
160	0.450	15700	82	93	46		6895	82	6895	6895	6895
170						34	6895		6895	6895	6895
180	0.438	2841	77	88	63	33	6895	77	6895	6895	6895
190	0.439	2750	77	88	64	34	6895	77	6895	6895	6895
200	0.429	2500	73	85	61	32	6895	73	6895	6895	6895
210	0.429	2499	73	85	61	32	6895	73	6895	6895	6895
220	0.413	2500	66	81	56	26	6895	66	6895	6895	6895
230	0.422	2499	70	83	59	29	6895	70	6895	6895	6895
240	0.422	2500	70	83	59	29	6895	70	6895	6895	6895
250	0.422	2500	70	83	59	29	6895	70	6895	6895	6895
260	0.404	2500	63	79	53	23	6895	63	6895	6895	6895
270							6895		6895	6895	6895

Allowable Stress Results for Case 16 : IP+VE+EW

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10	0.522	15700	108	114	71		6895	108	6895	6895	6895
20	0.512	15700	103	109	65		6895	103	6895	6895	6895
30	0.517	15700	104	110	68		6895	104	6895	6895	6895
40						71	6895		6895	6895	6895
50	0.616	19646	157	157	92	49	6895	157	6895	6895	6895
60	0.616	19646	157	157	92	49	6895	157	6895	6895	6895
70	0.614	19646	156	156	91	48	6895	156	6895	6895	6895
80	0.614	19646	156	157	92	49	6895	156	6895	6895	6895
90	0.614	19646	156	157	92	49	6895	156	6895	6895	6895
100	0.614	19646	156	157	92	49	6895	156	6895	6895	6895
110	0.614	19646	156	157	92	49	6895	156	6895	6895	6895
120							6895		6895	6895	6895
130	0.478	15700	86	96	50		6895	86	6895	6895	6895
140	0.462	15700	83	94	46		6895	83	6895	6895	6895



150	0.453	15700	82	93	46	6895	82	6895	6895	6895
160	0.450	15700	82	93	46	6895	82	6895	6895	6895
170						34	6895	6895	6895	6895
180	0.438	2841	77	88	63	33	6895	77	6895	6895
190	0.439	2750	77	88	64	34	6895	77	6895	6895
200	0.429	2500	73	85	61	32	6895	73	6895	6895
210	0.429	2499	73	85	61	32	6895	73	6895	6895
220	0.413	2500	66	81	56	26	6895	66	6895	6895
230	0.422	2499	70	83	59	29	6895	70	6895	6895
240	0.422	2500	70	83	59	29	6895	70	6895	6895
250	0.422	2500	70	83	59	29	6895	70	6895	6895
260	0.404	2500	63	79	53	23	6895	63	6895	6895
270						6895		6895	6895	6895

Allowable Stress Results for Case 17 : NP+VO+OW

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10	0.522	15700	108	114	71		6895	108	6895	6895	6895
20	0.512	15700	103	109	65		6895	103	6895	6895	6895
30	0.517	15700	104	110	68		6895	104	6895	6895	6895
40						71	6895		6895	6895	6895
50	0.616	19646	157	157	92	49	6895	157	6895	6895	6895
60	0.616	19646	157	157	92	49	6895	157	6895	6895	6895
70	0.614	19646	156	156	91	48	6895	156	6895	6895	6895
80	0.614	19646	156	157	92	49	6895	156	6895	6895	6895
90	0.614	19646	156	157	92	49	6895	156	6895	6895	6895
100	0.614	19646	156	157	92	49	6895	156	6895	6895	6895
110	0.614	19646	156	157	92	49	6895	156	6895	6895	6895
120						133	6895		6895	6895	6895
130	0.478	15700	86	96	50		6895	86	6895	6895	6895
140	0.462	15700	83	94	46		6895	83	6895	6895	6895
150	0.453	15700	82	93	46		6895	82	6895	6895	6895
160	0.450	15700	82	93	46		6895	82	6895	6895	6895
170						34	6895		6895	6895	6895
180	0.438	2841	77	88	63	33	6895	77	6895	6895	6895
190	0.439	2750	77	88	64	34	6895	77	6895	6895	6895
200	0.429	2500	73	85	61	32	6895	73	6895	6895	6895
210	0.429	2499	73	85	61	32	6895	73	6895	6895	6895
220	0.413	2500	66	81	56	26	6895	66	6895	6895	6895
230	0.422	2499	70	83	59	29	6895	70	6895	6895	6895
240	0.422	2500	70	83	59	29	6895	70	6895	6895	6895
250	0.422	2500	70	83	59	29	6895	70	6895	6895	6895
260	0.404	2500	63	79	53	23	6895	63	6895	6895	6895
270						17	6895		6895	6895	6895

Allowable Stress Results for Case 18 : FS+BS+IP+OW

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10	0.522	15700	108	114	71		6895	108	6895	6895	6895
20	0.512	15700	103	109	65		6895	103	6895	6895	6895
30	0.517	15700	104	110	68		6895	104	6895	6895	6895
40						71	6895		6895	6895	6895
50	0.616	19646	157	157	92	49	6895	157	6895	6895	6895
60	0.616	19646	157	157	92	49	6895	157	6895	6895	6895
70	0.614	19646	156	156	91	48	6895	156	6895	6895	6895
80	0.614	19646	156	157	92	49	6895	156	6895	6895	6895
90	0.614	19646	156	157	92	49	6895	156	6895	6895	6895
100	0.614	19646	156	157	92	49	6895	156	6895	6895	6895
110	0.614	19646	156	157	92	49	6895	156	6895	6895	6895
120							6895		6895	6895	6895
130	0.478	15700	86	96	50		6895	86	6895	6895	6895
140	0.462	15700	83	94	46		6895	83	6895	6895	6895
150	0.453	15700	82	93	46		6895	82	6895	6895	6895
160	0.450	15700	82	93	46		6895	82	6895	6895	6895
170						34	6895		6895	6895	6895
180	0.438	2841	77	88	63	33	6895	77	6895	6895	6895
190	0.439	2750	77	88	64	34	6895	77	6895	6895	6895
200	0.429	2500	73	85	61	32	6895	73	6895	6895	6895
210	0.429	2499	73	85	61	32	6895	73	6895	6895	6895
220	0.413	2500	66	81	56	26	6895	66	6895	6895	6895
230	0.422	2499	70	83	59	29	6895	70	6895	6895	6895
240	0.422	2500	70	83	59	29	6895	70	6895	6895	6895



250	0.422	2500	70	83	59	29	6895	70	6895	6895	6895
260	0.404	2500	63	79	53	23	6895	63	6895	6895	6895
270							6895		6895	6895	6895

Allowable Stress Results for Case 19 : FS+BS+EP+OW

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10	0.516	15700	108	114	72		6895	108	6895	6895	6895
20	0.507	15700	103	110	66		6895	103	6895	6895	6895
30	0.507	15700	104	110	69		6895	104	6895	6895	6895
40						71	6895		6895	6895	6895
50	0.607	19646	158	158	92	50	6895	158	6895	6895	6895
60	0.607	19646	158	158	92	50	6895	158	6895	6895	6895
70	0.605	19646	157	157	92	49	6895	157	6895	6895	6895
80	0.605	19646	158	158	92	50	6895	158	6895	6895	6895
90	0.605	19646	158	158	92	50	6895	158	6895	6895	6895
100	0.605	19646	158	158	92	50	6895	158	6895	6895	6895
110	0.605	19646	158	158	92	50	6895	158	6895	6895	6895
120						135	6895		6895	6895	6895
130	0.478	15700	86	96	50		6895	86	6895	6895	6895
140	0.462	15700	83	94	46		6895	83	6895	6895	6895
150	0.453	15700	82	93	46		6895	82	6895	6895	6895
160	0.450	15700	82	93	46		6895	82	6895	6895	6895
170						34	6895		6895	6895	6895
180	0.438	2841	77	88	63	33	48	25	43	11	33
190	0.439	2750	77	88	64	34	47	26	43	10	34
200	0.429	2500	73	85	61	32	43	24	39	9	32
210	0.429	2499	73	85	61	32	40	26	36	8	32
220	0.413	2500	66	81	56	26	29	22	27	5	26
230	0.422	2499	70	83	59	29	30	25	27	5	29
240	0.422	2500	70	83	59	29	27	26	25	4	29
250	0.422	2500	70	83	59	29	26	26	24	3	29
260	0.404	2500	63	79	53	23	16	22	14	2	23
270						41	6895		6895	6895	6895

Allowable Stress Results for Case 20 : BL+IP+OW

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10	0.522	15700	108	114	71		6895	108	6895	6895	6895
20	0.512	15700	103	109	65		6895	103	6895	6895	6895
30	0.517	15700	104	110	68		6895	104	6895	6895	6895
40						71	6895		6895	6895	6895
50	0.616	19646	157	157	92	49	6895	157	6895	6895	6895
60	0.616	19646	157	157	92	49	6895	157	6895	6895	6895
70	0.614	19646	156	156	91	48	6895	156	6895	6895	6895
80	0.614	19646	156	157	92	49	6895	156	6895	6895	6895
90	0.614	19646	156	157	92	49	6895	156	6895	6895	6895
100	0.614	19646	156	157	92	49	6895	156	6895	6895	6895
110	0.614	19646	156	157	92	49	6895	156	6895	6895	6895
120							6895		6895	6895	6895
130	0.478	15700	86	96	50		6895	86	6895	6895	6895
140	0.462	15700	83	94	46		6895	83	6895	6895	6895
150	0.453	15700	82	93	46		6895	82	6895	6895	6895
160	0.450	15700	82	93	46		6895	82	6895	6895	6895
170						34	6895		6895	6895	6895
180	0.438	2841	77	88	63	33	6895	77	6895	6895	6895
190	0.439	2750	77	88	64	34	6895	77	6895	6895	6895
200	0.429	2500	73	85	61	32	6895	73	6895	6895	6895
210	0.429	2499	73	85	61	32	6895	73	6895	6895	6895
220	0.413	2500	66	81	56	26	6895	66	6895	6895	6895
230	0.422	2499	70	83	59	29	6895	70	6895	6895	6895
240	0.422	2500	70	83	59	29	6895	70	6895	6895	6895
250	0.422	2500	70	83	59	29	6895	70	6895	6895	6895
260	0.404	2500	63	79	53	23	6895	63	6895	6895	6895
270							6895		6895	6895	6895

Note:

Load cases involving Wind and Seismic are assumed to cause a net section bending moment. These load case components are: EQ, EE, WI, WE, VO, VE, VF, HE, HI, WF. Static bending moments due to the weights of nozzles etc. are considered to be local in nature and will not cause a net section bending moment. This is an important consideration when computing the unity check per section 4.4.12 equations 4.4.86, 4.4.87, 4.4.90,



4.4.91 and 4.4.92.

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Skirt Support Analysis Per EN 13445-3 Section 16.12:

The analysis of this skirt has been conducted referencing the EN 13445-3 design code. It is important to note that the selected design code does not explicitly cover the analysis of this component. Therefore, the results provided herein should be regarded as informational only.

Item: Int skirt 4

Mean skirt diameter at the critical height is computed later.
 This is a Conical type Support Skirt , Wider at the Top

Half Apex Angle of the skirt a 1.100 deg

Total vessel (empty) weight	Fg	11828639.000 N
Head Weight below section 2-2	DeltaFg	769874.312 N
Weight of vessel contents (liquid)	Ff	6266327.000 N
Vessel Gross Weight	Fg + Ff	18094966.000 N

Skirt Mean Diameter Top End	DzTop	7127.604 mm
Skirt Mean Diameter Bottom End	DzBot	7079.604 mm
Skirt Analysis Thickness (Corroded)	ez	22.400 mm
Skirt Height	H	1250.000 mm

Additional Axial Force at section 1-1	F1	-106384.188 N
Moment at section 1-1	M1	22072449024.000 N-mm
Applied Moment - Bottom of Skirt	Mb	24105875456.000 N-mm
Weight at Bottom of Skirt	W	9826193.000 N

Structural Type B Skirt per EN 13445 Figure 16.12-2:

Type of Head:	Elliptical
Head Mean Diameter at Skirt Top	Db 7142.000 mm
Head Corroded Thickness	eb 30.000 mm

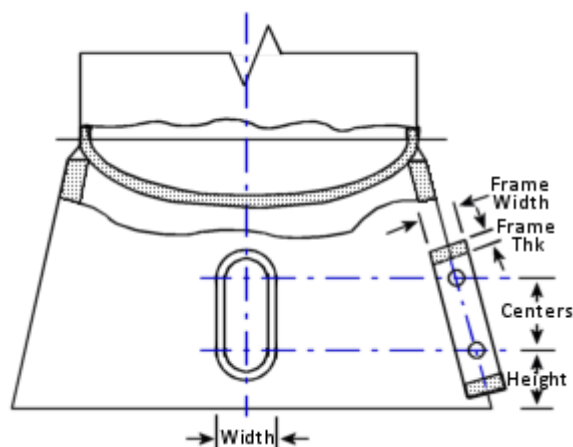
Per EN 13445 Section 7.5.4 [R, r]:

$$h_i = 1781.0 \text{ mm}$$

$$K = D_i / (2 * h_i) = 7112.0 / (2 * 1781.0) = 1.9966$$

$$R = D_i (0.44 * K + 0.02) = 7112.0 * (0.44 * 1.997 + 0.02) = 6390.2573 \text{ mm}$$

$$r = D_i ((0.5 / K) - 0.08) = 7112.0 * ((0.5 / 1.997) - 0.08) = 1212.0399 \text{ mm}$$



There are no Access Openings in the Skirt - Plain Skirt Considered

Moment of Inertia	Ixx = 3121867653120.000 mm ⁴
Moment of Inertia	Iyy = 3121867653120.000 mm ⁴
Section Modulus	W = 881933568.000 mm ³
Section Metal Area	A = 498295.438 mm ²

Analysis of the Section at weakest section Modulus [Z min]:



Radius of Gyration of Skirt at 4-4 [rg]:
 $= \sqrt{(\min(lxx, lyy) / \text{Area})}$
 $= \sqrt{(\min(0.31219E+13, 0.31219E+13) / 498295.438)}$
 $= 2503.018 \text{ mm}$

Skirt Slenderness Ratio [rg/L]:
 $= rg / L$
 $= 2503.018 / 1250.0$
 $= 2.002$

Correct the Moment [M4]:
top of skirt is the critical end
Controlling Moment [M4] = 0.22072E+11 N-mm

Axial Stress at Residual (Weakest) Surface Area [Sigma]:
Downwind Side:
 $= -M / Z - W / A$
 $= -0.22072E+11 / 35277E+10 - 9826193 / 498295.43750$
 $= -25.976 \text{ MPa}$

Upwind Side:
 $= +(M) / Z - W / A$
 $= +0.22072E+11 / 35277E+10 - 9826193 / 498295.43750$
 $= -13.466 \text{ MPa}$

16.12.3.4 Checking at connection areas:

Force at the joint, 16.12-1 [Fzp]:
 $= F1 - Fg - Ff + 4 \cdot M1 / Dz$
 $= -106384.188 - 11828639 - 6266327 + 4 \cdot 0.22072E+11 / 7127.604$
 $= -5819347.5 \text{ N}$

Force at the joint, 16.12-2 [Fzq]:
 $= F1 - Fg - Ff - 4 \cdot M1 / Dz$
 $= -106384.188 - 11828639 - 6266327 - 4 \cdot 0.22072E+11 / 7127.604$
 $= -30583352.0 \text{ N}$

16.12.3.4.1 Membrane stresses:

Membrane stress at 1-1, 16.12-3 & 4:

Stress [σ_{1pm}]:
 $= (Fzp + \Delta Fg) / (\pi \cdot Db \cdot eb) + P \cdot Db / (4 \cdot eb)$
 $= (-0.6E+07 + 769874 + 6266327) / (\pi \cdot 7142.0 \cdot 30.0) + 0.384 \cdot 7142.0 / (4 \cdot 30.0)$
 $= 24.66 \text{ MPa}$

Stress [σ_{1qm}]:
 $= (Fzq + \Delta Fg + Ff) / (\pi \cdot Db \cdot eb) + P \cdot Db / (4 \cdot eb)$
 $= (-0.3E+08 + 769874 + 6266327) / (\pi \cdot 7142.0 \cdot 30.0) + 0.384 \cdot 7142.0 / (4 \cdot 30.0)$
 $= -12.13 \text{ MPa}$

Membrane stress at 2-2, 16.12-9:

Stress [σ_{2pm} & σ_{2qm}]:
 $= (Ff + \Delta Fg) / (\pi \cdot Db \cdot eb) + P \cdot Db / (4 \cdot eb)$
 $= (6266327 + 769874) / (\pi \cdot 7142.0 \cdot 30.0) + 0.384 \cdot 7142.0 / (4 \cdot 30.0)$
 $= 33.31 \text{ MPa}$

Membrane stress at 3-3, 16.12 12 & 13:

Stress [σ_{3pm}]:
 $= Fzp / (\pi \cdot Dz \cdot ez)$
 $= -5819347.500 / (\pi \cdot 7127.604 \cdot 22.4)$
 $= -11.60 \text{ MPa}$

Stress [σ_{3qm}]:
 $= Fzq / (\pi \cdot Dz \cdot ez)$
 $= -0.3E+08 / (\pi \cdot 7127.604 \cdot 22.4)$
 $= -60.98 \text{ MPa}$



16.12.3.4.2 Bending stresses:

Cosine of angle, 16.12-25 [cos(γ)]:
 $= 1 - 0.5 (Db + eb - Dz + ez) / (r + eb)$
 $= 1 - 0.5 (7142.0 + 30.0 - 7079.6 + 22.4) / (1212.04 + 30.0)$
 $= 0.9731$

$\gamma = \arccos(\cos(\gamma)) = \arccos(0.973) = 13.317 \text{ deg}$

Dimension [a]:
 $= 0.5 \cdot \sqrt{(eb^2 + ez^2 + 2 \cdot eb \cdot ez \cdot \cos(\gamma))}$
 $= 0.5 \cdot \sqrt{(30.0^2 + 22.4^2 + 2 \cdot 30.0 \cdot 22.4 \cdot 0.973)}$
 $= 26.027 \text{ mm}$

Bending moment at verification points p and q [Mp & Mq]:
 $M_p = a \cdot F_{zp} = 26.02 \cdot -5819347.5 = \text{*****} \times 10^6 \text{ N-mm}$
 $M_q = a \cdot F_{zq} = 26.02 \cdot -30583352.0 = \text{*****} \times 10^6 \text{ N-mm}$

Correction factor [C]:
 $= 0.63 - 0.057 (eb/ez)^2$
 $= 0.63 - 0.057 (30.0/22.4)^2$
 $= 0.528$

Bending stresses in sections 1-1 to 3-3 at the outer surface (o), 16.12-26 to 29:

Stress [$\sigma_{1pb(o)}$ & $\sigma_{2pb(o)}$]:
 $= C \cdot 6 \cdot M_p / (\pi \cdot Db \cdot eb^2)$
 $= 0.528 \cdot 6 \cdot -0.2E+09 / (\pi \cdot 7142.0 \cdot 30.0^2)$
 $= -23.75 \text{ MPa}$

Stress [$\sigma_{1qb(o)}$ & $\sigma_{2qb(o)}$]:
 $= C \cdot 6 \cdot M_q / (\pi \cdot Db \cdot eb^2)$
 $= 0.528 \cdot 6 \cdot -0.8E+09 / (\pi \cdot 7142.0 \cdot 30.0^2)$
 $= -124.83 \text{ MPa}$

Stress [$\sigma_{3pb(o)}$]:
 $= C \cdot 6 \cdot M_p / (\pi \cdot Dz \cdot ez^2)$
 $= 0.528 \cdot 6 \cdot -0.2E+09 / (\pi \cdot 7127.604 \cdot 22.4^2)$
 $= -42.69 \text{ MPa}$

Stress [$\sigma_{3qb(o)}$]:
 $= C \cdot 6 \cdot M_q / (\pi \cdot Dz \cdot ez^2)$
 $= 0.528 \cdot 6 \cdot -0.8E+09 / (\pi \cdot 7127.604 \cdot 22.4^2)$
 $= -224.36 \text{ MPa}$

Geometry variables, 16.12-32 to 36:
 $eb/Db = 1.181/281.181 = 0.0042$
 $y = 125 \cdot eb/Db = 125 \cdot 30.0/7142.0 = 0.5251$

As this an elliptical head, it will be treated a a Korbbogen type head

Geometric value [α]:
 $= 1.51861 \cdot 2.71828 - 4.2335 \cdot RYY + 3.994$
 $= 1.51861 \cdot 2.71828 - 4.2335 \cdot 0.525 + 3.994 = 4.1585$

Bending stress in sections 1-1 and 2-2 Equation 16.12-31 [$\sigma_{1b(p)}$ & $\sigma_{2b(p)}$]:
 $= (P + Ph) \cdot Db ((\gamma / \gamma A) \cdot \alpha - 1) / (4 \cdot eb)$
 $= (0.384 + 0.0) \cdot 7142.0 ((13.32/40.0) \cdot 4.158 - 1) / (4 \cdot 30.0)$
 $= 8.79 \text{ MPa}$

16.12.3.4.3 Total stresses and strength conditions:

Structural shape B and C:

Stress [σ_{1pitot}]:
 $= \sigma_{1pm} - \sigma_{1pb(o)} + \sigma_{1b(p)}$
 $= 24.662 - -23.753 + 8.786$
 $= 57.20 \text{ MPa}$

Stress [σ_{1potot}]:



$$\begin{aligned} &= \sigma_{1pm} + \sigma_{1pb(o)} - \sigma_{1b(p)} \\ &= 24.662 + -23.753 - 8.786 \\ &= -7.88 \text{ MPa} \end{aligned}$$

$$\begin{aligned} \text{Stress } [\sigma_{1qitot}]: \\ &= \sigma_{1qm} - \sigma_{1qb(o)} + \sigma_{1b(p)} \\ &= -12.131 - -124.831 + 8.786 \\ &= 121.49 \text{ MPa} \end{aligned}$$

$$\begin{aligned} \text{Stress } [\sigma_{1qotot}]: \\ &= \sigma_{1qm} + \sigma_{1qb(o)} - \sigma_{1b(p)} \\ &= -12.131 + -124.831 - 8.786 \\ &= -145.75 \text{ MPa} \end{aligned}$$

$$\begin{aligned} \text{Stress } [\sigma_{2pitot}]: \\ &= \sigma_{2pm} + \sigma_{2pb(o)} + \sigma_{2b(p)} \\ &= 33.308 + -23.753 + 8.786 \\ &= 18.34 \text{ MPa} \end{aligned}$$

$$\begin{aligned} \text{Stress } [\sigma_{2potot}]: \\ &= \sigma_{2pm} - \sigma_{2pb(o)} - \sigma_{2b(p)} \\ &= 33.308 - -23.753 - 8.786 \\ &= 48.27 \text{ MPa} \end{aligned}$$

$$\begin{aligned} \text{Stress } [\sigma_{2qitot}]: \\ &= \sigma_{2qm} + \sigma_{2qb(o)} + \sigma_{2b(p)} \\ &= 33.308 + -124.831 + 8.786 \\ &= -82.74 \text{ MPa} \end{aligned}$$

$$\begin{aligned} \text{Stress } [\sigma_{2qotot}]: \\ &= \sigma_{2pm} - \sigma_{2pb(o)} - \sigma_{2b(p)} \\ &= 33.308 - -124.831 - 8.786 \\ &= 149.35 \text{ MPa} \end{aligned}$$

$$\begin{aligned} \text{Stress } [\sigma_{3pitot}]: \\ &= \sigma_{3pm} - \sigma_{3pb} \\ &= -11.603 - -42.691 \\ &= 31.09 \text{ MPa} \end{aligned}$$

$$\begin{aligned} \text{Stress } [\sigma_{3potot}]: \\ &= \sigma_{3pm} + \sigma_{3pb} \\ &= -11.603 + -42.691 \\ &= -54.29 \text{ MPa} \end{aligned}$$

$$\begin{aligned} \text{Stress } [\sigma_{3qitot}]: \\ &= \sigma_{3qm} - \sigma_{3qb} \\ &= -60.979 - -224.359 \\ &= 163.38 \text{ MPa} \end{aligned}$$

$$\begin{aligned} \text{Stress } [\sigma_{3qotot}]: \\ &= \sigma_{3qm} + \sigma_{3qb} \\ &= -60.979 + -224.359 \\ &= -285.34 \text{ MPa} \end{aligned}$$

Allowable stresses:

$$\begin{aligned} \text{For } \sigma_{1pitot} \text{ and } \sigma_{1potot}, \text{ the stress must not exceed:} \\ &= fs(3 - (1 / 1.5)) \cdot (\sigma_{1pm} / f)^2 \\ &= 156.4(3 - (1 / 1.5)) \cdot (24.662 / 156.4)^2 \\ &= 466.61 \text{ MPa} \end{aligned}$$

$$\begin{aligned} \text{For } \sigma_{1qitot} \text{ and } \sigma_{1qotot}, \text{ the stress must not exceed:} \\ &= fs(3 - (1 / 1.5)) \cdot (\sigma_{1qm} / f)^2 \\ &= 156.4(3 - (1 / 1.5)) \cdot (-12.131 / 156.4)^2 \\ &= 468.57 \text{ MPa} \end{aligned}$$

$$\begin{aligned} \text{For } \sigma_{2pitot} \text{ and } \sigma_{2potot}, \text{ the stress must not exceed:} \\ &= fs(3 - (1 / 1.5)) \cdot (\sigma_{2pm} / f)^2 \\ &= 156.4(3 - (1 / 1.5)) \cdot (33.308 / 156.4)^2 \\ &= 464.47 \text{ MPa} \end{aligned}$$



For σ_{2qitot} and σ_{2qotot} , the stress must not exceed:

$$\begin{aligned} &= fs(3 - (1 / 1.5) \cdot (\sigma_{2qm} / f)^2) \\ &= 156.4(3 - (1/1.5) \cdot (33.308/156.4)^2) \\ &= 464.47 \text{ MPa} \end{aligned}$$

For σ_{3pitot} & σ_{3potot} , the stress must not exceed:

$$\begin{aligned} &= fs(3 - (1 / 1.5) \cdot (\sigma_{3pm} / f)^2) \\ &= 156.4(3 - (1/1.5) \cdot (-11.603/156.4)^2) \\ &= 468.63 \text{ MPa} \end{aligned}$$

For σ_{3qitot} & σ_{3qotot} , the stress must not exceed:

$$\begin{aligned} &= fs(3 - (1 / 1.5) \cdot (\sigma_{3qm} / f)^2) \\ &= 156.4(3 - (1/1.5) \cdot (-60.979/156.4)^2) \\ &= 453.35 \text{ MPa} \end{aligned}$$

Please note: None of the following stresses in the table below are in the region where there is an access opening.

Summary of Stresses: (MPa)

Stress	Actual	Allowed	Result	Equation
Stot1pi	57.20	466.61	Pass	46
Stot1po	-7.88	466.61	Pass	47
Stot1qi	121.49	468.57	Pass	48
Stot1qo	-145.75	468.57	Pass	49
Stot2pi	18.34	464.47	Pass	50
Stot2po	48.27	464.47	Pass	51
Stot2qi	-82.74	464.47	Pass	52
Stot2qo	149.35	464.47	Pass	53
Stot3pi	31.09	468.63	Pass	54
Stot3po	-54.29	468.63	Pass	55
Stot3qi	163.38	453.35	Pass	56
Stot3qo	-285.34	453.35	Pass	57

The stresses are satisfactory in the region without openings.

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Shop/Field Installation Options :

Platform(s) installed in the Shop.
 Packing is installed in the Shop.
 Trays are installed in the Shop.
 Insulation is installed in the Shop.
 Lining is installed in the Shop.

Note : The CG is computed from the first Element From Node

Center of Gravity of the Platforms	36104.535 mm
Center of Gravity of the Packing	28588.305 mm
Center of Gravity of the Liquid	21887.191 mm
Center of Gravity of the Insulation	27308.736 mm
Center of Gravity of the Lining	39922.914 mm
Center of Gravity of the Stiffening Rings	46416.555 mm
Center of Gravity of the Nozzles	25718.602 mm
Center of Gravity of the Trays	55500.051 mm
Center of Gravity of the Added Weights (Operating)	43743.617 mm
Center of Gravity of the Added Weights (Empty)	39780.188 mm
Center of Gravity of Bare Shell New and Cold	21467.217 mm
Center of Gravity of Bare Shell Corroded	20978.830 mm
Vessel CG in the Operating Condition	28273.031 mm
Vessel CG in the Fabricated (Shop/Empty) Condition	26714.848 mm
Vessel CG in the Test Condition	31834.822 mm

Rigging Analysis Results:

• Lifting weight based on the user selected calculated erected weight

Total Effective Length of Vessel for this analysis	59900.01 mm
Total vessel weight (No Liquid) Twt	8805671.00 N
Impact weight multiplication factor Imp	1.20
Design lifting weight, DWT = Imp * Twt	10566805.00 N
Elevation of the Tailing Lug (bottom)	0.00 mm
Elevation of the Lifting Lug (top)	47000.00 mm
Design Reaction force at the tailing lug	4560623.00 N
Design Reaction force at the lifting lug	6006182.00 N
CG distance during field lifting operation	26714.85 mm
CG distance from tailing lug	26714.85 mm
CG distance from the nearer lifting lug/trunnion	20285.15 mm
Compressive Allowable Increase Factor	1.20

Critical Values: (MPa, mm)

	Max Stress	Elevation	Allowables
Bending	48.86	25850.01	96.58
Shear	9.26	46830.02	122.50 (0.7*S)

The compressive bending allowable stress is the minimum of the tensile allowable and the compressive allowable times the specified compressive allowable increase factor.

Forces and Moments at selected elevations (not all analysis points shown):

Distance mm	Bending Moment N-mm	Bending Stress MPa	Shear Force N	Shear Stress MPa
0.00	0.0	0.0	3141312.8	3.5
10800.00	37206876160.0	10.6	1378451.2	0.7
23700.01	45170196480.0	13.7	1711408.2	0.9
32600.02	30400294912.0	32.0	2572680.0	4.8
44550.02	5668897792.0	6.5	3825193.0	7.8
59850.04	351707488.0	0.6	65389.2	0.2



Unity Check (Actual Stress / Allowable Stress):

Maximum Unity Check is 0.5059 at elevation 25850.0098 mm - Must be ≤ 1

Note: The rigging analysis is performed using a uniformly distributed load.

--- Plot data successfully generated ...----

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Skirt Data :

Skirt Outside Diameter at Base	SOD, D6	6846.0000	mm
Skirt Thickness	STHK	42.0000	mm
Skirt Internal Corrosion Allowance	SCA	1.6000	mm
Skirt External Corrosion Allowance		0.0000	mm
Skirt Material		SA-516 70	

Basing Input: Type of Geometry: Continuous Top Ring W/Gussets

Thickness of Basing	TBA, ea4	56.0000	mm
Design Temperature of the Basing		38.00	°C
Basing Material		SA-516 70	
Basing Ope. All. Stress	BASOPE, f4	175.00	MPa
Basing Yield Stress		262.00	MPa
Inside Diameter of Basing	DI, D4	6678.0000	mm
Outside Diameter of Basing	DOU, D5	7278.0000	mm
Nominal Diameter of Bolts	BND, d(b0)	64.0000	mm
Bolt Corrosion Allowance	BCA	1.5000	mm
Bolt Material		SA-36	
Bolt Operating Allowable Stress	SA, fB	284.00	MPa
Number of Bolts	RN, nb	48	
Diameter of Bolt Circle	DC, DBC	7062.0000	mm
Bolt Allowable Shear Stress		170.400	MPa
Ultimate Comp. Strength of Concrete	FPC	41.0	MPa
Allowable Comp. Strength of Concrete	FC	22.5	MPa
Modular ratio Steel/Concrete		9.833	
Thickness of Gusset Plates	TGA, e7	28.0000	mm
Width of Gussets at Top Plate	TWDT	206.0000	mm
Width of Gussets at Base Plate	BWDT	206.0000	mm
Gusset Plate Elastic Modulus	E	201746.0	MPa
Gusset Plate Yield Stress	SY	262.0	MPa
Height of Chair	h1	400.0000	mm
Distance between Gussets	RG, b6	120.0000	mm
Dist. from Bolt Center to Gusset (Rg/2)	CG	60.0000	mm
Number of Gussets per bolt	NG	2	
Thickness of Top Plate or Ring	TTA, e5	56.0000	mm
Radial Width of the Top Plate	TOPWTH, b5	216.0000	mm
Anchor Bolt Hole Dia. in Top Plate	dia,H	90.0000	mm
External Corrosion Allowance	CA	1.6000	mm
Dead Weight of Vessel	DW	8912054.0	N
Operating Weight of Vessel	ROW	19070104.0	N
Earthquake Moment on Basing	EQMOM	82723504128.0	N-mm
Wind Moment on Basing	WIMOM	16481172480.0	N-mm
User Moment at the Base [Earthquake Case]		5854182912.0	N-mm (UEQMOM)
User Moment at the Base [WindLoad Case]		5854182912.0	N-mm (UWIMOM)
Percent Bolt Preload	ppl	100.0	
Use AISC A5.2 Increase in Fc and Bolt Stress		No	
Use Allowable Weld Stress per AISC J2.5		No	
Factor for Increase of Allowables	Fact	1.0000	

Results for Brownell and Young Basing Analysis : Analyze Option

Note: This analysis is based on Neutral Axis shift method for Steel on Concrete (or a material with significantly different Young's modulus).

Calculation of Bolt Loads for Multiple Load Conditions:

Bolt Load, Earthquake Operating Case:

$$= ((4 \cdot M/DC) - W) / nb \text{ per Jawad \& Farr, Eq. 12.3}$$

$$= ((4 \cdot 0.88578E+11/7062.0) - 18475402) / 48$$



= 659910.7 N

Bolt Load, Windload Empty Case:

$$= ((4 \cdot M/DC) - W) / nb \quad \text{per Jawad \& Farr, Eq. 12.3}$$

$$= ((4 \cdot 0.22335E+11/7062.0) - 8912054) / 48$$

$$= 77788.1 \text{ N}$$

Bolt Load, Windload Vortex Shedding Case:

$$= ((4 \cdot M/DC) - W) / nb \quad \text{per Jawad \& Farr, Eq. 12.3}$$

$$= ((4 \cdot 0/7062.0) - 19070104) / 48$$

$$= -397293.8 \text{ N}$$

Governing Bolt Load Condition, Earthquake + Operating Condition:

Area Available in 1 Bolt (corroded) Abss : 2210.1030 mm²
 Area Available in all the Bolts Abss * RN : 106084.9375 mm²

Trial#	k	knew	Cc	Ct	z	j	Ft	Fc	Fs
2	0.211	0.317	1.253	2.636	0.457	0.776	5283555.5	23758956.0	76592.9
4	0.370	0.396	1.690	2.290	0.422	0.784	6041331.5	24516734.0	100789.9
6	0.409	0.403	1.787	2.203	0.414	0.785	6242764.0	24718166.0	108259.8
8	0.406	0.404	1.779	2.211	0.414	0.784	6225776.5	24701178.0	107610.0
10	0.405	0.405	1.777	2.213	0.414	0.784	6221536.5	24696938.0	107448.3
11	0.405	0.405	1.776	2.213	0.415	0.784	6219416.5	24694818.0	107367.6

The Actual Stress in a Single Bolt [Sbolt]:

$$= 2 \cdot Ft / (T1 \cdot Dc \cdot Ct)$$

$$= 2 \cdot 6219417 / (4.782 \cdot 7062.0 \cdot 2.213)$$

$$= 166.434 \text{ MPa, Should be } \leq 284.0 \text{ MPa}$$

Thickness of the Band of Bolting Steel [T1]

$$= RN \cdot \text{Bolt Area} / (\pi \cdot Dc)$$

$$= 48 \cdot 2210.103 / (\pi \cdot 7062.0)$$

$$= 4.782 \text{ mm}$$

Check the Bearing Stress in the Concrete [fc(max)]

$$= fc[(2kd + t3) / (2kd)]$$

$$= 11.507[(2 \cdot 0.4047 \cdot 7062.0 + 300.0) / (2 \cdot 0.4047 \cdot 7062.0)]$$

$$= 12.111 \text{ MPa, Should be } \leq 22.5 \text{ MPa}$$

Values for table 10.3, l = 216.0, b = 231.103, l/b = 0.934646

Maximum Moment per unit width [Mmax]:

$$= \max(Mx, My) = \max(57615.977, 79197.125) = 79197.125 \text{ N}$$

Reqd Thickness of Basing, Brownell & Young Method [T]:

$$= \sqrt{(6 \cdot Mmax / fallow)} + Ca$$

$$= \sqrt{(6 \cdot 79197.125 / 172.9)} + 1.6$$

$$= 54.024 \text{ mm}$$

Nomenclature:

a = (Dc - Ds) / 2	Skirt Distance to Bolt Circle
P = Sa * Abss	Maximum Load on one Bolt
l	Radial Top Plate Width
lavg	Average Gusset Plate Width
g1 = Gamma 1	Constant Term f(b/l)
g2 = Gamma 2	Constant Term f(b/l)
g = Flat distance / 2	Nut 1/2 Dimension (from Tema)
Fb	Allowable Bending Stress

Values for table 10.6, l = 216.0, b = 120.0, b/l = 0.555556

As b/l (0.556) is less than 1, inverting b/l = 1.8.

Moment Term, based on geometry [Mo]:

$$= P / (4 \cdot \pi) [1.3(\ln((2l \sin(\pi \cdot a/l)) / (\pi \cdot g))) + 1] - [(0.7 - g2)P / (4\pi)]$$

$$= 627616 / (4 \cdot \pi) [1.3(\ln((2 \cdot 216.0 \cdot \sin(\pi \cdot 108.0 / 216.0)) / (\pi \cdot 50.0))) + 1] - [(0.7 - 0.04) \cdot 627616 / (4 \cdot \pi)]$$

$$= 82650.8203 \text{ N}$$

Required Thickness of Continuous Top Ring [Tc]:



$$= \sqrt{(6 \cdot \text{Abs}(\text{Mo}) / \text{Fb}) + \text{Ca}}$$

$$= \sqrt{(6 \cdot \text{Abs}(82650.82) / 174.66) + 1.6}$$

$$= 54.8865 \text{ mm}$$

Required Gusset Plate Thickness [tg]:

$$= P / (\text{Stress Term} \cdot \text{lavg}) + \text{Ca}$$

$$= 627616 / (124.11 \cdot 206.0) + 1.6$$

$$= 26.15 \text{ (not less than } 9.525 + 1.6 \text{) mm}$$

Bolt spacing [m]:

$$= \pi \cdot \text{Bolt Circle Diameter} / \text{number of Bolts}$$

$$= \pi \cdot 7062.0 / 48 = 462.207 \text{ mm}$$

Req. Skirt Thk. to withstand Local Bending, (Brownell and Young) [t]:

$$= 1.76 (P \cdot a / (m \cdot (h + tba) \cdot 1.5 \cdot \text{Sktope}))^{2/3} \cdot r^{1/3} + \text{Ca}$$

$$= 1.76 (627616 \cdot 108.0 / (462.21 \cdot 344.0 \cdot 242))^{2/3} \cdot 3423.01^{1/3} + \text{Ca}$$

$$= 38.653 + 1.6 = 40.253 \text{ mm}$$

Shear Stress in a Single Bolt [tb]:

$$= \text{Shear Force} / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 1981687 / (2210.103 \cdot 48)$$

$$= 18.7 \text{ MPa. Must be less than } 170.4 \text{ MPa.}$$

Summary of Basing Thickness Calculations

Required Basing Thickness (tension)	54.0240	mm
Actual Basing Thickness as entered by user	56.0000	mm
Required Thickness of Chair Cap	54.8865	mm
Actual Top Ring Thickness as entered by user	56.0000	mm
Required Gusset thickness, + CA	26.1503	mm
Actual Gusset Thickness as entered by user	28.0000	mm
Required Thickness of Skirt for Local Stress	40.2534	mm
Given Thickness of Skirt	42.0000	mm
Required Gusset Height to meet local stress	265.9346	mm

Weld Size Calculations per Steel Plate Engineering Data - Vol. 2

Compute the Weld load at the Skirt/Base Junction [W]:

$$= \text{SkirtStress} \cdot (\text{SkirtThickness} - \text{CA})$$

$$= 86.289 \cdot (42.0 - 1.6)$$

$$= 3485.79 \text{ N/mm}$$

Results for Computed Minimum Basing Weld Size [BWeld]:

$$= W / [(0.4 \cdot \text{Yield}) \cdot 2 \cdot 0.707]$$

$$= 3486 / [(0.4 \cdot 243) \cdot 2 \cdot 0.707]$$

$$= 25.406 \text{ mm}$$

Results for Computed Minimum Gusset and Top Plate to Skirt Weld Size

Vertical Plate Load [Wv]:

$$= \text{Bolt Load} / (\text{Cmwth} + 2 \cdot (\text{Hg} + \text{Tta}))$$

$$= 627616 / (137.0 + 2 \cdot (288.0 + 56.0))$$

$$= 760.747 \text{ N/mm}$$

Horizontal Plate Load [Wh]:

$$= \text{Bolt Load} \cdot e / (\text{Cmwth} \cdot (\text{Hg} + \text{Tta}) + 0.6667 \cdot (\text{Hg} + \text{Tta})^2)$$

$$= 627616 \cdot 108.0 / (137.0 \cdot (344.0) + 0.6667 \cdot (344.0)^2)$$

$$= 537.875 \text{ N/mm}$$

Resultant Weld Load [Wr]:

$$= \sqrt{Wv^2 + Wh^2}$$

$$= \sqrt{760.75^2 + 537.88^2}$$

$$= 931.689 \text{ N/mm}$$

Results for Computed Min Gusset and Top Plate to Skirt Weld Size [GsWeld]:

$$= Wr / [(0.4 \cdot \text{Yield}) \cdot 2 \cdot 0.707]$$

$$= 931.69 / [(0.4 \cdot 243) \cdot 2 \cdot 0.707]$$



= 6.791 mm

Results for Computed Minimum Gusset to Top Plate Weld Size

Weld Load [Wv]:

$$\begin{aligned} &= \text{Bolt Load} / (2 \cdot \text{TopWth}) \\ &= 627616 / (2 \cdot 206.0) \\ &= 1523.339 \text{ N/mm} \end{aligned}$$

Weld Load [Wh]:

$$\begin{aligned} &= \text{Bolt Load} \cdot e / (2 \cdot \text{Hgt} \cdot \text{TopWth}) \\ &= 627616 \cdot 108.0 / (2 \cdot 344.0 \cdot 206.0) \\ &= 478.258 \text{ N/mm} \end{aligned}$$

Resultant Weld Load [Wr]:

$$\begin{aligned} &= \sqrt{Wv^2 + Wh^2} \\ &= \sqrt{1523.34^2 + 478.26^2} \\ &= 1596.651 \text{ N/mm} \end{aligned}$$

Results for Computed Min Gusset to Top Plate Weld Size [GtpWeld]:

$$\begin{aligned} &= Wr / [(0.4 \cdot \text{Yield}) \cdot 2 \cdot 0.707] \\ &= 1596.65 / [(0.4 \cdot 243) \cdot 2 \cdot 0.707] \\ &= 11.637 \text{ mm} \end{aligned}$$

Note: The calculated weld sizes need not exceed the component thickness framing into the weld. At the same time, the weld must meet a minimum size specification which is 3/16 in. (4.76 mm) or 1/4 in. (6.35 mm), depending on the component thickness.

Summary of Required Weld Sizes:

Required Basing to Skirt Double Fillet Weld Size	25.4064	mm
Required Gusset to Skirt Double Fillet Weld Size	6.7907	mm
Required Top Plate to Skirt Weld Size	11.6373	mm
Required Gusset to Top Plate Double Fillet Weld Size	11.6373	mm

Review notes about nozzle loadings and supports in the Warnings report.

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Input, Nozzle Desc: D 36"

From: 40

Pressure for Reinforcement Calculations	P	4.2123	MPa
Temperature for Internal Pressure	Temp	130	°C
Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	156.40	MPa
Parent Allowable Stress At Ambient	Sva	175.00	MPa
Inside Diameter of Hemispherical Head	D	6738.00	mm
Head Finished (Minimum) Thickness	t	60.0000	mm
Head Internal Corrosion Allowance	cas	6.0000	mm
Head External Corrosion Allowance	co	0.0000	mm
Distance from Head Centerline	L1	0.0000	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

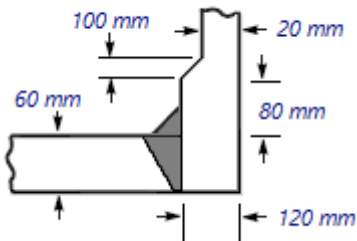
Material		SA-266 2	+IT
Material UNS Number		K03506	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	147.60	MPa
Allowable Stress At Ambient	Sna	165.00	MPa
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		36.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	20.0000	mm
Flange Material		SA-105	+IT
Hub Height of Integral Nozzle	h	80.0000	mm
Height of Beveled Transition	L'	100.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		120.0000	mm
Corrosion Allowance	can	6.0000	mm
Outside Projection	ho	200.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	53.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.
Flange Class		600	
Flange Grade		GR 1.1	
Flange Series		B	
Piping Exit Angle [North Clockwise reference]		90.0	deg
Bend Radius Multiplier		1.50	
Horizontal Run Length		4200.0	mm
Centerline Distance to Tangent		5400.0	mm

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: D 36"

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	36.000	in.
Actual Thickness Used in Calculation	0.787	in.



Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 926.4 \cdot (\exp(4.212/(147.6 \cdot 1.0)) - 1) + 6.0 + 0.0$$

$$= 13.4095 + 6.0 + 0.0 = 19.4095 \text{ mm Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 147.6 \cdot 1.0 \cdot \ln((2 \cdot 14.0 + 926.4) / 926.4) - 0.0$$

$$= 4.395 - 0.0 = 4.3951 \text{ MPa}$$

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c : 14.340 mm
 Min. required shell thickness, internal p + c : 51.757 mm

Nozzle neck minimum required thickness : 14.340 mm

Minimum nozzle neck thickness 19.990 mm, is greater than required thickness 14.340 mm

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 94.5,	Allowable	: 147.6 MPa	Passed
Expansion	: 0.0,	Allowable	: 296.2 MPa	Passed
Occasional	: 68.6,	Allowable	: 196.3 MPa	Passed
Shear	: 13.9,	Allowable	: 103.3 MPa	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

$$= \min[S_n/S, 1]$$

$$= \min[147.6/156.4, 1]$$

$$= 0.944$$

Thickness of Nozzle at Shell [tn]:

$$= \text{hub thickness} - \text{corrosion allowance}$$

$$= 120.0 - 6.0$$

$$= 114.000 \text{ mm}$$

Thickness of Nozzle at Top [tn2]:

$$= \text{thickness} - \text{corrosion allowance}$$

$$= 20.0 - 6.0$$

$$= 14.000 \text{ mm}$$

Effective Pressure Radius [Reff]:

$$= D_i/2 + \text{corrosion allowance}$$

$$= 6738.0/2 + 6.0$$

$$= 3375.000 \text{ mm}$$

Effective Length of Vessel Wall [LR]:

$$= \min(\sqrt{Reff \cdot t}, 2 \cdot R_n)$$



$$= \min(\sqrt{(3375.0 \cdot 54.0)}, 2 \cdot 463.2)$$
$$= 426.907 \text{ mm}$$

Intermediate value [Cn]:

$$= \min((t + t_e)/t_n)^{0.35}, 1)$$
$$= \min((54.0 + 0.0)/114.0)^{0.35}, 1)$$
$$= \min(0.77, 1)$$
$$= 0.770$$

Intermediate value [Cp]:

$$= 9999999.000$$

Effective Nozzle Wall Length Outside the Vessel [LH]:

$$= \min(t + t_e + F_p \cdot \sqrt{(R_n \cdot t_n)}, L_{pr1} + t)$$
$$= \min(54.0 + 0.0 + 0.77 \cdot \sqrt{(463.2 \cdot 114.0)}, 200.0 + 54.0)$$
$$= 230.912 \text{ mm}$$

Effective Vessel Thickness [teff]:

$$= t + ((A_5 \cdot f_{rp}) / LR)$$
$$= 54.0 + ((0.0 \cdot 1.0) / 426.907)$$
$$= 54.000 \text{ mm}$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$= t \cdot LR$$
$$= 54.0 \cdot 426.907$$
$$= 23053.002 \text{ mm}^2$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$= A_{2A} + A_{2B}$$
$$= 15276.0 + 6352.005$$
$$= 21628.004 \text{ mm}^2$$

Area Contributed by the Outside Fillet Weld [A41]:

$$= 0.5 \cdot \text{Leg}^2$$
$$= 0.5 \cdot 10.0^2$$
$$= 50.000 \text{ mm}^2$$

The total area contributed by A1 through A43 [AT]:

$$= A_1 + f_{rn}(A_2 + A_3) + A_{41} + A_{42} + A_{43}$$
$$= 23053.002 + 0.944(21628.004 + 0.0) + 50.0 + 0.0 + 0.0$$
$$= 43514.090 \text{ mm}^2$$

Nozzle Radius for Force Calculation [Rxn]:

$$= t_n / \ln[1 + t_n/R_n]$$
$$= 114.0 / \ln[1 + 114.0/463.2]$$
$$= 518.111 \text{ mm}$$

Shell Radius for Force Calculation [Rxs]:

$$= t_{eff} / \ln[1 + t_{eff}/R_{eff}]$$
$$= 54.0 / \ln[1 + 54.0/3375.0]$$
$$= 3401.921 \text{ mm}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$= 1.5 \cdot S \cdot E$$
$$= 1.5 \cdot 156.4 \cdot 1.0$$
$$= 234.6 \text{ MPa}$$

Force acting on the Nozzle [fN]:

$$= P \cdot R_{xn} \cdot LH$$
$$= 4.212 \cdot 518.111 \cdot 230.912$$
$$= 503907.6 \text{ N}$$

Force acting on the Shell [fS]:

$$= P \cdot R_{xs} \cdot (LR + t_n)/2$$
$$= 4.212 \cdot 3401.921 \cdot (426.907 + 114.0)/2$$
$$= 3875235.8 \text{ N}$$

Discontinuity Force from Internal Pressure [fY]:



$$= (P \cdot Rxs \cdot Rnc) / 2$$

$$= (4.212 \cdot 3401.921 \cdot 463.2) / 2$$

$$= 3318514.2 \text{ N}$$

Area Resisting Internal Pressure [Ap]:

$$= (fS + fY + fN) / P$$

$$= (3875236 + 3318514 + 503908) / 4.212$$

$$= 1827585.8 \text{ mm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$= Sallow / (2 \cdot Ap / AT - Rxs / (2 \cdot teff))$$

$$= 234.6 / (2 \cdot 1827586 / 43514.09 - 3401.921 / (2 \cdot 54.0))$$

$$= 4.5 \text{ MPa}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$= 2 \cdot S[t/Rxs]$$

$$= 2 \cdot 156.4[54.0/3401.921]$$

$$= 5.0 \text{ MPa}$$

Maximum Allowable Working Pressure [Pmax]:

$$= \min(Pmax1, Pmax2)$$

$$= \min(4.469, 4.965)$$

$$= 4.469 \text{ MPa}$$

Average Primary Membrane Stress [σAvg]:

$$= (fN + fS + fY) / AT$$

$$= (503908 + 3875236 + 3318514) / 43514.09$$

$$= 176.915 \text{ MPa}$$

General Primary Membrane Stress [σCirc]:

$$= P \cdot Rxs / (2 \cdot teff)$$

$$= 4.212 \cdot 3401.921 / (2 \cdot 54.0)$$

$$= 132.7 \text{ MPa}$$

Maximum Local Primary Membrane Stress [PL]:

$$= \max(2 \cdot \sigma Avg - \sigma Circ, \sigma Circ)$$

$$= \max(2 \cdot 176.915 - 132.684, 132.684)$$

$$= 221.1 \text{ MPa}$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	234.60	MPa
Local Primary Membrane Stress	PL	221.15	MPa
Maximum Allowable Working Pressure	Pmax	4.47	MPa

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$= (Rnc + tn) / Rnc$$

$$= (463.2 + 114.0) / 463.2$$

$$= 1.246 \text{ For set-in Nozzles}$$

Weld Length of Nozzle to Shell Weld [Lt]:

$$= \pi / 2 \cdot (Rn + tn)$$

$$= \pi / 2 \cdot (463.2 + 114.0)$$

$$= 906.664 \text{ mm}$$

Weld Throat Dimensions, (0.7071•Leg Dimensions) [L41T, L42T, L43T]:

$$= 7.071, 0.000, 0.000, \text{ mm}$$

Required Weld Throat Dimensions :

$$tc, L41T = 7.071, \text{ must be } \geq \min(0.7 \cdot 114.000, 6.000) \text{ Passed}$$

Weld Load Value [fwelds]:

$$= \min(fy \cdot ky, 1.5 \cdot Sn(A2 + A3), \pi / 4 \cdot P \cdot Rn^2 \cdot ky^2)$$

$$= \min(3318514 \cdot 1.25, 1.5 \cdot 147.6(21756.002 + 0.0), \pi / 4 \cdot 4.2 \cdot 463.2^2 \cdot 1.25^2)$$

$$= 1102106.125 \text{ N}$$

Weld Stress Value [τ]:

$$= fwelds / (Lt(0.49 \cdot L41T + 0.6 \cdot tw1 + 0.49 \cdot L43T))$$

$$= 1102106 / (906.664(0.49 \cdot 7.071 + 0.6 \cdot 53.0 + 0.49 \cdot 0.0))$$



= 34.473 < or = to 156.400 Weld Stress Passed

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Flange MDMT including temperature reduction per 3.11:

Flange MDMT with Temp reduction -65 °C

The Drop for this Nozzle is : 49.8132 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 309.8130 mm

Input Echo, WRC107/537 Item 1, Description: D 36" :

Diameter Basis for Vessel	Vbasis	ID	
Cylindrical or Spherical Vessel	Cylsph	Spherical	
Internal Corrosion Allowance	Cas	6.0000	mm
Vessel Diameter	Dv	6738.000	mm
Vessel Thickness	Tv	60.000	mm
Design Temperature	T1	130.0	C
Vessel Material		SA-516 70	
Vessel UNS Number		K02700	
Vessel Cold S.I. Allowable	Smc	175.00	MPa
Vessel Hot S.I. Allowable	Smh	156.40	MPa
Attachment Type	Type	Round	
WRC107 Attachment Classification	Holsol	Hollow	
Diameter Basis for Nozzle	Nbasis	ID	
Corrosion Allowance for Nozzle	Can	6.0000	mm
Nozzle Diameter	Dn	914.400	mm
Nozzle Thickness	Tn	120.000	mm
Nozzle Material		SA-266 2	
Nozzle UNS Number		K03506	
Nozzle Cold S.I. Allowable	SNmc	165.00	MPa
Nozzle Hot S.I. Allowable	SNmh	147.60	MPa
Design Internal Pressure	Dp	4.212	MPa
Include Pressure Thrust		No	

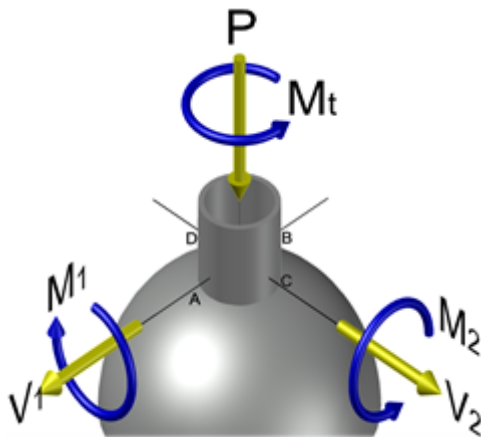
External Forces and Moments in WRC 107/537 Convention:

Radial Load (SUS)	P	80968.7	N
Longitudinal Shear (SUS)	(V1) V1	58477.4	N
Circumferential Shear (SUS)	(Vc) V2	58477.4	N
Circumferential Moment (SUS)	(Mc) M1	162008688.0	N-mm
Longitudinal Moment (SUS)	(Ml) M2	162008688.0	N-mm
Torsional Moment (SUS)	Mt	186309504.0	N-mm

Use Interactive Control No
 WRC107 Version Version March 1979

Include Pressure Stress Indices per Div. 2 No
 Compute Pressure Stress per WRC-368 No
 Local Loads applied at end of Nozzle/Attachment No

WRC Bulletin 537 provides equations for the dimensionless curves found in bulletin 107. As noted in the foreword to bulletin 537, "537 is equivalent to WRC 107". Where 107 is printed in the results below, "537" can be interchanged with "107".



WRC 107/537 has dimensional limitations that can be found in the bulletin's Foreword and which can also be inferred by review of the non-dimensional curve limits.

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$$

$$= 1154.4 + 2 \cdot 1.65 \cdot \sqrt{(3402.0 (60.0 - 6.0))}$$

$$= 2568.819 \text{ mm}$$

WRC 107 Stress Calculation for SUSTAINED loads:

Radial Load		P	80968.7	N
Circumferential Shear	(VC)	V2	58477.4	N
Longitudinal Shear	(VL)	V1	58477.4	N
Circumferential Moment	(MC)	M1	162008688.0	N-mm
Longitudinal Moment	(ML)	M2	162008688.0	N-mm
Torsional Moment		MT	186309504.0	N-mm

Dimensionless Param: U = 1.35 TAU = 5.00 (4.56) RHO = 0.47

Dimensionless Loads for Spherical Shells at Attachment Junction:

Curves read for 1979	Figure	Value	Location
$N(x) * T / P$	SP 1	0.02979	(A,B,C,D)
$M(x) / P$	SP 1	0.02854	(A,B,C,D)
$N(x) * T * \text{SQRT}(R_m * T) / MC$	SM 1	0.03108	(A,B,C,D)
$M(x) * \text{SQRT}(R_m * T) / MC$	SM 1	0.04132	(A,B,C,D)
$N(x) * T * \text{SQRT}(R_m * T) / ML$	SM 1	0.03108	(A,B,C,D)
$M(x) * \text{SQRT}(R_m * T) / ML$	SM 1	0.04132	(A,B,C,D)
$N(y) * T / P$	SP 1	0.04493	(A,B,C,D)
$M(y) / P$	SP 1	0.00811	(A,B,C,D)
$N(y) * T * \text{SQRT}(R_m * T) / MC$	SM 1	0.04264	(A,B,C,D)
$M(y) * \text{SQRT}(R_m * T) / MC$	SM 1	0.01333	(A,B,C,D)
$N(y) * T * \text{SQRT}(R_m * T) / ML$	SM 1	0.04264	(A,B,C,D)
$M(y) * \text{SQRT}(R_m * T) / ML$	SM 1	0.01333	(A,B,C,D)

Stress Concentration Factors: Kn = 1.00, Kb = 1.00

Stresses in the Vessel at the Attachment Junction (MPa)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Memb.	P	-0.8	-0.8	-0.8	-0.8	-0.8	-0.8	-0.8	-0.8
Rad. Bend.	P	-4.8	4.8	-4.8	4.8	-4.8	4.8	-4.8	4.8
Rad. Memb.	MC	0.0	0.0	0.0	0.0	-4.0	-4.0	4.0	4.0
Rad. Bend.	MC	0.0	0.0	0.0	0.0	-32.1	32.1	32.1	-32.1
Rad. Memb.	ML	-4.0	-4.0	4.0	4.0	0.0	0.0	0.0	0.0
Rad. Bend.	ML	-32.1	32.1	32.1	-32.1	0.0	0.0	0.0	0.0
Tot. Rad. Str.		-41.7	32.0	30.6	-24.2	-41.7	32.0	30.6	-24.2

Tang. Memb. P	-1.2	-1.2	-1.2	-1.2	-1.2	-1.2	-1.2	-1.2
Tang. Bend. P	-1.4	1.4	-1.4	1.4	-1.4	1.4	-1.4	1.4
Tang. Memb. MC	0.0	0.0	0.0	0.0	-5.5	-5.5	5.5	5.5
Tang. Bend. MC	0.0	0.0	0.0	0.0	-10.4	10.4	10.4	-10.4
Tang. Memb. ML	-5.5	-5.5	5.5	5.5	0.0	0.0	0.0	0.0
Tang. Bend. ML	-10.4	10.4	10.4	-10.4	0.0	0.0	0.0	0.0
Tot. Tang. Str.	-18.5	4.9	13.3	-4.7	-18.5	4.9	13.3	-4.7
Shear VC	0.6	0.6	-0.6	-0.6	0.0	0.0	0.0	0.0
Shear VL	0.0	0.0	0.0	0.0	-0.6	-0.6	0.6	0.6
Shear MT	1.6	1.6	1.6	1.6	1.6	1.6	1.6	1.6
Tot. Shear	2.2	2.2	1.1	1.1	1.1	1.1	2.2	2.2
Str. Int.	41.9	32.2	30.6	24.2	41.8	32.1	30.9	24.4

WRC 107/537 Stress Summations:

Vessel Stress Summation at Attachment Junction (MPa)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Pm (SUS)		132.7	132.7	132.7	132.7	132.7	132.7	132.7	132.7
Rad. Pl (SUS)		-4.9	-4.9	3.2	3.2	-4.9	-4.9	3.2	3.2
Rad. Q (SUS)		-36.9	36.9	27.4	-27.4	-36.9	36.9	27.4	-27.4
Long. Pm (SUS)		132.7	132.7	132.7	132.7	132.7	132.7	132.7	132.7
Long. Pl (SUS)		-6.8	-6.8	4.3	4.3	-6.8	-6.8	4.3	4.3
Long. Q (SUS)		-11.7	11.7	9.0	-9.0	-11.7	11.7	9.0	-9.0
Shear Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pl (SUS)		0.6	0.6	-0.6	-0.6	-0.6	-0.6	0.6	0.6
Shear Q (SUS)		1.6	1.6	1.6	1.6	1.6	1.6	1.6	1.6
Pm (SUS)		132.7	132.7	132.7	132.7	132.7	132.7	132.7	132.7
Pm+Pl (SUS)		128.0	128.0	137.2	137.2	128.0	128.0	137.2	137.2
Pm+Pl+Q (Total)		114.4	164.9	163.3	128.0	114.2	164.8	163.5	128.2

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	132.68	156.40	Passed
Pm+Pl (SUS)	137.23	234.60	Passed
Pm+Pl+Q (TOTAL)	164.89	469.20	Passed

The Pm+Pl+Q allowable above was based on the rules of ASME VIII-2 Part 4, 4.1.6.3 (a)&(b) or per ASME VIII-2, 5.5.6.1(1)&(2) Part 5 as applicable.

The Pm+Pl+Q allowable was based on a temperature range cycling from ambient to design temperature.

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Input, Nozzle Desc: L12 2"

From: 50

Pressure for Reinforcement Calculations	P	4.1717	MPa
Temperature for Internal Pressure	Temp	130	°C
Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	156.40	MPa
Parent Allowable Stress At Ambient	Sva	175.00	MPa
Inside Diameter of Cylindrical Shell	D	6700.00	mm
Shell Finished (Minimum) Thickness	t	98.0000	mm
Shell Internal Corrosion Allowance	cas	6.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		150.00	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

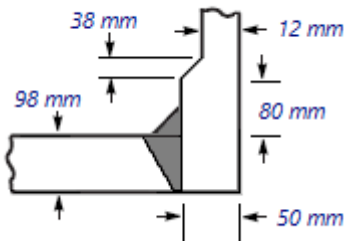
Material		SA-266 2	+IT
Material UNS Number		K03506	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	147.60	MPa
Allowable Stress At Ambient	Sna	165.00	MPa
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	12.0000	mm
Flange Material		SA-105	+IT
Hub Height of Integral Nozzle	h	80.0000	mm
Height of Beveled Transition	L'	38.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		50.0000	mm
Corrosion Allowance	can	6.0000	mm
Outside Projection	ho	196.8500	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	96.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.
Flange Class		600	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: L12 2"

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	2.000	in.
Actual Thickness Used in Calculation	0.472	in.



Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 62.8 \cdot (\exp(4.172/(147.6 \cdot 1.0)) - 1) + 6.0 + 0.0$$

$$= 0.9001 + 6.0 + 0.0 = 6.9001 \text{ mm Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 147.6 \cdot 1.0 \cdot \ln((2 \cdot 6.0 + 62.8) / 62.8) - 0.0$$

$$= 25.81 - 0.0 = 25.8098 \text{ MPa}$$

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c : 10.800 mm

Min. required shell thickness, internal p + c : 96.726 mm

Nozzle neck minimum required thickness : 10.800 mm

Minimum nozzle neck thickness 11.989 mm , is greater than required thickness 10.800 mm

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

$$= \min[S_n/S, 1]$$

$$= \min[147.6/156.4, 1]$$

$$= 0.944$$

Thickness of Nozzle at Shell [tn]:

$$= \text{hub thickness} - \text{corrosion allowance}$$

$$= 50.0 - 6.0$$

$$= 44.000 \text{ mm}$$

Thickness of Nozzle at Top [tn2]:

$$= \text{thickness} - \text{corrosion allowance}$$

$$= 12.0 - 6.0$$

$$= 6.000 \text{ mm}$$

Shell Diameter to Thickness ratio [D/t]:

$$= D_i/t$$

$$= 6712.0/92.0$$

$$= 72.957 \text{ must be less than 400.}$$

Effective Pressure Radius [Reff]:

$$= D_i/2 + \text{corrosion allowance}$$

$$= 6700.0/2 + 6.0$$

$$= 3356.000 \text{ mm}$$

Effective Length of Vessel Wall [LR]:

$$= \min(\sqrt{Reff \cdot t}, 2 \cdot R_n)$$

$$= \min(\sqrt{3356.0 \cdot 92.0}, 2 \cdot 31.4)$$

$$= 62.800 \text{ mm}$$

Thickness Limit Candidate [LH1]:

$$= \min[1.5t, t_e] + \sqrt{R_n \cdot t_n}$$

$$= \min[1.5 \cdot 92.0, 0.0] + \sqrt{31.4 \cdot 44.0}$$



= 37.170 mm

Thickness Limit Candidate [LH2]:

= Lpr1
= 196.85
= 196.850 mm

Thickness Limit Candidate [LH3]:

= $8(t + t_e)$
= $8(92.0 + 0.0)$
= 736.000 mm

Effective Nozzle Wall Length Outside the Vessel [LH]:

= $\min[LH1, LH2, LH3] + t$ for set-in nozzles
= $\min[37.17, 196.85, 736.0] + 92.0$
= 129.170 mm

Effective Vessel Thickness [teff]:

= $t + ((A5 \cdot frp) / LR)$
= $92.0 + ((0.0 \cdot 1.0) / 62.8)$
= 92.000 mm

Parameter [λ]:

= $\min(12, (2 \cdot Rn + tn) / (\sqrt{(Di + teff) \cdot teff}))$
= $\min(12, (2 \cdot 31.4 + 44.0) / (\sqrt{(6712.0 + 92.0) \cdot 92.0}))$
= 0.135

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

= $t \cdot LR \cdot \max((\lambda/5)^{0.85}, 1)$
= $92.0 \cdot 62.8 \cdot \max((0.135/5)^{0.85}, 1)$
= 5777.600 mm²

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

= $tn \cdot LH$
= $44.0 \cdot 129.17$
= 5683.475 mm²

Area Contributed by the Outside Fillet Weld [A41]:

= $0.5 \cdot Leg^2$
= $0.5 \cdot 10.0^2$
= 50.000 mm²

The total area contributed by A1 through A43 [AT]:

= $A1 + frn(A2 + A3) + A41 + A42 + A43$
= $5777.6 + 0.944(5683.475 + 0.0) + 50.0 + 0.0 + 0.0$
= 11191.287 mm²

Nozzle Radius for Force Calculation [Rxn]:

= $tn / \ln[1 + tn/Rn]$
= $44.0 / \ln[1 + 44.0/31.4]$
= 50.228 mm

Shell Radius for Force Calculation [Rxs]:

= $teff / \ln[1 + teff/Reff]$
= $92.0 / \ln[1 + 92.0/3356.0]$
= 3401.790 mm

Allowable Local Primary Membrane Stress [Sallow]:

= $1.5 \cdot S \cdot E$
= $1.5 \cdot 156.4 \cdot 1.0$
= 234.6 MPa

Force acting on the Nozzle [fN]:

= $P \cdot Rxn \cdot LH$
= $4.172 \cdot 50.228 \cdot 129.17$
= 27063.8 N

Force acting on the Shell [fS]:



$$\begin{aligned} &= P \cdot Rxs(LR + tn) \\ &= 4.172 \cdot 3401.79(62.8 + 44.0) \\ &= 1515503.8 \text{ N} \end{aligned}$$

Discontinuity Force from Internal Pressure [fY]:

$$\begin{aligned} &= P \cdot Rxs \cdot Rnc \\ &= 4.172 \cdot 3401.79 \cdot 31.4 \\ &= 445569.5 \text{ N} \end{aligned}$$

Area Resisting Internal Pressure [Ap]:

$$\begin{aligned} &= (fS + fY + fN)/P \\ &= (1515504 + 445569.5 + 27063.8)/4.172 \\ &= 476615.4 \text{ mm}^2 \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$\begin{aligned} &= Sallow / (2 \cdot Ap/AT - Rxs/teff) \\ &= 234.6 / (2 \cdot 476615.406/11191.287 - 3401.79/92.0) \\ &= 4.9 \text{ MPa} \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$\begin{aligned} &= S[t/Rxs] \\ &= 156.4[92.0/3401.79] \\ &= 4.2 \text{ MPa} \end{aligned}$$

Maximum Allowable Working Pressure [Pmax]:

$$\begin{aligned} &= \min(Pmax1, Pmax2) \\ &= \min(4.867, 4.23) \\ &= 4.230 \text{ MPa} \end{aligned}$$

Average Primary Membrane Stress [σAvg]:

$$\begin{aligned} &= (fN + fS + fY) / AT \\ &= (27063.773 + 1515504 + 445569.469)/11191.287 \\ &= 177.666 \text{ MPa} \end{aligned}$$

General Primary Membrane Stress [σCirc]:

$$\begin{aligned} &= P \cdot Rxs / teff \\ &= 4.172 \cdot 3401.79/92.0 \\ &= 154.3 \text{ MPa} \end{aligned}$$

Maximum Local Primary Membrane Stress [PL]:

$$\begin{aligned} &= \max(2 \cdot \sigma Avg - \sigma Circ, \sigma Circ) \\ &= \max(2 \cdot 177.666 - 154.253, 154.253) \\ &= 201.1 \text{ MPa} \end{aligned}$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	234.60	MPa
Local Primary Membrane Stress	PL	201.08	MPa
Maximum Allowable Working Pressure	Pmax	4.23	MPa

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$\begin{aligned} &= (Rnc + tn) / Rnc \\ &= (31.4 + 44.0)/31.4 \\ &= 2.401 \text{ For set-in Nozzles} \end{aligned}$$

Weld Length of Nozzle to Shell Weld [Lt]:

$$\begin{aligned} &= \pi/2 \cdot (Rn + tn) \\ &= \pi/2 \cdot (31.4 + 44.0) \\ &= 118.438 \text{ mm} \end{aligned}$$

Weld Throat Dimensions, (0.7071•Leg Dimensions) [L41T, L42T, L43T]:

$$= 7.071, 0.000, 0.000, \text{ mm}$$

Required Weld Throat Dimensions :

$$tc, L41T = 7.071, \text{ must be } \geq \min(0.7 \cdot 44.000, 6.000) \text{ Passed}$$

Weld Load Value [fwelds]:

$$\begin{aligned} &= \min(fy \cdot ky, 1.5 \cdot Sn(A2 + A3), \pi/4 \cdot P \cdot Rn^2 \cdot ky^2) \\ &= \min(445569 \cdot 2.4, 1.5 \cdot 147.6(5683.475+0.0), \pi/4 \cdot 4.2 \cdot 31.4^2 \cdot 2.4^2) \end{aligned}$$



= 18625.621 N

Weld Stress Value [τ]:

$$\begin{aligned} &= fwelds / (L_T(0.49 \cdot L_{41T} + 0.6 \cdot tw_1 + 0.49 \cdot L_{43T})) \\ &= 18625.621 / (118.438(0.49 \cdot 7.071 + 0.6 \cdot 92.0 + 0.49 \cdot 0.0)) \\ &= 2.681 < \text{or} = \text{to } 156.400 \text{ Weld Stress Passed} \end{aligned}$$

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Flange MDMT including temperature reduction per 3.11:

Flange MDMT with Temp reduction -66 °C

The Drop for this Nozzle is : 0.8488 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 295.6986 mm

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Input, Nozzle Desc: L14 1.5"

From: 50

Pressure for Reinforcement Calculations	P	4.1717	MPa
Temperature for Internal Pressure	Temp	130	°C
Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	156.40	MPa
Parent Allowable Stress At Ambient	Sva	175.00	MPa
Inside Diameter of Cylindrical Shell	D	6700.00	mm
Shell Finished (Minimum) Thickness	t	98.0000	mm
Shell Internal Corrosion Allowance	cas	6.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		150.00	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

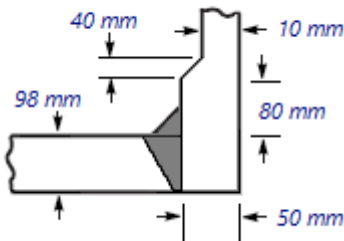
Material		SA-266 2	+IT
Material UNS Number		K03506	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	147.60	MPa
Allowable Stress At Ambient	Sna	165.00	MPa
Diameter Basis (for tr calc only)		Inside	
Layout Angle		30.00	deg
Diameter		1.5000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	10.0000	mm
Flange Material		SA-105	+IT
Hub Height of Integral Nozzle	h	80.0000	mm
Height of Beveled Transition	L'	40.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		50.0000	mm
Corrosion Allowance	can	6.0000	mm
Outside Projection	ho	250.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	96.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.
Flange Class		600	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: L14 1.5"

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	1.500	in.
Actual Thickness Used in Calculation	0.394	in.



Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 50.1 \cdot (\exp(4.172/(147.6 \cdot 1.0)) - 1) + 6.0 + 0.0$$

$$= 0.7181 + 6.0 + 0.0 = 6.7181 \text{ mm Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 147.6 \cdot 1.0 \cdot \ln((2 \cdot 4.0 + 50.1) / 50.1) - 0.0$$

$$= 21.866 - 0.0 = 21.8662 \text{ MPa}$$

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c : 9.420 mm
 Min. required shell thickness, internal p + c : 96.726 mm

Nozzle neck minimum required thickness : 9.420 mm

Minimum nozzle neck thickness 10.008 mm , is greater than required thickness 9.420 mm

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

$$= \min[S_n/S, 1]$$

$$= \min[147.6/156.4, 1]$$

$$= 0.944$$

Thickness of Nozzle at Shell [tn]:

$$= \text{hub thickness} - \text{corrosion allowance}$$

$$= 50.0 - 6.0$$

$$= 44.000 \text{ mm}$$

Thickness of Nozzle at Top [tn2]:

$$= \text{thickness} - \text{corrosion allowance}$$

$$= 10.0 - 6.0$$

$$= 4.000 \text{ mm}$$

Shell Diameter to Thickness ratio [D/t]:

$$= D_i/t$$

$$= 6712.0/92.0$$

$$= 72.957 \text{ must be less than } 400.$$

Effective Pressure Radius [Reff]:

$$= D_i/2 + \text{corrosion allowance}$$

$$= 6700.0/2 + 6.0$$

$$= 3356.000 \text{ mm}$$

Effective Length of Vessel Wall [LR]:

$$= \min(\sqrt{Reff \cdot t}, 2 \cdot R_n)$$

$$= \min(\sqrt{3356.0 \cdot 92.0}, 2 \cdot 25.05)$$

$$= 50.100 \text{ mm}$$

Thickness Limit Candidate [LH1]:

$$= \min[1.5t_e, \sqrt{R_n \cdot t_n}]$$

$$= \min[1.5 \cdot 92.0, 0.0] + \sqrt{25.05 \cdot 44.0}$$



$$= 33.199 \text{ mm}$$

Thickness Limit Candidate [LH2]:

$$\begin{aligned} &= L_{pr1} \\ &= 250.0 \\ &= 250.000 \text{ mm} \end{aligned}$$

Thickness Limit Candidate [LH3]:

$$\begin{aligned} &= 8(t + t_e) \\ &= 8(92.0 + 0.0) \\ &= 736.000 \text{ mm} \end{aligned}$$

Effective Nozzle Wall Length Outside the Vessel [LH]:

$$\begin{aligned} &= \min[LH1, LH2, LH3] + t \quad \text{for set-in nozzles} \\ &= \min[33.199, 250.0, 736.0] + 92.0 \\ &= 125.199 \text{ mm} \end{aligned}$$

Effective Vessel Thickness [teff]:

$$\begin{aligned} &= t + ((A5 \cdot frp) / LR) \\ &= 92.0 + ((0.0 \cdot 1.0) / 50.1) \\ &= 92.000 \text{ mm} \end{aligned}$$

Parameter [λ]:

$$\begin{aligned} &= \min\left(12, \left(2 \cdot R_n + t_n\right) / \sqrt{\left(\left(D_i + t_{eff}\right) \cdot t_{eff}\right)}\right) \\ &= \min\left(12, \left(2 \cdot 25.05 + 44.0\right) / \sqrt{\left(\left(6712.0 + 92.0\right) \cdot 92.0\right)}\right) \\ &= 0.119 \end{aligned}$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$\begin{aligned} &= t \cdot LR \cdot \max\left(\left(\lambda/5\right)^{0.85}, 1\right) \\ &= 92.0 \cdot 50.1 \cdot \max\left(\left(0.119/5\right)^{0.85}, 1\right) \\ &= 4609.200 \text{ mm}^2 \end{aligned}$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$\begin{aligned} &= t_n \cdot LH \\ &= 44.0 \cdot 125.199 \\ &= 5508.774 \text{ mm}^2 \end{aligned}$$

Area Contributed by the Outside Fillet Weld [A41]:

$$\begin{aligned} &= 0.5 \cdot \text{Leg}^2 \\ &= 0.5 \cdot 10.0^2 \\ &= 50.000 \text{ mm}^2 \end{aligned}$$

The total area contributed by A1 through A43 [AT]:

$$\begin{aligned} &= A1 + frn(A2 + A3) + A41 + A42 + A43 \\ &= 4609.2 + 0.944(5508.774 + 0.0) + 50.0 + 0.0 + 0.0 \\ &= 9858.019 \text{ mm}^2 \end{aligned}$$

Nozzle Radius for Force Calculation [Rxn]:

$$\begin{aligned} &= t_n / \ln[1 + t_n/R_n] \\ &= 44.0 / \ln[1 + 44.0/25.05] \\ &= 43.394 \text{ mm} \end{aligned}$$

Shell Radius for Force Calculation [Rxs]:

$$\begin{aligned} &= t_{eff} / \ln[1 + t_{eff}/R_{eff}] \\ &= 92.0 / \ln[1 + 92.0/3356.0] \\ &= 3401.790 \text{ mm} \end{aligned}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$\begin{aligned} &= 1.5 \cdot S \cdot E \\ &= 1.5 \cdot 156.4 \cdot 1.0 \\ &= 234.6 \text{ MPa} \end{aligned}$$

Force acting on the Nozzle [fN]:

$$\begin{aligned} &= P \cdot R_{xn} \cdot LH \\ &= 4.172 \cdot 43.394 \cdot 125.199 \\ &= 22662.8 \text{ N} \end{aligned}$$

Force acting on the Shell [fS]:



$$\begin{aligned} &= P \cdot Rxs(LR + tn) \\ &= 4.172 \cdot 3401.79(50.1 + 44.0) \\ &= 1335289.2 \text{ N} \end{aligned}$$

Discontinuity Force from Internal Pressure [fY]:

$$\begin{aligned} &= P \cdot Rxs \cdot Rnc \\ &= 4.172 \cdot 3401.79 \cdot 25.05 \\ &= 355462.2 \text{ N} \end{aligned}$$

Area Resisting Internal Pressure [Ap]:

$$\begin{aligned} &= (fS + fY + fN)/P \\ &= (1335289 + 355462.2 + 22662.8)/4.172 \\ &= 410756.2 \text{ mm}^2 \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$\begin{aligned} &= Sallow / (2 \cdot Ap/AT - Rxs/teff) \\ &= 234.6 / (2 \cdot 410756.219/9858.019 - 3401.79/92.0) \\ &= 5.1 \text{ MPa} \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$\begin{aligned} &= S[t/Rxs] \\ &= 156.4[92.0/3401.79] \\ &= 4.2 \text{ MPa} \end{aligned}$$

Maximum Allowable Working Pressure [Pmax]:

$$\begin{aligned} &= \min(Pmax1, Pmax2) \\ &= \min(5.061, 4.23) \\ &= 4.230 \text{ MPa} \end{aligned}$$

Average Primary Membrane Stress [σ Avg]:

$$\begin{aligned} &= (fN + fS + fY) / AT \\ &= (22662.803 + 1335289 + 355462.25)/9858.019 \\ &= 173.824 \text{ MPa} \end{aligned}$$

General Primary Membrane Stress [σ Circ]:

$$\begin{aligned} &= P \cdot Rxs / teff \\ &= 4.172 \cdot 3401.79/92.0 \\ &= 154.3 \text{ MPa} \end{aligned}$$

Maximum Local Primary Membrane Stress [PL]:

$$\begin{aligned} &= \max(2 \cdot \sigma Avg - \sigma Circ, \sigma Circ) \\ &= \max(2 \cdot 173.824 - 154.253, 154.253) \\ &= 193.4 \text{ MPa} \end{aligned}$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	234.60	MPa
Local Primary Membrane Stress	PL	193.39	MPa
Maximum Allowable Working Pressure	Pmax	4.23	MPa

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$\begin{aligned} &= (Rnc + tn) / Rnc \\ &= (25.05 + 44.0)/25.05 \\ &= 2.756 \text{ For set-in Nozzles} \end{aligned}$$

Weld Length of Nozzle to Shell Weld [Lt]:

$$\begin{aligned} &= \pi/2 \cdot (Rn + tn) \\ &= \pi/2 \cdot (25.05 + 44.0) \\ &= 108.463 \text{ mm} \end{aligned}$$

Weld Throat Dimensions, (0.7071•Leg Dimensions) [L41T, L42T, L43T]:

$$= 7.071, 0.000, 0.000, \text{ mm}$$

Required Weld Throat Dimensions :

$$tc, L41T = 7.071, \text{ must be } \geq \min(0.7 \cdot 44.000, 6.000) \text{ Passed}$$

Weld Load Value [fwelds]:

$$\begin{aligned} &= \min(fy \cdot ky, 1.5 \cdot Sn(A2 + A3), \pi/4 \cdot P \cdot Rn^2 \cdot ky^2) \\ &= \min(355462 \cdot 2.76, 1.5 \cdot 147.6(5508.774 + 0.0), \pi/4 \cdot 4.2 \cdot 25.05^2 \cdot 2.76^2) \end{aligned}$$



= 15620.521 N

Weld Stress Value [τ]:

$$\begin{aligned} &= fwelds / (L_T(0.49 \cdot L_{41T} + 0.6 \cdot tw_1 + 0.49 \cdot L_{43T})) \\ &= 15620.521 / (108.463(0.49 \cdot 7.071 + 0.6 \cdot 92.0 + 0.49 \cdot 0.0)) \\ &= 2.455 < \text{or} = \text{to } 156.400 \text{ Weld Stress Passed} \end{aligned}$$

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Flange MDMT including temperature reduction per 3.11:

Flange MDMT with Temp reduction -66 °C

The Drop for this Nozzle is : 0.7116 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 348.7114 mm

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Input, Nozzle Desc: MH-6 24"

From: 60

Pressure for Reinforcement Calculations	P	4.1276	MPa
Temperature for Internal Pressure	Temp	130	°C
Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	156.40	MPa
Parent Allowable Stress At Ambient	Sva	175.00	MPa
Inside Diameter of Cylindrical Shell	D	6700.00	mm
Shell Finished (Minimum) Thickness	t	98.0000	mm
Shell Internal Corrosion Allowance	cas	6.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		4399.98	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

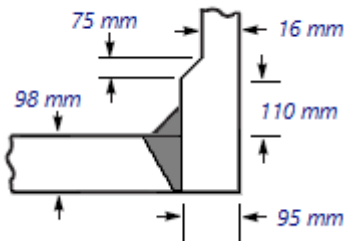
Material		SA-266 2	+IT
Material UNS Number		K03506	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	147.60	MPa
Allowable Stress At Ambient	Sna	165.00	MPa
Diameter Basis (for tr calc only)		Inside	
Layout Angle		225.00	deg
Diameter		24.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.0000	mm
Flange Material		SA-105	+IT
Hub Height of Integral Nozzle	h	110.0000	mm
Height of Beveled Transition	L'	75.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		95.0000	mm
Corrosion Allowance	can	0.0000	mm
Outside Projection	ho	310.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	94.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.
This is a Manway, Coupling or Access Opening.			
Flange Class		600	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: MH-6 24"

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	24.000	in.
Actual Thickness Used in Calculation	0.630	in.



Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 609.6 \cdot (\exp(4.128/(147.6 \cdot 1.0)) - 1) + 0.0 + 0.0$$

$$= 8.6439 + 0.0 + 0.0 = 8.6439 \text{ mm Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 147.6 \cdot 1.0 \cdot \ln((2 \cdot 16.0 + 609.6) / 609.6) - 0.0$$

$$= 7.552 - 0.0 = 7.5515 \text{ MPa}$$

This nozzle or its diameter limit intersects weld seam # [2] on this element. The distance between the nozzle and seam is, 351.43 mm, while the distance from the center of the nozzle to the edge of the diameter limit is: 955.45 mm

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

$$= \min[S_n/S, 1]$$

$$= \min[147.6/156.4, 1]$$

$$= 0.944$$

Thickness of Nozzle at Shell [tn]:

$$= \text{hub thickness} - \text{corrosion allowance}$$

$$= 95.0 - 0.0$$

$$= 95.000 \text{ mm}$$

Thickness of Nozzle at Top [tn2]:

$$= \text{thickness} - \text{corrosion allowance}$$

$$= 16.0 - 0.0$$

$$= 16.000 \text{ mm}$$

Shell Diameter to Thickness ratio [D/t]:

$$= D_i/t$$

$$= 6712.0/92.0$$

$$= 72.957 \text{ must be less than 400.}$$

Effective Pressure Radius [Reff]:

$$= D_i/2 + \text{corrosion allowance}$$

$$= 6700.0/2 + 6.0$$

$$= 3356.000 \text{ mm}$$

Effective Length of Vessel Wall [LR]:

$$= \min(\sqrt{Reff \cdot t}, 2 \cdot R_n)$$

$$= \min(\sqrt{3356.0 \cdot 92.0}, 2 \cdot 304.8)$$

$$= 555.655 \text{ mm}$$

Thickness Limit Candidate [LH1]:

$$= \min[1.5t, t_e] + \sqrt{R_n \cdot t_n}$$

$$= \min[1.5 \cdot 92.0, 0.0] + \sqrt{304.8 \cdot 95.0}$$

$$= 170.165 \text{ mm}$$

Thickness Limit Candidate [LH2]:

$$= L_{pr1}$$



$$= 310.0$$
$$= 310.000 \text{ mm}$$

Thickness Limit Candidate [LH3]:

$$= 8(t + t_e)$$
$$= 8(92.0 + 0.0)$$
$$= 736.000 \text{ mm}$$

Effective Nozzle Wall Length Outside the Vessel [LH]:

$$= \min[LH1, LH2, LH3] + t \quad \text{for set-in nozzles}$$
$$= \min[170.165, 310.0, 736.0] + 92.0$$
$$= 262.165 \text{ mm}$$

Effective Vessel Thickness [teff]:

$$= t + ((A5 \cdot frp) / LR)$$
$$= 92.0 + ((0.0 \cdot 1.0) / 555.655)$$
$$= 92.000 \text{ mm}$$

Parameter [λ]:

$$= \min(12, (2 \cdot Rn + t_n) / (\sqrt{ (D_i + t_{eff}) \cdot t_{eff} }))$$
$$= \min(12, (2 \cdot 304.8 + 95.0) / (\sqrt{ (6712.0 + 92.0) \cdot 92.0 }))$$
$$= 0.891$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$= t \cdot LR \cdot \max((\lambda/5)^{0.85}, 1)$$
$$= 92.0 \cdot 555.655 \cdot \max((0.891/5)^{0.85}, 1)$$
$$= 51120.227 \text{ mm}^2$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$= A2A + A2B$$
$$= 19190.0 + 3809.221$$
$$= 22999.221 \text{ mm}^2$$

Area Contributed by the Outside Fillet Weld [A41]:

$$= 0.5 \cdot \text{Leg}^2$$
$$= 0.5 \cdot 10.0^2$$
$$= 50.000 \text{ mm}^2$$

The total area contributed by A1 through A43 [AT]:

$$= A1 + frn(A2 + A3) + A41 + A42 + A43$$
$$= 51120.227 + 0.944(22999.221 + 0.0) + 50.0 + 0.0 + 0.0$$
$$= 72875.375 \text{ mm}^2$$

Nozzle Radius for Force Calculation [Rxn]:

$$= t_n / \ln[1 + t_n / Rn]$$
$$= 95.0 / \ln[1 + 95.0 / 304.8]$$
$$= 350.155 \text{ mm}$$

Shell Radius for Force Calculation [Rxs]:

$$= t_{eff} / \ln[1 + t_{eff} / R_{eff}]$$
$$= 92.0 / \ln[1 + 92.0 / 3356.0]$$
$$= 3401.790 \text{ mm}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$= 1.5 \cdot S \cdot E$$
$$= 1.5 \cdot 156.4 \cdot 1.0$$
$$= 234.6 \text{ MPa}$$

Force acting on the Nozzle [fN]:

$$= P \cdot Rxn \cdot LH$$
$$= 4.128 \cdot 350.155 \cdot 262.165$$
$$= 378870.3 \text{ N}$$

Force acting on the Shell [fS]:

$$= P \cdot Rxs(LR + t_n)$$
$$= 4.128 \cdot 3401.79(555.655 + 95.0)$$
$$= 9135127.0 \text{ N}$$



Discontinuity Force from Internal Pressure [fY]:

$$= P \cdot Rxs \cdot Rnc$$

$$= 4.128 \cdot 3401.79 \cdot 304.8$$

$$= 4279362.0 \text{ N}$$

Area Resisting Internal Pressure [Ap]:

$$= (fS + fY + fN) / P$$

$$= (9135127 + 4279362 + 378870.3) / 4.128$$

$$= 3342054.8 \text{ mm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$= Sallow / (2 \cdot Ap / AT - Rxs / teff)$$

$$= 234.6 / (2 \cdot 3342055 / 72875.375 - 3401.79 / 92.0)$$

$$= 4.3 \text{ MPa}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$= S[t / Rxs]$$

$$= 156.4 [92.0 / 3401.79]$$

$$= 4.2 \text{ MPa}$$

Maximum Allowable Working Pressure [Pmax]:

$$= \min(Pmax1, Pmax2)$$

$$= \min(4.285, 4.23)$$

$$= 4.230 \text{ MPa}$$

Average Primary Membrane Stress [σAvg]:

$$= (fN + fS + fY) / AT$$

$$= (378870.312 + 9135127 + 4279362) / 72875.375$$

$$= 189.289 \text{ MPa}$$

General Primary Membrane Stress [σCirc]:

$$= P \cdot Rxs / teff$$

$$= 4.128 \cdot 3401.79 / 92.0$$

$$= 152.6 \text{ MPa}$$

Maximum Local Primary Membrane Stress [PL]:

$$= \max(2 \cdot \sigma Avg - \sigma Circ, \sigma Circ)$$

$$= \max(2 \cdot 189.289 - 152.621, 152.621)$$

$$= 226.0 \text{ MPa}$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	234.60	MPa
Local Primary Membrane Stress	PL	225.96	MPa
Maximum Allowable Working Pressure	Pmax	4.23	MPa

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$= (Rnc + tn) / Rnc$$

$$= (304.8 + 95.0) / 304.8$$

$$= 1.312 \text{ For set-in Nozzles}$$

Weld Length of Nozzle to Shell Weld [Lt]:

$$= \pi / 2 \cdot (Rn + tn)$$

$$= \pi / 2 \cdot (304.8 + 95.0)$$

$$= 628.004 \text{ mm}$$

Weld Throat Dimensions, (0.7071•Leg Dimensions) [L41T, L42T, L43T]:

$$= 7.071, 0.000, 0.000, \text{ mm}$$

Required Weld Throat Dimensions :

$$tc, L41T = 7.071, \text{ must be } \geq \min(0.7 \cdot 95.000, 6.000) \text{ Passed}$$

Weld Load Value [fwelds]:

$$= \min(fy \cdot ky, 1.5 \cdot Sn(A2 + A3), \pi / 4 \cdot P \cdot Rn^2 \cdot ky^2)$$

$$= \min(4279362 \cdot 1.31, 1.5 \cdot 147.6(22999.221 + 0.0), \pi / 4 \cdot 4.1 \cdot 304.8^2 \cdot 1.31^2)$$

$$= 518121.938 \text{ N}$$

Weld Stress Value [τ]:

$$= fwelds / (Lt(0.49 \cdot L41T + 0.6 \cdot tw1 + 0.49 \cdot L43T))$$

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Note: This Element/Detail was specified as being Impact Tested.

Note: This Element/Detail was specified as being Impact Tested.

Flange MDMT with Temp reduction	-67 °C
---------------------------------	--------

The Cut Length for this Nozzle is, Drop + Ho + H + T : 431.9421 mm

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Input, Nozzle Desc: L11 2"

From: 70

Pressure for Reinforcement Calculations	P	4.1047	MPa
Temperature for Internal Pressure	Temp	130	°C
Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	156.40	MPa
Parent Allowable Stress At Ambient	Sva	175.00	MPa
Inside Diameter of Cylindrical Shell	D	6700.00	mm
Shell Finished (Minimum) Thickness	t	96.0000	mm
Shell Internal Corrosion Allowance	cas	6.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		6750.01	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

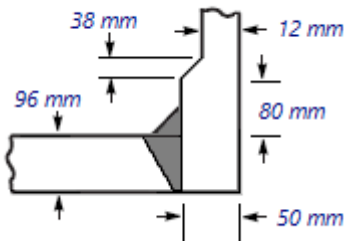
Material		SA-266 2	+IT
Material UNS Number		K03506	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	147.60	MPa
Allowable Stress At Ambient	Sna	165.00	MPa
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	12.0000	mm
Flange Material		SA-105	+IT
Flange Type		N/A	
Hub Height of Integral Nozzle	h	80.0000	mm
Height of Beveled Transition	L'	38.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		50.0000	mm
Style of FVC Nozzle		Long Weld Neck	
Corrosion Allowance	can	6.0000	mm
Outside Projection	ho	196.8500	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel Wgnv		96.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.
Flange Class		600	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: L11 2"

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	2.000	in.
Actual Thickness Used in Calculation	0.472	in.



Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 62.8 \cdot (\exp(4.105/(147.6 \cdot 1.0)) - 1) + 6.0 + 0.0$$

$$= 0.8855 + 6.0 + 0.0 = 6.8855 \text{ mm Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 147.6 \cdot 1.0 \cdot \ln((2 \cdot 6.0 + 62.8) / 62.8) - 0.0$$

$$= 25.81 - 0.0 = 25.8098 \text{ MPa}$$

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c : 10.800 mm

Min. required shell thickness, internal p + c : 95.249 mm

Nozzle neck minimum required thickness : 10.800 mm

Minimum nozzle neck thickness 11.989 mm , is greater than required thickness 10.800 mm

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

$$= \min[S_n/S, 1]$$

$$= \min[147.6/156.4, 1]$$

$$= 0.944$$

Thickness of Nozzle at Shell [tn]:

$$= \text{hub thickness} - \text{corrosion allowance}$$

$$= 50.0 - 6.0$$

$$= 44.000 \text{ mm}$$

Thickness of Nozzle at Top [tn2]:

$$= \text{thickness} - \text{corrosion allowance}$$

$$= 12.0 - 6.0$$

$$= 6.000 \text{ mm}$$

Shell Diameter to Thickness ratio [D/t]:

$$= D_i/t$$

$$= 6712.0/90.0$$

$$= 74.578 \text{ must be less than } 400.$$

Effective Pressure Radius [Reff]:

$$= D_i/2 + \text{corrosion allowance}$$

$$= 6700.0/2 + 6.0$$

$$= 3356.000 \text{ mm}$$

Effective Length of Vessel Wall [LR]:

$$= \min(\sqrt{Reff \cdot t}, 2 \cdot R_n)$$

$$= \min(\sqrt{3356.0 \cdot 90.0}, 2 \cdot 31.4)$$

$$= 62.800 \text{ mm}$$

Thickness Limit Candidate [LH1]:

$$= \min[1.5t, t_e] + \sqrt{R_n \cdot t_n}$$

$$= \min[1.5 \cdot 90.0, 0.0] + \sqrt{31.4 \cdot 44.0}$$



$$= 37.170 \text{ mm}$$

Thickness Limit Candidate [LH2]:

$$= L_{pr1}$$

$$= 196.85$$

$$= 196.850 \text{ mm}$$

Thickness Limit Candidate [LH3]:

$$= 8(t + t_e)$$

$$= 8(90.0 + 0.0)$$

$$= 720.000 \text{ mm}$$

Effective Nozzle Wall Length Outside the Vessel [LH]:

$$= \min[LH1, LH2, LH3] + t \quad \text{for set-in nozzles}$$

$$= \min[37.17, 196.85, 720.0] + 90.0$$

$$= 127.170 \text{ mm}$$

Effective Vessel Thickness [teff]:

$$= t + ((A5 \cdot frp) / LR)$$

$$= 90.0 + ((0.0 \cdot 1.0) / 62.8)$$

$$= 90.000 \text{ mm}$$

Parameter [λ]:

$$= \min\left(12, \left(2 \cdot R_n + t_n\right) / \sqrt{\left(\left(D_i + t_{eff}\right) \cdot t_{eff}\right)}\right)$$

$$= \min\left(12, \left(2 \cdot 31.4 + 44.0\right) / \sqrt{\left(\left(6712.0 + 90.0\right) \cdot 90.0\right)}\right)$$

$$= 0.136$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$= t \cdot LR \cdot \max\left(\left(\lambda/5\right)^{0.85}, 1\right)$$

$$= 90.0 \cdot 62.8 \cdot \max\left(\left(0.136/5\right)^{0.85}, 1\right)$$

$$= 5652.000 \text{ mm}^2$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$= t_n \cdot LH$$

$$= 44.0 \cdot 127.17$$

$$= 5595.475 \text{ mm}^2$$

Area Contributed by the Outside Fillet Weld [A41]:

$$= 0.5 \cdot \text{Leg}^2$$

$$= 0.5 \cdot 10.0^2$$

$$= 50.000 \text{ mm}^2$$

The total area contributed by A1 through A43 [AT]:

$$= A1 + \text{frn}(A2 + A3) + A41 + A42 + A43$$

$$= 5652.0 + 0.944(5595.475 + 0.0) + 50.0 + 0.0 + 0.0$$

$$= 10982.640 \text{ mm}^2$$

Nozzle Radius for Force Calculation [Rxn]:

$$= t_n / \ln[1 + t_n/R_n]$$

$$= 44.0 / \ln[1 + 44.0/31.4]$$

$$= 50.228 \text{ mm}$$

Shell Radius for Force Calculation [Rxs]:

$$= t_{eff} / \ln[1 + t_{eff}/R_{eff}]$$

$$= 90.0 / \ln[1 + 90.0/3356.0]$$

$$= 3400.797 \text{ mm}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$= 1.5 \cdot S \cdot E$$

$$= 1.5 \cdot 156.4 \cdot 1.0$$

$$= 234.6 \text{ MPa}$$

Force acting on the Nozzle [fN]:

$$= P \cdot R_{xn} \cdot LH$$

$$= 4.105 \cdot 50.228 \cdot 127.17$$

$$= 26216.7 \text{ N}$$

Force acting on the Shell [fS]:



$$\begin{aligned} &= P \cdot Rxs(LR + tn) \\ &= 4.105 \cdot 3400.797(62.8 + 44.0) \\ &= 1490721.0 \text{ N} \end{aligned}$$

Discontinuity Force from Internal Pressure [fY]:

$$\begin{aligned} &= P \cdot Rxs \cdot Rnc \\ &= 4.105 \cdot 3400.797 \cdot 31.4 \\ &= 438283.2 \text{ N} \end{aligned}$$

Area Resisting Internal Pressure [Ap]:

$$\begin{aligned} &= (fS + fY + fN)/P \\ &= (1490721 + 438283.2 + 26216.7)/4.105 \\ &= 476377.6 \text{ mm}^2 \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$\begin{aligned} &= Sallow / (2 \cdot Ap/AT - Rxs/teff) \\ &= 234.6 / (2 \cdot 476377.594/10982.64 - 3400.797/90.0) \\ &= 4.8 \text{ MPa} \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$\begin{aligned} &= S[t/Rxs] \\ &= 156.4[90.0/3400.797] \\ &= 4.1 \text{ MPa} \end{aligned}$$

Maximum Allowable Working Pressure [Pmax]:

$$\begin{aligned} &= \min(Pmax1, Pmax2) \\ &= \min(4.791, 4.139) \\ &= 4.139 \text{ MPa} \end{aligned}$$

Average Primary Membrane Stress [σAvg]:

$$\begin{aligned} &= (fN + fS + fY) / AT \\ &= (26216.674 + 1490721 + 438283.156)/10982.64 \\ &= 178.043 \text{ MPa} \end{aligned}$$

General Primary Membrane Stress [σCirc]:

$$\begin{aligned} &= P \cdot Rxs / teff \\ &= 4.105 \cdot 3400.797/90.0 \\ &= 155.1 \text{ MPa} \end{aligned}$$

Maximum Local Primary Membrane Stress [PL]:

$$\begin{aligned} &= \max(2 \cdot \sigma Avg - \sigma Circ, \sigma Circ) \\ &= \max(2 \cdot 178.043 - 155.103, 155.103) \\ &= 201.0 \text{ MPa} \end{aligned}$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	234.60	MPa
Local Primary Membrane Stress	PL	200.98	MPa
Maximum Allowable Working Pressure	Pmax	4.14	MPa

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$\begin{aligned} &= (Rnc + tn) / Rnc \\ &= (31.4 + 44.0)/31.4 \\ &= 2.401 \text{ For set-in Nozzles} \end{aligned}$$

Weld Length of Nozzle to Shell Weld [Lt]:

$$\begin{aligned} &= \pi/2 \cdot (Rn + tn) \\ &= \pi/2 \cdot (31.4 + 44.0) \\ &= 118.438 \text{ mm} \end{aligned}$$

Weld Throat Dimensions, (0.7071•Leg Dimensions) [L41T, L42T, L43T]:

$$= 7.071, 0.000, 0.000, \text{ mm}$$

Required Weld Throat Dimensions :

$$tc, L41T = 7.071, \text{ must be } \geq \min(0.7 \cdot 44.000, 6.000) \text{ Passed}$$

Weld Load Value [fwelds]:

$$\begin{aligned} &= \min(fy \cdot ky, 1.5 \cdot Sn(A2 + A3), \pi/4 \cdot P \cdot Rn^2 \cdot ky^2) \\ &= \min(438283 \cdot 2.4, 1.5 \cdot 147.6(5595.475+0.0), \pi/4 \cdot 4.1 \cdot 31.4^2 \cdot 2.4^2) \end{aligned}$$



= 18326.398 N

Weld Stress Value [τ]:

$$\begin{aligned} &= fwelds / (L_T(0.49 \cdot L_{41T} + 0.6 \cdot tw1 + 0.49 \cdot L_{43T})) \\ &= 18326.398 / (118.438(0.49 \cdot 7.071 + 0.6 \cdot 90.0 + 0.49 \cdot 0.0)) \\ &= 2.693 < \text{or} = \text{to } 156.400 \text{ Weld Stress Passed} \end{aligned}$$

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

The Drop for this Nozzle is : 0.8488 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 293.6987 mm

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Input, Nozzle Desc: L13 1.5"

From: 70

Pressure for Reinforcement Calculations	P	4.1047	MPa
Temperature for Internal Pressure	Temp	130	°C
Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	156.40	MPa
Parent Allowable Stress At Ambient	Sva	175.00	MPa
Inside Diameter of Cylindrical Shell	D	6700.00	mm
Shell Finished (Minimum) Thickness	t	96.0000	mm
Shell Internal Corrosion Allowance	cas	6.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		6750.01	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

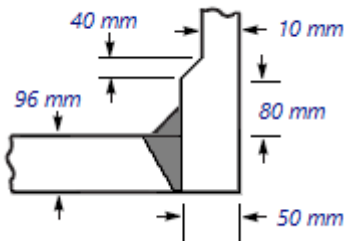
Material		SA-266 2	+IT
Material UNS Number		K03506	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	147.60	MPa
Allowable Stress At Ambient	Sna	165.00	MPa
Diameter Basis (for tr calc only)		Inside	
Layout Angle		30.00	deg
Diameter		1.5000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	10.0000	mm
Flange Material		SA-105	+IT
Flange Type		N/A	
Hub Height of Integral Nozzle	h	80.0000	mm
Height of Beveled Transition	L'	40.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		50.0000	mm
Style of FVC Nozzle		Long Weld Neck	
Corrosion Allowance	can	6.0000	mm
Outside Projection	ho	250.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel Wgnv		96.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.
Flange Class		600	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: L13 1.5"

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	1.500	in.
Actual Thickness Used in Calculation	0.394	in.



Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 50.1 \cdot (\exp(4.105/(147.6 \cdot 1.0)) - 1) + 6.0 + 0.0$$

$$= 0.7064 + 6.0 + 0.0 = 6.7064 \text{ mm Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 147.6 \cdot 1.0 \cdot \ln((2 \cdot 4.0 + 50.1) / 50.1) - 0.0$$

$$= 21.866 - 0.0 = 21.8662 \text{ MPa}$$

This nozzle or its diameter limit intersects weld seam # [6] on this element. The distance between the nozzle and seam is, 117.14 mm , while the distance from the center of the nozzle to the edge of the diameter limit is: 119.15 mm

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c : 9.420 mm
 Min. required shell thickness, internal p + c : 95.249 mm

Nozzle neck minimum required thickness : 9.420 mm

Minimum nozzle neck thickness 10.008 mm , is greater than required thickness 9.420 mm

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

$$= \min[S_n/S, 1]$$

$$= \min[147.6/156.4, 1]$$

$$= 0.944$$

Thickness of Nozzle at Shell [tn]:

$$= \text{hub thickness} - \text{corrosion allowance}$$

$$= 50.0 - 6.0$$

$$= 44.000 \text{ mm}$$

Thickness of Nozzle at Top [tn2]:

$$= \text{thickness} - \text{corrosion allowance}$$

$$= 10.0 - 6.0$$

$$= 4.000 \text{ mm}$$

Shell Diameter to Thickness ratio [D/t]:

$$= D_i/t$$

$$= 6712.0/90.0$$

$$= 74.578 \text{ must be less than 400.}$$

Effective Pressure Radius [Reff]:

$$= D_i/2 + \text{corrosion allowance}$$

$$= 6700.0/2 + 6.0$$

$$= 3356.000 \text{ mm}$$

Effective Length of Vessel Wall [LR]:

$$= \min(\sqrt{Reff \cdot t}, 2 \cdot R_n)$$

$$= \min(\sqrt{3356.0 \cdot 90.0}, 2 \cdot 25.05)$$

$$= 50.100 \text{ mm}$$



Thickness Limit Candidate [LH1]:

$$= \min[1.5t, t_e] + \sqrt{(R_n \cdot t_n)}$$

$$= \min[1.5 \cdot 90.0, 0.0] + \sqrt{(25.05 \cdot 44.0)}$$

$$= 33.199 \text{ mm}$$

Thickness Limit Candidate [LH2]:

$$= L_{pr1}$$

$$= 250.0$$

$$= 250.000 \text{ mm}$$

Thickness Limit Candidate [LH3]:

$$= 8(t + t_e)$$

$$= 8(90.0 + 0.0)$$

$$= 720.000 \text{ mm}$$

Effective Nozzle Wall Length Outside the Vessel [LH]:

$$= \min[LH1, LH2, LH3] + t \text{ for set-in nozzles}$$

$$= \min[33.199, 250.0, 720.0] + 90.0$$

$$= 123.199 \text{ mm}$$

Effective Vessel Thickness [teff]:

$$= t + ((A5 \cdot frp) / LR)$$

$$= 90.0 + ((0.0 \cdot 1.0) / 50.1)$$

$$= 90.000 \text{ mm}$$

Parameter [λ]:

$$= \min(12, (2 \cdot R_n + t_n) / (\sqrt{(D_i + t_{eff}) \cdot t_{eff}}))$$

$$= \min(12, (2 \cdot 25.05 + 44.0) / (\sqrt{((6712.0 + 90.0) \cdot 90.0)}))$$

$$= 0.120$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$= t \cdot LR \cdot \max((\lambda/5)^{0.85}, 1)$$

$$= 90.0 \cdot 50.1 \cdot \max((0.12/5)^{0.85}, 1)$$

$$= 4509.000 \text{ mm}^2$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$= t_n \cdot LH$$

$$= 44.0 \cdot 123.199$$

$$= 5420.774 \text{ mm}^2$$

Area Contributed by the Outside Fillet Weld [A41]:

$$= 0.5 \cdot \text{Leg}^2$$

$$= 0.5 \cdot 10.0^2$$

$$= 50.000 \text{ mm}^2$$

The total area contributed by A1 through A43 [AT]:

$$= A1 + \text{frn}(A2 + A3) + A41 + A42 + A43$$

$$= 4509.0 + 0.944(5420.774 + 0.0) + 50.0 + 0.0 + 0.0$$

$$= 9674.770 \text{ mm}^2$$

Nozzle Radius for Force Calculation [Rxn]:

$$= t_n / \ln[1 + t_n/R_n]$$

$$= 44.0 / \ln[1 + 44.0/25.05]$$

$$= 43.394 \text{ mm}$$

Shell Radius for Force Calculation [Rxs]:

$$= t_{eff} / \ln[1 + t_{eff}/R_{eff}]$$

$$= 90.0 / \ln[1 + 90.0/3356.0]$$

$$= 3400.797 \text{ mm}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$= 1.5 \cdot S \cdot E$$

$$= 1.5 \cdot 156.4 \cdot 1.0$$

$$= 234.6 \text{ MPa}$$

Force acting on the Nozzle [fN]:

$$= P \cdot R_{xn} \cdot LH$$



$$= 4.105 \cdot 43.394 \cdot 123.199$$
$$= 21942.5 \text{ N}$$

Force acting on the Shell [fS]:

$$= P \cdot Rxs(LR + tn)$$
$$= 4.105 \cdot 3400.797(50.1 + 44.0)$$
$$= 1313453.6 \text{ N}$$

Discontinuity Force from Internal Pressure [fY]:

$$= P \cdot Rxs \cdot Rnc$$
$$= 4.105 \cdot 3400.797 \cdot 25.05$$
$$= 349649.5 \text{ N}$$

Area Resisting Internal Pressure [Ap]:

$$= (fS + fY + fN)/P$$
$$= (1313454 + 349649.5 + 21942.5)/4.105$$
$$= 410551.1 \text{ mm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$= Sallow / (2 \cdot Ap/AT - Rxs/teff)$$
$$= 234.6 / (2 \cdot 410551.062/9674.77 - 3400.797/90.0)$$
$$= 5.0 \text{ MPa}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$= S[t/Rxs]$$
$$= 156.4[90.0/3400.797]$$
$$= 4.1 \text{ MPa}$$

Maximum Allowable Working Pressure [Pmax]:

$$= \min(Pmax1, Pmax2)$$
$$= \min(4.983, 4.139)$$
$$= 4.139 \text{ MPa}$$

Average Primary Membrane Stress [σAvg]:

$$= (fN + fS + fY) / AT$$
$$= (21942.506 + 1313454 + 349649.469)/9674.77$$
$$= 174.184 \text{ MPa}$$

General Primary Membrane Stress [σCirc]:

$$= P \cdot Rxs / teff$$
$$= 4.105 \cdot 3400.797/90.0$$
$$= 155.1 \text{ MPa}$$

Maximum Local Primary Membrane Stress [PL]:

$$= \max(2 \cdot \sigma Avg - \sigma Circ, \sigma Circ)$$
$$= \max(2 \cdot 174.184 - 155.103, 155.103)$$
$$= 193.3 \text{ MPa}$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	234.60	MPa
Local Primary Membrane Stress	PL	193.27	MPa
Maximum Allowable Working Pressure	Pmax	4.14	MPa

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$= (Rnc + tn) / Rnc$$
$$= (25.05 + 44.0)/25.05$$
$$= 2.756 \text{ For set-in Nozzles}$$

Weld Length of Nozzle to Shell Weld [Lt]:

$$= \pi/2 \cdot (Rn + tn)$$
$$= \pi/2 \cdot (25.05 + 44.0)$$
$$= 108.463 \text{ mm}$$

Weld Throat Dimensions, (0.7071•Leg Dimensions) [L41T, L42T, L43T]:

$$= 7.071, 0.000, 0.000, \text{ mm}$$

Required Weld Throat Dimensions :

$$tc, L41T = 7.071, \text{ must be } \geq \min(0.7 \cdot 44.000, 6.000) \text{ Passed}$$



Weld Load Value [fwelds]:

$$\begin{aligned} &= \min(f_y \cdot k_y, 1.5 \cdot S_n(A_2 + A_3), \pi/4 \cdot P \cdot R_n^2 \cdot k_y^2) \\ &= \min(349649 \cdot 2.76, 1.5 \cdot 147.6(5420.774 + 0.0), \pi/4 \cdot 4.1 \cdot 25.05^2 \cdot 2.76^2) \\ &= 15369.573 \text{ N} \end{aligned}$$

Weld Stress Value [τ]:

$$\begin{aligned} &= fwelds / (L_T(0.49 \cdot L_{41T} + 0.6 \cdot tw_1 + 0.49 \cdot L_{43T})) \\ &= 15369.573 / (108.463(0.49 \cdot 7.071 + 0.6 \cdot 90.0 + 0.49 \cdot 0.0)) \\ &= 2.466 < \text{or} = \text{to } 156.400 \text{ Weld Stress Passed} \end{aligned}$$

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

The Drop for this Nozzle is : 0.7116 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 346.7115 mm

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Input, Nozzle Desc: UC2 24"

From: 70

Pressure for Reinforcement Calculations	P	4.1047	MPa
Temperature for Internal Pressure	Temp	130	°C
Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	156.40	MPa
Parent Allowable Stress At Ambient	Sva	175.00	MPa
Inside Diameter of Cylindrical Shell	D	6700.00	mm
Shell Finished (Minimum) Thickness	t	96.0000	mm
Shell Internal Corrosion Allowance	cas	6.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		6899.99	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

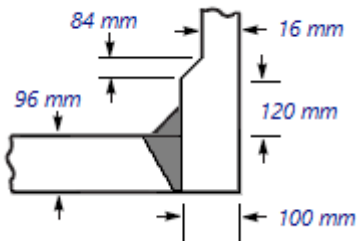
Material		SA-516 70	+IT
Material UNS Number		K02700	
Material Specification/Type		Plate	
Allowable Stress at Temperature	Sn	156.40	MPa
Allowable Stress At Ambient	Sna	175.00	MPa
Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		24.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.0000	mm
Flange Material		SA-105	+IT
Hub Height of Integral Nozzle	h	120.0000	mm
Height of Beveled Transition	L'	84.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		100.0000	mm
Corrosion Allowance	can	6.0000	mm
Outside Projection	ho	500.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	96.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.
Flange Class		600	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: UC2 24"

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	24.000	in.
Actual Thickness Used in Calculation	0.630	in.



Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 621.6 \cdot (\exp(4.105/(156.4 \cdot 1.0)) - 1) + 6.0 + 0.0$$

$$= 8.2649 + 6.0 + 0.0 = 14.2649 \text{ mm Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 156.4 \cdot 1.0 \cdot \ln((2 \cdot 10.0 + 621.6) / 621.6) - 0.0$$

$$= 4.953 - 0.0 = 4.9529 \text{ MPa}$$

This nozzle or its diameter limit intersects weld seam # [7] on this element. The distance between the nozzle and seam is, 410.01 mm , while the distance from the center of the nozzle to the edge of the diameter limit is: 954.38 mm

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c : 14.340 mm
 Min. required shell thickness, internal p + c : 95.249 mm

Nozzle neck minimum required thickness : 14.340 mm

Minimum nozzle neck thickness 16.002 mm , is greater than required thickness 14.340 mm

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

$$= \min[S_n/S, 1]$$

$$= \min[156.4/156.4, 1]$$

$$= 1.000$$

Thickness of Nozzle at Shell [tn]:

$$= \text{hub thickness} - \text{corrosion allowance}$$

$$= 100.0 - 6.0$$

$$= 94.000 \text{ mm}$$

Thickness of Nozzle at Top [tn2]:

$$= \text{thickness} - \text{corrosion allowance}$$

$$= 16.0 - 6.0$$

$$= 10.000 \text{ mm}$$

Shell Diameter to Thickness ratio [D/t]:

$$= D_i/t$$

$$= 6712.0/90.0$$

$$= 74.578 \text{ must be less than 400.}$$

Effective Pressure Radius [Reff]:

$$= D_i/2 + \text{corrosion allowance}$$

$$= 6700.0/2 + 6.0$$

$$= 3356.000 \text{ mm}$$

Effective Length of Vessel Wall [LR]:

$$= \min(\sqrt{Reff \cdot t}, 2 \cdot R_n)$$

$$= \min(\sqrt{3356.0 \cdot 90.0}, 2 \cdot 310.8)$$

$$= 549.582 \text{ mm}$$



Thickness Limit Candidate [LH1]:

$$\begin{aligned} &= \min[1.5t, t_e] + \sqrt{(R_n \cdot t_n)} \\ &= \min[1.5 \cdot 90.0, 0.0] + \sqrt{(310.8 \cdot 94.0)} \\ &= 170.925 \text{ mm} \end{aligned}$$

Thickness Limit Candidate [LH2]:

$$\begin{aligned} &= L_{pr1} \\ &= 500.0 \\ &= 500.000 \text{ mm} \end{aligned}$$

Thickness Limit Candidate [LH3]:

$$\begin{aligned} &= 8(t + t_e) \\ &= 8(90.0 + 0.0) \\ &= 720.000 \text{ mm} \end{aligned}$$

Effective Nozzle Wall Length Outside the Vessel [LH]:

$$\begin{aligned} &= \min[LH1, LH2, LH3] + t \quad \text{for set-in nozzles} \\ &= \min[170.925, 500.0, 720.0] + 90.0 \\ &= 260.925 \text{ mm} \end{aligned}$$

Effective Vessel Thickness [teff]:

$$\begin{aligned} &= t + ((A_5 \cdot frp) / LR) \\ &= 90.0 + ((0.0 \cdot 1.0) / 549.582) \\ &= 90.000 \text{ mm} \end{aligned}$$

Parameter [λ]:

$$\begin{aligned} &= \min(12, (2 \cdot R_n + t_n) / (\sqrt{(D_i + teff) \cdot teff})) \\ &= \min(12, (2 \cdot 310.8 + 94.0) / (\sqrt{(6712.0 + 90.0) \cdot 90.0})) \\ &= 0.915 \end{aligned}$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$\begin{aligned} &= t \cdot LR \cdot \max((\lambda/5)^{0.85}, 1) \\ &= 90.0 \cdot 549.582 \cdot \max((0.915/5)^{0.85}, 1) \\ &= 49462.355 \text{ mm}^2 \end{aligned}$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$\begin{aligned} &= A_{2A} + A_{2B} \\ &= 19740.002 + 3490.252 \\ &= 23230.254 \text{ mm}^2 \end{aligned}$$

Area Contributed by the Outside Fillet Weld [A41]:

$$\begin{aligned} &= 0.5 \cdot \text{Leg}^2 \\ &= 0.5 \cdot 10.0^2 \\ &= 50.000 \text{ mm}^2 \end{aligned}$$

The total area contributed by A1 through A43 [AT]:

$$\begin{aligned} &= A1 + \text{frn}(A2 + A3) + A41 + A42 + A43 \\ &= 49462.355 + 1.0(23230.254 + 0.0) + 50.0 + 0.0 + 0.0 \\ &= 72742.609 \text{ mm}^2 \end{aligned}$$

Nozzle Radius for Force Calculation [Rxn]:

$$\begin{aligned} &= t_n / \ln[1 + t_n/R_n] \\ &= 94.0 / \ln[1 + 94.0/310.8] \\ &= 355.733 \text{ mm} \end{aligned}$$

Shell Radius for Force Calculation [Rxs]:

$$\begin{aligned} &= teff / \ln[1 + teff/Reff] \\ &= 90.0 / \ln[1 + 90.0/3356.0] \\ &= 3400.797 \text{ mm} \end{aligned}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$\begin{aligned} &= 1.5 \cdot S \cdot E \\ &= 1.5 \cdot 156.4 \cdot 1.0 \\ &= 234.6 \text{ MPa} \end{aligned}$$

Force acting on the Nozzle [fN]:

$$= P \cdot R_{xn} \cdot L_H$$



$$= 4.105 \cdot 355.733 \cdot 260.925$$

$$= 380963.2 \text{ N}$$

Force acting on the Shell [fS]:

$$= P \cdot Rxs(LR + tn)$$

$$= 4.105 \cdot 3400.797(549.582 + 94.0)$$

$$= 8983154.0 \text{ N}$$

Discontinuity Force from Internal Pressure [fY]:

$$= P \cdot Rxs \cdot Rnc$$

$$= 4.105 \cdot 3400.797 \cdot 310.8$$

$$= 4338166.0 \text{ N}$$

Area Resisting Internal Pressure [Ap]:

$$= (fS + fY + fN)/P$$

$$= (8983154 + 4338166 + 380963.2)/4.105$$

$$= 3338477.2 \text{ mm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$= Sallow / (2 \cdot Ap/AT - Rxs/teff)$$

$$= 234.6 / (2 \cdot 3338477/72742.609 - 3400.797/90.0)$$

$$= 4.3 \text{ MPa}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$= S[t/Rxs]$$

$$= 156.4[90.0/3400.797]$$

$$= 4.1 \text{ MPa}$$

Maximum Allowable Working Pressure [Pmax]:

$$= \min(Pmax1, Pmax2)$$

$$= \min(4.344, 4.139)$$

$$= 4.139 \text{ MPa}$$

Average Primary Membrane Stress [σAvg]:

$$= (fN + fS + fY) / AT$$

$$= (380963.219 + 8983154 + 4338166)/72742.609$$

$$= 188.383 \text{ MPa}$$

General Primary Membrane Stress [σCirc]:

$$= P \cdot Rxs / teff$$

$$= 4.105 \cdot 3400.797/90.0$$

$$= 155.1 \text{ MPa}$$

Maximum Local Primary Membrane Stress [PL]:

$$= \max(2 \cdot \sigma Avg - \sigma Circ, \sigma Circ)$$

$$= \max(2 \cdot 188.383 - 155.103, 155.103)$$

$$= 221.7 \text{ MPa}$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	234.60	MPa
Local Primary Membrane Stress	PL	221.66	MPa
Maximum Allowable Working Pressure	Pmax	4.14	MPa

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$= (Rnc + tn) / Rnc$$

$$= (310.8 + 94.0)/310.8$$

$$= 1.302 \text{ For set-in Nozzles}$$

Weld Length of Nozzle to Shell Weld [Lt]:

$$= \pi/2 \cdot (Rn + tn)$$

$$= \pi/2 \cdot (310.8 + 94.0)$$

$$= 635.858 \text{ mm}$$

Weld Throat Dimensions, (0.7071•Leg Dimensions) [L41T, L42T, L43T]:

$$= 7.071, 0.000, 0.000, \text{ mm}$$

Required Weld Throat Dimensions :

$$tc, L41T = 7.071, \text{ must be } \geq \min(0.7 \cdot 94.000, 6.000) \text{ Passed}$$



Weld Load Value [fwelds]:

$$\begin{aligned} &= \min(f_y \cdot k_y, 1.5 \cdot S_n(A_2 + A_3), \pi/4 \cdot P \cdot R_n^2 \cdot k_y^2) \\ &= \min(4338166 \cdot 1.3, 1.5 \cdot 156.4(23230.254 + 0.0), \pi/4 \cdot 4.1 \cdot 310.8^2 \cdot 1.3^2) \\ &= 528220.625 \text{ N} \end{aligned}$$

Weld Stress Value [τ]:

$$\begin{aligned} &= \text{fwelds} / (L_T(0.49 \cdot L_{41T} + 0.6 \cdot t_{w1} + 0.49 \cdot L_{43T})) \\ &= 528221 / (635.858(0.49 \cdot 7.071 + 0.6 \cdot 90.0 + 0.49 \cdot 0.0)) \\ &= 14.457 < \text{or} = \text{to } 156.400 \text{ Weld Stress Passed} \end{aligned}$$

Nozzle/Structural Discontinuity Proximity Warning:

This nozzle's diameter limit intersects a major structural discontinuity. Reduce the diameter limit in the nozzle dialog so that there is no interference or move the nozzle away from the intersecting weld by 55.911 mm.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Flange MDMT including temperature reduction per 3.11:

Flange MDMT with Temp reduction -68 °C

The Drop for this Nozzle is : 24.5470 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 620.5469 mm

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Input, Nozzle Desc: MH-5 24"

From: 90

Pressure for Reinforcement Calculations	P	4.1047	MPa
Temperature for Internal Pressure	Temp	130	°C
Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	156.40	MPa
Parent Allowable Stress At Ambient	Sva	175.00	MPa
Inside Diameter of Cylindrical Shell	D	6706.00	mm
Shell Finished (Minimum) Thickness	t	92.0000	mm
Shell Internal Corrosion Allowance	cas	1.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		12399.98	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

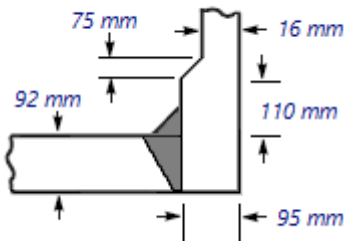
Material		SA-266 2	+IT
Material UNS Number		K03506	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	147.60	MPa
Allowable Stress At Ambient	Sna	165.00	MPa
Diameter Basis (for tr calc only)		Inside	
Layout Angle		225.00	deg
Diameter		24.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.0000	mm
Flange Material		SA-105	+IT
Hub Height of Integral Nozzle	h	110.0000	mm
Height of Beveled Transition	L'	75.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		95.0000	mm
Corrosion Allowance	can	0.0000	mm
Outside Projection	ho	310.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	92.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.
This is a Manway, Coupling or Access Opening.			
Flange Class		600	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: MH-5 24"

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	24.000	in.
Actual Thickness Used in Calculation	0.630	in.



Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 609.6 \cdot (\exp(4.105/(147.6 \cdot 1.0)) - 1) + 0.0 + 0.0$$

$$= 8.5953 + 0.0 + 0.0 = 8.5953 \text{ mm Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 147.6 \cdot 1.0 \cdot \ln((2 \cdot 16.0 + 609.6) / 609.6) - 0.0$$

$$= 7.552 - 0.0 = 7.5515 \text{ MPa}$$

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

$$= \min[S_n/S, 1]$$

$$= \min[147.6/156.4, 1]$$

$$= 0.944$$

Thickness of Nozzle at Shell [tn]:

$$= \text{hub thickness} - \text{corrosion allowance}$$

$$= 95.0 - 0.0$$

$$= 95.000 \text{ mm}$$

Thickness of Nozzle at Top [tn2]:

$$= \text{thickness} - \text{corrosion allowance}$$

$$= 16.0 - 0.0$$

$$= 16.000 \text{ mm}$$

Shell Diameter to Thickness ratio [D/t]:

$$= D_i/t$$

$$= 6708.0/91.0$$

$$= 73.714 \text{ must be less than 400.}$$

Effective Pressure Radius [Reff]:

$$= D_i/2 + \text{corrosion allowance}$$

$$= 6706.0/2 + 1.0$$

$$= 3354.000 \text{ mm}$$

Effective Length of Vessel Wall [LR]:

$$= \min(\sqrt{Reff \cdot t}, 2 \cdot R_n)$$

$$= \min(\sqrt{3354.0 \cdot 91.0}, 2 \cdot 304.8)$$

$$= 552.462 \text{ mm}$$

Thickness Limit Candidate [LH1]:

$$= \min[1.5t, te] + \sqrt{R_n \cdot tn}$$

$$= \min[1.5 \cdot 91.0, 0.0] + \sqrt{304.8 \cdot 95.0}$$

$$= 170.165 \text{ mm}$$

Thickness Limit Candidate [LH2]:

$$= L_{pr1}$$

$$= 310.0$$

$$= 310.000 \text{ mm}$$

Thickness Limit Candidate [LH3]:



$$\begin{aligned}
 &= 8(t + t_e) \\
 &= 8(91.0 + 0.0) \\
 &= 728.000 \text{ mm}
 \end{aligned}$$

Effective Nozzle Wall Length Outside the Vessel [LH]:

$$\begin{aligned}
 &= \min[LH1, LH2, LH3] + t \quad \text{for set-in nozzles} \\
 &= \min[170.165, 310.0, 728.0] + 91.0 \\
 &= 261.165 \text{ mm}
 \end{aligned}$$

Effective Vessel Thickness [teff]:

$$\begin{aligned}
 &= t + ((A5 \cdot frp) / LR) \\
 &= 91.0 + ((0.0 \cdot 1.0) / 552.462) \\
 &= 91.000 \text{ mm}
 \end{aligned}$$

Parameter [λ]:

$$\begin{aligned}
 &= \min(12, (2 \cdot Rn + tn) / (\sqrt{ (Di + teff) \cdot teff })) \\
 &= \min(12, (2 \cdot 304.8 + 95.0) / (\sqrt{ (6708.0 + 91.0) \cdot 91.0 })) \\
 &= 0.896
 \end{aligned}$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$\begin{aligned}
 &= t \cdot LR \cdot \max((\lambda/5)^{0.85}, 1) \\
 &= 91.0 \cdot 552.462 \cdot \max((0.896/5)^{0.85}, 1) \\
 &= 50274.020 \text{ mm}^2
 \end{aligned}$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$\begin{aligned}
 &= A2A + A2B \\
 &= 19095.0 + 3809.221 \\
 &= 22904.221 \text{ mm}^2
 \end{aligned}$$

Area Contributed by the Outside Fillet Weld [A41]:

$$\begin{aligned}
 &= 0.5 \cdot Leg^2 \\
 &= 0.5 \cdot 10.0^2 \\
 &= 50.000 \text{ mm}^2
 \end{aligned}$$

The total area contributed by A1 through A43 [AT]:

$$\begin{aligned}
 &= A1 + frn(A2 + A3) + A41 + A42 + A43 \\
 &= 50274.02 + 0.944(22904.221 + 0.0) + 50.0 + 0.0 + 0.0 \\
 &= 71939.508 \text{ mm}^2
 \end{aligned}$$

Nozzle Radius for Force Calculation [Rxn]:

$$\begin{aligned}
 &= tn / \ln[1 + tn/Rn] \\
 &= 95.0 / \ln[1 + 95.0/304.8] \\
 &= 350.155 \text{ mm}
 \end{aligned}$$

Shell Radius for Force Calculation [Rxs]:

$$\begin{aligned}
 &= teff / \ln[1 + teff/Reff] \\
 &= 91.0 / \ln[1 + 91.0/3354.0] \\
 &= 3399.296 \text{ mm}
 \end{aligned}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$\begin{aligned}
 &= 1.5 \cdot S \cdot E \\
 &= 1.5 \cdot 156.4 \cdot 1.0 \\
 &= 234.6 \text{ MPa}
 \end{aligned}$$

Force acting on the Nozzle [fN]:

$$\begin{aligned}
 &= P \cdot Rxn \cdot LH \\
 &= 4.105 \cdot 350.155 \cdot 261.165 \\
 &= 375334.8 \text{ N}
 \end{aligned}$$

Force acting on the Shell [fS]:

$$\begin{aligned}
 &= P \cdot Rxs(LR + tn) \\
 &= 4.105 \cdot 3399.296(552.462 + 95.0) \\
 &= 9033324.0 \text{ N}
 \end{aligned}$$

Discontinuity Force from Internal Pressure [fY]:

$$\begin{aligned}
 &= P \cdot Rxs \cdot Rnc \\
 &= 4.105 \cdot 3399.296 \cdot 304.8 \\
 &= 4252540.0 \text{ N}
 \end{aligned}$$



Area Resisting Internal Pressure [Ap]:

$$\begin{aligned} &= (fS + fY + fN) / P \\ &= (9033324 + 4252540 + 375334.8) / 4.105 \\ &= 3328467.2 \text{ mm}^2 \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$\begin{aligned} &= Sallow / (2 \cdot Ap / AT - Rxs / teff) \\ &= 234.6 / (2 \cdot 3328467 / 71939.508 - 3399.296 / 91.0) \\ &= 4.3 \text{ MPa} \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$\begin{aligned} &= S[t / Rxs] \\ &= 156.4[91.0 / 3399.296] \\ &= 4.2 \text{ MPa} \end{aligned}$$

Maximum Allowable Working Pressure [Pmax]:

$$\begin{aligned} &= \min(Pmax1, Pmax2) \\ &= \min(4.252, 4.187) \\ &= 4.187 \text{ MPa} \end{aligned}$$

Average Primary Membrane Stress [σ_{Avg}]:

$$\begin{aligned} &= (fN + fS + fY) / AT \\ &= (375334.812 + 9033324 + 4252540) / 71939.508 \\ &= 189.915 \text{ MPa} \end{aligned}$$

General Primary Membrane Stress [σ_{Circ}]:

$$\begin{aligned} &= P \cdot Rxs / teff \\ &= 4.105 \cdot 3399.296 / 91.0 \\ &= 153.3 \text{ MPa} \end{aligned}$$

Maximum Local Primary Membrane Stress [PL]:

$$\begin{aligned} &= \max(2 \cdot \sigma_{Avg} - \sigma_{Circ}, \sigma_{Circ}) \\ &= \max(2 \cdot 189.915 - 153.331, 153.331) \\ &= 226.5 \text{ MPa} \end{aligned}$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	234.60	MPa
Local Primary Membrane Stress	PL	226.50	MPa
Maximum Allowable Working Pressure	Pmax	4.19	MPa

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$\begin{aligned} &= (Rnc + tn) / Rnc \\ &= (304.8 + 95.0) / 304.8 \\ &= 1.312 \text{ For set-in Nozzles} \end{aligned}$$

Weld Length of Nozzle to Shell Weld [Lt]:

$$\begin{aligned} &= \pi / 2 \cdot (Rn + tn) \\ &= \pi / 2 \cdot (304.8 + 95.0) \\ &= 628.004 \text{ mm} \end{aligned}$$

Weld Throat Dimensions, (0.7071 • Leg Dimensions) [L41T, L42T, L43T]:

$$= 7.071, 0.000, 0.000, \text{ mm}$$

Required Weld Throat Dimensions :

$$tc, L41T = 7.071, \text{ must be } \geq \min(0.7 \cdot 95.000, 6.000) \text{ Passed}$$

Weld Load Value [fwelds]:

$$\begin{aligned} &= \min(fy \cdot ky, 1.5 \cdot Sn(A2 + A3), \pi / 4 \cdot P \cdot Rn^2 \cdot ky^2) \\ &= \min(4252540 \cdot 1.31, 1.5 \cdot 147.6(22904.221 + 0.0), \pi / 4 \cdot 4.1 \cdot 304.8^2 \cdot 1.31^2) \\ &= 515252.375 \text{ N} \end{aligned}$$

Weld Stress Value [τ]:

$$\begin{aligned} &= fwelds / (Lt(0.49 \cdot L41T + 0.6 \cdot tw1 + 0.49 \cdot L43T)) \\ &= 515252 / (628.004(0.49 \cdot 7.071 + 0.6 \cdot 91.0 + 0.49 \cdot 0.0)) \\ &= 14.131 < \text{ or } = \text{ to } 156.400 \text{ Weld Stress Passed} \end{aligned}$$

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

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Input, Nozzle Desc: MH-4 24"

From: 100

Pressure for Reinforcement Calculations	P	4.1047	MPa
Temperature for Internal Pressure	Temp	130	°C
Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	156.40	MPa
Parent Allowable Stress At Ambient	Sva	175.00	MPa
Inside Diameter of Cylindrical Shell	D	6706.00	mm
Shell Finished (Minimum) Thickness	t	92.0000	mm
Shell Internal Corrosion Allowance	cas	1.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		14899.98	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

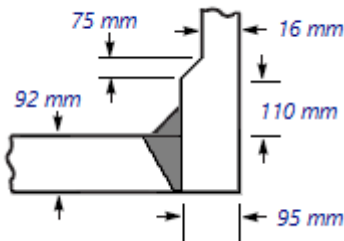
Material		SA-266 2	+IT
Material UNS Number		K03506	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	147.60	MPa
Allowable Stress At Ambient	Sna	165.00	MPa
Diameter Basis (for tr calc only)		Inside	
Layout Angle		225.00	deg
Diameter		24.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.0000	mm
Flange Material		SA-105	+IT
Hub Height of Integral Nozzle	h	110.0000	mm
Height of Beveled Transition	L'	75.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		95.0000	mm
Corrosion Allowance	can	0.0000	mm
Outside Projection	ho	310.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	92.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.
This is a Manway, Coupling or Access Opening.			
Flange Class		600	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: MH-4 24"

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	24.000	in.
Actual Thickness Used in Calculation	0.630	in.



Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 609.6 \cdot (\exp(4.105/(147.6 \cdot 1.0)) - 1) + 0.0 + 0.0$$

$$= 8.5953 + 0.0 + 0.0 = 8.5953 \text{ mm Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 147.6 \cdot 1.0 \cdot \ln((2 \cdot 16.0 + 609.6) / 609.6) - 0.0$$

$$= 7.552 - 0.0 = 7.5515 \text{ MPa}$$

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

$$= \min[S_n/S, 1]$$

$$= \min[147.6/156.4, 1]$$

$$= 0.944$$

Thickness of Nozzle at Shell [tn]:

$$= \text{hub thickness} - \text{corrosion allowance}$$

$$= 95.0 - 0.0$$

$$= 95.000 \text{ mm}$$

Thickness of Nozzle at Top [tn2]:

$$= \text{thickness} - \text{corrosion allowance}$$

$$= 16.0 - 0.0$$

$$= 16.000 \text{ mm}$$

Shell Diameter to Thickness ratio [D/t]:

$$= D_i/t$$

$$= 6708.0/91.0$$

$$= 73.714 \text{ must be less than 400.}$$

Effective Pressure Radius [Reff]:

$$= D_i/2 + \text{corrosion allowance}$$

$$= 6706.0/2 + 1.0$$

$$= 3354.000 \text{ mm}$$

Effective Length of Vessel Wall [LR]:

$$= \min(\sqrt{Reff \cdot t}, 2 \cdot R_n)$$

$$= \min(\sqrt{3354.0 \cdot 91.0}, 2 \cdot 304.8)$$

$$= 552.462 \text{ mm}$$

Thickness Limit Candidate [LH1]:

$$= \min[1.5t, te] + \sqrt{R_n \cdot tn}$$

$$= \min[1.5 \cdot 91.0, 0.0] + \sqrt{304.8 \cdot 95.0}$$

$$= 170.165 \text{ mm}$$

Thickness Limit Candidate [LH2]:

$$= L_{pr1}$$

$$= 310.0$$

$$= 310.000 \text{ mm}$$

Thickness Limit Candidate [LH3]:



$$\begin{aligned}
 &= 8(t + t_e) \\
 &= 8(91.0 + 0.0) \\
 &= 728.000 \text{ mm}
 \end{aligned}$$

Effective Nozzle Wall Length Outside the Vessel [LH]:

$$\begin{aligned}
 &= \min[LH1, LH2, LH3] + t \quad \text{for set-in nozzles} \\
 &= \min[170.165, 310.0, 728.0] + 91.0 \\
 &= 261.165 \text{ mm}
 \end{aligned}$$

Effective Vessel Thickness [teff]:

$$\begin{aligned}
 &= t + ((A5 \cdot frp) / LR) \\
 &= 91.0 + ((0.0 \cdot 1.0) / 552.462) \\
 &= 91.000 \text{ mm}
 \end{aligned}$$

Parameter [λ]:

$$\begin{aligned}
 &= \min(12, (2 \cdot Rn + tn) / (\sqrt{ (Di + teff) \cdot teff })) \\
 &= \min(12, (2 \cdot 304.8 + 95.0) / (\sqrt{ (6708.0 + 91.0) \cdot 91.0 })) \\
 &= 0.896
 \end{aligned}$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$\begin{aligned}
 &= t \cdot LR \cdot \max((\lambda/5)^{0.85}, 1) \\
 &= 91.0 \cdot 552.462 \cdot \max((0.896/5)^{0.85}, 1) \\
 &= 50274.020 \text{ mm}^2
 \end{aligned}$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$\begin{aligned}
 &= A2A + A2B \\
 &= 19095.0 + 3809.221 \\
 &= 22904.221 \text{ mm}^2
 \end{aligned}$$

Area Contributed by the Outside Fillet Weld [A41]:

$$\begin{aligned}
 &= 0.5 \cdot Leg^2 \\
 &= 0.5 \cdot 10.0^2 \\
 &= 50.000 \text{ mm}^2
 \end{aligned}$$

The total area contributed by A1 through A43 [AT]:

$$\begin{aligned}
 &= A1 + \text{frn}(A2 + A3) + A41 + A42 + A43 \\
 &= 50274.02 + 0.944(22904.221 + 0.0) + 50.0 + 0.0 + 0.0 \\
 &= 71939.508 \text{ mm}^2
 \end{aligned}$$

Nozzle Radius for Force Calculation [Rxn]:

$$\begin{aligned}
 &= tn / \ln[1 + tn/Rn] \\
 &= 95.0 / \ln[1 + 95.0/304.8] \\
 &= 350.155 \text{ mm}
 \end{aligned}$$

Shell Radius for Force Calculation [Rxs]:

$$\begin{aligned}
 &= teff / \ln[1 + teff/Reff] \\
 &= 91.0 / \ln[1 + 91.0/3354.0] \\
 &= 3399.296 \text{ mm}
 \end{aligned}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$\begin{aligned}
 &= 1.5 \cdot S \cdot E \\
 &= 1.5 \cdot 156.4 \cdot 1.0 \\
 &= 234.6 \text{ MPa}
 \end{aligned}$$

Force acting on the Nozzle [fN]:

$$\begin{aligned}
 &= P \cdot Rxn \cdot LH \\
 &= 4.105 \cdot 350.155 \cdot 261.165 \\
 &= 375334.8 \text{ N}
 \end{aligned}$$

Force acting on the Shell [fS]:

$$\begin{aligned}
 &= P \cdot Rxs(LR + tn) \\
 &= 4.105 \cdot 3399.296(552.462 + 95.0) \\
 &= 9033324.0 \text{ N}
 \end{aligned}$$

Discontinuity Force from Internal Pressure [fY]:

$$\begin{aligned}
 &= P \cdot Rxs \cdot Rnc \\
 &= 4.105 \cdot 3399.296 \cdot 304.8 \\
 &= 4252540.0 \text{ N}
 \end{aligned}$$



Area Resisting Internal Pressure [Ap]:

$$\begin{aligned} &= (fS + fY + fN) / P \\ &= (9033324 + 4252540 + 375334.8) / 4.105 \\ &= 3328467.2 \text{ mm}^2 \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$\begin{aligned} &= Sallow / (2 \cdot Ap / AT - Rxs / teff) \\ &= 234.6 / (2 \cdot 3328467 / 71939.508 - 3399.296 / 91.0) \\ &= 4.3 \text{ MPa} \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$\begin{aligned} &= S[t / Rxs] \\ &= 156.4[91.0 / 3399.296] \\ &= 4.2 \text{ MPa} \end{aligned}$$

Maximum Allowable Working Pressure [Pmax]:

$$\begin{aligned} &= \min(Pmax1, Pmax2) \\ &= \min(4.252, 4.187) \\ &= 4.187 \text{ MPa} \end{aligned}$$

Average Primary Membrane Stress [σ_{Avg}]:

$$\begin{aligned} &= (fN + fS + fY) / AT \\ &= (375334.812 + 9033324 + 4252540) / 71939.508 \\ &= 189.915 \text{ MPa} \end{aligned}$$

General Primary Membrane Stress [σ_{Circ}]:

$$\begin{aligned} &= P \cdot Rxs / teff \\ &= 4.105 \cdot 3399.296 / 91.0 \\ &= 153.3 \text{ MPa} \end{aligned}$$

Maximum Local Primary Membrane Stress [PL]:

$$\begin{aligned} &= \max(2 \cdot \sigma_{Avg} - \sigma_{Circ}, \sigma_{Circ}) \\ &= \max(2 \cdot 189.915 - 153.331, 153.331) \\ &= 226.5 \text{ MPa} \end{aligned}$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	234.60	MPa
Local Primary Membrane Stress	PL	226.50	MPa
Maximum Allowable Working Pressure	Pmax	4.19	MPa

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$\begin{aligned} &= (Rnc + tn) / Rnc \\ &= (304.8 + 95.0) / 304.8 \\ &= 1.312 \text{ For set-in Nozzles} \end{aligned}$$

Weld Length of Nozzle to Shell Weld [Lt]:

$$\begin{aligned} &= \pi / 2 \cdot (Rn + tn) \\ &= \pi / 2 \cdot (304.8 + 95.0) \\ &= 628.004 \text{ mm} \end{aligned}$$

Weld Throat Dimensions, (0.7071 • Leg Dimensions) [L41T, L42T, L43T]:

$$= 7.071, 0.000, 0.000, \text{ mm}$$

Required Weld Throat Dimensions :

$$tc, L41T = 7.071, \text{ must be } \geq \min(0.7 \cdot 95.000, 6.000) \text{ Passed}$$

Weld Load Value [fwelds]:

$$\begin{aligned} &= \min(fy \cdot ky, 1.5 \cdot Sn(A2 + A3), \pi / 4 \cdot P \cdot Rn^2 \cdot ky^2) \\ &= \min(4252540 \cdot 1.31, 1.5 \cdot 147.6(22904.221 + 0.0), \pi / 4 \cdot 4.1 \cdot 304.8^2 \cdot 1.31^2) \\ &= 515252.375 \text{ N} \end{aligned}$$

Weld Stress Value [τ]:

$$\begin{aligned} &= fwelds / (Lt(0.49 \cdot L41T + 0.6 \cdot tw1 + 0.49 \cdot L43T)) \\ &= 515252 / (628.004(0.49 \cdot 7.071 + 0.6 \cdot 91.0 + 0.49 \cdot 0.0)) \\ &= 14.131 < \text{ or } = \text{ to } 156.400 \text{ Weld Stress Passed} \end{aligned}$$

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

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Input, Nozzle Desc: A 36"

From: 110

Pressure for Reinforcement Calculations	P	4.1047	MPa
Temperature for Internal Pressure	Temp	130	°C
Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	156.40	MPa
Parent Allowable Stress At Ambient	Sva	175.00	MPa
Inside Diameter of Cylindrical Shell	D	6706.00	mm
Shell Finished (Minimum) Thickness	t	92.0000	mm
Shell Internal Corrosion Allowance	cas	1.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		16649.98	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

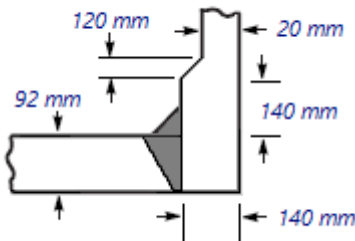
Material		SA-266 2	+IT
Material UNS Number		K03506	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	147.60	MPa
Allowable Stress At Ambient	Sna	165.00	MPa
Diameter Basis (for tr calc only)		Inside	
Layout Angle		180.00	deg
Diameter		36.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	20.0000	mm
Flange Material		SA-105	+IT
Hub Height of Integral Nozzle	h	140.0000	mm
Height of Beveled Transition	L'	120.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		140.0000	mm
Corrosion Allowance	can	6.0000	mm
Outside Projection	ho	360.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	10.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.
Flange Class		600	
Flange Grade		GR 1.1	
Flange Series		B	

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: A 36"

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	36.000	in.
Actual Thickness Used in Calculation	0.787	in.



Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 926.4 \cdot (\exp(4.105/(147.6 \cdot 1.0)) - 1) + 6.0 + 0.0$$

$$= 13.0622 + 6.0 + 0.0 = 19.0622 \text{ mm Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 147.6 \cdot 1.0 \cdot \ln((2 \cdot 14.0 + 926.4) / 926.4) - 0.0$$

$$= 4.395 - 0.0 = 4.3951 \text{ MPa}$$

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c : 14.340 mm
 Min. required shell thickness, internal p + c : 90.196 mm

Nozzle neck minimum required thickness : 14.340 mm

Minimum nozzle neck thickness 19.990 mm , is greater than required thickness 14.340 mm

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 78.8,	Allowable	: 147.6 MPa	Passed
Expansion	: 0.0,	Allowable	: 311.9 MPa	Passed
Occasional	: 66.9,	Allowable	: 196.3 MPa	Passed
Shear	: 7.1,	Allowable	: 103.3 MPa	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

$$= \min[S_n/S, 1]$$

$$= \min[147.6/156.4, 1]$$

$$= 0.944$$

Thickness of Nozzle at Shell [tn]:

$$= \text{hub thickness} - \text{corrosion allowance}$$

$$= 140.0 - 6.0$$

$$= 134.000 \text{ mm}$$

Thickness of Nozzle at Top [tn2]:

$$= \text{thickness} - \text{corrosion allowance}$$

$$= 20.0 - 6.0$$

$$= 14.000 \text{ mm}$$

Shell Diameter to Thickness ratio [D/t]:

$$= D_i/t$$

$$= 6708.0/91.0$$

$$= 73.714 \text{ must be less than 400.}$$

Effective Pressure Radius [Reff]:

$$= D_i/2 + \text{corrosion allowance}$$



$$= 6706.0/2 + 1.0$$
$$= 3354.000 \text{ mm}$$

Effective Length of Vessel Wall [LR]:

$$= \min(\sqrt{\text{Reff} \cdot t}, 2 \cdot R_n)$$
$$= \min(\sqrt{3354.0 \cdot 91.0}, 2 \cdot 463.2)$$
$$= 552.462 \text{ mm}$$

Thickness Limit Candidate [LH1]:

$$= \min[1.5t, t_e] + \sqrt{R_n \cdot t_n}$$
$$= \min[1.5 \cdot 91.0, 0.0] + \sqrt{463.2 \cdot 134.0}$$
$$= 249.136 \text{ mm}$$

Thickness Limit Candidate [LH2]:

$$= L_{pr1}$$
$$= 360.0$$
$$= 360.000 \text{ mm}$$

Thickness Limit Candidate [LH3]:

$$= 8(t + t_e)$$
$$= 8(91.0 + 0.0)$$
$$= 728.000 \text{ mm}$$

Effective Nozzle Wall Length Outside the Vessel [LH]:

$$= \min[LH1, LH2, LH3] + t \quad \text{for set-in nozzles}$$
$$= \min[249.136, 360.0, 728.0] + 91.0$$
$$= 340.136 \text{ mm}$$

Effective Vessel Thickness [teff]:

$$= t + ((A5 \cdot \text{frp}) / LR)$$
$$= 91.0 + ((0.0 \cdot 1.0) / 552.462)$$
$$= 91.000 \text{ mm}$$

Parameter [λ]:

$$= \min(12, (2 \cdot R_n + t_n) / (\sqrt{(D_i + \text{teff}) \cdot \text{teff}}))$$
$$= \min(12, (2 \cdot 463.2 + 134.0) / (\sqrt{(6708.0 + 91.0) \cdot 91.0}))$$
$$= 1.348$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$= t \cdot LR \cdot \max((\lambda/5)^{0.85}, 1)$$
$$= 91.0 \cdot 552.462 \cdot \max((1.348/5)^{0.85}, 1)$$
$$= 50274.020 \text{ mm}^2$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$= A2A + A2B$$
$$= 30954.0 + 8668.895$$
$$= 39622.895 \text{ mm}^2$$

Area Contributed by the Outside Fillet Weld [A41]:

$$= 0.5 \cdot \text{Leg}^2$$
$$= 0.5 \cdot 10.0^2$$
$$= 50.000 \text{ mm}^2$$

The total area contributed by A1 through A43 [AT]:

$$= A1 + \text{frn}(A2 + A3) + A41 + A42 + A43$$
$$= 50274.02 + 0.944(39622.895 + 0.0) + 50.0 + 0.0 + 0.0$$
$$= 87717.492 \text{ mm}^2$$

Nozzle Radius for Force Calculation [Rxn]:

$$= t_n / \ln[1 + t_n/R_n]$$
$$= 134.0 / \ln[1 + 134.0/463.2]$$
$$= 527.366 \text{ mm}$$

Shell Radius for Force Calculation [Rxs]:

$$= \text{teff} / \ln[1 + \text{teff}/\text{Reff}]$$
$$= 91.0 / \ln[1 + 91.0/3354.0]$$
$$= 3399.296 \text{ mm}$$



Allowable Local Primary Membrane Stress [Sallow]:

$$= 1.5 \cdot S \cdot E$$

$$= 1.5 \cdot 156.4 \cdot 1.0$$

$$= 234.6 \text{ MPa}$$

Force acting on the Nozzle [fN]:

$$= P \cdot R_{xn} \cdot L_H$$

$$= 4.105 \cdot 527.366 \cdot 340.136$$

$$= 736222.7 \text{ N}$$

Force acting on the Shell [fS]:

$$= P \cdot R_{xs} (L_R + t_n)$$

$$= 4.105 \cdot 3399.296 (552.462 + 134.0)$$

$$= 9577447.0 \text{ N}$$

Discontinuity Force from Internal Pressure [fY]:

$$= P \cdot R_{xs} \cdot R_{nc}$$

$$= 4.105 \cdot 3399.296 \cdot 463.2$$

$$= 6462521.5 \text{ N}$$

Area Resisting Internal Pressure [Ap]:

$$= (f_S + f_Y + f_N) / P$$

$$= (9577447 + 6462522 + 736223) / 4.105$$

$$= 4087416.2 \text{ mm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$= Sallow / (2 \cdot A_p / A_T - R_{xs} / t_{eff})$$

$$= 234.6 / (2 \cdot 4087416 / 87717.492 - 3399.296 / 91.0)$$

$$= 4.2 \text{ MPa}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$= S / [t / R_{xs}]$$

$$= 156.4 [91.0 / 3399.296]$$

$$= 4.2 \text{ MPa}$$

Maximum Allowable Working Pressure [Pmax]:

$$= \min(P_{max1}, P_{max2})$$

$$= \min(4.201, 4.187)$$

$$= 4.187 \text{ MPa}$$

Average Primary Membrane Stress [σAvg]:

$$= (f_N + f_S + f_Y) / A_T$$

$$= (736223 + 9577447 + 6462522) / 87717.492$$

$$= 191.269 \text{ MPa}$$

General Primary Membrane Stress [σCirc]:

$$= P \cdot R_{xs} / t_{eff}$$

$$= 4.105 \cdot 3399.296 / 91.0$$

$$= 153.3 \text{ MPa}$$

Maximum Local Primary Membrane Stress [PL]:

$$= \max(2 \cdot \sigma_{Avg} - \sigma_{Circ}, \sigma_{Circ})$$

$$= \max(2 \cdot 191.269 - 153.331, 153.331)$$

$$= 229.2 \text{ MPa}$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	234.60	MPa
Local Primary Membrane Stress	PL	229.21	MPa
Maximum Allowable Working Pressure	Pmax	4.19	MPa

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$= (R_{nc} + t_n) / R_{nc}$$

$$= (463.2 + 134.0) / 463.2$$

$$= 1.289 \text{ For set-in Nozzles}$$

Weld Length of Nozzle to Shell Weld [Lt]:

$$= \pi / 2 \cdot (R_n + t_n)$$

$$= \pi / 2 \cdot (463.2 + 134.0)$$



= 938.080 mm

Weld Throat Dimensions, (0.7071•Leg Dimensions) [L41T, L42T, L43T]:
 = 7.071, 0.000, 0.000, mm

Required Weld Throat Dimensions :
 tc, L41T = 7.071, must be $\geq \min(0.7 * 134.000, 6.000)$ Passed

Weld Load Value [fwelds]:
 = $\min(f_y \cdot k_y, 1.5 \cdot S_n(A_2 + A_3), \pi/4 \cdot P \cdot R_n^2 \cdot k_y^2)$
 = $\min(6462522 \cdot 1.29, 1.5 \cdot 147.6(39622.895 + 0.0), \pi/4 \cdot 4.1 \cdot 463.2^2 \cdot 1.29^2)$
 = 1149672.125 N

Weld Stress Value [τ]:
 = $fwelds / (L_t(0.49 \cdot L_{41T} + 0.6 \cdot tw_1 + 0.49 \cdot L_{43T}))$
 = $1149672 / (938.08(0.49 \cdot 7.071 + 0.6 \cdot 10.0 + 0.49 \cdot 0.0))$
 = 129.497 < or = to 156.400 Weld Stress Passed

Nozzle/Structural Discontinuity Proximity Warning:

This nozzle's diameter limit intersects a major structural discontinuity. Reduce the diameter limit in the nozzle dialog so that there is no interference or move the nozzle away from the intersecting weld by 401.162 mm.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Flange MDMT including temperature reduction per 3.11:
 Flange MDMT with Temp reduction -68 °C

The Drop for this Nozzle is : 53.6118 mm
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 505.6117 mm

Input Echo, WRC297 Item 1, Description: A 36" :

Diameter Basis for Cylindrical Shell	ID		
Shell Internal Corrosion Allowance	1.0000	mm	
Shell External Corrosion Allowance	0.0000	mm	
Shell Diameter	6706.000	mm	
Shell Thickness	92.0000	mm	
Shell Stress Concentration Factor	1.000		
Vessel Material	SA-516 70		
Vessel Cold S.I. Allowable	Smc	175.00	MPa
Vessel Hot S.I. Allowable	Smh	156.40	MPa
Diameter Basis for Nozzle	ID		
Nozzle Corrosion Allowance	6.0000	mm	
Nozzle Diameter	914.400	mm	
Nozzle Thickness	140.0000	mm	
Nozzle Stress Concentration Factor	1.000		
Nozzle Material	SA-266 2		
Nozzle Cold S.I. Allowable	SNmc	165.00	MPa
Nozzle Hot S.I. Allowable	SNmh	147.60	MPa

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Note:
 External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUSained loads.



Design Internal Pressure	Dp	4.10	MPa
Radial Load	P	53979.15	N
Circumferential Shear	Vc	38984.99	N
Longitudinal Shear	Vl	38984.99	N
Circumferential Moment	Mc	0.720E+08	N-mm
Longitudinal Moment	Ml	0.720E+08	N-mm
Torsional Moment	Mt	0.828E+08	N-mm

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

*Warning - The ratio, Dn/Tn (8.531) is < 10
 check the Limitations of WRC bulletin 297.*

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$
 $= 1194.4 + 2 \cdot 1.65 \cdot \sqrt{(3399.5(92.0 - 1.0))}$
 $= 3029.848 \text{ mm}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	6799.000 mm
Nozzle Outside Diameter	(d) =	1194.400 mm
Vessel Thickness used	(T) =	91.000 mm
Nozzle Thickness used	(t) =	134.000 mm
T / t	=	0.679
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.518$$

Nr/P	=	0.088
Mr/P	=	0.161
M0/P	=	0.049
N0/P	=	0.110
MrD/Mc	=	0.284
NrDL/Mc	=	0.115
M0d/Mc	=	0.094
N0DL/Mc	=	0.056
MrD/Ml	=	0.144
NrDL/Ml	=	0.081
M0D/Ml	=	0.049
N0DL/Ml	=	0.133

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	0	0	0	0
Outplane Bending (P)	-1	1	-1	1
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (Ml)	0	0	0	0
Outplane Bending (Ml)	-2	2	2	-2
Normal Pressure Stress	149	153	149	153
Outplane Stress Summary	143	155	149	153

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	-6	6	-6	6
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0



Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	-6	6	6	-6
Inplane Pressure Stress	74	74	74	74
Inplane Stress Summary	60	86	74	74

Vessel Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress normal to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	0	0	0	0
Shear Stress Summary	0	0	0	0

Vessel Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Two * Max Shear Stress	143	155	149	153

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Membrane (P)	0	0	0	0
Outplane Bending (P)	-1	1	-1	1
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	-4	4	4	-4
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	74	74	74	74
Outplane Stress Summary	67	79	76	72

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses parallel to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside
Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	-6	6	-6	6
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	-12	12	12	-12
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	149	153	149	153
Inplane Stress Summary	129	170	155	147

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Torsional Shear (Mt)	0	0	0	0
Shear Stress Summary	0	0	0	0

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities)	Left Outside	Left Inside	Right Outside	Right Inside

Two * Max Shear Stress 129 170 155 147

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au	Al	Bu	Bl
	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Hoop Membrane (P)	0	0	0	0
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	14	19	14	19
Hoop Stress Summary	13	18	15	20

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au	Al	Bu	Bl
	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	-2	2	-2	2
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	-2	2	2	-2
Axial Pressure Stress	7	7	7	7
Axial Stress Summary	2	10	7	7

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Shear stress)	Au	Al	Bu	Bl
	Outside	Inside	Outside	Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	0	0	0	0
Shear Stress Summary	0	0	0	0

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stress Intensities)	Au	Al	Bu	Bl
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	13	18	15	20

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Hoop Membrane (P)	0	0	0	0
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	14	19	14	19
Hoop Stress Summary	13	18	14	19

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	-2	2	-2	2

Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	-4	4	4	-4
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	7	7	7	7

Axial Stress Summary	0	13	10	5
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Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu	Cl	Du	Dl
	Outside	Inside	Outside	Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	0	0	0	0
Shear Stress Summary	0	0	0	0

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu	Cl	Du	Dl
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	13	18	14	19

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	149	153	149	153	149	153	149	153
Circ. Pl (SUS)	0	0	0	0	0	0	0	0
Circ. Q (SUS)	-3	3	1	-1	-18	18	6	-6
Long. Pm (SUS)	74	74	74	74	74	74	74	74
Long. Pl (SUS)	0	0	0	0	0	0	0	0
Long. Q (SUS)	-12	12	0	0	-5	5	3	-3
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	149.0	153.0	149.0	153.0	149.0	153.0	149.0	153.0
Pm+Pl (SUS)	149.0	153.0	149.0	153.0	149.0	153.0	149.0	153.0
Pm+Pl+Q (Total)	146.0	156.0	150.0	152.0	131.0	171.0	155.0	147.0

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	153.00	156.40	Passed
Pm+Pl (SUS)	153.00	234.60	Passed
Pm+Pl+Q (TOTAL)	171.00	469.20	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl



Circ. Pm (SUS)	14	19	14	19	14	19	14	19
Circ. Pl (SUS)	0	0	0	0	0	0	0	0
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	7	7	7	7	7	7	7	7
Long. Pl (SUS)	0	0	0	0	0	0	0	0
Long. Q (SUS)	-4	4	0	0	-6	6	2	-2
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	14.0	19.0	14.0	19.0	14.0	19.0	14.0	19.0
Pm+Pl (SUS)	14.0	19.0	14.0	19.0	14.0	19.0	14.0	19.0
Pm+Pl+Q (Total)	14.0	19.0	14.0	19.0	14.0	19.0	14.0	19.0

Nozzle Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	19.00	147.60	Passed
Pm+Pl (SUS)	19.00	221.40	Passed
Pm+Pl+Q (TOTAL)	19.00	442.80	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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Input, Nozzle Desc: E 8"

From: 120

Pressure for Reinforcement Calculations	P	4.1007	MPa
Temperature for Internal Pressure	Temp	130	°C
Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	156.40	MPa
Parent Allowable Stress At Ambient	Sva	175.00	MPa
Inside Diameter of Hemispherical Head	D	6740.00	mm
Head Finished (Minimum) Thickness	t	52.0000	mm
Head Internal Corrosion Allowance	cas	1.0000	mm
Head External Corrosion Allowance	co	0.0000	mm
Distance from Head Centerline	L1	0.0000	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

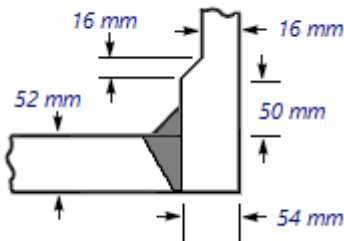
Material		SA-266 2	+IT
Material UNS Number		K03506	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	MPa
Allowable Stress At Ambient	Sna	138.00	MPa
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		6.7402	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.0000	mm
Flange Material		SA-105	+IT
Hub Height of Integral Nozzle	h	50.0000	mm
Height of Beveled Transition	L'	16.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		54.0000	mm
Corrosion Allowance	can	3.0000	mm
Outside Projection	ho	260.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	20.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	14.5000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.
Flange Class		600	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: E 8"

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	6.740	in.
Actual Thickness Used in Calculation	0.630	in.



Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 177.2 \cdot (\exp(4.101/(138.0 \cdot 1.0)) - 1) + 3.0 + 0.0$$

$$= 2.6723 + 3.0 + 0.0 = 5.6723 \text{ mm Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 138.0 \cdot 1.0 \cdot \ln((2 \cdot 13.0 + 177.2) / 177.2) - 0.0$$

$$= 18.894 - 0.0 = 18.8938 \text{ MPa}$$

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c : 10.160 mm
 Min. required shell thickness, internal p + c : 45.484 mm

Nozzle neck minimum required thickness : 10.160 mm

Minimum nozzle neck thickness 16.002 mm, is greater than required thickness 10.160 mm

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 47.9	Allowable	: 138.0 MPa	Passed
Expansion	: 0.0	Allowable	: 297.1 MPa	Passed
Occasional	: 13.0	Allowable	: 183.5 MPa	Passed
Shear	: 19.4	Allowable	: 96.6 MPa	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

$$= \min[S_n/S, 1]$$

$$= \min[138.0/156.4, 1]$$

$$= 0.882$$

Thickness of Nozzle at Shell [tn]:

$$= \text{hub thickness} - \text{corrosion allowance}$$

$$= 54.0 - 3.0$$

$$= 51.000 \text{ mm}$$

Thickness of Nozzle at Top [tn2]:

$$= \text{thickness} - \text{corrosion allowance}$$

$$= 16.0 - 3.0$$

$$= 13.000 \text{ mm}$$

Effective Pressure Radius [Reff]:

$$= D_i/2 + \text{corrosion allowance}$$

$$= 6740.0/2 + 1.0$$

$$= 3371.000 \text{ mm}$$

Effective Length of Vessel Wall [LR]:

$$= \min(\sqrt{Reff \cdot t}, 2 \cdot R_n)$$



$$= \min(\sqrt{(3371.0 \cdot 51.0)}, 2 \cdot 88.6)$$
$$= 177.200 \text{ mm}$$

Intermediate value [Cn]:

$$= \min((t + t_e)/t_n)^{0.35}, 1)$$
$$= \min((51.0 + 0.0)/51.0)^{0.35}, 1)$$
$$= \min(1.0, 1)$$
$$= 1.000$$

Intermediate value [Cp]:

$$= 9999999.000$$

Effective Nozzle Wall Length Outside the Vessel [LH]:

$$= \min(t + t_e + F_p \cdot \sqrt{(R_n \cdot t_n)}, L_{pr1} + t)$$
$$= \min(51.0 + 0.0 + 1.0 \cdot \sqrt{(88.6 \cdot 51.0)}, 260.0 + 51.0)$$
$$= 118.221 \text{ mm}$$

Effective Vessel Thickness [teff]:

$$= t + ((A_5 \cdot f_{rp}) / LR)$$
$$= 51.0 + ((0.0 \cdot 1.0) / 177.2)$$
$$= 51.000 \text{ mm}$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$= t \cdot LR$$
$$= 51.0 \cdot 177.2$$
$$= 9037.199 \text{ mm}^2$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$= A_{2A} + A_{2C}$$
$$= 5151.0 + 344.133$$
$$= 5495.133 \text{ mm}^2$$

Area Contributed by the Outside Fillet Weld [A41]:

$$= 0.5 \cdot \text{Leg}^2$$
$$= 0.5 \cdot 20.0^2$$
$$= 200.000 \text{ mm}^2$$

The total area contributed by A1 through A43 [AT]:

$$= A_1 + f_{rn}(A_2 + A_3) + A_{41} + A_{42} + A_{43}$$
$$= 9037.199 + 0.882(5495.133 + 0.0) + 200.0 + 0.0 + 0.0$$
$$= 14085.847 \text{ mm}^2$$

Nozzle Radius for Force Calculation [Rxn]:

$$= t_n / \ln[1 + t_n/R_n]$$
$$= 51.0 / \ln[1 + 51.0/88.6]$$
$$= 112.174 \text{ mm}$$

Shell Radius for Force Calculation [Rxs]:

$$= t_{eff} / \ln[1 + t_{eff}/R_{eff}]$$
$$= 51.0 / \ln[1 + 51.0/3371.0]$$
$$= 3396.426 \text{ mm}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$= 1.5 \cdot S \cdot E$$
$$= 1.5 \cdot 156.4 \cdot 1.0$$
$$= 234.6 \text{ MPa}$$

Force acting on the Nozzle [fN]:

$$= P \cdot R_{xn} \cdot LH$$
$$= 4.101 \cdot 112.174 \cdot 118.221$$
$$= 54376.0 \text{ N}$$

Force acting on the Shell [fS]:

$$= P \cdot R_{xs} \cdot (LR + t_n)/2$$
$$= 4.101 \cdot 3396.426 \cdot (177.2 + 51.0)/2$$
$$= 1589017.9 \text{ N}$$

Discontinuity Force from Internal Pressure [fY]:



$$= (P \cdot Rxs \cdot Rnc) / 2$$

$$= (4.101 \cdot 3396.426 \cdot 88.6) / 2$$

$$= 616945.6 \text{ N}$$

Area Resisting Internal Pressure [Ap]:

$$= (fS + fY + fN) / P$$

$$= (1589018 + 616946 + 54376.0) / 4.101$$

$$= 551255.1 \text{ mm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$= Sallow / (2 \cdot Ap / AT - Rxs / (2 \cdot teff))$$

$$= 234.6 / (2 \cdot 551255 / 14085.847 - 3396.426 / (2 \cdot 51.0))$$

$$= 5.2 \text{ MPa}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$= 2 \cdot S[t/Rxs]$$

$$= 2 \cdot 156.4[51.0/3396.426]$$

$$= 4.7 \text{ MPa}$$

Maximum Allowable Working Pressure [Pmax]:

$$= \min(Pmax1, Pmax2)$$

$$= \min(5.217, 4.697)$$

$$= 4.697 \text{ MPa}$$

Average Primary Membrane Stress [σAvg]:

$$= (fN + fS + fY) / AT$$

$$= (54376.043 + 1589018 + 616946) / 14085.847$$

$$= 160.482 \text{ MPa}$$

General Primary Membrane Stress [σCirc]:

$$= P \cdot Rxs / (2 \cdot teff)$$

$$= 4.101 \cdot 3396.426 / (2 \cdot 51.0)$$

$$= 136.5 \text{ MPa}$$

Maximum Local Primary Membrane Stress [PL]:

$$= \max(2 \cdot \sigma Avg - \sigma Circ, \sigma Circ)$$

$$= \max(2 \cdot 160.482 - 136.546, 136.546)$$

$$= 184.4 \text{ MPa}$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	234.60	MPa
Local Primary Membrane Stress	PL	184.42	MPa
Maximum Allowable Working Pressure	Pmax	4.70	MPa

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$= (Rnc + tn) / Rnc$$

$$= (88.6 + 51.0) / 88.6$$

$$= 1.576 \text{ For set-in Nozzles}$$

Weld Length of Nozzle to Shell Weld [Lt]:

$$= \pi / 2 \cdot (Rn + tn)$$

$$= \pi / 2 \cdot (88.6 + 51.0)$$

$$= 219.283 \text{ mm}$$

Weld Throat Dimensions, (0.7071•Leg Dimensions) [L41T, L42T, L43T]:

$$= 14.142, 0.000, 0.000, \text{ mm}$$

Required Weld Throat Dimensions :

$$tc, L41T = 14.142, \text{ must be } \geq \min(0.7 \cdot 51.000, 6.000) \text{ Passed}$$

Weld Load Value [fwelds]:

$$= \min(fy \cdot ky, 1.5 \cdot Sn(A2 + A3), \pi / 4 \cdot P \cdot Rn^2 \cdot ky^2)$$

$$= \min(616946 \cdot 1.58, 1.5 \cdot 138.0(5495.133 + 0.0), \pi / 4 \cdot 4.1 \cdot 88.6^2 \cdot 1.58^2)$$

$$= 62759.836 \text{ N}$$

Weld Stress Value [τ]:

$$= fwelds / (Lt(0.49 \cdot L41T + 0.6 \cdot tw1 + 0.49 \cdot L43T))$$

$$= 62759.836 / (219.283(0.49 \cdot 14.142 + 0.6 \cdot 14.5 + 0.49 \cdot 0.0))$$



= 18.313 < or = to 156.400 Weld Stress Passed

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Flange MDMT including temperature reduction per 3.11:

Flange MDMT with Temp reduction -68 °C

The Drop for this Nozzle is : 2.8928 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 314.8928 mm

Input Echo, WRC107/537 Item 1, Description: E 8" :

Diameter Basis for Vessel	Vbasis	ID	
Cylindrical or Spherical Vessel	Cylsph	Spherical	
Internal Corrosion Allowance	Cas	1.0000	mm
Vessel Diameter	Dv	6740.000	mm
Vessel Thickness	Tv	52.000	mm
Design Temperature	T1	130.0	C
Vessel Material		SA-516 70	
Vessel UNS Number		K02700	
Vessel Cold S.I. Allowable	Smc	175.00	MPa
Vessel Hot S.I. Allowable	Smh	156.40	MPa
Attachment Type	Type	Round	
WRC107 Attachment Classification	Holsol	Hollow	
Diameter Basis for Nozzle	Nbasis	ID	
Corrosion Allowance for Nozzle	Can	3.0000	mm
Nozzle Diameter	Dn	171.200	mm
Nozzle Thickness	Tn	54.000	mm
Nozzle Material		SA-266 2	
Nozzle UNS Number		K03506	
Nozzle Cold S.I. Allowable	SNmc	138.00	MPa
Nozzle Hot S.I. Allowable	SNmh	138.00	MPa
Design Internal Pressure	Dp	4.101	MPa
Include Pressure Thrust		No	

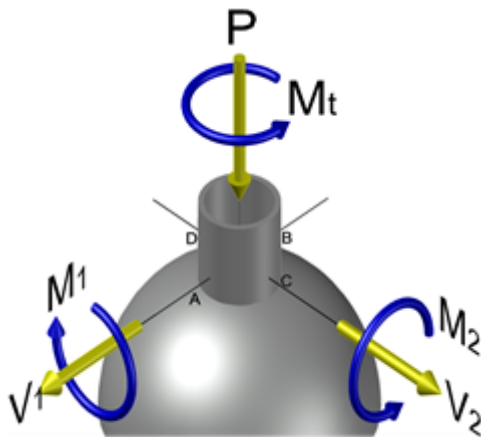
External Forces and Moments in WRC 107/537 Convention:

Radial Load (SUS)	P	17993.1	N
Longitudinal Shear (SUS)	(V1) V1	12995.0	N
Circumferential Shear (SUS)	(Vc) V2	12995.0	N
Circumferential Moment (SUS)	(Mc) M1	8000420.0	N-mm
Longitudinal Moment (SUS)	(Ml) M2	8000420.0	N-mm
Torsional Moment (SUS)	Mt	9200489.0	N-mm

Use Interactive Control No
 WRC107 Version Version March 1979

Include Pressure Stress Indices per Div. 2 No
 Compute Pressure Stress per WRC-368 No
 Local Loads applied at end of Nozzle/Attachment No

WRC Bulletin 537 provides equations for the dimensionless curves found in bulletin 107. As noted in the foreword to bulletin 537, "537 is equivalent to WRC 107". Where 107 is printed in the results below, "537" can be interchanged with "107".



WRC 107/537 has dimensional limitations that can be found in the bulletin's Foreword and which can also be inferred by review of the non-dimensional curve limits.

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$$

$$= 279.2 + 2 \cdot 1.65 \cdot \sqrt{(3396.5 (52.0 - 1.0))}$$

$$= 1652.656 \text{ mm}$$

WRC 107 Stress Calculation for SUSTAINED loads:

Radial Load		P	17993.1	N
Circumferential Shear	(VC)	V2	12995.0	N
Longitudinal Shear	(VL)	V1	12995.0	N
Circumferential Moment	(MC)	M1	8000420.0	N-mm
Longitudinal Moment	(ML)	M2	8000420.0	N-mm
Torsional Moment		MT	9200489.0	N-mm

Dimensionless Param: U = 0.34 TAU = 5.00 (2.24) RHO = 1.00

Dimensionless Loads for Spherical Shells at Attachment Junction:

Curves read for 1979	Figure	Value	Location
$N(x) * T / P$	SP 2	0.09554	(A,B,C,D)
$M(x) / P$	SP 2	0.12482	(A,B,C,D)
$N(x) * T * \text{SQRT}(R_m * T) / MC$	SM 2	0.20660	(A,B,C,D)
$M(x) * \text{SQRT}(R_m * T) / MC$	SM 2	0.38715	(A,B,C,D)
$N(x) * T * \text{SQRT}(R_m * T) / ML$	SM 2	0.20660	(A,B,C,D)
$M(x) * \text{SQRT}(R_m * T) / ML$	SM 2	0.38715	(A,B,C,D)
$N(y) * T / P$	SP 2	0.16626	(A,B,C,D)
$M(y) / P$	SP 2	0.04861	(A,B,C,D)
$N(y) * T * \text{SQRT}(R_m * T) / MC$	SM 2	0.04057	(A,B,C,D)
$M(y) * \text{SQRT}(R_m * T) / MC$	SM 2	0.17369	(A,B,C,D)
$N(y) * T * \text{SQRT}(R_m * T) / ML$	SM 2	0.04057	(A,B,C,D)
$M(y) * \text{SQRT}(R_m * T) / ML$	SM 2	0.17369	(A,B,C,D)

Stress Concentration Factors: Kn = 1.00, Kb = 1.00

Stresses in the Vessel at the Attachment Junction (MPa)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Memb.	P	-0.7	-0.7	-0.7	-0.7	-0.7	-0.7	-0.7	-0.7
Rad. Bend.	P	-5.2	5.2	-5.2	5.2	-5.2	5.2	-5.2	5.2
Rad. Memb.	MC	0.0	0.0	0.0	0.0	-1.5	-1.5	1.5	1.5
Rad. Bend.	MC	0.0	0.0	0.0	0.0	-17.2	17.2	17.2	-17.2
Rad. Memb.	ML	-1.5	-1.5	1.5	1.5	0.0	0.0	0.0	0.0
Rad. Bend.	ML	-17.2	17.2	17.2	-17.2	0.0	0.0	0.0	0.0
Tot. Rad. Str.		-24.5	20.2	12.8	-11.1	-24.5	20.2	12.8	-11.1

Tang. Memb. P	-1.2	-1.2	-1.2	-1.2	-1.2	-1.2	-1.2	-1.2
Tang. Bend. P	-2.0	2.0	-2.0	2.0	-2.0	2.0	-2.0	2.0
Tang. Memb. MC	0.0	0.0	0.0	0.0	-0.3	-0.3	0.3	0.3
Tang. Bend. MC	0.0	0.0	0.0	0.0	-7.7	7.7	7.7	-7.7
Tang. Memb. ML	-0.3	-0.3	0.3	0.3	0.0	0.0	0.0	0.0
Tang. Bend. ML	-7.7	7.7	7.7	-7.7	0.0	0.0	0.0	0.0
Tot. Tang. Str.	-11.2	8.3	4.8	-6.5	-11.2	8.3	4.8	-6.5
Shear VC	0.6	0.6	-0.6	-0.6	0.0	0.0	0.0	0.0
Shear VL	0.0	0.0	0.0	0.0	-0.6	-0.6	0.6	0.6
Shear MT	1.5	1.5	1.5	1.5	1.5	1.5	1.5	1.5
Tot. Shear	2.1	2.1	0.9	0.9	0.9	0.9	2.1	2.1
Str. Int.	24.8	20.5	12.9	11.3	24.6	20.2	13.3	11.9

WRC 107/537 Stress Summations:

Vessel Stress Summation at Attachment Junction (MPa)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Pm (SUS)		136.5	136.5	136.5	136.5	136.5	136.5	136.5	136.5
Rad. Pl (SUS)		-2.2	-2.2	0.9	0.9	-2.2	-2.2	0.9	0.9
Rad. Q (SUS)		-22.3	22.3	12.0	-12.0	-22.3	22.3	12.0	-12.0
Long. Pm (SUS)		136.5	136.5	136.5	136.5	136.5	136.5	136.5	136.5
Long. Pl (SUS)		-1.5	-1.5	-0.9	-0.9	-1.5	-1.5	-0.9	-0.9
Long. Q (SUS)		-9.7	9.7	5.7	-5.7	-9.7	9.7	5.7	-5.7
Shear Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pl (SUS)		0.6	0.6	-0.6	-0.6	-0.6	-0.6	0.6	0.6
Shear Q (SUS)		1.5	1.5	1.5	1.5	1.5	1.5	1.5	1.5
Pm (SUS)		136.5	136.5	136.5	136.5	136.5	136.5	136.5	136.5
Pm+Pl (SUS)		135.4	135.4	137.6	137.6	135.4	135.4	137.6	137.6
Pm+Pl+Q (Total)		125.7	157.0	149.5	130.2	125.4	156.8	149.9	130.8

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	136.55	156.40	Passed
Pm+Pl (SUS)	137.59	234.60	Passed
Pm+Pl+Q (TOTAL)	157.05	469.20	Passed

The Pm+Pl+Q allowable above was based on the rules of ASME VIII-2 Part 4, 4.1.6.3 (a)&(b) or per ASME VIII-2, 5.5.6.1(1)&(2) Part 5 as applicable.

The Pm+Pl+Q allowable was based on a temperature range cycling from ambient to design temperature.

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Input, Nozzle Desc: C 48"

From: 170

Pressure for Reinforcement Calculations	P	0.4743	MPa
Temperature for Internal Pressure	Temp	130	°C
Design External Pressure	Pext	0.05	MPa
Temperature for External Pressure	Tempex	130	°C
Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	156.40	MPa
Parent Allowable Stress At Ambient	Sva	175.00	MPa
Inside Diameter of Elliptical Head	D	7100.00	mm
Aspect Ratio of Elliptical Head	Ar	2.00	
Head Finished (Minimum) Thickness	t	36.0000	mm
Head Internal Corrosion Allowance	cas	6.0000	mm
Head External Corrosion Allowance	co	0.0000	mm
Distance from Head Centerline	L1	0.0000	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

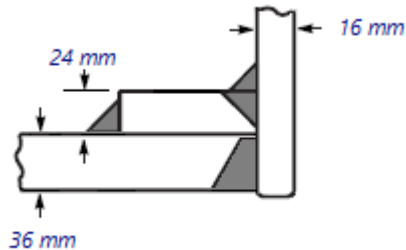
Material		SA-516 70	+IT
Material UNS Number		K02700	
Material Specification/Type		Plate	
Allowable Stress at Temperature	Sn	156.40	MPa
Allowable Stress At Ambient	Sna	175.00	MPa
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		46.7402	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.0000	mm
Flange Material		SA-105	+IT
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	6.0000	mm
Outside Projection	ho	310.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	16.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	16.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Pad Material		SA-516 70	+IT
Pad Allowable Stress at Temperature	Sp	156.40	MPa
Pad Allowable Stress At Ambient	Spa	175.00	MPa
Diameter of Pad along vessel surface	Dp	2439.2000	mm
Thickness of Pad	te	24.0000	mm
Weld leg size between Pad and Shell	Wp	16.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	24.0000	mm
Reinforcing Pad Width		609.9999	mm
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.
Flange Class		150	
Flange Grade		GR 1.1	
Flange Series		A	
Piping Exit Angle [North Clockwise reference]		90.0	deg
Bend Radius Multiplier		1.00	
Horizontal Run Length		4200.0	mm
Centerline Distance to Tangent		3800.0	mm

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: C 48"

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation 46.740 in.
 Actual Thickness Used in Calculation 0.630 in.



Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:
 $= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$
 $= 0.5 \cdot 1199.2 \cdot (\exp(0.474/(156.4 \cdot 1.0)) - 1) + 6.0 + 0.0$
 $= 1.8213 + 6.0 + 0.0 = 7.8213 \text{ mm Equation 4.3.1}$

Computed Maximum Allowable Working Pressure [MAWP]:
 $= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$
 $= 156.4 \cdot 1.0 \cdot \ln((2 \cdot 10.0 + 1199.2) / 1199.2) - 0.0$
 $= 2.587 - 0.0 = 2.5869 \text{ MPa}$

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c :	14.340 mm
Min. required shell thickness, internal p + c :	21.234 mm
Min. required shell thickness, external p + c :	23.988 mm

Nozzle neck minimum required thickness	: 14.340 mm
--	-------------

Minimum nozzle neck thickness 16.002 mm , is greater than required thickness 14.340 mm

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained :	15.4,	Allowable :	156.4 MPa	Passed
Expansion :	0.0,	Allowable :	398.8 MPa	Passed
Occasional :	14.1,	Allowable :	208.0 MPa	Passed
Shear :	17.3,	Allowable :	109.5 MPa	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:
 $= \min[S_n/S, 1]$
 $= \min[156.4/156.4, 1]$
 $= 1.000$

Pad Material Factor [frp]:
 $= \min[S_p/S, 1]$
 $= \min[156.4/156.4, 1]$
 $= 1.000$

Thickness of Nozzle [tn]:
 $= \text{thickness} - \text{corrosion allowance}$
 $= 16.0 - 6.0$
 $= 10.000 \text{ mm}$



Effective Pressure Radius [Reff]:
$$= 0.9/6 \cdot Di [2 + (Di/(2h))^2]$$
$$= 0.9/6 \cdot 7112.0 [2 + (7112.0/(2 \cdot 1781.0))^2]$$
$$= 6386.436 \text{ mm}$$

Effective Length of Vessel Wall [LR]:

For Nozzles with reinforcing pads, Lr is min[LR1, LR2, LR3]:

Limit [LR1]:
$$= \sqrt{(Reff \cdot t) + W}$$
$$= \sqrt{(6386.436 \cdot 30.0) + 610.0}$$
$$= 1047.714 \text{ mm}$$

Limit [LR2]:
$$= \sqrt{((Reff + t) \cdot (t + te))}$$
$$= \sqrt{((6386.436 + 30.0) \cdot (30.0 + 24.0))}$$
$$= 588.632 \text{ mm}$$

Limit [LR3]:
$$= 2 \cdot Rn$$
$$= 2 \cdot 599.6$$
$$= 1199.200 \text{ mm}$$

Final value of [LR]:
$$= \min(LR1, LR2, LR3)$$
$$= \min(1047.714, 588.632, 1199.2)$$
$$= 588.632 \text{ mm}$$

Intermediate value [Cn]:
$$= \min(((t + te)/tn)^{0.35}, 1)$$
$$= \min(((30.0 + 24.0)/10.0)^{0.35}, 1)$$
$$= \min(1.804, 1)$$
$$= 1.000$$

Intermediate value [Cp]:
$$= \exp((0.35 \cdot di - Xo)/(16t))$$
$$= \exp((0.35 \cdot 7112.0 - 609.6)/(16 \cdot 30.0))$$
$$= 50.191$$

Effective Nozzle Wall Length Outside the Vessel [LH]:
$$= \min(t + te + Fp \cdot \sqrt{Rn \cdot tn}, Lpr1 + t)$$
$$= \min(30.0 + 24.0 + 1.0 \cdot \sqrt{599.6 \cdot 10.0}, 310.0 + 30.0)$$
$$= 131.434 \text{ mm}$$

Effective Vessel Thickness [teff]:
$$= t + ((A5 \cdot frp) / LR)$$
$$= 30.0 + ((14127.166 \cdot 1.0) / 588.632)$$
$$= 54.000 \text{ mm}$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:
$$= t \cdot LR$$
$$= 30.0 \cdot 588.632$$
$$= 17658.957 \text{ mm}^2$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:
$$= tn \cdot LH$$
$$= 10.0 \cdot 131.434$$
$$= 1314.338 \text{ mm}^2$$

Area Contributed by the Outside Fillet Weld [A41]:
$$= 0.5 \cdot Leg41^2$$
$$= 0.5 \cdot 16.0^2$$
$$= 128.000 \text{ mm}^2$$

Area Contributed by the Reinforcing Pad [A5]:
$$= \min(W \cdot te, LR \cdot te)$$
$$= \min(610.0 \cdot 24.0, 588.632 \cdot 24.0)$$



$$= 14127.166 \text{ mm}^2$$

The total area contributed by A1 through A5 [AT]:
 $= A1 + \text{frn}(A2 + A3) + A41 + A42 + A43 + \text{frp}(A5)$
 $= 17658.957 + 1.0(1314.338 + 0.0) + 128.0 + 0.0 + 0.0 + 1.0(14127.166)$
 $= 33228.461 \text{ mm}^2$

Nozzle Radius for Force Calculation [Rxn]:
 $= t_n / \ln[1 + t_n/R_n]$
 $= 10.0 / \ln[1 + 10.0/599.6]$
 $= 604.588 \text{ mm}$

Shell Radius for Force Calculation [Rxs]:
 $= t_{eff} / \ln[1 + t_{eff}/R_{eff}]$
 $= 54.0 / \ln[1 + 54.0/6386.436]$
 $= 6413.417 \text{ mm}$

Allowable Local Primary Membrane Stress [Sallow]:
 $= 1.5 \cdot S \cdot E$
 $= 1.5 \cdot 156.4 \cdot 1.0$
 $= 234.6 \text{ MPa}$

Force acting on the Nozzle [fN]:
 $= P \cdot R_{xn} \cdot L_H$
 $= 0.474 \cdot 604.588 \cdot 131.434$
 $= 37690.0 \text{ N}$

Force acting on the Shell [fS]:
 $= P \cdot R_{xs} \cdot (L_R + t_n)/2$
 $= 0.474 \cdot 6413.417 \cdot (588.632 + 10.0)/2$
 $= 910497.1 \text{ N}$

Discontinuity Force from Internal Pressure [fY]:
 $= (P \cdot R_{xs} \cdot R_{nc})/2$
 $= (0.474 \cdot 6413.417 \cdot 599.6)/2$
 $= 911969.5 \text{ N}$

Area Resisting Internal Pressure [Ap]:
 $= (f_S + f_Y + f_N)/P$
 $= (910497 + 911970 + 37690.0)/0.474$
 $= 3921844.0 \text{ mm}^2$

Maximum Allowable Working Pressure Candidate [Pmax1]:
 $= S_{allow} / (2 \cdot A_p / AT - R_{xs} / (2 \cdot t_{eff}))$
 $= 234.6 / (2 \cdot 3921844 / 33228.461 - 6413.417 / (2 \cdot 54.0))$
 $= 1.3 \text{ MPa}$

Maximum Allowable Working Pressure Candidate [Pmax2]:
 $= 2 \cdot S[t/R_{xs}]$
 $= 2 \cdot 156.4[30.0/6413.417]$
 $= 1.5 \text{ MPa}$

Maximum Allowable Working Pressure [Pmax]:
 $= \min(P_{max1}, P_{max2})$
 $= \min(1.328, 1.463)$
 $= 1.328 \text{ MPa}$

Average Primary Membrane Stress [σ_{Avg}]:
 $= (f_N + f_S + f_Y) / AT$
 $= (37689.977 + 910497 + 911970) / 33228.461$
 $= 55.986 \text{ MPa}$

General Primary Membrane Stress [σ_{Circ}]:
 $= P \cdot R_{xs} / (2 \cdot t_{eff})$
 $= 0.474 \cdot 6413.417 / (2 \cdot 54.0)$
 $= 28.2 \text{ MPa}$

Maximum Local Primary Membrane Stress [PL]:
 $= \max(2 \cdot \sigma_{Avg} - \sigma_{Circ}, \sigma_{Circ})$
 $= \max(2 \cdot 55.986 - 28.168, 28.168)$



= 83.8 MPa

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	234.60	MPa
Local Primary Membrane Stress	PL	83.80	MPa
Maximum Allowable Working Pressure	Pmax	1.33	MPa

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$= (R_{nc} + t_n) / R_{nc}$$

$$= (599.6 + 10.0) / 599.6$$

$$= 1.017 \text{ For set-in Nozzles}$$

Weld Length of Nozzle to Shell Weld [L_T]:

$$= \pi/2 \cdot (R_n + t_n)$$

$$= \pi/2 \cdot (599.6 + 10.0)$$

$$= 957.558 \text{ mm}$$

Weld Length of Pad to Shell Weld [L_{TP}]:

$$= \pi/2 \cdot (R_n + t_n + W)$$

$$= \pi/2 \cdot (599.6 + 10.0 + 610.0)$$

$$= 1915.743 \text{ mm}$$

Weld Throat Dimensions, (0.7071•Leg Dimensions) [L_{41T}, L_{42T}, L_{43T}]:

$$= 11.314, 11.314, 0.000, \text{ mm}$$

Required Weld Throat Dimensions :

$$t_c, L_{41T} = 11.314, \text{ must be } \geq \min(0.7 \cdot 10.000, 6.000) \text{ Passed}$$

$$t_{f1}, L_{42T} = 11.314, \text{ must be } \geq \min(0.6 \cdot 24.000, 0.6 \cdot 30.000) \text{ Failed}$$

Weld Load Value [fwelds]:

$$= \min(f_y \cdot k_y, 1.5 \cdot S_n(A_2 + A_3), \pi/4 \cdot P \cdot R_n^2 \cdot k_y^2)$$

$$= \min(911970 \cdot 1.02, 1.5 \cdot 156.4(1314.338 + 0.0), \pi/4 \cdot 0.5 \cdot 599.6^2 \cdot 1.02^2)$$

$$= 138432.828 \text{ N}$$

Discontinuity Force [fws]:

$$= fwelds \cdot t \cdot S / (t \cdot S + t_e \cdot S_p)$$

$$= 138432.8 \cdot 30.0 \cdot 156 / (30.0 \cdot 156 + 24.0 \cdot 156)$$

$$= 76907.125 \text{ N}$$

Discontinuity Force [fwp]:

$$= fwelds \cdot t_e \cdot S_p / (t \cdot S + t_e \cdot S_p)$$

$$= 138432.8 \cdot 24.0 \cdot 156 / (30.0 \cdot 156 + 24.0 \cdot 156)$$

$$= 61525.703 \text{ N}$$

Shear Stress [τ₁]:

$$= fws / (L_T \cdot (0.6 \cdot t_{w1} + 0.49 \cdot L_{43T}))$$

$$= 76907.125 / (957.558 \cdot (0.6 \cdot 16.0 + 0.49 \cdot 0.0))$$

$$= 8.367 \text{ MPa}$$

Shear Stress [τ₂]:

$$= fwp / (L_T \cdot (0.6 \cdot t_{w2} + 0.49 \cdot L_{41T}))$$

$$= 61525.703 / (957.558 \cdot (0.6 \cdot 24.0 + 0.49 \cdot 11.314))$$

$$= 3.222 \text{ MPa}$$

Shear Stress [τ₃]:

$$= fwp / (L_{TP} \cdot (0.49 \cdot L_{42T}))$$

$$= 61525.703 / (1915.743 \cdot (0.49 \cdot 11.314))$$

$$= 5.794 \text{ MPa}$$

Maximum Shear Stress in the Welds:

$$= \max(\tau_1, \tau_2, \tau_3)$$

$$= \max(8.367, 3.222, 5.794)$$

$$= 8.4 \text{ must be less than or equal to } 156.4 \text{ MPa}$$

Nozzle Calculations per Section 4.5: External Pressure Case:

Nozzle Radius for Force Calculation [R_{xn}]:

$$= t_n / \ln[1 + t_n/R_n]$$



$$= 10.0 / \ln[1 + 10.0/599.6]$$
$$= 604.588 \text{ mm}$$

Shell Radius for Force Calculation [Rxs]:

$$= t_{eff} / \ln[1 + t_{eff}/R_{eff}]$$
$$= 54.0 / \ln[1 + 54.0/6386.436]$$
$$= 6413.417 \text{ mm}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$= 1.5 \cdot S$$
$$= 1.5 \cdot 156.4$$
$$= 234.6 \text{ MPa}$$

Force acting on the Nozzle [fN]:

$$= P \cdot R_{xn} \cdot L_H$$
$$= 0.052 \cdot 604.588 \cdot 131.434$$
$$= 4110.5 \text{ N}$$

Force acting on the Shell [fS]:

$$= P \cdot R_{xs} \cdot (L_R + t_n)/2$$
$$= 0.052 \cdot 6413.417 \cdot (588.632 + 10.0)/2$$
$$= 99300.2 \text{ N}$$

Discontinuity Force from External Pressure [fY]:

$$= (P \cdot R_{xs} \cdot R_{nc})/2$$
$$= (0.052 \cdot 6413.417 \cdot 599.6)/2$$
$$= 99460.8 \text{ N}$$

Area Resisting External Pressure [Ap]:

$$= (f_S + f_Y + f_N)/P$$
$$= (99300.2 + 99460.8 + 4110.5)/0.052$$
$$= 3921844.0 \text{ mm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$= S_{allow} / (2 \cdot A_p / A_T - R_{xs} / (2 \cdot t_{eff}))$$
$$= 234.6 / (2 \cdot 3921844 / 33228.461 - 6413.417 / (2 \cdot 54.0))$$
$$= 1.3 \text{ MPa}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$= 2 \cdot S / (t / R_{xs})$$
$$= 2 \cdot 156.4 / (30.0 / 6413.417)$$
$$= 1.5 \text{ MPa}$$

Maximum Allowable Working Pressure [Pmax]:

$$= \min(P_{max1}, P_{max2})$$
$$= \min(1.328, 1.463)$$
$$= 1.328 \text{ MPa}$$

Average Primary Membrane Stress [σ_{Avg}]:

$$= (f_N + f_S + f_Y) / A_T$$
$$= (4110.526 + 99300.203 + 99460.789) / 33228.461$$
$$= 6.106 \text{ MPa}$$

General Primary Membrane Stress [σ_{Circ}]:

$$= P \cdot R_{xs} / (2 \cdot t_{eff})$$
$$= 0.052 \cdot 6413.417 / (2 \cdot 54.0)$$
$$= 3.1 \text{ MPa}$$

Maximum Local Primary Membrane Stress [PL]:

$$= \max(2 \cdot \sigma_{Avg} - \sigma_{Circ}, \sigma_{Circ})$$
$$= \max(2 \cdot 6.106 - 3.072, 3.072)$$
$$= 9.1 \text{ MPa}$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	234.60	MPa
Local Primary Membrane Stress	PL	9.14	MPa
Maximum Allowable External Pressure	Pmax	1.33	MPa

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:



Nozzle Neck to Flange Weld (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Nozzle Neck to Pad Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Nozzle Neck to Pad Weld for Reinf. pad (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Shell to Pad Weld at Pad OD for pad (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Gov. MDMT of the Nozzle : -12 °C
 Gov. MDMT of the Reinforcement Pad : -12 °C

Flange MDMT including temperature reduction per 3.11:
 Flange MDMT with Temp reduction -104 °C

The Drop for this Nozzle is : 29.0130 mm
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 375.0131 mm

Input Echo, WRC107/537 Item 1, Description: C 48" :

Diameter Basis for Vessel	Vbasis	ID	
Cylindrical or Spherical Vessel	Cylsph	Spherical	
Internal Corrosion Allowance	Cas	6.0000	mm
Vessel Diameter	Dv	12780.000	mm
Vessel Thickness	Tv	36.000	mm
Design Temperature	T1	130.0	C
Vessel Material		SA-516 70	
Vessel UNS Number		K02700	
Vessel Cold S.I. Allowable	Smc	175.00	MPa
Vessel Hot S.I. Allowable	Smh	156.40	MPa
Attachment Type	Type	Round	
WRC107 Attachment Classification	Holsol	Hollow	
Diameter Basis for Nozzle	Nbasis	ID	
Corrosion Allowance for Nozzle	Can	6.0000	mm
Nozzle Diameter	Dn	1187.200	mm
Nozzle Thickness	Tn	16.000	mm
Nozzle Material		SA-516 70	
Nozzle UNS Number		K02700	
Nozzle Cold S.I. Allowable	SNmc	175.00	MPa
Nozzle Hot S.I. Allowable	SNmh	156.40	MPa
Thickness of Reinforcing Pad	Tpad	24.000	mm
Diameter of Reinforcing Pad	Dpad	2439.200	mm
Design Internal Pressure	Dp	0.474	MPa
Include Pressure Thrust		No	

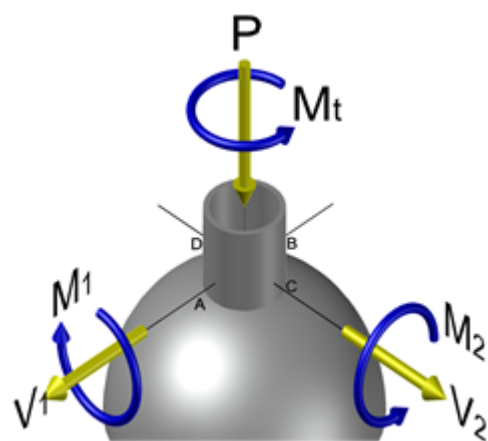
External Forces and Moments in WRC 107/537 Convention:

Radial Load (SUS)	P	-899999.4	N
Longitudinal Shear (SUS)	(V1) V1	60000.0	N
Circumferential Shear (SUS)	(Vc) V2	60000.0	N
Circumferential Moment (SUS)	(Mc) M1	201600448.0	N-mm
Longitudinal Moment (SUS)	(Ml) M2	201600448.0	N-mm
Torsional Moment (SUS)	Mt	287999488.0	N-mm

Use Interactive Control No
 WRC107 Version Version March 1979

Include Pressure Stress Indices per Div. 2 No
 Compute Pressure Stress per WRC-368 No
 Local Loads applied at end of Nozzle/Attachment No

WRC Bulletin 537 provides equations for the dimensionless curves found in bulletin 107. As noted in the foreword to bulletin 537, "537 is equivalent to WRC 107". Where 107 is printed in the results below, "537" can be interchanged with "107".



WRC 107/537 has dimensional limitations that can be found in the bulletin's Foreword and which can also be inferred by review of the non-dimensional curve limits.

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$$

$$= 1219.2 + 2 \cdot 1.65 \cdot \sqrt{(6411.0 (36.0 - 6.0))}$$

$$= 2666.430 \text{ mm}$$

WRC 107 Stress Calculation for SUSTained loads:

Radial Load	P	-899999.4	N
Circumferential Shear	(VC) V2	60000.0	N
Longitudinal Shear	(VL) V1	60000.0	N
Circumferential Moment	(MC) M1	201600448.0	N-mm
Longitudinal Moment	(ML) M2	201600448.0	N-mm
Torsional Moment	MT	287999488.0	N-mm

Dimensionless Param: U = 1.04 TAU = 50.00 (60.46) RHO = 5.40

Dimensionless Loads for Spherical Shells at Attachment Junction:

Curves read for 1979	Figure	Value	Location
$N(x) * T / P$	SP 9	0.01218	(A,B,C,D)
$M(x) / P$	SP 9	0.00908	(A,B,C,D)
$N(x) * T * \text{SQRT}(R_m * T) / MC$	SM 9	0.02209	(A,B,C,D)
$M(x) * \text{SQRT}(R_m * T) / MC$	SM 9	0.01698	(A,B,C,D)
$N(x) * T * \text{SQRT}(R_m * T) / ML$	SM 9	0.02209	(A,B,C,D)
$M(x) * \text{SQRT}(R_m * T) / ML$	SM 9	0.01698	(A,B,C,D)
$N(y) * T / P$	SP 9	0.23263	(A,B,C,D)
$M(y) / P$	SP 9	0.02681	(A,B,C,D)
$N(y) * T * \text{SQRT}(R_m * T) / MC$	SM 9	0.29570	(A,B,C,D)
$M(y) * \text{SQRT}(R_m * T) / MC$	SM 9	0.06077	(A,B,C,D)
$N(y) * T * \text{SQRT}(R_m * T) / ML$	SM 9	0.29570	(A,B,C,D)
$M(y) * \text{SQRT}(R_m * T) / ML$	SM 9	0.06077	(A,B,C,D)

Stress Concentration Factors: Kn = 1.00, Kb = 1.00

Stresses in the Vessel at the Attachment Junction (MPa)



Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Memb. P		3.8	3.8	3.8	3.8	3.8	3.8	3.8	3.8
Rad. Bend. P		16.8	-16.8	16.8	-16.8	16.8	-16.8	16.8	-16.8
Rad. Memb. MC		0.0	0.0	0.0	0.0	-2.6	-2.6	2.6	2.6
Rad. Bend. MC		0.0	0.0	0.0	0.0	-12.0	12.0	12.0	-12.0
Rad. Memb. ML		-2.6	-2.6	2.6	2.6	0.0	0.0	0.0	0.0
Rad. Bend. ML		-12.0	12.0	12.0	-12.0	0.0	0.0	0.0	0.0
Tot. Rad. Str.		6.0	-3.7	35.1	-22.4	6.0	-3.7	35.1	-22.4
Tang. Memb. P		71.8	71.8	71.8	71.8	71.8	71.8	71.8	71.8
Tang. Bend. P		49.6	-49.6	49.6	-49.6	49.6	-49.6	49.6	-49.6
Tang. Memb. MC		0.0	0.0	0.0	0.0	-34.7	-34.7	34.7	34.7
Tang. Bend. MC		0.0	0.0	0.0	0.0	-42.8	42.8	42.8	-42.8
Tang. Memb. ML		-34.7	-34.7	34.7	34.7	0.0	0.0	0.0	0.0
Tang. Bend. ML		-42.8	42.8	42.8	-42.8	0.0	0.0	0.0	0.0
Tot. Tang. Str.		44.0	30.2	198.9	14.1	44.0	30.2	198.9	14.1
Shear VC		0.6	0.6	-0.6	-0.6	0.0	0.0	0.0	0.0
Shear VL		0.0	0.0	0.0	0.0	-0.6	-0.6	0.6	0.6
Shear MT		2.3	2.3	2.3	2.3	2.3	2.3	2.3	2.3
Tot. Shear		2.9	2.9	1.7	1.7	1.7	1.7	2.9	2.9
Str. Int.		44.2	34.4	199.0	36.6	44.0	34.1	199.0	36.9

Unitless Prm: U = 2.78 TAU = 0.00 (121.46) RHO = 0.00 (3.00)

Dimensionless Loads for Spherical Shells at Pad edge:

Curves read for 1979	Figure	Value	Location
N(x) * T / P	SR 2	0.02175	(A,B,C,D)
M(x) / P	SR 2	0.00990	(A,B,C,D)
N(x) * T * SQRT(Rm * T) / MC	SR 3	0.01691	(A,B,C,D)
M(x) * SQRT(Rm * T) / MC	SR 3	0.01053	(A,B,C,D)
N(x) * T * SQRT(Rm * T) / ML	SR 3	0.01691	(A,B,C,D)
M(x) * SQRT(Rm * T) / ML	SR 3	0.01053	(A,B,C,D)
N(y) * T / P	SR 2	0.00650	(A,B,C,D)
M(y) / P	SR 2	0.00300	(A,B,C,D)
N(y) * T * SQRT(Rm * T) / MC	SR 3	0.00507	(A,B,C,D)
M(y) * SQRT(Rm * T) / MC	SR 3	0.00317	(A,B,C,D)
N(y) * T * SQRT(Rm * T) / ML	SR 3	0.00507	(A,B,C,D)
M(y) * SQRT(Rm * T) / ML	SR 3	0.00317	(A,B,C,D)

Stress Concentration Factors: Kn = 1.00, Kb = 1.00

Stresses in the Vessel at the Edge of Reinforcing Pad (MPa)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Memb. P		21.8	21.8	21.8	21.8	21.8	21.8	21.8	21.8
Rad. Bend. P		59.4	-59.4	59.4	-59.4	59.4	-59.4	59.4	-59.4
Rad. Memb. MC		0.0	0.0	0.0	0.0	-8.6	-8.6	8.6	8.6
Rad. Bend. MC		0.0	0.0	0.0	0.0	-32.3	32.3	32.3	-32.3
Rad. Memb. ML		-8.6	-8.6	8.6	8.6	0.0	0.0	0.0	0.0
Rad. Bend. ML		-32.3	32.3	32.3	-32.3	0.0	0.0	0.0	0.0
Tot. Rad. Str.		40.3	-14.0	122.0	-61.3	40.3	-14.0	122.0	-61.3
Tang. Memb. P		6.5	6.5	6.5	6.5	6.5	6.5	6.5	6.5
Tang. Bend. P		18.0	-18.0	18.0	-18.0	18.0	-18.0	18.0	-18.0
Tang. Memb. MC		0.0	0.0	0.0	0.0	-2.6	-2.6	2.6	2.6



Tang. Bend. MC	0.0	0.0	0.0	0.0	-9.7	9.7	9.7	-9.7
Tang. Memb. ML	-2.6	-2.6	2.6	2.6	0.0	0.0	0.0	0.0
Tang. Bend. ML	-9.7	9.7	9.7	-9.7	0.0	0.0	0.0	0.0
Tot. Tang. Str.	12.2	-4.4	36.8	-18.6	12.2	-4.4	36.8	-18.6
Shear VC	0.5	0.5	-0.5	-0.5	0.0	0.0	0.0	0.0
Shear VL	0.0	0.0	0.0	0.0	-0.5	-0.5	0.5	0.5
Shear MT	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0
Tot. Shear	1.5	1.5	0.5	0.5	0.5	0.5	1.5	1.5
Str. Int.	40.4	14.3	122.0	61.3	40.3	14.1	122.1	61.3

WRC 107/537 Stress Summations:

Vessel Stress Summation at Attachment Junction (MPa)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Pm (SUS)		28.2	28.2	28.2	28.2	28.2	28.2	28.2	28.2
Rad. Pl (SUS)		1.2	1.2	6.4	6.4	1.2	1.2	6.4	6.4
Rad. Q (SUS)		4.9	-4.9	28.8	-28.8	4.9	-4.9	28.8	-28.8
Long. Pm (SUS)		28.2	28.2	28.2	28.2	28.2	28.2	28.2	28.2
Long. Pl (SUS)		37.1	37.1	106.5	106.5	37.1	37.1	106.5	106.5
Long. Q (SUS)		6.9	-6.9	92.4	-92.4	6.9	-6.9	92.4	-92.4
Shear Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pl (SUS)		0.6	0.6	-0.6	-0.6	-0.6	-0.6	0.6	0.6
Shear Q (SUS)		2.3	2.3	2.3	2.3	2.3	2.3	2.3	2.3
Pm (SUS)		28.2	28.2	28.2	28.2	28.2	28.2	28.2	28.2
Pm+Pl (SUS)		65.3	65.3	134.7	134.7	65.3	65.3	134.7	134.7
Pm+Pl+Q (Total)		72.4	58.7	227.2	42.4	72.2	58.5	227.2	42.5

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	28.21	156.40	Passed
Pm+Pl (SUS)	134.72	234.60	Passed
Pm+Pl+Q (TOTAL)	227.21	469.20	Passed

The Pm+Pl+Q allowable above was based on the rules of ASME VIII-2 Part 4, 4.1.6.3 (a)&(b) or per ASME VIII-2, 5.5.6.1(1)&(2) Part 5 as applicable.

The Pm+Pl+Q allowable was based on a temperature range cycling from ambient to design temperature.

WRC 107/537 Stress Summations:

Vessel Stress Summation at Reinforcing Pad Edge (MPa)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Pm (SUS)		28.2	28.2	28.2	28.2	28.2	28.2	28.2	28.2
Rad. Pl (SUS)		13.1	13.1	30.4	30.4	13.1	13.1	30.4	30.4
Rad. Q (SUS)		27.2	-27.2	91.7	-91.7	27.2	-27.2	91.7	-91.7



Long. Pm (SUS)	28.2	28.2	28.2	28.2	28.2	28.2	28.2	28.2
Long. Pl (SUS)	3.9	3.9	9.1	9.1	3.9	3.9	9.1	9.1
Long. Q (SUS)	8.3	-8.3	27.7	-27.7	8.3	-8.3	27.7	-27.7
Shear Pm (SUS)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pl (SUS)	0.5	0.5	-0.5	-0.5	-0.5	-0.5	0.5	0.5
Shear Q (SUS)	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0
Pm (SUS)	28.2	28.2	28.2	28.2	28.2	28.2	28.2	28.2
Pm+Pl (SUS)	41.4	41.4	58.6	58.6	41.4	41.4	58.6	58.6
Pm+Pl+Q (Total)	68.6	24.1	150.3	42.6	68.5	23.9	150.3	42.7

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	28.21	156.40	Passed
Pm+Pl (SUS)	58.61	234.60	Passed
Pm+Pl+Q (TOTAL)	150.29	469.20	Passed

The Pm+Pl+Q allowable above was based on the rules of ASME VIII-2 Part 4, 4.1.6.3 (a)&(b) or per ASME VIII-2, 5.5.6.1(1)&(2) Part 5 as applicable.

The Pm+Pl+Q allowable was based on a temperature range cycling from ambient to design temperature.

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Input, Nozzle Desc: MH-3 24"

From: 190

Pressure for Reinforcement Calculations	P	0.4140	MPa
Temperature for Internal Pressure	Temp	130	°C
Design External Pressure	Pext	0.05	MPa
Temperature for External Pressure	Tempex	130	°C

Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	156.40	MPa
Parent Allowable Stress At Ambient	Sva	175.00	MPa

Inside Diameter of Cylindrical Shell	D	7100.00	mm
Design Length of Section	L	2750.0002	mm
Shell Finished (Minimum) Thickness	t	24.0000	mm
Shell Internal Corrosion Allowance	cas	6.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		29550.05	mm
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User Entered Minimum Design Metal Temperature		-12.00	°C
---	--	--------	----

Type of Element Connected to the Parent : Nozzle

Material		SA-516 70	+IT
Material UNS Number		K02700	
Material Specification/Type		Plate	
Allowable Stress at Temperature	Sn	156.40	MPa
Allowable Stress At Ambient	Sna	175.00	MPa

Diameter Basis (for tr calc only)		Inside	
Layout Angle		225.00	deg
Diameter		22.7402	in.

Size and Thickness Basis		Actual	
Actual Thickness	tn	16.0000	mm

Flange Material		SA-105	+IT
Flange Type		Weld Neck	Flange

Corrosion Allowance	can	6.0000	mm
---------------------	-----	--------	----

Outside Projection	ho	310.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	16.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	16.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Pad Material		SA-516 70	+IT
Pad Allowable Stress at Temperature	Sp	156.40	MPa
Pad Allowable Stress At Ambient	Spa	175.00	MPa
Diameter of Pad along vessel surface	Dp	1209.6000	mm
Thickness of Pad	te	20.0000	mm
Weld leg size between Pad and Shell	Wp	16.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	20.0000	mm
Reinforcing Pad Width		300.0000	mm
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.
This is a Manway, Coupling or Access Opening.			

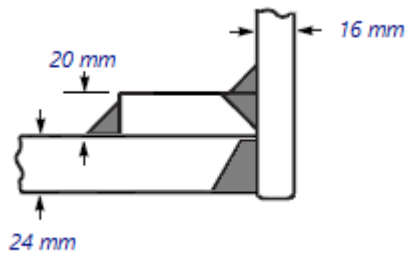
Flange Class		150
Flange Grade		GR 1.1

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: MH-3 24"

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	22.740	in.
Actual Thickness Used in Calculation	0.630	in.



Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:
 $= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$
 $= 0.5 \cdot 589.6 \cdot (\exp(0.414/(156.4 \cdot 1.0)) - 1) + 6.0 + 0.0$
 $= 0.7813 + 6.0 + 0.0 = 6.7813 \text{ mm Equation 4.3.1}$

Computed Maximum Allowable Working Pressure [MAWP]:
 $= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$
 $= 156.4 \cdot 1.0 \cdot \ln((2 \cdot 10.0 + 589.6) / 589.6) - 0.0$
 $= 5.217 - 0.0 = 5.2173 \text{ MPa}$

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:
 $= \min[S_n/S, 1]$
 $= \min[156.4/156.4, 1]$
 $= 1.000$

Pad Material Factor [frp]:
 $= \min[S_p/S, 1]$
 $= \min[156.4/156.4, 1]$
 $= 1.000$

Thickness of Nozzle [tn]:
 $= \text{thickness} - \text{corrosion allowance}$
 $= 16.0 - 6.0$
 $= 10.000 \text{ mm}$

Shell Diameter to Thickness ratio [D/t]:
 $= D_i/t$
 $= 7112.0/18.0$
 $= 395.111 \text{ must be less than 400.}$

Effective Pressure Radius [Reff]:
 $= D_i/2 + \text{corrosion allowance}$
 $= 7100.0/2 + 6.0$
 $= 3556.000 \text{ mm}$

Effective Length of Vessel Wall [LR]:

For Nozzles with reinforcing pads, Lr is min[LR1, LR2, LR3]:

Limit [LR1]:
 $= \sqrt{(Reff \cdot t) + W}$
 $= \sqrt{(3556.0 \cdot 18.0) + 300.0}$
 $= 552.998 \text{ mm}$

Limit [LR2]:
 $= \sqrt{((Reff + t) \cdot (t + t_e))}$
 $= \sqrt{((3556.0 + 18.0) \cdot (18.0 + 20.0))}$
 $= 368.527 \text{ mm}$

Limit [LR3]:



$$\begin{aligned} &= 2 \cdot R_n \\ &= 2 \cdot 294.8 \\ &= 589.600 \text{ mm} \end{aligned}$$

Final value of [LR]:

$$\begin{aligned} &= \min(LR1, LR2, LR3) \\ &= \min(552.998, 368.527, 589.6) \\ &= 368.527 \text{ mm} \end{aligned}$$

Thickness Limit Candidate [LH1]:

$$\begin{aligned} &= \min[1.5t, t_e] + \sqrt{(R_n \cdot t_n)} \\ &= \min[1.5 \cdot 18.0, 20.0] + \sqrt{(294.8 \cdot 10.0)} \\ &= 74.295 \text{ mm} \end{aligned}$$

Thickness Limit Candidate [LH2]:

$$\begin{aligned} &= L_{pr1} \\ &= 310.0 \\ &= 310.000 \text{ mm} \end{aligned}$$

Thickness Limit Candidate [LH3]:

$$\begin{aligned} &= 8(t + t_e) \\ &= 8(18.0 + 20.0) \\ &= 304.000 \text{ mm} \end{aligned}$$

Effective Nozzle Wall Length Outside the Vessel [LH]:

$$\begin{aligned} &= \min[LH1, LH2, LH3] + t \quad \text{for set-in nozzles} \\ &= \min[74.295, 310.0, 304.0] + 18.0 \\ &= 92.295 \text{ mm} \end{aligned}$$

Effective Vessel Thickness [teff]:

$$\begin{aligned} &= t + ((A5 \cdot frp) / LR) \\ &= 18.0 + ((6000.0 \cdot 1.0) / 368.527) \\ &= 34.281 \text{ mm} \end{aligned}$$

Parameter [λ]:

$$\begin{aligned} &= \min(12, (2 \cdot R_n + t_n) / (\sqrt{(D_i + t_{eff}) \cdot t_{eff} })) \\ &= \min(12, (2 \cdot 294.8 + 10.0) / (\sqrt{(7112.0 + 34.281) \cdot 34.281 })) \\ &= 1.211 \end{aligned}$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$\begin{aligned} &= t \cdot LR \cdot \max((\lambda/5)^{0.85}, 1) \\ &= 18.0 \cdot 368.527 \cdot \max((1.211/5)^{0.85}, 1) \\ &= 6633.482 \text{ mm}^2 \end{aligned}$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$\begin{aligned} &= t_n \cdot LH \\ &= 10.0 \cdot 92.295 \\ &= 922.955 \text{ mm}^2 \end{aligned}$$

Area Contributed by the Pad Fillet Weld [A42]:

$$\begin{aligned} &= 0.5 \cdot Leg42^2 \\ &= 0.5 \cdot 16.0^2 \\ &= 128.000 \text{ mm}^2 \end{aligned}$$

Area Contributed by the Outside Fillet Weld [A41]:

$$\begin{aligned} &= 0.5 \cdot Leg41^2 \\ &= 0.5 \cdot 16.0^2 \\ &= 128.000 \text{ mm}^2 \end{aligned}$$

Area Contributed by the Reinforcing Pad [A5]:

$$\begin{aligned} &= \min(W \cdot t_e, LR \cdot t_e) \\ &= \min(300.0 \cdot 20.0, 368.527 \cdot 20.0) \\ &= 6000.000 \text{ mm}^2 \end{aligned}$$

The total area contributed by A1 through A5 [AT]:

$$\begin{aligned} &= A1 + frn(A2 + A3) + A41 + A42 + A43 + frp(A5) \\ &= 6633.482 + 1.0(922.955 + 0.0) + 128.0 + 128.0 + 0.0 + 1.0(6000.0) \\ &= 13812.437 \text{ mm}^2 \end{aligned}$$



Nozzle Radius for Force Calculation [Rxn]:

$$= t_n / \ln[1 + t_n/R_n]$$

$$= 10.0/\ln[1 + 10.0/294.8]$$

$$= 299.773 \text{ mm}$$

Shell Radius for Force Calculation [Rxs]:

$$= t_{eff} / \ln[1 + t_{eff}/R_{eff}]$$

$$= 34.281/\ln[1 + 34.281/3556.0]$$

$$= 3573.114 \text{ mm}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$= 1.5 \cdot S \cdot E$$

$$= 1.5 \cdot 156.4 \cdot 1.0$$

$$= 234.6 \text{ MPa}$$

Force acting on the Nozzle [fN]:

$$= P \cdot R_{xn} \cdot L_H$$

$$= 0.414 \cdot 299.773 \cdot 92.295$$

$$= 11452.8 \text{ N}$$

Force acting on the Shell [fS]:

$$= P \cdot R_{xs} (L_R + t_n)$$

$$= 0.414 \cdot 3573.114 (368.527 + 10.0)$$

$$= 559863.6 \text{ N}$$

Discontinuity Force from Internal Pressure [fY]:

$$= P \cdot R_{xs} \cdot R_{nc}$$

$$= 0.414 \cdot 3573.114 \cdot 294.8$$

$$= 436026.7 \text{ N}$$

Area Resisting Internal Pressure [Ap]:

$$= (f_S + f_Y + f_N)/P$$

$$= (559864 + 436026.7 + 11452.8)/0.414$$

$$= 2433540.8 \text{ mm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$= S_{allow} / (2 \cdot A_p/AT - R_{xs}/t_{eff})$$

$$= 234.6 / (2 \cdot 2433541/13812.437 - 3573.114/34.281)$$

$$= 0.9 \text{ MPa}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$= S[t/R_{xs}]$$

$$= 156.4[18.0/3573.114]$$

$$= 0.8 \text{ MPa}$$

Maximum Allowable Working Pressure [Pmax]:

$$= \min(P_{max1}, P_{max2})$$

$$= \min(0.945, 0.788)$$

$$= 0.788 \text{ MPa}$$

Average Primary Membrane Stress [σAvg]:

$$= (f_N + f_S + f_Y) / AT$$

$$= (11452.793 + 559864 + 436026.688) / 13812.437$$

$$= 72.936 \text{ MPa}$$

General Primary Membrane Stress [σCirc]:

$$= P \cdot R_{xs} / t_{eff}$$

$$= 0.414 \cdot 3573.114 / 34.281$$

$$= 43.1 \text{ MPa}$$

Maximum Local Primary Membrane Stress [PL]:

$$= \max(2 \cdot \sigma_{Avg} - \sigma_{Circ}, \sigma_{Circ})$$

$$= \max(2 \cdot 72.936 - 43.149, 43.149)$$

$$= 102.7 \text{ MPa}$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	234.60	MPa
Local Primary Membrane Stress	PL	102.72	MPa
Maximum Allowable Working Pressure	Pmax	0.79	MPa



Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$\begin{aligned} &= (R_{nc} + t_n) / R_{nc} \\ &= (294.8 + 10.0) / 294.8 \\ &= 1.034 \text{ For set-in Nozzles} \end{aligned}$$

Weld Length of Nozzle to Shell Weld [L_T]:

$$\begin{aligned} &= \pi/2 \cdot (R_n + t_n) \\ &= \pi/2 \cdot (294.8 + 10.0) \\ &= 478.779 \text{ mm} \end{aligned}$$

Weld Length of Pad to Shell Weld [L_{TP}]:

$$\begin{aligned} &= \pi/2 \cdot (R_n + t_n + W) \\ &= \pi/2 \cdot (294.8 + 10.0 + 300.0) \\ &= 950.018 \text{ mm} \end{aligned}$$

Weld Throat Dimensions, (0.7071•Leg Dimensions) [L_{41T}, L_{42T}, L_{43T}]:

$$= 11.314, 11.314, 0.000, \text{ mm}$$

Required Weld Throat Dimensions :

$$\begin{aligned} t_c, L_{41T} &= 11.314, \text{ must be } \geq \min(0.7 \cdot 10.000, 6.000) \text{ Passed} \\ t_{f1}, L_{42T} &= 11.314, \text{ must be } \geq \min(0.6 \cdot 20.000, 0.6 \cdot 18.000) \text{ Passed} \end{aligned}$$

Weld Load Value [fwlds]:

$$\begin{aligned} &= \min(f_y \cdot k_y, 1.5 \cdot S_n(A_2 + A_3), \pi/4 \cdot P \cdot R_n^2 \cdot k_y^2) \\ &= \min(436027 \cdot 1.03, 1.5 \cdot 156.4(922.955 + 0.0), \pi/4 \cdot 0.4 \cdot 294.8^2 \cdot 1.03^2) \\ &= 30203.588 \text{ N} \end{aligned}$$

Discontinuity Force [fws]:

$$\begin{aligned} &= fwlds \cdot t \cdot S / (t \cdot S + t_e \cdot S_p) \\ &= 30203.6 \cdot 18.0 \cdot 156 / (18.0 \cdot 156 + 20.0 \cdot 156) \\ &= 14306.964 \text{ N} \end{aligned}$$

Discontinuity Force [fwp]:

$$\begin{aligned} &= fwlds \cdot t_e \cdot S_p / (t \cdot S + t_e \cdot S_p) \\ &= 30203.6 \cdot 20.0 \cdot 156 / (18.0 \cdot 156 + 20.0 \cdot 156) \\ &= 15896.626 \text{ N} \end{aligned}$$

Shear Stress [τ_1]:

$$\begin{aligned} &= fws / (L_T \cdot (0.6 \cdot t_{w1} + 0.49 \cdot L_{43T})) \\ &= 14306.964 / (478.779 \cdot (0.6 \cdot 16.0 + 0.49 \cdot 0.0)) \\ &= 3.113 \text{ MPa} \end{aligned}$$

Shear Stress [τ_2]:

$$\begin{aligned} &= fwp / (L_T \cdot (0.6 \cdot t_{w2} + 0.49 \cdot L_{41T})) \\ &= 15896.626 / (478.779 \cdot (0.6 \cdot 20.0 + 0.49 \cdot 11.314)) \\ &= 1.893 \text{ MPa} \end{aligned}$$

Shear Stress [τ_3]:

$$\begin{aligned} &= fwp / (L_{TP} \cdot (0.49 \cdot L_{42T})) \\ &= 15896.626 / (950.018 \cdot (0.49 \cdot 11.314)) \\ &= 3.019 \text{ MPa} \end{aligned}$$

Maximum Shear Stress in the Welds:

$$\begin{aligned} &= \max(\tau_1, \tau_2, \tau_3) \\ &= \max(3.113, 1.893, 3.019) \\ &= 3.1 \text{ must be less than or equal to } 156.4 \text{ MPa} \end{aligned}$$

Nozzle Calculations per Section 4.5: External Pressure Case:

Nozzle Radius for Force Calculation [R_{xn}]:

$$\begin{aligned} &= t_n / \ln[1 + t_n/R_n] \\ &= 10.0 / \ln[1 + 10.0/294.8] \\ &= 299.773 \text{ mm} \end{aligned}$$

Shell Radius for Force Calculation [R_{xs}]:

$$\begin{aligned} &= t_{eff} / \ln[1 + t_{eff}/R_{eff}] \\ &= 34.281 / \ln[1 + 34.281/3556.0] \end{aligned}$$



= 3573.114 mm

Allowable Local Primary Membrane Stress [Sallow]:

= $1.5 \cdot S$
= $1.5 \cdot 156.4$
= 234.6 MPa

Force acting on the Nozzle [fN]:

= $P \cdot Rxn \cdot LH$
= $0.052 \cdot 299.773 \cdot 92.295$
= 1431.2 N

Force acting on the Shell [fS]:

= $P \cdot Rxs(LR + tn)$
= $0.052 \cdot 3573.114(368.527 + 10.0)$
= 69963.9 N

Discontinuity Force from External Pressure [fY]:

= $P \cdot Rxs \cdot Rnc$
= $0.052 \cdot 3573.114 \cdot 294.8$
= 54488.5 N

Area Resisting External Pressure [Ap]:

= $(fS + fY + fN) / P$
= $(69963.9 + 54488.5 + 1431.2) / 0.052$
= 2433541.0 mm²

Maximum Allowable Working Pressure Candidate [Pmax1]:

= $Sallow / (2 \cdot Ap / AT - Rxs / teff)$
= $234.6 / (2 \cdot 2433541 / 13812.437 - 3573.114 / 34.281)$
= 0.9 MPa

Maximum Allowable Working Pressure Candidate [Pmax2]:

= $S[t / Rxs]$
= $156.4[18.0 / 3573.114]$
= 0.8 MPa

Maximum Allowable Working Pressure [Pmax]:

= $\min(Pmax1, Pmax2)$
= $\min(0.945, 0.788)$
= 0.788 MPa

Average Primary Membrane Stress [σ_{Avg}]:

= $(fN + fS + fY) / AT$
= $(1431.21 + 69963.93 + 54488.531) / 13812.437$
= 9.115 MPa

General Primary Membrane Stress [σ_{Circ}]:

= $P \cdot Rxs / teff$
= $0.052 \cdot 3573.114 / 34.281$
= 5.4 MPa

Maximum Local Primary Membrane Stress [PL]:

= $\max(2 \cdot \sigma_{Avg} - \sigma_{Circ}, \sigma_{Circ})$
= $\max(2 \cdot 9.115 - 5.392, 5.392)$
= 12.8 MPa

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	234.60	MPa
Local Primary Membrane Stress	PL	12.84	MPa
Maximum Allowable External Pressure	Pmax	0.79	MPa

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Nozzle Neck to Pad Weld for the Nozzle (Impact tested) :



Note: This Element/Detail was specified as being Impact Tested.

Nozzle Neck to Pad Weld for Reinf. pad (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Shell to Pad Weld at Pad OD for pad (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Gov. MDMT of the Nozzle	:	-12 °C
Gov. MDMT of the Reinforcement Pad	:	-12 °C

Flange MDMT including temperature reduction per 3.11:
Flange MDMT with Temp reduction -104 °C

The Drop for this Nozzle is : 13.1093 mm
The Cut Length for this Nozzle is, Drop + Ho + H + T : 347.1093 mm

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Input, Nozzle Desc: L2 2"

From: 200

Pressure for Reinforcement Calculations	P	0.3880	MPa
Temperature for Internal Pressure	Temp	130	°C
Design External Pressure	Pext	0.05	MPa
Temperature for External Pressure	Tempex	130	°C

Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	156.40	MPa
Parent Allowable Stress At Ambient	Sva	175.00	MPa

Inside Diameter of Cylindrical Shell	D	7100.00	mm
Design Length of Section	L	2500.0000	mm
Shell Finished (Minimum) Thickness	t	22.0000	mm
Shell Internal Corrosion Allowance	cas	6.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		33249.95	mm
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User Entered Minimum Design Metal Temperature		-12.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-105	+IT
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	MPa
Allowable Stress At Ambient	Sna	138.00	MPa

Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		2.0000	in.

Size and Thickness Basis		Actual	
Actual Thickness	tn	16.7640	mm

Flange Material		SA-105	+IT
Flange Type		Long Weld Neck	

Corrosion Allowance	can	3.0000	mm
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Outside Projection	ho	206.3750	mm
Weld leg size between Nozzle and Pad/Shell	Wo	16.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	16.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.

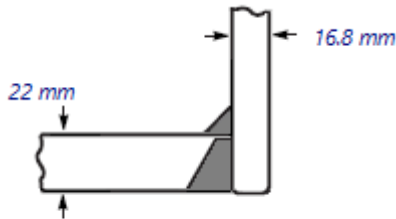
Flange Class		300
Flange Grade		GR 1.1

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: L2 2"

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation		2.000	in.
Actual Thickness Used in Calculation		0.660	in.



Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 56.8 \cdot (\exp(0.388/(138.0 \cdot 1.0)) - 1) + 3.0 + 0.0$$

$$= 0.08 + 3.0 + 0.0 = 3.08 \text{ mm Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 138.0 \cdot 1.0 \cdot \ln((2 \cdot 13.764 + 56.8) / 56.8) - 0.0$$

$$= 54.535 - 0.0 = 54.5345 \text{ MPa}$$

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c :	7.800 mm
Min. required shell thickness, internal p + c :	14.833 mm
Min. required shell thickness, external p + c :	22.592 mm

Nozzle neck minimum required thickness	: 7.800 mm
--	------------

Minimum nozzle neck thickness 16.764 mm , is greater than required thickness 7.800 mm

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

$$= \min[S_n/S, 1]$$

$$= \min[138.0/156.4, 1]$$

$$= 0.882$$

Thickness of Nozzle [tn]:

$$= \text{thickness} - \text{corrosion allowance}$$

$$= 16.764 - 3.0$$

$$= 13.764 \text{ mm}$$

Shell Diameter to Thickness ratio [D/t]:

$$= D_i/t$$

$$= 7112.0/16.0$$

$$= 444.500 \text{ must be less than 400; Calculation not possible.}$$

Nozzle Calculations per Section 4.5: External Pressure Case:

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Note: Impact testing of LWN flanges meeting requirements of UCS-66(c)(4) is not required.

The Drop for this Nozzle is : 0.2504 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 228.6252 mm

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Input, Nozzle Desc: L4 2"

From: 200

Pressure for Reinforcement Calculations	P	0.3880	MPa
Temperature for Internal Pressure	Temp	130	°C
Design External Pressure	Pext	0.05	MPa
Temperature for External Pressure	Tempex	130	°C

Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	156.40	MPa
Parent Allowable Stress At Ambient	Sva	175.00	MPa

Inside Diameter of Cylindrical Shell	D	7100.00	mm
Design Length of Section	L	2500.0000	mm
Shell Finished (Minimum) Thickness	t	22.0000	mm
Shell Internal Corrosion Allowance	cas	6.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		33249.95	mm
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User Entered Minimum Design Metal Temperature		-12.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-105	+IT
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	MPa
Allowable Stress At Ambient	Sna	138.00	MPa

Diameter Basis (for tr calc only)		Inside	
Layout Angle		30.00	deg
Diameter		2.0000	in.

Size and Thickness Basis		Actual	
Actual Thickness	tn	16.7640	mm

Flange Material		SA-105	+IT
Flange Type		Long Weld Neck	

Corrosion Allowance	can	3.0000	mm
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Outside Projection	ho	206.3750	mm
Weld leg size between Nozzle and Pad/Shell	Wo	16.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	16.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.

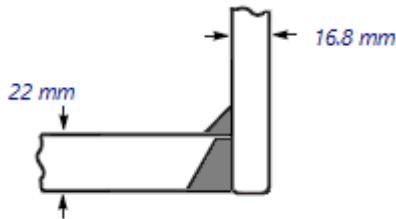
Flange Class		300
Flange Grade		GR 1.1

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: L4 2"

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation		2.000	in.
Actual Thickness Used in Calculation		0.660	in.



Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 56.8 \cdot (\exp(0.388/(138.0 \cdot 1.0)) - 1) + 3.0 + 0.0$$

$$= 0.08 + 3.0 + 0.0 = 3.08 \text{ mm Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 138.0 \cdot 1.0 \cdot \ln((2 \cdot 13.764 + 56.8) / 56.8) - 0.0$$

$$= 54.535 - 0.0 = 54.5345 \text{ MPa}$$

This nozzle or its diameter limit intersects weld seam # [7] on this element. The distance between the nozzle and seam is, 554.67 mm , while the distance from the center of the nozzle to the edge of the diameter limit is: 673.33 mm

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c :	7.800 mm
Min. required shell thickness, internal p + c	14.833 mm
Min. required shell thickness, external p + c	22.592 mm

Nozzle neck minimum required thickness	7.800 mm
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Minimum nozzle neck thickness 16.764 mm , is greater than required thickness 7.800 mm

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

$$= \min[S_n/S, 1]$$

$$= \min[138.0/156.4, 1]$$

$$= 0.882$$

Thickness of Nozzle [tn]:

$$= \text{thickness} - \text{corrosion allowance}$$

$$= 16.764 - 3.0$$

$$= 13.764 \text{ mm}$$

Shell Diameter to Thickness ratio [D/t]:

$$= D_i/t$$

$$= 7112.0/16.0$$

$$= 444.500 \text{ must be less than 400; Calculation not possible.}$$

Nozzle Calculations per Section 4.5: External Pressure Case:

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Note: Impact testing of LWN flanges meeting requirements of UCS-66(c)(4) is not required.

The Drop for this Nozzle is : 0.2504 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 228.6252 mm





Input, Nozzle Desc: L6 2"

From: 200

Pressure for Reinforcement Calculations	P	0.3880	MPa
Temperature for Internal Pressure	Temp	130	°C
Design External Pressure	Pext	0.05	MPa
Temperature for External Pressure	Tempex	130	°C

Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	156.40	MPa
Parent Allowable Stress At Ambient	Sva	175.00	MPa

Inside Diameter of Cylindrical Shell	D	7100.00	mm
Design Length of Section	L	2500.0000	mm
Shell Finished (Minimum) Thickness	t	22.0000	mm
Shell Internal Corrosion Allowance	cas	6.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		33249.95	mm
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User Entered Minimum Design Metal Temperature		-12.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-105	+IT
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	MPa
Allowable Stress At Ambient	Sna	138.00	MPa

Diameter Basis (for tr calc only)		Inside	
Layout Angle		60.00	deg
Diameter		2.0000	in.

Size and Thickness Basis		Actual	
Actual Thickness	tn	16.7640	mm

Flange Material		SA-105	+IT
Flange Type		Long Weld Neck	

Corrosion Allowance	can	3.0000	mm
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Outside Projection	ho	206.3750	mm
Weld leg size between Nozzle and Pad/Shell	Wo	16.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	16.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.

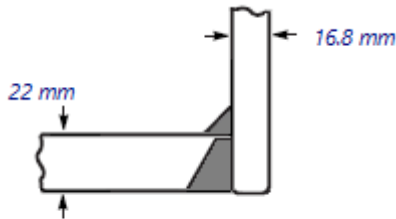
Flange Class		300
Flange Grade		GR 1.1

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: L6 2"

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation		2.000	in.
Actual Thickness Used in Calculation		0.660	in.



Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 56.8 \cdot (\exp(0.388/(138.0 \cdot 1.0)) - 1) + 3.0 + 0.0$$

$$= 0.08 + 3.0 + 0.0 = 3.08 \text{ mm Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 138.0 \cdot 1.0 \cdot \ln((2 \cdot 13.764 + 56.8) / 56.8) - 0.0$$

$$= 54.535 - 0.0 = 54.5345 \text{ MPa}$$

This nozzle or its diameter limit intersects weld seam # [8] on this element. The distance between the nozzle and seam is, 594.14 mm , while the distance from the center of the nozzle to the edge of the diameter limit is: 673.33 mm

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c :	7.800 mm
Min. required shell thickness, internal p + c	14.833 mm
Min. required shell thickness, external p + c	22.592 mm

Nozzle neck minimum required thickness	7.800 mm
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Minimum nozzle neck thickness 16.764 mm , is greater than required thickness 7.800 mm

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

$$= \min[S_n/S, 1]$$

$$= \min[138.0/156.4, 1]$$

$$= 0.882$$

Thickness of Nozzle [tn]:

$$= \text{thickness} - \text{corrosion allowance}$$

$$= 16.764 - 3.0$$

$$= 13.764 \text{ mm}$$

Shell Diameter to Thickness ratio [D/t]:

$$= D_i/t$$

$$= 7112.0/16.0$$

$$= 444.500 \text{ must be less than 400; Calculation not possible.}$$

Nozzle Calculations per Section 4.5: External Pressure Case:

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Note: Impact testing of LWN flanges meeting requirements of UCS-66(c)(4) is not required.

The Drop for this Nozzle is : 0.2504 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 228.6252 mm





Input, Nozzle Desc: L8 2"

From: 200

Pressure for Reinforcement Calculations	P	0.3880	MPa
Temperature for Internal Pressure	Temp	130	°C
Design External Pressure	Pext	0.05	MPa
Temperature for External Pressure	Tempex	130	°C

Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	156.40	MPa
Parent Allowable Stress At Ambient	Sva	175.00	MPa

Inside Diameter of Cylindrical Shell	D	7100.00	mm
Design Length of Section	L	2500.0000	mm
Shell Finished (Minimum) Thickness	t	22.0000	mm
Shell Internal Corrosion Allowance	cas	6.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		33249.95	mm
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User Entered Minimum Design Metal Temperature		-12.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-105	+IT
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	MPa
Allowable Stress At Ambient	Sna	138.00	MPa

Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		2.0000	in.

Size and Thickness Basis		Actual	
Actual Thickness	tn	16.7640	mm

Flange Material		SA-105	+IT
Flange Type		Long Weld Neck	

Corrosion Allowance	can	3.0000	mm
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Outside Projection	ho	206.3750	mm
Weld leg size between Nozzle and Pad/Shell	Wo	16.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	16.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.

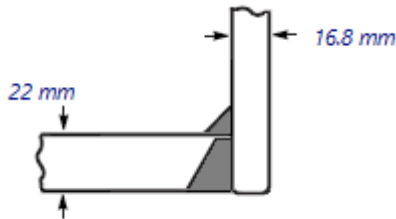
Flange Class		300
Flange Grade		GR 1.1

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: L8 2"

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation		2.000	in.
Actual Thickness Used in Calculation		0.660	in.



Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 56.8 \cdot (\exp(0.388/(138.0 \cdot 1.0)) - 1) + 3.0 + 0.0$$

$$= 0.08 + 3.0 + 0.0 = 3.08 \text{ mm Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 138.0 \cdot 1.0 \cdot \ln((2 \cdot 13.764 + 56.8) / 56.8) - 0.0$$

$$= 54.535 - 0.0 = 54.5345 \text{ MPa}$$

This nozzle or its diameter limit intersects weld seam # [1] on this element. The distance between the nozzle and seam is, 0.00 mm , while the distance from the center of the nozzle to the edge of the diameter limit is: 673.33 mm

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c :	7.800 mm
Min. required shell thickness, internal p + c	14.833 mm
Min. required shell thickness, external p + c	22.592 mm

Nozzle neck minimum required thickness	7.800 mm
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Minimum nozzle neck thickness 16.764 mm , is greater than required thickness 7.800 mm

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

$$= \min[S_n/S, 1]$$

$$= \min[138.0/156.4, 1]$$

$$= 0.882$$

Thickness of Nozzle [tn]:

$$= \text{thickness} - \text{corrosion allowance}$$

$$= 16.764 - 3.0$$

$$= 13.764 \text{ mm}$$

Shell Diameter to Thickness ratio [D/t]:

$$= D_i/t$$

$$= 7112.0/16.0$$

$$= 444.500 \text{ must be less than 400; Calculation not possible.}$$

Nozzle Calculations per Section 4.5: External Pressure Case:

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Note: Impact testing of LWN flanges meeting requirements of UCS-66(c)(4) is not required.

The Drop for this Nozzle is : 0.2504 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 228.6252 mm





Input, Nozzle Desc: L10 2"

From: 200

Pressure for Reinforcement Calculations	P	0.3880	MPa
Temperature for Internal Pressure	Temp	130	°C
Design External Pressure	Pext	0.05	MPa
Temperature for External Pressure	Tempex	130	°C

Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	156.40	MPa
Parent Allowable Stress At Ambient	Sva	175.00	MPa

Inside Diameter of Cylindrical Shell	D	7100.00	mm
Design Length of Section	L	2500.0000	mm
Shell Finished (Minimum) Thickness	t	22.0000	mm
Shell Internal Corrosion Allowance	cas	6.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		33249.95	mm
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User Entered Minimum Design Metal Temperature		-12.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-105	+IT
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	MPa
Allowable Stress At Ambient	Sna	138.00	MPa

Diameter Basis (for tr calc only)		Inside	
Layout Angle		120.00	deg
Diameter		2.0000	in.

Size and Thickness Basis		Actual	
Actual Thickness	tn	16.7640	mm

Flange Material		SA-105	+IT
Flange Type		Long Weld Neck	

Corrosion Allowance	can	3.0000	mm
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Outside Projection	ho	206.3750	mm
Weld leg size between Nozzle and Pad/Shell	Wo	16.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	16.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.

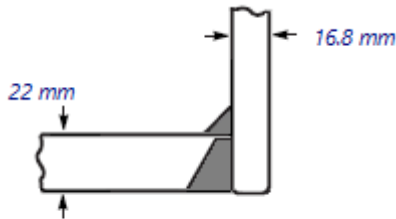
Flange Class		300
Flange Grade		GR 1.1

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: L10 2"

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation		2.000	in.
Actual Thickness Used in Calculation		0.660	in.



Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 56.8 \cdot (\exp(0.388/(138.0 \cdot 1.0)) - 1) + 3.0 + 0.0$$

$$= 0.08 + 3.0 + 0.0 = 3.08 \text{ mm Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 138.0 \cdot 1.0 \cdot \ln((2 \cdot 13.764 + 56.8) / 56.8) - 0.0$$

$$= 54.535 - 0.0 = 54.5345 \text{ MPa}$$

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c :	7.800 mm
Min. required shell thickness, internal p + c :	14.833 mm
Min. required shell thickness, external p + c :	22.592 mm

Nozzle neck minimum required thickness	: 7.800 mm
--	------------

Minimum nozzle neck thickness 16.764 mm , is greater than required thickness 7.800 mm

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

$$= \min[S_n/S, 1]$$

$$= \min[138.0/156.4, 1]$$

$$= 0.882$$

Thickness of Nozzle [tn]:

$$= \text{thickness} - \text{corrosion allowance}$$

$$= 16.764 - 3.0$$

$$= 13.764 \text{ mm}$$

Shell Diameter to Thickness ratio [D/t]:

$$= D_i/t$$

$$= 7112.0/16.0$$

$$= 444.500 \text{ must be less than 400; Calculation not possible.}$$

Nozzle Calculations per Section 4.5: External Pressure Case:

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Note: Impact testing of LWN flanges meeting requirements of UCS-66(c)(4) is not required.

The Drop for this Nozzle is : 0.2504 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 228.6252 mm

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Input, Nozzle Desc: B 36"

From: 200

Pressure for Reinforcement Calculations	P	0.3880	MPa
Temperature for Internal Pressure	Temp	130	°C
Design External Pressure	Pext	0.05	MPa
Temperature for External Pressure	Tempex	130	°C

Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	156.40	MPa
Parent Allowable Stress At Ambient	Sva	175.00	MPa

Inside Diameter of Cylindrical Shell	D	7100.00	mm
Design Length of Section	L	2499.8989	mm
Shell Finished (Minimum) Thickness	t	26.0000	mm
Shell Internal Corrosion Allowance	cas	6.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		34250.05	mm
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User Entered Minimum Design Metal Temperature		-12.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-516 70	+IT
Material UNS Number		K02700	
Material Specification/Type		Plate	
Allowable Stress at Temperature	Sn	156.40	MPa
Allowable Stress At Ambient	Sna	175.00	MPa

Diameter Basis (for tr calc only)		Inside	
Layout Angle		180.00	deg
Diameter		34.1102	in.

Size and Thickness Basis		Actual	
Actual Thickness	tn	24.0000	mm

Flange Material		SA-105	+IT
Flange Type		Weld Neck	Flange

Corrosion Allowance	can	6.0000	mm
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Outside Projection	ho	310.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	16.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	16.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Pad Material		SA-516 70	+IT
Pad Allowable Stress at Temperature	Sp	156.40	MPa
Pad Allowable Stress At Ambient	Spa	175.00	MPa
Diameter of Pad along vessel surface	Dp	1614.4000	mm
Thickness of Pad	te	18.0000	mm
Weld leg size between Pad and Shell	Wp	16.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	18.0000	mm
Reinforcing Pad Width		350.0000	mm
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.

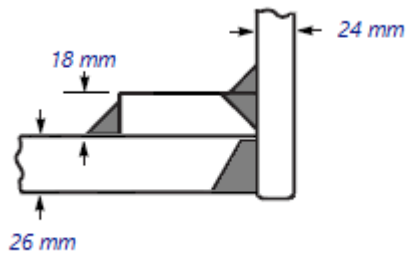
Flange Class		150
Flange Grade		GR 1.1
Flange Series		A

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: B 36"

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	34.110	in.
Actual Thickness Used in Calculation	0.945	in.



Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:
 $= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$
 $= 0.5 \cdot 878.4 \cdot (\exp(0.388/(156.4 \cdot 1.0)) - 1) + 6.0 + 0.0$
 $= 1.0909 + 6.0 + 0.0 = 7.0909 \text{ mm Equation 4.3.1}$

Computed Maximum Allowable Working Pressure [MAWP]:
 $= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$
 $= 156.4 \cdot 1.0 \cdot \ln((2 \cdot 18.0 + 878.4) / 878.4) - 0.0$
 $= 6.282 - 0.0 = 6.282 \text{ MPa}$

This nozzle or its diameter limit intersects weld seam # [3] on this element. The distance between the nozzle and seam is, 435.74 mm , while the distance from the center of the nozzle to the edge of the diameter limit is: 825.83 mm

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c : 14.340 mm
 Min. required shell thickness, internal p + c : 14.833 mm
 Min. required shell thickness, external p + c : 22.592 mm

Nozzle neck minimum required thickness : 14.340 mm

Minimum nozzle neck thickness 24.003 mm , is greater than required thickness 14.340 mm

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 17.7,	Allowable	: 156.4 MPa	Passed
Expansion	: 0.0,	Allowable	: 396.5 MPa	Passed
Occasional	: 4.6,	Allowable	: 208.0 MPa	Passed
Shear	: 10.0,	Allowable	: 109.5 MPa	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:
 $= \min[S_n/S, 1]$
 $= \min[156.4/156.4, 1]$
 $= 1.000$

Pad Material Factor [frp]:
 $= \min[S_p/S, 1]$
 $= \min[156.4/156.4, 1]$
 $= 1.000$

Thickness of Nozzle [tn]:
 $= \text{thickness} - \text{corrosion allowance}$
 $= 24.0 - 6.0$
 $= 18.000 \text{ mm}$

Shell Diameter to Thickness ratio [D/t]:



$$\begin{aligned}
 &= D_i/t \\
 &= 7112.0/20.0 \\
 &= 355.600 \text{ must be less than } 400.
 \end{aligned}$$

Effective Pressure Radius [Reff]:

$$\begin{aligned}
 &= D_i/2 + \text{corrosion allowance} \\
 &= 7100.0/2 + 6.0 \\
 &= 3556.000 \text{ mm}
 \end{aligned}$$

Effective Length of Vessel Wall [LR]:

For Nozzles with reinforcing pads, Lr is min[LR1, LR2, LR3]:

Limit [LR1]:

$$\begin{aligned}
 &= \sqrt{(Reff \cdot t) + W} \\
 &= \sqrt{(3556.0 \cdot 20.0) + 350.0} \\
 &= 616.683 \text{ mm}
 \end{aligned}$$

Limit [LR2]:

$$\begin{aligned}
 &= \sqrt{((Reff + t) \cdot (t + t_e))} \\
 &= \sqrt{((3556.0 + 20.0) \cdot (20.0 + 18.0))} \\
 &= 368.630 \text{ mm}
 \end{aligned}$$

Limit [LR3]:

$$\begin{aligned}
 &= 2 \cdot R_n \\
 &= 2 \cdot 439.2 \\
 &= 878.400 \text{ mm}
 \end{aligned}$$

Final value of [LR]:

$$\begin{aligned}
 &= \min(LR1, LR2, LR3) \\
 &= \min(616.683, 368.63, 878.4) \\
 &= 368.630 \text{ mm}
 \end{aligned}$$

Thickness Limit Candidate [LH1]:

$$\begin{aligned}
 &= \min[1.5t, t_e] + \sqrt{(R_n \cdot t_n)} \\
 &= \min[1.5 \cdot 20.0, 18.0] + \sqrt{(439.2 \cdot 18.0)} \\
 &= 106.913 \text{ mm}
 \end{aligned}$$

Thickness Limit Candidate [LH2]:

$$\begin{aligned}
 &= L_{pr1} \\
 &= 310.0 \\
 &= 310.000 \text{ mm}
 \end{aligned}$$

Thickness Limit Candidate [LH3]:

$$\begin{aligned}
 &= 8(t + t_e) \\
 &= 8(20.0 + 18.0) \\
 &= 304.000 \text{ mm}
 \end{aligned}$$

Effective Nozzle Wall Length Outside the Vessel [LH]:

$$\begin{aligned}
 &= \min[LH1, LH2, LH3] + t \text{ for set-in nozzles} \\
 &= \min[106.913, 310.0, 304.0] + 20.0 \\
 &= 126.913 \text{ mm}
 \end{aligned}$$

Effective Vessel Thickness [teff]:

$$\begin{aligned}
 &= t + ((A5 \cdot frp) / LR) \\
 &= 20.0 + ((6300.0 \cdot 1.0) / 368.63) \\
 &= 37.090 \text{ mm}
 \end{aligned}$$

Parameter [λ]:

$$\begin{aligned}
 &= \min(12, (2 \cdot R_n + t_n) / (\sqrt{(D_i + teff) \cdot teff})) \\
 &= \min(12, (2 \cdot 439.2 + 18.0) / (\sqrt{(7112.0 + 37.09) \cdot 37.09})) \\
 &= 1.741
 \end{aligned}$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$\begin{aligned}
 &= t \cdot LR \cdot \max((\lambda/5)^{0.85}, 1) \\
 &= 20.0 \cdot 368.63 \cdot \max((1.741/5)^{0.85}, 1) \\
 &= 7372.598 \text{ mm}^2
 \end{aligned}$$



Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$\begin{aligned} &= t_n \cdot L_H \\ &= 18.0 \cdot 126.913 \\ &= 2284.442 \text{ mm}^2 \end{aligned}$$

Area Contributed by the Pad Fillet Weld [A42]:

$$\begin{aligned} &= 0.5 \cdot \text{Leg}^2 \\ &= 0.5 \cdot 16.0^2 \\ &= 128.000 \text{ mm}^2 \end{aligned}$$

Area Contributed by the Outside Fillet Weld [A41]:

$$\begin{aligned} &= 0.5 \cdot \text{Leg}^2 \\ &= 0.5 \cdot 16.0^2 \\ &= 128.000 \text{ mm}^2 \end{aligned}$$

Area Contributed by the Reinforcing Pad [A5]:

$$\begin{aligned} &= \min(W \cdot t_e, LR \cdot t_e) \\ &= \min(350.0 \cdot 18.0, 368.63 \cdot 18.0) \\ &= 6300.000 \text{ mm}^2 \end{aligned}$$

The total area contributed by A1 through A5 [AT]:

$$\begin{aligned} &= A1 + \text{frn}(A2 + A3) + A41 + A42 + A43 + \text{frp}(A5) \\ &= 7372.598 + 1.0(2284.442 + 0.0) + 128.0 + 128.0 + 0.0 + 1.0(6300.0) \\ &= 16213.039 \text{ mm}^2 \end{aligned}$$

Nozzle Radius for Force Calculation [Rxn]:

$$\begin{aligned} &= t_n / \ln[1 + t_n/R_n] \\ &= 18.0 / \ln[1 + 18.0/439.2] \\ &= 448.140 \text{ mm} \end{aligned}$$

Shell Radius for Force Calculation [Rxs]:

$$\begin{aligned} &= t_{eff} / \ln[1 + t_{eff}/R_{eff}] \\ &= 37.09 / \ln[1 + 37.09/3556.0] \\ &= 3574.516 \text{ mm} \end{aligned}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$\begin{aligned} &= 1.5 \cdot S \cdot E \\ &= 1.5 \cdot 156.4 \cdot 1.0 \\ &= 234.6 \text{ MPa} \end{aligned}$$

Force acting on the Nozzle [fN]:

$$\begin{aligned} &= P \cdot R_{xn} \cdot L_H \\ &= 0.388 \cdot 448.14 \cdot 126.913 \\ &= 22065.6 \text{ N} \end{aligned}$$

Force acting on the Shell [fS]:

$$\begin{aligned} &= P \cdot R_{xs}(LR + t_n) \\ &= 0.388 \cdot 3574.516(368.63 + 18.0) \\ &= 536176.5 \text{ N} \end{aligned}$$

Discontinuity Force from Internal Pressure [fY]:

$$\begin{aligned} &= P \cdot R_{xs} \cdot R_{nc} \\ &= 0.388 \cdot 3574.516 \cdot 439.2 \\ &= 609080.6 \text{ N} \end{aligned}$$

Area Resisting Internal Pressure [Ap]:

$$\begin{aligned} &= (f_S + f_Y + f_N) / P \\ &= (536177 + 609081 + 22065.6) / 0.388 \\ &= 3008817.2 \text{ mm}^2 \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$\begin{aligned} &= S_{allow} / (2 \cdot A_p / AT - R_{xs} / t_{eff}) \\ &= 234.6 / (2 \cdot 3008817 / 16213.039 - 3574.516 / 37.09) \\ &= 0.9 \text{ MPa} \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$\begin{aligned} &= S / t / R_{xs} \\ &= 156.4 / 20.0 / 3574.516 \\ &= 0.9 \text{ MPa} \end{aligned}$$



Maximum Allowable Working Pressure [Pmax]:

$$\begin{aligned} &= \min(P_{max1}, P_{max2}) \\ &= \min(0.854, 0.875) \\ &= 0.854 \text{ MPa} \end{aligned}$$

Average Primary Membrane Stress [σ_{Avg}]:

$$\begin{aligned} &= (f_N + f_S + f_Y) / A_T \\ &= (22065.648 + 536177 + 609081) / 16213.039 \\ &= 72.005 \text{ MPa} \end{aligned}$$

General Primary Membrane Stress [σ_{Circ}]:

$$\begin{aligned} &= P \cdot R_{xs} / t_{eff} \\ &= 0.388 \cdot 3574.516 / 37.09 \\ &= 37.4 \text{ MPa} \end{aligned}$$

Maximum Local Primary Membrane Stress [PL]:

$$\begin{aligned} &= \max(2 \cdot \sigma_{Avg} - \sigma_{Circ}, \sigma_{Circ}) \\ &= \max(2 \cdot 72.005 - 37.393, 37.393) \\ &= 106.6 \text{ MPa} \end{aligned}$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	234.60	MPa
Local Primary Membrane Stress	PL	106.62	MPa
Maximum Allowable Working Pressure	Pmax	0.85	MPa

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$\begin{aligned} &= (R_{nc} + t_n) / R_{nc} \\ &= (439.2 + 18.0) / 439.2 \\ &= 1.041 \text{ For set-in Nozzles} \end{aligned}$$

Weld Length of Nozzle to Shell Weld [LT]:

$$\begin{aligned} &= \pi/2 \cdot (R_n + t_n) \\ &= \pi/2 \cdot (439.2 + 18.0) \\ &= 718.168 \text{ mm} \end{aligned}$$

Weld Length of Pad to Shell Weld [LTp]:

$$\begin{aligned} &= \pi/2 \cdot (R_n + t_n + W) \\ &= \pi/2 \cdot (439.2 + 18.0 + 350.0) \\ &= 1267.947 \text{ mm} \end{aligned}$$

Weld Throat Dimensions, (0.7071•Leg Dimensions) [L41T, L42T, L43T]:

$$= 11.314, 11.314, 0.000, \text{ mm}$$

Required Weld Throat Dimensions :

$$\begin{aligned} t_c, L41T &= 11.314, \text{ must be } \geq \min(0.7 \cdot 18.000, 6.000) \text{ Passed} \\ t_{f1}, L42T &= 11.314, \text{ must be } \geq \min(0.6 \cdot 18.000, 0.6 \cdot 20.000) \text{ Passed} \end{aligned}$$

Weld Load Value [fwelds]:

$$\begin{aligned} &= \min(f_y \cdot k_y, 1.5 \cdot S_n(A_2 + A_3), \pi/4 \cdot P \cdot R_n^2 \cdot k_y^2) \\ &= \min(609081 \cdot 1.04, 1.5 \cdot 156.4(2284.442 + 0.0), \pi/4 \cdot 0.4 \cdot 439.2^2 \cdot 1.04^2) \\ &= 63693.859 \text{ N} \end{aligned}$$

Discontinuity Force [fws]:

$$\begin{aligned} &= f_{welds} \cdot t \cdot S / (t \cdot S + t_e \cdot S_p) \\ &= 63693.9 \cdot 20.0 \cdot 156 / (20.0 \cdot 156 + 18.0 \cdot 156) \\ &= 33523.086 \text{ N} \end{aligned}$$

Discontinuity Force [fwp]:

$$\begin{aligned} &= f_{welds} \cdot t_e \cdot S_p / (t \cdot S + t_e \cdot S_p) \\ &= 63693.9 \cdot 18.0 \cdot 156 / (20.0 \cdot 156 + 18.0 \cdot 156) \\ &= 30170.779 \text{ N} \end{aligned}$$

Shear Stress [τ_1]:

$$\begin{aligned} &= f_{ws} / (L_T \cdot (0.6 \cdot t_{w1} + 0.49 \cdot L_{43T})) \\ &= 33523.086 / (718.168 \cdot (0.6 \cdot 16.0 + 0.49 \cdot 0.0)) \\ &= 4.863 \text{ MPa} \end{aligned}$$

Shear Stress [τ_2]:



$$\begin{aligned} &= fwp / (Lr \cdot (0.6 \cdot tw2 + 0.49 \cdot L41T)) \\ &= 30170.779 / (718.168 \cdot (0.6 \cdot 18.0 + 0.49 \cdot 11.314)) \\ &= 2.571 \text{ MPa} \end{aligned}$$

Shear Stress [τ_3]:

$$\begin{aligned} &= fwp / (LrP \cdot (0.49 \cdot L42T)) \\ &= 30170.779 / (1267.947 \cdot (0.49 \cdot 11.314)) \\ &= 4.293 \text{ MPa} \end{aligned}$$

Maximum Shear Stress in the Welds:

$$\begin{aligned} &= \max(\tau_1, \tau_2, \tau_3) \\ &= \max(4.863, 2.571, 4.293) \\ &= 4.9 \text{ must be less than or equal to } 156.4 \text{ MPa} \end{aligned}$$

Nozzle Calculations per Section 4.5: External Pressure Case:

Nozzle Radius for Force Calculation [R_{xn}]:

$$\begin{aligned} &= tn / \ln[1 + tn/Rn] \\ &= 18.0 / \ln[1 + 18.0/439.2] \\ &= 448.140 \text{ mm} \end{aligned}$$

Shell Radius for Force Calculation [R_{xs}]:

$$\begin{aligned} &= teff / \ln[1 + teff/Reff] \\ &= 37.09 / \ln[1 + 37.09/3556.0] \\ &= 3574.516 \text{ mm} \end{aligned}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$\begin{aligned} &= 1.5 \cdot S \\ &= 1.5 \cdot 156.4 \\ &= 234.6 \text{ MPa} \end{aligned}$$

Force acting on the Nozzle [f_N]:

$$\begin{aligned} &= P \cdot R_{xn} \cdot LH \\ &= 0.052 \cdot 448.14 \cdot 126.913 \\ &= 2942.1 \text{ N} \end{aligned}$$

Force acting on the Shell [f_S]:

$$\begin{aligned} &= P \cdot R_{xs} (LR + tn) \\ &= 0.052 \cdot 3574.516 (368.63 + 18.0) \\ &= 71489.7 \text{ N} \end{aligned}$$

Discontinuity Force from External Pressure [f_Y]:

$$\begin{aligned} &= P \cdot R_{xs} \cdot R_{nc} \\ &= 0.052 \cdot 3574.516 \cdot 439.2 \\ &= 81210.2 \text{ N} \end{aligned}$$

Area Resisting External Pressure [A_p]:

$$\begin{aligned} &= (f_S + f_Y + f_N) / P \\ &= (71489.7 + 81210.2 + 2942.1) / 0.052 \\ &= 3008818.0 \text{ mm}^2 \end{aligned}$$

Maximum Allowable Working Pressure Candidate [P_{max1}]:

$$\begin{aligned} &= Sallow / (2 \cdot A_p / AT - R_{xs} / teff) \\ &= 234.6 / (2 \cdot 3008818 / 16213.039 - 3574.516 / 37.09) \\ &= 0.9 \text{ MPa} \end{aligned}$$

Maximum Allowable Working Pressure Candidate [P_{max2}]:

$$\begin{aligned} &= S / [t / R_{xs}] \\ &= 156.4 [20.0 / 3574.516] \\ &= 0.9 \text{ MPa} \end{aligned}$$

Maximum Allowable Working Pressure [P_{max}]:

$$\begin{aligned} &= \min(P_{max1}, P_{max2}) \\ &= \min(0.854, 0.875) \\ &= 0.854 \text{ MPa} \end{aligned}$$

Average Primary Membrane Stress [σ_{Avg}]:

$$\begin{aligned} &= (f_N + f_S + f_Y) / AT \\ &= (2942.066 + 71489.695 + 81210.164) / 16213.039 \\ &= 9.601 \text{ MPa} \end{aligned}$$



General Primary Membrane Stress [σ_{Circ}]:

$$= P \cdot R_s / t_{eff}$$

$$= 0.052 \cdot 3574.516 / 37.09$$

$$= 5.0 \text{ MPa}$$

Maximum Local Primary Membrane Stress [PL]:

$$= \max(2 \cdot \sigma_{Avg} - \sigma_{Circ}, \sigma_{Circ})$$

$$= \max(2 \cdot 9.601 - 4.986, 4.986)$$

$$= 14.2 \text{ MPa}$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	234.60	MPa
Local Primary Membrane Stress	PL	14.22	MPa
Maximum Allowable External Pressure	Pmax	0.85	MPa

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Nozzle Neck to Pad Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Nozzle Neck to Pad Weld for Reinf. pad (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Shell to Pad Weld at Pad OD for pad (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Gov. MDMT of the Nozzle	:	-12 °C
Gov. MDMT of the Reinforcement Pad	:	-12 °C

Flange MDMT including temperature reduction per 3.11:

Flange MDMT with Temp reduction	-104 °C
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The Drop for this Nozzle is : 29.5641 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 365.5641 mm

Input Echo, WRC297 Item 1, Description: B 36" :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	6.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	7100.000	mm
Shell Thickness	26.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc	175.00 MPa
Vessel Hot S.I. Allowable	Smh	156.40 MPa

Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	6.0000	mm
Nozzle Diameter	866.400	mm
Nozzle Thickness	24.0000	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-516 70	
Nozzle Cold S.I. Allowable	SNmc	175.00 MPa
Nozzle Hot S.I. Allowable	SNmh	156.40 MPa

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.



Diameter of Reinforcing Pad 1614.400 mm
 Thickness of Reinforcing Pad 18.0000 mm

Note:

*External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUSstained loads.*

Design Internal Pressure	Dp	0.39	MPa
Radial Load	P	-67500.18	N
Circumferential Shear	Vc	44999.97	N
Longitudinal Shear	Vl	44999.97	N
Circumferential Moment	Mc	0.113E+09	N-mm
Longitudinal Moment	Ml	0.113E+09	N-mm
Torsional Moment	Mt	0.162E+09	N-mm

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - c_a))}$
 $= 914.4 + 2 \cdot 1.65 \cdot \sqrt{(3575.0(26.0 - 6.0))}$
 $= 1796.803 \text{ mm}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	7150.000 mm
Nozzle Outside Diameter	(d) =	914.400 mm
Vessel Thickness used	(T) =	38.000 mm
Nozzle Thickness used	(t) =	18.000 mm
T / t	=	2.111
d / t	=	50.800

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.754$$

Nr/P	=	0.050
Mr/P	=	0.095
M0/P	=	0.055
N0/P	=	0.163
MrD/Mc	=	0.191
NrDL/Mc	=	0.092
M0d/Mc	=	0.127
N0DL/Mc	=	0.206
MrD/Ml	=	0.113
NrDL/Ml	=	0.061
M0D/Ml	=	0.065
N0DL/Ml	=	0.308

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	7	7	7	7
Outplane Bending (P)	15	-15	15	-15
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (Ml)	-26	-26	26	26
Outplane Bending (Ml)	-33	33	33	-33
Normal Pressure Stress	36	36	36	36
Outplane Stress Summary	0	35	119	21

Vessel Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
--------------------	----	----	----	----



(Stresses parallel to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Inplane Membrane (P)	2	2	2	2
Inplane Bending (P)	26	-26	26	-26
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-5	-5	5	5
Inplane Bending (ML)	-58	58	58	-58
Inplane Pressure Stress	18	18	18	18
Inplane Stress Summary	-16	46	110	-59

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	3	3	3	3
Shear Stress Summary	4	4	2	2

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	17	48	119	81

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	7	7	7	7
Outplane Bending (P)	15	-15	15	-15
Outplane Membrane (Mc)	-17	-17	17	17
Outplane Bending (Mc)	-65	65	65	-65
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	18	18	18	18
Outplane Stress Summary	-42	58	124	-37

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Inplane Membrane (P)	2	2	2	2
Inplane Bending (P)	26	-26	26	-26
Inplane Membrane (Mc)	-7	-7	7	7
Inplane Bending (Mc)	-98	98	98	-98
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	36	36	36	36
Inplane Stress Summary	-40	102	171	-78

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Torsional Shear (Mt)	3	3	3	3



Shear Stress Summary 2 2 4 4

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stress Intensities	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress	44	102	171	78

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Hoop Membrane (P)	7	7	7	7
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-26	-26	26	26
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	11	11	11	11

Hoop Stress Summary -7 -7 45 45

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Axial Membrane (P)	1	1	1	1
Axial Bending (P)	88	-88	88	-88
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-10	-10	10	10
Axial Bending (ML)	-190	190	190	-190
Axial Pressure Stress	5	5	5	5

Axial Stress Summary -105 98 295 -260

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	1	1	-1	-1
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	6	6	6	6

Shear Stress Summary 8 8 5 5

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stress Intensities	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	106	107	295	306

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	7	7	7	7
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-17	-17	17	17
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	11	11	11	11

Hoop Stress Summary 1 1 36 36



Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	1	1	1	1
Axial Bending (P)	88	-88	88	-88
Axial Membrane (Mc)	-10	-10	10	10
Axial Bending (Mc)	-332	332	332	-332
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	5	5	5	5
Axial Stress Summary	-247	240	437	-402

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-1	-1	1	1
Shear due to Torsion	6	6	6	6
Shear Stress Summary	5	5	8	8

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	248	240	437	440

Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - c_a)}$
 $= 914.4 + 2 \cdot 1.65 \cdot \sqrt{3566.0(26.0 - 6.0)}$
 $= 1795.692 \text{ mm}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter (D) = 7132.000 mm
 Nozzle Outside Diameter (d) = 1614.400 mm
 Vessel Thickness used (T) = 20.000 mm
 Nozzle Thickness used (t) = 368.000 mm
 $T / t = 0.054$
 $d / t = 10.000$

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 4.275$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

Nr/P = 0.051
 Mr/P = 0.079
 M0/P = 0.022
 N0/P = 0.025
 MrD/Mc = 0.248
 NrDL/Mc = 0.108
 M0d/Mc = 0.074
 N0DL/Mc = 0.049
 MrD/Ml = 0.055
 NrDL/Ml = 0.061
 M0D/Ml = 0.017
 N0DL/Ml = 0.028

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
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Outplane Membrane (P)	4	4	4	4
Outplane Bending (P)	22	-22	22	-22
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-4	-4	4	4
Outplane Bending (ML)	-17	17	17	-17
Normal Pressure Stress	68	69	68	69

Outplane Stress Summary	72	64	117	38
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Vessel Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses parallel to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Inplane Membrane (P)	8	8	8	8
Inplane Bending (P)	80	-80	80	-80
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-10	-10	10	10
Inplane Bending (ML)	-58	58	58	-58
Inplane Pressure Stress	34	34	34	34

Inplane Stress Summary	54	10	192	-84
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Vessel Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress normal to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	1	1	1	1

Shear Stress Summary	2	2	1	1
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Vessel Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Two * Max Shear Stress	72	64	192	123

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Membrane (P)	4	4	4	4
Outplane Bending (P)	22	-22	22	-22
Outplane Membrane (Mc)	-8	-8	8	8
Outplane Bending (Mc)	-78	78	78	-78
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	34	34	34	34

Outplane Stress Summary	-25	86	147	-52
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Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses parallel to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside
Inplane Membrane (P)	8	8	8	8
Inplane Bending (P)	80	-80	80	-80
Inplane Membrane (Mc)	-18	-18	18	18
Inplane Bending (Mc)	-261	261	261	-261
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	68	69	68	69

Inplane Stress Summary -122 239 437 -244

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE
 (Shear stress normal to
 circumferential plane)

	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Torsional Shear (Mt)	1	1	1	1

Shear Stress Summary 1 1 2 2

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE
 (Stress Intensities)

	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress	122	239	437	244

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	36	36	36	36	36	36	36	36
Circ. Pl (SUS)	-19	-19	33	33	-5	-5	9	9
Circ. Q (SUS)	-18	18	48	-48	-72	72	124	-124
Long. Pm (SUS)	18	18	18	18	18	18	18	18
Long. Pl (SUS)	-3	-3	7	7	-10	-10	24	24
Long. Q (SUS)	-32	32	84	-84	-50	50	80	-80
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	3	3	3	3	3	3	3	3
Pm (SUS)	36.0	36.0	36.0	36.0	36.0	36.0	36.0	36.0
Pm+Pl (SUS)	17.0	17.0	69.0	69.0	31.0	31.0	45.0	45.0
Pm+Pl+Q (Total)	17.5	47.7	118.0	80.2	44.5	103.2	169.2	79.2

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	36.00	156.40	Passed
Pm+Pl (SUS)	69.00	234.60	Passed
Pm+Pl+Q (TOTAL)	169.19	469.20	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Reinforcing Pad Edge (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	68	69	68	69	68	69	68	69
Circ. Pl (SUS)	0	0	8	8	-10	-10	26	26
Circ. Q (SUS)	5	-5	39	-39	-181	181	341	-341

Long. Pm (SUS)	34	34	34	34	34	34	34	34
Long. Pl (SUS)	-2	-2	18	18	-4	-4	12	12
Long. Q (SUS)	22	-22	138	-138	-56	56	100	-100
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	1	1	1	1	1	1	1	1
Pm (SUS)	68.0	69.0	68.0	69.0	68.0	69.0	68.0	69.0
Pm+Pl (SUS)	68.0	69.0	76.0	77.0	58.0	59.0	94.0	95.0
Pm+Pl+Q (Total)	73.1	64.0	190.0	124.0	123.0	240.0	435.0	246.0

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	69.00	156.40	Passed
Pm+Pl (SUS)	95.00	234.60	Passed
Pm+Pl+Q (TOTAL)	435.00	469.20	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	11	11	11	11	11	11	11	11
Circ. Pl (SUS)	-19	-19	33	33	-10	-10	24	24
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	5	5	5	5	5	5	5	5
Long. Pl (SUS)	-9	-9	11	11	-9	-9	11	11
Long. Q (SUS)	-102	102	278	-278	-244	244	420	-420
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	6	6	6	6	6	6	6	6
Pm (SUS)	11.0	11.0	11.0	11.0	11.0	11.0	11.0	11.0
Pm+Pl (SUS)	8.2	8.2	44.0	44.0	5.4	5.4	35.1	35.1
Pm+Pl+Q (Total)	106.5	106.9	294.1	306.2	249.2	240.1	436.1	439.2

Nozzle Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	11.00	156.40	Passed
Pm+Pl (SUS)	44.04	234.60	Passed
Pm+Pl+Q (TOTAL)	439.22	469.20	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.



Input, Nozzle Desc: K 6"

From: 200

Pressure for Reinforcement Calculations	P	0.3880	MPa
Temperature for Internal Pressure	Temp	130	°C
Design External Pressure	Pext	0.05	MPa
Temperature for External Pressure	Tempex	130	°C

Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	156.40	MPa
Parent Allowable Stress At Ambient	Sva	175.00	MPa

Inside Diameter of Cylindrical Shell	D	7100.00	mm
Design Length of Section	L	2499.8989	mm
Shell Finished (Minimum) Thickness	t	22.0000	mm
Shell Internal Corrosion Allowance	cas	6.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		34450.05	mm
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User Entered Minimum Design Metal Temperature		-12.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	+IT
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	143.60	MPa
Allowable Stress At Ambient	Sna	161.00	MPa

Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		6.0000	in.

Size and Thickness Basis		Minimum	
Nominal Thickness		160	

Flange Material		SA-105	+IT
Flange Type		Weld Neck Flange	

Corrosion Allowance	can	3.0000	mm
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Outside Projection	ho	210.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	16.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	16.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Pad Material		SA-516 70	+IT
Pad Allowable Stress at Temperature	Sp	156.40	MPa
Pad Allowable Stress At Ambient	Spa	175.00	MPa
Diameter of Pad along vessel surface	Dp	322.4000	mm
Thickness of Pad	te	18.0000	mm
Weld leg size between Pad and Shell	Wp	16.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	18.0000	mm
Reinforcing Pad Width		77.0625	mm
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.

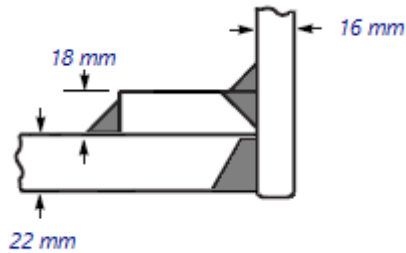
Flange Class		150	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: K 6"

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation		5.368	in.
Actual Thickness Used in Calculation		0.628	in.



Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 142.36 \cdot (\exp(0.388/(143.6 \cdot 1.0)) - 1) + 3.0 + 0.0$$

$$= 0.1926 + 3.0 + 0.0 = 3.1926 \text{ mm Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 143.6 \cdot 1.0 \cdot \ln((2 \cdot 12.958 + 142.36) / 142.36) - 0.0$$

$$= 24.016 - 0.0 = 24.0158 \text{ MPa}$$

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c :	9.220 mm
Min. required shell thickness, internal p + c :	14.833 mm
Min. required shell thickness, external p + c :	22.592 mm

Nozzle neck minimum required thickness	: 9.220 mm
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Minimum nozzle neck thickness 15.951 mm , is greater than required thickness 9.220 mm

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 22.3,	Allowable	: 143.6 MPa	Passed
Expansion	: 0.0,	Allowable	: 358.5 MPa	Passed
Occasional	: 1.0,	Allowable	: 191.0 MPa	Passed
Shear	: 14.6,	Allowable	: 100.5 MPa	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

$$= \min[S_n/S, 1]$$

$$= \min[143.6/156.4, 1]$$

$$= 0.918$$

Pad Material Factor [frp]:

$$= \min[S_p/S, 1]$$

$$= \min[156.4/156.4, 1]$$

$$= 1.000$$

Thickness of Nozzle [tn]:

$$= \text{thickness} - \text{corrosion allowance}$$

$$= 15.958 - 3.0$$

$$= 12.958 \text{ mm}$$

Shell Diameter to Thickness ratio [D/t]:

$$= D_i/t$$

$$= 7112.0/16.0$$

$$= 444.500 \text{ must be less than 400; Calculation not possible.}$$

Nozzle Calculations per Section 4.5: External Pressure Case:



Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Nozzle Neck to Pad Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Nozzle Neck to Pad Weld for Reinf. pad (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Shell to Pad Weld at Pad OD for pad (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Gov. MDMT of the Nozzle : -12 °C
Gov. MDMT of the Reinforcement Pad : -12 °C

Flange MDMT including temperature reduction per 3.11:
Flange MDMT with Temp reduction -104 °C

The Drop for this Nozzle is : 0.9972 mm
The Cut Length for this Nozzle is, Drop + Ho + H + T : 232.9970 mm

Input Echo, WRC297 Item 1, Description: K 6" :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	6.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	7100.000	mm
Shell Thickness	22.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc 175.00	MPa
Vessel Hot S.I. Allowable	Smh 156.40	MPa

Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	3.0000	mm
Nozzle Diameter	136.360	mm
Nozzle Thickness	15.9576	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-106 B	
Nozzle Cold S.I. Allowable	SNmc 161.00	MPa
Nozzle Hot S.I. Allowable	SNmh 143.60	MPa

Diameter of Reinforcing Pad	322.400	mm
Thickness of Reinforcing Pad	18.0000	mm

Note:
External Forces and Moments in WRC 107/537 Convention:
These loads are assumed to be SUStained loads.

Design Internal Pressure	Dp	0.39	MPa
Radial Load	P	11250.01	N
Circumferential Shear	Vc	7500.00	N
Longitudinal Shear	Vl	7500.00	N
Circumferential Moment	Mc	0.315E+07	N-mm
Longitudinal Moment	Ml	0.315E+07	N-mm
Torsional Moment	Mt	0.450E+07	N-mm

Include Axial Pressure Thrust No

Include Pressure Stress Indices per Div. 2 No
 Local Loads applied at end of Nozzle/Attachment No

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$
 $= 168.275 + 2 \cdot 1.65 \cdot \sqrt{(3573.0(22.0 - 6.0))}$
 $= 957.299 \text{ mm}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter (D) = 7146.000 mm
 Nozzle Outside Diameter (d) = 168.275 mm
 Vessel Thickness used (T) = 34.000 mm
 Nozzle Thickness used (t) = 12.958 mm
 $T / t = 2.624$
 $d / t = 12.987$

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 0.341$$

Nr/P = 0.209
 Mr/P = 0.207
 M0/P = 0.172
 N0/P = 0.212
 MrD/Mc = 0.206
 NrDL/Mc = 0.180
 M0d/Mc = 0.204
 N0DL/Mc = -0.126
 MrD/Ml = 0.200
 NrDL/Ml = 0.179
 M0D/Ml = 0.196
 N0DL/Ml = -0.060

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	-2	-2	-2	-2
Outplane Bending (P)	-10	10	-10	10
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	-18	18	18	-18
Normal Pressure Stress	40	40	40	40
Outplane Stress Summary	10	68	46	28

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	-2	-2	-2	-2
Inplane Bending (P)	-12	12	-12	12
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-2	-2	2	2
Inplane Bending (ML)	-19	19	19	-19
Inplane Pressure Stress	20	20	20	20
Inplane Stress Summary	-16	46	28	13

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
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Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	2	2	2	2
Shear Stress Summary	3	3	2	2

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	27	69	46	29

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	-2	-2	-2	-2
Outplane Bending (P)	-10	10	-10	10
Outplane Membrane (Mc)	2	2	-2	-2
Outplane Bending (Mc)	-19	19	19	-19
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	20	20	20	20
Outplane Stress Summary	-9	49	25	6

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Inplane Membrane (P)	-2	-2	-2	-2
Inplane Bending (P)	-12	12	-12	12
Inplane Membrane (Mc)	-2	-2	2	2
Inplane Bending (Mc)	-19	19	19	-19
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	40	40	40	40
Inplane Stress Summary	3	67	49	33

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Torsional Shear (Mt)	2	2	2	2
Shear Stress Summary	2	2	3	3

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress	13	68	49	34

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Hoop Membrane (P)	-2	-2	-2	-2
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0



Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	2	2	2	2
Hoop Stress Summary	1	1	0	0

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Axial Membrane (P)	-1	-1	-1	-1
Axial Bending (P)	-41	41	-41	41
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-13	-13	13	13
Axial Bending (ML)	-73	73	73	-73
Axial Pressure Stress	1	1	1	1
Axial Stress Summary	-129	100	45	-19

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	2	2	-2	-2
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	7	7	7	7
Shear Stress Summary	9	9	5	5

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	132	101	47	22

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	-2	-2	-2	-2
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	2	2	-2	-2
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	2	2	2	2
Hoop Stress Summary	2	2	-1	-1

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	-1	-1	-1	-1
Axial Bending (P)	-41	41	-41	41
Axial Membrane (Mc)	-13	-13	13	13
Axial Bending (Mc)	-77	77	77	-77
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	1	1	1	1
Axial Stress Summary	-133	104	49	-23

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
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Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-2	-2	2	2
Shear due to Torsion	7	7	7	7

Shear Stress Summary	5	5	9	9
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Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stress Intensities	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	135	104	55	29

Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$$

$$= 168.275 + 2 \cdot 1.65 \cdot \sqrt{(3564.0(22.0 - 6.0))}$$

$$= 956.305 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	7128.000 mm
Nozzle Outside Diameter	(d) =	322.400 mm
Vessel Thickness used	(T) =	16.000 mm
Nozzle Thickness used	(t) =	90.020 mm
T / t	=	0.178
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 0.955$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

Nr/P	=	0.116
Mr/P	=	0.235
M0/P	=	0.068
N0/P	=	0.111
MrD/Mc	=	0.313
NrDL/Mc	=	0.071
M0d/Mc	=	0.095
N0DL/Mc	=	0.042
MrD/Ml	=	0.200
NrDL/Ml	=	0.099
M0D/Ml	=	0.062
N0DL/Ml	=	0.097

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	-4	-4	-4	-4
Outplane Bending (P)	-17	17	-17	17
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (Ml)	-3	-3	3	3
Outplane Bending (Ml)	-14	14	14	-14
Normal Pressure Stress	86	86	86	86
Outplane Stress Summary	45	109	81	88

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	-5	-5	-5	-5
Inplane Bending (P)	-61	61	-61	61
Inplane Membrane (Mc)	0	0	0	0



Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-3	-3	3	3
Inplane Bending (ML)	-45	45	45	-45
Inplane Pressure Stress	43	43	43	43

Inplane Stress Summary	-73	141	25	57
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Vessel Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	B1
(Shear stress normal to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	1	1	1	1

Shear Stress Summary	2	2	0	0
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Vessel Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	B1
(Stress Intensities)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Two * Max Shear Stress	119	142	81	88
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Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	D1
(Stresses Normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Outplane Membrane (P)	-4	-4	-4	-4
Outplane Bending (P)	-17	17	-17	17
Outplane Membrane (Mc)	-1	-1	1	1
Outplane Bending (Mc)	-21	21	21	-21
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	43	43	43	43

Outplane Stress Summary	-3	76	43	35
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Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	D1
(Stresses parallel to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Inplane Membrane (P)	-5	-5	-5	-5
Inplane Bending (P)	-61	61	-61	61
Inplane Membrane (Mc)	-2	-2	2	2
Inplane Bending (Mc)	-71	71	71	-71
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	86	86	86	86

Inplane Stress Summary	-55	212	93	74
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Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	D1
(Shear stress normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Torsional Shear (Mt)	1	1	1	1

Shear Stress Summary	0	0	2	2
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Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	D1
(Stress Intensities)	Left Outside	Left Inside	Right Outside	Right Inside



Two * Max Shear Stress 55 212 93 74

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	40	40	40	40	40	40	40	40
Circ. Pl (SUS)	-2	-2	-2	-2	-4	-4	0	0
Circ. Q (SUS)	-28	28	8	-8	-31	31	7	-7
Long. Pm (SUS)	20	20	20	20	20	20	20	20
Long. Pl (SUS)	-4	-4	0	0	0	0	-4	-4
Long. Q (SUS)	-31	31	7	-7	-29	29	9	-9
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	2	2	2	2	2	2	2	2
Pm (SUS)	40.0	40.0	40.0	40.0	40.0	40.0	40.0	40.0
Pm+Pl (SUS)	38.0	38.0	38.0	38.0	36.0	36.0	40.0	40.0
Pm+Pl+Q (Total)	25.3	66.2	46.2	30.2	14.6	67.2	47.2	33.2

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	40.00	156.40	Passed
Pm+Pl (SUS)	40.00	234.60	Passed
Pm+Pl+Q (TOTAL)	67.22	469.20	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Reinforcing Pad Edge (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	86	86	86	86	86	86	86	86
Circ. Pl (SUS)	-7	-7	-1	-1	-7	-7	-3	-3
Circ. Q (SUS)	-31	31	-3	3	-132	132	10	-10
Long. Pm (SUS)	43	43	43	43	43	43	43	43
Long. Pl (SUS)	-8	-8	-2	-2	-5	-5	-3	-3
Long. Q (SUS)	-106	106	-16	16	-38	38	4	-4
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	1	1	1	1	1	1	1	1
Pm (SUS)	86.0	86.0	86.0	86.0	86.0	86.0	86.0	86.0
Pm+Pl (SUS)	79.0	79.0	85.0	85.0	79.0	79.0	83.0	83.0
Pm+Pl+Q (Total)	119.0	141.0	82.0	88.0	53.0	211.0	93.0	73.0

Vessel Stress Summation Comparison (MPa):

Type of	Max. S.I.	S.I. Allowable	Result
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Stress Int.			
Pm (SUS)	86.00	156.40	Passed
Pm+Pl (SUS)	85.00	234.60	Passed
Pm+Pl+Q (TOTAL)	211.01	469.20	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	2	2	2	2	2	2	2	2
Circ. Pl (SUS)	-2	-2	-2	-2	0	0	-4	-4
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	1	1	1	1	1	1	1	1
Long. Pl (SUS)	-14	-14	12	12	-14	-14	12	12
Long. Q (SUS)	-114	114	32	-32	-118	118	36	-36
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	2	2	-2	-2	-2	-2	2	2
Shear Q (SUS)	7	7	7	7	7	7	7	7
Pm (SUS)	2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0
Pm+Pl (SUS)	13.6	13.6	13.6	13.6	15.5	15.5	15.5	15.5
Pm+Pl+Q (Total)	128.3	102.6	46.1	21.5	133.4	105.2	54.1	27.7

Nozzle Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	2.00	143.60	Passed
Pm+Pl (SUS)	15.52	215.40	Passed
Pm+Pl+Q (TOTAL)	133.38	430.80	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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Input, Nozzle Desc: I 28"

From: 210

Pressure for Reinforcement Calculations	P	0.3880	MPa
Temperature for Internal Pressure	Temp	130	°C
Design External Pressure	Pext	0.05	MPa
Temperature for External Pressure	Tempex	130	°C

Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	156.40	MPa
Parent Allowable Stress At Ambient	Sva	175.00	MPa

Inside Diameter of Cylindrical Shell	D	7100.00	mm
Design Length of Section	L	2500.0049	mm
Shell Finished (Minimum) Thickness	t	24.0000	mm
Shell Internal Corrosion Allowance	cas	6.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		36450.05	mm
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User Entered Minimum Design Metal Temperature		-12.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-516 70	+IT
Material UNS Number		K02700	
Material Specification/Type		Plate	
Allowable Stress at Temperature	Sn	156.40	MPa
Allowable Stress At Ambient	Sna	175.00	MPa

Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		28.0000	in.

Size and Thickness Basis		Actual	
Actual Thickness	tn	22.0000	mm

Flange Material		SA-105	+IT
Flange Type		Weld Neck	Flange

Corrosion Allowance	can	6.0000	mm
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Outside Projection	ho	300.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	22.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Pad Material		SA-516 70	+IT
Pad Allowable Stress at Temperature	Sp	156.40	MPa
Pad Allowable Stress At Ambient	Spa	175.00	MPa
Diameter of Pad along vessel surface	Dp	1395.2000	mm
Thickness of Pad	te	22.0000	mm
Weld leg size between Pad and Shell	Wp	18.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	22.0000	mm
Reinforcing Pad Width		320.0000	mm
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.

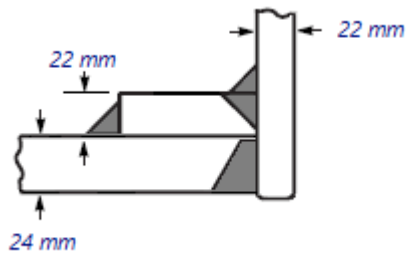
Flange Class		150
Flange Grade		GR 1.1
Flange Series		B

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: I 28"

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	28.000	in.
Actual Thickness Used in Calculation	0.866	in.



Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:
 $= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$
 $= 0.5 \cdot 723.2 \cdot (\exp(0.388/(156.4 \cdot 1.0)) - 1) + 6.0 + 0.0$
 $= 0.8982 + 6.0 + 0.0 = 6.8982 \text{ mm Equation 4.3.1}$

Computed Maximum Allowable Working Pressure [MAWP]:
 $= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$
 $= 156.4 \cdot 1.0 \cdot \ln((2 \cdot 16.0 + 723.2) / 723.2) - 0.0$
 $= 6.772 - 0.0 = 6.7716 \text{ MPa}$

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c :	14.340 mm
Min. required shell thickness, internal p + c :	14.833 mm
Min. required shell thickness, external p + c :	22.592 mm

Nozzle neck minimum required thickness	: 14.340 mm
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Minimum nozzle neck thickness 21.996 mm , is greater than required thickness 14.340 mm

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	17.3,	Allowable	:	156.4 MPa	Passed
Expansion	:	0.0,	Allowable	:	397.0 MPa	Passed
Occasional	:	4.3,	Allowable	:	208.0 MPa	Passed
Shear	:	10.2,	Allowable	:	109.5 MPa	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:
 $= \min[S_n/S, 1]$
 $= \min[156.4/156.4, 1]$
 $= 1.000$

Pad Material Factor [frp]:
 $= \min[S_p/S, 1]$
 $= \min[156.4/156.4, 1]$
 $= 1.000$

Thickness of Nozzle [tn]:
 $= \text{thickness} - \text{corrosion allowance}$
 $= 22.0 - 6.0$
 $= 16.000 \text{ mm}$

Shell Diameter to Thickness ratio [D/t]:
 $= D_i/t$
 $= 7112.0/18.0$
 $= 395.111 \text{ must be less than 400.}$



Effective Pressure Radius [Reff]:

$$\begin{aligned} &= Di/2 + \text{corrosion allowance} \\ &= 7100.0/2 + 6.0 \\ &= 3556.000 \text{ mm} \end{aligned}$$

Effective Length of Vessel Wall [LR]:

For Nozzles with reinforcing pads, Lr is min[LR1, LR2, LR3]:

Limit [LR1]:

$$\begin{aligned} &= \sqrt{(Reff \cdot t) + W} \\ &= \sqrt{(3556.0 \cdot 18.0) + 320.0} \\ &= 572.998 \text{ mm} \end{aligned}$$

Limit [LR2]:

$$\begin{aligned} &= \sqrt{((Reff + t) \cdot (t + te))} \\ &= \sqrt{((3556.0 + 18.0) \cdot (18.0 + 22.0))} \\ &= 378.101 \text{ mm} \end{aligned}$$

Limit [LR3]:

$$\begin{aligned} &= 2 \cdot Rn \\ &= 2 \cdot 361.6 \\ &= 723.200 \text{ mm} \end{aligned}$$

Final value of [LR]:

$$\begin{aligned} &= \min(LR1, LR2, LR3) \\ &= \min(572.998, 378.101, 723.2) \\ &= 378.101 \text{ mm} \end{aligned}$$

Thickness Limit Candidate [LH1]:

$$\begin{aligned} &= \min[1.5t, te] + \sqrt{(Rn \cdot tn)} \\ &= \min[1.5 \cdot 18.0, 22.0] + \sqrt{(361.6 \cdot 16.0)} \\ &= 98.063 \text{ mm} \end{aligned}$$

Thickness Limit Candidate [LH2]:

$$\begin{aligned} &= Lpr1 \\ &= 300.0 \\ &= 300.000 \text{ mm} \end{aligned}$$

Thickness Limit Candidate [LH3]:

$$\begin{aligned} &= 8(t + te) \\ &= 8(18.0 + 22.0) \\ &= 320.000 \text{ mm} \end{aligned}$$

Effective Nozzle Wall Length Outside the Vessel [LH]:

$$\begin{aligned} &= \min[LH1, LH2, LH3] + t \quad \text{for set-in nozzles} \\ &= \min[98.063, 300.0, 320.0] + 18.0 \\ &= 116.063 \text{ mm} \end{aligned}$$

Effective Vessel Thickness [teff]:

$$\begin{aligned} &= t + ((A5 \cdot frp) / LR) \\ &= 18.0 + ((7040.0 \cdot 1.0) / 378.101) \\ &= 36.619 \text{ mm} \end{aligned}$$

Parameter [λ]:

$$\begin{aligned} &= \min(12, (2 \cdot Rn + tn) / \sqrt{((Di + teff) \cdot teff)}) \\ &= \min(12, (2 \cdot 361.6 + 16.0) / \sqrt{((7112.0 + 36.619) \cdot 36.619)}) \\ &= 1.445 \end{aligned}$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$\begin{aligned} &= t \cdot LR \cdot \max((\lambda/5)^{0.85}, 1) \\ &= 18.0 \cdot 378.101 \cdot \max((1.445/5)^{0.85}, 1) \\ &= 6805.809 \text{ mm}^2 \end{aligned}$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$\begin{aligned} &= tn \cdot LH \\ &= 16.0 \cdot 116.063 \\ &= 1857.010 \text{ mm}^2 \end{aligned}$$



Area Contributed by the Pad Fillet Weld [A42]:

$$\begin{aligned} &= 0.5 \cdot \text{Leg42}^2 \\ &= 0.5 \cdot 18.0^2 \\ &= 162.000 \text{ mm}^2 \end{aligned}$$

Area Contributed by the Outside Fillet Weld [A41]:

$$\begin{aligned} &= 0.5 \cdot \text{Leg41}^2 \\ &= 0.5 \cdot 10.0^2 \\ &= 50.000 \text{ mm}^2 \end{aligned}$$

Area Contributed by the Reinforcing Pad [A5]:

$$\begin{aligned} &= \min(W \cdot t_e, LR \cdot t_e) \\ &= \min(320.0 \cdot 22.0, 378.101 \cdot 22.0) \\ &= 7040.000 \text{ mm}^2 \end{aligned}$$

The total area contributed by A1 through A5 [AT]:

$$\begin{aligned} &= A1 + \text{frn}(A2 + A3) + A41 + A42 + A43 + \text{frp}(A5) \\ &= 6805.809 + 1.0(1857.01 + 0.0) + 50.0 + 162.0 + 0.0 + 1.0(7040.0) \\ &= 15914.819 \text{ mm}^2 \end{aligned}$$

Nozzle Radius for Force Calculation [Rxn]:

$$\begin{aligned} &= t_n / \ln[1 + t_n/R_n] \\ &= 16.0 / \ln[1 + 16.0/361.6] \\ &= 369.543 \text{ mm} \end{aligned}$$

Shell Radius for Force Calculation [Rxs]:

$$\begin{aligned} &= t_{eff} / \ln[1 + t_{eff}/R_{eff}] \\ &= 36.619 / \ln[1 + 36.619/3556.0] \\ &= 3574.286 \text{ mm} \end{aligned}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$\begin{aligned} &= 1.5 \cdot S \cdot E \\ &= 1.5 \cdot 156.4 \cdot 1.0 \\ &= 234.6 \text{ MPa} \end{aligned}$$

Force acting on the Nozzle [fN]:

$$\begin{aligned} &= P \cdot R_{xn} \cdot L_H \\ &= 0.388 \cdot 369.543 \cdot 116.063 \\ &= 16640.0 \text{ N} \end{aligned}$$

Force acting on the Shell [fS]:

$$\begin{aligned} &= P \cdot R_{xs} (LR + t_n) \\ &= 0.388 \cdot 3574.286 (378.101 + 16.0) \\ &= 546501.6 \text{ N} \end{aligned}$$

Discontinuity Force from Internal Pressure [fY]:

$$\begin{aligned} &= P \cdot R_{xs} \cdot R_{nc} \\ &= 0.388 \cdot 3574.286 \cdot 361.6 \\ &= 501432.9 \text{ N} \end{aligned}$$

Area Resisting Internal Pressure [Ap]:

$$\begin{aligned} &= (f_S + f_Y + f_N) / P \\ &= (546502 + 501433 + 16640.0) / 0.388 \\ &= 2743980.0 \text{ mm}^2 \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$\begin{aligned} &= S_{allow} / (2 \cdot A_p / AT - R_{xs} / t_{eff}) \\ &= 234.6 / (2 \cdot 2743980 / 15914.819 - 3574.286 / 36.619) \\ &= 0.9 \text{ MPa} \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$\begin{aligned} &= S / [t / R_{xs}] \\ &= 156.4 [18.0 / 3574.286] \\ &= 0.8 \text{ MPa} \end{aligned}$$

Maximum Allowable Working Pressure [Pmax]:

$$\begin{aligned} &= \min(P_{max1}, P_{max2}) \\ &= \min(0.949, 0.788) \\ &= 0.788 \text{ MPa} \end{aligned}$$



Average Primary Membrane Stress [σ_{Avg}]:

$$\begin{aligned} &= (fN + fS + fY) / AT \\ &= (16640.021 + 546502 + 501433) / 15914.819 \\ &= 66.898 \text{ MPa} \end{aligned}$$

General Primary Membrane Stress [σ_{Circ}]:

$$\begin{aligned} &= P \cdot Rxs / teff \\ &= 0.388 \cdot 3574.286 / 36.619 \\ &= 37.9 \text{ MPa} \end{aligned}$$

Maximum Local Primary Membrane Stress [PL]:

$$\begin{aligned} &= \max(2 \cdot \sigma_{Avg} - \sigma_{Circ}, \sigma_{Circ}) \\ &= \max(2 \cdot 66.898 - 37.871, 37.871) \\ &= 95.9 \text{ MPa} \end{aligned}$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	234.60	MPa
Local Primary Membrane Stress	PL	95.92	MPa
Maximum Allowable Working Pressure	Pmax	0.79	MPa

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$\begin{aligned} &= (Rnc + tn) / Rnc \\ &= (361.6 + 16.0) / 361.6 \\ &= 1.044 \text{ For set-in Nozzles} \end{aligned}$$

Weld Length of Nozzle to Shell Weld [LT]:

$$\begin{aligned} &= \pi/2 \cdot (Rn + tn) \\ &= \pi/2 \cdot (361.6 + 16.0) \\ &= 593.133 \text{ mm} \end{aligned}$$

Weld Length of Pad to Shell Weld [LTP]:

$$\begin{aligned} &= \pi/2 \cdot (Rn + tn + W) \\ &= \pi/2 \cdot (361.6 + 16.0 + 320.0) \\ &= 1095.788 \text{ mm} \end{aligned}$$

Weld Throat Dimensions, (0.7071•Leg Dimensions) [L41T, L42T, L43T]:

$$= 7.071, 12.728, 0.000, \text{ mm}$$

Required Weld Throat Dimensions :

$$\begin{aligned} &tc, L41T = 7.071, \text{ must be } \geq \min(0.7 \cdot 16.000, 6.000) \text{ Passed} \\ &tf1, L42T = 12.728, \text{ must be } \geq \min(0.6 \cdot 22.000, 0.6 \cdot 18.000) \text{ Passed} \end{aligned}$$

Weld Load Value [fwelds]:

$$\begin{aligned} &= \min(fy \cdot ky, 1.5 \cdot Sn(A2 + A3), \pi/4 \cdot P \cdot Rn^2 \cdot ky^2) \\ &= \min(501433 \cdot 1.04, 1.5 \cdot 156.4(1857.01 + 0.0), \pi/4 \cdot 0.4 \cdot 361.6^2 \cdot 1.04^2) \\ &= 43445.910 \text{ N} \end{aligned}$$

Discontinuity Force [fws]:

$$\begin{aligned} &= fwelds \cdot t \cdot S / (t \cdot S + te \cdot Sp) \\ &= 43445.9 \cdot 18.0 \cdot 156 / (18.0 \cdot 156 + 22.0 \cdot 156) \\ &= 19550.658 \text{ N} \end{aligned}$$

Discontinuity Force [fwp]:

$$\begin{aligned} &= fwelds \cdot te \cdot Sp / (t \cdot S + te \cdot Sp) \\ &= 43445.9 \cdot 22.0 \cdot 156 / (18.0 \cdot 156 + 22.0 \cdot 156) \\ &= 23895.248 \text{ N} \end{aligned}$$

Shear Stress [τ_1]:

$$\begin{aligned} &= fws / (LT \cdot (0.6 \cdot tw1 + 0.49 \cdot L43T)) \\ &= 19550.658 / (593.133 \cdot (0.6 \cdot 18.0 + 0.49 \cdot 0.0)) \\ &= 3.052 \text{ MPa} \end{aligned}$$

Shear Stress [τ_2]:

$$\begin{aligned} &= fwp / (LT \cdot (0.6 \cdot tw2 + 0.49 \cdot L41T)) \\ &= 23895.248 / (593.133 \cdot (0.6 \cdot 22.0 + 0.49 \cdot 7.071)) \\ &= 2.418 \text{ MPa} \end{aligned}$$



Shear Stress [τ_3]:

$$\begin{aligned} &= fwp / (LTP \cdot (0.49 \cdot L42T)) \\ &= 23895.248 / (1095.788 \cdot (0.49 \cdot 12.728)) \\ &= 3.497 \text{ MPa} \end{aligned}$$

Maximum Shear Stress in the Welds:

$$\begin{aligned} &= \max(\tau_1, \tau_2, \tau_3) \\ &= \max(3.052, 2.418, 3.497) \\ &= 3.5 \text{ must be less than or equal to } 156.4 \text{ MPa} \end{aligned}$$

Nozzle Calculations per Section 4.5: External Pressure Case:

Nozzle Radius for Force Calculation [R_{xn}]:

$$\begin{aligned} &= t_n / \ln[1 + t_n/R_n] \\ &= 16.0 / \ln[1 + 16.0/361.6] \\ &= 369.543 \text{ mm} \end{aligned}$$

Shell Radius for Force Calculation [R_{xs}]:

$$\begin{aligned} &= t_{eff} / \ln[1 + t_{eff}/R_{eff}] \\ &= 36.619 / \ln[1 + 36.619/3556.0] \\ &= 3574.286 \text{ mm} \end{aligned}$$

Allowable Local Primary Membrane Stress [S_{allow}]:

$$\begin{aligned} &= 1.5 \cdot S \\ &= 1.5 \cdot 156.4 \\ &= 234.6 \text{ MPa} \end{aligned}$$

Force acting on the Nozzle [f_N]:

$$\begin{aligned} &= P \cdot R_{xn} \cdot L_H \\ &= 0.052 \cdot 369.543 \cdot 116.063 \\ &= 2218.7 \text{ N} \end{aligned}$$

Force acting on the Shell [f_S]:

$$\begin{aligned} &= P \cdot R_{xs} (L_R + t_n) \\ &= 0.052 \cdot 3574.286 (378.101 + 16.0) \\ &= 72866.4 \text{ N} \end{aligned}$$

Discontinuity Force from External Pressure [f_Y]:

$$\begin{aligned} &= P \cdot R_{xs} \cdot R_{nc} \\ &= 0.052 \cdot 3574.286 \cdot 361.6 \\ &= 66857.2 \text{ N} \end{aligned}$$

Area Resisting External Pressure [A_p]:

$$\begin{aligned} &= (f_S + f_Y + f_N) / P \\ &= (72866.4 + 66857.2 + 2218.7) / 0.052 \\ &= 2743980.0 \text{ mm}^2 \end{aligned}$$

Maximum Allowable Working Pressure Candidate [P_{max1}]:

$$\begin{aligned} &= S_{allow} / (2 \cdot A_p / AT - R_{xs} / t_{eff}) \\ &= 234.6 / (2 \cdot 2743980 / 15914.819 - 3574.286 / 36.619) \\ &= 0.9 \text{ MPa} \end{aligned}$$

Maximum Allowable Working Pressure Candidate [P_{max2}]:

$$\begin{aligned} &= S / [t / R_{xs}] \\ &= 156.4 [18.0 / 3574.286] \\ &= 0.8 \text{ MPa} \end{aligned}$$

Maximum Allowable Working Pressure [P_{max}]:

$$\begin{aligned} &= \min(P_{max1}, P_{max2}) \\ &= \min(0.949, 0.788) \\ &= 0.788 \text{ MPa} \end{aligned}$$

Average Primary Membrane Stress [σ_{Avg}]:

$$\begin{aligned} &= (f_N + f_S + f_Y) / AT \\ &= (2218.654 + 72866.359 + 66857.242) / 15914.819 \\ &= 8.920 \text{ MPa} \end{aligned}$$

General Primary Membrane Stress [σ_{Circ}]:

$$\begin{aligned} &= P \cdot R_{xs} / t_{eff} \\ &= 0.052 \cdot 3574.286 / 36.619 \end{aligned}$$



= 5.0 MPa

Maximum Local Primary Membrane Stress [PL]:

= max($2 \cdot \sigma_{Avg} - \sigma_{Circ}$, σ_{Circ})

= max($2 \cdot 8.92 - 5.049$, 5.049)

= 12.8 MPa

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	234.60	MPa
Local Primary Membrane Stress	PL	12.79	MPa
Maximum Allowable External Pressure	Pmax	0.79	MPa

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Nozzle Neck to Pad Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Nozzle Neck to Pad Weld for Reinf. pad (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Shell to Pad Weld at Pad OD for pad (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Gov. MDMT of the Nozzle	:	-12 °C
Gov. MDMT of the Reinforcement Pad	:	-12 °C

Flange MDMT including temperature reduction per 3.11:

Flange MDMT with Temp reduction	-104 °C
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The Drop for this Nozzle is : 20.1391 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 344.1392 mm

Input Echo, WRC297 Item 1, Description: I 28" :

Diameter Basis for Cylindrical Shell	ID		
Shell Internal Corrosion Allowance	6.0000	mm	
Shell External Corrosion Allowance	0.0000	mm	
Shell Diameter	7100.000	mm	
Shell Thickness	24.0000	mm	
Shell Stress Concentration Factor	1.000		
Vessel Material	SA-516 70		
Vessel Cold S.I. Allowable	Smc	175.00	MPa
Vessel Hot S.I. Allowable	Smh	156.40	MPa
Diameter Basis for Nozzle	ID		
Nozzle Corrosion Allowance	6.0000	mm	
Nozzle Diameter	711.200	mm	
Nozzle Thickness	22.0000	mm	
Nozzle Stress Concentration Factor	1.000		
Nozzle Material	SA-516 70		
Nozzle Cold S.I. Allowable	SNmc	175.00	MPa
Nozzle Hot S.I. Allowable	SNmh	156.40	MPa

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Diameter of Reinforcing Pad	1395.200	mm
Thickness of Reinforcing Pad	22.0000	mm



Note:

*External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUstained loads.*

Design Internal Pressure	Dp	0.39	MPa
Radial Load	P	-52500.19	N
Circumferential Shear	Vc	35000.02	N
Longitudinal Shear	Vl	35000.02	N
Circumferential Moment	Mc	0.686E+08	N-mm
Longitudinal Moment	Ml	0.686E+08	N-mm
Torsional Moment	Mt	0.980E+08	N-mm

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - c_a)}$
 $= 755.2 + 2 \cdot 1.65 \cdot \sqrt{3576.0(24.0 - 6.0)}$
 $= 1592.438 \text{ mm}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	7152.000 mm
Nozzle Outside Diameter	(d) =	755.200 mm
Vessel Thickness used	(T) =	40.000 mm
Nozzle Thickness used	(t) =	16.000 mm
T / t	=	2.500
d / t	=	47.200

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.412$$

Nr/P	=	0.060
Mr/P	=	0.101
M0/P	=	0.069
N0/P	=	0.193
MrD/Mc	=	0.185
NrDL/Mc	=	0.098
M0d/Mc	=	0.151
N0DL/Mc	=	0.202
MrD/Ml	=	0.126
NrDL/Ml	=	0.073
M0D/Ml	=	0.091
N0DL/Ml	=	0.335

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	6	6	6	6
Outplane Bending (P)	13	-13	13	-13
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (Ml)	-18	-18	18	18
Outplane Bending (Ml)	-30	30	30	-30
Normal Pressure Stress	34	34	34	34
Outplane Stress Summary	4	39	104	15

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	1	1	1	1



Inplane Bending (P)	19	-19	19	-19
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-4	-4	4	4
Inplane Bending (ML)	-43	43	43	-43
Inplane Pressure Stress	17	17	17	17

Inplane Stress Summary	-8	38	86	-39
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Vessel Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress normal to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	2	2	2	2

Shear Stress Summary	3	3	1	1
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Vessel Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Two * Max Shear Stress	14	42	104	55
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Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Outplane Membrane (P)	6	6	6	6
Outplane Bending (P)	13	-13	13	-13
Outplane Membrane (Mc)	-11	-11	11	11
Outplane Bending (Mc)	-51	51	51	-51
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	17	17	17	17

Outplane Stress Summary	-25	49	99	-30
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Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses parallel to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Inplane Membrane (P)	1	1	1	1
Inplane Bending (P)	19	-19	19	-19
Inplane Membrane (Mc)	-5	-5	5	5
Inplane Bending (Mc)	-63	63	63	-63
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	34	34	34	34

Inplane Stress Summary	-12	74	124	-40
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Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Torsional Shear (Mt)	2	2	2	2

Shear Stress Summary	1	1	3	3
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Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
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(Stress Intensities	Left Outside	Left Inside	Right Outside	Right Inside
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Two * Max Shear Stress	26	74	125	41
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Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
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Hoop Membrane (P)	6	6	6	6
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-18	-18	18	18
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	10	10	10	10

Hoop Stress Summary	-2	-1	35	36
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Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
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Axial Membrane (P)	1	1	1	1
Axial Bending (P)	87	-87	87	-87
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-10	-10	10	10
Axial Bending (ML)	-191	191	191	-191
Axial Pressure Stress	5	5	5	5

Axial Stress Summary	-106	99	295	-262
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Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
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Shear due to (Vc)	1	1	-1	-1
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	6	6	6	6

Shear Stress Summary	8	8	4	4
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Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stress Intensities	Au Outside	Al Inside	Bu Outside	Bl Inside
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Two * Max Shear Stress	107	103	296	298
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Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
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Hoop Membrane (P)	6	6	6	6
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-11	-11	11	11
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	10	10	10	10

Hoop Stress Summary	5	5	28	28
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Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
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Axial Membrane (P)	1	1	1	1
Axial Bending (P)	87	-87	87	-87
Axial Membrane (Mc)	-10	-10	10	10
Axial Bending (Mc)	-289	289	289	-289
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	5	5	5	5
Axial Stress Summary	-205	198	394	-361

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-1	-1	1	1
Shear due to Torsion	6	6	6	6
Shear Stress Summary	4	4	8	8

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	210	198	394	389

Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$
 $= 755.2 + 2 \cdot 1.65 \cdot \sqrt{(3565.0(24.0 - 6.0))}$
 $= 1591.149 \text{ mm}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter (D)	=	7130.000 mm
Nozzle Outside Diameter (d)	=	1395.200 mm
Vessel Thickness used (T)	=	18.000 mm
Nozzle Thickness used (t)	=	336.000 mm
T / t	=	0.054
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 3.895$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

Nr/P	=	0.054
Mr/P	=	0.086
M0/P	=	0.024
N0/P	=	0.027
MrD/Mc	=	0.254
NrDL/Mc	=	0.112
M0d/Mc	=	0.076
N0DL/Mc	=	0.050
MrD/Ml	=	0.060
NrDL/Ml	=	0.068
M0D/Ml	=	0.018
N0DL/Ml	=	0.031

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	4	4	4	4
Outplane Bending (P)	23	-23	23	-23
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0



Outplane Membrane (ML)	-4	-4	4	4
Outplane Bending (ML)	-16	16	16	-16
Normal Pressure Stress	76	76	76	76
Outplane Stress Summary	82	69	125	46

Vessel Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses parallel to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Inplane Membrane (P)	8	8	8	8
Inplane Bending (P)	84	-84	84	-84
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-10	-10	10	10
Inplane Bending (ML)	-54	54	54	-54
Inplane Pressure Stress	38	38	38	38
Inplane Stress Summary	66	7	196	-81

Vessel Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress normal to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	1	1	1	1
Shear Stress Summary	2	2	0	0

Vessel Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Two * Max Shear Stress	82	69	196	127

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Membrane (P)	4	4	4	4
Outplane Bending (P)	23	-23	23	-23
Outplane Membrane (Mc)	-7	-7	7	7
Outplane Bending (Mc)	-68	68	68	-68
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	38	38	38	38
Outplane Stress Summary	-10	80	142	-42

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses parallel to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside
Inplane Membrane (P)	8	8	8	8
Inplane Bending (P)	84	-84	84	-84
Inplane Membrane (Mc)	-16	-16	16	16
Inplane Bending (Mc)	-230	230	230	-230
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	76	76	76	76
Inplane Stress Summary	-78	215	416	-212

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)			Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Shear (Vc)			0	0	0	0
Outplane Shear (Vl)			0	0	0	0
Torsional Shear (Mt)			1	1	1	1
Shear Stress Summary			0	0	2	2

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stress Intensities)			Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress			78	215	417	212

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	34	34	34	34	34	34	34	34
Circ. Pl (SUS)	-12	-12	24	24	-4	-4	6	6
Circ. Q (SUS)	-17	17	43	-43	-44	44	82	-82
Long. Pm (SUS)	17	17	17	17	17	17	17	17
Long. Pl (SUS)	-3	-3	5	5	-5	-5	17	17
Long. Q (SUS)	-24	24	62	-62	-38	38	64	-64
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	2	2	2	2	2	2	2	2
Pm (SUS)	34.0	34.0	34.0	34.0	34.0	34.0	34.0	34.0
Pm+Pl (SUS)	22.0	22.0	58.0	58.0	30.0	30.0	40.0	40.0
Pm+Pl+Q (Total)	15.5	40.6	101.2	55.1	26.3	74.2	122.2	42.3

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	34.00	156.40	Passed
Pm+Pl (SUS)	58.00	234.60	Passed
Pm+Pl+Q (TOTAL)	122.17	469.20	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Reinforcing Pad Edge (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	76	76	76	76	76	76	76	76
Circ. Pl (SUS)	0	0	8	8	-8	-8	24	24
Circ. Q (SUS)	7	-7	39	-39	-146	146	314	-314
Long. Pm (SUS)	38	38	38	38	38	38	38	38
Long. Pl (SUS)	-2	-2	18	18	-3	-3	11	11
Long. Q (SUS)	30	-30	138	-138	-45	45	91	-91

Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear P1 (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	1	1	1	1	1	1	1	1
Pm (SUS)	76.0	76.0	76.0	76.0	76.0	76.0	76.0	76.0
Pm+P1 (SUS)	76.0	76.0	84.0	84.0	68.0	68.0	100.0	100.0
Pm+P1+Q (Total)	83.1	69.0	194.0	127.0	78.0	214.0	414.0	214.0

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	76.00	156.40	Passed
Pm+P1 (SUS)	100.00	234.60	Passed
Pm+P1+Q (TOTAL)	414.00	469.20	Passed

Because only sustained loads were specified, the Pm+P1+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	10	10	10	10	10	10	10	10
Circ. P1 (SUS)	-12	-12	24	24	-5	-5	17	17
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	5	5	5	5	5	5	5	5
Long. P1 (SUS)	-9	-9	11	11	-9	-9	11	11
Long. Q (SUS)	-104	104	278	-278	-202	202	376	-376
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear P1 (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	6	6	6	6	6	6	6	6
Pm (SUS)	10.0	10.0	10.0	10.0	10.0	10.0	10.0	10.0
Pm+P1 (SUS)	4.4	4.4	34.1	34.1	9.2	9.2	27.1	27.1
Pm+P1+Q (Total)	108.5	103.0	294.1	296.2	211.2	198.1	392.1	387.3

Nozzle Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	10.00	156.40	Passed
Pm+P1 (SUS)	34.06	234.60	Passed
Pm+P1+Q (TOTAL)	392.13	469.20	Passed

Because only sustained loads were specified, the Pm+P1+Q allowable was 3 * Smh.



Input, Nozzle Desc: L1 2"

From: 220

Pressure for Reinforcement Calculations	P	0.3880	MPa
Temperature for Internal Pressure	Temp	130	°C
Design External Pressure	Pext	0.05	MPa
Temperature for External Pressure	Tempex	130	°C

Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	156.40	MPa
Parent Allowable Stress At Ambient	Sva	175.00	MPa

Inside Diameter of Cylindrical Shell	D	7100.00	mm
Design Length of Section	L	2499.9951	mm
Shell Finished (Minimum) Thickness	t	20.0000	mm
Shell Internal Corrosion Allowance	cas	6.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		39349.96	mm
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User Entered Minimum Design Metal Temperature		-12.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-105	+IT
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	MPa
Allowable Stress At Ambient	Sna	138.00	MPa

Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		2.0000	in.

Size and Thickness Basis		Actual	
Actual Thickness	tn	16.7640	mm

Flange Material		SA-105	+IT
Flange Type		Long Weld Neck	

Corrosion Allowance	can	3.0000	mm
---------------------	-----	--------	----

Outside Projection	ho	206.3750	mm
Weld leg size between Nozzle and Pad/Shell	Wo	16.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	16.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.

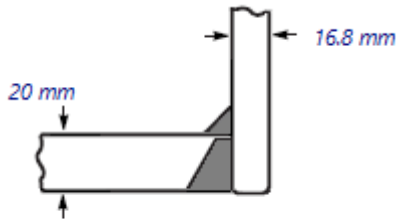
Flange Class		300
Flange Grade		GR 1.1

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: L1 2"

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation		2.000	in.
Actual Thickness Used in Calculation		0.660	in.



Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 56.8 \cdot (\exp(0.388/(138.0 \cdot 1.0)) - 1) + 3.0 + 0.0$$

$$= 0.08 + 3.0 + 0.0 = 3.08 \text{ mm Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 138.0 \cdot 1.0 \cdot \ln((2 \cdot 13.764 + 56.8) / 56.8) - 0.0$$

$$= 54.535 - 0.0 = 54.5345 \text{ MPa}$$

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c :	7.800 mm
Min. required shell thickness, internal p + c :	14.833 mm
Min. required shell thickness, external p + c :	22.589 mm

Nozzle neck minimum required thickness	: 7.800 mm
--	------------

Minimum nozzle neck thickness 16.764 mm , is greater than required thickness 7.800 mm

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

$$= \min[S_n/S, 1]$$

$$= \min[138.0/156.4, 1]$$

$$= 0.882$$

Thickness of Nozzle [tn]:

$$= \text{thickness} - \text{corrosion allowance}$$

$$= 16.764 - 3.0$$

$$= 13.764 \text{ mm}$$

Shell Diameter to Thickness ratio [D/t]:

$$= D_i/t$$

$$= 7112.0/14.0$$

$$= 508.000 \text{ must be less than 400; Calculation not possible.}$$

Nozzle Calculations per Section 4.5: External Pressure Case:

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Note: Impact testing of LWN flanges meeting requirements of UCS-66(c)(4) is not required.

The Drop for this Nozzle is : 0.2504 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 226.6253 mm

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Input, Nozzle Desc: L3 2"

From: 220

Pressure for Reinforcement Calculations	P	0.3880	MPa
Temperature for Internal Pressure	Temp	130	°C
Design External Pressure	Pext	0.05	MPa
Temperature for External Pressure	Tempex	130	°C

Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	156.40	MPa
Parent Allowable Stress At Ambient	Sva	175.00	MPa

Inside Diameter of Cylindrical Shell	D	7100.00	mm
Design Length of Section	L	2499.9951	mm
Shell Finished (Minimum) Thickness	t	20.0000	mm
Shell Internal Corrosion Allowance	cas	6.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		39349.96	mm
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User Entered Minimum Design Metal Temperature		-12.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-105	+IT
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	MPa
Allowable Stress At Ambient	Sna	138.00	MPa

Diameter Basis (for tr calc only)		Inside	
Layout Angle		30.00	deg
Diameter		2.0000	in.

Size and Thickness Basis		Actual	
Actual Thickness	tn	16.7640	mm

Flange Material		SA-105	+IT
Flange Type		Long Weld Neck	

Corrosion Allowance	can	3.0000	mm
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Outside Projection	ho	206.3750	mm
Weld leg size between Nozzle and Pad/Shell	Wo	16.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	16.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.

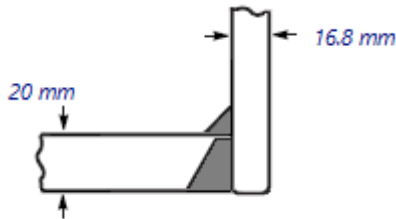
Flange Class		300
Flange Grade		GR 1.1

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: L3 2"

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation		2.000	in.
Actual Thickness Used in Calculation		0.660	in.



Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 56.8 \cdot (\exp(0.388/(138.0 \cdot 1.0)) - 1) + 3.0 + 0.0$$

$$= 0.08 + 3.0 + 0.0 = 3.08 \text{ mm Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 138.0 \cdot 1.0 \cdot \ln((2 \cdot 13.764 + 56.8) / 56.8) - 0.0$$

$$= 54.535 - 0.0 = 54.5345 \text{ MPa}$$

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c :	7.800 mm
Min. required shell thickness, internal p + c :	14.833 mm
Min. required shell thickness, external p + c :	22.589 mm

Nozzle neck minimum required thickness	: 7.800 mm
--	------------

Minimum nozzle neck thickness 16.764 mm , is greater than required thickness 7.800 mm

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

$$= \min[S_n/S, 1]$$

$$= \min[138.0/156.4, 1]$$

$$= 0.882$$

Thickness of Nozzle [tn]:

$$= \text{thickness} - \text{corrosion allowance}$$

$$= 16.764 - 3.0$$

$$= 13.764 \text{ mm}$$

Shell Diameter to Thickness ratio [D/t]:

$$= D_i/t$$

$$= 7112.0/14.0$$

$$= 508.000 \text{ must be less than 400; Calculation not possible.}$$

Nozzle Calculations per Section 4.5: External Pressure Case:

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Note: Impact testing of LWN flanges meeting requirements of UCS-66(c)(4) is not required.

The Drop for this Nozzle is : 0.2504 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 226.6253 mm

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Input, Nozzle Desc: L5 2"

From: 220

Pressure for Reinforcement Calculations	P	0.3880	MPa
Temperature for Internal Pressure	Temp	130	°C
Design External Pressure	Pext	0.05	MPa
Temperature for External Pressure	Tempex	130	°C

Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	156.40	MPa
Parent Allowable Stress At Ambient	Sva	175.00	MPa

Inside Diameter of Cylindrical Shell	D	7100.00	mm
Design Length of Section	L	2499.9951	mm
Shell Finished (Minimum) Thickness	t	20.0000	mm
Shell Internal Corrosion Allowance	cas	6.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		39349.96	mm
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User Entered Minimum Design Metal Temperature		-12.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-105	+IT
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	MPa
Allowable Stress At Ambient	Sna	138.00	MPa

Diameter Basis (for tr calc only)		Inside	
Layout Angle		60.00	deg
Diameter		2.0000	in.

Size and Thickness Basis		Actual	
Actual Thickness	tn	16.7640	mm

Flange Material		SA-105	+IT
Flange Type		Long Weld Neck	

Corrosion Allowance	can	3.0000	mm
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Outside Projection	ho	206.3750	mm
Weld leg size between Nozzle and Pad/Shell	Wo	16.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	16.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.

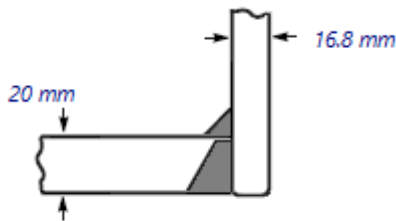
Flange Class		300
Flange Grade		GR 1.1

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: L5 2"

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation		2.000	in.
Actual Thickness Used in Calculation		0.660	in.



Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 56.8 \cdot (\exp(0.388/(138.0 \cdot 1.0)) - 1) + 3.0 + 0.0$$

$$= 0.08 + 3.0 + 0.0 = 3.08 \text{ mm Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 138.0 \cdot 1.0 \cdot \ln((2 \cdot 13.764 + 56.8) / 56.8) - 0.0$$

$$= 54.535 - 0.0 = 54.5345 \text{ MPa}$$

This nozzle or its diameter limit intersects weld seam # [6] on this element. The distance between the nozzle and seam is, 158.41 mm , while the distance from the center of the nozzle to the edge of the diameter limit is: 755.70 mm

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c :	7.800 mm
Min. required shell thickness, internal p + c	14.833 mm
Min. required shell thickness, external p + c	22.589 mm

Nozzle neck minimum required thickness	: 7.800 mm
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Minimum nozzle neck thickness 16.764 mm , is greater than required thickness 7.800 mm

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

$$= \min[S_n/S, 1]$$

$$= \min[138.0/156.4, 1]$$

$$= 0.882$$

Thickness of Nozzle [tn]:

$$= \text{thickness} - \text{corrosion allowance}$$

$$= 16.764 - 3.0$$

$$= 13.764 \text{ mm}$$

Shell Diameter to Thickness ratio [D/t]:

$$= D_i/t$$

$$= 7112.0/14.0$$

$$= 508.000 \text{ must be less than 400; Calculation not possible.}$$

Nozzle Calculations per Section 4.5: External Pressure Case:

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Note: Impact testing of LWN flanges meeting requirements of UCS-66(c)(4) is not required.

The Drop for this Nozzle is : 0.2504 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 226.6253 mm





Input, Nozzle Desc: L7 2"

From: 220

Pressure for Reinforcement Calculations	P	0.3880	MPa
Temperature for Internal Pressure	Temp	130	°C
Design External Pressure	Pext	0.05	MPa
Temperature for External Pressure	Tempex	130	°C

Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	156.40	MPa
Parent Allowable Stress At Ambient	Sva	175.00	MPa

Inside Diameter of Cylindrical Shell	D	7100.00	mm
Design Length of Section	L	2499.9951	mm
Shell Finished (Minimum) Thickness	t	20.0000	mm
Shell Internal Corrosion Allowance	cas	6.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		39349.96	mm
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User Entered Minimum Design Metal Temperature		-12.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-105	+IT
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	MPa
Allowable Stress At Ambient	Sna	138.00	MPa

Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		2.0000	in.

Size and Thickness Basis		Actual	
Actual Thickness	tn	16.7640	mm

Flange Material		SA-105	+IT
Flange Type		Long Weld Neck	

Corrosion Allowance	can	3.0000	mm
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Outside Projection	ho	206.3750	mm
Weld leg size between Nozzle and Pad/Shell	Wo	16.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	16.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.

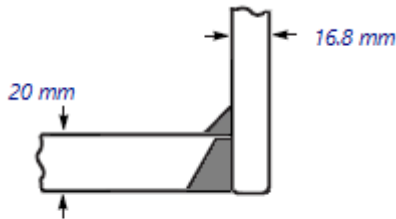
Flange Class		300
Flange Grade		GR 1.1

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: L7 2"

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation		2.000	in.
Actual Thickness Used in Calculation		0.660	in.



Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 56.8 \cdot (\exp(0.388/(138.0 \cdot 1.0)) - 1) + 3.0 + 0.0$$

$$= 0.08 + 3.0 + 0.0 = 3.08 \text{ mm Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 138.0 \cdot 1.0 \cdot \ln((2 \cdot 13.764 + 56.8) / 56.8) - 0.0$$

$$= 54.535 - 0.0 = 54.5345 \text{ MPa}$$

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c :	7.800 mm
Min. required shell thickness, internal p + c :	14.833 mm
Min. required shell thickness, external p + c :	22.589 mm

Nozzle neck minimum required thickness	: 7.800 mm
--	------------

Minimum nozzle neck thickness 16.764 mm , is greater than required thickness 7.800 mm

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

$$= \min[S_n/S, 1]$$

$$= \min[138.0/156.4, 1]$$

$$= 0.882$$

Thickness of Nozzle [tn]:

$$= \text{thickness} - \text{corrosion allowance}$$

$$= 16.764 - 3.0$$

$$= 13.764 \text{ mm}$$

Shell Diameter to Thickness ratio [D/t]:

$$= D_i/t$$

$$= 7112.0/14.0$$

$$= 508.000 \text{ must be less than 400; Calculation not possible.}$$

Nozzle Calculations per Section 4.5: External Pressure Case:

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Note: Impact testing of LWN flanges meeting requirements of UCS-66(c)(4) is not required.

The Drop for this Nozzle is : 0.2504 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 226.6253 mm

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Input, Nozzle Desc: L9 2"

From: 220

Pressure for Reinforcement Calculations	P	0.3880	MPa
Temperature for Internal Pressure	Temp	130	°C
Design External Pressure	Pext	0.05	MPa
Temperature for External Pressure	Tempex	130	°C

Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	156.40	MPa
Parent Allowable Stress At Ambient	Sva	175.00	MPa

Inside Diameter of Cylindrical Shell	D	7100.00	mm
Design Length of Section	L	2499.9951	mm
Shell Finished (Minimum) Thickness	t	20.0000	mm
Shell Internal Corrosion Allowance	cas	6.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		39349.96	mm
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User Entered Minimum Design Metal Temperature		-12.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-105	+IT
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	MPa
Allowable Stress At Ambient	Sna	138.00	MPa

Diameter Basis (for tr calc only)		Inside	
Layout Angle		120.00	deg
Diameter		2.0000	in.

Size and Thickness Basis		Actual	
Actual Thickness	tn	16.7640	mm

Flange Material		SA-105	+IT
Flange Type		Long Weld Neck	

Corrosion Allowance	can	3.0000	mm
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Outside Projection	ho	206.3750	mm
Weld leg size between Nozzle and Pad/Shell	Wo	16.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	16.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.

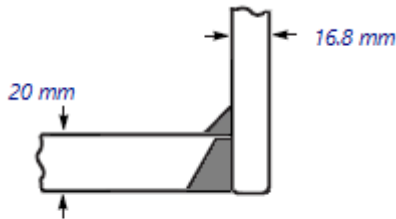
Flange Class		300
Flange Grade		GR 1.1

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: L9 2"

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation		2.000	in.
Actual Thickness Used in Calculation		0.660	in.



Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 56.8 \cdot (\exp(0.388/(138.0 \cdot 1.0)) - 1) + 3.0 + 0.0$$

$$= 0.08 + 3.0 + 0.0 = 3.08 \text{ mm Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 138.0 \cdot 1.0 \cdot \ln((2 \cdot 13.764 + 56.8) / 56.8) - 0.0$$

$$= 54.535 - 0.0 = 54.5345 \text{ MPa}$$

This nozzle or its diameter limit intersects weld seam # [7] on this element. The distance between the nozzle and seam is, 554.67 mm , while the distance from the center of the nozzle to the edge of the diameter limit is: 755.70 mm

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c :	7.800 mm
Min. required shell thickness, internal p + c	14.833 mm
Min. required shell thickness, external p + c	22.589 mm

Nozzle neck minimum required thickness	7.800 mm
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Minimum nozzle neck thickness 16.764 mm , is greater than required thickness 7.800 mm

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

$$= \min[S_n/S, 1]$$

$$= \min[138.0/156.4, 1]$$

$$= 0.882$$

Thickness of Nozzle [tn]:

$$= \text{thickness} - \text{corrosion allowance}$$

$$= 16.764 - 3.0$$

$$= 13.764 \text{ mm}$$

Shell Diameter to Thickness ratio [D/t]:

$$= D_i/t$$

$$= 7112.0/14.0$$

$$= 508.000 \text{ must be less than 400; Calculation not possible.}$$

Nozzle Calculations per Section 4.5: External Pressure Case:

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Note: Impact testing of LWN flanges meeting requirements of UCS-66(c)(4) is not required.

The Drop for this Nozzle is : 0.2504 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 226.6253 mm





Input, Nozzle Desc: MH-2 24"

From: 230

Pressure for Reinforcement Calculations	P	0.3880	MPa
Temperature for Internal Pressure	Temp	130	°C
Design External Pressure	Pext	0.05	MPa
Temperature for External Pressure	Tempex	130	°C

Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	156.40	MPa
Parent Allowable Stress At Ambient	Sva	175.00	MPa

Inside Diameter of Cylindrical Shell	D	7100.00	mm
Design Length of Section	L	2500.0012	mm
Shell Finished (Minimum) Thickness	t	16.0000	mm
Shell Internal Corrosion Allowance	cas	1.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		42850.05	mm
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User Entered Minimum Design Metal Temperature		-12.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-516 70	+IT
Material UNS Number		K02700	
Material Specification/Type		Plate	
Allowable Stress at Temperature	Sn	156.40	MPa
Allowable Stress At Ambient	Sna	175.00	MPa

Diameter Basis (for tr calc only)		Inside	
Layout Angle		225.00	deg
Diameter		22.7402	in.

Size and Thickness Basis		Actual	
Actual Thickness	tn	16.0000	mm

Flange Material		SA-105	+IT
Flange Type		Weld Neck	Flange

Corrosion Allowance	can	6.0000	mm
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Outside Projection	ho	310.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	16.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	12.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Pad Material		SA-516 70	+IT
Pad Allowable Stress at Temperature	Sp	156.40	MPa
Pad Allowable Stress At Ambient	Spa	175.00	MPa
Diameter of Pad along vessel surface	Dp	1209.6000	mm
Thickness of Pad	te	12.0000	mm
Weld leg size between Pad and Shell	Wp	12.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	12.0000	mm
Reinforcing Pad Width		300.0000	mm
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.
This is a Manway, Coupling or Access Opening.			

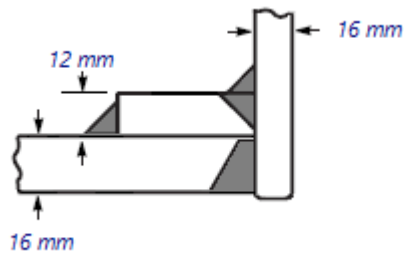
Flange Class		150
Flange Grade		GR 1.1

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: MH-2 24"

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	22.740	in.
Actual Thickness Used in Calculation	0.630	in.



Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:
 $= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$
 $= 0.5 \cdot 589.6 \cdot (\exp(0.388/(156.4 \cdot 1.0)) - 1) + 6.0 + 0.0$
 $= 0.7322 + 6.0 + 0.0 = 6.7322 \text{ mm Equation 4.3.1}$

Computed Maximum Allowable Working Pressure [MAWP]:
 $= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$
 $= 156.4 \cdot 1.0 \cdot \ln((2 \cdot 10.0 + 589.6) / 589.6) - 0.0$
 $= 5.217 - 0.0 = 5.2173 \text{ MPa}$

This nozzle or its diameter limit intersects weld seam # [1] on this element. The distance between the nozzle and seam is, 0.00 mm , while the distance from the center of the nozzle to the edge of the diameter limit is: 755.70 mm

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:
 $= \min[S_n/S, 1]$
 $= \min[156.4/156.4, 1]$
 $= 1.000$

Pad Material Factor [frp]:
 $= \min[S_p/S, 1]$
 $= \min[156.4/156.4, 1]$
 $= 1.000$

Thickness of Nozzle [tn]:
 $= \text{thickness} - \text{corrosion allowance}$
 $= 16.0 - 6.0$
 $= 10.000 \text{ mm}$

Shell Diameter to Thickness ratio [D/t]:
 $= D_i/t$
 $= 7102.0/15.0$
 $= 473.467 \text{ must be less than 400; Calculation not possible.}$

Nozzle Calculations per Section 4.5: External Pressure Case:

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Nozzle Neck to Pad Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Nozzle Neck to Pad Weld for Reinf. pad (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.



Shell to Pad Weld at Pad OD for pad (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Gov. MDMT of the Nozzle	:	-12 °C
Gov. MDMT of the Reinforcement Pad	:	-12 °C

Flange MDMT including temperature reduction per 3.11:
Flange MDMT with Temp reduction -104 °C

The Drop for this Nozzle is : 13.1093 mm
The Cut Length for this Nozzle is, Drop + Ho + H + T : 339.1095 mm

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Input, Nozzle Desc: UC1 24"

From: 240

Pressure for Reinforcement Calculations	P	0.3880	MPa
Temperature for Internal Pressure	Temp	130	°C
Design External Pressure	Pext	0.05	MPa
Temperature for External Pressure	Tempex	130	°C

Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	156.40	MPa
Parent Allowable Stress At Ambient	Sva	175.00	MPa

Inside Diameter of Cylindrical Shell	D	7100.00	mm
Design Length of Section	L	2500.0098	mm
Shell Finished (Minimum) Thickness	t	24.0000	mm
Shell Internal Corrosion Allowance	cas	1.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		43949.96	mm
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User Entered Minimum Design Metal Temperature		-12.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-516 70	+IT
Material UNS Number		K02700	
Material Specification/Type		Plate	
Allowable Stress at Temperature	Sn	156.40	MPa
Allowable Stress At Ambient	Sna	175.00	MPa

Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		24.0000	in.

Size and Thickness Basis		Actual	
Actual Thickness	tn	16.0000	mm

Flange Material		SA-105	+IT
Flange Type		Weld Neck	Flange

Corrosion Allowance	can	6.0000	mm
---------------------	-----	--------	----

Outside Projection	ho	300.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	18.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Pad Material		SA-516 70	+IT
Pad Allowable Stress at Temperature	Sp	156.40	MPa
Pad Allowable Stress At Ambient	Spa	175.00	MPa
Diameter of Pad along vessel surface	Dp	1041.6000	mm
Thickness of Pad	te	20.0000	mm
Weld leg size between Pad and Shell	Wp	18.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	20.0000	mm
Reinforcing Pad Width		200.0000	mm
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.

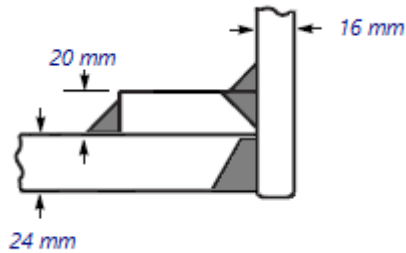
Flange Class		150
Flange Grade		GR 1.1

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: UC1 24"

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	24.000	in.
Actual Thickness Used in Calculation	0.630	in.



Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 621.6 \cdot (\exp(0.388/(156.4 \cdot 1.0)) - 1) + 6.0 + 0.0$$

$$= 0.772 + 6.0 + 0.0 = 6.772 \text{ mm Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 156.4 \cdot 1.0 \cdot \ln((2 \cdot 10.0 + 621.6) / 621.6) - 0.0$$

$$= 4.953 - 0.0 = 4.9529 \text{ MPa}$$

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c :	14.340 mm
Min. required shell thickness, internal p + c :	9.820 mm
Min. required shell thickness, external p + c :	12.582 mm

Nozzle neck minimum required thickness	: 12.582 mm
--	-------------

Minimum nozzle neck thickness 16.002 mm , is greater than required thickness 12.582 mm

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

$$= \min[S_n/S, 1]$$

$$= \min[156.4/156.4, 1]$$

$$= 1.000$$

Pad Material Factor [frp]:

$$= \min[S_p/S, 1]$$

$$= \min[156.4/156.4, 1]$$

$$= 1.000$$

Thickness of Nozzle [tn]:

$$= \text{thickness} - \text{corrosion allowance}$$

$$= 16.0 - 6.0$$

$$= 10.000 \text{ mm}$$

Shell Diameter to Thickness ratio [D/t]:

$$= D_i/t$$

$$= 7102.0/23.0$$

$$= 308.783 \text{ must be less than 400.}$$

Effective Pressure Radius [Reff]:

$$= D_i/2 + \text{corrosion allowance}$$

$$= 7100.0/2 + 1.0$$

$$= 3551.000 \text{ mm}$$

Effective Length of Vessel Wall [LR]:

For Nozzles with reinforcing pads, Lr is min[LR1, LR2, LR3]:

Limit [LR1]:

$$= \sqrt{(Reff \cdot t)} + W$$

$$= \sqrt{(3551.0 \cdot 23.0)} + 200.0$$



$$= 485.785 \text{ mm}$$

Limit [LR2]:

$$\begin{aligned} &= \sqrt{((R_{eff} + t) \cdot (t + t_e))} \\ &= \sqrt{((3551.0 + 23.0) \cdot (23.0 + 20.0))} \\ &= 392.023 \text{ mm} \end{aligned}$$

Limit [LR3]:

$$\begin{aligned} &= 2 \cdot R_n \\ &= 2 \cdot 310.8 \\ &= 621.600 \text{ mm} \end{aligned}$$

Final value of [LR]:

$$\begin{aligned} &= \min(LR1, LR2, LR3) \\ &= \min(485.785, 392.023, 621.6) \\ &= 392.023 \text{ mm} \end{aligned}$$

Thickness Limit Candidate [LH1]:

$$\begin{aligned} &= \min[1.5t, t_e] + \sqrt{(R_n \cdot t_n)} \\ &= \min[1.5 \cdot 23.0, 20.0] + \sqrt{(310.8 \cdot 10.0)} \\ &= 75.749 \text{ mm} \end{aligned}$$

Thickness Limit Candidate [LH2]:

$$\begin{aligned} &= L_{pr1} \\ &= 300.0 \\ &= 300.000 \text{ mm} \end{aligned}$$

Thickness Limit Candidate [LH3]:

$$\begin{aligned} &= 8(t + t_e) \\ &= 8(23.0 + 20.0) \\ &= 344.000 \text{ mm} \end{aligned}$$

Effective Nozzle Wall Length Outside the Vessel [LH]:

$$\begin{aligned} &= \min[LH1, LH2, LH3] + t \quad \text{for set-in nozzles} \\ &= \min[75.749, 300.0, 344.0] + 23.0 \\ &= 98.749 \text{ mm} \end{aligned}$$

Effective Vessel Thickness [teff]:

$$\begin{aligned} &= t + ((A5 \cdot frp) / LR) \\ &= 23.0 + ((4000.0 \cdot 1.0) / 392.023) \\ &= 33.203 \text{ mm} \end{aligned}$$

Parameter [λ]:

$$\begin{aligned} &= \min(12, (2 \cdot R_n + t_n) / (\sqrt{(D_i + t_{eff}) \cdot t_{eff}})) \\ &= \min(12, (2 \cdot 310.8 + 10.0) / (\sqrt{(7102.0 + 33.203) \cdot 33.203})) \\ &= 1.298 \end{aligned}$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$\begin{aligned} &= t \cdot LR \cdot \max((\lambda/5)^{0.85}, 1) \\ &= 23.0 \cdot 392.023 \cdot \max((1.298/5)^{0.85}, 1) \\ &= 9016.528 \text{ mm}^2 \end{aligned}$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$\begin{aligned} &= t_n \cdot L_H \\ &= 10.0 \cdot 98.749 \\ &= 987.494 \text{ mm}^2 \end{aligned}$$

Area Contributed by the Pad Fillet Weld [A42]:

$$\begin{aligned} &= 0.5 \cdot \text{Leg4}^2 \\ &= 0.5 \cdot 18.0^2 \\ &= 162.000 \text{ mm}^2 \end{aligned}$$

Area Contributed by the Outside Fillet Weld [A41]:

$$\begin{aligned} &= 0.5 \cdot \text{Leg4}^2 \\ &= 0.5 \cdot 10.0^2 \\ &= 50.000 \text{ mm}^2 \end{aligned}$$

Area Contributed by the Reinforcing Pad [A5]:



$$\begin{aligned} &= \min(W \cdot te , LR \cdot te) \\ &= \min(200.0 \cdot 20.0, 392.023 \cdot 20.0) \\ &= 4000.000 \text{ mm}^2 \end{aligned}$$

The total area contributed by A1 through A5 [AT]:

$$\begin{aligned} &= A1 + frn(A2 + A3) + A41 + A42 + A43 + frp(A5) \\ &= 9016.528 + 1.0(987.494 + 0.0) + 50.0 + 162.0 + 0.0 + 1.0(4000.0) \\ &= 14216.023 \text{ mm}^2 \end{aligned}$$

Nozzle Radius for Force Calculation [Rxn]:

$$\begin{aligned} &= tn / \ln[1 + tn/Rn] \\ &= 10.0 / \ln[1 + 10.0/310.8] \\ &= 315.773 \text{ mm} \end{aligned}$$

Shell Radius for Force Calculation [Rxs]:

$$\begin{aligned} &= teff / \ln[1 + teff/Reff] \\ &= 33.203 / \ln[1 + 33.203/3551.0] \\ &= 3567.592 \text{ mm} \end{aligned}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$\begin{aligned} &= 1.5 \cdot S \cdot E \\ &= 1.5 \cdot 156.4 \cdot 1.0 \\ &= 234.6 \text{ MPa} \end{aligned}$$

Force acting on the Nozzle [fN]:

$$\begin{aligned} &= P \cdot Rxn \cdot LH \\ &= 0.388 \cdot 315.773 \cdot 98.749 \\ &= 12097.8 \text{ N} \end{aligned}$$

Force acting on the Shell [fS]:

$$\begin{aligned} &= P \cdot Rxs(LR + tn) \\ &= 0.388 \cdot 3567.592(392.023 + 10.0) \\ &= 556443.5 \text{ N} \end{aligned}$$

Discontinuity Force from Internal Pressure [fY]:

$$\begin{aligned} &= P \cdot Rxs \cdot Rnc \\ &= 0.388 \cdot 3567.592 \cdot 310.8 \\ &= 430181.0 \text{ N} \end{aligned}$$

Area Resisting Internal Pressure [Ap]:

$$\begin{aligned} &= (fS + fY + fN)/P \\ &= (556444 + 430181.0 + 12097.8)/0.388 \\ &= 2574243.5 \text{ mm}^2 \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$\begin{aligned} &= Sallow / (2 \cdot Ap/AT - Rxs/teff) \\ &= 234.6 / (2 \cdot 2574244/14216.023 - 3567.592/33.203) \\ &= 0.9 \text{ MPa} \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$\begin{aligned} &= S[t/Rxs] \\ &= 156.4[23.0/3567.592] \\ &= 1.0 \text{ MPa} \end{aligned}$$

Maximum Allowable Working Pressure [Pmax]:

$$\begin{aligned} &= \min(Pmax1, Pmax2) \\ &= \min(0.921, 1.008) \\ &= 0.921 \text{ MPa} \end{aligned}$$

Average Primary Membrane Stress [σ Avg]:

$$\begin{aligned} &= (fN + fS + fY) / AT \\ &= (12097.764 + 556444 + 430180.969)/14216.023 \\ &= 70.259 \text{ MPa} \end{aligned}$$

General Primary Membrane Stress [σ Circ]:

$$\begin{aligned} &= P \cdot Rxs / teff \\ &= 0.388 \cdot 3567.592/33.203 \\ &= 41.7 \text{ MPa} \end{aligned}$$

Maximum Local Primary Membrane Stress [PL]:



= max(2 • σ_{Avg} - σ_{Circ} , σ_{Circ})
 = max(2 • 70.259 - 41.689, 41.689)
 = 98.8 MPa

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	234.60	MPa
Local Primary Membrane Stress	PL	98.83	MPa
Maximum Allowable Working Pressure	Pmax	0.92	MPa

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

= ($R_{nc} + t_n$) / R_{nc}
 = (310.8 + 10.0) / 310.8
 = 1.032 For set-in Nozzles

Weld Length of Nozzle to Shell Weld [L_T]:

= $\pi/2 \cdot (R_n + t_n)$
 = $\pi/2 \cdot (310.8 + 10.0)$
 = 503.911 mm

Weld Length of Pad to Shell Weld [L_{TP}]:

= $\pi/2 \cdot (R_n + t_n + W)$
 = $\pi/2 \cdot (310.8 + 10.0 + 200.0)$
 = 818.071 mm

Weld Throat Dimensions, (0.7071 • Leg Dimensions) [L_{41T} , L_{42T} , L_{43T}]:

= 7.071, 12.728, 0.000, mm

Required Weld Throat Dimensions :

t_c , L_{41T} = 7.071, must be $\geq \min(0.7 \cdot 10.000, 6.000)$ Passed
 t_{f1} , L_{42T} = 12.728, must be $\geq \min(0.6 \cdot 20.000, 0.6 \cdot 23.000)$ Passed

Weld Load Value [fwelds]:

= min($f_y \cdot k_y$, $1.5 \cdot S_n(A_2 + A_3)$, $\pi/4 \cdot P \cdot R_n^2 \cdot k_y^2$)
 = min(430181 • 1.03, $1.5 \cdot 156.4(987.494 + 0.0)$, $\pi/4 \cdot 0.4 \cdot 310.8^2 \cdot 1.03^2$)
 = 31358.385 N

Discontinuity Force [fws]:

= fwelds • $t \cdot S / (t \cdot S + t_e \cdot S_p)$
 = $31358.4 \cdot 23.0 \cdot 156 / (23.0 \cdot 156 + 20.0 \cdot 156)$
 = 16773.090 N

Discontinuity Force [fwp]:

= fwelds • $t_e \cdot S_p / (t \cdot S + t_e \cdot S_p)$
 = $31358.4 \cdot 20.0 \cdot 156 / (23.0 \cdot 156 + 20.0 \cdot 156)$
 = 14585.294 N

Shear Stress [τ_1]:

= $fws / (L_T \cdot (0.6 \cdot t_{w1} + 0.49 \cdot L_{43T}))$
 = $16773.09 / (503.911 \cdot (0.6 \cdot 18.0 + 0.49 \cdot 0.0))$
 = 3.082 MPa

Shear Stress [τ_2]:

= $fwp / (L_T \cdot (0.6 \cdot t_{w2} + 0.49 \cdot L_{41T}))$
 = $14585.294 / (503.911 \cdot (0.6 \cdot 20.0 + 0.49 \cdot 7.071))$
 = 1.872 MPa

Shear Stress [τ_3]:

= $fwp / (L_{TP} \cdot (0.49 \cdot L_{42T}))$
 = $14585.294 / (818.071 \cdot (0.49 \cdot 12.728))$
 = 2.859 MPa

Maximum Shear Stress in the Welds:

= max(τ_1 , τ_2 , τ_3)
 = max(3.082, 1.872, 2.859)
 = 3.1 must be less than or equal to 156.4 MPa

Nozzle Calculations per Section 4.5: External Pressure Case:



Nozzle Radius for Force Calculation [Rxn]:

$$\begin{aligned} &= t_n / \ln[1 + t_n/R_n] \\ &= 10.0/\ln[1 + 10.0/310.8] \\ &= 315.773 \text{ mm} \end{aligned}$$

Shell Radius for Force Calculation [Rxs]:

$$\begin{aligned} &= t_{eff} / \ln[1 + t_{eff}/R_{eff}] \\ &= 33.203/\ln[1 + 33.203/3551.0] \\ &= 3567.592 \text{ mm} \end{aligned}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$\begin{aligned} &= 1.5 \cdot S \\ &= 1.5 \cdot 156.4 \\ &= 234.6 \text{ MPa} \end{aligned}$$

Force acting on the Nozzle [fN]:

$$\begin{aligned} &= P \cdot R_{xn} \cdot L_H \\ &= 0.052 \cdot 315.773 \cdot 98.749 \\ &= 1613.0 \text{ N} \end{aligned}$$

Force acting on the Shell [fS]:

$$\begin{aligned} &= P \cdot R_{xs} (L_R + t_n) \\ &= 0.052 \cdot 3567.592 (392.023 + 10.0) \\ &= 74191.9 \text{ N} \end{aligned}$$

Discontinuity Force from External Pressure [fY]:

$$\begin{aligned} &= P \cdot R_{xs} \cdot R_{nc} \\ &= 0.052 \cdot 3567.592 \cdot 310.8 \\ &= 57357.1 \text{ N} \end{aligned}$$

Area Resisting External Pressure [Ap]:

$$\begin{aligned} &= (f_N + f_Y + f_N)/P \\ &= (74191.9 + 57357.1 + 1613.0)/0.052 \\ &= 2574243.5 \text{ mm}^2 \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$\begin{aligned} &= S_{allow} / (2 \cdot A_p/AT - R_{xs}/t_{eff}) \\ &= 234.6 / (2 \cdot 2574244/14216.023 - 3567.592/33.203) \\ &= 0.9 \text{ MPa} \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$\begin{aligned} &= S[t/R_{xs}] \\ &= 156.4[23.0/3567.592] \\ &= 1.0 \text{ MPa} \end{aligned}$$

Maximum Allowable Working Pressure [Pmax]:

$$\begin{aligned} &= \min(P_{max1}, P_{max2}) \\ &= \min(0.921, 1.008) \\ &= 0.921 \text{ MPa} \end{aligned}$$

Average Primary Membrane Stress [σAvg]:

$$\begin{aligned} &= (f_N + f_S + f_Y) / AT \\ &= (1613.024 + 74191.945 + 57357.059)/14216.023 \\ &= 9.368 \text{ MPa} \end{aligned}$$

General Primary Membrane Stress [σCirc]:

$$\begin{aligned} &= P \cdot R_{xs} / t_{eff} \\ &= 0.052 \cdot 3567.592/33.203 \\ &= 5.6 \text{ MPa} \end{aligned}$$

Maximum Local Primary Membrane Stress [PL]:

$$\begin{aligned} &= \max(2 \cdot \sigma_{Avg} - \sigma_{Circ}, \sigma_{Circ}) \\ &= \max(2 \cdot 9.368 - 5.559, 5.559) \\ &= 13.2 \text{ MPa} \end{aligned}$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	234.60	MPa
Local Primary Membrane Stress	PL	13.18	MPa
Maximum Allowable External Pressure	Pmax	0.92	MPa

Noz1: 94 5:17pm Aug 25, 2026

Nozzle Neck to Flange Weld (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Note: This Element/Detail was specified as being Impact Tested.

Note: This Element/Detail was specified as being Impact Tested.

Note: This Element/Detail was specified as being Impact Tested.

Note: This Element/Detail was specified as being Impact Tested.

Gov. MDMT of the Nozzle	: -12 °C
Gov. MDMT of the Reinforcement Pad	: -12 °C

Flange MDMT including temperature reduction per 3.11:
 Flange MDMT with Temp reduction -104 °C

The Drop for this Nozzle is : 14.5243 mm
The Cut Length for this Nozzle is, Drop + Ho + H + T : 338.5244 mm



Input, Nozzle Desc: MH-1 24"

From: 240

Pressure for Reinforcement Calculations	P	0.3880	MPa
Temperature for Internal Pressure	Temp	130	°C
Design External Pressure	Pext	0.05	MPa
Temperature for External Pressure	Tempex	130	°C

Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	156.40	MPa
Parent Allowable Stress At Ambient	Sva	175.00	MPa

Inside Diameter of Cylindrical Shell	D	7100.00	mm
Design Length of Section	L	2500.0098	mm
Shell Finished (Minimum) Thickness	t	16.0000	mm
Shell Internal Corrosion Allowance	cas	1.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		45149.96	mm
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User Entered Minimum Design Metal Temperature		-12.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-516 70	+IT
Material UNS Number		K02700	
Material Specification/Type		Plate	
Allowable Stress at Temperature	Sn	156.40	MPa
Allowable Stress At Ambient	Sna	175.00	MPa

Diameter Basis (for tr calc only)		Inside	
Layout Angle		225.00	deg
Diameter		22.7402	in.

Size and Thickness Basis		Actual	
Actual Thickness	tn	16.0000	mm

Flange Material		SA-105	+IT
Flange Type		Weld Neck	Flange

Corrosion Allowance	can	6.0000	mm
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Outside Projection	ho	310.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	16.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	14.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Pad Material		SA-516 70	+IT
Pad Allowable Stress at Temperature	Sp	156.40	MPa
Pad Allowable Stress At Ambient	Spa	175.00	MPa
Diameter of Pad along vessel surface	Dp	1209.6000	mm
Thickness of Pad	te	16.0000	mm
Weld leg size between Pad and Shell	Wp	16.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	16.0000	mm
Reinforcing Pad Width		300.0000	mm
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.
This is a Manway, Coupling or Access Opening.			

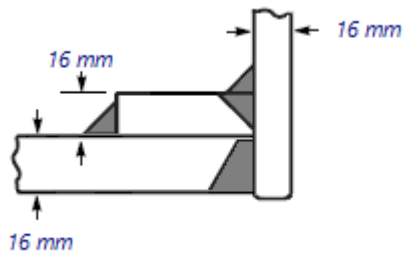
Flange Class		150
Flange Grade		GR 1.1

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: MH-1 24"

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	22.740	in.
Actual Thickness Used in Calculation	0.630	in.



Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:
 $= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$
 $= 0.5 \cdot 589.6 \cdot (\exp(0.388/(156.4 \cdot 1.0)) - 1) + 6.0 + 0.0$
 $= 0.7322 + 6.0 + 0.0 = 6.7322 \text{ mm Equation 4.3.1}$

Computed Maximum Allowable Working Pressure [MAWP]:
 $= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$
 $= 156.4 \cdot 1.0 \cdot \ln((2 \cdot 10.0 + 589.6) / 589.6) - 0.0$
 $= 5.217 - 0.0 = 5.2173 \text{ MPa}$

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:
 $= \min[S_n/S, 1]$
 $= \min[156.4/156.4, 1]$
 $= 1.000$

Pad Material Factor [frp]:
 $= \min[S_p/S, 1]$
 $= \min[156.4/156.4, 1]$
 $= 1.000$

Thickness of Nozzle [tn]:
 $= \text{thickness} - \text{corrosion allowance}$
 $= 16.0 - 6.0$
 $= 10.000 \text{ mm}$

Shell Diameter to Thickness ratio [D/t]:
 $= D_i/t$
 $= 7102.0/15.0$
 $= 473.467 \text{ must be less than 400; Calculation not possible.}$

Nozzle Calculations per Section 4.5: External Pressure Case:

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Nozzle Neck to Pad Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Nozzle Neck to Pad Weld for Reinf. pad (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Shell to Pad Weld at Pad OD for pad (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.



Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Gov. MDMT of the Nozzle : -12 °C
Gov. MDMT of the Reinforcement Pad : -12 °C

Flange MDMT including temperature reduction per 3.11:
Flange MDMT with Temp reduction -104 °C

The Drop for this Nozzle is : 13.1093 mm
The Cut Length for this Nozzle is, Drop + Ho + H + T : 339.1095 mm

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Input, Nozzle Desc: S1 1"

From: 250

Pressure for Reinforcement Calculations	P	0.3880	MPa
Temperature for Internal Pressure	Temp	130	°C
Design External Pressure	Pext	0.05	MPa
Temperature for External Pressure	Tempex	130	°C

Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa

Inside Diameter of Cylindrical Shell	D	7100.00	mm
Design Length of Section	L	2500.0000	mm
Shell Finished (Minimum) Thickness	t	16.0000	mm
Shell Internal Corrosion Allowance	cas	1.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		47949.95	mm
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User Entered Minimum Design Metal Temperature		-12.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-105	+IT
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	MPa
Allowable Stress At Ambient	Sna	138.00	MPa

Diameter Basis (for tr calc only)		Inside	
Layout Angle		225.00	deg
Diameter		1.0000	in.

Size and Thickness Basis		Actual	
Actual Thickness	tn	12.7000	mm

Flange Material		SA-105	+IT
Flange Type		Long Weld Neck	

Corrosion Allowance	can	6.0000	mm
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Outside Projection	ho	214.3130	mm
Weld leg size between Nozzle and Pad/Shell	Wo	16.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	12.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.

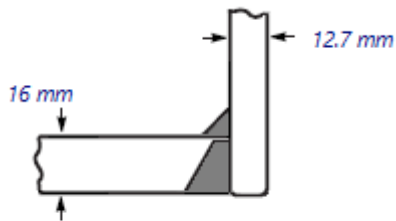
Flange Class		150
Flange Grade		GR 1.1

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: S1 1"

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation		1.000	in.
Actual Thickness Used in Calculation		0.500	in.



Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$\begin{aligned} &= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o \\ &= 0.5 \cdot 37.4 \cdot (\exp(0.388/(138.0 \cdot 1.0)) - 1) + 6.0 + 0.0 \\ &= 0.0527 + 6.0 + 0.0 = 6.0527 \text{ mm Equation 4.3.1} \end{aligned}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$\begin{aligned} &= S \cdot E \cdot \ln((2 \cdot t + D)/D) - P_{liq} \\ &= 138.0 \cdot 1.0 \cdot \ln((2 \cdot 6.7 + 37.4)/37.4) - 0.0 \\ &= 42.259 - 0.0 = 42.2591 \text{ MPa} \end{aligned}$$

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c :	9.420 mm
Min. required shell thickness, internal p + c	10.998 mm
Min. required shell thickness, external p + c	12.582 mm

Nozzle neck minimum required thickness	: 9.420 mm
--	------------

Minimum nozzle neck thickness 12.700 mm , is greater than required thickness 9.420 mm

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

$$\begin{aligned} &= \min[S_n/S, 1] \\ &= \min[138.0/138.0, 1] \\ &= 1.000 \end{aligned}$$

Thickness of Nozzle [tn]:

$$\begin{aligned} &= \text{thickness} - \text{corrosion allowance} \\ &= 12.7 - 6.0 \\ &= 6.700 \text{ mm} \end{aligned}$$

Shell Diameter to Thickness ratio [D/t]:

$$\begin{aligned} &= D_i/t \\ &= 7102.0/15.0 \\ &= 473.467 \text{ must be less than 400; Calculation not possible.} \end{aligned}$$

Nozzle Calculations per Section 4.5: External Pressure Case:

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Note: Impact testing of LWN flanges meeting requirements of UCS-66(c)(4) is not required.

The Drop for this Nozzle is : 0.0907 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 230.4039 mm

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Input, Nozzle Desc: H1 42"

From: 250

Pressure for Reinforcement Calculations	P	0.3880	MPa
Temperature for Internal Pressure	Temp	130	°C
Design External Pressure	Pext	0.05	MPa
Temperature for External Pressure	Tempex	130	°C

Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa

Inside Diameter of Cylindrical Shell	D	7100.00	mm
Design Length of Section	L	2500.0000	mm
Shell Finished (Minimum) Thickness	t	26.0000	mm
Shell Internal Corrosion Allowance	cas	1.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		47550.05	mm
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User Entered Minimum Design Metal Temperature		-12.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-516 70	+IT
Material UNS Number		K02700	
Material Specification/Type		Plate	
Allowable Stress at Temperature	Sn	156.40	MPa
Allowable Stress At Ambient	Sna	175.00	MPa

Diameter Basis (for tr calc only)		Inside	
Layout Angle		180.00	deg
Diameter		39.9528	in.

Size and Thickness Basis		Actual	
Actual Thickness	tn	26.0000	mm

Flange Material		SA-105	+IT
Flange Type		Weld Neck Flange	

Corrosion Allowance	can	6.0000	mm
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Outside Projection	ho	310.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	16.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	16.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Pad Material		SA-516 70	+IT
Pad Allowable Stress at Temperature	Sp	156.40	MPa
Pad Allowable Stress At Ambient	Spa	175.00	MPa
Diameter of Pad along vessel surface	Dp	1766.8000	mm
Thickness of Pad	te	16.0000	mm
Weld leg size between Pad and Shell	Wp	16.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	16.0000	mm
Reinforcing Pad Width		350.0001	mm
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.

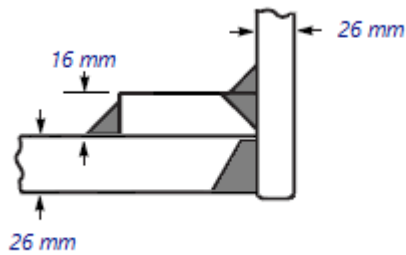
Flange Class		150
Flange Grade		GR 1.1
Flange Series		B

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: H1 42"

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation		39.953	in.
Actual Thickness Used in Calculation		1.024	in.



Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:
 $= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$
 $= 0.5 \cdot 1026.8 \cdot (\exp(0.388/(156.4 \cdot 1.0)) - 1) + 6.0 + 0.0$
 $= 1.2752 + 6.0 + 0.0 = 7.2752 \text{ mm Equation 4.3.1}$

Computed Maximum Allowable Working Pressure [MAWP]:
 $= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$
 $= 156.4 \cdot 1.0 \cdot \ln((2 \cdot 20.0 + 1026.8) / 1026.8) - 0.0$
 $= 5.977 - 0.0 = 5.977 \text{ MPa}$

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c :	14.340 mm
Min. required shell thickness, internal p + c	10.998 mm
Min. required shell thickness, external p + c	12.582 mm

Nozzle neck minimum required thickness	: 12.582 mm
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Minimum nozzle neck thickness 26.010 mm , is greater than required thickness 12.582 mm

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	16.6,	Allowable	:	156.4 MPa	Passed
Expansion	:	0.0,	Allowable	:	397.6 MPa	Passed
Occasional	:	4.9,	Allowable	:	208.0 MPa	Passed
Shear	:	9.0,	Allowable	:	109.5 MPa	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:
 $= \min[S_n/S, 1]$
 $= \min[156.4/138.0, 1]$
 $= 1.000$

Pad Material Factor [frp]:
 $= \min[S_p/S, 1]$
 $= \min[156.4/138.0, 1]$
 $= 1.000$

Thickness of Nozzle [tn]:
 $= \text{thickness} - \text{corrosion allowance}$
 $= 26.0 - 6.0$
 $= 20.000 \text{ mm}$

Shell Diameter to Thickness ratio [D/t]:
 $= D_i/t$
 $= 7102.0/25.0$
 $= 284.080 \text{ must be less than } 400.$



Effective Pressure Radius [Reff]:

$$\begin{aligned}
 &= Di/2 + \text{corrosion allowance} \\
 &= 7100.0/2 + 1.0 \\
 &= 3551.000 \text{ mm}
 \end{aligned}$$

Effective Length of Vessel Wall [LR]:

For Nozzles with reinforcing pads, Lr is min[LR1, LR2, LR3]:

Limit [LR1]:

$$\begin{aligned}
 &= \sqrt{(Reff \cdot t) + W} \\
 &= \sqrt{(3551.0 \cdot 25.0) + 350.0} \\
 &= 647.951 \text{ mm}
 \end{aligned}$$

Limit [LR2]:

$$\begin{aligned}
 &= \sqrt{((Reff + t) \cdot (t + te))} \\
 &= \sqrt{((3551.0 + 25.0) \cdot (25.0 + 16.0))} \\
 &= 382.905 \text{ mm}
 \end{aligned}$$

Limit [LR3]:

$$\begin{aligned}
 &= 2 \cdot Rn \\
 &= 2 \cdot 513.4 \\
 &= 1026.800 \text{ mm}
 \end{aligned}$$

Final value of [LR]:

$$\begin{aligned}
 &= \min(LR1, LR2, LR3) \\
 &= \min(647.951, 382.905, 1026.8) \\
 &= 382.905 \text{ mm}
 \end{aligned}$$

Thickness Limit Candidate [LH1]:

$$\begin{aligned}
 &= \min[1.5t, te] + \sqrt{(Rn \cdot tn)} \\
 &= \min[1.5 \cdot 25.0, 16.0] + \sqrt{(513.4 \cdot 20.0)} \\
 &= 117.331 \text{ mm}
 \end{aligned}$$

Thickness Limit Candidate [LH2]:

$$\begin{aligned}
 &= Lpr1 \\
 &= 310.0 \\
 &= 310.000 \text{ mm}
 \end{aligned}$$

Thickness Limit Candidate [LH3]:

$$\begin{aligned}
 &= 8(t + te) \\
 &= 8(25.0 + 16.0) \\
 &= 328.000 \text{ mm}
 \end{aligned}$$

Effective Nozzle Wall Length Outside the Vessel [LH]:

$$\begin{aligned}
 &= \min[LH1, LH2, LH3] + t \quad \text{for set-in nozzles} \\
 &= \min[117.331, 310.0, 328.0] + 25.0 \\
 &= 142.331 \text{ mm}
 \end{aligned}$$

Effective Vessel Thickness [teff]:

$$\begin{aligned}
 &= t + ((A5 \cdot frp) / LR) \\
 &= 25.0 + ((5600.0 \cdot 1.0) / 382.905) \\
 &= 39.625 \text{ mm}
 \end{aligned}$$

Parameter [λ]:

$$\begin{aligned}
 &= \min(12, (2 \cdot Rn + tn) / \sqrt{((Di + teff) \cdot teff)}) \\
 &= \min(12, (2 \cdot 513.4 + 20.0) / \sqrt{((7102.0 + 39.625) \cdot 39.625)}) \\
 &= 1.968
 \end{aligned}$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$\begin{aligned}
 &= t \cdot LR \cdot \max((\lambda/5)^{0.85}, 1) \\
 &= 25.0 \cdot 382.905 \cdot \max((1.968/5)^{0.85}, 1) \\
 &= 9572.617 \text{ mm}^2
 \end{aligned}$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$\begin{aligned}
 &= tn \cdot LH \\
 &= 20.0 \cdot 142.331 \\
 &= 2846.623 \text{ mm}^2
 \end{aligned}$$



Area Contributed by the Pad Fillet Weld [A42]:

$$\begin{aligned} &= 0.5 \cdot \text{Leg42}^2 \\ &= 0.5 \cdot 16.0^2 \\ &= 128.000 \text{ mm}^2 \end{aligned}$$

Area Contributed by the Outside Fillet Weld [A41]:

$$\begin{aligned} &= 0.5 \cdot \text{Leg41}^2 \\ &= 0.5 \cdot 16.0^2 \\ &= 128.000 \text{ mm}^2 \end{aligned}$$

Area Contributed by the Reinforcing Pad [A5]:

$$\begin{aligned} &= \min(W \cdot t_e, LR \cdot t_e) \\ &= \min(350.0 \cdot 16.0, 382.905 \cdot 16.0) \\ &= 5600.000 \text{ mm}^2 \end{aligned}$$

The total area contributed by A1 through A5 [AT]:

$$\begin{aligned} &= A1 + \text{frn}(A2 + A3) + A41 + A42 + A43 + \text{frp}(A5) \\ &= 9572.617 + 1.0(2846.623 + 0.0) + 128.0 + 128.0 + 0.0 + 1.0(5600.0) \\ &= 18275.240 \text{ mm}^2 \end{aligned}$$

Nozzle Radius for Force Calculation [Rxn]:

$$\begin{aligned} &= t_n / \ln[1 + t_n/R_n] \\ &= 20.0 / \ln[1 + 20.0/513.4] \\ &= 523.337 \text{ mm} \end{aligned}$$

Shell Radius for Force Calculation [Rxs]:

$$\begin{aligned} &= t_{eff} / \ln[1 + t_{eff}/R_{eff}] \\ &= 39.625 / \ln[1 + 39.625/3551.0] \\ &= 3570.782 \text{ mm} \end{aligned}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$\begin{aligned} &= 1.5 \cdot S \cdot E \\ &= 1.5 \cdot 138.0 \cdot 1.0 \\ &= 207.0 \text{ MPa} \end{aligned}$$

Force acting on the Nozzle [fN]:

$$\begin{aligned} &= P \cdot R_{xn} \cdot L_H \\ &= 0.388 \cdot 523.337 \cdot 142.331 \\ &= 28898.6 \text{ N} \end{aligned}$$

Force acting on the Shell [fS]:

$$\begin{aligned} &= P \cdot R_{xs} (LR + t_n) \\ &= 0.388 \cdot 3570.782 (382.905 + 20.0) \\ &= 558162.6 \text{ N} \end{aligned}$$

Discontinuity Force from Internal Pressure [fY]:

$$\begin{aligned} &= P \cdot R_{xs} \cdot R_{nc} \\ &= 0.388 \cdot 3570.782 \cdot 513.4 \\ &= 711236.9 \text{ N} \end{aligned}$$

Area Resisting Internal Pressure [Ap]:

$$\begin{aligned} &= (f_S + f_Y + f_N) / P \\ &= (558163 + 711237 + 28898.6) / 0.388 \\ &= 3346411.5 \text{ mm}^2 \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$\begin{aligned} &= S_{allow} / (2 \cdot A_p / AT - R_{xs} / t_{eff}) \\ &= 207.0 / (2 \cdot 3346412 / 18275.24 - 3570.782 / 39.625) \\ &= 0.7 \text{ MPa} \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$\begin{aligned} &= S / t / R_{xs} \\ &= 138.0 / 25.0 / 3570.782 \\ &= 1.0 \text{ MPa} \end{aligned}$$

Maximum Allowable Working Pressure [Pmax]:

$$\begin{aligned} &= \min(P_{max1}, P_{max2}) \\ &= \min(0.75, 0.966) \\ &= 0.750 \text{ MPa} \end{aligned}$$



Average Primary Membrane Stress [σ_{Avg}]:

$$\begin{aligned} &= (fN + fS + fY) / AT \\ &= (28898.576 + 558163 + 711237) / 18275.24 \\ &= 71.047 \text{ MPa} \end{aligned}$$

General Primary Membrane Stress [σ_{Circ}]:

$$\begin{aligned} &= P \cdot Rxs / teff \\ &= 0.388 \cdot 3570.782 / 39.625 \\ &= 35.0 \text{ MPa} \end{aligned}$$

Maximum Local Primary Membrane Stress [PL]:

$$\begin{aligned} &= \max(2 \cdot \sigma_{Avg} - \sigma_{Circ}, \sigma_{Circ}) \\ &= \max(2 \cdot 71.047 - 34.964, 34.964) \\ &= 107.1 \text{ MPa} \end{aligned}$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	207.00	MPa
Local Primary Membrane Stress	PL	107.13	MPa
Maximum Allowable Working Pressure	Pmax	0.75	MPa

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$\begin{aligned} &= (Rnc + tn) / Rnc \\ &= (513.4 + 20.0) / 513.4 \\ &= 1.039 \text{ For set-in Nozzles} \end{aligned}$$

Weld Length of Nozzle to Shell Weld [L_T]:

$$\begin{aligned} &= \pi/2 \cdot (Rn + tn) \\ &= \pi/2 \cdot (513.4 + 20.0) \\ &= 837.863 \text{ mm} \end{aligned}$$

Weld Length of Pad to Shell Weld [L_{TP}]:

$$\begin{aligned} &= \pi/2 \cdot (Rn + tn + W) \\ &= \pi/2 \cdot (513.4 + 20.0 + 350.0) \\ &= 1387.641 \text{ mm} \end{aligned}$$

Weld Throat Dimensions, (0.7071 • Leg Dimensions) [L_{41T} , L_{42T} , L_{43T}]:

$$= 11.314, 11.314, 0.000, \text{ mm}$$

Required Weld Throat Dimensions :

$$\begin{aligned} &tc, L_{41T} = 11.314, \text{ must be } \geq \min(0.7 \cdot 20.000, 6.000) \text{ Passed} \\ &tf1, L_{42T} = 11.314, \text{ must be } \geq \min(0.6 \cdot 16.000, 0.6 \cdot 25.000) \text{ Passed} \end{aligned}$$

Weld Load Value [fwlds]:

$$\begin{aligned} &= \min(fy \cdot ky, 1.5 \cdot Sn(A2 + A3), \pi/4 \cdot P \cdot Rn^2 \cdot ky^2) \\ &= \min(711237 \cdot 1.04, 1.5 \cdot 156.4(2846.623 + 0.0), \pi/4 \cdot 0.4 \cdot 513.4^2 \cdot 1.04^2) \\ &= 86694.375 \text{ N} \end{aligned}$$

Discontinuity Force [fws]:

$$\begin{aligned} &= fwlds \cdot t \cdot S / (t \cdot S + te \cdot Sp) \\ &= 86694.4 \cdot 25.0 \cdot 138 / (25.0 \cdot 138 + 16.0 \cdot 156) \\ &= 50247.906 \text{ N} \end{aligned}$$

Discontinuity Force [fwp]:

$$\begin{aligned} &= fwlds \cdot te \cdot Sp / (t \cdot S + te \cdot Sp) \\ &= 86694.4 \cdot 16.0 \cdot 156 / (25.0 \cdot 138 + 16.0 \cdot 156) \\ &= 36446.477 \text{ N} \end{aligned}$$

Shear Stress [τ_1]:

$$\begin{aligned} &= fws / (L_T \cdot (0.6 \cdot tw1 + 0.49 \cdot L_{43T})) \\ &= 50247.906 / (837.863 \cdot (0.6 \cdot 16.0 + 0.49 \cdot 0.0)) \\ &= 6.248 \text{ MPa} \end{aligned}$$

Shear Stress [τ_2]:

$$\begin{aligned} &= fwp / (L_T \cdot (0.6 \cdot tw2 + 0.49 \cdot L_{41T})) \\ &= 36446.477 / (837.863 \cdot (0.6 \cdot 16.0 + 0.49 \cdot 11.314)) \\ &= 2.873 \text{ MPa} \end{aligned}$$



Shear Stress [τ_3]:

$$\begin{aligned} &= fwp / (LTP \cdot (0.49 \cdot L42T)) \\ &= 36446.477 / (1387.641 \cdot (0.49 \cdot 11.314)) \\ &= 4.738 \text{ MPa} \end{aligned}$$

Maximum Shear Stress in the Welds:

$$\begin{aligned} &= \max(\tau_1, \tau_2, \tau_3) \\ &= \max(6.248, 2.873, 4.738) \\ &= 6.2 \text{ must be less than or equal to } 138.0 \text{ MPa} \end{aligned}$$

Nozzle Calculations per Section 4.5: External Pressure Case:

Nozzle Radius for Force Calculation [R_{xn}]:

$$\begin{aligned} &= t_n / \ln[1 + t_n/R_n] \\ &= 20.0 / \ln[1 + 20.0/513.4] \\ &= 523.337 \text{ mm} \end{aligned}$$

Shell Radius for Force Calculation [R_{xs}]:

$$\begin{aligned} &= t_{eff} / \ln[1 + t_{eff}/R_{eff}] \\ &= 39.625 / \ln[1 + 39.625/3551.0] \\ &= 3570.782 \text{ mm} \end{aligned}$$

Allowable Local Primary Membrane Stress [S_{allow}]:

$$\begin{aligned} &= 1.5 \cdot S \\ &= 1.5 \cdot 138.0 \\ &= 207.0 \text{ MPa} \end{aligned}$$

Force acting on the Nozzle [f_N]:

$$\begin{aligned} &= P \cdot R_{xn} \cdot L_H \\ &= 0.052 \cdot 523.337 \cdot 142.331 \\ &= 3853.1 \text{ N} \end{aligned}$$

Force acting on the Shell [f_S]:

$$\begin{aligned} &= P \cdot R_{xs} (L_R + t_n) \\ &= 0.052 \cdot 3570.782 (382.905 + 20.0) \\ &= 74421.2 \text{ N} \end{aligned}$$

Discontinuity Force from External Pressure [f_Y]:

$$\begin{aligned} &= P \cdot R_{xs} \cdot R_{nc} \\ &= 0.052 \cdot 3570.782 \cdot 513.4 \\ &= 94830.9 \text{ N} \end{aligned}$$

Area Resisting External Pressure [A_p]:

$$\begin{aligned} &= (f_S + f_Y + f_N) / P \\ &= (74421.2 + 94830.9 + 3853.1) / 0.052 \\ &= 3346411.5 \text{ mm}^2 \end{aligned}$$

Maximum Allowable Working Pressure Candidate [P_{max1}]:

$$\begin{aligned} &= S_{allow} / (2 \cdot A_p / AT - R_{xs} / t_{eff}) \\ &= 207.0 / (2 \cdot 3346412 / 18275.24 - 3570.782 / 39.625) \\ &= 0.7 \text{ MPa} \end{aligned}$$

Maximum Allowable Working Pressure Candidate [P_{max2}]:

$$\begin{aligned} &= S / R_{xs} \\ &= 138.0 / [25.0 / 3570.782] \\ &= 1.0 \text{ MPa} \end{aligned}$$

Maximum Allowable Working Pressure [P_{max}]:

$$\begin{aligned} &= \min(P_{max1}, P_{max2}) \\ &= \min(0.75, 0.966) \\ &= 0.750 \text{ MPa} \end{aligned}$$

Average Primary Membrane Stress [σ_{Avg}]:

$$\begin{aligned} &= (f_N + f_S + f_Y) / AT \\ &= (3853.116 + 74421.156 + 94830.922) / 18275.24 \\ &= 9.473 \text{ MPa} \end{aligned}$$

General Primary Membrane Stress [σ_{Circ}]:

$$\begin{aligned} &= P \cdot R_{xs} / t_{eff} \\ &= 0.052 \cdot 3570.782 / 39.625 \end{aligned}$$



= 4.7 MPa

Maximum Local Primary Membrane Stress [PL]:

= max(2 • σ_{Avg} - σ_{Circ} , σ_{Circ})
 = max(2 • 9.473 - 4.662, 4.662)
 = 14.3 MPa

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	207.00	MPa
Local Primary Membrane Stress	PL	14.28	MPa
Maximum Allowable External Pressure	Pmax	0.75	MPa

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Nozzle Neck to Pad Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Nozzle Neck to Pad Weld for Reinf. pad (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Shell to Pad Weld at Pad OD for pad (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Gov. MDMT of the Nozzle	:	-12 °C
Gov. MDMT of the Reinforcement Pad	:	-12 °C

Flange MDMT including temperature reduction per 3.11:

Flange MDMT with Temp reduction	-104 °C
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The Drop for this Nozzle is : 40.3014 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 376.3014 mm

Input Echo, WRC297 Item 1, Description: H1 42" :

Diameter Basis for Cylindrical Shell	ID		
Shell Internal Corrosion Allowance	1.0000	mm	
Shell External Corrosion Allowance	0.0000	mm	
Shell Diameter	7100.000	mm	
Shell Thickness	26.0000	mm	
Shell Stress Concentration Factor	1.000		
Vessel Material	SA-516 70		
Vessel Cold S.I. Allowable	Smc	138.00	MPa
Vessel Hot S.I. Allowable	Smh	138.00	MPa
Diameter Basis for Nozzle	ID		
Nozzle Corrosion Allowance	6.0000	mm	
Nozzle Diameter	1014.800	mm	
Nozzle Thickness	26.0000	mm	
Nozzle Stress Concentration Factor	1.000		
Nozzle Material	SA-516 70		
Nozzle Cold S.I. Allowable	SNmc	175.00	MPa
Nozzle Hot S.I. Allowable	SNmh	156.40	MPa

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Diameter of Reinforcing Pad	1766.800	mm
Thickness of Reinforcing Pad	16.0000	mm



Note:

*External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUstained loads.*

Design Internal Pressure	Dp	0.39	MPa
Radial Load	P	-78750.05	N
Circumferential Shear	Vc	52500.19	N
Longitudinal Shear	Vl	52500.19	N
Circumferential Moment	Mc	0.154E+09	N-mm
Longitudinal Moment	Ml	0.154E+09	N-mm
Torsional Moment	Mt	0.221E+09	N-mm

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - ca)}$$

$$= 1066.8 + 2 \cdot 1.65 \cdot \sqrt{3571.5(26.0 - 1.0)}$$

$$= 2052.874 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	7143.000 mm
Nozzle Outside Diameter	(d) =	1066.800 mm
Vessel Thickness used	(T) =	41.000 mm
Nozzle Thickness used	(t) =	20.000 mm
T / t	=	2.050
d / t	=	53.340

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.971$$

Nr/P	=	0.046
Mr/P	=	0.089
M0/P	=	0.050
N0/P	=	0.154
MrD/Mc	=	0.187
NrDL/Mc	=	0.089
M0d/Mc	=	0.120
N0DL/Mc	=	0.210
MrD/Ml	=	0.103
NrDL/Ml	=	0.055
M0D/Ml	=	0.056
N0DL/Ml	=	0.296

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	7	7	7	7
Outplane Bending (P)	13	-13	13	-13
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (Ml)	-25	-25	25	25
Outplane Bending (Ml)	-29	29	29	-29
Normal Pressure Stress	33	33	33	33
Outplane Stress Summary	0	30	109	23

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	2	2	2	2



Inplane Bending (P)	24	-24	24	-24
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-4	-4	4	4
Inplane Bending (ML)	-53	53	53	-53
Inplane Pressure Stress	16	16	16	16

Inplane Stress Summary	-14	42	101	-54
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Vessel Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress normal to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	3	3	3	3

Shear Stress Summary	3	3	2	2
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Vessel Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Two * Max Shear Stress	16	43	109	78
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Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Outplane Membrane (P)	7	7	7	7
Outplane Bending (P)	13	-13	13	-13
Outplane Membrane (Mc)	-18	-18	18	18
Outplane Bending (Mc)	-62	62	62	-62
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	16	16	16	16

Outplane Stress Summary	-42	53	117	-33
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Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses parallel to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Inplane Membrane (P)	2	2	2	2
Inplane Bending (P)	24	-24	24	-24
Inplane Membrane (Mc)	-7	-7	7	7
Inplane Bending (Mc)	-96	96	96	-96
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	33	33	33	33

Inplane Stress Summary	-43	100	164	-78
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Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Torsional Shear (Mt)	3	3	3	3

Shear Stress Summary	2	2	3	3
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Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
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(Stress Intensities	Left Outside	Left Inside	Right Outside	Right Inside
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Two * Max Shear Stress	45	100	165	78
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Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
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Hoop Membrane (P)	7	7	7	7
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-25	-25	25	25
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	11	12	11	12

Hoop Stress Summary	-6	-6	44	44
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Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
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Axial Membrane (P)	1	1	1	1
Axial Bending (P)	77	-77	77	-77
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-9	-9	9	9
Axial Bending (ML)	-164	164	164	-164
Axial Pressure Stress	5	5	5	5

Axial Stress Summary	-88	84	257	-225
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Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
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Shear due to (Vc)	1	1	-1	-1
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	6	6	6	6

Shear Stress Summary	7	7	4	4
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Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stress Intensities	Au Outside	Al Inside	Bu Outside	Bl Inside
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Two * Max Shear Stress	89	91	258	270
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Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
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Hoop Membrane (P)	7	7	7	7
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-18	-18	18	18
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	11	12	11	12

Hoop Stress Summary	0	1	37	37
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Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
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Axial Membrane (P)	1	1	1	1
Axial Bending (P)	77	-77	77	-77
Axial Membrane (Mc)	-9	-9	9	9
Axial Bending (Mc)	-310	310	310	-310
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	5	5	5	5
Axial Stress Summary	-234	230	404	-371

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-1	-1	1	1
Shear due to Torsion	6	6	6	6
Shear Stress Summary	4	4	7	7

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	235	230	404	409

Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$
 $= 1066.8 + 2 \cdot 1.65 \cdot \sqrt{(3563.5(26.0 - 1.0))}$
 $= 2051.769 \text{ mm}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter (D)	=	7127.000 mm
Nozzle Outside Diameter (d)	=	1766.800 mm
Vessel Thickness used (T)	=	25.000 mm
Nozzle Thickness used (t)	=	370.000 mm
T / t	=	0.068
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 4.186$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

Nr/P	=	0.051
Mr/P	=	0.081
M0/P	=	0.022
N0/P	=	0.028
MrD/Mc	=	0.250
NrDL/Mc	=	0.107
M0d/Mc	=	0.075
N0DL/Mc	=	0.052
MrD/Ml	=	0.057
NrDL/Ml	=	0.061
M0D/Ml	=	0.017
N0DL/Ml	=	0.031

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	3	3	3	3
Outplane Bending (P)	16	-16	16	-16
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0



Outplane Membrane (ML)	-4	-4	4	4
Outplane Bending (ML)	-14	14	14	-14
Normal Pressure Stress	54	55	54	55
Outplane Stress Summary	56	52	94	31

Vessel Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	B1
(Stresses parallel to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Inplane Membrane (P)	6	6	6	6
Inplane Bending (P)	61	-61	61	-61
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-8	-8	8	8
Inplane Bending (ML)	-47	47	47	-47
Inplane Pressure Stress	27	27	27	27
Inplane Stress Summary	38	12	151	-66

Vessel Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	B1
(Shear stress normal to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	1	1	1	1
Shear Stress Summary	2	2	1	1

Vessel Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	B1
(Stress Intensities)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Two * Max Shear Stress	56	52	151	98

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	D1
(Stresses Normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Membrane (P)	3	3	3	3
Outplane Bending (P)	16	-16	16	-16
Outplane Membrane (Mc)	-7	-7	7	7
Outplane Bending (Mc)	-62	62	62	-62
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	27	27	27	27
Outplane Stress Summary	-21	69	117	-41

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	D1
(Stresses parallel to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside
Inplane Membrane (P)	6	6	6	6
Inplane Bending (P)	61	-61	61	-61
Inplane Membrane (Mc)	-14	-14	14	14
Inplane Bending (Mc)	-209	209	209	-209
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	54	55	54	55
Inplane Stress Summary	-101	195	346	-193

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)			Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Shear (Vc)			0	0	0	0
Outplane Shear (Vl)			0	0	0	0
Torsional Shear (Mt)			1	1	1	1
Shear Stress Summary			1	1	2	2

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stress Intensities)			Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress			101	195	346	193

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	33	33	33	33	33	33	33	33
Circ. Pl (SUS)	-18	-18	32	32	-5	-5	9	9
Circ. Q (SUS)	-16	16	42	-42	-72	72	120	-120
Long. Pm (SUS)	16	16	16	16	16	16	16	16
Long. Pl (SUS)	-2	-2	6	6	-11	-11	25	25
Long. Q (SUS)	-29	29	77	-77	-49	49	75	-75
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	3	3	3	3	3	3	3	3
Pm (SUS)	33.0	33.0	33.0	33.0	33.0	33.0	33.0	33.0
Pm+Pl (SUS)	15.0	15.0	65.0	65.0	28.0	28.0	42.0	42.0
Pm+Pl+Q (Total)	15.6	43.7	108.0	78.2	47.0	100.2	162.2	78.2

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	33.00	138.00	Passed
Pm+Pl (SUS)	65.00	207.00	Passed
Pm+Pl+Q (TOTAL)	162.19	414.00	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Reinforcing Pad Edge (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	54	55	54	55	54	55	54	55
Circ. Pl (SUS)	-1	-1	7	7	-8	-8	20	20
Circ. Q (SUS)	2	-2	30	-30	-148	148	270	-270
Long. Pm (SUS)	27	27	27	27	27	27	27	27
Long. Pl (SUS)	-2	-2	14	14	-4	-4	10	10
Long. Q (SUS)	14	-14	108	-108	-46	46	78	-78

Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear P1 (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	1	1	1	1	1	1	1	1
Pm (SUS)	54.0	55.0	54.0	55.0	54.0	55.0	54.0	55.0
Pm+P1 (SUS)	53.0	54.0	61.0	62.0	46.0	47.0	74.0	75.0
Pm+P1+Q (Total)	55.1	52.0	149.0	99.0	102.0	195.0	344.0	195.0

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	55.00	138.00	Passed
Pm+P1 (SUS)	75.00	207.00	Passed
Pm+P1+Q (TOTAL)	344.00	414.00	Passed

Because only sustained loads were specified, the Pm+P1+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	11	12	11	12	11	12	11	12
Circ. P1 (SUS)	-18	-18	32	32	-11	-11	25	25
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	5	5	5	5	5	5	5	5
Long. P1 (SUS)	-8	-8	10	10	-8	-8	10	10
Long. Q (SUS)	-87	87	241	-241	-233	233	387	-387
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear P1 (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	6	6	6	6	6	6	6	6
Pm (SUS)	11.0	12.0	11.0	12.0	11.0	12.0	11.0	12.0
Pm+P1 (SUS)	7.2	6.3	43.0	44.0	3.6	4.5	36.0	37.0
Pm+P1+Q (Total)	90.6	91.1	256.1	270.2	236.2	230.1	402.1	409.2

Nozzle Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	12.00	156.40	Passed
Pm+P1 (SUS)	44.03	234.60	Passed
Pm+P1+Q (TOTAL)	409.24	469.20	Passed

Because only sustained loads were specified, the Pm+P1+Q allowable was 3 * Smh.



Input, Nozzle Desc: H2 42"

From: 250

Pressure for Reinforcement Calculations	P	0.3880	MPa
Temperature for Internal Pressure	Temp	130	°C
Design External Pressure	Pext	0.05	MPa
Temperature for External Pressure	Tempex	130	°C

Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa

Inside Diameter of Cylindrical Shell	D	7100.00	mm
Design Length of Section	L	2500.0000	mm
Shell Finished (Minimum) Thickness	t	26.0000	mm
Shell Internal Corrosion Allowance	cas	1.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		47550.05	mm
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User Entered Minimum Design Metal Temperature		-12.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-516 70	+IT
Material UNS Number		K02700	
Material Specification/Type		Plate	
Allowable Stress at Temperature	Sn	156.40	MPa
Allowable Stress At Ambient	Sna	175.00	MPa

Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		39.9528	in.

Size and Thickness Basis		Actual	
Actual Thickness	tn	26.0000	mm

Flange Material		SA-105	+IT
Flange Type		Weld Neck	Flange

Corrosion Allowance	can	6.0000	mm
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Outside Projection	ho	310.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	16.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	16.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Pad Material		SA-516 70	+IT
Pad Allowable Stress at Temperature	Sp	156.40	MPa
Pad Allowable Stress At Ambient	Spa	175.00	MPa
Diameter of Pad along vessel surface	Dp	1824.4000	mm
Thickness of Pad	te	16.0000	mm
Weld leg size between Pad and Shell	Wp	16.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	16.0000	mm
Reinforcing Pad Width		378.8000	mm
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.

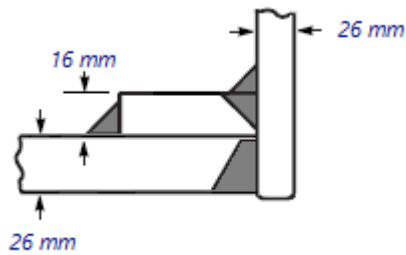
Flange Class		150
Flange Grade		GR 1.1
Flange Series		A

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: H2 42"

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation		39.953	in.
Actual Thickness Used in Calculation		1.024	in.



Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:
 $= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$
 $= 0.5 \cdot 1026.8 \cdot (\exp(0.388/(156.4 \cdot 1.0)) - 1) + 6.0 + 0.0$
 $= 1.2752 + 6.0 + 0.0 = 7.2752 \text{ mm Equation 4.3.1}$

Computed Maximum Allowable Working Pressure [MAWP]:
 $= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$
 $= 156.4 \cdot 1.0 \cdot \ln((2 \cdot 20.0 + 1026.8) / 1026.8) - 0.0$
 $= 5.977 - 0.0 = 5.977 \text{ MPa}$

This nozzle or its diameter limit intersects weld seam # [2] on this element. The distance between the nozzle and seam is, 217.56 mm , while the distance from the center of the nozzle to the edge of the diameter limit is: 916.30 mm

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c : 14.340 mm
 Min. required shell thickness, internal p + c : 10.998 mm
 Min. required shell thickness, external p + c : 12.582 mm

Nozzle neck minimum required thickness : 12.582 mm

Minimum nozzle neck thickness 26.010 mm , is greater than required thickness 12.582 mm

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	16.6,	Allowable	:	156.4 MPa	Passed
Expansion	:	0.0,	Allowable	:	397.6 MPa	Passed
Occasional	:	4.9,	Allowable	:	208.0 MPa	Passed
Shear	:	9.0,	Allowable	:	109.5 MPa	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:
 $= \min[S_n/S, 1]$
 $= \min[156.4/138.0, 1]$
 $= 1.000$

Pad Material Factor [frp]:
 $= \min[S_p/S, 1]$
 $= \min[156.4/138.0, 1]$
 $= 1.000$

Thickness of Nozzle [tn]:
 $= \text{thickness} - \text{corrosion allowance}$
 $= 26.0 - 6.0$
 $= 20.000 \text{ mm}$

Shell Diameter to Thickness ratio [D/t]:



$$\begin{aligned}
 &= D_i/t \\
 &= 7102.0/25.0 \\
 &= 284.080 \text{ must be less than } 400.
 \end{aligned}$$

Effective Pressure Radius [Reff]:

$$\begin{aligned}
 &= D_i/2 + \text{corrosion allowance} \\
 &= 7100.0/2 + 1.0 \\
 &= 3551.000 \text{ mm}
 \end{aligned}$$

Effective Length of Vessel Wall [LR]:

For Nozzles with reinforcing pads, Lr is min[LR1, LR2, LR3]:

Limit [LR1]:

$$\begin{aligned}
 &= \sqrt{(R_{eff} \cdot t) + W} \\
 &= \sqrt{(3551.0 \cdot 25.0) + 378.8} \\
 &= 676.751 \text{ mm}
 \end{aligned}$$

Limit [LR2]:

$$\begin{aligned}
 &= \sqrt{((R_{eff} + t) \cdot (t + t_e))} \\
 &= \sqrt{((3551.0 + 25.0) \cdot (25.0 + 16.0))} \\
 &= 382.905 \text{ mm}
 \end{aligned}$$

Limit [LR3]:

$$\begin{aligned}
 &= 2 \cdot R_n \\
 &= 2 \cdot 513.4 \\
 &= 1026.800 \text{ mm}
 \end{aligned}$$

Final value of [LR]:

$$\begin{aligned}
 &= \min(LR1, LR2, LR3) \\
 &= \min(676.751, 382.905, 1026.8) \\
 &= 382.905 \text{ mm}
 \end{aligned}$$

Thickness Limit Candidate [LH1]:

$$\begin{aligned}
 &= \min[1.5t, t_e] + \sqrt{(R_n \cdot t_n)} \\
 &= \min[1.5 \cdot 25.0, 16.0] + \sqrt{(513.4 \cdot 20.0)} \\
 &= 117.331 \text{ mm}
 \end{aligned}$$

Thickness Limit Candidate [LH2]:

$$\begin{aligned}
 &= L_{pr1} \\
 &= 310.0 \\
 &= 310.000 \text{ mm}
 \end{aligned}$$

Thickness Limit Candidate [LH3]:

$$\begin{aligned}
 &= 8(t + t_e) \\
 &= 8(25.0 + 16.0) \\
 &= 328.000 \text{ mm}
 \end{aligned}$$

Effective Nozzle Wall Length Outside the Vessel [LH]:

$$\begin{aligned}
 &= \min[LH1, LH2, LH3] + t \text{ for set-in nozzles} \\
 &= \min[117.331, 310.0, 328.0] + 25.0 \\
 &= 142.331 \text{ mm}
 \end{aligned}$$

Effective Vessel Thickness [teff]:

$$\begin{aligned}
 &= t + ((A5 \cdot frp) / LR) \\
 &= 25.0 + ((6060.8 \cdot 1.0) / 382.905) \\
 &= 40.828 \text{ mm}
 \end{aligned}$$

Parameter [λ]:

$$\begin{aligned}
 &= \min(12, (2 \cdot R_n + t_n) / (\sqrt{(D_i + t_{eff}) \cdot t_{eff}})) \\
 &= \min(12, (2 \cdot 513.4 + 20.0) / (\sqrt{(7102.0 + 40.828) \cdot 40.828})) \\
 &= 1.938
 \end{aligned}$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$\begin{aligned}
 &= t \cdot LR \cdot \max((\lambda/5)^{0.85}, 1) \\
 &= 25.0 \cdot 382.905 \cdot \max((1.938/5)^{0.85}, 1) \\
 &= 9572.617 \text{ mm}^2
 \end{aligned}$$



Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$\begin{aligned} &= t_n \cdot L_H \\ &= 20.0 \cdot 142.331 \\ &= 2846.623 \text{ mm}^2 \end{aligned}$$

Area Contributed by the Pad Fillet Weld [A42]:

$$\begin{aligned} &= 0.5 \cdot \text{Leg42}^2 - \text{Area Cut by Reinforcement Limit} \\ &= 0.5 \cdot 16.0^2 - 70.75 \\ &= 57.250 \text{ mm}^2 \end{aligned}$$

Area Contributed by the Outside Fillet Weld [A41]:

$$\begin{aligned} &= 0.5 \cdot \text{Leg41}^2 \\ &= 0.5 \cdot 16.0^2 \\ &= 128.000 \text{ mm}^2 \end{aligned}$$

Area Contributed by the Reinforcing Pad [A5]:

$$\begin{aligned} &= \min(W \cdot t_e, LR \cdot t_e) \\ &= \min(378.8 \cdot 16.0, 382.905 \cdot 16.0) \\ &= 6060.800 \text{ mm}^2 \end{aligned}$$

The total area contributed by A1 through A5 [AT]:

$$\begin{aligned} &= A1 + \text{frn}(A2 + A3) + A41 + A42 + A43 + \text{frp}(A5) \\ &= 9572.617 + 1.0(2846.623 + 0.0) + 128.0 + 57.25 + 0.0 + 1.0(6060.8) \\ &= 18665.291 \text{ mm}^2 \end{aligned}$$

Nozzle Radius for Force Calculation [Rxn]:

$$\begin{aligned} &= t_n / \ln[1 + t_n/R_n] \\ &= 20.0 / \ln[1 + 20.0/513.4] \\ &= 523.337 \text{ mm} \end{aligned}$$

Shell Radius for Force Calculation [Rxs]:

$$\begin{aligned} &= t_{eff} / \ln[1 + t_{eff}/R_{eff}] \\ &= 40.828 / \ln[1 + 40.828/3551.0] \\ &= 3571.378 \text{ mm} \end{aligned}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$\begin{aligned} &= 1.5 \cdot S \cdot E \\ &= 1.5 \cdot 138.0 \cdot 1.0 \\ &= 207.0 \text{ MPa} \end{aligned}$$

Force acting on the Nozzle [fN]:

$$\begin{aligned} &= P \cdot R_{xn} \cdot L_H \\ &= 0.388 \cdot 523.337 \cdot 142.331 \\ &= 28898.6 \text{ N} \end{aligned}$$

Force acting on the Shell [fS]:

$$\begin{aligned} &= P \cdot R_{xs}(LR + t_n) \\ &= 0.388 \cdot 3571.378(382.905 + 20.0) \\ &= 558255.8 \text{ N} \end{aligned}$$

Discontinuity Force from Internal Pressure [fY]:

$$\begin{aligned} &= P \cdot R_{xs} \cdot R_{nc} \\ &= 0.388 \cdot 3571.378 \cdot 513.4 \\ &= 711355.6 \text{ N} \end{aligned}$$

Area Resisting Internal Pressure [Ap]:

$$\begin{aligned} &= (f_S + f_Y + f_N) / P \\ &= (558256 + 711356 + 28898.6) / 0.388 \\ &= 3346957.0 \text{ mm}^2 \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$\begin{aligned} &= S_{allow} / (2 \cdot A_p / AT - R_{xs} / t_{eff}) \\ &= 207.0 / (2 \cdot 3346957 / 18665.291 - 3571.378 / 40.828) \\ &= 0.8 \text{ MPa} \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$\begin{aligned} &= S / t / R_{xs} \\ &= 138.0 / 25.0 / 3571.378 \\ &= 1.0 \text{ MPa} \end{aligned}$$



Maximum Allowable Working Pressure [Pmax]:

$$\begin{aligned} &= \min(P_{max1}, P_{max2}) \\ &= \min(0.763, 0.966) \\ &= 0.763 \text{ MPa} \end{aligned}$$

Average Primary Membrane Stress [σ_{Avg}]:

$$\begin{aligned} &= (f_N + f_S + f_Y) / A_T \\ &= (28898.576 + 558256 + 711356) / 18665.291 \\ &= 69.574 \text{ MPa} \end{aligned}$$

General Primary Membrane Stress [σ_{Circ}]:

$$\begin{aligned} &= P \cdot R_{xs} / t_{eff} \\ &= 0.388 \cdot 3571.378 / 40.828 \\ &= 33.9 \text{ MPa} \end{aligned}$$

Maximum Local Primary Membrane Stress [PL]:

$$\begin{aligned} &= \max(2 \cdot \sigma_{Avg} - \sigma_{Circ}, \sigma_{Circ}) \\ &= \max(2 \cdot 69.574 - 33.939, 33.939) \\ &= 105.2 \text{ MPa} \end{aligned}$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	207.00	MPa
Local Primary Membrane Stress	PL	105.21	MPa
Maximum Allowable Working Pressure	Pmax	0.76	MPa

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$\begin{aligned} &= (R_{nc} + t_n) / R_{nc} \\ &= (513.4 + 20.0) / 513.4 \\ &= 1.039 \text{ For set-in Nozzles} \end{aligned}$$

Weld Length of Nozzle to Shell Weld [LT]:

$$\begin{aligned} &= \pi/2 \cdot (R_n + t_n) \\ &= \pi/2 \cdot (513.4 + 20.0) \\ &= 837.863 \text{ mm} \end{aligned}$$

Weld Length of Pad to Shell Weld [LTp]:

$$\begin{aligned} &= \pi/2 \cdot (R_n + t_n + W) \\ &= \pi/2 \cdot (513.4 + 20.0 + 378.8) \\ &= 1432.880 \text{ mm} \end{aligned}$$

Weld Throat Dimensions, (0.7071•Leg Dimensions) [L41T, L42T, L43T]:

$$= 11.314, 11.314, 0.000, \text{ mm}$$

Required Weld Throat Dimensions :

$$\begin{aligned} t_c, L41T &= 11.314, \text{ must be } \geq \min(0.7 \cdot 20.000, 6.000) \text{ Passed} \\ t_{f1}, L42T &= 11.314, \text{ must be } \geq \min(0.6 \cdot 16.000, 0.6 \cdot 25.000) \text{ Passed} \end{aligned}$$

Weld Load Value [fwelds]:

$$\begin{aligned} &= \min(f_y \cdot k_y, 1.5 \cdot S_n(A_2 + A_3), \pi/4 \cdot P \cdot R_n^2 \cdot k_y^2) \\ &= \min(711356 \cdot 1.04, 1.5 \cdot 156.4(2846.623 + 0.0), \pi/4 \cdot 0.4 \cdot 513.4^2 \cdot 1.04^2) \\ &= 86694.375 \text{ N} \end{aligned}$$

Discontinuity Force [fws]:

$$\begin{aligned} &= f_{welds} \cdot t \cdot S / (t \cdot S + t_e \cdot S_p) \\ &= 86694.4 \cdot 25.0 \cdot 138 / (25.0 \cdot 138 + 16.0 \cdot 156) \\ &= 50247.906 \text{ N} \end{aligned}$$

Discontinuity Force [fwp]:

$$\begin{aligned} &= f_{welds} \cdot t_e \cdot S_p / (t \cdot S + t_e \cdot S_p) \\ &= 86694.4 \cdot 16.0 \cdot 156 / (25.0 \cdot 138 + 16.0 \cdot 156) \\ &= 36446.477 \text{ N} \end{aligned}$$

Shear Stress [τ_1]:

$$\begin{aligned} &= f_{ws} / (L_T \cdot (0.6 \cdot t_{w1} + 0.49 \cdot L_{43T})) \\ &= 50247.906 / (837.863 \cdot (0.6 \cdot 16.0 + 0.49 \cdot 0.0)) \\ &= 6.248 \text{ MPa} \end{aligned}$$

Shear Stress [τ_2]:



$$\begin{aligned} &= fwp / (Lr \cdot (0.6 \cdot tw2 + 0.49 \cdot L41T)) \\ &= 36446.477 / (837.863 \cdot (0.6 \cdot 16.0 + 0.49 \cdot 11.314)) \\ &= 2.873 \text{ MPa} \end{aligned}$$

Shear Stress [τ_3]:

$$\begin{aligned} &= fwp / (LrP \cdot (0.49 \cdot L42T)) \\ &= 36446.477 / (1432.88 \cdot (0.49 \cdot 11.314)) \\ &= 4.589 \text{ MPa} \end{aligned}$$

Maximum Shear Stress in the Welds:

$$\begin{aligned} &= \max(\tau_1, \tau_2, \tau_3) \\ &= \max(6.248, 2.873, 4.589) \\ &= 6.2 \text{ must be less than or equal to } 138.0 \text{ MPa} \end{aligned}$$

Nozzle Calculations per Section 4.5: External Pressure Case:

Nozzle Radius for Force Calculation [R_{xn}]:

$$\begin{aligned} &= tn / \ln[1 + tn/Rn] \\ &= 20.0 / \ln[1 + 20.0/513.4] \\ &= 523.337 \text{ mm} \end{aligned}$$

Shell Radius for Force Calculation [R_{xs}]:

$$\begin{aligned} &= teff / \ln[1 + teff/Reff] \\ &= 40.828 / \ln[1 + 40.828/3551.0] \\ &= 3571.378 \text{ mm} \end{aligned}$$

Allowable Local Primary Membrane Stress [S_{allow}]:

$$\begin{aligned} &= 1.5 \cdot S \\ &= 1.5 \cdot 138.0 \\ &= 207.0 \text{ MPa} \end{aligned}$$

Force acting on the Nozzle [f_N]:

$$\begin{aligned} &= P \cdot R_{xn} \cdot LH \\ &= 0.052 \cdot 523.337 \cdot 142.331 \\ &= 3853.1 \text{ N} \end{aligned}$$

Force acting on the Shell [f_S]:

$$\begin{aligned} &= P \cdot R_{xs} (LR + tn) \\ &= 0.052 \cdot 3571.378 (382.905 + 20.0) \\ &= 74433.6 \text{ N} \end{aligned}$$

Discontinuity Force from External Pressure [f_Y]:

$$\begin{aligned} &= P \cdot R_{xs} \cdot R_{nc} \\ &= 0.052 \cdot 3571.378 \cdot 513.4 \\ &= 94846.7 \text{ N} \end{aligned}$$

Area Resisting External Pressure [A_p]:

$$\begin{aligned} &= (f_S + f_Y + f_N) / P \\ &= (74433.6 + 94846.7 + 3853.1) / 0.052 \\ &= 3346957.2 \text{ mm}^2 \end{aligned}$$

Maximum Allowable Working Pressure Candidate [P_{max1}]:

$$\begin{aligned} &= S_{allow} / (2 \cdot A_p / AT - R_{xs} / teff) \\ &= 207.0 / (2 \cdot 3346957 / 18665.291 - 3571.378 / 40.828) \\ &= 0.8 \text{ MPa} \end{aligned}$$

Maximum Allowable Working Pressure Candidate [P_{max2}]:

$$\begin{aligned} &= S / [t / R_{xs}] \\ &= 138.0 [25.0 / 3571.378] \\ &= 1.0 \text{ MPa} \end{aligned}$$

Maximum Allowable Working Pressure [P_{max}]:

$$\begin{aligned} &= \min(P_{max1}, P_{max2}) \\ &= \min(0.763, 0.966) \\ &= 0.763 \text{ MPa} \end{aligned}$$

Average Primary Membrane Stress [σ_{Avg}]:

$$\begin{aligned} &= (f_N + f_S + f_Y) / AT \\ &= (3853.116 + 74433.57 + 94846.734) / 18665.291 \\ &= 9.276 \text{ MPa} \end{aligned}$$



General Primary Membrane Stress [σ_{Circ}]:

$$= P \cdot R_s / t_{eff}$$

$$= 0.052 \cdot 3571.378 / 40.828$$

$$= 4.5 \text{ MPa}$$

Maximum Local Primary Membrane Stress [PL]:

$$= \max(2 \cdot \sigma_{Avg} - \sigma_{Circ}, \sigma_{Circ})$$

$$= \max(2 \cdot 9.276 - 4.525, 4.525)$$

$$= 14.0 \text{ MPa}$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	207.00	MPa
Local Primary Membrane Stress	PL	14.03	MPa
Maximum Allowable External Pressure	Pmax	0.76	MPa

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Nozzle Neck to Pad Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Nozzle Neck to Pad Weld for Reinf. pad (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Shell to Pad Weld at Pad OD for pad (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Gov. MDMT of the Nozzle	:	-12 °C
Gov. MDMT of the Reinforcement Pad	:	-12 °C

Flange MDMT including temperature reduction per 3.11:

Flange MDMT with Temp reduction	-104 °C
---------------------------------	---------

The Drop for this Nozzle is : 40.3014 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 376.3014 mm

Input Echo, WRC297 Item 1, Description: H2 42" :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	1.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	7100.000	mm
Shell Thickness	26.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc	138.00 MPa
Vessel Hot S.I. Allowable	Smh	138.00 MPa

Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	6.0000	mm
Nozzle Diameter	1014.800	mm
Nozzle Thickness	26.0000	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-516 70	
Nozzle Cold S.I. Allowable	SNmc	175.00 MPa
Nozzle Hot S.I. Allowable	SNmh	156.40 MPa

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.



Diameter of Reinforcing Pad 1824.400 mm
 Thickness of Reinforcing Pad 16.0000 mm

Note:

*External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUSTAINED loads.*

Design Internal Pressure	Dp	0.39	MPa
Radial Load	P	-78750.05	N
Circumferential Shear	Vc	52500.19	N
Longitudinal Shear	Vl	52500.19	N
Circumferential Moment	Mc	0.154E+09	N-mm
Longitudinal Moment	Ml	0.154E+09	N-mm
Torsional Moment	Mt	0.221E+09	N-mm

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - c_a))}$
 $= 1066.8 + 2 \cdot 1.65 \cdot \sqrt{(3571.5(26.0 - 1.0))}$
 $= 2052.874 \text{ mm}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	7143.000 mm
Nozzle Outside Diameter	(d) =	1066.800 mm
Vessel Thickness used	(T) =	41.000 mm
Nozzle Thickness used	(t) =	20.000 mm
T / t	=	2.050
d / t	=	53.340

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.971$$

Nr/P	=	0.046
Mr/P	=	0.089
M0/P	=	0.050
N0/P	=	0.154
MrD/Mc	=	0.187
NrDL/Mc	=	0.089
M0d/Mc	=	0.120
N0DL/Mc	=	0.210
MrD/Ml	=	0.103
NrDL/Ml	=	0.055
M0D/Ml	=	0.056
N0DL/Ml	=	0.296

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	7	7	7	7
Outplane Bending (P)	13	-13	13	-13
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (Ml)	-25	-25	25	25
Outplane Bending (Ml)	-29	29	29	-29
Normal Pressure Stress	33	33	33	33
Outplane Stress Summary	0	30	109	23

Vessel Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
--------------------	----	----	----	----



(Stresses parallel to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Inplane Membrane (P)	2	2	2	2
Inplane Bending (P)	24	-24	24	-24
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-4	-4	4	4
Inplane Bending (ML)	-53	53	53	-53
Inplane Pressure Stress	16	16	16	16
Inplane Stress Summary	-14	42	101	-54

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	3	3	3	3
Shear Stress Summary	3	3	2	2

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	16	43	109	78

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	7	7	7	7
Outplane Bending (P)	13	-13	13	-13
Outplane Membrane (Mc)	-18	-18	18	18
Outplane Bending (Mc)	-62	62	62	-62
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	16	16	16	16
Outplane Stress Summary	-42	53	117	-33

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Inplane Membrane (P)	2	2	2	2
Inplane Bending (P)	24	-24	24	-24
Inplane Membrane (Mc)	-7	-7	7	7
Inplane Bending (Mc)	-96	96	96	-96
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	33	33	33	33
Inplane Stress Summary	-43	100	164	-78

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Torsional Shear (Mt)	3	3	3	3



Shear Stress Summary 2 2 3 3

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stress Intensities	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	45	100	165	78

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au	Al	Bu	Bl
	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside
Hoop Membrane (P)	7	7	7	7
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-25	-25	25	25
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	11	12	11	12
Hoop Stress Summary	-6	-6	44	44

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au	Al	Bu	Bl
	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside
Axial Membrane (P)	1	1	1	1
Axial Bending (P)	77	-77	77	-77
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-9	-9	9	9
Axial Bending (ML)	-164	164	164	-164
Axial Pressure Stress	5	5	5	5
Axial Stress Summary	-88	84	257	-225

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Shear stress)	Au	Al	Bu	Bl
	Outside	Inside	Outside	Inside
Shear due to (Vc)	1	1	-1	-1
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	6	6	6	6
Shear Stress Summary	7	7	4	4

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stress Intensities	Au	Al	Bu	Bl
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	89	91	258	270

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Hoop Membrane (P)	7	7	7	7
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-18	-18	18	18
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	11	12	11	12
Hoop Stress Summary	0	1	37	37



Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	1	1	1	1
Axial Bending (P)	77	-77	77	-77
Axial Membrane (Mc)	-9	-9	9	9
Axial Bending (Mc)	-310	310	310	-310
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	5	5	5	5
Axial Stress Summary	-234	230	404	-371

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-1	-1	1	1
Shear due to Torsion	6	6	6	6
Shear Stress Summary	4	4	7	7

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	235	230	404	409

Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - c_a)}$
 $= 1066.8 + 2 \cdot 1.65 \cdot \sqrt{3563.5(26.0 - 1.0)}$
 $= 2051.769 \text{ mm}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter (D) = 7127.000 mm
 Nozzle Outside Diameter (d) = 1824.400 mm
 Vessel Thickness used (T) = 25.000 mm
 Nozzle Thickness used (t) = 398.800 mm
 $T / t = 0.063$
 $d / t = 10.000$

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 4.322$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

Nr/P = 0.050
 Mr/P = 0.078
 M0/P = 0.022
 N0/P = 0.027
 MrD/Mc = 0.248
 NrDL/Mc = 0.106
 M0d/Mc = 0.074
 N0DL/Mc = 0.050
 MrD/Ml = 0.055
 NrDL/Ml = 0.059
 M0D/Ml = 0.017
 N0DL/Ml = 0.029

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
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Outplane Membrane (P)	3	3	3	3
Outplane Bending (P)	16	-16	16	-16
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-3	-3	3	3
Outplane Bending (ML)	-13	13	13	-13
Normal Pressure Stress	54	55	54	55

Outplane Stress Summary	57	51	92	32
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Vessel Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses parallel to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Inplane Membrane (P)	6	6	6	6
Inplane Bending (P)	59	-59	59	-59
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-7	-7	7	7
Inplane Bending (ML)	-44	44	44	-44
Inplane Pressure Stress	27	27	27	27

Inplane Stress Summary	40	11	145	-62
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Vessel Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress normal to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	1	1	1	1

Shear Stress Summary	2	2	0	0
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Vessel Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Two * Max Shear Stress	57	52	145	94

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Membrane (P)	3	3	3	3
Outplane Bending (P)	16	-16	16	-16
Outplane Membrane (Mc)	-6	-6	6	6
Outplane Bending (Mc)	-60	60	60	-60
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	27	27	27	27

Outplane Stress Summary	-19	67	114	-38
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Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses parallel to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside
Inplane Membrane (P)	6	6	6	6
Inplane Bending (P)	59	-59	59	-59
Inplane Membrane (Mc)	-14	-14	14	14
Inplane Bending (Mc)	-200	200	200	-200
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	54	55	54	55

Inplane Stress Summary -94 189 335 -184

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)		Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Shear (Vc)		0	0	0	0
Outplane Shear (Vl)		0	0	0	0
Torsional Shear (Mt)		1	1	1	1

Shear Stress Summary 0 0 2 2

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stress Intensities)		Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
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Two * Max Shear Stress 94 189 335 184

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	33	33	33	33	33	33	33	33
Circ. Pl (SUS)	-18	-18	32	32	-5	-5	9	9
Circ. Q (SUS)	-16	16	42	-42	-72	72	120	-120
Long. Pm (SUS)	16	16	16	16	16	16	16	16
Long. Pl (SUS)	-2	-2	6	6	-11	-11	25	25
Long. Q (SUS)	-29	29	77	-77	-49	49	75	-75
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	3	3	3	3	3	3	3	3
Pm (SUS)	33.0	33.0	33.0	33.0	33.0	33.0	33.0	33.0
Pm+Pl (SUS)	15.0	15.0	65.0	65.0	28.0	28.0	42.0	42.0
Pm+Pl+Q (Total)	15.6	43.7	108.0	78.2	47.0	100.2	162.2	78.2

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	33.00	138.00	Passed
Pm+Pl (SUS)	65.00	207.00	Passed
Pm+Pl+Q (TOTAL)	162.19	414.00	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Reinforcing Pad Edge (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	54	55	54	55	54	55	54	55
Circ. Pl (SUS)	0	0	6	6	-8	-8	20	20
Circ. Q (SUS)	3	-3	29	-29	-141	141	259	-259

Long. Pm (SUS)	27	27	27	27	27	27	27	27
Long. Pl (SUS)	-1	-1	13	13	-3	-3	9	9
Long. Q (SUS)	15	-15	103	-103	-44	44	76	-76
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	1	1	1	1	1	1	1	1
Pm (SUS)	54.0	55.0	54.0	55.0	54.0	55.0	54.0	55.0
Pm+Pl (SUS)	54.0	55.0	60.0	61.0	46.0	47.0	74.0	75.0
Pm+Pl+Q (Total)	57.1	52.0	143.0	95.0	95.0	188.0	333.0	184.0

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	55.00	138.00	Passed
Pm+Pl (SUS)	75.00	207.00	Passed
Pm+Pl+Q (TOTAL)	333.00	414.00	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	11	12	11	12	11	12	11	12
Circ. Pl (SUS)	-18	-18	32	32	-11	-11	25	25
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	5	5	5	5	5	5	5	5
Long. Pl (SUS)	-8	-8	10	10	-8	-8	10	10
Long. Q (SUS)	-87	87	241	-241	-233	233	387	-387
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	6	6	6	6	6	6	6	6
Pm (SUS)	11.0	12.0	11.0	12.0	11.0	12.0	11.0	12.0
Pm+Pl (SUS)	7.2	6.3	43.0	44.0	3.6	4.5	36.0	37.0
Pm+Pl+Q (Total)	90.6	91.1	256.1	270.2	236.2	230.1	402.1	409.2

Nozzle Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	12.00	156.40	Passed
Pm+Pl (SUS)	44.03	234.60	Passed
Pm+Pl+Q (TOTAL)	409.24	469.20	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.



Input, Nozzle Desc: F 42"

From: 270

Pressure for Reinforcement Calculations	P	0.3840	MPa
Temperature for Internal Pressure	Temp	130	°C
Design External Pressure	Pext	0.05	MPa
Temperature for External Pressure	Tempex	130	°C

Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	156.40	MPa
Parent Allowable Stress At Ambient	Sva	175.00	MPa

Inside Diameter of Elliptical Head	D	7100.00	mm
Aspect Ratio of Elliptical Head	Ar	2.00	
Head Finished (Minimum) Thickness	t	16.0000	mm
Head Internal Corrosion Allowance	cas	1.0000	mm
Head External Corrosion Allowance	co	0.0000	mm

Distance from Head Centerline	L1	0.0000	mm
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User Entered Minimum Design Metal Temperature		-12.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-516 70	+IT
Material UNS Number		K02700	
Material Specification/Type		Plate	
Allowable Stress at Temperature	Sn	156.40	MPa
Allowable Stress At Ambient	Sna	175.00	MPa

Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		40.1102	in.

Size and Thickness Basis		Actual	
Actual Thickness	tn	24.0000	mm

Flange Material		SA-105	+IT
Flange Type		Weld Neck	Flange

Corrosion Allowance	can	6.0000	mm
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Outside Projection	ho	500.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	16.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	14.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Pad Material		SA-516 70	+IT
Pad Allowable Stress at Temperature	Sp	156.40	MPa
Pad Allowable Stress At Ambient	Spa	175.00	MPa
Diameter of Pad along vessel surface	Dp	2126.8000	mm
Thickness of Pad	te	16.0000	mm
Weld leg size between Pad and Shell	Wp	16.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	16.0000	mm
Reinforcing Pad Width		530.0000	mm
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.

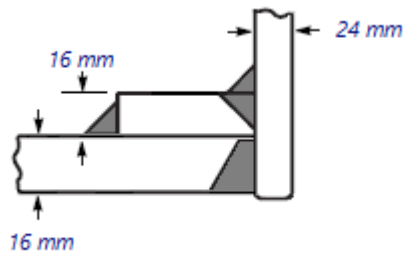
Flange Class		150
Flange Grade		GR 1.1
Flange Series		B

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: F 42"

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	40.110	in.
Actual Thickness Used in Calculation	0.945	in.



Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:
 $= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$
 $= 0.5 \cdot 1030.8 \cdot (\exp(0.384/(156.4 \cdot 1.0)) - 1) + 6.0 + 0.0$
 $= 1.267 + 6.0 + 0.0 = 7.267 \text{ mm Equation 4.3.1}$

Computed Maximum Allowable Working Pressure [MAWP]:
 $= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$
 $= 156.4 \cdot 1.0 \cdot \ln((2 \cdot 18.0 + 1030.8) / 1030.8) - 0.0$
 $= 5.369 - 0.0 = 5.369 \text{ MPa}$

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c :	14.340 mm
Min. required shell thickness, internal p + c :	13.515 mm
Min. required shell thickness, external p + c :	13.986 mm

Nozzle neck minimum required thickness	: 13.986 mm
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Minimum nozzle neck thickness 24.003 mm , is greater than required thickness 13.986 mm

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	18.3,	Allowable	:	156.4 MPa	Passed
Expansion	:	0.0,	Allowable	:	395.9 MPa	Passed
Occasional	:	5.4,	Allowable	:	208.0 MPa	Passed
Shear	:	9.9,	Allowable	:	109.5 MPa	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:
 $= \min[S_n/S, 1]$
 $= \min[156.4/156.4, 1]$
 $= 1.000$

Pad Material Factor [frp]:
 $= \min[S_p/S, 1]$
 $= \min[156.4/156.4, 1]$
 $= 1.000$

Thickness of Nozzle [tn]:
 $= \text{thickness} - \text{corrosion allowance}$
 $= 24.0 - 6.0$
 $= 18.000 \text{ mm}$

Effective Pressure Radius [Reff]:
 $= 0.9/6 \cdot D_i [2 + (D_i/(2h))^2]$
 $= 0.9/6 \cdot 7102.0 [2 + (7102.0/(2 \cdot 1776.0))^2]$
 $= 6389.400 \text{ mm}$

Effective Length of Vessel Wall [LR]:

For Nozzles with reinforcing pads, Lr is min[LR1, LR2, LR3]:

$$\begin{aligned}\text{Limit [LR1]:} \\ &= \sqrt{(\text{Reff} \cdot t) + W} \\ &= \sqrt{(6389.4 \cdot 15.0) + 530.0} \\ &= 839.582 \text{ mm}\end{aligned}$$

$$\begin{aligned}\text{Limit [LR2]:} \\ &= \sqrt{((\text{Reff} + t) \cdot (t + t_e))} \\ &= \sqrt{((6389.4 + 15.0) \cdot (15.0 + 16.0))} \\ &= 445.574 \text{ mm}\end{aligned}$$

$$\begin{aligned}\text{Limit [LR3]:} \\ &= 2 \cdot R_n \\ &= 2 \cdot 515.4 \\ &= 1030.800 \text{ mm}\end{aligned}$$

$$\begin{aligned}\text{Final value of [LR]:} \\ &= \min(\text{LR1}, \text{LR2}, \text{LR3}) \\ &= \min(839.582, 445.574, 1030.8) \\ &= 445.574 \text{ mm}\end{aligned}$$

$$\begin{aligned}\text{Intermediate value [Cn]:} \\ &= \min(((t + t_e)/t_n)^{0.35}, 1) \\ &= \min(((15.0 + 16.0)/18.0)^{0.35}, 1) \\ &= \min(1.21, 1) \\ &= 1.000\end{aligned}$$

$$\begin{aligned}\text{Intermediate value [Cp]:} \\ &= \exp((0.35 \cdot d_i - X_o)/(16t)) \\ &= \exp((0.35 \cdot 7102.0 - 533.4)/(16 \cdot 15.0)) \\ &= 3410.392\end{aligned}$$

$$\begin{aligned}\text{Effective Nozzle Wall Length Outside the Vessel [LH]:} \\ &= \min(t + t_e + F_p \cdot \sqrt{(R_n \cdot t_n)}, L_{pr1} + t) \\ &= \min(15.0 + 16.0 + 1.0 \cdot \sqrt{(515.4 \cdot 18.0)}, 500.0 + 15.0) \\ &= 127.318 \text{ mm}\end{aligned}$$

$$\begin{aligned}\text{Effective Vessel Thickness [teff]:} \\ &= t + ((A_5 \cdot \text{frp}) / \text{LR}) \\ &= 15.0 + ((7129.188 \cdot 1.0) / 445.574) \\ &= 31.000 \text{ mm}\end{aligned}$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

$$\begin{aligned}\text{Area Contributed by the Vessel Wall [A1]:} \\ &= t \cdot \text{LR} \\ &= 15.0 \cdot 445.574 \\ &= 6683.613 \text{ mm}^2\end{aligned}$$

$$\begin{aligned}\text{Area Contributed by the Nozzle Outside the Vessel Wall [A2]:} \\ &= t_n \cdot \text{LH} \\ &= 18.0 \cdot 127.318 \\ &= 2291.728 \text{ mm}^2\end{aligned}$$

$$\begin{aligned}\text{Area Contributed by the Outside Fillet Weld [A41]:} \\ &= 0.5 \cdot \text{Leg}^2 \\ &= 0.5 \cdot 16.0^2 \\ &= 128.000 \text{ mm}^2\end{aligned}$$

$$\begin{aligned}\text{Area Contributed by the Reinforcing Pad [A5]:} \\ &= \min(W \cdot t_e, \text{LR} \cdot t_e) \\ &= \min(530.0 \cdot 16.0, 445.574 \cdot 16.0) \\ &= 7129.188 \text{ mm}^2\end{aligned}$$

$$\begin{aligned}\text{The total area contributed by A1 through A5 [AT]:} \\ &= A_1 + \text{frn}(A_2 + A_3) + A_{41} + A_{42} + A_{43} + \text{frp}(A_5) \\ &= 6683.613 + 1.0(2291.728 + 0.0) + 128.0 + 0.0 + 0.0 + 1.0(7129.188)\end{aligned}$$



$$= 16232.530 \text{ mm}^2$$

Nozzle Radius for Force Calculation [Rxn]:

$$\begin{aligned} &= t_n / \ln[1 + t_n/R_n] \\ &= 18.0 / \ln[1 + 18.0/515.4] \\ &= 524.349 \text{ mm} \end{aligned}$$

Shell Radius for Force Calculation [Rxs]:

$$\begin{aligned} &= t_{eff} / \ln[1 + t_{eff}/R_{eff}] \\ &= 31.0 / \ln[1 + 31.0/6389.4] \\ &= 6404.845 \text{ mm} \end{aligned}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$\begin{aligned} &= 1.5 \cdot S \cdot E \\ &= 1.5 \cdot 156.4 \cdot 1.0 \\ &= 234.6 \text{ MPa} \end{aligned}$$

Force acting on the Nozzle [fN]:

$$\begin{aligned} &= P \cdot R_{xn} \cdot L_H \\ &= 0.384 \cdot 524.349 \cdot 127.318 \\ &= 25633.4 \text{ N} \end{aligned}$$

Force acting on the Shell [fS]:

$$\begin{aligned} &= P \cdot R_{xs} \cdot (L_R + t_n) / 2 \\ &= 0.384 \cdot 6404.845 \cdot (445.574 + 18.0) / 2 \\ &= 570022.9 \text{ N} \end{aligned}$$

Discontinuity Force from Internal Pressure [fY]:

$$\begin{aligned} &= (P \cdot R_{xs} \cdot R_{nc}) / 2 \\ &= (0.384 \cdot 6404.845 \cdot 515.4) / 2 \\ &= 633749.2 \text{ N} \end{aligned}$$

Area Resisting Internal Pressure [Ap]:

$$\begin{aligned} &= (f_S + f_Y + f_N) / P \\ &= (570023 + 633749 + 25633.4) / 0.384 \\ &= 3201848.5 \text{ mm}^2 \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$\begin{aligned} &= S_{allow} / (2 \cdot A_p / A_T - R_{xs} / (2 \cdot t_{eff})) \\ &= 234.6 / (2 \cdot 3201849 / 16232.53 - 6404.845 / (2 \cdot 31.0)) \\ &= 0.8 \text{ MPa} \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$\begin{aligned} &= 2 \cdot S / [t / R_{xs}] \\ &= 2 \cdot 156.4 [15.0 / 6404.845] \\ &= 0.7 \text{ MPa} \end{aligned}$$

Maximum Allowable Working Pressure [Pmax]:

$$\begin{aligned} &= \min(P_{max1}, P_{max2}) \\ &= \min(0.806, 0.733) \\ &= 0.733 \text{ MPa} \end{aligned}$$

Average Primary Membrane Stress [σAvg]:

$$\begin{aligned} &= (f_N + f_S + f_Y) / A_T \\ &= (25633.367 + 570023 + 633749) / 16232.53 \\ &= 75.744 \text{ MPa} \end{aligned}$$

General Primary Membrane Stress [σCirc]:

$$\begin{aligned} &= P \cdot R_{xs} / (2 \cdot t_{eff}) \\ &= 0.384 \cdot 6404.845 / (2 \cdot 31.0) \\ &= 39.7 \text{ MPa} \end{aligned}$$

Maximum Local Primary Membrane Stress [PL]:

$$\begin{aligned} &= \max(2 \cdot \sigma_{Avg} - \sigma_{Circ}, \sigma_{Circ}) \\ &= \max(2 \cdot 75.744 - 39.669, 39.669) \\ &= 111.8 \text{ MPa} \end{aligned}$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	234.60	MPa
Local Primary Membrane Stress	PL	111.82	MPa



Maximum Allowable Working Pressure Pmax 0.73 MPa

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$\begin{aligned} &= (R_{nc} + t_n) / R_{nc} \\ &= (515.4 + 18.0) / 515.4 \\ &= 1.035 \text{ For set-in Nozzles} \end{aligned}$$

Weld Length of Nozzle to Shell Weld [LT]:

$$\begin{aligned} &= \pi/2 \cdot (R_n + t_n) \\ &= \pi/2 \cdot (515.4 + 18.0) \\ &= 837.863 \text{ mm} \end{aligned}$$

Weld Length of Pad to Shell Weld [LTP]:

$$\begin{aligned} &= \pi/2 \cdot (R_n + t_n + W) \\ &= \pi/2 \cdot (515.4 + 18.0 + 530.0) \\ &= 1670.385 \text{ mm} \end{aligned}$$

Weld Throat Dimensions, (0.7071 • Leg Dimensions) [L41T, L42T, L43T]:

$$= 11.314, 11.314, 0.000, \text{ mm}$$

Required Weld Throat Dimensions :

$$\begin{aligned} t_c, L41T &= 11.314, \text{ must be } \geq \min(0.7 \cdot 18.000, 6.000) \text{ Passed} \\ t_f1, L42T &= 11.314, \text{ must be } \geq \min(0.6 \cdot 16.000, 0.6 \cdot 15.000) \text{ Passed} \end{aligned}$$

Weld Load Value [fwelds]:

$$\begin{aligned} &= \min(f_y \cdot k_y, 1.5 \cdot S_n(A_2 + A_3), \pi/4 \cdot P \cdot R_n^2 \cdot k_y^2) \\ &= \min(633749 \cdot 1.03, 1.5 \cdot 156.4(2291.728 + 0.0), \pi/4 \cdot 0.4 \cdot 515.4^2 \cdot 1.03^2) \\ &= 85800.609 \text{ N} \end{aligned}$$

Discontinuity Force [fws]:

$$\begin{aligned} &= fwelds \cdot t \cdot S / (t \cdot S + t_e \cdot S_p) \\ &= 85800.6 \cdot 15.0 \cdot 156 / (15.0 \cdot 156 + 16.0 \cdot 156) \\ &= 41516.426 \text{ N} \end{aligned}$$

Discontinuity Force [fwp]:

$$\begin{aligned} &= fwelds \cdot t_e \cdot S_p / (t \cdot S + t_e \cdot S_p) \\ &= 85800.6 \cdot 16.0 \cdot 156 / (15.0 \cdot 156 + 16.0 \cdot 156) \\ &= 44284.184 \text{ N} \end{aligned}$$

Shear Stress [τ1]:

$$\begin{aligned} &= fws / (L_T \cdot (0.6 \cdot t_{w1} + 0.49 \cdot L_{43T})) \\ &= 41516.426 / (837.863 \cdot (0.6 \cdot 14.0 + 0.49 \cdot 0.0)) \\ &= 5.899 \text{ MPa} \end{aligned}$$

Shear Stress [τ2]:

$$\begin{aligned} &= fwp / (L_T \cdot (0.6 \cdot t_{w2} + 0.49 \cdot L_{41T})) \\ &= 44284.184 / (837.863 \cdot (0.6 \cdot 16.0 + 0.49 \cdot 11.314)) \\ &= 3.490 \text{ MPa} \end{aligned}$$

Shear Stress [τ3]:

$$\begin{aligned} &= fwp / (L_{TP} \cdot (0.49 \cdot L_{42T})) \\ &= 44284.184 / (1670.385 \cdot (0.49 \cdot 11.314)) \\ &= 4.783 \text{ MPa} \end{aligned}$$

Maximum Shear Stress in the Welds:

$$\begin{aligned} &= \max(\tau_1, \tau_2, \tau_3) \\ &= \max(5.899, 3.49, 4.783) \\ &= 5.9 \text{ must be less than or equal to } 156.4 \text{ MPa} \end{aligned}$$

Nozzle Calculations per Section 4.5: External Pressure Case:

Nozzle Radius for Force Calculation [Rxn]:

$$\begin{aligned} &= t_n / \ln[1 + t_n/R_n] \\ &= 18.0 / \ln[1 + 18.0/515.4] \\ &= 524.349 \text{ mm} \end{aligned}$$

Shell Radius for Force Calculation [Rxs]:

$$= t_{eff} / \ln[1 + t_{eff}/R_{eff}]$$



$$= 31.0 / \ln[1 + 31.0 / 6389.4]$$
$$= 6404.845 \text{ mm}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$= 1.5 \cdot S$$
$$= 1.5 \cdot 156.4$$
$$= 234.6 \text{ MPa}$$

Force acting on the Nozzle [fN]:

$$= P \cdot R_{xn} \cdot L_H$$
$$= 0.052 \cdot 524.349 \cdot 127.318$$
$$= 3453.4 \text{ N}$$

Force acting on the Shell [fS]:

$$= P \cdot R_{xs} \cdot (L_R + t_n) / 2$$
$$= 0.052 \cdot 6404.845 \cdot (445.574 + 18.0) / 2$$
$$= 76794.2 \text{ N}$$

Discontinuity Force from External Pressure [fY]:

$$= (P \cdot R_{xs} \cdot R_{nc}) / 2$$
$$= (0.052 \cdot 6404.845 \cdot 515.4) / 2$$
$$= 85379.5 \text{ N}$$

Area Resisting External Pressure [Ap]:

$$= (f_S + f_Y + f_N) / P$$
$$= (76794.2 + 85379.5 + 3453.4) / 0.052$$
$$= 3201848.5 \text{ mm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$= S_{allow} / (2 \cdot A_p / A_T - R_{xs} / (2 \cdot t_{eff}))$$
$$= 234.6 / (2 \cdot 3201849 / 16232.53 - 6404.845 / (2 \cdot 31.0))$$
$$= 0.8 \text{ MPa}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$= 2 \cdot S / [t / R_{xs}]$$
$$= 2 \cdot 156.4 [15.0 / 6404.845]$$
$$= 0.7 \text{ MPa}$$

Maximum Allowable Working Pressure [Pmax]:

$$= \min(P_{max1}, P_{max2})$$
$$= \min(0.806, 0.733)$$
$$= 0.733 \text{ MPa}$$

Average Primary Membrane Stress [σ_{Avg}]:

$$= (f_N + f_S + f_Y) / A_T$$
$$= (3453.362 + 76794.25 + 85379.539) / 16232.53$$
$$= 10.204 \text{ MPa}$$

General Primary Membrane Stress [σ_{Circ}]:

$$= P \cdot R_{xs} / (2 \cdot t_{eff})$$
$$= 0.052 \cdot 6404.845 / (2 \cdot 31.0)$$
$$= 5.3 \text{ MPa}$$

Maximum Local Primary Membrane Stress [PL]:

$$= \max(2 \cdot \sigma_{Avg} - \sigma_{Circ}, \sigma_{Circ})$$
$$= \max(2 \cdot 10.204 - 5.344, 5.344)$$
$$= 15.1 \text{ MPa}$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	234.60	MPa
Local Primary Membrane Stress	PL	15.06	MPa
Maximum Allowable External Pressure	Pmax	0.73	MPa

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Nozzle Neck to Pad Weld for the Nozzle (Impact tested) :



Note: This Element/Detail was specified as being Impact Tested.

Nozzle Neck to Pad Weld for Reinf. pad (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Shell to Pad Weld at Pad OD for pad (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Gov. MDMT of the Nozzle : -12 °C
 Gov. MDMT of the Reinforcement Pad : -12 °C

Flange MDMT including temperature reduction per 3.11:
 Flange MDMT with Temp reduction -104 °C

The Drop for this Nozzle is : 22.2568 mm
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 538.2570 mm

Input Echo, WRC107/537 Item 1, Description: F 42" :

Diameter Basis for Vessel	Vbasis	ID	
Cylindrical or Spherical Vessel	Cylsph	Spherical	
Internal Corrosion Allowance	Cas	1.0000	mm
Vessel Diameter	Dv	12780.000	mm
Vessel Thickness	Tv	16.000	mm
Design Temperature	T1	130.0	C
Vessel Material		SA-516 70	
Vessel UNS Number		K02700	
Vessel Cold S.I. Allowable	Smc	175.00	MPa
Vessel Hot S.I. Allowable	Smh	156.40	MPa
Attachment Type	Type	Round	
WRC107 Attachment Classification	Holsol	Hollow	
Diameter Basis for Nozzle	Nbasis	ID	
Corrosion Allowance for Nozzle	Can	6.0000	mm
Nozzle Diameter	Dn	1018.800	mm
Nozzle Thickness	Tn	24.000	mm
Nozzle Material		SA-516 70	
Nozzle UNS Number		K02700	
Nozzle Cold S.I. Allowable	SNmc	175.00	MPa
Nozzle Hot S.I. Allowable	SNmh	156.40	MPa
Thickness of Reinforcing Pad	Tpad	16.000	mm
Diameter of Reinforcing Pad	Dpad	2126.800	mm
Design Internal Pressure	Dp	0.384	MPa
Include Pressure Thrust		No	

External Forces and Moments in WRC 107/537 Convention:

Radial Load (SUS)	P	-78750.1	N
Longitudinal Shear (SUS)	(V1) V1	52500.2	N
Circumferential Shear (SUS)	(Vc) V2	52500.2	N
Circumferential Moment (SUS)	(Mc) M1	154357792.0	N-mm
Longitudinal Moment (SUS)	(Ml) M2	154357792.0	N-mm
Torsional Moment (SUS)	Mt	220511344.0	N-mm

Use Interactive Control No
 WRC107 Version Version March 1979

Include Pressure Stress Indices per Div. 2 No



Rad. Bend. P	3.7	-3.7	3.7	-3.7	3.7	-3.7	3.7	-3.7
Rad. Memb. MC	0.0	0.0	0.0	0.0	-10.5	-10.5	10.5	10.5
Rad. Bend. MC	0.0	0.0	0.0	0.0	-26.4	26.4	26.4	-26.4
Rad. Memb. ML	-10.5	-10.5	10.5	10.5	0.0	0.0	0.0	0.0
Rad. Bend. ML	-26.4	26.4	26.4	-26.4	0.0	0.0	0.0	0.0
Tot. Rad. Str.	-31.5	13.9	42.3	-17.9	-31.5	13.9	42.3	-17.9
Tang. Memb. P	14.5	14.5	14.5	14.5	14.5	14.5	14.5	14.5
Tang. Bend. P	9.1	-9.1	9.1	-9.1	9.1	-9.1	9.1	-9.1
Tang. Memb. MC	0.0	0.0	0.0	0.0	-74.5	-74.5	74.5	74.5
Tang. Bend. MC	0.0	0.0	0.0	0.0	-79.9	79.9	79.9	-79.9
Tang. Memb. ML	-74.5	-74.5	74.5	74.5	0.0	0.0	0.0	0.0
Tang. Bend. ML	-79.9	79.9	79.9	-79.9	0.0	0.0	0.0	0.0
Tot. Tang. Str.	-130.8	10.7	178.0	0.1	-130.8	10.7	178.0	0.1
Shear VC	1.0	1.0	-1.0	-1.0	0.0	0.0	0.0	0.0
Shear VL	0.0	0.0	0.0	0.0	-1.0	-1.0	1.0	1.0
Shear MT	4.0	4.0	4.0	4.0	4.0	4.0	4.0	4.0
Tot. Shear	5.0	5.0	3.0	3.0	3.0	3.0	5.0	5.0
Str. Int.	131.1	17.5	178.1	18.9	130.9	15.6	178.2	20.6

Unitless Prm: U = 3.43 TAU = 0.00 (58.58) RHO = 0.00 (0.83)

Dimensionless Loads for Spherical Shells at Pad edge:

Curves read for 1979	Figure	Value	Location
N(x) * T / P	SR 2	0.02175	(A,B,C,D)
M(x) / P	SR 2	0.00990	(A,B,C,D)
N(x) * T * SQRT(Rm * T) / MC	SR 3	0.01691	(A,B,C,D)
M(x) * SQRT(Rm * T) / MC	SR 3	0.01053	(A,B,C,D)
N(x) * T * SQRT(Rm * T) / ML	SR 3	0.01691	(A,B,C,D)
M(x) * SQRT(Rm * T) / ML	SR 3	0.01053	(A,B,C,D)
N(y) * T / P	SR 2	0.00650	(A,B,C,D)
M(y) / P	SR 2	0.00300	(A,B,C,D)
N(y) * T * SQRT(Rm * T) / MC	SR 3	0.00507	(A,B,C,D)
M(y) * SQRT(Rm * T) / MC	SR 3	0.00317	(A,B,C,D)
N(y) * T * SQRT(Rm * T) / ML	SR 3	0.00507	(A,B,C,D)
M(y) * SQRT(Rm * T) / ML	SR 3	0.00317	(A,B,C,D)

Stress Concentration Factors: Kn = 1.00, Kb = 1.00

Stresses in the Vessel at the Edge of Reinforcing Pad (MPa)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Memb. P		7.6	7.6	7.6	7.6	7.6	7.6	7.6	7.6
Rad. Bend. P		20.8	-20.8	20.8	-20.8	20.8	-20.8	20.8	-20.8
Rad. Memb. MC		0.0	0.0	0.0	0.0	-37.4	-37.4	37.4	37.4
Rad. Bend. MC		0.0	0.0	0.0	0.0	-139.8	139.8	139.8	-139.8
Rad. Memb. ML		-37.4	-37.4	37.4	37.4	0.0	0.0	0.0	0.0
Rad. Bend. ML		-139.8	139.8	139.8	-139.8	0.0	0.0	0.0	0.0
Tot. Rad. Str.		-148.9	89.2	205.7	-115.6	-148.9	89.2	205.7	-115.6
Tang. Memb. P		2.3	2.3	2.3	2.3	2.3	2.3	2.3	2.3
Tang. Bend. P		6.3	-6.3	6.3	-6.3	6.3	-6.3	6.3	-6.3
Tang. Memb. MC		0.0	0.0	0.0	0.0	-11.2	-11.2	11.2	11.2
Tang. Bend. MC		0.0	0.0	0.0	0.0	-42.2	42.2	42.2	-42.2
Tang. Memb. ML		-11.2	-11.2	11.2	11.2	0.0	0.0	0.0	0.0
Tang. Bend. ML		-42.2	42.2	42.2	-42.2	0.0	0.0	0.0	0.0
Tot. Tang. Str.		-44.8	26.9	62.0	-35.0	-44.8	26.9	62.0	-35.0

Vessel Stress Summation at Attachment Junction (MPa)

Vessel Stress Summation Comparison (MPa):

The Pm+Pl+Q allowable above was based on the rules of ASME VIII-2 Part 4, 4.1.6.3 (a)&(b) or per ASME VIII-2, 5.5.6.1(1)&(2) Part 5 as applicable.

The P_m+P_l+Q allowable was based on a temperature range cycling from ambient to design temperature.

Vessel Stress Summation at Reinforcing Pad Edge (MPa)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad.	Pm (SUS)	39.7	39.7	39.7	39.7	39.7	39.7	39.7	39.7
Rad.	Pl (SUS)	-29.8	-29.8	45.1	45.1	-29.8	-29.8	45.1	45.1
Rad.	Q (SUS)	-119.0	119.0	160.6	-160.6	-119.0	119.0	160.6	-160.6
Long.	Pm (SUS)	39.7	39.7	39.7	39.7	39.7	39.7	39.7	39.7
Long.	Pl (SUS)	-8.9	-8.9	13.5	13.5	-8.9	-8.9	13.5	13.5
Long.	Q (SUS)	-35.9	35.9	48.5	-48.5	-35.9	35.9	48.5	-48.5
Shear	Pm (SUS)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0



Shear P1 (SUS)	1.0	1.0	-1.0	-1.0	-1.0	-1.0	1.0	1.0
Shear Q (SUS)	2.1	2.1	2.1	2.1	2.1	2.1	2.1	2.1
Pm (SUS)	39.7	39.7	39.7	39.7	39.7	39.7	39.7	39.7
Pm+P1 (SUS)	30.8	30.8	84.8	84.8	30.8	30.8	84.8	84.8
Pm+P1+Q (Total)	109.3	129.0	245.4	80.6	109.2	128.9	245.4	80.8

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	39.68	156.40	Passed
Pm+P1 (SUS)	84.77	234.60	Passed
Pm+P1+Q (TOTAL)	245.42	469.20	Passed

The Pm+P1+Q allowable above was based on the rules of ASME VIII-2 Part 4, 4.1.6.3 (a)&(b) or per ASME VIII-2, 5.5.6.1(1)&(2) Part 5 as applicable.

The Pm+P1+Q allowable was based on a temperature range cycling from ambient to design temperature.

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Noted issues for the current model:

Listed below are the known problem areas for the current design. If one or more of the design flags are turned on, please re-run the analysis. Some of these issues may be resolved when using updated input values.

**** Warning: One or more nozzles do not meet proper Spacing Requirements!**

**** Warning: The computed vessel MAWP seems to be less than the design pressure!**

Please review all reports carefully!

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